

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 7 PREFABRICATED CONCRETE CONSTRUCTION

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Section 7) gives recommendations regarding modular planning, component sizes, prefabrication systems, design considerations, joints and manufacture, storage, transportation and erection of prefabricated concrete elements for use in buildings and such related requirements for prefabricated concrete.

Prefabrication, though desirable for large-scale building activities, has been gaining importance in the country over the past decade. Two key aspects of prefabrication that need to be considered are the selection of an appropriate system for different categories of buildings and the sizing of their components. In this context, the principle of modular coordination is of significant value and its use is recommended.

Advantages of recent trends in prefabrication have been taken note of and also the hazards attended to such construction. Some of the essential requirements for the manufacture of the prefabricated components and elements are also included in this section.

Since the aim of prefabrication is to effect economy and improve quality and speed of construction, the selection of proper materials for prefabrication is also an important factor in the popularization of this technique. The use of locally available materials with required characteristics and the materials which, due to their innate characteristics such as lightweight, easy workability, thermal insulation, non-combustibility, etc, that effect economy and improve quality, may also be tried. However, this Section pertains to prefabricated elements with cementitious materials.

The design of prefabricated buildings shall include provision for installations of all services and required piping, wiring and accessories to be installed in the building.

This Section was first published in 1970 and was subsequently revised in 1983, 2005 and 2016 as a chapter of National Building Code of India. In the first revision in 1983, the main changes made included: inclusion of brief provision regarding importance of architectural treatment and finishes as applicable to prefabricated buildings; addition of a brief clause on the requirements of materials for use in prefabrication; elaboration of the clause on prefabricating systems and structural elements; revision of the clause on testing of components to include testing of structure or part of structure; and addition of a brief clause on the manufacture of cellular concrete.

In the second revision in 2005, this Section, earlier named as 'Prefabrication and systems building', was renamed and restructured as follows:

Section 7 Prefabrication, systems buildings and mixed/composite construction

Subsection 7A Prefabricated concrete

Subsection 7B Systems buildings and mixed/composite construction

In the second revision in 2005, several significant changes were introduced to enhance the standard. The modular coordination and modular dimensions of components were revised to allow greater flexibility in planning. Provisions related to tolerances were updated to address various types of prefabricated components. Additionally, a detailed clause on design requirements was incorporated to ensure the safety of prefabricated buildings against progressive collapse. Furthermore, a new clause specifying the sampling procedure for testing components was added.

In the third revision in 2016, in the Subsections 7A and 7B, several significant modifications had been incorporated. Definitions for new terms had been added and existing terminologies had been modified where necessary. Provisions related to prefabricated systems had been updated and detailed guidelines on the diaphragm action of floor systems had been included. The emulative system had been categorized and elaborated on, covering its definition, analysis, design and detailing. Design considerations had been revised to account for accidental impacts due to vehicles. Provisions and testing procedures for the water tightness of joints had been introduced, along with illustrations of typical precast joint details. Fire resistance testing of prototypes under sustained loads had been addressed and reliance solely on frictional resistance for connections between adjacent prefabricated

members had been replaced with recommendations for resistance using shear. Updates have also been made to provisions on site prefabrication and vertical stacking guidelines, including restrictions on the number of components stacked on the ground. The time duration for load testing selected components from a lot had been modified. Additionally, new provisions for tolerances during erection and associated design considerations had been added. An annex had been included to address common defects, their causes and remedies, covering issues such as dimensional deviations, cracks, honeycombing, damages, strand slippage and alignment problems.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

In this revision, it has been decided to merge subsections 7A and 7B into Section 7, 'Prefabricated Concrete Construction'. The following major modifications have been incorporated in this revision:

- a) Definitions of some new terms have been added and existing terminologies modified, wherever required;
- b) The emulative design approach has been adopted wherein precast elements are connected at site using wet concrete or grout;
- c) Emulative system has been categorized and detailed with respect to its definition, analysis, design, detailing, etc;
- d) Provisions relating to prefabricated systems namely moment resisting frame system, braced frame system, load bearing wall system, cell system and hybrid system are given along with their types and their applications;
- e) Provisions relating to the design of corbel, bearing including dapped bearing are given in detail;
- f) Provisions relating to quality control and assurance have been included;
- g) Provisions for tolerances in erection and associated design considerations have been updated;
- h) Provisions and testing procedures for water tightness of joints have been included;
- j) A guideline for 3D printed concrete structures has been included in [Annex B](#);
- k) A comprehensive guideline on prefabricated prefinished volumetric construction (PPVC) methods, materials, design standards and performance criteria for guidance to building professionals and stakeholders involved in modular construction has been included in [Annex C](#);
- m) Guidelines for design for manufacturing and assembly has been included in [Annex D](#);
- n) [Annex E](#) on common precast connection has been modified based on the current practices in precast industry; and
- p) Sustainability in precast concrete construction has been included in [Annex G](#).

This information contained in this section is largely based on IS 15916 : 2026, 'Prefabricated concrete construction — Code of practice (*second revision*)' (*under publication*).

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

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Important Explanatory Note for Users of the NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to ‘**accepted standard**’ in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Section. For example:

- a) Accepted standard [C-7(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part C/Section 7, that is, IS 13920 : 2016 ‘Ductile design and detailing of reinforced concrete structures subjected to seismic forces — Code of practice (*first revision*)’.
- b) Accepted standard [C-7(9)] refers to the Indian Standard(s) given at serial number (9) of the list of standards given at the end of this Part C/Section 7, that is, IS 16172 : 2023 ‘Reinforcement couplers for mechanical splices of steel bars in concrete — Specification (*first revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 7 PREFABRICATED CONCRETE CONSTRUCTION

1 SCOPE

1.1 This NBCS (Part C/Section 7) covers provisions for prefabricated concrete construction in buildings, including prefabrication systems, design considerations, element tolerances, joints and requirements for manufacture, storage, transportation and erection.

1.2 This Section is limited to the emulative detailing approach, wherein precast elements are connected at site using wet concrete or grout with reinforcement detailing in accordance with this Section and relevant design standards.

1.3 Structural analysis shall consider appropriate computational models, including connections, and the overall performance shall be verified in terms of strength, stiffness, and ductility. The design intent shall be to achieve performance comparable to cast in-situ concrete construction. The design and detailing provisions specified for cast in-situ concrete in good practice [C-7(1)] shall form the basis, and the design engineer shall provide similar detailing to achieve equivalent behaviour compare to cast in situ practice. If required by the contract statement, this can be demonstrated by appropriate testing as specified by the Competent Authority and [12](#) in agreement with the design team and client.

1.4 Where exact replication of cast in-situ detailing is not feasible, the minimum requirements specified in this Section shall be followed. In cases not covered by this Section, alternative detailing approaches may be adopted, provided their performance is validated through testing (in terms of strength, stiffness, and ductility) as specified and approved by the competent authority in agreement with the design team and client.

2 —

NOTE — This clause is intentionally left blank.

3 TERMINOLOGY

For the purpose of this section, the following definitions shall apply.

3.1 Authority Having Jurisdiction — The authority which has been created by a statute and which, for the purpose of administering the NBCS/part, may authorize a committee or an official or an agency to act on its behalf; hereinafter called the ‘Authority’.

3.2 Basic Module — The fundamental module used in modular coordination, the size of which is selected for general application to building and its components.

NOTE — The value of the basic module has been chosen as 50 mm for the maximum flexibility and convenience. The symbol for the basic module is *M*.

3.3 Cellular Concrete — The material consisting of an inorganic binder (such as lime or cement or both) in combination with a finely ground material containing siliceous material (such as sand), gas generating material (for example, aluminium powder), water and harmless additives (optional); and/or steam cured under high pressure in autoclaves.

3.4 Components — A building product formed as a distinct unit having specified sizes in three dimensions.

3.5 Composite Members — Structural members comprising of prefabricated structural units of steel, prestressed concrete or reinforced concrete and cast in-situ concrete connected together in such a manner that they act monolithically.

3.6 Diaphragm — It is a horizontal or nearly horizontal structural system (for example, reinforced concrete floors and horizontal bracing systems), which transmits lateral forces to vertical elements that resist earthquake-induced inertia effects.

3.7 Increments — Difference between two homologous dimensions of components of successive sizes.

3.8 Light-Weight Concrete — Concrete of substantially lower unit weight ($1\,400\text{ kg/m}^3$ to $1\,800\text{ kg/m}^3$) than that made with gravel or crushed stone.

3.9 Module — A unit of size used in dimensional coordination.

3.10 Modular Coordination — Dimensional coordination employing the basic module or a multimodule.

NOTE — The purposes of modular co-ordination are:

- a) to reduce the variety of component sizes produced; and
- b) to allow the building designer greater flexibility in the arrangement of components.

3.11 Modular Grid — A rectangular coordinate reference system in which the distance between consecutive lines is the basic module or a multimodule. This multimodule may differ for each of the two dimensions of the grid.

3.12 Multimodule — A module whose size is a selected multiple of the basic module.

3.13 Prefabricate — To fabricate concrete components or assembled units prior to erection or installation in a building.

3.14 Prefabricated Building — The partly/fully assembled and erected building, of which the structural parts consist of prefabricated individual units or assemblies using ordinary or controlled materials, including service facilities; and in which the service equipment may be either prefabricated or constructed in-situ.

3.15 Sandwich Concrete Panels — Panels made by sandwiching an insulation material between two layers of reinforced/prestressed concrete to act as insulation for composite or non-composite concrete panels.

3.16 Self Compacting Concrete — Concrete that is able to flow (650 mm in diameter) under its own weight and completely fill the voids within the formwork, even in the presence of dense reinforcement without any vibration, whilst maintaining homogeneity without segregation.

3.17 Shear Connectors — Structural elements, such as anchors, studs, channels, loops and spirals, intended to transmit the shear between the prefabricated member and the cast in-situ concrete and also to prevent separation at the interface (to transmit the interfacial shear).

3.18 System — It is a particular method of construction of buildings with certain order and discipline using the prefabricated components which are inter-related in functions and are produced based on a set of instructions.

3.19 Unit — Building component formed as a simple article with all three dimensions specified, complete in itself but intended to be part of a larger, compound unit or complete building. Examples are brick, block, floor panel, wall panel and other individual components.

3.20 Emulative Detailing System — A connection detailing system for precast concrete structures that provides a similar level of ductility equivalent to that of a conventionally designed, cast in-situ, monolithic concrete structure subjected to earthquake.

3.21 Jointed Detailing System — A connection detailing system for precast concrete structures that has individual precast components separated from each other but connected using special connections such as welded or bolted plates. These connections are not allowed as per this Section, if adopted appropriate testing needs to be carried out as per [10](#).

3.22 Terminology Relating to 3D Printing

3.22.1 Extrusion-Based Concrete 3D Printing — Extrusion based printing refers to the delivery of the concrete through a nozzle in the required shape and dimension, in a layer-by-layer deposition process. The nozzle can either be placed on a gantry or attached to a central robot or a rail mounted mechanism.

3.22.2 Binder Jetting-Based 3D Printing — Binder jetting is a 3D printing technology in which a liquid binding agent (water) is sprayed on the surface of a thin layer of the binder (cement and aggregate) to attain the required shape and dimensions. A new layer of the binder is then placed on the top of the existing layer, and the process of spraying the water is repeated. This process is continued until reaching the required height of the object. The formed object is allowed to harden before removing the excess unreacted binder.

3.22.3 Print Plane — The plane on which the material is deposited.

3.22.4 Directions in the Printed Specimens — The performance of the printed specimen varies with the principal directions of the printed concrete. The directions of the printed concrete are designated as X, Y and Z with reference to the print pathway, where:

- a) X direction is parallel to the print direction and in the print plane;
- b) Y direction is perpendicular to the print direction in the print plane and could cross different layers; and
- c) Z direction is perpendicular to the print direction and print plane, crossing many printed layers.

3.22.5 Concrete for 3D Printing — Concrete is a composite material with cement (ordinary Portland cement blended with or without supplementary cementitious materials), water, fine aggregate, coarse aggregate and admixtures. Fine aggregates and coarse aggregate need to comply with the requirements of good practices [C- 7(2)] and [C-7(3)]. Aggregates can be used in the mix subjected to testing and validation, confirming to the required printable properties like pumpability, buildability and extrudability.

3.22.6 Printable Concrete — Printable concrete is a mixture of cement (with or without supplementary cementitious materials), water, aggregates and admixtures that can be extruded through the nozzle of the printer and attain the designed dimension within certain tolerance zone on placing additional layers.

3.22.7 Buildability — Buildability is the ability of the printed layers to develop adequate green strength to withstand the self-weight as well as load due to subsequent layers without significant geometrical deformation or collapse of the freshly printed component.

3.22.8 Extrudability — Extrudability is the ability of the mixture to pass through the nozzle under pressure and attain the designed shape after the extrusion.

3.22.9 Pumpability — Pumpability is the ease of transporting concrete mixture from the pump to the nozzle.

3.22.10 Printability — The printability of the mixture is a combination of the pumpability and extrudability.

3.22.11 Open Time — Open time represents the time over which the 3D printing mixture remains printable. This defines the time period in which the mixture can be delivered through the pumping system and dispensed through the nozzle without clogging of the extrusion system.

3.22.12 Filament — The extruded shape is termed as the filament. The cross section of the filament can be circular, rectangular, or square, depending on the shape of the nozzle. The filament is characterised by its diameter (for circular nozzle) or by its orthogonal dimensions – width and thickness. However, it can be of any shape.

3.22.13 Layer — A layer is the object formed with a single pass of the printer. Multiple layers are stacked one over the other to build up the full object. Each layer is characterized by the height or thickness, for rectangular or square nozzles.

3.22.14 Bond — Bond is an interface between successive layers or filaments of a 3D printed concrete specimen. The interface between layers can be called as ‘horizontal’ bond, while the interface between filaments side by side can be called as ‘vertical’ bond.

3.22.15 Shape Stability — Shape stability is the ability of the printed component to maintain the dimension as subsequent layers are placed.

3.22.16 Base Layer — A base layer is the first printed layer on which all following layers are built on.

3.22.17 Dead Zone Length — The dead zone length is defined as the length of static deposition of concrete or where the material flow stops such as near the nozzle with rapid change in cross section.

3.22.18 Extracted Specimen — Specimen of the required dimension which is extracted (cored/sawn) from the 3D printed concrete elements in any orientation.

3.22.19 Speed of Extrusion (Extrusion Speed) — The rate at which the extrusion system moves while extruding the concrete. Generally expressed in length/unit time, volume/unit time, or mass/unit time.

3.22.20 Admixtures

3.22.20.1 Superplasticizers — Superplasticizers are required in 3D printed mixes in order to control the workability of concrete and make it suitable for pumping and extrusion.

3.22.20.2 Accelerators — Accelerators are admixtures added to the concrete to decrease the initial setting time and increase the early-age strength.

3.22.21 Viscosity Modifier — An admixture that when added to the concrete, modifies the viscosity of the mix.

3.22.22 Out-of-Plane Deformation — Distortion/bending of the printed layers from the predefined (vertical) plane of the specimen. Intentional cantilevering or offset shall be allowed and is not considered as out-of-plane deformation.

4 NOTATIONS AND SYMBOLS

The notations used in this section are given at [Annex A](#).

5 MATERIALS

The relevant materials for precast structures shall satisfy the requirements given in the following sections. For addressing sustainability aspects in selecting materials, a reference may be made to [Annex G](#) (see [G-2](#) and [G-5](#)).

5.1 Concrete Ingredients

5.1.1 Cement

The cement shall be selected in accordance with good practices [C-7(2)] to provide minimum required strength for lifting, transportation and durability. Each shipment of cement shall be referenced to a certified mill test report that indicates compliance with the specified type of cement. However, accelerated tests shall be carried out before application.

5.1.2 Mineral Admixtures

For mineral admixtures, the provisions of Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS shall apply, unless otherwise specified by the client or the design.

5.1.3 Aggregates

Aggregates used in concrete shall comply with accepted standard [C-7(4)]. All aggregates must be sourced from approved suppliers, with representative samples tested according to the requirements specified in the governing standards. The desired concrete properties must be demonstrated through trial mixes to ensure compliance and satisfactory performance.

The provisions of good practice [C-7(2)] shall apply while deciding the maximum and minimum size of coarse aggregate.

For stitch joints between precast elements with thickness less than 75 mm, the maximum size of aggregate shall be limited to 10 mm.

5.1.4 Water

Water used for concrete mixtures shall conform to Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS. In addition, mixing water shall be free from deleterious matter that may interfere with the setting or strength development of the concrete. Water shall be tested based on the batching requirements at least once a month.

5.1.5 Chemical Admixtures

5.1.5.1 All chemical admixtures such as accelerating admixtures, retarding admixtures, water-reducing admixtures, air-entraining admixtures and super-plasticizing admixtures shall comply with accepted standards [C-7(5)].

5.1.5.2 Admixtures containing chloride ions shall not be used.

5.1.5.3 Admixtures must be checked for compatibility with cement and other admixtures to ensure proper performance. A trial mix program shall evaluate slump, stability, workability, air content and strength under typical conditions, including temperature and humidity. The effect of dosage variations and the order of admixture addition shall be determined from supplier recommendations or trial.

5.1.5.4 Synthetic or natural pigments may be used in precast facade or decorative elements to achieve desired colours. The pigments must be insoluble in water, free of soluble salts, acids, calcium sulphate and resistant to sunlight, alkalis and weak acids. The pigment type and amount must not affect concrete setting time or strength.

5.2 Concrete Mix Design

Qualification of a concrete mix for use in precast concrete production shall be through laboratory testing, production trials and quality control testing. The minimum grades of concrete shall be as per good practices [C-7(2)].

5.2.1 Mix Development

5.2.1.1 Concrete mixes for precast elements shall be determined through laboratory methods, complying with Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS exposure class requirements. The mix shall meet project needs, service conditions and desired surface finish (colour and texture), with adequate workability for proper placement and consolidation. Water use shall be minimized to the amount necessary for proper placing and vibration consolidation.

5.2.1.2 For standard concrete, the mix design procedure shall be as per good practice [C-7(6)]. The good practices [C-7(2)] shall be used for the correlation between compressive strength and tensile strength, modulus of elasticity, shrinkage, creep coefficient, co-efficient of thermal expansion, etc.

5.2.1.3 Special concretes (for example, high-performance, ultra-high performance) shall be designed based on specialized literature and approved by the competent authority. The literature shall also correlate compressive strength with other properties such as tensile strength, modulus of elasticity, shrinkage, creep and thermal expansion.

5.2.1.4 Steel fibre reinforced concrete, specified in structural elements to enhance mechanical and durability properties, shall be designed using hooked, crimped or straight steel fibres as per specialized literature and with appropriate approvals from competent authority. Any construction joint in precast steel fibre reinforced concrete elements shall have reinforcement bar across the joint to transfer stresses.

5.2.1.5 Lightweight concrete may be used for non-structural precast elements (such as wall panels, facade units, and sandwich panels) to reduce the self-weight, enhance ease of handling and transportation, thermal insulation, and sound absorption. For lightweight concrete, adjustments are required during mix design to account for the lower density and altered workability. Specialized literature shall be referred and appropriate approvals from competent authority shall be obtained.

5.2.2 Miscellaneous

5.2.2.1 All the concrete mixes to be used in the production of precast elements shall be evaluated as trial batches, prepared in accordance with good practice [C-7(7)] under conditions that simulate the actual production and finishing as closely as possible.

5.2.2.2 If the source/manufacturer of cement, source/gradation of aggregates or source/manufacturer of admixture are changed after mix finalization, the proportions of the mix shall be validated and re-designed, if required.

5.2.2.3 When accelerated curing is to be used, the mix proportions shall be based on similarly cured test specimens.

5.2.2.4 When self-compacting concrete is used, the mix proportion shall meet the requirement of good practices [C-7(8)].

5.3 Reinforcement

5.3.1 Reinforcing bars and prestressing strands shall be in accordance with the provisions of Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS, respectively. Reinforcing steel deliveries shall be identified with a heat number least that can be correlated with a mill certificate.

5.3.2 For welding of reinforcement, the good practice [C-7(9)] and Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS shall be referred.

5.3.3 Reinforcement couplers shall comply with the accepted standard [C-7(10)].

5.3.4 If glass fibre reinforced polymer (GFRP) bars are used, the material shall conform to the requirements of accepted standard [C-7(11)]. Such bars shall be used only in precast elements with low risk, based on the decision of engineer-in-charge and client.

5.3.5 Welded wire reinforcement shall conform to accepted standard [C-7(12)].

5.3.6 Welding shall not be performed near any prestressing strand, whether tensioned or not tensioned. The prestressing strand shall not be exposed to spatter, direct heat, or short-circuited current flow.

5.3.7 Prestressing accessories shall comply with the requirements of good practice [C-7(3)].

5.3.8 Plastics for welded wire and bar supports shall be composed of polyethylene, styrene copolymer rubber-resin blends, polyvinyl chlorides and polytetrafluoroethylene. Plastics shall be alkali-resistant and have thermal properties compatible with concrete.

5.4 Grout

5.4.1 Cementitious Grout

5.4.1.1 Cementitious grout used in the precast application shall be as per good practice [C-7(2)].

5.4.1.2 Non-shrink non-metallic high-strength and flowable cementitious grouts shall be used for all structural joints between two precast elements having gap less than 50 mm. The grout strength shall exceed the strength of the joining member by at least 5 MPa. If the strength of the members varies, the cementitious grout with the highest required strength shall be used. For special connection, strength of the grout shall be as per [8](#).

5.4.1.3 For joint width more than that recommended in the grout manufacturer's guidelines or 50 mm, the grout shall be mixed with 10 mm down aggregate or self-compacting concrete or micro concrete shall be used.

5.4.2 Chemical Grouts

Chemical grouts shall be used as per manufacturer's guidelines and its properties shall comply with accepted standard [C-7(13)].

5.5 Mould

5.5.1 All moulds, regardless of material, shall conform to the profiles, dimensions and tolerances of precast concrete product indicated by the contract documents and the approved shop drawings.

5.5.2 Repeated use of moulds shall not cause any dimensional or planar changes that exceed allowable tolerances of the member. When exposed to temperature fluctuations or moisture, the warping or buckling of moulds shall not exceed to allowable tolerance limits.

5.5.3 Moulds exposed to external vibrations must be capable of transmitting these vibrations evenly across a sufficient area, without flexing, warping or causing plate flutter.

5.5.4 If it is planned to apply a prestressing force by jacking against the mould, the mould shall be sufficiently strong to withstand the force without buckling or wrinkling and maintain the required dimensional tolerances.

5.5.5 When different materials are used to construct the mould, they shall not affect the colour and texture of the precast concrete product intended to be installed unrendered. The mould materials shall be non-absorbent or properly sealed to prevent excessive moisture absorption to minimize variations in finish.

5.5.6 Wooden moulds shall be sealed with suitable materials to prevent moisture absorption.

5.5.7 Steel moulds shall be visually inspected before each use for rust, distortion and tightness of steel sheet joints.

5.5.8 Fibre reinforced polymer moulds shall be used for achieving localized architectural finishes.

5.5.9 Moulds shall be coated with mould release agents that permit its release without damaging or staining the concrete and without affecting subsequent coating, painting, or caulking operations. Debonding agents or wax shall be used during match casting.

5.5.10 Whenever concrete is to be painted or stained, only mould release agents compatible with the coating shall be permitted, unless surface preparation is performed to ensure good adhesion between the coating/stain and the concrete.

5.6 Hardware, Lifting Devices and Miscellaneous Materials

5.6.1 Hardware

5.6.1.1 Hardware refers to items:

- a) To be placed on or in the structure to receive the precast concrete units (anchor bolts, angles, or plates with suitable anchors);
- b) To be embedded in the concrete units for connection or for other mechanical, plumbing, glazing, miscellaneous iron, masonry, roofing works, etc; and
- c) Necessary for the installation of the precast concrete units.

5.6.1.2 Hardware shall be made from ductile materials or as specified in the design. For high-strength materials, the design must ensure elastic behaviour. For testing, [12](#) shall be referred.

5.6.1.3 Welding of steel plates, angles, and other shapes shall be carried out by certified welders in conformance with standard procedures. The size, length, type and location of all welds shall conform to the approved shop drawings, and no unspecified welds shall be added without the approval of the designer.

5.6.1.4 Electrodes used for welding operations shall be bought in sealed (airtight) containers. The selection of a suitable electrode shall be made based on the material to be welded, intricacies based on the geometry of the job, thickness, weld profile and weld position. The maintenance of any electrode used shall be as prescribed by the manufacturer.

5.6.1.5 All metallic hardware surfaces that are exposed to or within 12 mm of concrete surfaces that are exposed to the weather, corrosive conditions, or condensation preferably be protected against corrosion or be made of non-corrosive materials. Hardware shall be properly cleaned before application of protective treatment. Corrosion protection shall be done as required. Care shall be taken to prevent chemicals, such as acids, from coming in contact with the hardware and causing corrosion.

5.6.1.6 Supplier of hardware shall furnish records of compliance to specification requirements and mill certificates for the material used. Review of hardware fabrication shall be performed by quality control personnel.

5.6.2 Lifting Devices

5.6.2.1 Lifting devices shall be capable of supporting the element in all positions planned during manufacture, storage, delivery and erection. Safe loads for lifting inserts or hooks or devices shall be supplied by the manufacturer of proprietary devices. Information on the use and installation of the devices to ensure proper performance shall be supplied by the manufacturer. For factor of safety and capacity of lifting inserts or hooks, [7](#) and [8](#) shall be referred, respectively. For testing, [12](#) shall be referred.

5.6.2.2 If the lifting device is subjected to dynamic loads, it shall preferably be fabricated using ductile materials or shall be designed to remain elastic.

5.6.2.3 Reinforcing bars provided as main reinforcement in the members shall not be used as lifting hooks. Separate reinforcing bars of required grade, bent to the correct size and shape, may be used exclusively as lifting hooks, with adequate embedment or anchorage within the precast member. Where necessary, smooth rebars with adequate embedment length shall be used as lifting hooks. Lifting hooks made of reinforcing bars shall not be used for multiple handling applications.

5.6.2.4 Each bar size and configuration shall be considered such that it meets load and handling requirements.

5.6.2.5 Coil rods and bolts shall not be welded when used in lifting operations.

5.6.2.6 Connection hardware shall not be used for lifting or handling unless carefully reviewed and approved by the site engineer/designer.

5.6.2.7 Shop drawings shall clearly define insert dimensions and locations for fabrication and placement or refer to standard details.

5.6.2.8 Members incorporating embedded hardware shall be provided with corrosion protection during long-term storage.

5.6.3 Surface Finishes and Coating

5.6.3.1 All surface finishes and coating on precast concrete units shall be as stated in the shop drawing.

5.6.3.2 When specified in the project requirements, for finishing processes such as sandblasting, acid etching, surface retardation, the use of mould liners, or the application of veneer facing materials may be employed to achieve the desired surface appearance. These methods shall help control form joints, form blemishes and air voids within acceptable limits.

5.6.3.3 Based on project requirements and client specifications, curing compounds for precast elements shall be chosen for compatibility with coatings, moisture retention, and ease of removal. Only membrane-forming compounds that meet standards for water retention and curing efficiency shall be used. Application should occur immediately after demoulding to prevent moisture loss and ensure proper strength. The surface must be free of dust or laitance, and curing must be uniform to avoid streaking. Curing compounds shall be non-residual or removable when subsequent coatings are applied.

5.6.3.4 Based on project requirements and client specifications, anti-carbonation coatings shall be selected for CO₂ diffusion resistance, breathability and UV stability, conforming to durability standards. They must be tested for permeability and adhesion on precast surfaces. Application should occur on clean, dry, fully cured concrete, free from laitance, dust and form oil. Coatings must be applied evenly to avoid streaks and ensure effective protection.

5.6.3.5 Based on project requirements and client specifications, paints used on precast elements shall be alkali-resistant, weatherproof, and compatible with the concrete substrate. Selection shall be based on exposure conditions, including moisture, UV and chemical resistance requirements. Before application, the surface must be properly primed and any efflorescence or moisture-related defects must be addressed to prevent peeling or discoloration. Roller or spray application shall ensure uniform coverage and multiple coats shall be applied as required to achieve the desired opacity and aesthetic consistency.

5.6.3.6 Based on project requirements and client specifications, epoxy coatings, selected for chemical resistance, adhesion, impact resistance and fire rating, are used for flooring, structural connections and corrosion protection. Surfaces must be clean and dry, with a primer applied for better bonding. Curing conditions should be controlled, and components must be mixed correctly and applied evenly to avoid uneven gloss or discoloration.

5.6.3.7 Based on project requirements and client specifications, polyurethane or hybrid sealants for external precast joints shall be compatible with concrete, offer high elongation (for grouted joint 200 percent or more, for non-grouted joints 400 percent or more) and resist UV and environmental degradation. Selection should consider joint movement, exposure and adhesion. Joints must be clean, dry and free of contaminants before application, with a primer used if needed. Proper tooling is essential to ensure seamless finishes and prevent air pockets. Sealants in grouted joints act as moisture barriers, so their lifespan shall be verified.

5.6.4 Insulation Material

5.6.4.1 Rigid foam insulation for precast elements shall be selected based on thermal resistance, compressive strength and moisture resistance. Expanded polystyrene, extruded polystyrene, polyurethane or polyisocyanurate, etc, are preferred for long-term efficiency. Panels must conform to accepted standard [C-7(14)] or relevant manufacturer guidelines for density, strength and thermal conductivity. Panels shall be precisely cut for a snug fit to minimize thermal bridging. Exposed foam surfaces must be protected with coatings for durability and UV resistance. Foam with strength more than 10 MPa or as per design may be used in flooring to reduce concrete weight and thickness.

5.6.4.2 Mineral wool insulation shall be selected for its fire resistance, acoustic properties, and vapour permeability. Rigid or semi-rigid panels shall be used to withstand the casting process without compression. Mineral wool products conforming to accepted standard [C-7(15)] or any relevant manufacturer's guidelines shall possess thermal conductivity, density, moisture absorption and fire resistance requirements. Before placement, mineral wool boards shall be dry, securely positioned and properly supported to prevent displacement when concrete is poured. Special care shall be taken to prevent fibre shedding, which can affect worker safety and surface finish. If exposed, mineral wool shall be coated or encased to prevent moisture absorption and fibre degradation over time.

5.6.4.3 Lightweight concrete used for insulation layers in precast elements shall be selected based on its density, thermal performance and compatibility with structural concrete. Autoclaved aerated concrete and non-autoclaved cellular concrete conforming to accepted standard [C-7(16)] shall be used or relevant manufacturer's guidelines shall be followed. Proper mix design and curing conditions shall be maintained to achieve uniform density and minimize shrinkage or cracking. During production, the interface between lightweight and normal concrete layers shall be well bonded to prevent delamination. Surface smoothness and uniformity shall be checked, as inconsistencies may affect finishing and overall insulation performance.

5.7 Connections

In addition to the requirements in [5.6.1](#), the materials of a connection shall be selected such that embrittlement of any part of the assembled connection do not occur. Non-ferrous inserts shall be resistant to electrolytic action and alkali attack. If more than one material is used in a connection, abutting materials shall be selected such that corrosion is not induced. Dissimilar metals shall not be embedded near or in direct contact in moist or saturated concrete unless it is ensured that detrimental chemical or electrochemical (galvanic) reactions do not occur. For continuity of rebar, grouted coupler, grout filled tube or lapping of rebar shall be used. However, depending on seismic requirements the type of continuity method shall be selected. For testing and performance evaluation [12](#) shall be referred.

5.7.1 Wire Loop System — Loop Box

5.7.1.1 The wire loop system (proprietary product) is a load-transferring, cast-in element consisting of a box and a U-shaped wire loop for connecting prefabricated concrete elements. Generally, the joint with wire loop boxes is considered non-structural and should only be used for durability and gravity load requirements.

5.7.1.2 Relevant manufacturer's guidelines shall be used for selecting the appropriate product. In addition to that, the following points shall be considered:

- a) The box and the two flexible wire loops shall be made of carbon steel sheet and high strength wire ropes, respectively. The surface of the box shall be equipped with a profiled surface to ensure sufficient bonding with the surrounding concrete;
- b) Proper selection involves ensuring compliance with tensile strength requirements and alignment accuracy to prevent misfits. High-strength, non-shrink non-metallic cementitious grout is recommended to encapsulate loops fully. Care shall be taken in application to maintain uniform joint appearance, especially in exposed architectural elements;
- c) The looped wire ropes shall not react with grout and form corrosion. Also, shall not be used if found having corrosion; and
- d) The manufacturer shall furnish the detailed technical datasheet along with the test results corroborating the values noted in the technical datasheet. Also, the same shall be verified by the engineer-in-charge by conducting appropriate testing per batch.

5.7.2 Grouted Coupler

5.7.2.1 Grouted couplers (proprietary product) provide mechanical splicing of rebars between precast elements, ensuring structural continuity.

5.7.2.2 Their selection shall consider compatibility with reinforcing bar sizes and compliance with applicable standards.

5.7.2.3 Reinforcing bar ends and coupler interiors shall be clean before grouting to ensure proper bonding. A high-performance cementitious grout, characterized by non-shrink non-metallic high strength properties, high early strength shall be used to ensure effective load transfer. The grout shall meet the specified compressive strength and flowability requirements for complete void filling. Misalignment or incomplete grouting shall be addressed through inspection to maintain both functionality and appearance.

5.7.2.4 Class H coupler as per accepted standard [C-7(10)] shall be used with appropriate testing as per [12.3.1](#).

5.7.3 Grout Filled Tubes (Dowel Tube)

5.7.3.1 The tubes shall be corrugated to have proper bond with concrete and grout.

5.7.3.2 The length of the tube shall be as per development length or design requirements as per [7](#) and [8](#).

5.7.3.3 The tube shall be free from oil, grease or loose particles prior to installation and grouting. A high-performance cementitious grout, characterized by non-shrink non-metallic high strength properties, high early strength shall be used to ensure effective load transfer. The grout shall meet the specified compressive strength and flowability requirements for complete void filling. Adequate curing time shall be allowed for strength development. Grout shall be placed using gravity feed to ensure complete filling and venting shall ensure proper air escape. Misalignment or incomplete grouting shall be addressed through inspection to maintain both functionality and appearance.

5.7.4 Terminators (Headed Bars)

5.7.4.1 Headed bars are used in precast concrete elements to enhance anchorage efficiency by providing an enlarged head at the end of the reinforcement bar, reducing the required development length.

5.7.4.2 These bars shall be manufactured from high-strength reinforcement steel to ensure adequate ductility and mechanical performance. The heads shall be formed through welding, forging, or mechanical attachment and shall comply with the specified geometric and strength requirements. The selection of headed bars shall be based on design specifications, including grade of bar, head dimensions and load transfer capacity.

5.7.5 Metal Inserts

5.7.5.1 Steel plates shall be cast into precast elements to facilitate connections with other structural components for non-structural or gravity load transfer. Plates shall be made from structural-grade steel with appropriate thickness, positioned accurately during casting and securely anchored to prevent displacement.

5.7.5.2 Welded studs shall provide reliable load transfer by creating strong mechanical bonds between steel inserts and concrete. Studs shall be manufactured from high-strength steel and shall conform to applicable material and welding standards. Proper fusion and alignment shall be ensured and weld quality shall be verified through inspection to prevent structural deficiencies.

5.7.5.3 Corrosion protection, such as galvanization or stainless-steel alternatives, shall be applied based on environmental exposure conditions. Protective measures shall not interfere with mechanical performance or bonding with concrete. In exposed applications, surface finishes shall be such as to maintain aesthetic quality. Fire safety requirements shall also be verified for exposed applications.

5.7.6 Shims

5.7.6.1 Plastic shims shall be composed of materials that provide sufficient compressive strength (not less than 10 MPa or as required by the design) and demonstrate resistance to moisture, UV exposure, and temperature variations. They shall be dimensionally stable, free from warping or defects and sized to ensure a precise and secure fit for their intended application. Plastic shims shall be used for thicknesses up to 40 mm or less.

5.7.6.2 Metal shims shall be used for any shim thickness; however, use is preferable for thicker shims (typically greater than 40 mm). These shims shall be having higher strength and load-bearing capacity, particularly where greater stability and structural support are needed. For multi-layer shims, tack welding shall be performed between the layers to ensure a strong bond and prevent shifting or separation of the individual layers. Metal shims that are exposed to the environment or are in contact with moisture or corrosive elements shall be provided with appropriate corrosion protection. Care must be taken during installation to ensure that the corrosion protection is intact and not compromised by any mechanical processes.

6 PRECAST CONCRETE SYSTEMS

6.1 General

In precast concrete construction, the design shall address additional requirements including selection of component types, appropriate connection detailing, structural continuity, flexibility and restraint. Structural connections between elements shall be explicitly classified and designed as pinned, rigid, or ductile based on seismic performance criteria. Horizontal elements such as slabs, beams and staircases shall be detailed as simply supported or made continuous using in-situ topping, integrity reinforcement and collector elements to provide diaphragm action for lateral load transfer to the lateral force resisting system (LFRS). Vertical elements (for example, columns and walls) shall be designed as continuous members unless specifically excluded from LFRS. Beam-to-column joints not forming part of LFRS shall be designed to resist gravity loads with compatible displacements, moments based on design requirements and erection eccentricity. Such frames shall incorporate structural bracing to resist lateral forces. For selection of precast concrete system [6.2](#) to [6.5](#) shall be referred.

6.2 System Considerations

6.2.1 Structural Concept

Precast concrete structural systems shall be explicitly designed to ensure continuity between elements and robustness of the overall system. The selected system shall be robust and shall be designed to withstand the following:

- a) Stability against progressive collapse;

- b) Prevention of structural failure under all design load combinations;
- c) Fire resistance as per applicable regulations;
- d) Adequate ductility and energy dissipation capacity;
- e) Redundancy and continuity in load paths;
- f) Control of cracking within serviceability limits;
- g) Acceptable deformation within serviceability limits; and
- h) Durability against environmental exposure.

6.2.1.1 The structural system and each of its components shall maintain stability during all stages, including transportation, erection and service life. The following factors shall be considered for the selection of precast system:

- a) Grid distances;
- b) Spanning direction of slab and beam;
- c) Position of columns and walls;
- d) Position of load bearing walls and/or facade and/or facade at stilt;
- e) Position and approximate requirements for stabilizing elements; and
- f) Position of expansion joints.

6.2.1.2 The appropriate precast structural system shall be selected based on [Table 1](#), considering building usage, architectural constraints, seismic performance requirements and constructability. The response reduction factor (R) shall be considered as per 5 of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS.

6.2.2 Standardization of Precast Components/Elements

The precast components/elements shall preferably be standardized based on the structural, architectural and/or aesthetic requirements. The precast components/elements given below but not limited to, shall be used in accordance with relevant Indian Standards (see Part B ‘Building Materials’ of this NBCS) and the accepted standards, where available:

- a) Reinforced/prestressed concrete channel unit;
- b) Reinforced/prestressed concrete slab unit;
- c) Reinforced/prestressed concrete beams;
- d) Reinforced/prestressed concrete columns;
- e) Reinforced/prestressed concrete hollow core slab;
- f) Reinforced concrete waffle slab/shells;
- g) Reinforced/prestressed concrete wall elements;
- h) Hollow/solid concrete blocks and battens;
- j) Precast planks and joists for flooring and roofing;
- k) Precast joists and trussed girders;
- m) Light weight/cellular concrete slabs/wall panels;
- n) Precast lintels and *Chajjas*;
- p) Large panel prefabricates;
- q) Reinforced/prestressed concrete trusses;
- r) Reinforced/prestressed roof purlins;
- s) Precast concrete L-panel units;
- t) Prefabricated concrete double-T units;

- u) Prefabricated sandwich concrete panels;
- v) Precast concrete foundations; and
- y) Precast concrete staircases.

Any other type of components shall be used with approval of the Competent Authority and appropriate testing.

Table 1 Precast Concrete (PC) Buildings with Different Structural Systems

(Clause [6.2.1.2](#))

Sl No.	Structural System
(1)	(2)
i)	<i>Building with Precast Concrete Moment Frames:</i> a) Buildings with single or multi-storey cantilever columns with simply supported beams (see Note 1) b) Building with ordinary moment resisting frames (OMRFs) (see Note 2) c) Buildings with special moment resisting frames (SMRFs)
ii)	<i>Buildings with Precast Concrete Structural Walls:</i> a) Ordinary structural walls (see Note 2) b) Ductile structural walls
iii)	<i>Buildings with Precast Concrete Moment Frames and Precast/RC Structural Walls (Dual System) (see Note 3):</i> a) Buildings with ordinary structural walls with OMRFs (see Note 2) b) Buildings with ordinary structural walls with SMRFs (see Note 2) c) Buildings with ductile structural walls with OMRFs (see Note 2) d) Buildings with ductile structural walls with SMRFs
iv)	<i>Precast Cell System:</i> a) Ordinary precast/RC structural wall system (see Note 2) b) Ductile precast/RC structural wall system
NOTES 1 Precast single-storey or multi-storey columns with simply supported beams shall be designed using a response reduction factor (R) of 2, as this system permits only one mode of plastic hinge formation. 2 Precast/RC emulative structures in Seismic Zones III and above shall be designed to be ductile. Hence, precast ordinary moment resisting frame system (OMRF) is not allowed in these seismic zones. 3 Buildings with precast RC structural walls also include buildings having precast RC structural walls and precast RC moment frames, but where: a) frames are not designed to carry design lateral loads, or b) frames are designed to carry design lateral loads, but do not fulfil the requirements of 'dual systems'. 4 Various types of moment resisting frame systems are explained in 6.3.4, the structural designer shall select the appropriate system based on the requirements of the structure. 5 For all structures, precautions shall be taken to ensure continuity of load path, adequacy of connections, and compliance with drift limitations. 6 For shear walls with coupling beams, CIS coupling beams shall be used. 7 For elements designed to transfer gravity loads and not part of LFRS, assemblage testing shall be carried out to verify the failure mode of gravity frame elements and to ensure compliance with displacement requirements (equivalent to 'R' times the design storey drift), also the performance criteria with an approval from institute of repute may be verified. 8 The use of precast structural systems in Seismic Zone IV and above shall be permitted subject to approval by institute of repute. The adequacy of such systems shall be established through testing and/or appropriate structural analysis, as deemed appropriate.	

6.3 Basic Precast Concrete Systems

Precast concrete building systems are composed of some basic types of structural systems. These systems shall be combined in various ways to obtain a suitable and effective structural concept that meets the needs of specific buildings. The most common systems are:

- a) Precast concrete buildings with moment frames;
- b) Precast concrete buildings with structural walls;
- c) Precast concrete buildings with moment frames and structural walls (or dual system);
- d) Precast concrete buildings with precast or cast-in-situ flat slab and precast concrete moment frames and precast concrete structural walls; and
- e) Cell/3D system.

6.3.1 The above list is not unique as many combinations are possible to achieve the same objectives, such as the use of arches and rigid portal frames.

6.3.2 When facade systems act as load-bearing walls, their design shall explicitly consider lateral stability. For non-load-bearing facades, beam deflection and frame movement shall be accommodated in the connection detailing to avoid unintended cracking or overstressing of facade panels.

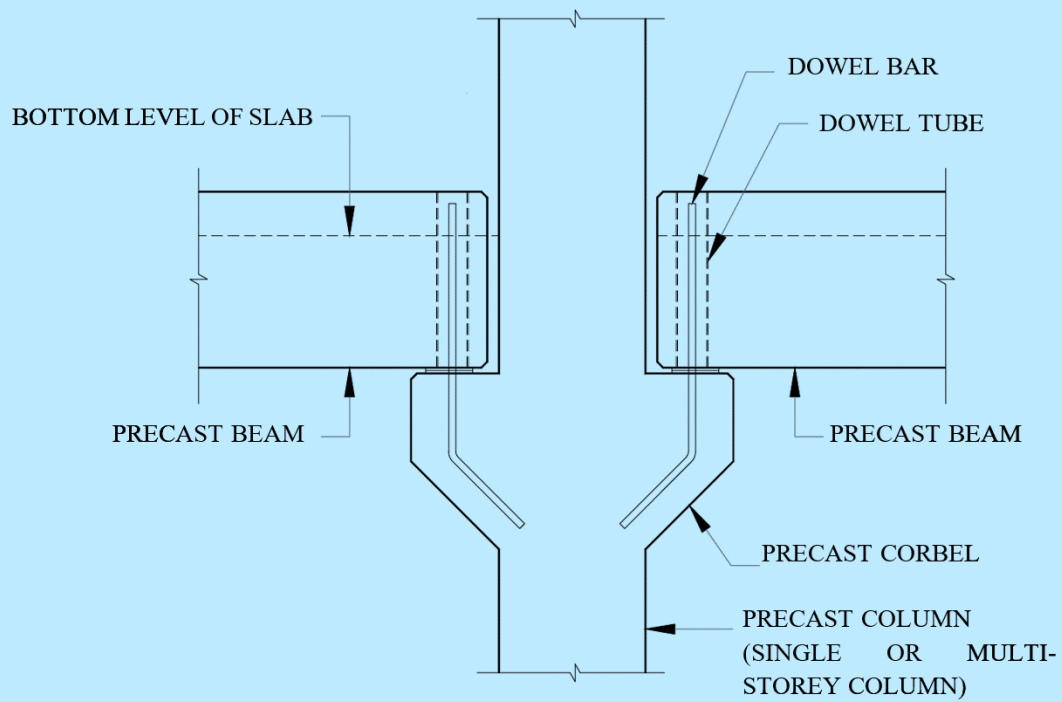
6.3.3 Where partial fixity is introduced by structural topping or dowels, the resulting flexural stiffness (EI) may be considered in serviceability limit state checks. The contribution of topping shall be analytically substantiated.

6.3.4 Skeletal Frame Systems

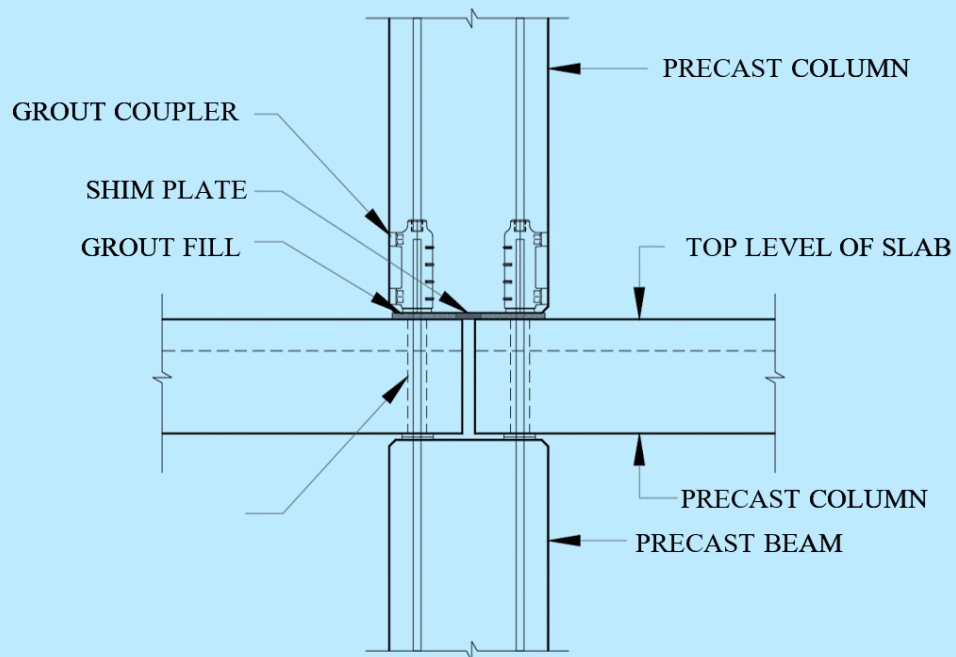
Skeletal frame systems composed of beams and columns shall be used where architectural flexibility, large spans, or column-free space is required, including in commercial, educational, healthcare and industrial facilities. The following skeletal systems shall be permitted:

- a) Moment resisting frames; and
- b) Moment frames with stabilizing cores (with precast structural walls).

Depending on the importance of the structure, structural framing system, connection detailing requirement and the level of ductility demanded by the building, the precast beam to column or beam to beam connection shall be selected from the five types as shown in [Fig. 1](#). Type 1 and Type 2 (with minor fixity) shall be adopted with cantilever with single or multi-storey columns with simply supported beam. Types 2, 3 and 4 with detailing as per the good practice [C-7(2)] shall be adopted with OMRF. Types 2, 3, and 4 with detailing as per the good practice [C-7(1)] and detailing of connection as per this NBCS shall be adopted with SMRF. In the case of Type 3 and Type 4, the location of beam-to-beam connection shall be selected such that plastic hinge formation zones are avoided (at least more than $0.15 L_{\text{beam}}$). In the case of secondary beams or short length beams, the location of the connection can be at the mid span with appropriate detailing.



1A) TYPE 1



1B) TYPE 2

FIG. 1 PRECAST BEAM-COLUMN CONNECTION (CONTINUED)

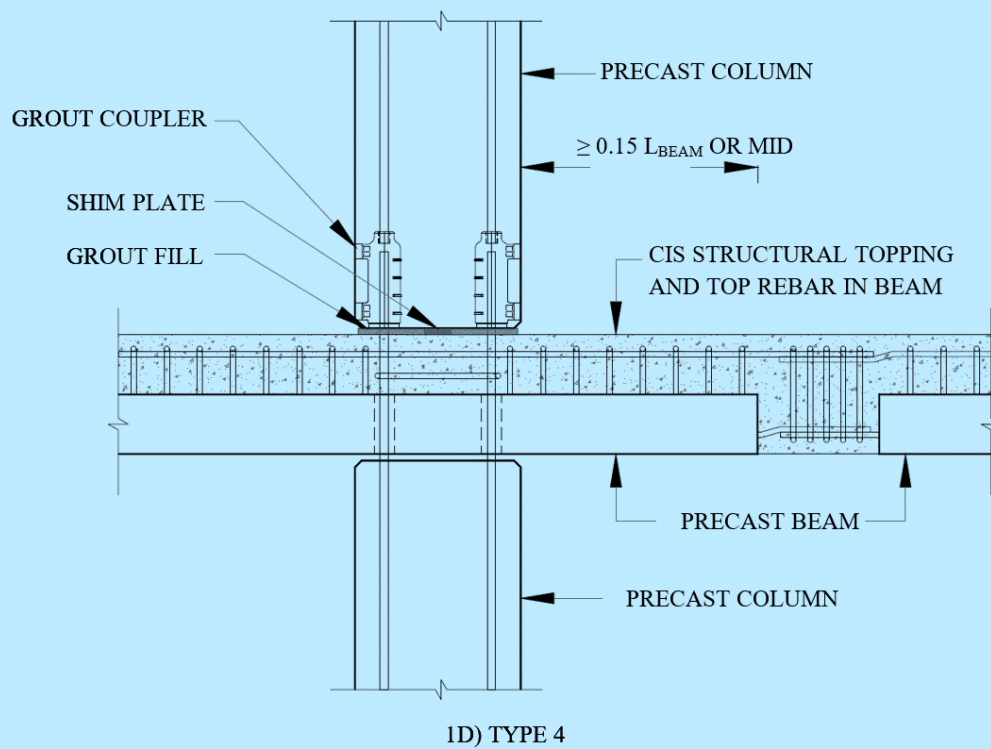
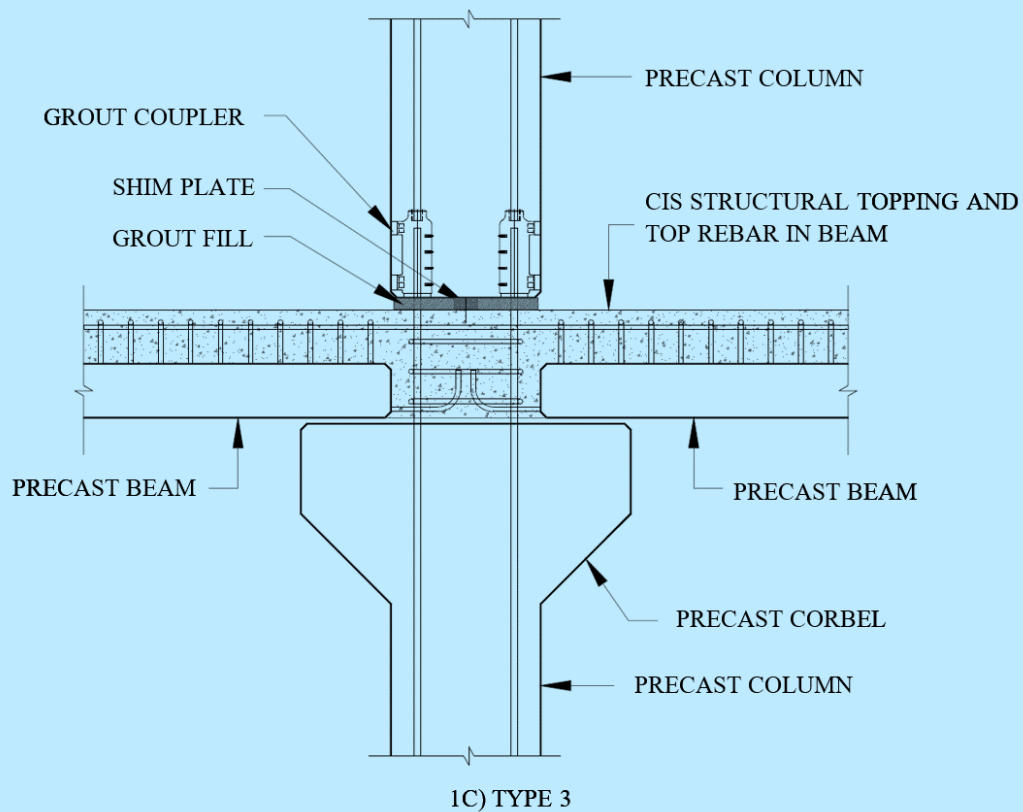
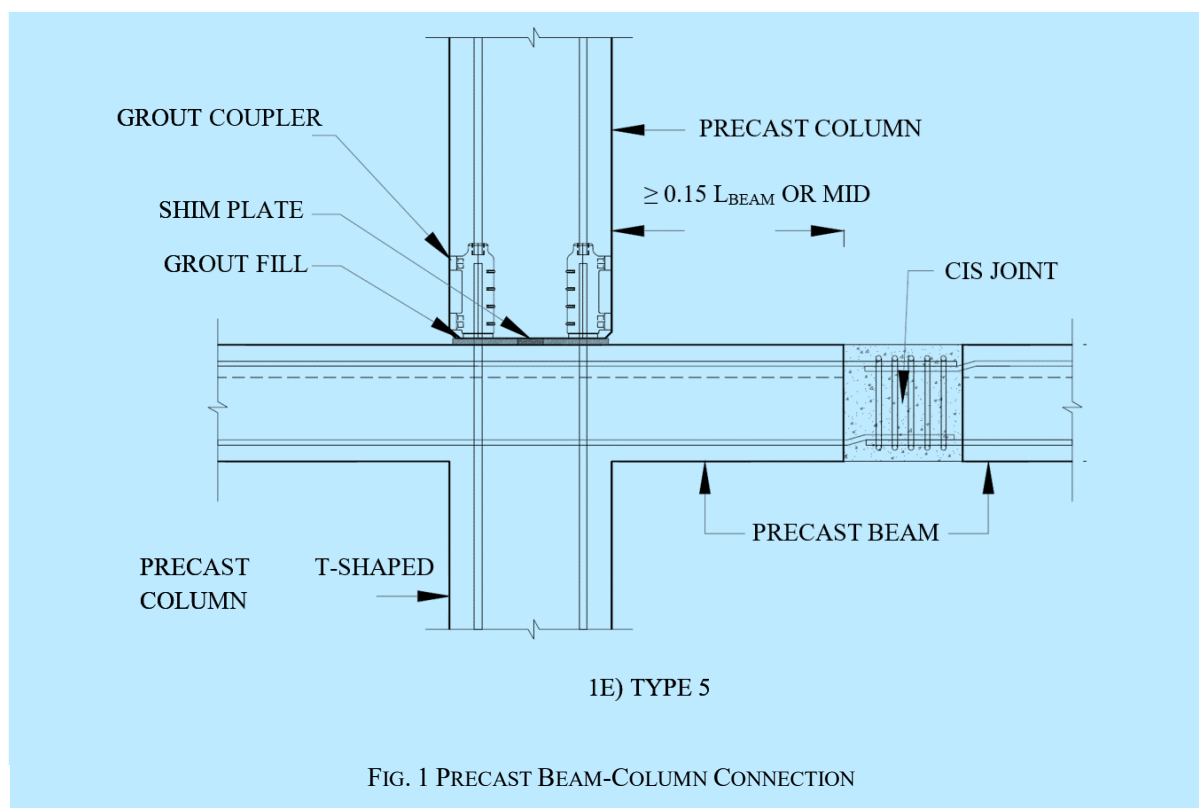


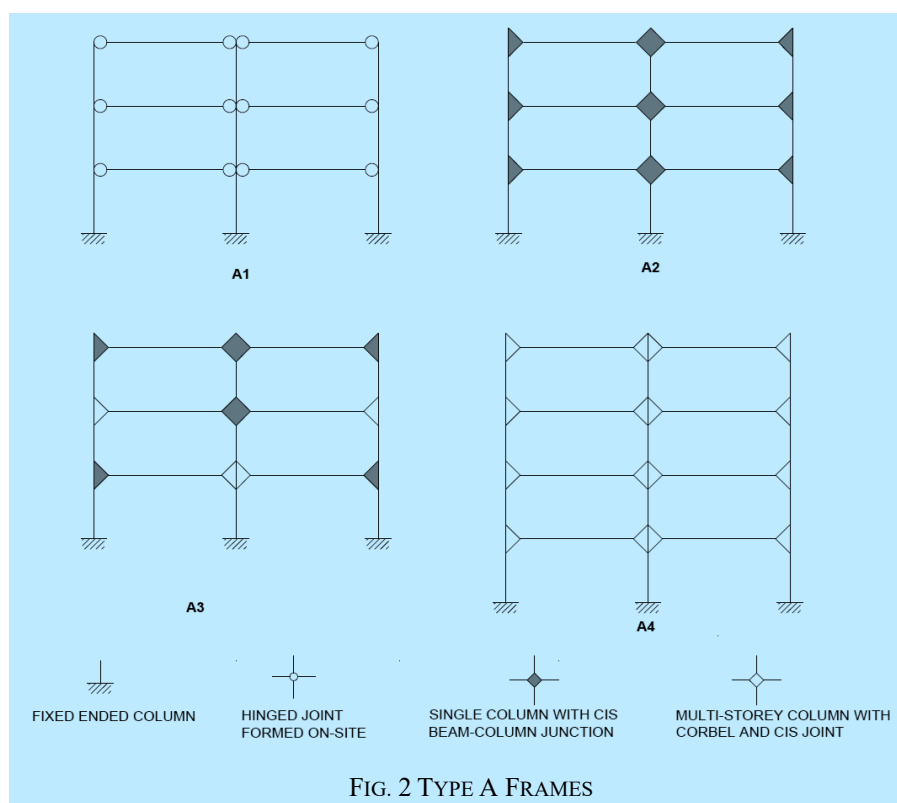
FIG. 1 PRECAST BEAM-COLUMN CONNECTION (CONTINUED)



6.3.4.1 Moment resisting frames

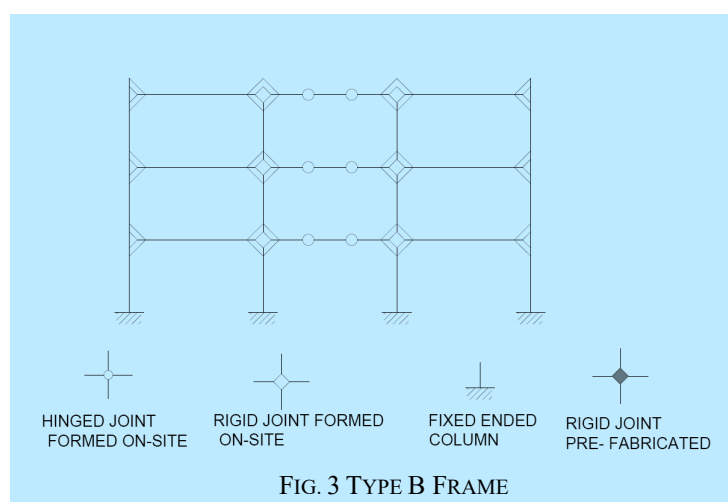
Rigid beam-column connections shall be formed using Type A, Type B, or Type C systems, depending on the required ductility level and building importance. All systems shall include structural topping for diaphragm action in accordance with 6.3.6:

- a) *Type-A frames* — Type A frames shall consist of precast beam units between columns/column corbels, where columns can be cast-in-place or precast (see Fig. 2). These shall be achieved by adopting either one or a combination of the following arrangements:
 - 1) System A1 — This system consists of single or multi-storey columns fixed at the base, supporting simply supported beams on corbels or directly above the columns. The slab units shall be simply supported between the ledges or above the beams;
 - 2) System A2 — This system consists of single storey columns fixed at the base with provision of seating notch or temporary supports or permanent corbels for beams. Depending on design requirements, beams and slabs shall be designed as simply supported for temporary loading and made continuous for permanent loading;
 - 3) System A3 — This system consists of single or multi-storey columns fixed at the base and staggered joints between the columns at other grid levels with corbels to support the beams. Staggering of columns shall be arranged such that only 50 percent of all columns are spliced at any given storey level. Depending on design requirements, beams and slabs shall be designed as simply supported for temporary loading and made continuous for permanent loading; and
 - 4) System A4 — This system consists of 3 or 4 storeys columns fixed at the base with corbels to support the beams. Depending on design requirements, beams and slabs shall be designed as simply supported for temporary loading and made continuous for permanent loading.



b) *Type-B frames* — Type B frames shall consist of precast beam units passing through columns (see Fig. 3). These shall be achieved by adopting the following arrangement:

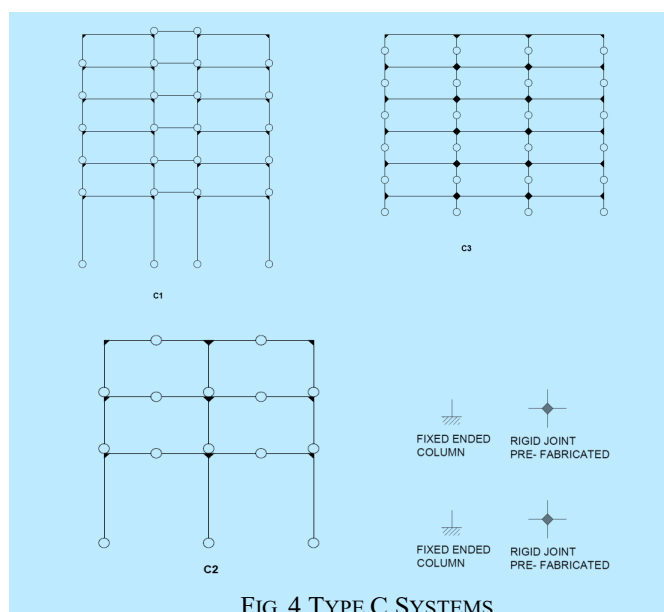
- 1) System B1 — Columns shall be 1 storey high and spliced at each floor level. Precast beams run through the columns and are spliced at point of contraflexure. It shall be ensured that beam splices are avoided at end spans. Slab units shall be designed as simply supported.



c) *Type-C frames* — Type C frames (see Fig. 4) shall consist of monolithically fabricated beam column units such as cruciform, T/L-units, H-frames, portal frames, etc. These shall be achieved by adopting either one or a combination of the following arrangements:

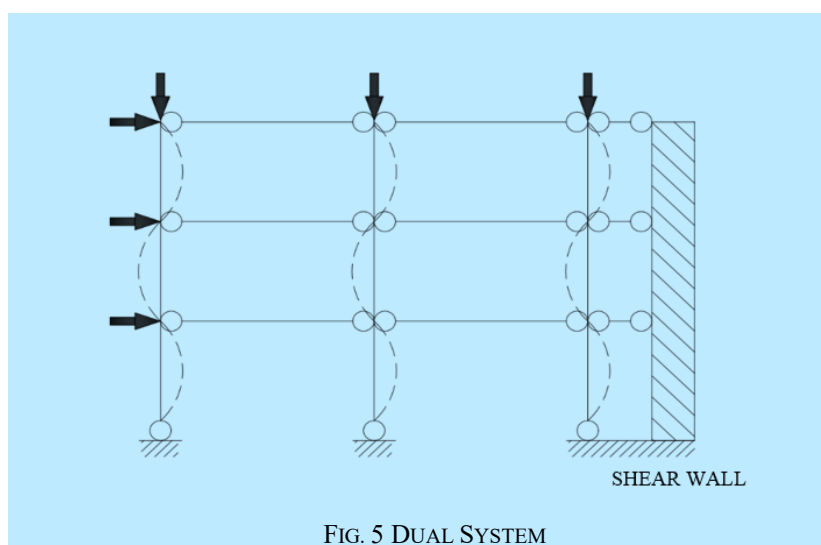
- 1) System C1 — Single storey portal frames shall be placed alternately and simply supported at floor levels. Orthogonal beams and beams connecting the portal shall be simply supported or continuous based on the design requirements. Slab units shall be designed as simply supported; and
- 2) System C2 — L-shaped and T-shaped frame units shall be placed simply supported on the base and on top of each other. Hinged or moment resisting connections shall be made between frame units at the mid-span of beams. Slab units shall be designed as simply supported.

- 3) System C3 — H-shaped frames with a cantilever beam shall be provided with rigid connection at the junction of beam and column. Hinge or moment resisting connections shall be provided at the mid-height of columns and centre of beams. Slab units shall be designed as simply supported.



6.3.4.2 Moment frames with structural wall

In this system, lateral stability shall be achieved through structural walls, structural cores, or bracing. Beam-column and column-base connections shall be designed as pinned or rigid in accordance with the required ductility and seismic performance level (see [Fig. 5](#)).



6.3.5 Wall Systems

Wall systems in precast construction shall be classified on the basis of the design of connections between individual concrete panels, which when joined together form the structural wall (see [Fig. 6](#)). The types of emulative wall systems shall be as follows:

- a) Wall systems without shear transferring vertical joints; and
- b) Monolithic wall systems.

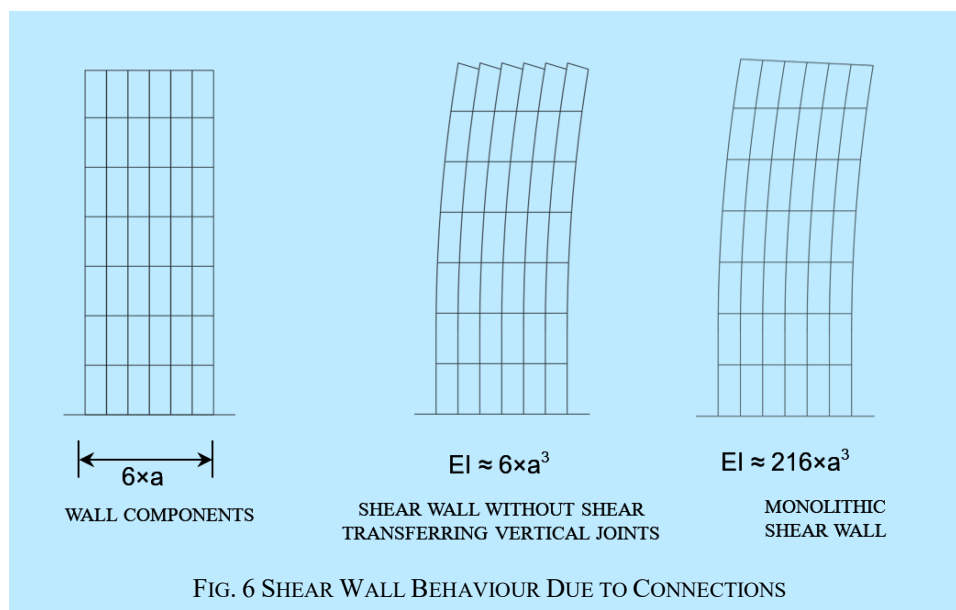
These classifications shall guide the detailing of connections and the overall performance of the system under gravity and lateral loads.

6.3.5.1 For systems without shear transferring vertical joints, the horizontal joints shall be designed to exhibit ductile behaviour. Vertical joints shall not be relied upon for structural continuity and shall be detailed to allow for movement or minimal shear capacity to be provided. Each panel shall be designed as an independent unit for gravity and lateral loads.

6.3.5.2 For monolithic wall systems, horizontal joints shall be detailed to permit ductile behaviour. Vertical joints shall be designed as strong connections with sufficient capacity to remain elastic under design-level lateral loads, preserving global ductility.

6.3.5.3 For jointed wall systems, both vertical and horizontal joints shall be detailed to provide ductile behaviour. Connections shall be capable of undergoing inelastic deformation with energy dissipation, contributing to system-level ductility.

6.3.5.4 Combination of monolithic and systems without shear transferring vertical joints shall be adopted based on the performance requirements of the building.



6.3.6 Floor/Roof Systems

Precast floor and roof systems shall be designed to function as horizontal diaphragms capable of transferring in-plane shear and distributing lateral loads to the lateral force resisting system (LFRS) under all applicable load combinations.

6.3.6.1 A minimum structural topping of 75 mm with reinforcing mesh or welded wire mesh shall be used in Seismic Zones III and above which can be reduced to 50 mm in Seismic Zone II. The diameter of re-bars used in both reinforcing mesh and welded wire mesh shall not be less than 6 mm. Welded wire mesh shall be avoided in Seismic Zones III and above. The minimum structural topping thickness shall be measured above the slab top level even at the highest camber location. Where service conduits of larger diameter are provided, the thickness of structural topping shall be such that a minimum cover of 25 mm is maintained above the conduit. The structural topping shall be provided on the entire floor.

6.3.6.2 Precast floor and roof systems shall be selected from the types defined in [6.3.6.3](#) to [6.3.6.6](#), based on span length, loading and diaphragm requirements.

6.3.6.3 Hollow-core slab (HCS) system

This system is made up of prestressed precast concrete slabs with hollow cores (see [Fig. 7](#)). These systems may be used for multi-story buildings with moderate spans. Structural topping shall be provided to ensure diaphragm action. The top surface of the slab shall be intentionally roughened to amplitude more than 6 mm. The

individual HCS units shall not be considered for diaphragm force transfer:

- a) The thickness of hollow core slabs varies depending on the span and load requirements, but typically range from 150 mm to 500 mm;
- b) Standard width of hollow core slab may be 1 200 mm or 2 400 mm; and
- c) Standard dimensions of hollow core slabs shall conform to good practice [C-7(17)].

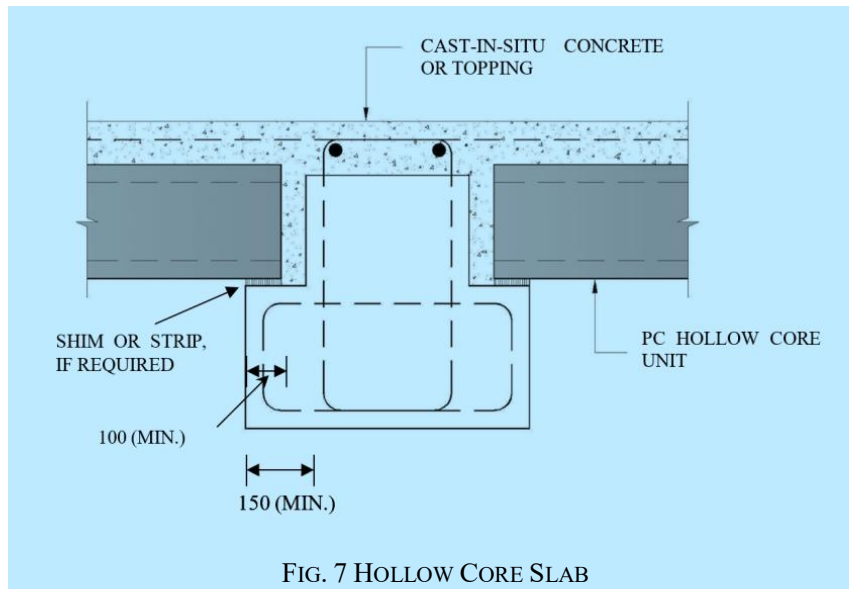


FIG. 7 HOLLOW CORE SLAB

6.3.6.4 Double tee slab system

This system is made up of precast concrete slabs with a double tee cross-section. The double tee shape provides greater strength and stiffness than a flat slab, making it suitable for larger spans and higher loads. These shall conform to good practice [C-7(17)]. Structural topping shall be provided to ensure diaphragm action. The double tee units shall not be considered in diaphragm action if the units are not connected for required force transfer, structural continuity and overall stability of the system.

6.3.6.5 Solid slab system

Solid precast concrete slab systems shall comprise precast elements that are thinner than equivalent hollow-core slabs. These slabs shall be used in residential or commercial buildings, where individual slab units are intended to span shorter distances.

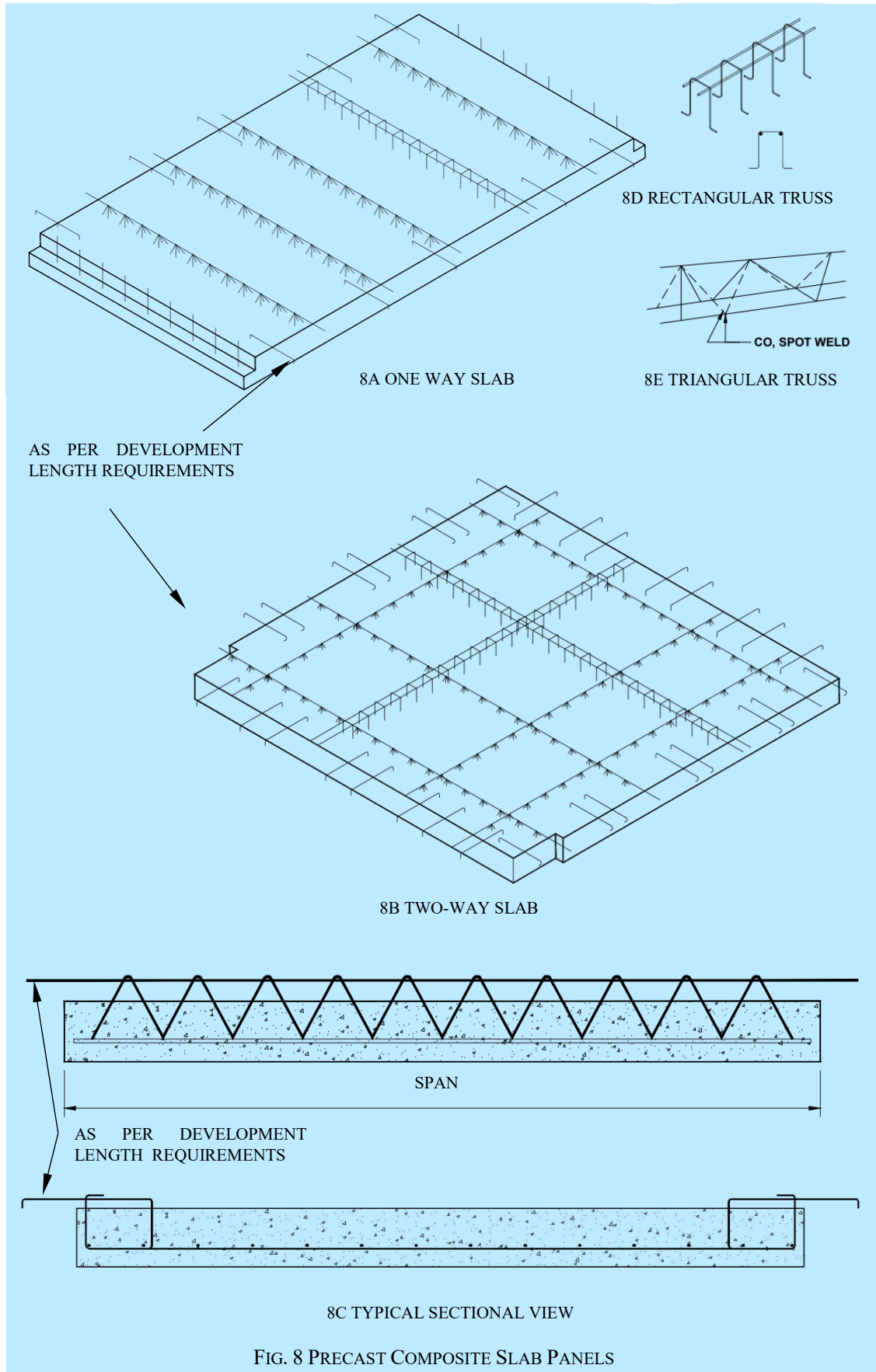
The slab system shall be considered to act as a structural diaphragm when adequate connections are provided between the slab units and the supporting structural elements and when a structural topping concrete is placed and properly bonded.

In cases where no structural topping is provided, the slab system may still be designed to act as a diaphragm, provided that the connection details ensure adequate transfer of in-plane forces, structural continuity and overall stability of the system.

6.3.6.6 One-way/two-way composite slab system

6.3.6.6.1 Composite slab (see Fig. 8), also termed as half slab is a precast slab system cast in two stages. The first stage is to manufacture thin precast concrete panels, typically of about 60 mm to 85 mm thick, with the slab's bottom reinforcement and projected shear reinforcement in the form of truss. The projected reinforcement facilitates for multiple point lifting/handling as well as acts as shear connectors for in-situ deck concreting.

6.3.6.6.2 The slab panels shall be designed for overall thickness (precast and in-situ deck) as one-way or two-way in accordance with provisions of 5 of Part C 'Structural Design, Section 1 Loads and Load Effects' of this NBCS, and Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS.



6.3.7 Facade Systems

Precast facade system shall be made up of following types of non-load bearing cladding panels:

- a) Precast insulated concrete panels;
- b) Precast solid cladding panels;
- c) Architectural precast decorative facade panels; and
- d) Precast concrete fins.

All façade connections shall comply with the requirements of the good practice [C-7(18)] and [6.3.2](#). Design shall account for differential movement, panel dead load and seismic restraint.

6.4 Hybrid System

The following systems are considered in precast hybrid system:

- a) Precast concrete seismic structural system (PRESSS) (utilization of bonded or non-bonded prestressing tendons); and
- b) Mixed system (utilization of cast-in-situ along with prestressed concrete (PC) members or steel members).

6.4.1 Precast Concrete Seismic Structural System (PRESSS)

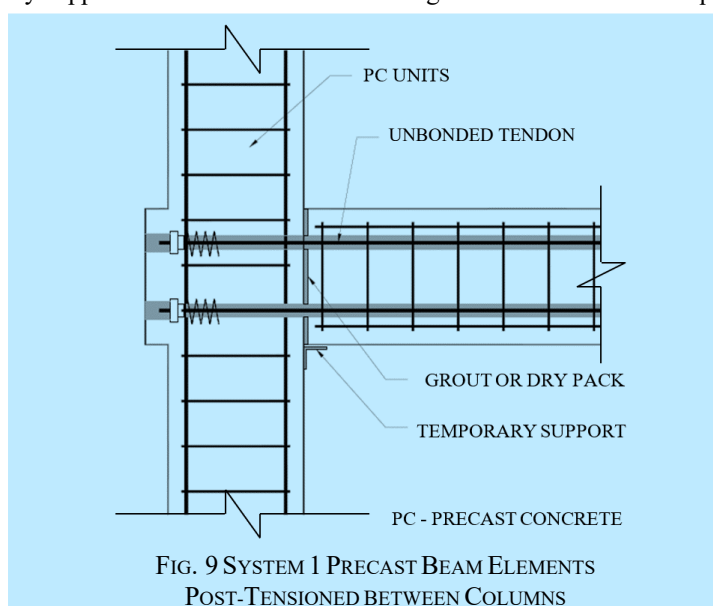
Precast concrete seismic structural system (PRESSS) shall be implemented as an equivalent monolithic system, consisting of precast elements with grouted joints and bonded post-tensioned tendons. The main advantages of post-tensioning are deflection recovery and immediate occupancy of the structure. Two different system of PRESSS are explained in [6.4.1.1](#) and [6.4.1.2](#) and application of this system shall be carried out by conducting successful full-scale testing, verification through nonlinear analysis and approvals from competent authorities or institutes of repute.

6.4.1.1 System 1 precast beam elements post-tensioned between columns

System 1 shall use precast beams post-tensioned between multi-storey columns. Beam length shall be shorter than clear span (20 mm or tolerance requirements, whichever is greater). Post-tensioning and joint grouting shall occur after beam placement (see [Fig. 9](#)).

6.4.1.1.1 The beams shall be temporarily seated on steel brackets or props. Grout shall then be placed in the gap between beam ends and column faces, followed by post-tensioning operation and grouting of tendons.

6.4.1.1.2 The temporary supports shall be removed once the grout has achieved the required design strength.



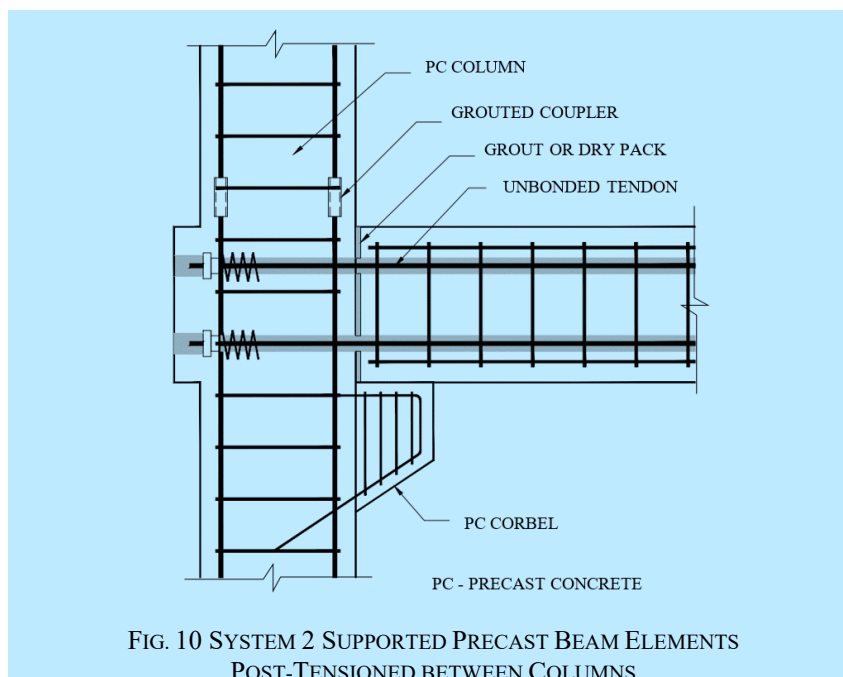
6.4.1.2 System 2 supported precast beam elements post-tensioned between columns

In this system, the beams shall be seated on corbels that transfer gravity loads to the adjacent columns. The beams shall be either built prismatic or with dapped ends, depending on the position of the corbel (see [Fig. 10](#)) and then post-tensioned.

6.4.1.2.1 The columns in this system shall normally be one-storey in height.

6.4.1.2.2 Columns shall be connected through a mechanical coupler or a grouted steel sleeve.

6.4.1.2.3 This system shall be adopted for buildings designed to demonstrate elastic response, with couplers giving continuity between the column elements, this is utilized for ductile moment resisting frames.



6.4.2 Mixed System (Cast In-Situ Precast or Structural Steel-Precast)

This system shall use pretensioned prestressed concrete or reinforced concrete beam shell units or prestressed concrete beams as permanent formwork for cast in-situ beams or structural steel beams (as shown in [Fig. 11](#)). Application of this system shall be carried out by conducting successful full-scale testing, nonlinear analysis and approvals from competent authorities or institutes of repute.

6.4.2.1 The precast shells shall typically be left permanently in position after the cast-in-place reinforced concrete core has been cast. The shells can be made from fibre reinforced concrete or reinforced concrete, depending on the design requirements.

6.4.2.2 The precast U-beams or structural steel beams shall be designed to carry self-weight and construction loads and act compositely with the reinforced concrete core when subjected to other loading in the completed structure.

6.4.2.3 Precast U-beams may not be connected by reinforcement to the cast-in-place concrete of the beam or column. If the interfacial shear is found to be not sufficient then shear connector (protruding stirrups or ties) shall be provided between the interfaces.

6.4.2.4 The design shall ensure that the composite beams shall develop plastic hinges during major earthquakes.

6.4.2.5 All precast shell units shall be free of laitance, oil, dust and loose debris before placement of cast in-situ concrete. Construction sequences shall ensure adequate bond between interfaces.

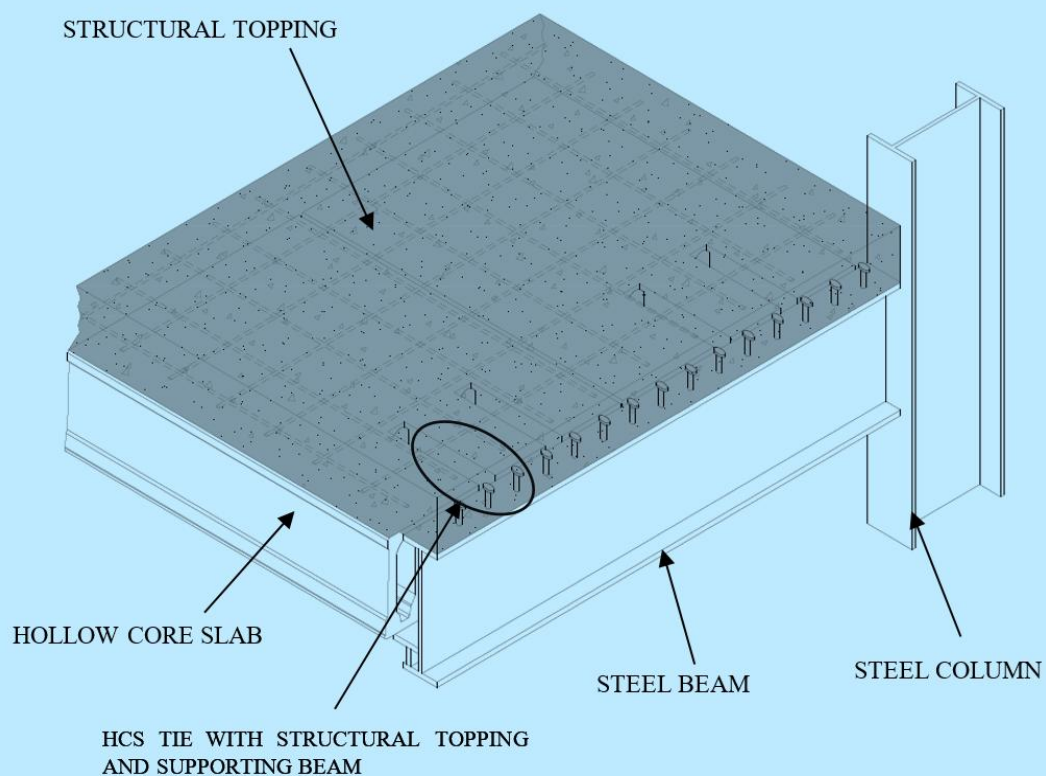
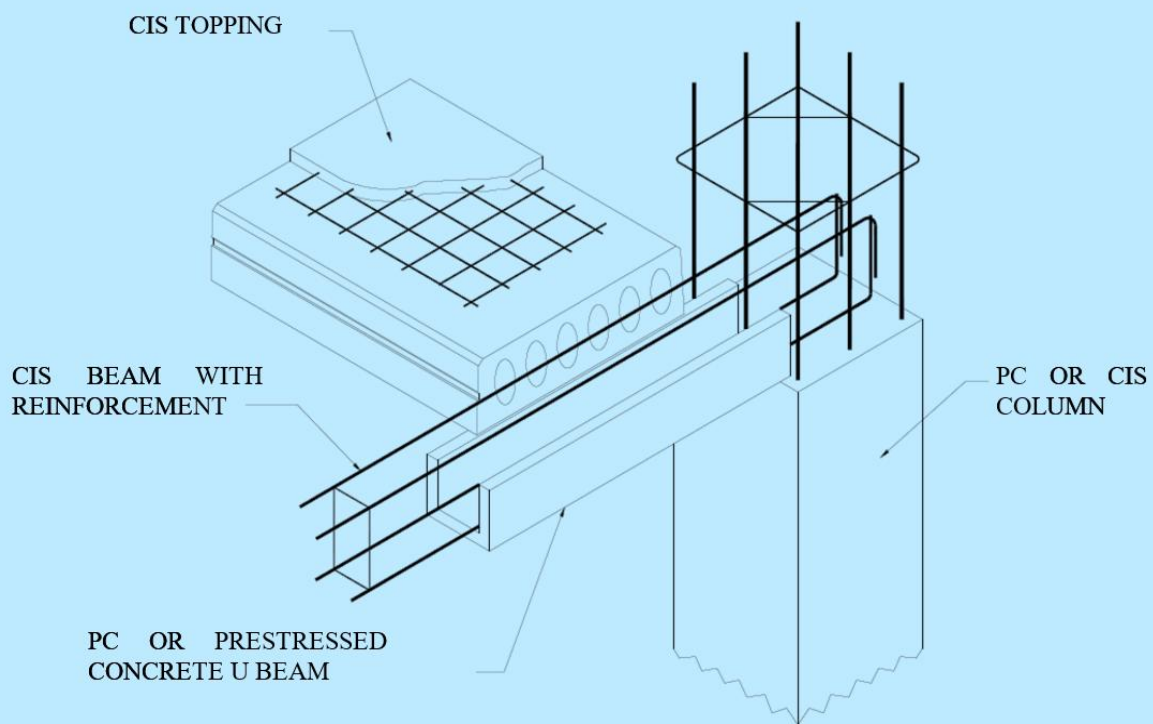


FIG. 11 MIXED SYSTEM

6.5 Advanced Precast Systems

Advanced precast systems shall include modular 3D precast elements, digitally fabricated units and prefabricated structural modules integrated via design for manufacturing and assembly (DFMA) principles.

6.5.1 3D Printed System

3D printed precast systems shall only be used upon approval from the competent authority and shall follow [Annex B](#) for design, durability and testing criteria.

6.5.2 Cell Systems

Cell systems shall consist of volumetric precast units fabricated as open or closed modules. Connections to surrounding walls and floors shall be verified for strength and serviceability. [Annex C](#) shall be referred for further details.

Cell elements for specific area of a structure, such as wet zones (for example, toilets and kitchens), shall be used and integrated with wall and floor systems with appropriate connections.

6.6 Design for Manufacturing and Assembly (DFMA) Method

DFMA-based methods shall conform to [Annex D](#). Designers shall ensure that precast elements are standardized, repeatable and modular to reduce rework and improve assembly efficiency.

7 STRUCTURAL ANALYSIS

7.1 General

Structural analysis shall be carried out to evaluate internal forces and deformations in members and the structural system. The analysis shall verify compliance with strength, serviceability, stability and robustness criteria. These effects include the distribution of forces and moments as well as the calculation of stresses, strains, curvatures, rotations and displacements under both static and dynamic conditions. Advanced analysis methods, such as nonlinear pushover analysis and time history analysis, may be adopted based on project requirements. However, these methods shall be mandatory for irregular structures, critical facilities and structures located in high seismic regions. For sustainable design aspects refer to [G-6](#).

7.1.1 All the members and structural system shall be analysed to determine the critical load effects as per methods given in Part C 'Structural Design, Section 5 Structural Concrete' and relevant clauses of this Section.

7.1.2 To carry out analysis, the geometry, boundary conditions and behaviour of the structure need to be idealized both for global and local behaviour. The structure is idealized by considering it as made up of elements/components which can be linear, two dimensional or three dimensional.

7.1.3 Structural analysis shall use methods capable of capturing action effects with sufficient accuracy. These may include classical mechanics or numerical methods such as finite element analysis, provided the model appropriately represents the actual structure.

7.1.4 Precast structures shall be analysed as either emulative systems, where precast performance matches that of monolithic CIS structures, or jointed systems with discrete joint behaviour. Emulative models shall assume monolithic behaviour with joints designed for equivalent force transfer. Lateral force resistance shall be achieved through moment-resisting joints or orthogonally placed structural walls or as per the design requirements.

7.1.5 For precast bearing wall buildings, rotational stiffness of floor-wall and floor-beam joints shall be neglected in structural design. Each component shall be independently designed considering the applicable end conditions, structural loads as specified in [7.2](#) and the various stages of construction.

7.1.6 Members shall also be designed to resist handling, erection and impact loads expected during the transportation and erection phases [7.14](#).

NOTES

1 Jointed construction using welded and bolted connections is not covered in this NBCS. This may be adopted with approval from competent authority and appropriate testing.

2 Rotational stiffness may be accounted for long/short term deflection and serviceability calculations as per [6.3.4](#), provided that the approach is mutually agreed by the design team and approved by the Authority.

3 Horizontal precast members that transfer gravity loads but are not part of the lateral force resisting system (LFRS) shall be designed assuming pinned end conditions, unless justified otherwise through analysis.

4 Linear model, non-linear model and P-Delta analyses may be used for design, provided the analysis method, assumptions and modelling parameters are clearly defined and justified and approved by the Competent Authority.

7.2 Loads and Forces

The design shall consider all the loading requirements as specified in Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS. Consideration shall also be given to construction loads, any special requirements, such as for plant loads or storage loads and the loading condition should be modified accordingly.

7.2.1 Structural design shall take into consideration the dead loads, live or imposed loads, wind loads, forces such as those caused by earthquake effects, prestressing effects, crane loads, vibration, impact, natural disasters (flood, landslide, etc), blast and effects due to shrinkage, creep, temperature, etc, where applicable.

7.2.1.1 Dead load

Dead loads shall be calculated based on unit weights which shall be established taking into consideration the materials specified for construction or **2** of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS.

7.2.1.2 Imposed load

Imposed loads shall be considered as per **3** of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS. A minimum construction load (as per design requirement) of 1.0 kN/m² or construction traffic load shall be adopted for design, if not specified in the contract document. However, due consideration shall be given to actual possible construction loads.

7.2.1.3 Wind load

Wind load shall be considered as per **4** of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS.

7.2.1.4 Construction traffic load

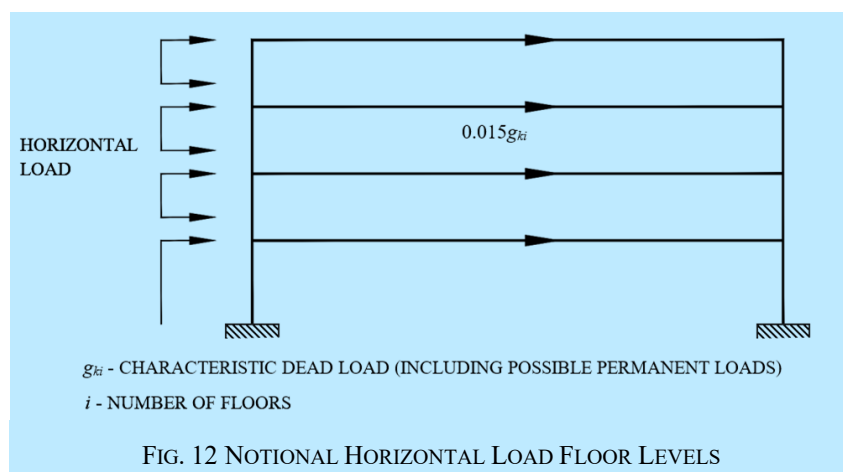
Construction traffic load shall be assumed in accordance with the project requirements. A minimum traffic load (as per design requirement) of 1.0 kN/m² or imposed load shall be adopted for design, if not specified in the contract document.

7.2.1.5 Seismic load

The seismic forces shall be calculated in accordance with **5** of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS.

7.2.1.6 Notional horizontal load

Notional horizontal load (as shown in [Fig. 12](#)) shall be taken as not less than 1.5 percent of the characteristic dead load and accidental loads such as from earth movement, impact of construction vehicles, etc.



7.2.1.7 Shrinkage, creep and temperature effects

The effects of shrinkage, creep and temperature shall be considered according to the good practice [C-7(2)]. Special literature can be referred to for critical structures.

7.2.1.8 Environmental loading

The effects of thermal loading and changes in values of creep and shrinkage due to environmental exposure shall be considered according to the good practice [C-7(2)] or as defined in design basis report.

7.2.1.9 Other forces and effects

Relevant clauses of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS shall be followed for additional forces such as snow load, liquid retaining loads, impact due to equipment, cranes, etc.

7.2.2 Load Combinations

Basic load combination shall be as per the good practice [C-7(2)] and Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS. In addition, precast specific load combinations specified by the competent authority or given in this NBCS shall be adopted.

7.2.2.1 Factor of safety/partial safety factor for load combinations

Load and material safety factors shall be as per the good practice [C-7(2)]. Precast specific partial safety factors given at various sections of this NBCS shall be adopted accordingly.

7.3 Effective Span

7.3.1 Except for the purely simply supported spans, the effective spans for all elements shall be estimated as per the relevant clauses of the good practice [C-7(2)].

7.3.2 Effective span for all elements during the temporary erection stage are to be considered based on the temporary support condition.

7.3.3 The effective span for simply supported elements shall be considered as the centre-to-centre distance between the bearing pads or shims at each end. If the multiple shims are used, only the shims or pads close to the end of the elements shall be considered as governing case.

7.4 Stiffness

7.4.1 Relative Stiffness

The relative stiffness of the member shall be calculated based on the moment of inertia of the section considering gross section, transformed section and cracked section as provided in the good practice [C-7(2)].

7.4.2 Stiffness Modifiers

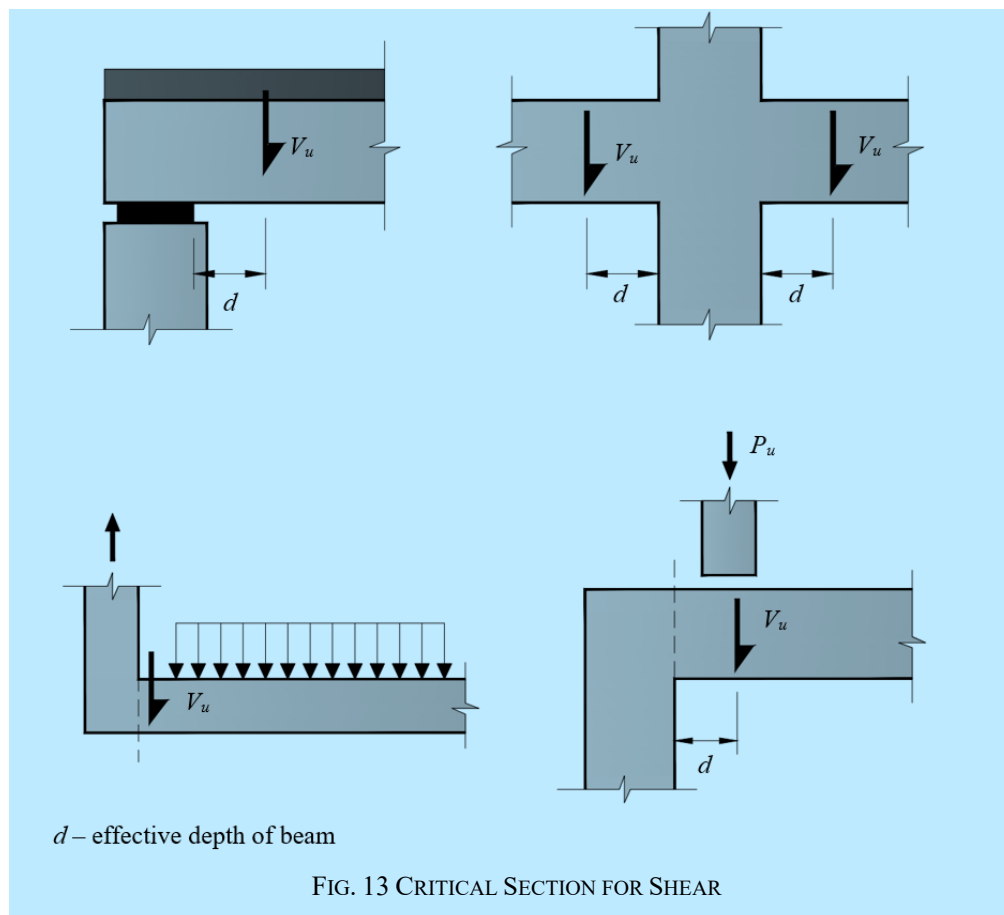
Stiffness reduction factors for members and connections shall be applied as per 5 of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS, considering structural system type and ductility requirements.

7.4.3 Critical Sections for Moment and Shear

The critical sections for moment and shear shall be identified as per the good practice [C-7(2)] and additionally as per [7.4.3.1](#) and [7.4.3.2](#).

7.4.3.1 Critical section for shear as per the good practice [C-7(2)] is shown in [Fig. 13](#).

7.4.3.2 For L-beams and inverted T-beams carrying loads within the bottom 30 percent of section depth (refer [Fig. 13](#)), the critical section for shear shall be taken at the face of support, unless a monolithic or emulative beam-column joint is specifically detailed to resist shear.



7.5 Concentrated Loads on Floor Slabs

7.5.1 Concentrated loads shall be applied as per Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS. Effective width for slabs carrying concentrated loads shall be calculated as per the good practice [C-7(2)] along with 7.5.2 and 7.5.3.

7.5.2 The load distribution for hollow core slab depends on the one-way action of the plank and the cast in place structural topping. The effective width for hollow core slabs shall be calculated as per [7.5.3](#).

7.5.3 Moments and shears induced by line and concentrated loads shall be resisted by an effective section as described in [Fig. 14](#). If the total deck width, perpendicular to the span, is less than the span, modification may be

required. In addition, the following limitations would apply:

- As the width of the system becomes narrower than the span length, the effective resisting widths will become narrower;
- For extremely high span-depth ratios (more than approximately 50), the effective section at mid-span shall be reduced by 10 percent to 20 percent;
- For spans less than about 3 m, the effective width at the support may become narrower; and
- Local load concentrations can cause longitudinal splitting failures due to transverse bending in the system and punching shear failure. The magnitude of concentrated loads shall be limited to minimise such failures.

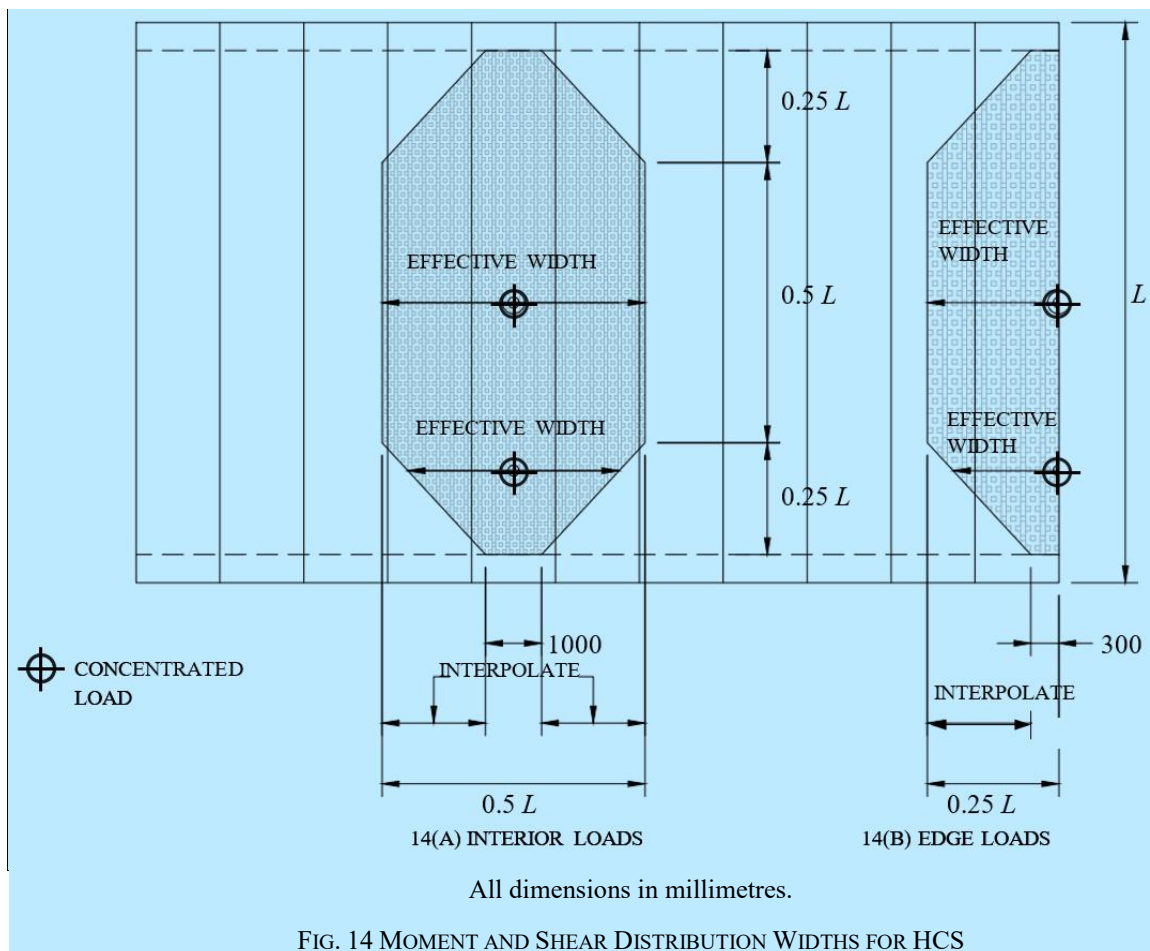


FIG. 14 MOMENT AND SHEAR DISTRIBUTION WIDTHS FOR HCS

7.6 Stability Design

7.6.1 Design Philosophy

The following factors shall be considered in designing for stability of a precast structure:

- The overall structural system shall be checked for global and local stability under all applicable loading conditions including temporary and final stages;
- Precast structural systems shall be designed to resist wind and all applicable lateral loads;
- Cross walls or sway frames shall be so arranged to provide lateral stability;
- When lateral loads are low, stability shall be verified using a notional horizontal force equal to 1.5 percent of total vertical load, as shown in [Fig. 12](#);
- Lateral stability shall be maintained during all construction and erection stages. Temporary bracing, shoring, or props shall be detailed and installed to match the staged loading conditions. A temporary

works scheme including propping layout, sequencing and dismantling plan shall be submitted and approved before execution;

- f) Inverted T and L-beams shall be analysed for asymmetric loading during both construction and service stages, using all applicable load combinations;
- g) The strength of grout shall be at least 50 percent of the design strength or as per design requirement before application of any other loads (temporary or permanent) on the element. Stress and cracking shall be evaluated and verified before proceeding;
- h) Long-span and high-rise precast structures shall include engineered bracing and verified stability measures during all construction and service stages; and
- j) Temporary anchors shall be designed for proper alignment and safety of the elements. Appropriate washers shall be used.

7.6.1.1 Overturning

Stability against overturning shall be checked in accordance with the requirements in Part C/Section 5 of this NBCS. For L and inverted T beams, lateral support shall be provided during construction to avoid stability related failures.

7.6.1.2 Sliding

Stability against sliding shall be checked in accordance with the requirements in the good practice [C-7(2)]. All the precast members during construction shall be propped or anchored to prevent sliding from the supporting structure. Minimum bearing as per [7.11](#) shall be followed.

7.6.1.3 Probable variation in dead load

Probable variations in dead load during casting of in-situ components shall be accounted for in the design. Pouring of structural topping above the long span elements (such as HCS) shall be monitored to have the required pouring depth measured from top of the highest point of the element (camber point) instead of global level.

7.6.1.4 Moment connection

Moment connection shall be used for lateral force resistance system frames. Where frames are pin-jointed, transverse stability shall be ensured through structural walls or bracing system. Integrity ties shall be provided to maintain system continuity.

7.6.1.5 Lateral sway

Stability against lateral sway shall be evaluated using the good practice [C-7(2)] and 5 of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS, including during temporary construction stages and final occupancy.

7.6.1.6 Lateral force resistance system design

The lateral force resistance system design shall be in accordance with the relevant provisions of Part C ‘Structural Design, Section 1 Loads and Load Effects’, Part C ‘Structural Design, Section 5 Structural Concrete’ and Part C ‘Structural Design, Section 6 Steel and Steel-Concrete Composite Construction, Subsection A Steel’ of this NBCS with the guidelines stipulated in 6. Additional impact of lateral loads such as earth fill, or liquid/snow retention shall be analysed as per the project requirements.

7.7 Durability Design

7.7.1 Design Philosophy

The following factors shall be considered for achieving the design life and durability of a precast structure:

- a) Geometry and detailing of precast elements;
- b) Concrete mix design, including constituent materials;

- c) Nominal and minimum cover as per environmental class;
- d) Environmental exposure classification in accordance with the good practice [C-7(2)];
- e) Fire resistance rating based on intended use;
- f) Access provisions for inspection and maintenance;
- g) Controlled production quality;
- h) Handling, transport and erection constraints; and
- j) Watertight and durable joint detailing.

In addition to above requirements, the recommendations specified in the good practice [C-7(2)] and relevant accepted standards shall be followed.

7.7.2 Service Life

7.7.2.1 The service life of any structure shall be specified in the design basis report (DBR) or considered based on the good practice [C-7(2)].

7.7.2.2 For temporary structure, the service life shall be considered as the period the structure is required to satisfy the intended function, or as mentioned in the contract document.

7.7.2.3 The service life specified for a structure does not apply to architectural finishes, replaceable materials such as sealants, paints, non-structural grout. The service life of such items shall be considered as per the manufacturer's guidelines and mutually agreed upon by all the stakeholders.

7.7.3 Shape of Precast Units

7.7.3.1 Sharp corners or sudden changes in geometry, re-entrant corners, or grooves in a section causes stress concentrations that may lead to cracking or spalling of concrete and therefore shall be avoided. For sections with practical constraints, stress concentrations shall be checked and strengthening or local detailing shall be provided, as necessary.

7.7.3.2 Extra reinforcements shall be provided around the cut-outs or opening, if any, present in an element. Larger cut-outs shall be considered in the analysis to characterize the stress concentration and its effects on the design as per the good practice [C-7(2)].

7.7.3.3 Buckling and instability must be prevented during the lifting and erection of long, slender precast units, as per the good practice [C-7(2)]. For thin sections (thickness less than 75 mm), the positioning of lifting inserts should be such that it prevents buckling of the compression flange, especially during the handling and manoeuvring of the elements.

7.7.3.4 The precast unit shall be designed and detailed to have sufficient drainage to avoid standing water or excessive moisture.

7.7.4 Concrete Cover

Concrete cover for precast concrete elements shall comply with the requirements of the good practice [C-7(2)], unless reduced values from [Table 2](#) are adopted. Reduced cover values shall only be permitted if the elements are produced in a controlled environment.

7.7.4.1 The minimum concrete cover for reinforcement to ensure durability against fire and corrosion shall comply with the provision of the good practice [C-7(2)]. Fire resistance of joint fillers and other components shall meet or exceed the fire resistance rating of the connected precast members, or as prescribed in [7.8](#).

7.7.4.2 The cover to all brackets, fixings, etc shall comply with the minimum cover requirements specified for reinforcement as per the above table or the good practice [C-7(2)].

Table 2 Nominal Cover in Precast Elements

(Clause 7.7.4)

Sl No.	Exposure	Elements	Nominal Concrete Cover, (mm, Min)
(1)	(2)	(3)	(4)
i)	Mild	All	20
ii)	Moderate	Slabs and walls	25
		For the ties/stirrups of beams and columns exposed to weather	25
		For ties/stirrups of beams and columns non-exposed to weather	Greater of d_b or 20
iii)	Severe	All	40
iv)	Very Severe	All	45
v)	Extreme	All	70
NOTES 1 The prescribed covers for precast elements are lesser than the covers noted in the good practice [C-7(2)] to account for the greater control for proportioning, placing and curing inherent to precast concrete produced in a controlled environment. 2 Controlled environment does not imply that the precast members shall be manufactured in a factory. Elements manufactured at the jobsite or in a temporary factory also qualify under these requirements, if the controls are equal to that normally adopted in a factory. The client and engineer-in-charge shall verify the temporary factory and certify the same, if the cover requirements of this Table are adopted. 3 d_b is bar diameter.			

7.7.5 Protection and Maintenance of Joints and Connections

7.7.5.1 The connection between the precast element shall be properly filled with high strength, non-shrink, non-metallic cementitious grout (minimum 50 MPa or 5 MPa more than the strength of the element, whichever higher) to prevent corrosion, cracking and spalling of concrete. Where the joint gap between the precast elements is more than 50 mm, the provisions of 5.4.1.3 shall apply.

7.7.5.2 Steel plates or inserts used in permanent connections shall be provided with adequate concrete cover or protected with corrosion-resistant coatings. Inserts for temporary use shall be removed or treated to prevent corrosion.

7.7.5.3 Where minimum cover cannot be ensured due to geometric or project constraints, only corrosion-resistant materials (such as stainless steel or galvanized steel) shall be used for connectors and fixings.

7.7.6 Other Effects

7.7.6.1 Joint spacing and width shall account for cumulative movements including production tolerances, creep, shrinkage, settlement and thermal expansion to prevent cracking and spalling.

7.7.6.2 In addition to satisfying general requirements for durability, strength, stability, control of cracking and limitation of deformation, the following provisions shall be met:

- Structural components shall be designed to limit cracking and deformation from early thermal movements, creep and shrinkage; and

- b) Bearings or movement joints shall be provided to minimize structural restraints. Where restraint is unavoidable, the design shall include measures to resist resulting stresses, with reinforcement provided as per as per the good practice [C-7(2)].

7.8 Fire Resistance

7.8.1 Fire Resistance Requirement

7.8.1.1 The fire resistance of a structural element shall be expressed in terms of time in hours in accordance with good practices [C-7(19)]. General requirements for fire protection are given in good practice Part F 'Fire and Life Safety' of this NBCS.

7.8.1.2 Elements and joints wider than 50 mm shall comply with fire resistance provisions of the good practice [C-7(2)], except where otherwise specified in this section.

7.8.1.3 All joints in fire-prone structures shall be filled with mortar, grout, or concrete in accordance with the design requirements.

7.8.1.4 The joints between slab elements or hollow core elements with structural topping of more than 50 mm (minimum thickness of structural topping concrete) may not require any additional treatment for fire rating.

7.8.1.5 For joints between structural wall elements with a width less than or equal to 50 mm shall be filled with grout of strength minimum 50 MPa or 5 MPa more than the strength of the element, whichever higher. For non-structural wall elements, the grout strength shall be at least equal to that of the element. For joints with a width exceeding 50 mm, the provisions of [5.4.1.3](#) shall apply.

7.9 Structural Design of Precast Concrete Components

7.9.1 Footings

7.9.1.1 All the design and detailing requirements of the good practice [C-7(2)] shall be followed for members/elements along with any specific criteria given in this Section.

7.9.1.2 Precast footing with multiple elements stitched using appropriate connection arrangement shall be adopted, wherever required. However, footing subjected to vibration shall be without any joints, preferably a single unit or cast in-situ or the system shall be tested prior to application for the required level of vibration.

7.9.1.3 The minimum thickness at the edge of a precast footing shall be as per the good practice [C-7(2)].

7.9.1.4 The connections shall be detailed as per [8](#) to ensure effective force transfer between elements.

7.9.1.5 Detailing of column, pedestal and wall connections to the footing elements shall be as per [8](#).

7.9.2 Walls

7.9.2.1 Design and detailing of wall elements shall comply with the good practice [C-7(2)], in addition to the specific requirements of this Section.

7.9.2.2 As the elements are manufactured in controlled environment, walls subjected to marginal shear forces, minimum longitudinal and transverse reinforcement shall be 0.1 percent of gross area. For other cases, reinforcement shall be as per the good practice [C-7(2)].

7.9.2.3 The connections shall be detailed as per [8](#) and shall ensure proper force transfer between the elements.

7.9.2.4 Walls shall be tied to intersecting elements such as floors, columns, other walls and footings using appropriate ties or mechanical connections that ensure force transfer and structural continuity.

7.9.3 Shear Wall

7.9.3.1 All the design and detailing requirements of the good practice [C-7(2)] shall be followed for members/elements along with any specific criteria given in this Section.

7.9.3.2 Minimum reinforcement for structural wall shall be as per the good practice [C-7(2)] and as per good practice [C-7(18)].

7.9.3.3 At least one continuous vertical tie of 12 mm diameter or higher shall be provided from foundation to terrace at each end of the structural wall using Class H coupler as per [C-7(9)] or any other connection with appropriate lapping or welding as per the design requirements.

7.9.3.4 The connections shall be detailed as per [8](#) and to ensure adequate transfer of axial, shear and bending forces between connected elements.

7.9.3.5 Shear walls shall be structurally tied to intersecting elements, including floors, columns, other walls and footings, using mechanical ties, dowels, or other engineered connections capable of transferring expected design loads.

7.9.4 Floors/Slabs

7.9.4.1 All the design and detailing requirements of the good practice of [C-7(1)] shall be followed for members/elements along with any specific criteria given in this Section.

7.9.4.2 Minimum reinforcement for slab elements and structural topping the minimum reinforcement shall be as per good practice [C-7(1)].

7.9.4.3 The thickness of structural topping shall be considered if the maximum shear stress at any location is less than 0.4 MPa. Also, the top surface of the slab shall be roughened to a minimum amplitude of 6 mm. Where the shear stress exceeds 0.4 MPa, adequate shear reinforcement or studs shall be provided to ensure the composite action between the topping and the floor slab elements.

7.9.4.4 The minimum grade of concrete for the structural topping shall be equal to M30 or as per design (whichever is higher).

7.9.4.5 The connections shall be detailed as per [8](#) and shall ensure proper force transfer between the elements.

7.9.4.6 For pre-tensioned and post-tensioned members, stress limits shall be as per good practice [C-7(3)].

7.9.4.7 For simply supported slab at least one third of the maximum positive reinforcement shall be extended till the centre of the bearing. For other cases, at least one fourth of the reinforcement shall extend into the support for a length of 150 mm. Adequate development length shall be provided to the bottom reinforcement as per good practice [C-7(2)]

7.9.4.8 For requirements relating to design and detailing for precast structures with hollow core slabs, [Annex H](#) may be referred.

7.9.5 Columns

7.9.5.1 All the design and detailing requirements of the good practice [C-7(1)] shall be followed for members/elements along with any specific criteria given in this NBCS.

7.9.5.2 Minimum reinforcement requirements shall be as per the good practice [C-7(2)].

7.9.5.3 The connections shall be detailed as per [8](#) and shall ensure proper force transfer between the elements.

7.9.5.4 At least four continuous vertical ties of 12 mm diameter or higher shall be provided from foundation till the roof using Class H coupler as per accepted standard [C-7(9)] or any other connection, with appropriate lapping or welding.

7.9.5.5 Transverse reinforcement as per the good practice [C-7(2)] shall be provided in the members as well as the splice locations (grouted coupler, grouted duct, mechanical connections, etc).

7.9.5.6 For connection using grouted coupler of height less than 250 mm, minimum cover to the confinement reinforcement shall be 25 mm at the location of coupler. For all other cases cover shall be as per the good practice [C-7(2)].

7.9.5.6 For connection using grouted duct/tube or any other, minimum cover shall be as per the good practice [C-7(2)].

7.9.6 Beams

7.9.6.1 All the design and detailing requirements of good practice [C-7(1)], [C-7 (2)] and [C-7(3)] shall be followed for members/elements along with any specific criteria given in this Section.

7.9.6.2 Minimum reinforcement requirements shall be as per the good practice [C-7(2)].

7.9.6.3 The connections shall be detailed as per [8](#) and shall ensure proper force transfer between the elements.

7.9.6.4 Transverse reinforcement as per good practice [C-7(2)] shall be provided in the members as well as the splice locations (grouted coupler, grouted duct, mechanical connections, etc).

7.9.6.5 For connection using mechanical coupler of length less than 500 mm, minimum cover of 20 mm shall be maintained to the transverse reinforcement at the coupler location. In all other cases the cover shall be as per the good practice [C-7(2)].

7.9.6.6 For beam made composite to the cast in-situ topping (T-action or L-action, in compression), the thickness of cast in-situ shall be considered in the height of the beam and the width of flange shall be calculated according to the good practice [C-7(2)].

7.9.6.7 Stress limits for pre-tensioned and post-tensioned beams shall be as per good practice [C-7(3)].

7.9.6.8 For simply supported beam at least one third of the maximum positive reinforcement shall be extended till the centre of the bearing. For other cases, the reinforcement shall extend to the other member or to the support and shall develop f_y at the tension face of the support.

7.9.7 Staircase/Landing

7.9.7.1 All the design and detailing requirements of Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS shall be followed for members along with any specific criteria given in this Section.

7.9.7.2 Minimum reinforcement requirements shall be as per Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS and shall ensure proper force transfer between the elements.

7.9.7.3 The connections shall be detailed as per [8](#).

7.9.8 Composite Concrete Construction

7.9.8.1 All the design and detailing requirements of Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS shall be followed for members along with any specific criteria given in this Section.

7.9.8.2 The connections shall be detailed as per [8](#) and shall ensure proper force transfer between the elements.

7.9.9 Insulated Panels/Thermal Comfort

7.9.9.1 All the design and detailing requirements of Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS shall be followed for members along with any specific criteria given in this Section.

7.9.9.2 Minimum thickness of the concrete wythe shall be 50 mm.

7.9.9.3 Individual wythe shall be connected to each using wythe connectors by providing solid zones or special reinforcements.

7.9.9.4 Maximum spacing between the reinforcements in the wythe shall be 250 mm or as per design.

7.9.9.5 For walls made from composite wythe, the solid zone shall be designed to transfer the shear demand on the wall and for moment transfer complete composite wythe section shall be considered.

7.9.9.6 Insulation material if provided shall have adequate bond with the concrete wythes to avoid splitting of junction during handling.

7.9.9.7 The connections with other elements shall be detailed as per [8](#) and shall ensure proper force transfer between the elements.

7.9.10 Composite Floors

7.9.10.1 All the design and detailing requirements of Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS shall be followed for members along with any specific criteria given in this Section.

7.9.10.2 No shear reinforcement/studs shall be provided if the maximum shear stress at any location is less than or equal to 0.4 MPa.

7.9.10.3 The minimum grade of concrete for the structural topping shall be equal to M30 or as per design (whichever is higher).

7.9.10.4 The connections shall be detailed as per [8](#) and shall ensure proper force transfer between the elements.

7.9.11 Others Structural Elements

7.9.11.1 Any structural element shall be designed and detailed as per Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS.

7.9.11.2 The connections shall be detailed as per [8](#) and shall ensure proper force transfer between the elements.

7.9.11.3 Any connection scheme found effective in force transfer shall be used, subjected to approval of competent authority.

7.10 Design of Diaphragm

For the stability of the precast concrete building horizontal loads due to wind and earthquake shall be transmitted to the lateral force resistance system through the floor acting as horizontal deep beam. However, in a precast slab system, the floor comprises individual units generally not connected to each other. A structural topping concrete is provided to act as the diaphragm, or the individual elements are connected. In this system, the diaphragm action shall ensure the following:

- a) Collect and transfer lateral forces at each level to the lateral force resisting elements such as shear walls, moment frames, etc; and
- b) Tie individual lateral force resisting elements into a single lateral force resisting system.

7.10.1 Diaphragm action arises from the in-plane stiffness inherent to the floor system. As a result, substantial in-plane forces can develop in the diaphragm during lateral loading. In the context of precast concrete construction, the panelled nature of floor systems can pose challenges in the seismic design of diaphragms. These challenges may involve ensuring sufficient strength, stiffness, and in some cases, ductility in both the diaphragm and its connections.

7.10.2 In case of buildings whose floors can provide rigid horizontal diaphragm action, the total shear in any horizontal plane shall be distributed to the various vertical elements of lateral forces resisting system, assuming the floors to be infinitely rigid in the horizontal plane.

7.10.3 In case of building whose floor diaphragms cannot be treated as infinitely rigid in their own plane, the lateral shear at each floor shall be distributed to the vertical elements resisting the lateral forces, considering the in-plane flexibility of the diaphragms.

NOTE — Reinforced concrete monolithic slab-beam floors or those consisting of prefabricated/precast elements such as solid slabs, hollow core slabs or double tees with reinforced topping shall satisfy the requirements of [7.10.10](#). In all cases, the precast elements need not be considered as part of the diaphragm except solid slabs with proper connectivity. The gravity load design, however, shall be composite including the structural topping as well as the precast element provided the interface shear at the topping and element face is analysed and designed for the same.

7.10.4 Rigid Diaphragms

A diaphragm is classified as rigid if it can distribute the horizontal forces to the vertical components in proportion to their relative stiffness. Diaphragms of concrete slabs of concrete-filled metal deck with span-to-depth ratios of 3 or less in structures that have no horizontal irregularities are permitted to be idealized as rigid. The minimum thickness of structural topping concrete shall be as per [7.10.10](#).

Where the diaphragm is very stiff compared to the vertical elements, as in a low aspect ratio, cast-in-place diaphragm supported by moment frames, it is acceptable to model the diaphragm as a completely rigid element.

7.10.5 Flexible Diaphragms

A floor diaphragm shall be considered to be flexible, if it deforms such that the maximum lateral displacement measured from the chord of the deformed shape at any point of the diaphragm is more than 1.5 times the average displacement of the entire diaphragm or as per good practice [C-7(18)] and 5 of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS. Flexible diaphragm can be adopted in modular structures having frequent openings such as substation, electrical rooms, industrial structures, etc.

7.10.5.1 Where the diaphragm is flexible compared with the vertical elements, as in some jointed precast systems supported by structural walls, it may be acceptable to model the diaphragm as a flexible beam spanning between rigid supports.

7.10.5.2 In other cases, it may be advisable to adopt a more detailed analytical model to account for the effects of diaphragm flexibility on the distribution of displacements and forces. Examples include buildings in which diaphragm and vertical element stiffness have approximately the same value, buildings with large force transfers, and parking structures in which ramps connect between floors and act essentially as bracing elements within the building.

7.10.6 Topped Diaphragms

Topped floor and roof diaphragms include systems with in-situ concrete slab over precast RC planks and joists. The in-situ concrete slab is considered to act composite with the precast RC planks with sufficient surface roughness to transfer horizontal shear, so that the flexural strength of the precast component is enhanced by the added structural depth of the in-situ slab.

NOTE — In the computational model of the structure, the diaphragm shall be modelled considering the thickness of the topping/structural topping with extra dead load applied on top of the diaphragm to consider the precast planks or elements.

7.10.7 In-plane Diaphragm Design Loads and Forces

Topped floor and roof diaphragms shall be designed to resist design in-plane inertial forces due to earthquake forces and wind loads from the structural analysis. Earthquake forces shall be calculated in accordance with good practices [C-7(1)] and [C-7(20)] and 5 of Part C ‘Structural design/Section 1 Loads and load effects’ of this NBCS. Wind loads shall be estimated in accordance with Part C ‘Structural design/Section 1 Loads and load effects’ of this NBCS.

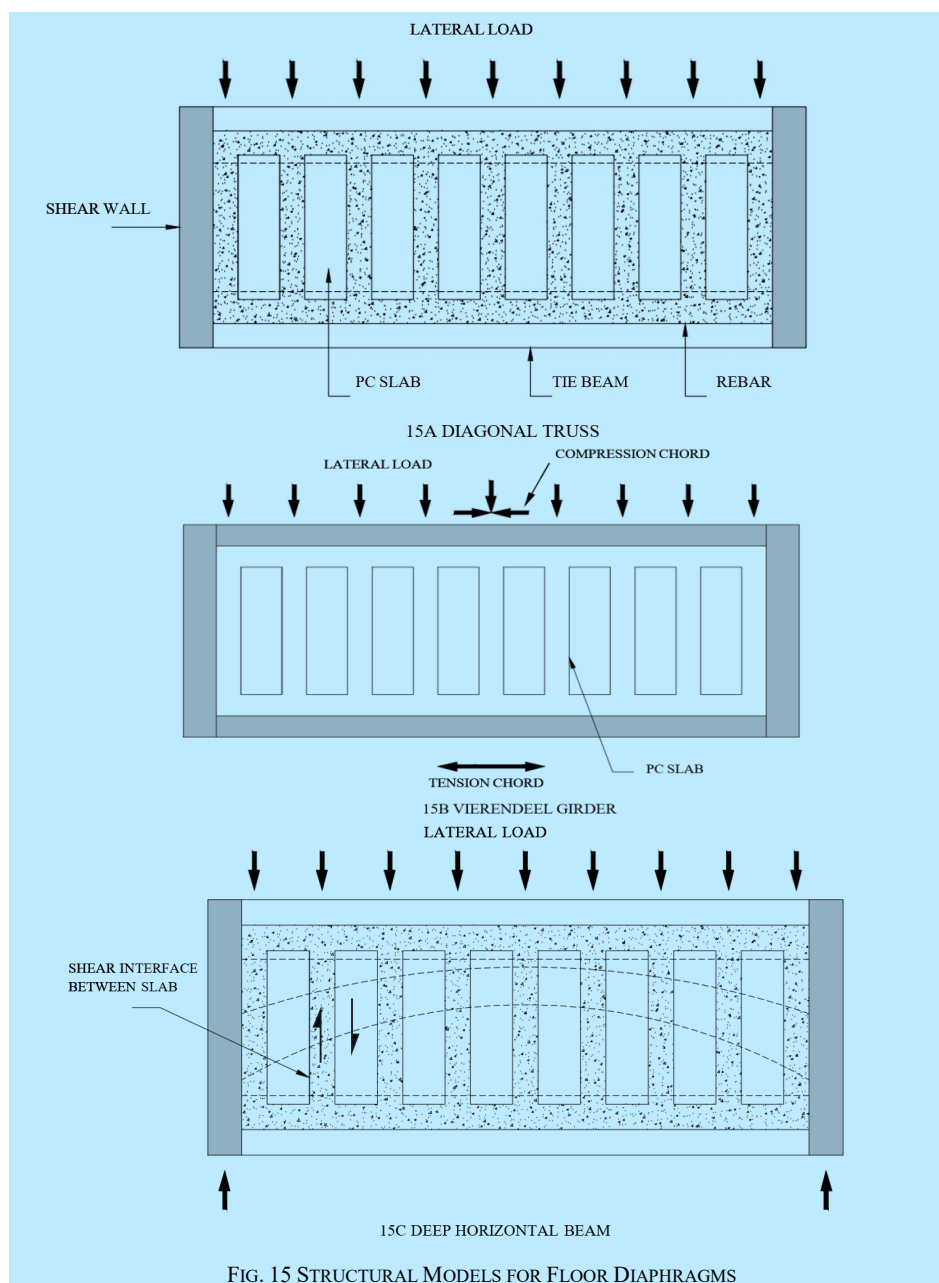
7.10.8 Standard Analysis Methods for Diaphragm Design

7.10.8.1 The magnitude of the inertia force at each floor level is estimated using a capacity design approach or a diaphragm design force distribution based on equivalent lateral loads. A key design step is to transform the diaphragm design forces into internal actions. For regular floor plans, a ‘horizontal plate girder’ analogy is typically used in design. In this approach, diaphragm design forces are typically applied at each level as a

distributed in-plane load along the length of the diaphragm. The resulting girder internal forces are used to determine the amount of flexural reinforcement (chord steel), shear (web) reinforcement, and lateral system reaction load path reinforcement (collector steel). An irregular mass distribution in plan may require the design forces to take on a smeared load configuration as the mass distribution with respect to diaphragm depth has relevance in determining the proper web reinforcement. For more complex diaphragm configurations, a strut-and-tie method can be used. The strut-and-tie method accurately predicts the behaviour of floor diaphragms, in general, and may become the preferred method of diaphragm analysis (see [Fig. 15](#) and [Fig. 16](#)).

7.10.8.2 Calculation of diaphragm in-plane design moments, shears and axial forces shall be consistent with requirements of equilibrium and with design boundary conditions. It shall be permitted to calculate design moments, shears, and axial forces in accordance with one of (a) through (d):

- A rigid diaphragm model if the diaphragm can be idealized as rigid;
- A flexible diaphragm model if the diaphragm can be idealized as flexible;
- A bounding analysis in which the design values are the envelope of values obtained by assuming upper bound and lower bound in-plane stiffness for the diaphragm in two or more separate analyses; and
- A finite element model considering diaphragm flexibility (semi rigid).



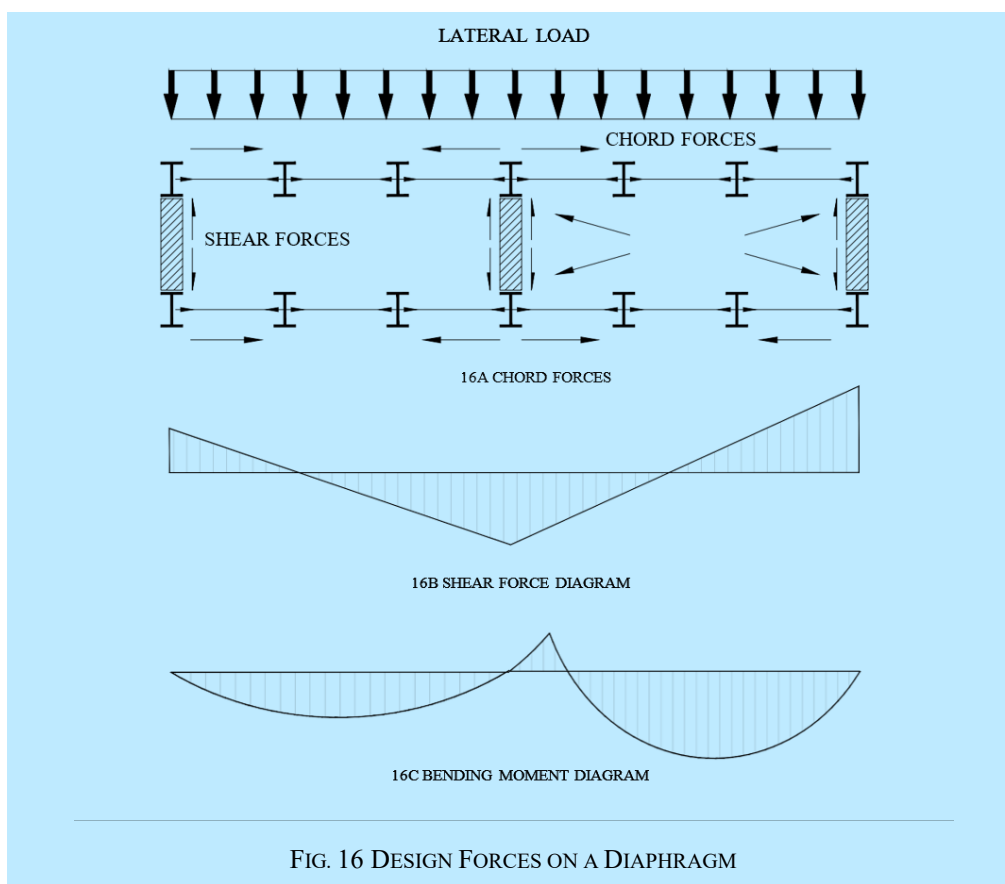


FIG. 16 DESIGN FORCES ON A DIAPHRAGM

7.10.9 Materials

7.10.9.1 Design properties of in-situ concrete for the topped diaphragms and diaphragm reinforcement shall be considered as per Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS.

7.10.9.2 Minimum grade of structural topping concrete shall be 40 MPa or as per design (whichever is higher).

7.10.10 Minimum thickness of topped diaphragms

7.10.10.1 Topped floor and roof diaphragms shall have thickness as required for stability, strength and stiffness under factored load combinations.

7.10.10.2 Topped floor and roof diaphragms that comprises RC planks composite with in-situ concrete slabs, the full thickness of the entire diaphragm must be sufficient to resist the design actions. In system with special precast slab elements such as hollow core slabs or double tees, the diaphragm thickness shall be limited to the in-situ structural topping concrete.

7.10.10.3 The minimum thickness of in-situ structural topping concrete shall be as per [6.3.6.1](#).

7.10.11 Diaphragm Connections

Connections between individual precast RC planks shall not be designed for any diaphragm forces in a topped diaphragm where the diaphragm thickness is limited to cast-in-place structural topping only. Nominal connections required to facilitate the vertical alignment of the RC planks and flanges shall be provided.

7.10.12 Opening and Voids in the In-situ Concrete Topping Slab

The effects of openings and voids in the in-situ concrete topping slab shall be considered in the diaphragm design. Detailing around the openings and voids must be done in accordance with Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS.

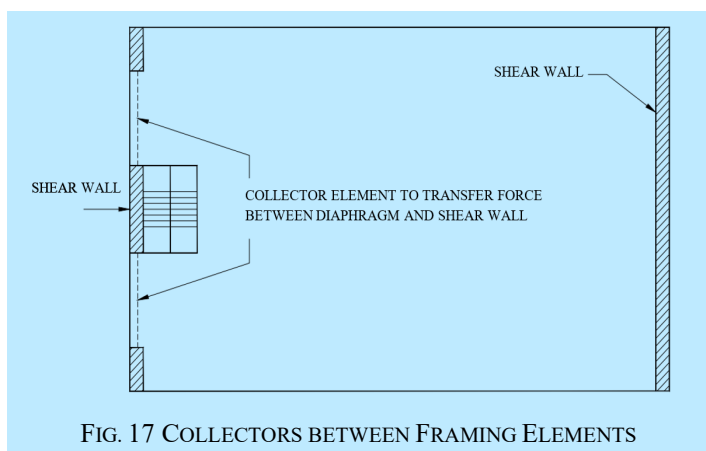
7.10.13 Collectors

7.10.13.1 Collectors shall extend from the vertical elements of the lateral-force-resisting system across all or part of the diaphragm depth as required to transfer shear from the diaphragm to the vertical element (as shown in Fig. 17). It shall be permitted to discontinue a collector along lengths of vertical elements of the lateral-force-resisting system where transfer of design collector forces is not required.

7.10.13.2 Collectors shall be designed as tension members, compression members, or both, in accordance with the criteria for axial member in Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS.

7.10.13.3 Where a collector is designed to transfer forces to a vertical element, collector reinforcement shall extend along the vertical element to at least the greater of (a) and (b):

- The length required to develop the reinforcement in tension; and
- The length required to transmit the design forces to the vertical element through shear-friction in accordance with 7.12, through mechanical connectors, or through other force transfer mechanisms.



7.10.14 Reinforcement Limits

7.10.14.1 Reinforcement to resist shrinkage and temperature stresses shall be estimated according to relevant clauses of Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS.

7.10.14.2 Reinforcement designed to resist diaphragm in-plane forces shall be in addition to reinforcement designed to resist other load effects, except that reinforcement designed to resist shrinkage and temperature effects shall be permitted to also resist the diaphragm in-plane forces.

7.10.15 Reinforcement Detailing

7.10.15.1 Concrete cover for reinforcement shall be in accordance with good practice [C-7(2)].

7.10.15.2 Development lengths of deformed and prestressed reinforcement shall be in accordance with good practice [C-7(2)] and [C-7(3)], respectively, unless longer lengths are required by high seismic zone requirements.

7.10.15.3 Splices of deformed reinforcement shall be in accordance with Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS.

7.10.15.4 Bundled bars shall be in accordance with Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS.

7.10.16 Reinforcement Spacing

7.10.16.1 Minimum spacing of reinforcement shall be in accordance with Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS.

7.10.16.2 Maximum spacing of deformed reinforcement shall be the lesser of five times the diaphragm thickness and 450 mm.

7.10.17 Diaphragm and Collector Reinforcement

7.10.17.1 Tensile or compressive force in reinforcement at each section of the diaphragm or collector shall be developed on each side of that section.

7.10.17.2 Reinforcement provided to resist tension shall extend beyond the point at which it is no longer required to resist tension for at least a length of L_d , except at diaphragm edges and at expansion joints.

7.10.17.3 Welded wire mesh of diameter 6 mm or higher shall be used in the structural topping concrete to achieve the diaphragm reinforcement.

7.10.18 Other Parameters

7.10.18.1 Heavy duty PVC or metal conduits with diameter up to 20 mm are preferable. Larger diameter conduits (25 mm or 32 mm or higher) may be permitted as per project requirements with proportionately higher thickness of structural topping.

7.10.18.2 Single junction box shall be used to conduct the conduits.

7.11 Designing of Bearing Region

7.11.1 General

The bearing region shall ensure structural integrity through the following measures:

- a) Overlapping reinforcement in reinforced bearing zones;
- b) Physical restraint to prevent loss of bearing due to movement; and
- c) Tolerance provisions to accommodate cumulative deviations due to production and erection.

7.11.2 Net Bearing Width of Non-Isolated Members

The net bearing width of a member shall not be less than 40 mm and shall be determined by:

$$\text{Net bearing width} = \frac{\text{ultimate support reaction}}{\text{effective bearing length} \times \text{ultimate bearing stress}}$$

The net bearing width should be increased to cater for any free movement permitted or rotation of the bearing about the support.

7.11.3 Effective Bearing Length

The effective bearing length of a member shall be taken as the least of:

- a) Actual bearing length;
- b) One-half of bearing length plus 100 mm; or
- c) 600 mm.

7.11.4 Design Ultimate Bearing Stress

The design ultimate bearing stress shall be based on the weaker bearing surface and shall be:

- a) $0.4f_{ck}$ for dry concrete bearings;
- b) $0.6f_{ck}$ for mortar-bedded concrete bearings; or
- c) $0.8f_{ck}$ for steel bearing plates embedded in concrete, provided the plate dimensions do not exceed 40 percent of the supporting concrete dimension.

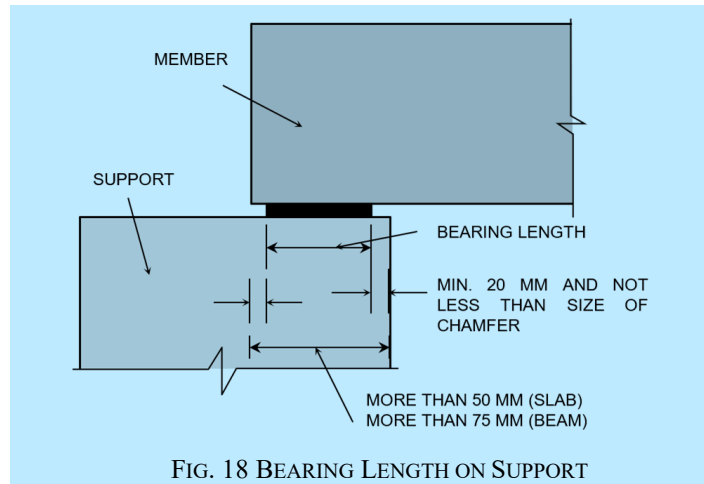
An intermediate value of bearing stress between dry and bedded bearings may be used for flexible bedding.

7.11.5 Net Bearing Width of Isolated Members

For isolated members, the net bearing width shall be increased by 20 mm over that required for non-isolated members.

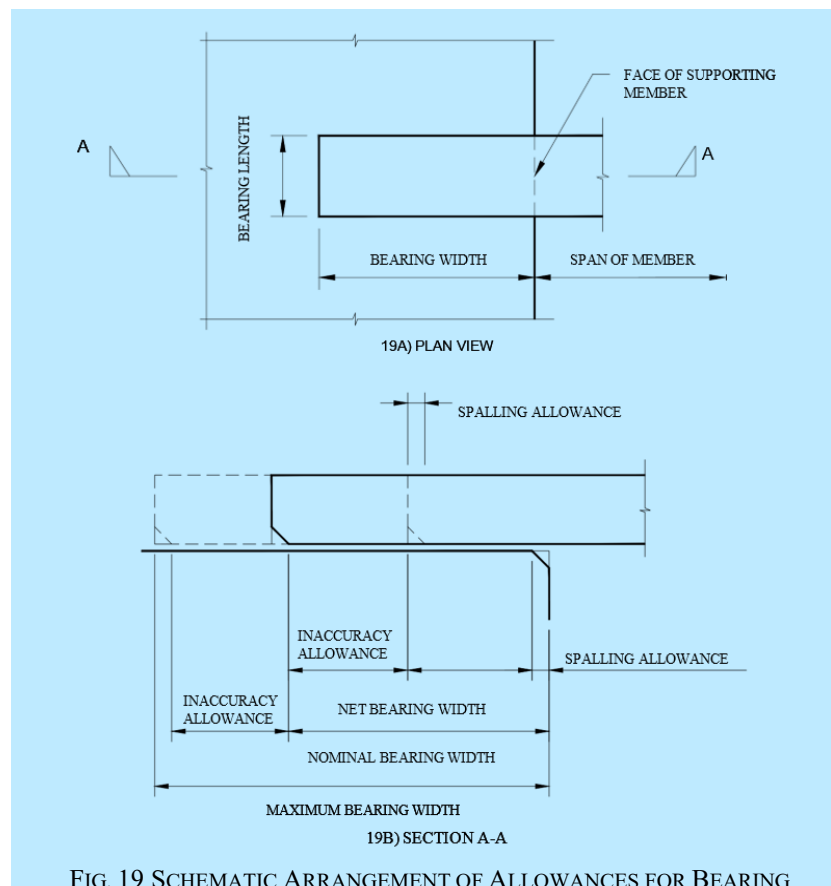
7.11.6 Minimum Dimensions of Bearing

The minimum requirement for bearing length shall be as per [Fig. 18](#).



7.11.7 Detailing for Simple Bearing

Over and above the net bearing width calculated above, allowances for spalling and constructional inaccuracies should be made in the nominal width selected (see [Fig. 19](#)). The effects of accidental displacement of a supported member during erection shall also be considered. Minimum anchorage lengths shall comply with requirements specified in [9](#).



7.11.8 Allowances for Spalling at Supports

The design shall account for reduction in effective bearing area due to potential spalling, in accordance with [Table 3](#) and [Table 4](#). When determining the outer edge of a support or the end of a supported member, chamfers occurring within the areas subjected to spalling may be excluded.

Table 3 Allowances for Effects of Spalling at Supports		
(Clause 7.11.8)		
Sl No.	Material of Support	Ineffective Distance, Measured from Outer Edge of Support
		mm
(1)	(2)	(3)
i)	Steel	0
ii)	Concrete $\geq$ M30	15
iii)	Concrete $<$ M30	25
iv)	RC with outer edge less than 300 mm deep and vertical loop bar not greater than 12 mm	Nominal cover to reinforcement on outer face of support
v)	RC with outer edge less than 300 mm deep and vertical loop bar is 16 mm or above	Nominal cover plus inner radius of bend of bars

Table 4 Allowances for Effects of Spalling at Supported Members		
(Clause 7.11.8)		
Sl No.	Reinforcement at Bearing of Support Member	Ineffective Distance, Measured from Outer Edge of Support
		mm
(1)	(2)	(3)
i)	Straight bar, horizontal loops or vertical loops of 12 mm diameter or less	Greater of nominal cover or 10 mm
ii)	Tendons or straight bars exposed at end of member	0
iii)	Vertical loops of 16 mm diameter or above	Nominal end cover plus inner radius of bend of bars

7.11.9 Shims

Plastic or metal shims shall be used as per structural and durability requirements and shall be specified as per design requirements.

7.11.9.1 Steel shims shall not be used at areas susceptible to spalling. Load transfer through shims shall be included in the structural design.

7.11.9.2 The total thickness of individual shims shall not exceed 40 mm.

7.11.9.3 For joint gaps exceeding 40 mm, steel shims shall be used and shall be tack welded when shim thickness exceeds 40 mm to ensure load transfer and stability. Minimum thickness of shim plates shall be 5 mm.

7.11.10 Allowance for Construction Inaccuracies

The design shall include allowances for construction inaccuracies due to deviations in layout, manufacturing and erection. In absence of statistical deviation data, the minimum allowances shall be:

- 15 mm or 3 mm/m for steel or precast concrete supports;

- b) 20 mm or 4 mm/m for masonry supports; and
- c) 25 mm or 5 mm/m for in-situ concrete supports.

7.11.11 Bearings Transmitting Compressive Forces

The horizontal joints between load-bearing walls or columns shall be designed to resist all the forces and moments implicit in the assumptions made in analysing the structure as a whole and in designing the individual member to be joined. In the absence of more accurate information derived from a comprehensive programme of suitable tests, the area of concrete to be considered in calculating the strength of the joint in a wall or column shall be as per [8.3.2](#). Joints and joint reinforcement shall be detailed to prevent premature splitting or spalling at member ends.

7.11.12 Other Forces at Bearings

- a) *Horizontal forces at bearing* — The horizontal forces at bearings may be induced by creep, shrinkage, temperature effects, misalignment, lack of plumb or other causes. When these forces are significant, the structural capacity of the supporting member may be impaired. A minimal force of $0.2 V_u$ (vertical or shear force on the bearing) shall be used in design. Allowances should be made by the provision of:
 - 1) Sliding bearings which allow longitudinal and lateral movement;
 - 2) Additional lateral reinforcement at the top of the supporting member; or
 - 3) Continuity reinforcement which ties the ends of the supported members together.
- b) *Rotation at bearing* — Suitable bearings should be used to accommodate the rotations at end supports in particular for flexural members. Allowances should be made for any consequential increases in bending moments or bearing stresses due to rotations.

7.11.13 Design of Bearing on Plain Concrete

7.11.13.1 Plain concrete bearing (see [Fig. 20](#)) shall be used in situations where the bearing is uniform and the bearing stresses are low.

7.11.13.2 In other situations, and in thin-stemmed components where the bearing area is less than $12\,500\text{ mm}^2$, a minimum reinforcement equal to $N_u/0.65f_y$ (but not less than one bar of 10 mm diameter) is recommended. If horizontal load is not present, a minimum recommended value of $N_u = 0.2 V_u$ shall be used.

where

N_u = factored axial force (minimum shall be $0.2 V_u$); and

V_u = factored shear force.

7.11.13.3 The design bearing strength of plain concrete shall be calculated based on the following equation as given below:

Supporting area is wider in all sides than the loaded area (lesser of)

$$V_{uR} = (0.45f_{ck}A_1) \sqrt{\frac{A_2}{A_1}}$$

$$\sqrt{\frac{A_2}{A_1}} \text{ shall not exceed } 2,$$

$$V_{uR} = 2(0.45f_{ck}A_1)$$

Other cases

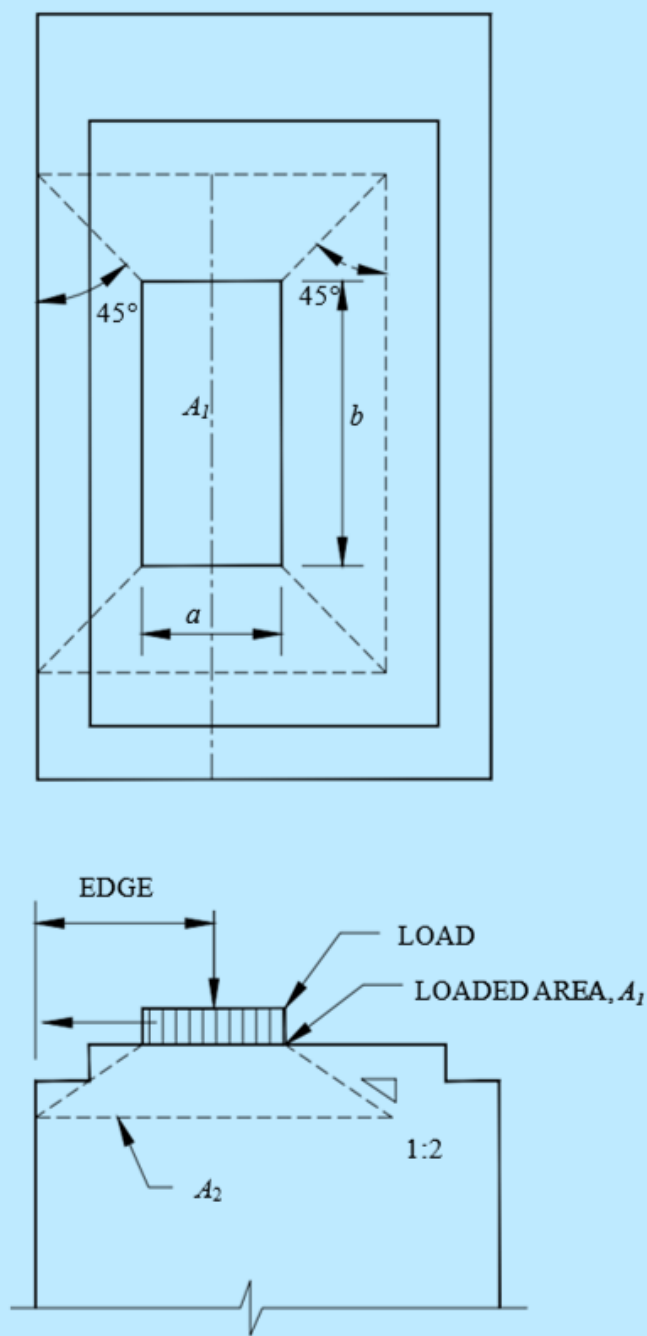
$$V_{uR} = 0.45f_{ck}A_1$$

where

V_{uR} = design shear resistance;

A_1 = loaded area (bearing constant area) (see [Fig. 20](#)); and

A_2 = area of the lower base of pyramid, cone or tapered wedge (see [Fig. 20](#)).



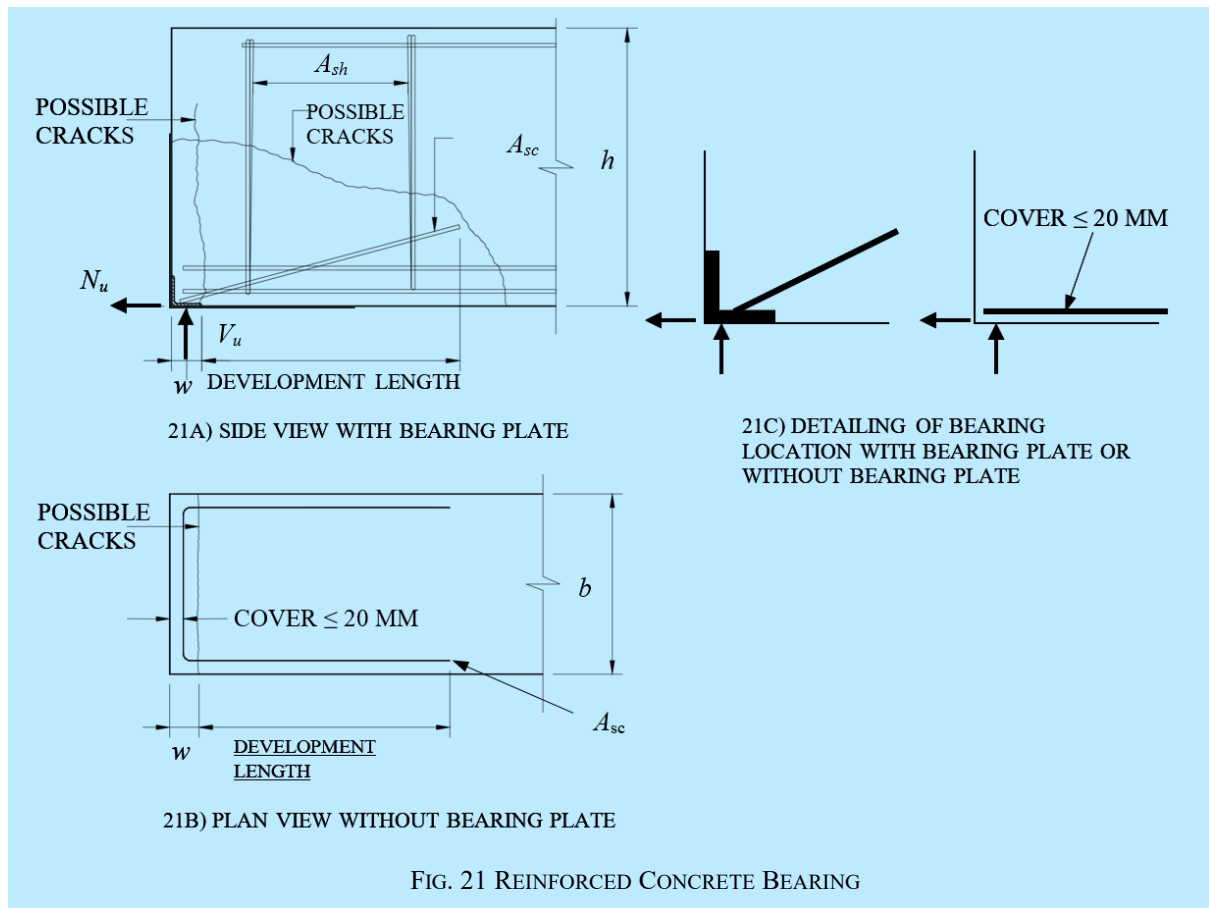
A_1 = LOADED AREA (BEARING CONTACT AREA)
 A_2 = AREA OF THE LOWER BASE OF PYRAMID, CONE OR TAPERED WEDGE

FIG. 20 BEARING ON PLAIN CONCRETE

7.11.14 Design of Reinforced Concrete Bearing (for thin Stem Elements and Rectangular Beams)

7.11.14.1 If the applied load V_u , exceeds the design bearing strength V_{uR} , as calculated by equation in [7.11.13.3](#), reinforcement shall be provided in the bearing area as per [Fig. 21](#).

7.11.14.2 The area of reinforcement $A_{vf} + A_n$ shall be calculated as per [7.12](#) and [7.11.16](#). This reinforcement shall be appropriately developed on each side of the vertical crack by development length, hooks, or welding to an anchor bar, bearing plate, or angle.



7.11.14.3 The area of vertical shear reinforcement shall be calculated as follows:

$$A_{sh} = \frac{(A_{vf} + A_n)f_y}{\mu f_{ys}}$$

where

A_{vf} = area of shear friction reinforcement;

A_n = area of tension reinforcement;

f_y = yield strength of tension reinforcement;

μ = coefficient of friction as per [Table 6](#); and

f_{ys} = yield strength of shear reinforcement.

7.11.14.4 Diagonal tension reinforcement such as stirrups or welded-wire mesh may be counted as A_{sh} but shall not be added to separately designed shear reinforcement.

7.11.14.5 When components are subjected to bearing force more than $0.45f_{ck}A_l$ confinement reinforcement in all directions shall be provided. A_l shall be considered as per [Fig. 20](#).

7.11.15 Dapped End Bearing (Thin Stem and Regular Beams)

7.11.15.1 Dapped end members shall be designed for the potential failure modes shown in [Fig. 22](#) to [Fig. 24](#) with the reinforcement required for each consideration.

7.11.15.2 Dap reinforcement shall be provided in all cases where one or more of the following conditions:

- The depth of the recess exceeds the lesser of 0.2 H or 200 mm;

- b) The length of the recess (L_p in Fig. 22) exceeds 300 mm for non-thin-stemmed components and the nib (the concrete section above the dap) is not designed for the full flexural and shear capacity; and
- c) For non-thin-stemmed components, less than 1/3 of the main flexural reinforcement extends to the end of the component above the dap.

7.11.15.3 Thin-stemmed components such as double tees, T beam, etc, shall be designed with the following additional consideration:

- a) Stem width is 125 mm or less;
- b) Vertical hanger reinforcement or inclined reinforcement at an angle greater than 90° measured from horizontal; and
- c) Dap height $\leq 0.5 h$.

7.11.15.4 The design equations given in this section shall be considered for cases where shear span-to-depth ratio (a_v/d) do not exceed 1.0.

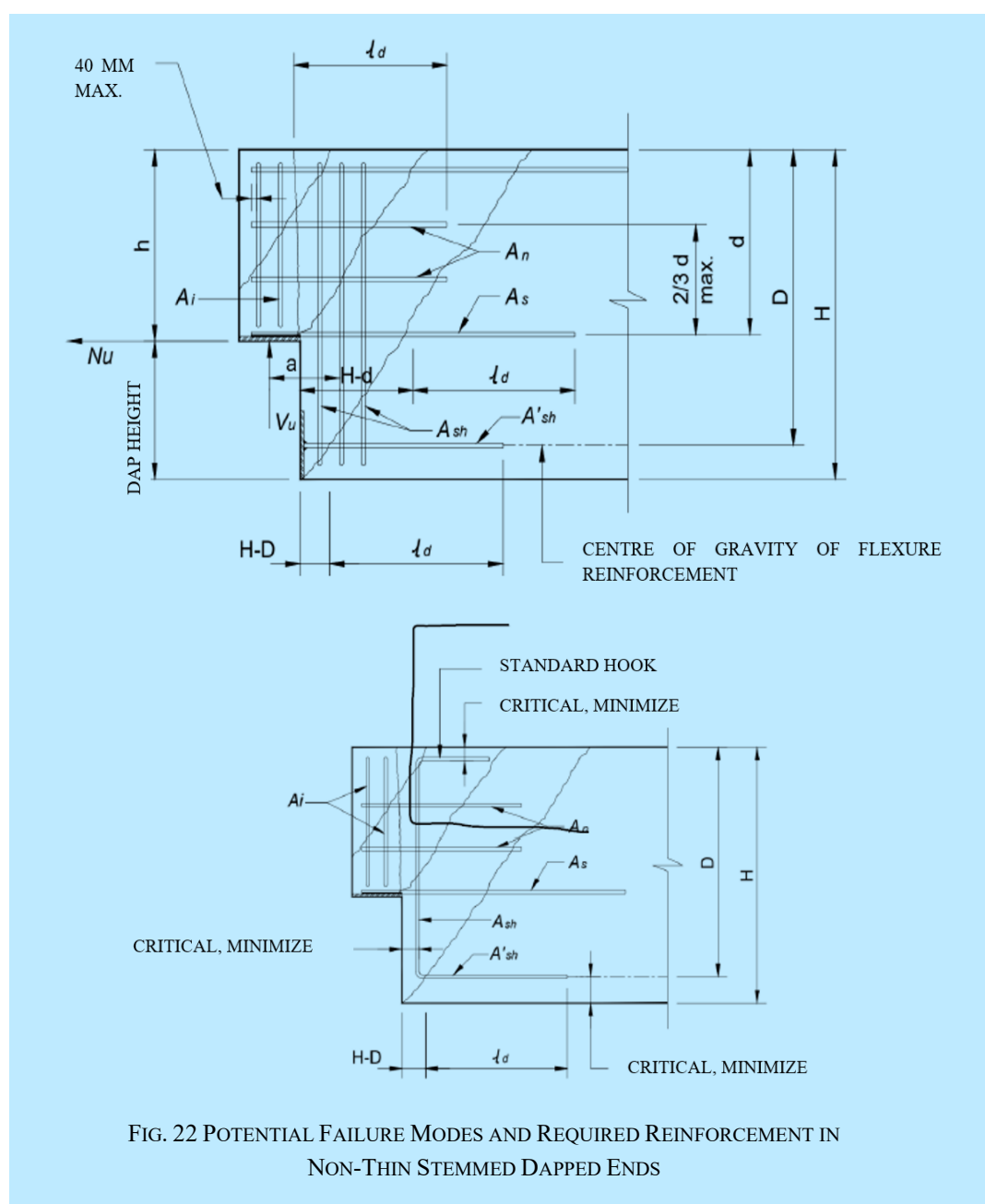


FIG. 22 POTENTIAL FAILURE MODES AND REQUIRED REINFORCEMENT IN NON-THIN STEMMED DAPPED ENDS

7.11.15.5 High strength rebars may be used, however, the yield strength of reinforcement shall be restricted to 415 MPa.

7.11.15.6 For short, shallow recesses, the hanger reinforcement A_{sh} , and A'_{sh} , shall not be necessary. Confinement reinforcement A_v and flexural reinforcement $A_{vf} + A_n$, in accordance with [7.11.14](#) shall be provided.

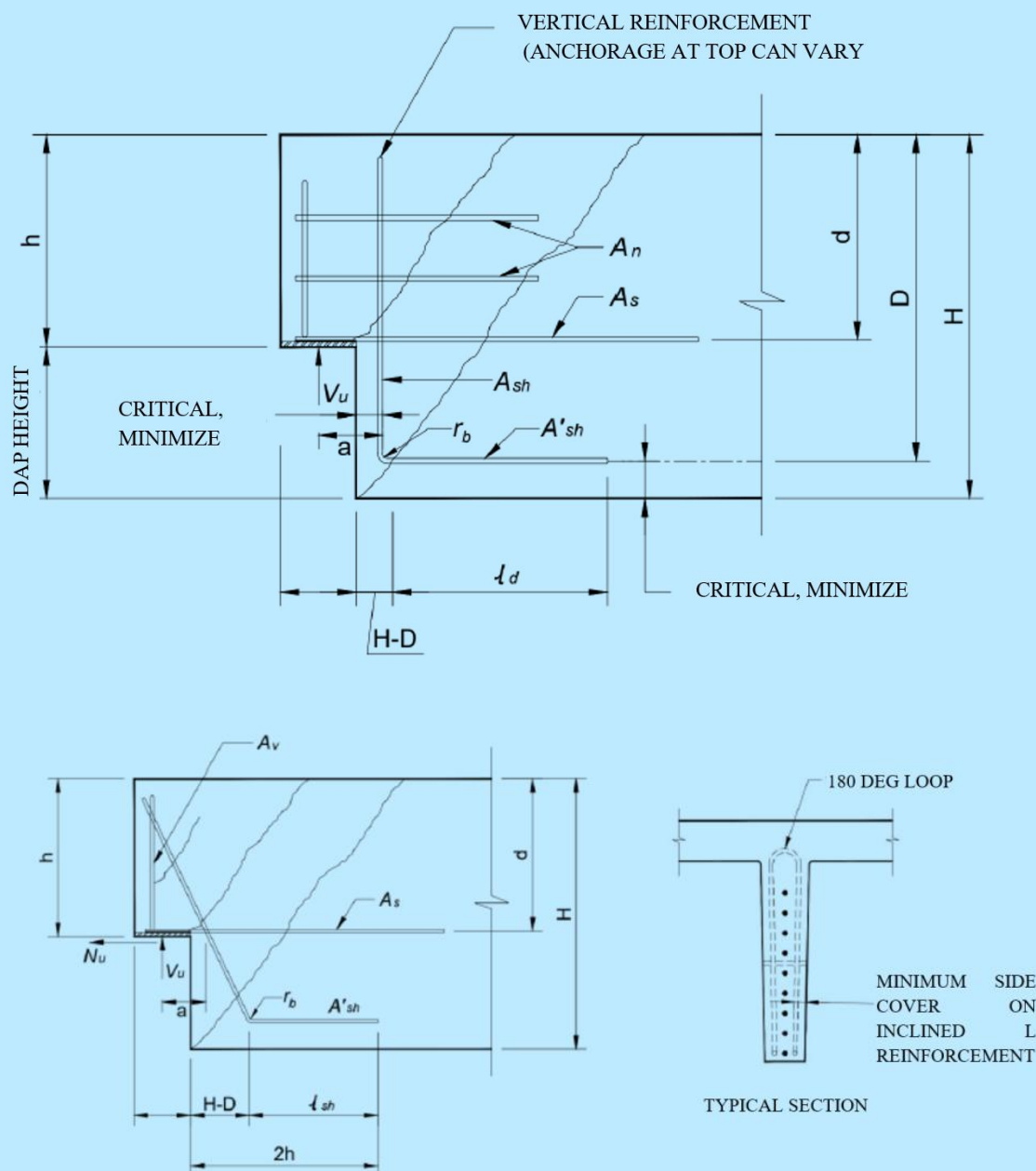
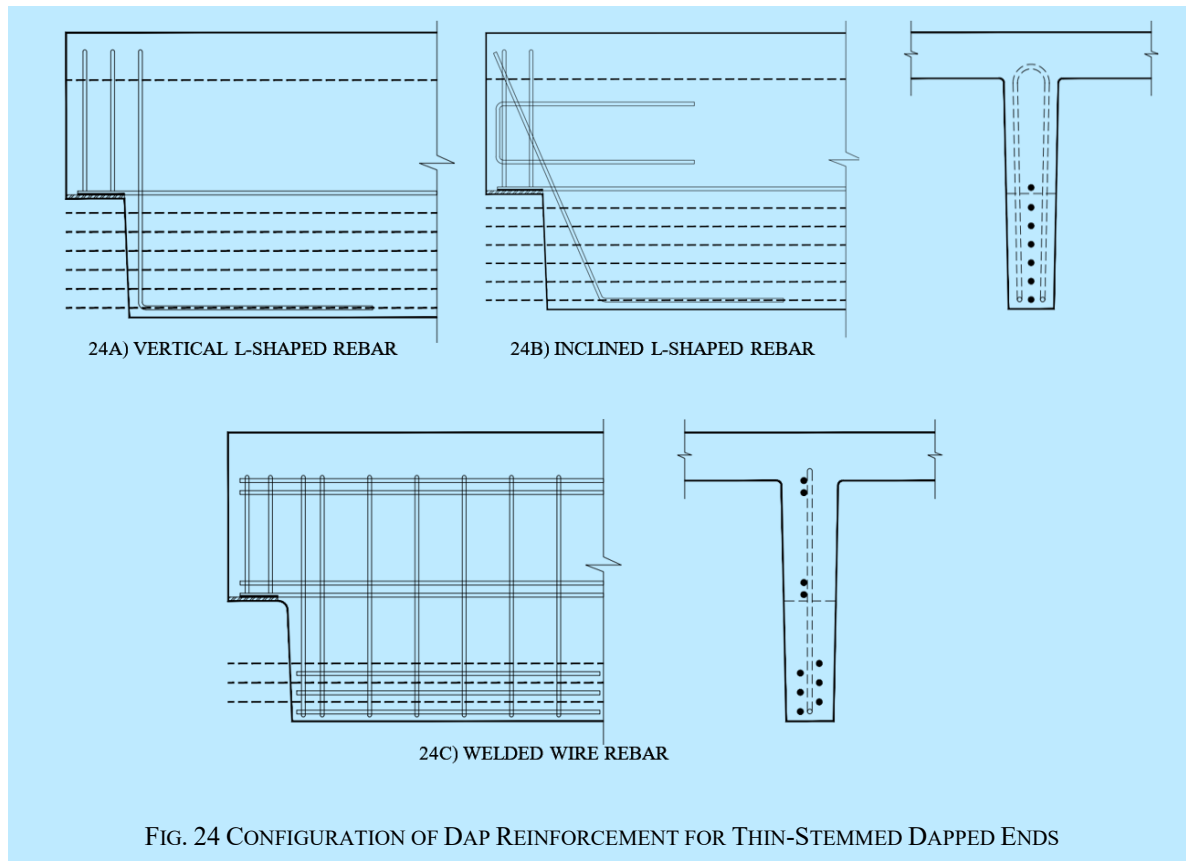


FIG. 23 POTENTIAL FAILURE MODES AND REQUIRED REINFORCEMENT IN THIN-STEMMED DAPPED ENDS



7.11.15.7 The area of horizontal reinforcement A_s (area of reinforcement for flexure and axial tension in the extended end) shall be the larger of the two values calculated using the equations given below. The values of A_f , A_n shall be calculated as per [7.11.17](#) and A_{vf} is as defined under [7.12](#):

$$A_s = A_f + A_n, \text{ or}$$

$$A_s = \frac{2}{3}A_{vf} + A_n$$

7.11.15.8 The area of horizontal reinforcement A_h (area of direct shear reinforcement) shall be provided in the nib to arrest Crack 2 as per the following and shall be distributed within $2/3 d$ measured from the primary tension reinforcement A_s :

$$A_h = 0.5(A_s - A_n)$$

7.11.15.9 The area of hanger reinforcement A_{sh} shall be provided to arrest Crack 3, shall be as per the following:

$$A_{sh} = \frac{V_u}{0.75f_y}$$

7.11.15.10 The area of vertical shear reinforcement A_v shall be provided to arrest Crack 4 as per the following:

$$V_{ur} = 0.75(A_v f_y + A_h f_y + 0.15\sqrt{f_{ck}}bd)$$

$$\text{Minimum } A_v = \frac{1}{2f_y}(1.34V_u - 0.15\sqrt{f_{ck}}bd)$$

7.11.15.11 The anchorage of reinforcement shall be as per the following:

- a) Horizontal bars A_s shall be extended a minimum of L_d beyond Crack 5 and anchored at the end of the component by welding to cross bars, plates, or angles;
- b) Horizontal bars A_h shall be extended a minimum of L_d , beyond Crack 2 and anchored at the end of the component by hooks or other suitable means;
- c) To ensure development of hanger reinforcement A_{sh} , the rebar shall be bent and continued parallel to the bottom of the component, or a separate horizontal anchored reinforcement with $A'_{sh} \geq A_{sh}$ shall be provided. The extension of reinforcement (A'_{sh}) at bottom of the component must be at least L_d beyond Crack 5. The A'_{sh} reinforcement may be anchored on the dap side by welding it to a plate, angle, or cross bar;
- d) The beam flexure reinforcement may also be used to ensure development of A_{sh} reinforcement, provided that the flexural reinforcement is adequately anchored on the dap side and the A_{sh} reinforcement is wrapped around the flexural reinforcement;
- e) In thin-stemmed components, it is recommended that the extension of the hanger reinforcement be greater than L_d . A'_{sh} shall extend a distance that is the greater of $1.5 L_d$ of the strand past the face of the dap or $2.0 L_d$ of A'_{sh} beyond the diagonal Crack 5;
- f) Vertical reinforcement A_v should be properly anchored by hooks, as required by Part C/Section 5 of this NBCS;
- g) Thin-stemmed components, except for the vertical 'Z' configuration, which has the bar welded to the plate, should have at least one 12 mm diameter rebar welded to the bearing plate or have equivalent development length (L_{rebar}); and
- h) Welded-wire reinforcement in place of bars may be used for reinforcement. It should be anchored in accordance with Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS. In thin-stemmed components the use of welded-wire reinforcement should be limited to situations where it is placed symmetrically in the width of the stem.

7.11.15.12 *Other considerations*

In addition to the reinforcement for the dapped end, it is important that the following be considered:

- a) The depth of the extended end should not be less than about one half the depth of the beam, unless the beam is deeper than necessary for other than structural reasons;
- b) The hanger reinforcement A_{sh} should be placed as close as practical to the re-entrant corner and as symmetrically as possible around the centroid of the section being considered. This reinforcement requirement is not additive to other shear-reinforcement requirements but should be in addition to torsion reinforcement;
- c) The shear strength of the full-depth beam behind the dap must be verified;
- d) For thin-stemmed components, for a distance of $2H$ beyond the dap the concrete contribution to shear strength should be limited to:

$$V_c = 0.15\sqrt{f_{ck}}b_wd_p$$

or

$$V_c = 0.23\sqrt{f_{ck}}b_wd_p, \text{ where at least two bonded strands in the extended end are present}$$

where

b_w = average width of the stem for the full depth of the section; and

d_p = distance from the top to the centroid of the prestressing strand for the full depth section and shall not be less than $0.8 H$.

- e) For thin-stemmed components, for a distance $2H$ beyond the dap, the reinforcement contribution to shear strength V_s shall be limited as per the following:

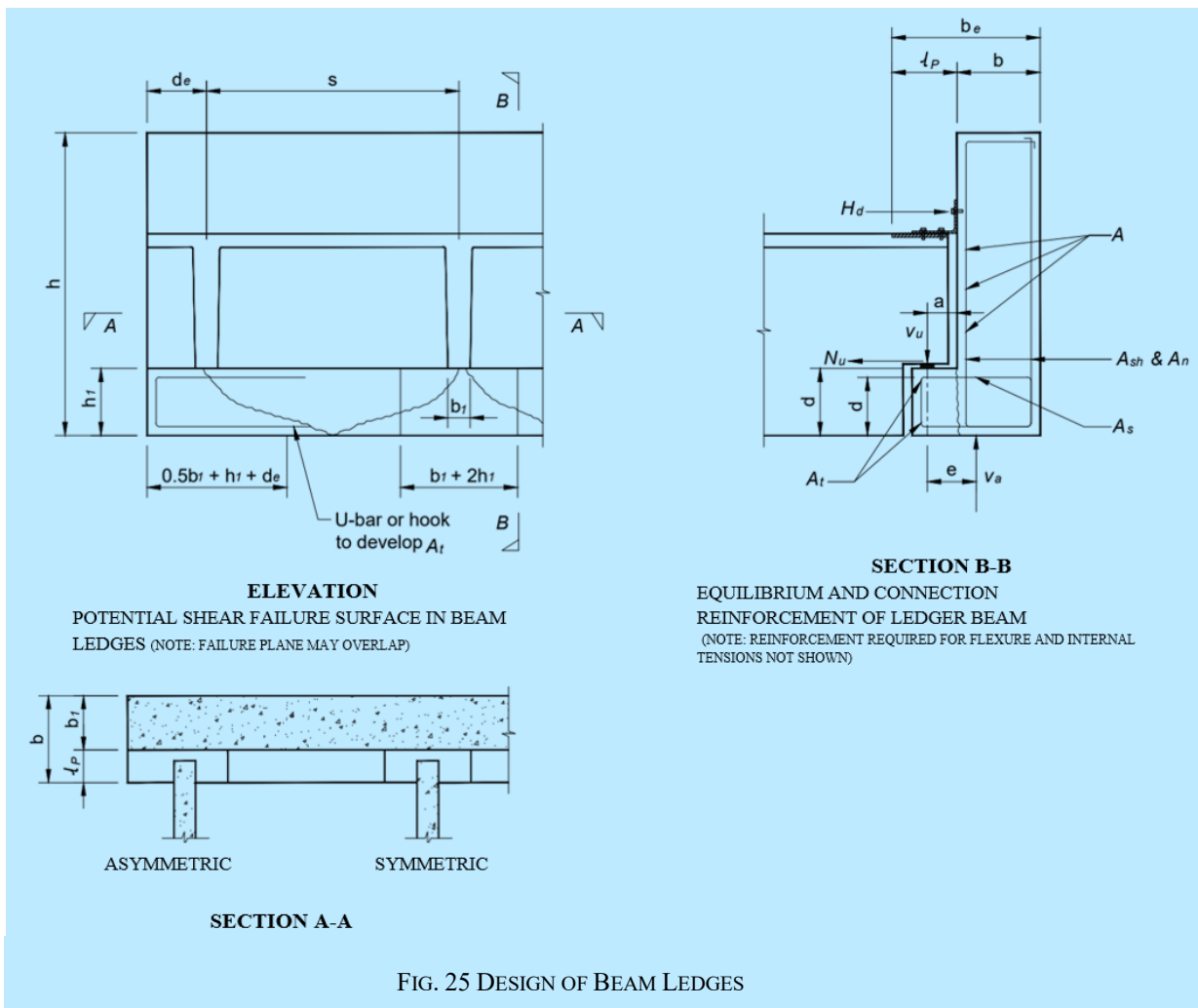
$$V_s = 0.15\sqrt{f_{ck}}b_wd_p$$

7.11.16 Ledges

7.11.16.1 General

Beam and wall ledges (as shown in Fig. 25) shall be designed as per the following guidelines. Hanger steel requirements for the ledge in a wall need not be satisfied except where penetrations occur just below ledges. The design procedure explained in this section shall be applicable for the following cases:

- $N_u \leq V_u$;
- Ledges with heights of 200 mm to 450 mm;
- Ledge projections of 150 mm to 250 mm;
- Bearing widths of components on the ledge ranging from 100 mm to 300 mm;
- Distances of end load to end of ledge ranging from 100 mm to 900 mm; and
- Design concrete compressive strengths f_{ck} of 30 MPa to 80 MPa.



7.11.16.2 For shear strength of ledge for concentrated loads, not near the ends, where $d_e \geq 0.5 b_t + h_l + l_p$, a symmetric failure controls the design punching shear strength of the ledge shall be determined as the lesser of following two equations:

$$V_{uR} = 0.056\gamma\beta\sqrt{f_{ck}}h_l(b_t + 2h_l + 2l_p)$$

or

$$V_{uR} = 0.028\gamma\beta\sqrt{f_{ck}}h_l(b_t + 2h_l + s + 2l_p)$$

For concentrated loads that are near ends, where $d_e \leq 0.5 b_t + h_l + l_p$, an asymmetric failure controls the design punching shear strength of the ledge and shall be as per the following:

$$V_{uR} = 0.056\gamma\beta\sqrt{f_{ck}}h_l(0.5b_t + h_l + d_e + l_p)$$

or

$$V_{uR} = 0.56\gamma\beta\sqrt{f_{ck}}h_l(0.5b_t + h_l + d_e + l_p)$$

where

$\gamma = \sqrt{1 + 12.5 \frac{f_{pc}}{f_{ck}}}$, a factor for the effect of prestressing, 1 for precast beams;

β = coefficient of shear strength based on R (as per Fig. 26);

h_l = height of beam ledge;

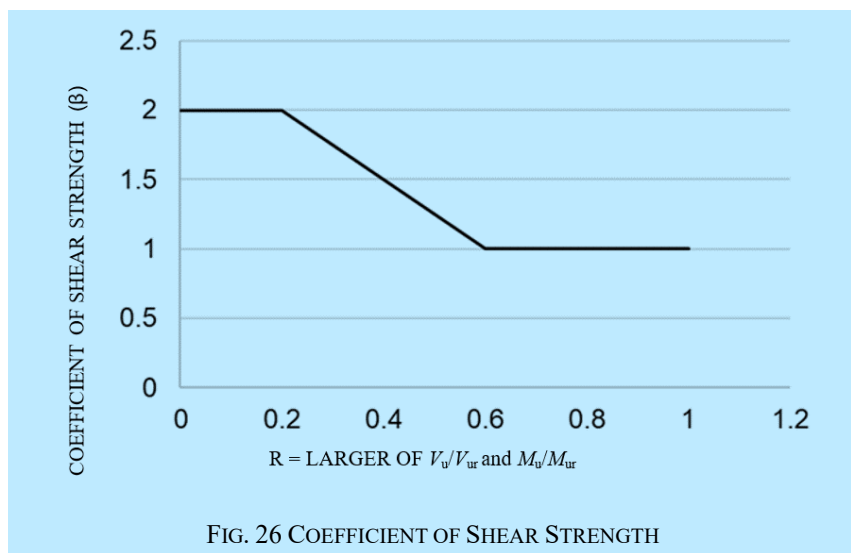
b_t = bearing width;

l_p = length of ledge projection;

s = minimum spacing of concentrated loads;

d_e = distance from centre of load to the end of the ledge; and

f_{pc} = average prestress force after losses divided by gross concrete area of beam.



7.11.16.4 For ledges supporting a continuous load or a series of closely spaced concentrated loads, punching shear strength per metre run shall be determined as per the following:

$$V_{uR} = 0.675\sqrt{f_{ck}}h_l$$

7.11.16.5 If the applied factored load exceeds the strength as determined above, the ledge shall be designed for shear transfer and diagonal tension in accordance with [7.11.15.8](#) through [7.11.15.10](#).

7.11.16.6 Transverse (cantilever) bending of the ledge requires flexural reinforcement ($A_s = A_f + A_n$), shall be computed using [7.11.17.5](#). Such reinforcement shall be uniformly spaced over a width of $6h_l$ on either side of the bearing, but not to exceed half the distance to the next load. Bar spacing shall not exceed the ledge depth h_l or 450 mm whichever is less.

7.11.16.7 Longitudinal reinforcements shall be placed in both the top and bottom of the ledge portion of the beam and calculated as per the following:

$$A_l = \frac{200l_p d_l}{0.75f_y}$$

where

d_l = design depth of A_l reinforcement in [Fig. 25](#).

7.11.16.8 Hanger steel A_{sh} shall be computed using following equation and distributed in the web of the beam. The reinforcement shall be uniformly spaced over a width of $6h_l$ on either side of the bearing, but shall not exceed half the distance to the next load. The area of reinforcement shall not be additive to shear and torsional reinforcement.

$$A_{sh} = \frac{Vu}{0.75f_y} (m)$$

where

m for L beam

$$m = \frac{(d_s + a) - \left(3 - 2\frac{h_l}{h}\right)\left(\frac{h_l}{h}\right)^2\left(\frac{b_l}{2}\right) - e\gamma_t \frac{x_l^2 y_l}{(x_l^2 y_l + x_w^2 y_w)}}{d_s} \geq 0.6, \text{ and}$$

m for inverted tee-beam

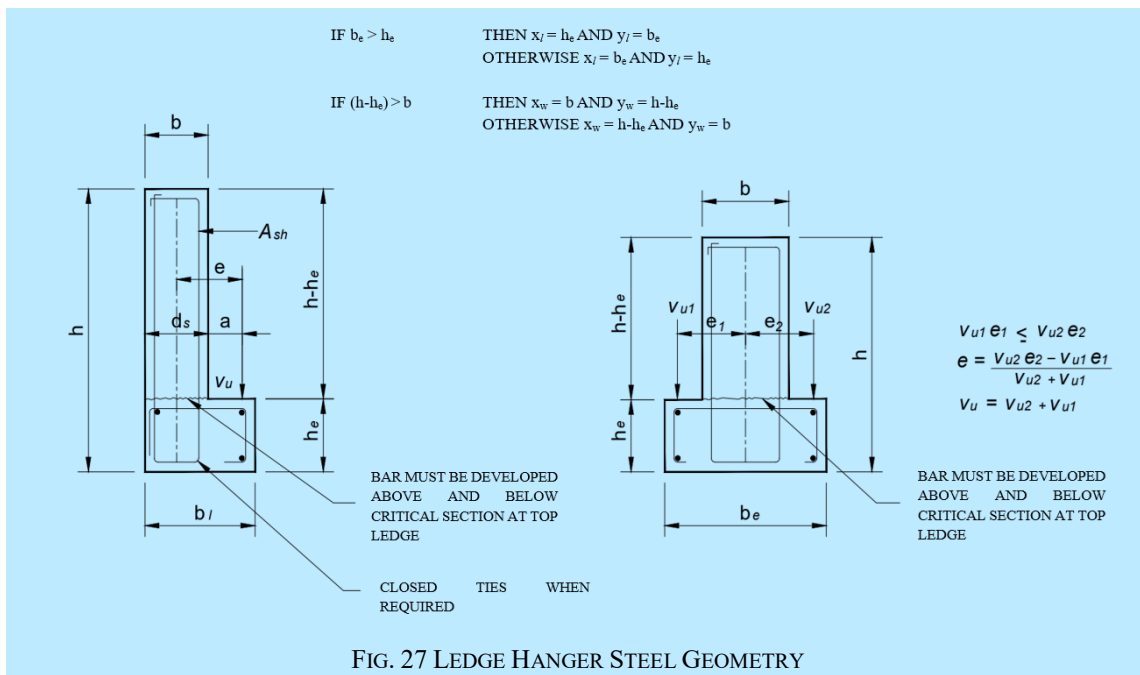
$$m = \frac{\left(\frac{b}{2} + e\right) - \left(3 - 2\frac{h_l}{h}\right)\left(\frac{h_l}{h}\right)^2\left(\frac{b}{2}\right) - e\gamma_t \frac{x_l^2 y_l}{(x_l^2 y_l + x_w^2 y_w)}}{d_s} \geq 0.4$$

where

d_s = distance from outside face of web to A_{sh} reinforcement;

b_l = width of web plus ledge being designed;

x_l, y_l, x_w, y_w , = defined in [Fig. 27](#).



7.11.17 Concrete Corbels

7.11.17.1 General

Concrete corbels are short cantilevers that support other components and transmit the load to the supporting component and tend to act as simple trusses or deep beams (see [Fig. 28](#)).

7.11.17.2 The design based on strut and-tie method shall be as per Part C/Section 5 of this NBCS. For steel brackets, Part C/Section 6A of this NBCS shall be referred.

7.11.17.3 The cantilever beam method of designing a corbel presented in this section shall be used according to following limitations:

- $a_v/d \leq 1$; and
- $N_u \leq V_u$.

7.11.17.4 The dimensions of a corbel shall be selected such that the nominal shear strength shall not exceed the strength calculated as per [Table 6](#).

7.11.17.5 The area of primary tension reinforcement A_{sc} shall be greater of $(A_f + A_n)$ and $(2/3) A_{vf} + A_n$

7.11.17.6 The area of reinforcement A_f shall be calculated as per the following equation:

$$A_f = \frac{V_u a_v + N_u (h - d)}{0.7 f_y d}$$

7.11.17.7 The area of reinforcement A_n shall be calculated as per the following equation:

$$A_n = \frac{N_u}{0.75 f_y}$$

7.11.17.8 The area of reinforcement A_{vf} shall be calculated as per shear friction (see [7.12](#)).

7.11.17.9 Unless tensile forces are prevented from being applied to the bracket or corbel, N_u shall be considered as $0.2V_u$ or as specified in the project.

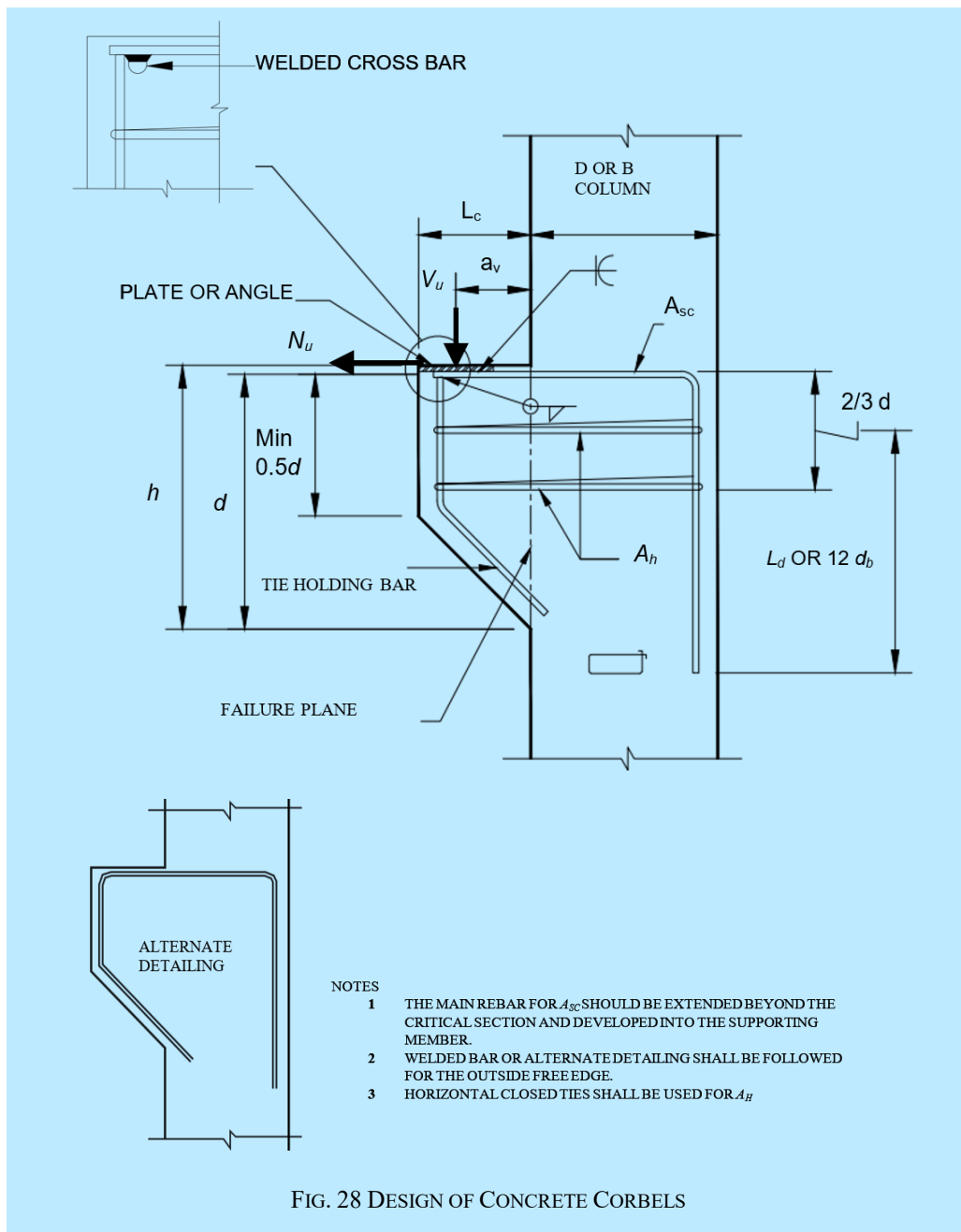


FIG. 28 DESIGN OF CONCRETE CORBELS

7.11.17.10 If bearing pads are used to avoid tensile forces, the value of N_u that corresponds to the force causing the pad to slip shall be considered.

7.11.17.11 Minimum reinforcement (A_{sc}) shall be calculated as per the following equation. Where the corbel depth is larger than required by design a reduced depth d may be used consistently in all the design equations:

$$A_{sc,min} = 0.032 \frac{f_{ck}}{f_y} b d$$

7.11.17.12 The area of stirrups (A_h) shall be calculated as per the following equation:

$$A_h \geq 0.5(A_{sc} - A_n)$$

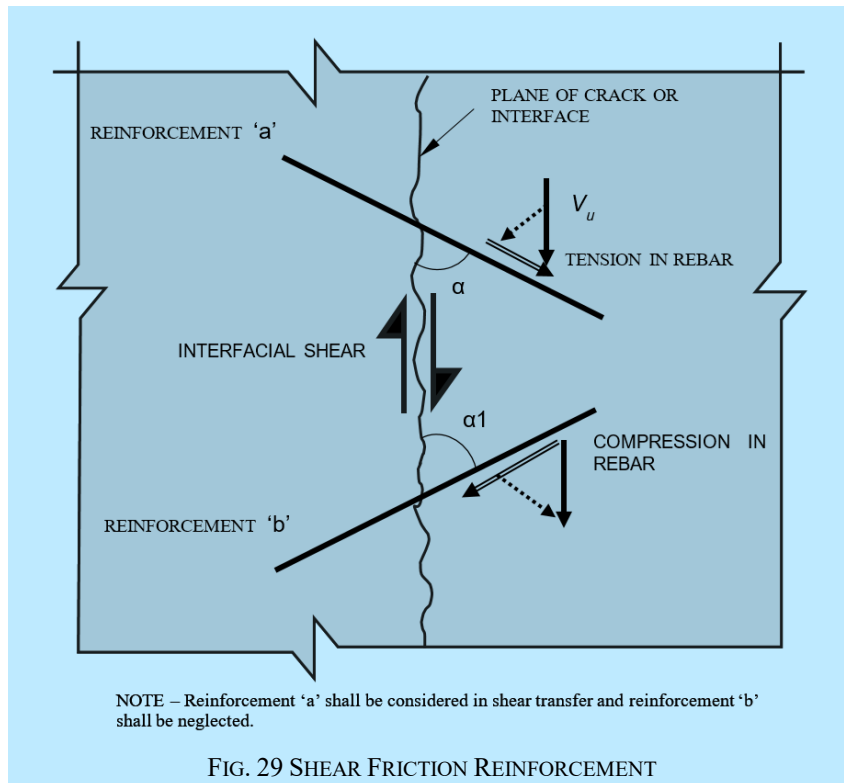
7.11.17.13 The shear reinforcement shall be provided as closed stirrups and uniformly distributed within $2/3 d$ measured from the primary tension reinforcement.

7.11.17.14 The development length for primary tension reinforcing bar shall be as per good practice [C-7(2)]. Development length for tension reinforcement shall be measured from the assumed failure plane (face of the supporting member). For member not having enough embedment length, hooked bars shall be used.

7.11.17.15 The main tension steel within the concrete corbel shall be developed at the outer edge of the corbel by welding to a cross bar, bearing plate with cross bar, angle section or bending along the depth of corbel.

7.12 Shear Friction

Shear friction shall be applied to consider shear transfer across interface between dissimilar materials or interface between concrete cast at different time or across a shear plane as shown in [Fig. 29](#), reinforcement 'a'. The following clauses shall be used.



7.12.1 Nominal Shear Strength

7.12.1.1 Value of V_{uR} across the assumed shear plane shall be calculated using equation in [7.12.1.2](#) or [7.12.1.3](#) depending on the angle of shear reinforcement.

7.12.1.2 If shear-friction reinforcement is perpendicular to the shear plane, nominal shear strength across the assumed shear plane shall be calculated by:

$$V_{uR} = 0.75\mu A_{vf}f_y$$

7.12.1.3 If shear-friction reinforcement is inclined to the shear plane and the shear force induces tension in the shear friction reinforcement, nominal shear strength across the assumed shear plane shall be calculated by:

$$V_{uR} = 0.75 A_{vf}f_y (\mu \sin \alpha + \cos \alpha)$$

where

A_{vf} = area of reinforcement crossing the assumed shear plane to resist shear;

μ = coefficient of friction in accordance with [Table 6](#); and

α = angle between shear-friction reinforcement and assumed shear plane.

7.12.2 The value of V_{uR} across the assumed shear plane shall not exceed the limits in [Table 5](#). Where concretes of different strengths are cast against each other, the lesser value of f_{ck} shall be used in [Table 5](#).

Table 5 Maximum V_{uR} Across the Assumed Shear Plane			
(Clause 7.12.2)			
Sl No.	Surface Condition	Maximum V_{uR}	
(1)	(2)	(3)	
i)	Concrete placed monolithically or placed against intentionally roughened surface (amplitude of 6 mm or more)	$0.12f_{ck}A_{cr}$, or $(2.475 + 0.049f_{ck}) A_{cr}$, or $8.25A_{cr}$	Equation given in 7.12.1.2 least of the three
ii)	Other cases	$0.12f_{ck}A_{cr}$ or $4.125A_{cr}$	
			Equation given in 7.12.1.3 least of the two

Table 6 Coefficient of Friction (Clauses 7.11.14.3 , 7.11.17.4 and 7.12.1.3)		
SI No.	Contact Surface Condition	Coefficient of Friction
(1)	(2)	(3)
i)	Concrete placed monolithically	1.4
ii)	Concrete placed against hardened concrete that is clean, free of laitance and intentionally roughened to a full amplitude of approximately 6 mm	1.0
iii)	Concrete placed against hardened concrete that is clean, free of laitance and not intentionally roughened	0.6
iv)	Concrete placed against as-rolled structural steel that is clean, free of paint and with shear transferred across the contact surface by headed studs or by welded deformed bars or wire	0.7

7.12.3 The frictional component of compression load (if any) present at the interface shall be considered along with the shear friction capacity.

7.12.4 Shear friction capacity shall not be considered for a joint producing compression in the joint reinforcement as shown in [Fig. 29](#) reinforcement 'b'.

7.13 Serviceability Design Checks

7.13.1 Deflection

Deflection calculations and permissible limits shall comply with the provisions of Part C/Section 5 of this NBCS. Partial fixity provided by the structural topping and nominal connections may be considered for serviceability checks.

7.13.2 Crack

The relevant provisions of good practice [C-7(2)] shall be followed for crack width calculations and limits. If structural cracks are observed on the tension face of the element, testing of the element shall be carried out in accordance with [10](#) or as directed by engineer-in-charge and the client.

7.13.3 In-Service Considerations

The relevant provisions of good practice [C-7(2)] shall be followed for service consideration calculations and limits.

7.14 Design of Demoulding, Handling Systems and Temporary Forces

7.14.1 Demoulding Forces

Forces on the elements during lifting from the casting bed due to suction or adhesion between the precast elements and the mould should be accounted while deciding the strength of concrete. These are accounted for by applying an equivalent load factor to the self-weight of the member and treating it as an equivalent static force to evaluate the stresses in the precast element. [Table 7](#) gives recommended minimum values of equivalent load factor for demoulding forces for different product types and finishes.

Table 7 Recommended Load Factors to Account for Forces in Demoulding of Elements (Clauses 7.14.1 , 10.8 and 10.8.3)			
Sl No.	Product Type	Finish	
		Exposed Aggregate with Retarder	Smooth Mould (form oil only)
(1)	(2)	(3)	(4)
i)	Flat with removable side forms. No formed rebates or reveals	1.2	1.3
ii)	Flat with removable side forms with formed rebates or reveals	1.3	1.4
iii)	Fluted with proper draft	1.4	1.6
iv)	Sculptured	1.5	1.7
NOTES 1 These factors are to be applied to the flexural design of precast elements only. For lifting inserts, see Table 10 . 2 The above values are recommended values only. Guidance should also be sought from the precast manufacturer based on their pre-casting experience if the load factors need to be increased. 3 The associated imposed loads or wind loads, if any, are to be assessed and considered under the appropriate load combination.			

7.14.2 Handling Stresses

Precast elements shall be designed and handled to prevent permanent damage during demoulding, storage, transportation, and erection. Design shall account for all loads acting during these stages. Minor cracks within acceptable design crack widths shall be repaired using methods approved by the contract or chemical supplier guidelines.

7.14.3 Handling and Transportation

An allowance should be made for dynamic loads and impact forces arising in handling, transportation and erection. Similar to demoulding force consideration, [Table 8](#) gives recommended values for equivalent load factor to be applied to the self-weight of members to allow for these forces.

Table 8 Recommended Load Factors to Account for Dynamic Forces Arising During Handling, Transportation and Erection of Elements
(Clause [7.14.3](#))

Sl No.	Stage	Load Factor
(1)	(2)	(3)
i)	Yard handling	1.2
ii)	Transportation	1.5
iii)	Erection	1.2
NOTES		
1 These factors are to be applied to the flexural design of precast elements only. For lifting inserts, refer 7.14.6 .		
2 The above values are recommended values only. Under certain conditions higher factors may apply that is certain unfavourable load conditions.		
3 The associated imposed loads or wind loads, if any, are to be assessed and considered with the appropriate load combination.		

7.14.4 Early Demoulding/Handling of Precast Element

For precast elements lifted and handled prior to gaining full strength the elements together with any lifting inserts should be designed accordingly. Recommended minimum concrete strengths for lifting and handling of precast elements are given in [Table 9](#). Precast elements should be designed, using these lower strengths, to span between lifting points without excessive cracking or deflection. For prestressed concrete, consideration should also be given to stresses resulting from transfer of prestressing forces.

Table 9 Recommended Minimum Concrete Strength for Lifting and Handling

(Clauses [7.14.4](#) and [10.8.4.1](#))

Sl No.	Application	Minimum Concrete Strength
(1)	(2)	N/mm ² (3)
i)	None specified, fine controlled crane, non-prestressed	10*
ii)	Lifting which involves significant impact or high acceleration	15*
iii)	All units where concrete strength for lifting is specified in the specification	As specified
iv)	Concentrically prestressed elements (piles, wall panels, or thin floor slabs)	20
v)	Eccentrically prestressed (T beam, deep flowing units)	25
vi)	Highly stressed prestressed elements	30 or as specified

*Depending on anchor length or as recommended by insert manufacturer.

7.14.5 Temporary Stages/Erection Sequence

The design of precast elements at temporary stages can be critical compared to the permanent stages. Consideration should be given to the loading imposed on the precast elements during each phase of construction. The following shall be verified:

- Precast sections of composite elements which are required to support self-weight plus construction load prior to casting of an in-situ topping;
- Lower precast floor slabs or precast stair flights which support propping to upper levels during installation; and
- Bearing or halving joints which support higher temporary construction loads because of back propping to upper levels.

The design should also take into consideration that the structural action and framing might be different during the temporary stages resulting in higher stresses in individual members.

7.14.6 Lifting Inserts

The number of lifting inserts and location of inserts shall be decided by the following:

- Lifting insert capacity (safe working load);
- Total weight of the element;
- Strength of concrete at age of lifting;
- Shape of the unit;
- Location of the inserts so that the failure of any one insert does not cause failure of the entire lifting system thereby ensuring the element can still be safely supported;
- Position of any cut-outs and/or openings;
- Rigging arrangement; and
- Anchorage of inserts.

All lifting inserts shall be designed by accounting the factors of safety given in [Table 10](#). Typically lifting inserts shall be designed with a factor of safety of 4.

Table 10 Recommended Factors of Safety for Lifting Inserts (Table 7 and Clause 7.14.6)		
SI No.	Item	Recommended Factor of Safety
(1)	(2)	(3)
i)	Bracing members	2
ii)	Bracing connections	3
iii)	Bracing inserts cast into precast members	3
iv)	Lifting inserts, normal circumstances	4
v)	Lifting inserts, multiple usage	5
vi)	Rebar inserts, single usage	2
vii)	Rebar inserts, multiple usage	4

7.14.7 Bracing Design

Bracing should be designed for both construction and wind loading. The construction load should be a minimum value of 1.5 kN/m² but should be increased if considered appropriate.

Wind loads shall be calculated in accordance Part C 'Structural Design, Section 1 Loads and Load Effects' of this NBCS and the bracing shall be designed for it.

8 JOINTS AND CONNECTIONS

8.1 Basic Design Considerations

A joint shall be considered based on the action of force transfer (tension, compression, shear, moment, etc) at an interface (one or many) between the precast components. A connection, which is a zone that includes various joints, shall be designed in combination of any of the following forces:

- a) Compressive force;
- b) Tensile force;
- c) Shear force; and
- d) Flexural and torsional moments, as applicable.

8.1.1 Transfer of Compressive Force

8.1.1.1 The precast or cast in-situ member supporting the other elements shall be designed to safely resist all the compressive forces generated by various load combination.

8.1.1.2 The joint shall be designed for all other forces arising from the external forces along with the predominant compression. Shear forces more than $0.1 P_u$ (axial load) shall not be neglected in the design of the joint.

8.1.1.3 The members may rotate because of temperature change, creep and shrinkage, hence bearing pads or strips shall be provided to account for the rotation. Minimum thickness of the joint shall be decided based on tolerance requirements, and shall not be less than 10 mm.

8.1.1.4 Appropriate edge distance for the bearing pad shall be provided to avoid crushing or cracking of edges.

8.1.1.5 Connection having bearing materials harder than concrete shall be checked for ultimate limit state as per the provisions of good practice [C-7(2)]. Busting reinforcement shall be provided in the supporting member, if required.

8.1.1.6 Connection having bearing materials softer than concrete shall be checked for serviceability limit states as per the provisions of good practice [C-7(2)]. Splitting reinforcement shall be provided in the supporting member, if required.

8.1.1.7 Joint using steel inserts for connection shall be properly designed and executed as per the design requirements. The steel components shall be anchored to the concrete members for proper transfer of compressive forces to the main reinforcement.

8.1.1.8 Connections shall be provided with reinforcement as follows or with equivalent connection:

- a) For walls elements, dowels shall be placed at a maximum spacing of $1/4^{\text{th}}$ of the length of the wall panels or 450 mm or as required by design. Also, at least two extreme edge dowels shall be continuous from foundation till roof using Class H coupler as per [C-7(9)] or any other connection;
- b) For column minimum four corner bars of 12 mm diameter or as per design requirements shall be continuous from foundation till roof using Class H coupler as per [C-7(9)] or any other connection;
- c) For beam at least one 12 mm dowel at centre of the bearing region of the beam shall be provided to avoid accidental failure; and
- d) For any other member appropriate reinforcement shall be provided to avoid accidental lateral loads.

8.1.2 Transfer of Tensile Forces

8.1.2.1 Tensile forces generated at the joint shall be resisted by tie bars sufficiently anchored at both the sides of the joint.

8.1.2.2 Adhesive bond between the members shall not be accounted in force transfer.

8.1.2.3 The tensile stresses generated in the connecting members shall also be linked to the reinforcement of the member by proper provision of ties or any other connection.

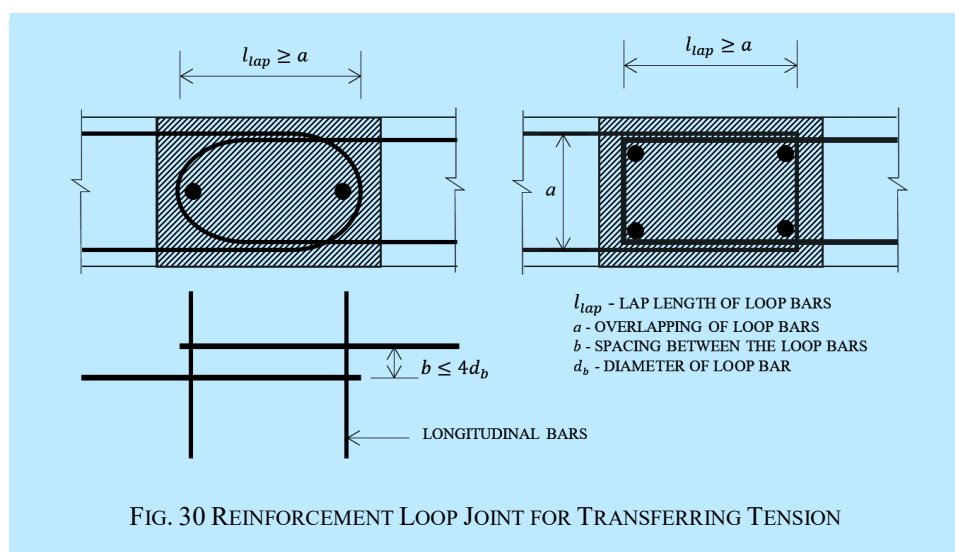
8.1.2.4 Premature brittle failure shall be avoided for tie members and material selected for making tie shall be ductile.

8.1.2.5 Joints with grout filled tubes or sleeves shall be utilised with adequate development length. International practice indicates that the confinement of dowel tube enhances the rebar anchorage performance, reducing the required development length. Hence, for determining development length reduced development length shall be adopted based on test results as per [12](#). In absence of test results, the bond strength of concrete shall be considered. Alternatively, other joint configurations may be adopted, provided their performance is evaluated through assemblage and system level testing along with nonlinear analysis and approved from competent authorities or institutes of repute.

8.1.2.6 A joint using bolting, welding or lap splicing shall be properly anchored to the precast components and transfer of forces in the member shall be verified.

8.1.2.7 Reinforcement loops (U-bars) protruding from precast elements may be used to provide tie between the elements as shown in [Fig. 30](#). Reinforcement of 8 mm or higher diameter shall be provided as longitudinal reinforcement (preferably four numbers with U-bars) as shown in the [Fig. 30](#). Lap length, spacing of loop bars and overlapping length shall be as per the [Fig. 30](#).

8.1.2.8 Anchoring of reinforcement shall be such that high stress concentration in the members are avoided.



8.1.3 Transfer of Shear Force

8.1.3.1 Shear forces may be transferred between concrete elements by adhesion or friction at joint interfaces, shear-key effect at indented joint faces, dowel action of transverse steel bars, pins and bolts, or by other mechanical connection devices. The frictional resistance can be enhanced by the pull-out resistance of tie bars properly placed across the joint.

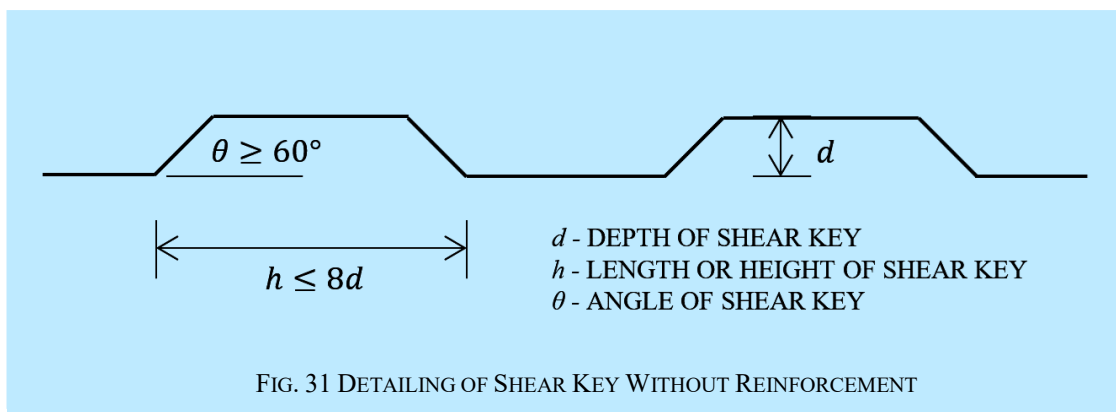
8.1.3.2 Adhesive bond or adhesion between the interfaces of elements shall not be considered in the calculation of shear strength of the joint.

8.1.3.3 Shear resistance of joint between two elements with reinforcement may be calculated using shear friction (see [7.12](#)) equation or as per good practice [C-7(2)]. Shear friction shall not be used if the reinforcement crossing the joint is subjected to compressive forces by the displacement of the members.

8.1.3.4 Permanent compressive stresses acting on the joint shall also be considered along with shear friction.

8.1.3.5 Unreinforced shear keys or castellation or indented joint face shall be used for members subjected to very high shear forces. This connection shall be designed to remain elastic under ultimate limit states to avoid sudden brittle failure of the joint. Transverse reinforcement across the interfaces shall be provided to prevent sudden separation of the members.

8.1.3.6 The dimension of shear keys shall be as per [Fig. 31](#).



8.1.4 Transfer of Bending and Torsional Moment

8.1.4.1 Strong connections shall be used possessing stiffness, strength and ductility confirming to that of cast in-situ construction.

8.1.4.2 Moment resisting connections shall be designed such that ductile failures occur without any local modes of brittle failure (shear, pull-out, etc).

8.1.4.3 Continuity within the joint may be obtained by any of the following or combination:

- a) Lapping of bars;
- b) Reinforcement grouted into apertures;
- c) Overlapping loop or U-bars;
- d) Reinforcement connected using threaded couplers; and
- e) Any other type of connection with appropriate testing of the connection.

8.2 Connection Based on Precast Concrete Seismic Systems

The connection of seismic moment resisting frames and structural walls incorporating precast concrete elements generally fall into two broad categories, either 'equivalent monolithic (emulative)' systems or, 'jointed' systems. The distinction between these types of construction is based on the design of the connections between the precast concrete elements as provided in the following sections.

8.2.1 Equivalent Monolithic Systems (Emulative)

A precast concrete structural system satisfying the requirements of this clause shall have strength and toughness equivalent to that provided by a comparable monolithic reinforced concrete structure.

8.2.1.1 Connections in monolithic systems

The connections between precast concrete elements of equivalent monolithic systems can be subdivided into two categories:

- a) *Strong connections of nominal ductility*

In moment resisting frames and structural walls these connections are protected by a capacity design approach which ensures that flexural yielding occurs away from the connection region.

b) *Ductile connections*

Ductile connections of equivalent monolithic systems typically comprise longitudinal reinforcing bars in the connection which are expected to enter the post-elastic range in a severe earthquake.

In moment resisting frames yield penetration may occur into the connection end-region. The potential plastic hinge region may extend a distance along the end of the member as in cast-in-place construction.

8.2.2 Jointed Systems

This system is not in the purview of this NBCS.

8.3 Design of Connections

The design of connections shall follow those design methods and considerations for reinforced concrete, prestressed concrete and structural steel. Proprietary connections shall be based on tests and approved from competent authority or institutes of repute. The following types are the broad category of the connections based on connecting members:

- a) Column to foundation;
- b) Column to column;
- c) Beam to column;
- d) Beam to beam;
- e) Beam to slab;
- f) Beam to wall;
- g) Wall to wall;
- h) Wall to slab; and
- j) Slab to slab.

Examples of typical connections are shown in [Annex E](#). These connections are for illustrative purposes only. In all instances, the design should demonstrate that the connections are able to resist the applied structural actions. Where appropriate, the suitability of the connection to act as a pinned or moment connection is highlighted. The location of the connections for a framed structure shall be as per [6](#) ([Fig. 2](#), [Fig. 3](#), [Fig. 4](#) and [Fig. 5](#)).

8.3.1 Detailing of Reinforcement

The following shall be considered:

- a) *Lapping of bars* — Where bars passing through the connection are lapped to provide continuity of reinforcement, the recommendations good practice [C-7(2)] and integrity tie requirements shall be followed;
- b) *Reinforcement grouted into aperture* — An adequate capacity should be provided for grouted reinforcing bars to prevent pull-out. Embedment length shall be as per manufacturer guidelines, however testing shall be carried out as per [12](#). If grouted tube is used, the embedment length of dowel shall be at least 300 mm or as required by design, whichever is higher;
- c) *Reinforcement loops* — Loop reinforcement shall be provided as per [8.1.2](#) and Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS;
- d) *Couplers* — Reinforcement may be connected by couplers. The concrete cover to the couplers (other than grouted tube or duct) shall be at least 25 mm or as per specifications of the project. A locking device should be used for threaded coupler connections where there is a risk of the threaded connection working loose, for example during vibration of in-situ concrete. Reference should be made to [C-7(9)] and the manufacturers’ technical specifications for guidance and acceptance criteria for the usage. Testing of couplers shall be as per [12](#) and relevant guidelines of [C-7(9)]. When the system is part of lateral force resisting system, the grout strength shall not be less than 90 MPa and testing shall be carried out as per [12](#) by the supplier of the coupler or as agreed upon by the project team. Modify to mention 12 for LFRS and 7(9) for regular coupler without lateral force; and

- e) *Welding of bars* — When bars are connected by welding, the connection shall ensure full capacity of the bar is achieved and failure of weld is avoided. Testing of the welded samples shall be as per relevant standards. Welding of rebar shall be avoided for normal circumstances, to be used for repair or strengthening of a connection, if required.

8.3.2 Connections in Compression

8.3.2.1 The design of connections in tension shall follow the guidelines of [8.1.1](#).

8.3.2.2 In normal circumstances, the area of concrete to be utilized for calculating the strength of the connection in a wall or column should be the greater of:

- a) 75 percent of the area of contact between the wall or column and connection; or
- b) The area of the in-situ concrete excluding the part of any intruding slab or beam units. However, those parts of the floor slab or beam units that are solid over the bearing may be included in calculating the strength and such units should be properly bedded on concrete or mortar of adequate quality. This area should not be taken as greater than 90 percent of the wall or column area.

8.3.3 Connections in Tension

8.3.3.1 The design of connection in tension shall follow the guidelines of [8.1.2](#).

8.3.3.2 Stirrups shall be placed along the lap length, for the connection utilizing lapping of bars, grouted duct, or grouted coupler for transfer of tensile forces.

8.3.3.3 Welding or bolting shall be used with appropriate design. The embedded inserts shall be appropriately anchored to the primary member.

8.3.4 Connection in Shear

8.3.4.1 The design of connection in shear shall follow the guidelines of [8.1.3](#).

8.3.4.2 A shear connection may be considered effective, if the connection is grouted with a suitable concrete or mortar mix of strength of at least 5 MPa higher than the connecting members and the criteria of [8.1.3](#) is satisfied.

8.3.4.3 For joints subjected to minimal shear between wall panels or thin members, looped wire ropes shall be used to connect between the members with a maximum spacing of 450 mm. These connections shall not be used in lateral force resisting system and in transfer of seismic forces.

8.3.4.4 Transfer of forces solely by friction caused due to gravity load shall not be used.

8.3.5 Bond, Anchorage, Bearing, Laps, Joints and Bends in Bars

8.3.5.1 Provision of bond, anchorage, bearing, laps, joints and bend in bars shall be as follows:

- a) *Avoidance of bond failure due to ultimate loads* — At both sides of any cross-section the force in each bar should be developed by an appropriate embedment length or other end anchorage;
- b) *Anchorage bond stress* — Anchorage bond stress (τ_b) is assumed to be constant over the effective anchorage length. It may be taken as the force in the bar divided by its effective surface anchorage area, as given below:

$$\tau_b = \frac{F_s}{\pi \phi_E L}$$

where

τ_b = bond stress;

F_s = force in the bar or group of bars;

ϕ_E = effective bar size which, for a single bar is equal to the bar size and, for a group of bars;
and

L = anchorage length.

- c) *Values for design ultimate anchorage bond stress* — Values for design ultimate anchorage bond stress τ_{bu} may be obtained from the equation. The anchorage bond stress (τ_b) shall not exceed the design ultimate anchorage bond stress:

$$\tau_{bu} = \beta \sqrt{f_{cu}}$$

where

τ_{bu} = design ultimate anchorage bond stress; and

β = a coefficient dependent on the bar type.

The values of β may be taken from [Table 11](#). These values include a partial safety factor (γ_m) of 1.4.

Table 11 Values of Bond Coefficient β			
[Clause 8.3.5.1(c)]			
Sl No.	Bar Type	β	
		Bar in Tension	Bar in Compression
(1)	(2)	(3)	(4)
i)	Plain bar	0.28	0.35
ii)	Deformed bar	0.4	0.63

8.4 Connections Other than Those Involving Continuity of Reinforcement

8.4.1 Joints with Structural Steel Inserts

Joints with structural steel inserts generally consist of a steel plate or rolled steel section projecting from the face of a column to support the end of a beam. The reinforcement in the ends of the supported member shall be designed in accordance with the provisions mentioned in Part C/Section 5 of this NBCS. Detailed design provisions shall include the following:

- The steel sections and any bolted or welded connections should be designed in accordance with the appropriate recommendations;
- Design ultimate bearing stresses shall be as per [7.11](#); and
- Consideration should be given in the design to the possibility of vertical splitting under the steel section because of shrinkage effects and localized bearing stresses, for example under narrow steel plate.

8.4.2 Resin Adhesives

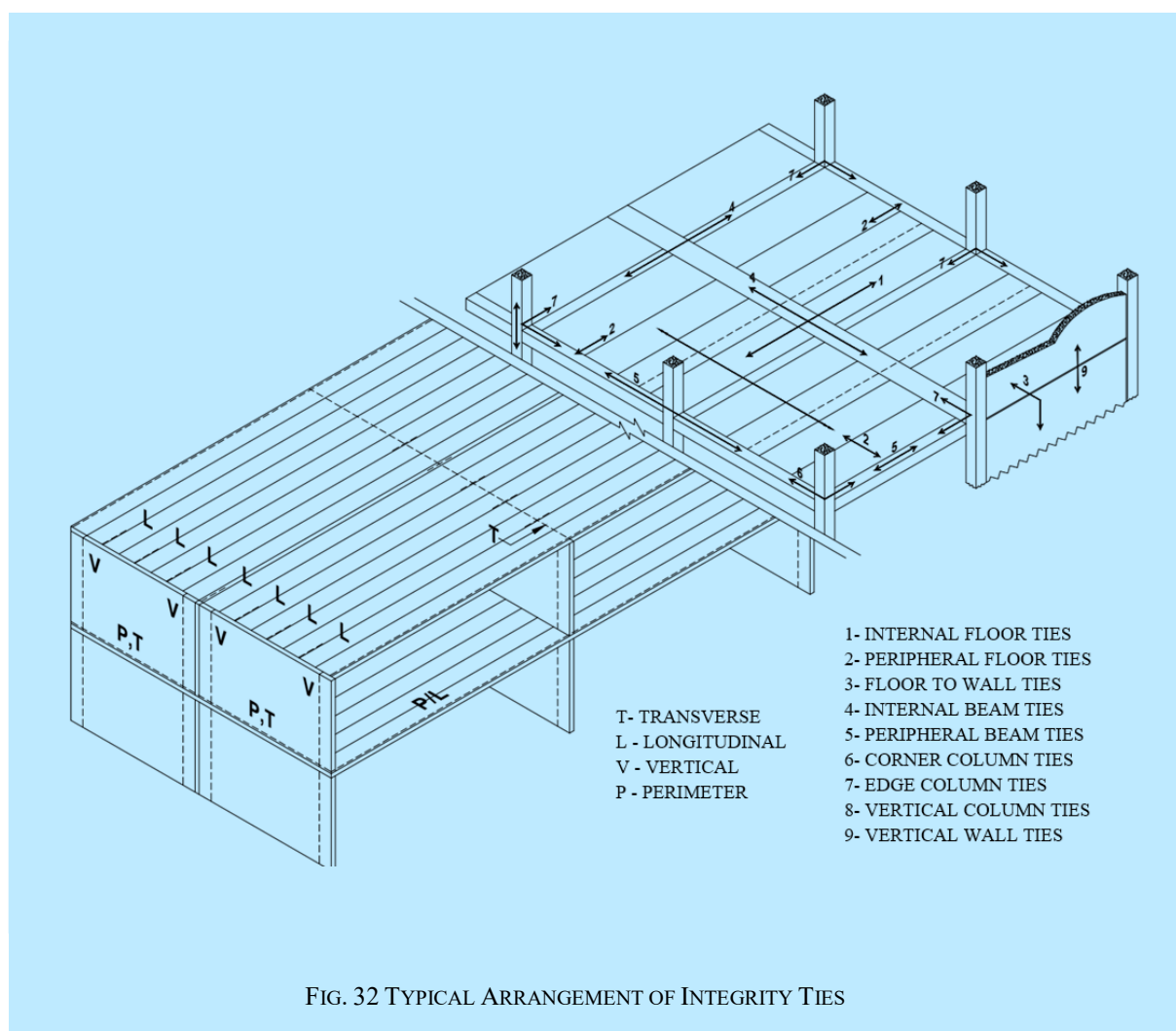
Resin adhesives may be used to form joints subjected to compression but not to resist tension or shear. These adhesives should only be used where they are adequately protected against the effects of fire.

8.4.3 Other Types of Connections

Any other type of connection that can be shown to be capable of carrying the ultimate loads acting on it may be used. Amongst those suitable for resisting shear and flexure are those made by prestressing across the joint.

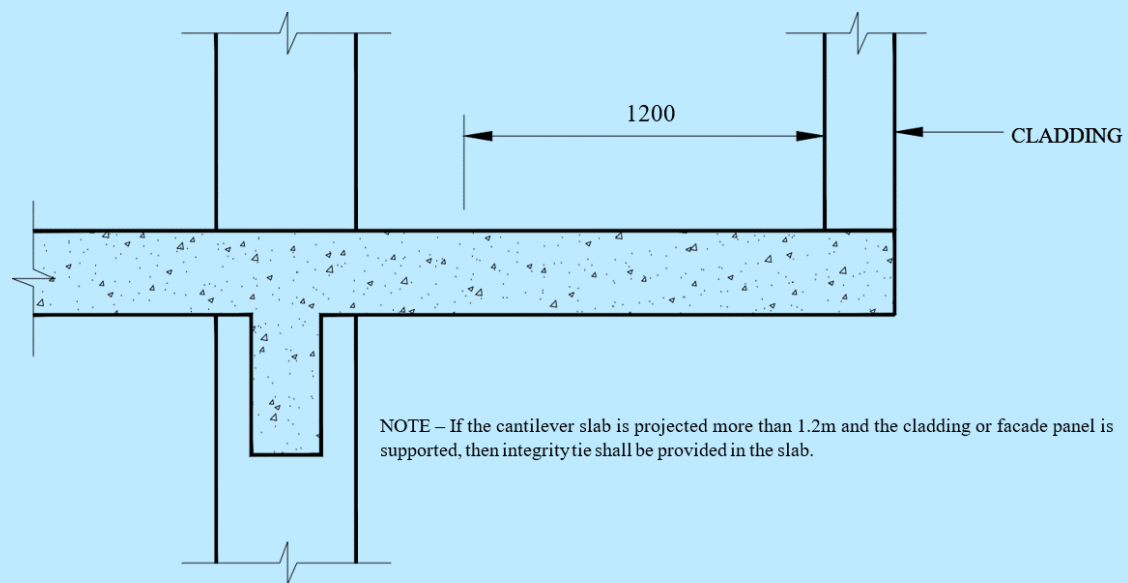
8.5 Structural Integrity and Progressive Collapse

Reinforcement and connections shall be detailed to tie the structure together effectively maintaining adequate structural integrity and to prevent progressive collapse. Integrity ties shall be followed in accordance with the following clauses. [Figure 32](#) shows the typical arrangements of integrity ties in a large panel structure and frame structure. Progressive collapse requirements shall be as per [8.5.10](#).

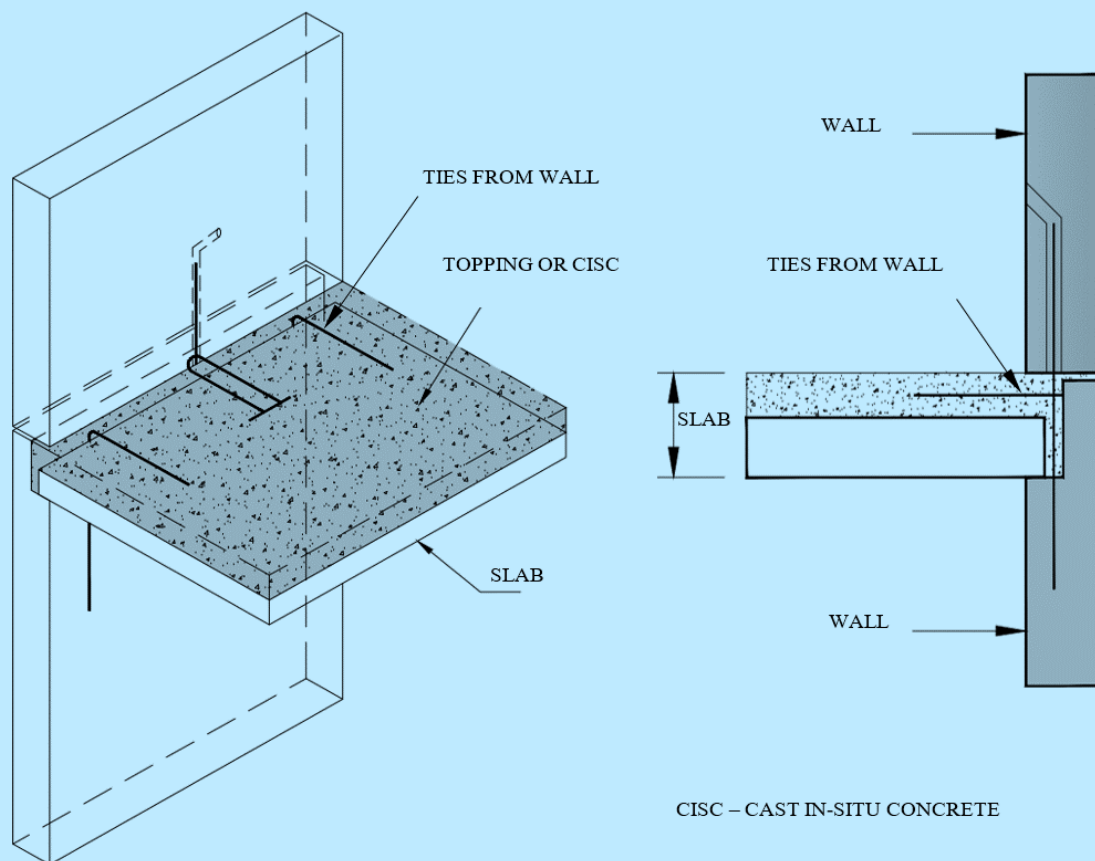


8.5.1 Peripheral Ties

At each floor and roof level an effectively continuous tie shall be provided within 1.2 m of the edge of the building or within the perimeter wall (see Fig. 33). The tie shall be capable to resist a tensile force of F_t equal to 60 kN or $(20 + 4N)$ kN, whichever is less, where N is the number of storeys (including basement).

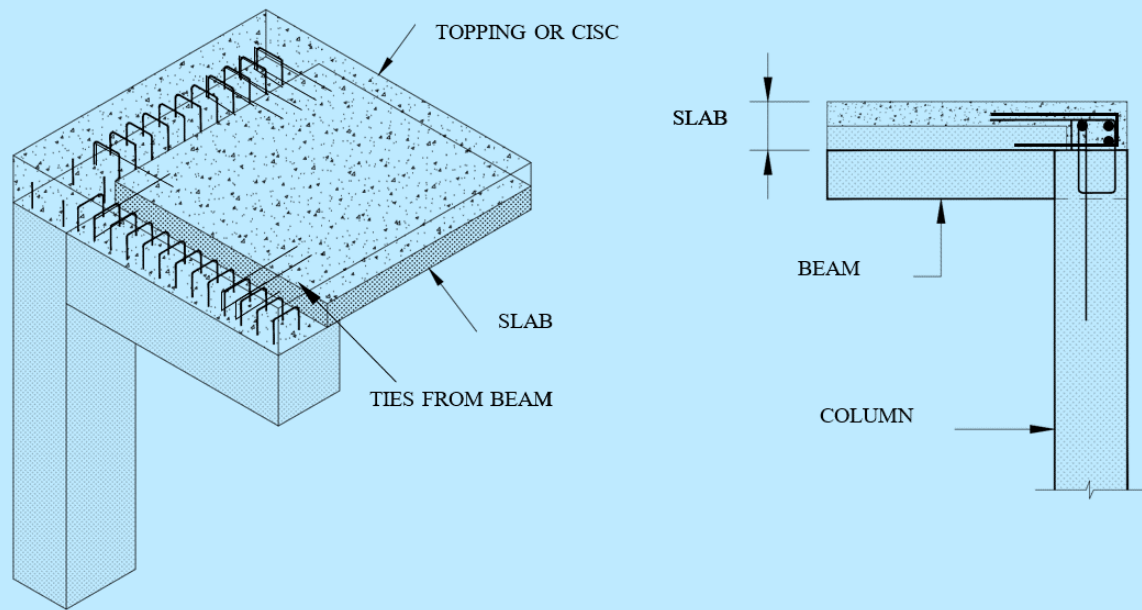


33A) POSITION OF PERIPHERAL TIE

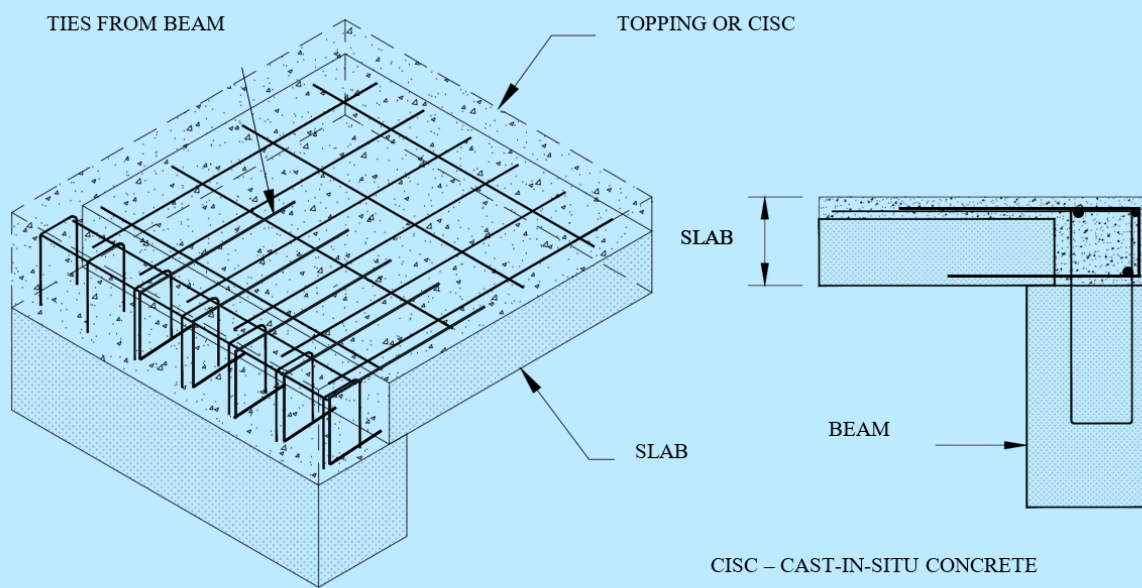


33B) PRECAST WALL TO WALL CONNECTION WITH SHEATHING PIPE

FIG. 33 POSITION FOR PERIPHERAL TIE (CONTINUED)

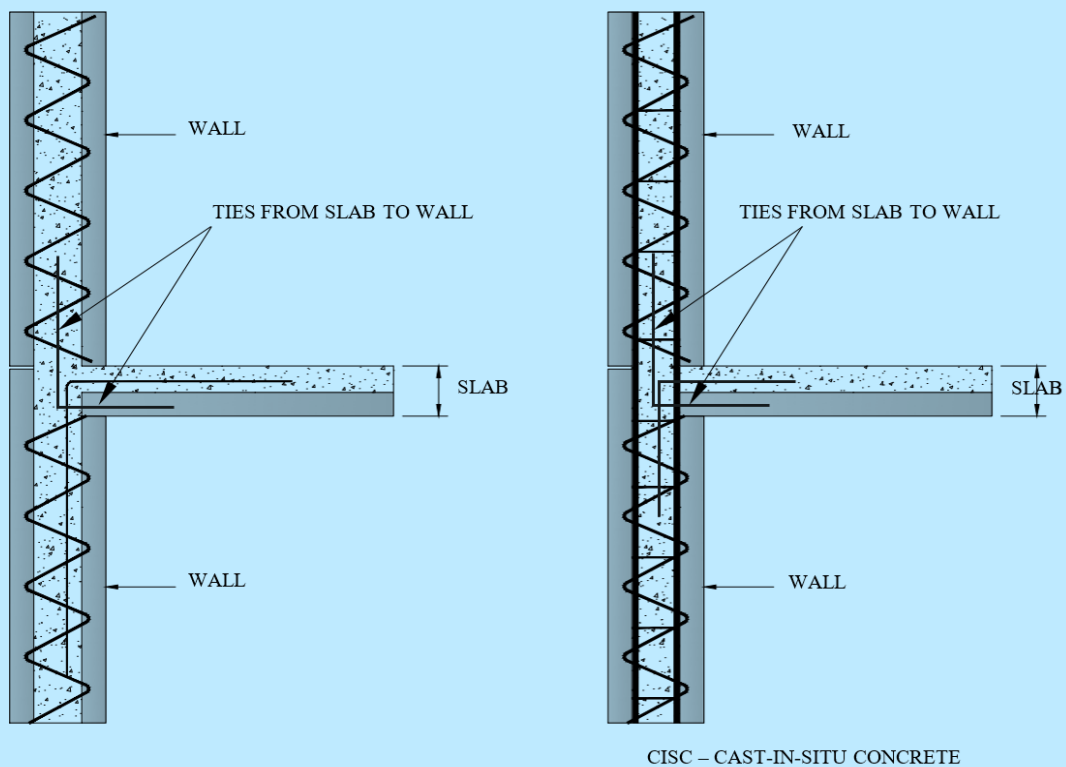
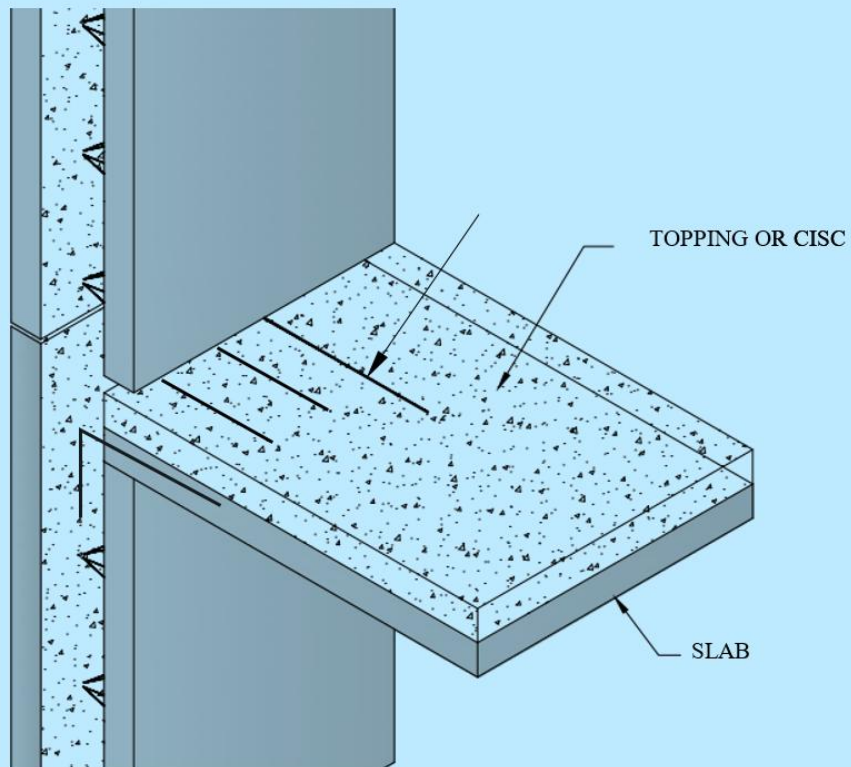


33C) PRECAST COLUMN, BEAM AND SLAB CONNECTION



33D) PRECAST BEAM SLAB CONNECTION

FIG. 33 POSITION FOR PERIPHERAL TIE (CONTINUED)



33E) PRECAST BEAM SLAB CONNECTION

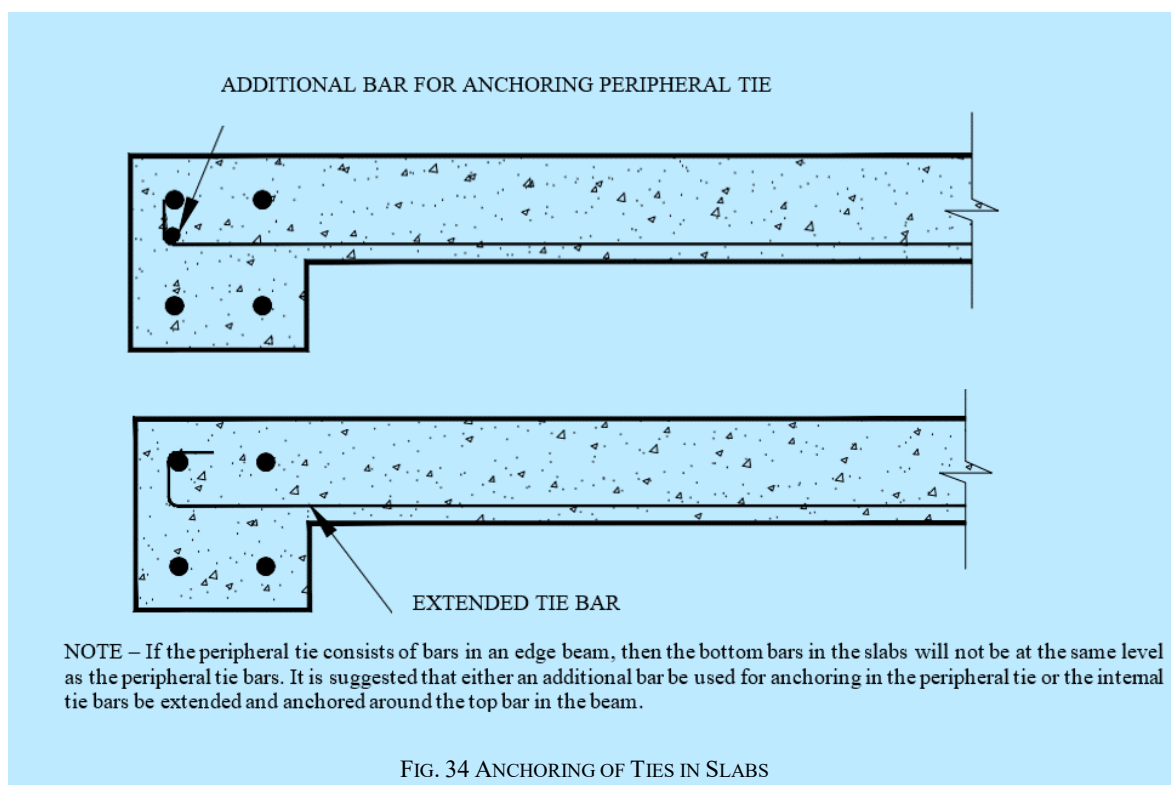
FIG. 33 POSITION FOR PERIPHERAL TIE

8.5.2 Internal Ties

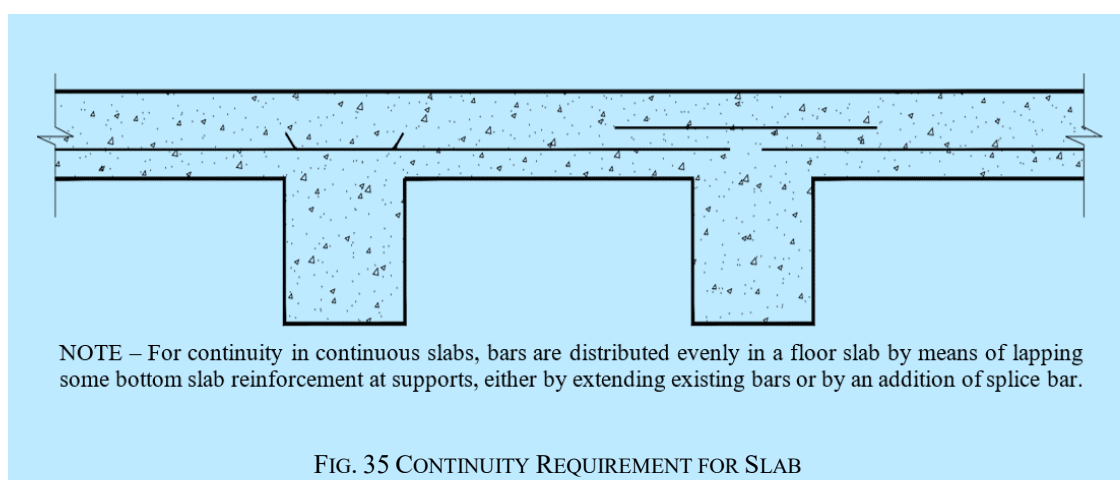
These shall be provided at each floor and roof level in two directions approximately at right angles. Ties shall be effectively continuous throughout their length and be anchored to the peripheral tie at both ends, unless continuing as horizontal ties to columns or walls (see Fig. 34). The tensile strength, in kN/m width shall be the greater of:

$$\left(\frac{g_k + q_k}{7.5}\right) \frac{L_r F_t}{5} \text{ and } F_t$$

where, $(g_k + q_k)$ is the sum of average characteristic dead and imposed floor loads, in kN/m² and L_r is the greater of the distance between the centre of columns, frames or walls supporting any two adjacent floor spans in the direction of the tie under consideration.



The bars providing these ties shall be distributed evenly in the slabs (see Fig. 35) or shall be grouped at or in the beams, walls or other appropriate positions but at spacing generally not greater than $1.5 L_r$.



8.5.3 Horizontal Ties to Column and Wall

All external load-bearing members such as columns and walls shall be anchored or tied horizontally into the structure at each floor and roof level. The design force for the tie shall be greater of:

- a) $2F_t$ kN or $L_s \times (F_t/2.5)$ kN, whichever is less for a column or for each metre length, if there is a wall (L_s is the floor to ceiling height in metre); and
- b) 3 percent of the total ultimate vertical load in the column or wall at that level.

For corner columns, this tie force should be provided in each of two directions approximately at right angles.

8.5.4 Vertical Ties

Each column and each wall carrying vertical load should be tied continuously from the foundation to the roof level.

8.5.4.1 The reinforcement provided is required to resist a nominal tensile force of at least 50 kN per horizontal metre of wall.

8.5.4.2 At least two integrity ties shall be provided in each wall panel.

8.5.4.3 Connection between precast column shall have vertical integrity ties, with a nominal tensile strength of $1.4A_g$, where A_g is the gross area of the column. For column with larger cross section reduced effective area equal to the actual cross section required shall be used. However, the reduced area shall be at least one-half of the gross area of column.

8.5.4.4 In situation where provision of vertical ties cannot be done, the element should be removed, and the surrounding members shall be designed to bridge the gap.

8.5.4.5 All members in the lateral force resisting system shall have one tie continuous from foundation to terrace at each end of the member with Class H coupler as per [C-7(9)] (ties as per design) or direct lapping (ties as per lapping requirements).

8.5.5 Tie Anchorage

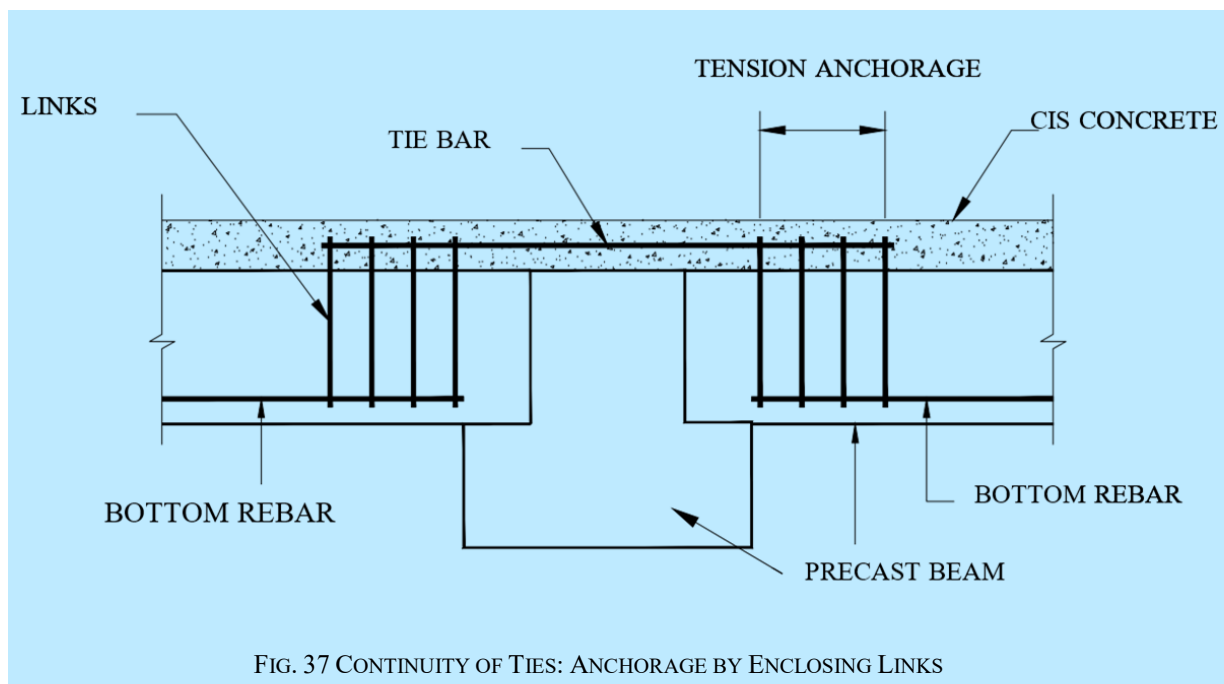
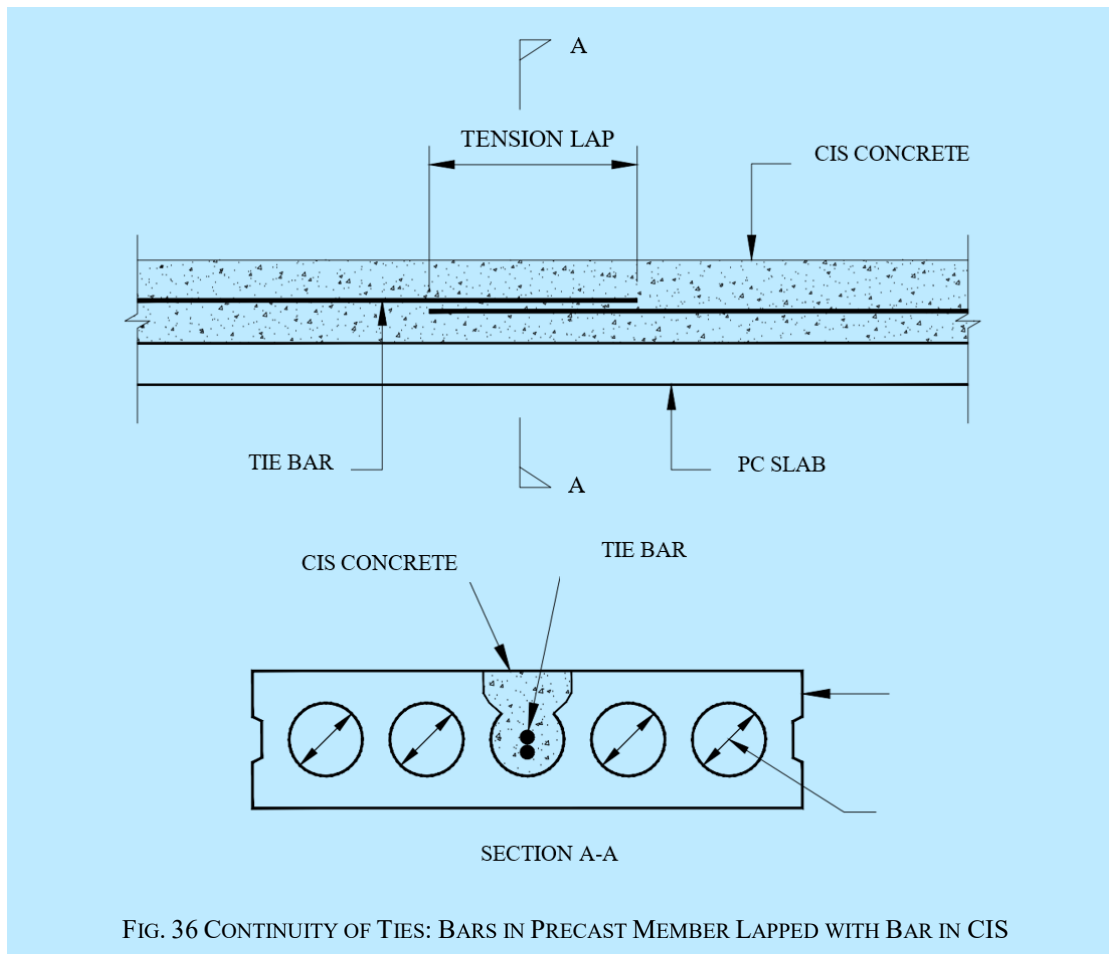
A tie crossing another tie at right angles may be considered anchored if the bars extend 12 times the bar diameter or an equivalent anchorage beyond the other tie; or an effective anchorage length, calculated from the tension force, beyond the centreline of the other tie. At abrupt changes in construction or at re-entrant corners, it shall be ensured that the ties are sufficiently anchored or to be made effective by other means.

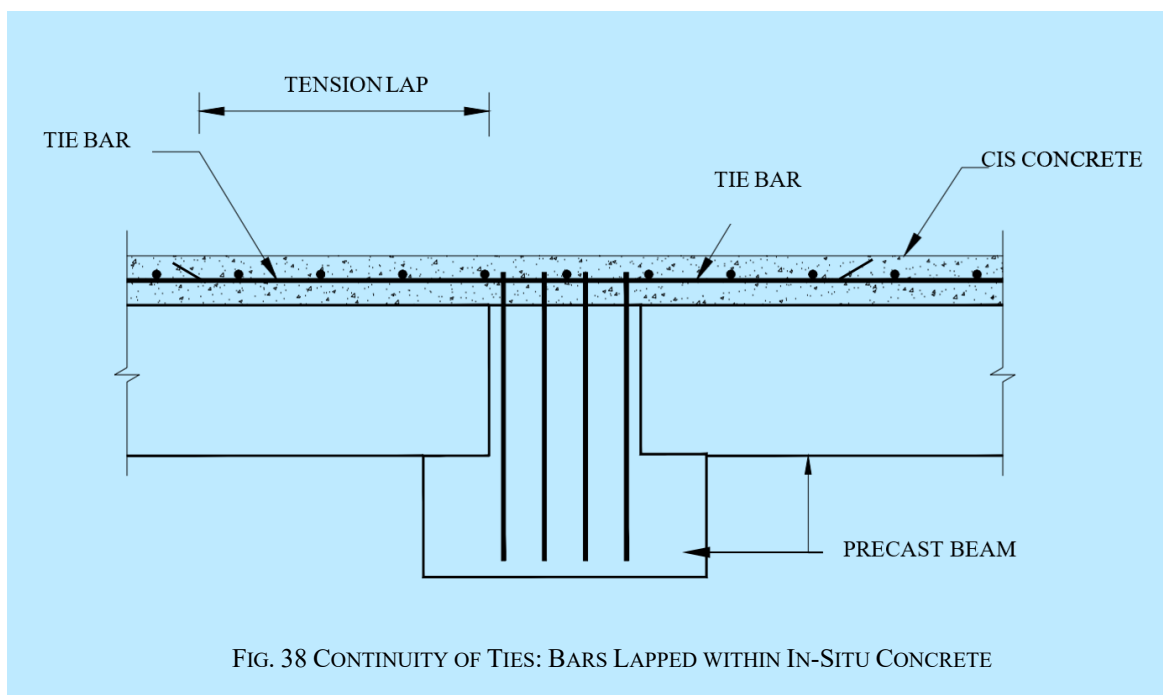
8.5.6 Continuity of Ties

A continuous tie shall satisfy [8.3.1](#). The minimum thickness of in-situ concrete section where tie bars are provided shall be at least the total dimension of the bar diameter (or two diameters at laps) and twice the maximum aggregate size plus 10 mm.

To provide continuity, the tie shall also satisfy one of the following:

- a) A bar in a precast concrete unit lapped with a bar encased in in-situ concrete bounded on two opposite sides by rough faces of the same precast unit (see [Fig. 36](#));
- b) A bar in a precast concrete unit lapped with a bar encased in the topping of in-situ concrete tied to the precast unit by links or stirrups. The tensile resistance of such links and stirrups should not be less than the designed tension in the tie (see [Fig. 37](#)); and
- c) Bars lapped within in-situ topping concrete with projecting links or stirrups from the supporting precast units, such as beams or slabs (see [Fig. 38](#)).





8.5.7 Ties between Precast Element and Topping

In potential plastic hinge regions, the consequences of delamination of the topping from the precast members shall be assessed. If the maximum shear stress at any section exceeds 0.4 MPa, connectors shall be provided between the topping and precast to achieve composite action as per the following guidelines:

- Ties with an effective area of 40 mm^2 per m^2 of floor area, or equivalent connectors, shall connect the topping to the precast element;
- Spacing of connectors shall not exceed 1 500 mm and the tributary area of topping reliant on each connector shall not exceed 2.25 m^2 ;
- Connectors shall engage horizontal reinforcement, or shall be otherwise effectively anchored into both the topping and the precast element, or into the joints between precast elements; and
- In HCS appropriate connection shall be placed in slab joints.

8.5.8 Anchorage in Structures

All precast floor and precast roof units should be effectively anchored if these units are not utilised to provide the ties. The anchorage shall be capable of supporting the dead weight of the precast unit to that part of the structure which contains the ties.

8.5.9 Eccentricity

Ties connecting precast floor or roof units shall be so designed and placed as to minimise eccentricity.

8.5.10 Anchorage at supports

In case reinforcing bars are used to provide the structural integrity of slab ends supported on corbels or nibs, great care shall be taken to ensure it is adequately lapped and anchored. Allowance should also be made for constructional inaccuracies.

8.5.11 Progressive Collapse

The following recommendations shall be followed for ensuring stability of prefabricated structures:

- Adequate strengthening of external joints of external wall panels is important since these elements are not fully restrained on both sides by floor panels. Adequate design precautions shall be taken.

Connections for the external wall panels are likely to be the weakest points of a precast panel building, if not designed properly;

- b) Restraint shall be provided to all load bearing elements at the corners of the building. These elements and the external ends of cross-wall units shall be stiffened either by introducing columns as connecting units or by jointing them to non-structural wall units, which in emergency shall act in force transfer;
- c) In prefabricated construction, the possibility of gas or other explosions which can damage primary structural elements leading to progressive collapse of the structure shall be considered. It is, therefore, necessary to consider the possibility of progressive collapse in which the failure or displacement of one element of a structure causes the failure or displacement of another element and results in the partial or total collapse of the building;
- d) Provision in the design to reduce the probability of progressive collapse is essential in buildings of height more than 15 m and is of relatively higher priority than for buildings of lower height. However, buildings of special importance shall be checked for progressive collapse irrespective of height guidelines. One way of preventing progressive collapse of external wall panels is to design the combined precast wall panels of entire width as a beam of depth equal to the floor height for the impact loading of upper floors and provide reinforcement accordingly at the top and bottom of all precast wall panels at each floor. To make reinforcement continuous across joints, top and bottom bars projected into joints between wall panels shall be joined with lapping bars;
- e) It is necessary to ensure that any local damage to a structure does not spread to other parts of the structure remote from the point of mishap and that the overall stability is not impaired, but it may not be necessary to stiffen all parts of the structure against local damage or collapse in the immediate vicinity of a mishap, unless the design briefs specially require this to be done. The requirements as specified at [8.5](#) shall be followed to prevent progressive collapse of the structure;
- f) Additional protection may be required in respect of damage from vehicles. Further, it is necessary to consider the effect of damage to or displacement of a load-bearing member by an uncontrolled vehicle. It is strongly recommended that important structural members are adequately protected by concrete kerbs or similar method. In areas with possibility of passenger vehicular impact, the precast components shall be designed for an un-factored load of 27 kN, in any direction, acting at heights between 450 mm and 700 mm on an area not larger than 300 mm × 300 mm to produce maximum load effect. This load is not required to act concurrently with any handrail or guard rail system loads; and
- g) In all aspects of erection that affect structural design, it is essential that the designer shall maintain a close liaison with the builder/contractor regarding the erection procedures to be followed.

8.5.12 Key Element

For buildings of five or more storeys, the layout should be checked to identify key elements. A key element is such that its failure would cause the collapse of more than a limited area close to it. The limited area defined above may be taken equal to 70 m² or 15 percent of the area of the storey, whichever is less. If key element exists, it is preferable to modify the layout so that the key element is avoided or design the element for a force of 36 kN/m².

8.6 Miscellaneous Items for Joints

8.6.1 Sealants for Joints

In choosing the appropriate type of sealant for the joints, consideration should be made in respect of the movements between the components to be joined, the bond that is achievable between the components and the sealant and the nature of the sealing material itself. Hybrid, PU (polyurethane, polysulphide), silicone based, or any other sealant shall be used depending on requirements.

8.6.2 Joint Width

Joint widths should be sized to accommodate construction tolerances and movements without overstressing the jointing material. Irrespective of movement, a joint width of 10 mm is the minimum practicable for sealant application.

8.6.2.1 Minimum width of joint between the erected elements or supporting elements shall be as follows:

- a) 10 mm, if not specified;
- b) 20 mm for wall elements;
- c) 30 mm for column elements; and
- d) For larger elements as per grouting requirements but shall preferably not exceed 50 mm.

8.6.3 Joint Preparation

Satisfactory performance is critically dependent on the satisfactory adhesion of the sealant to the joint surfaces. Primers may be recommended by the manufacturer for some materials. Formwork oils, curing compounds, silicone waterproofing admixtures and surface coating materials may reduce bond and require special precautions.

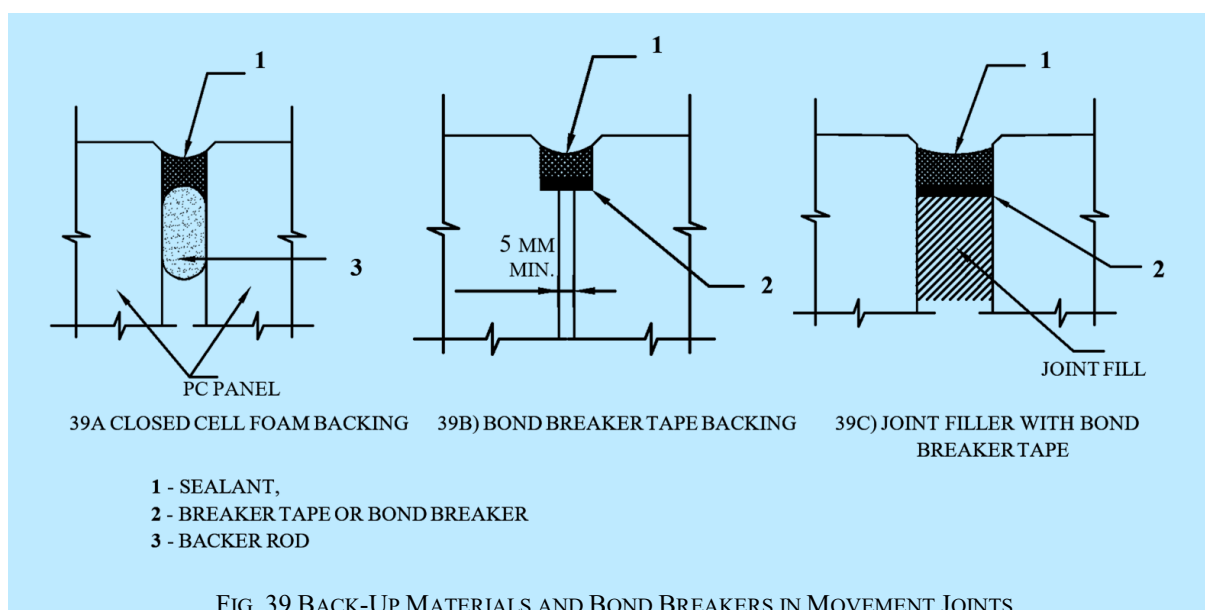
Water jetting, sand blasting, wire brushing or the use of retarder may be required to prepare the concrete joint surfaces in certain instances. Consideration should be given to the use of chamfers to reduce edge damage.

8.6.4 Back up Material and Bond Breaker

To ensure good adhesion, a firm backing should be provided for the application of sealants. The required shape and proportions of the sealant are achieved by the correct installation of the back up material.

Sealants in movement joints should not adhere to the backing material to avoid any unnecessary restraint. To ensure that the sealant and back up materials, bond breakers and joints fillers are compatible and appropriate for the proposed end use, advice from the manufacturer should be obtained as necessary.

The sealant backing may be provided by a closed cell foam back up material alone (see Fig. 39A), a thin self-adhesive bond breaker tape (see Fig. 39B) or a joint filler separated from the sealant by a thin self-adhesive bond breaking tape (see Fig. 39C).



8.6.4.1 Minimum joint gap widths

For elastic sealants, the minimum thickness shall be 5 mm, the ratio of width to thickness should never exceed 1 : 1 and for optimum performance 2 : 1.

8.6.4.2 Sealant application

High moisture content may be detrimental to the adhesion of the sealant. Sealing should not be undertaken if there is free water present on the surface of the concrete.

8.6.4.3 Performance, choice and failure of sealant

The magnitude of the movement, and also the mode, frequency and rate of movement affect the performance of a sealed joint.

There are many factors affecting the choice of sealant suitable for different types of movement. However, as a general rule, joints which have to accommodate frequent and rapid movement need an elastic sealant, while joints in massive components, with high thermal inertia resulting in much slower movement, may be satisfactorily sealed with an elastoplastic, plasto-elastic or a plastic sealant.

The different ways in which sealants can fail as a consequence of different factors including climatic conditions, environmental factors, substrate incompatibility, abrasion and traffic loading need to be considered when selecting a sealant.

8.6.5 Sealing Strips

Preformed sealing strips of appropriate sizes and sections shall be used as required. There are two basic types:

- a) Mastic strips; and
- b) Impregnated or coated cellular strips.

8.6.5.1 Mastic strips

These should be normally installed during the assembly of components. They require an initial compression to ensure proper adhesion to the components forming the joint. As a degree of compression is also required during in service, mastic strips should not be used for joints which open beyond their assembled size.

8.6.5.2 Impregnated or coated cellular strips

Impregnated or coated cellular strips may be supplied in a pre-compressed form, and one face may be adhesive coated. When pre-compressed, they can readily be installed within the joint gaps, but should be of a suitable size to be maintained under the degree of compression specified by the manufacturer throughout the range of joint movements. The degree of compression may be varied according to the level of sealing performance required.

To ensure uniformity of compression and stability of the seal, it is important that the joint faces are parallel. Sufficient joint depth shall be provided to accommodate the seal with these depths to width ratios. For external applications, sealing strips should have adequate exposure resistance for the proposed service conditions.

8.6.6 Joint Fillers

Fillers for movement joints should be carefully selected to suit their intended use. To ensure that the sealant and filler are compatible, advice from the manufacturer should be sought. Fillers should be non-absorbent with appropriate elastic properties and fire resistant.

8.6.7 Overlapping of Precast Facade

The resistance to water penetration between the upper and lower panel of precast facade is commonly provided by an upstand profile. A minimum overlap of 75 mm between the upper and lower panel is recommended.

9 CONSTRUCTION PLANNING

9.1 Pre-Engineering and Design Coordination

9.1.1 Importance of Pre-Engineering

Pre-engineering plays a critical role in the planning and execution of any precast concrete project. To ensure a successful outcome, the planning engineer should begin working from the design stage to ensure proper production, installation, connections, and logistics. Key resources must be identified and prepared for each phase of the project, including:

- a) Availability land/approvals/water and electricity for a site-based factory;

- b) Suitable plant and machinery;
- c) Steel moulds;
- d) Lifting machinery at factory;
- e) Trailers;
- f) Lifting machinery on site (tower cranes/mobile cranes); and
- g) Site access for transportation of large elements as well as route studies and local regulations and hours for operating large trailers.

9.1.2 Resource Planning and Feasibility Factors

The resources should be carefully planned based on the types of project elements and their quantities. The engineer is also responsible for reserving these resources throughout the project duration. Key feasibility considerations include:

- a) *Building geometry* — Ensure shapes and geometry are as symmetrical and regular as possible;
- b) *Grid-lines* — Column and wall locations should, preferably, follow a standardized pattern;
- c) *Height clearances* — Consideration of clear height expectations between beams and slabs;
- d) *Grid spacing* — Optimize grid spacing to take full advantage of precast elements; and
- e) *Standardization* — Standardize elements within one floor and across multiple floors when feasible.

9.1.3 Pre-Engineering Design Stage Process

The pre-engineering design stage should include the following steps:

- a) Study of building functionality;
- b) Load assumptions and calculations;
- c) Preparation of the design basis report (DBR);
- d) Structural analysis;
- e) Preliminary framing plan selection for precast elements;
- f) Approval of preliminary framing plans by the architect/client;
- g) Review meetings with architects and consultants;
- h) Finalization of the structural framing plan after incorporating feedback and value engineering;
- j) Release of the good for construction (GFC) and structural framing plan (R0), including:
 - 1) Geometry of precast elements;
 - 2) General arrangement (GA) drawings with reinforcement details; and
 - 3) Connections and inserts.
- k) Coordination with mechanical, electrical, plumbing and other service consultants should occur during the early stages.

9.2 Project Planning, Scheduling and Monitoring

9.2.1 Production Planning

Careful planning is essential for ensuring that precast concrete systems are produced efficiently. Key factors to address include:

- a) Synchronization of manufacturing schedules and resources to meet construction and storage needs;
- b) Anticipation of any special transportation or access challenges;
- c) Use of detailed production shop drawings to facilitate smooth execution; and
- d) Factory curing periods for high strength concrete elements.

9.2.1.1 Shop drawings

Shop drawings should contain:

- a) Member shapes (elevations and sections) and dimensions;
- b) Details of reinforcement, anchors, inserts, etc;
- c) Connection details;
- d) Production tolerances;
- e) Handling and lifting devices;
- f) Finishing and bearing seat details;
- g) Transportation and storage methods; and
- h) Unique identification of elements (ID, location and orientation).

9.2.1.2 Factory vs on-site production

A decision whether precast elements should be produced in an off-site factory or on-site, based on factors such as:

- a) Project scale and volume;
- b) Availability of factory space near the site;
- c) Contractual agreements regarding space provision;
- d) Element type, weight, and size; and
- e) Cost-benefit analysis.

9.2.2 Production Schedule

9.2.2.1 Cycle time and throughout rate and the key aspects

- a) *Cycle time* — refers to the total time to complete one production cycle of an element, from mould preparation to quality checks and delivery; and
- b) *Throughput rate* — refers to the rate of production (units per day, week, or month).

9.2.2.2 Developing the production schedule

A production schedule developed based on the following:

- a) Splitting the floor plan into various ‘installation parts’, typically aligned with crane locations and capacities;
- b) Estimation the number of elements per part and the installation duration for each part based on element type and connection complexity;
- c) Using historical data and experience to estimate installation cycle times for each element;
- d) Assigning unique element IDs to each installation part and sequence them accordingly;
- e) Publishing a sequence chart that includes:
 - 1) Planned drawing release date;
 - 2) Planned casting date;
 - 3) Planned transportation date; and
 - 4) Planned installation date.

9.3 Project and Factory Logistics Planning

9.3.1 Haulage Planning

Transportation constraints influence the size and shape of precast elements. The following factors should be considered in haulage planning:

- a) Trailer lengths (standard lengths of 6 m, 12 m, with special cases up to 30 m);
- b) Trailer width (typically between 2.4 m to 2.7 m);
- c) Height clearance (3.6 m to 4 m, adjustable with low-bed trailers);
- d) Weight capacity (25 MT to 35 MT);
- e) Special permits for oversized loads that may be needed;
- f) Route surveys and permissions to be obtained if crossing bridges or navigating urban areas; and
- g) Large element transportation time slots through day/night as per local regulations.

9.3.2 Factory Stockyard and Logistics Coordination

Precast elements in the factory stockyard should be stored using first in, first out (FIFO) or last in, first out (LIFO) methods, depending on site requirements. Proper stock registers and inventory management software should be used for efficient tracking and organization. Storage size of the yards should be defined by the minimum concrete strength and curing periods.

9.3.3 Crane Planning

A comprehensive crane layout should be developed in collaboration with the installation planning engineers considering the following:

- a) Types of cranes (tower cranes, mobile cranes);
- b) Site constraints (space, access);
- c) Ground strata for crane footing;
- d) Lifting capacities and feeding locations;
- e) Crane removal and repositioning areas; and
- f) Double handling constraints due to storage at site or crane reach.

9.4 Installation Sequence

Once the casting sequence is set, installation dates to each element should be assigned based on factors such as:

- a) Crane type and operational speed;
- b) Reduced crane efficiency where multiple cranes are deployed with overlapping zones;
- c) Element type and installation cycle time;
- d) Connection complexity;
- e) Grouting and curing durations;
- f) The number of shifts and crane operation hours; and
- g) Adjustment of installation dates and resources as needed based on available machinery and logistics.

9.5 On Site Work Planning

9.5.1 Surveying

The first activity on-site is surveying and line-out. It should be ensured that key points are marked, including:

- a) Building corners and faces;

- b) Compound wall boundaries;
- c) Grid lines and column centres;
- d) Footing corners;
- e) Shear wall grid lines; and
- f) Site constraints such as high-tension lines or existing canals or water bodies or filled up areas or marsh lands.

9.5.1.1 Drone surveys are recommended for more accurate and automated site measurements, which also help in tracking progress.

9.5.2 Cast-In Situ Works

Cast in-situ (CIS) work is crucial in precast projects. The interface between precast and CIS activities typically occurs during installation and connection phases. The interfaces of the joints shall be roughened to ensure proper bonding with the CIS concrete or grout. Minimizing inter-trade dependencies improves project success in terms of both time and cost. This interface also needs careful attention, as the dowels projecting from precast elements may have higher variation that affects precast installation. Detailed as-built surveys must be completed well in advance to avoid last minute repairs and long delays.

9.6 Digital Tracking and Monitoring Tools

To enhance precision and efficiency, building information modelling (BIM) may be implemented for both production and assembly of precast elements. This technology supports:

- a) Access to real-time drawing by all engineers, reducing errors;
- b) Seamless use across production, installation, and progress monitoring stages; and
- c) Required infrastructure such as Wi-Fi network and system access between various agencies are be in place.

10 CONSTRUCTION, MANUFACTURE, STORAGE, TRANSPORT AND ERECTION OF PRECAST ELEMENTS

10.1 Manufacture of Precast Concrete Elements

10.1.1 A precast facility shall be located based on proximity to raw materials, logistics infrastructure, storage capacity, and crane access to ensure economic production and streamlined erection.

10.1.2 Manufacture

Precast elements may be manufactured at a permanent factory or temporary site facility, selected based on project scale, logistics, and environmental goals. For reduction in site pollution and wastage to achieve sustainability in precast construction a reference may be made to [G-4](#).

10.1.2.1 Factory prefabrication

Factory prefabrication involves the commercial production of standardized components in a factory setting, typically for long-term manufacturing. It is a capital-intensive process where work is carried out year-round, ideally under a covered shed to minimize the impact of seasonal variations. This type of facility allows for high levels of mechanization, with production organized in a factory-like manner, supported by a consistent team of workers.

10.1.2.2 Site prefabrication

Site prefabrication refers to the production of precast components at or near the construction site and shall be adopted for large-scale or long-duration projects where it offers measurable advantages in transportation cost and

productivity. Although temporary, the site factory should be comparable to a permanent facility with strict quality control. There are the following two types of site prefabrication:

- a) *Semi-mechanized* — Carried out in open areas using primarily local labour, with minimal equipment; moulds may be stationary or mobile, and the level of skill shall be appropriate to the component complexity; and
- b) *Fully-mechanized* — Conducted within covered sheds using skilled labour and factory-grade machinery, designed for high-volume and high-quality precast production.

10.1.2.3 Precast factory quality audits

All precast production facilities shall undergo quality audits at least annually. Audits shall be conducted by a competent team jointly approved by the contractor and client. Audit scope shall include production quality, safety and documentation.

10.1.3 The manufacturing of precast elements shall involve three categories of processes: core, auxiliary and subsidiary. Each category shall be planned and coordinated to ensure continuous, high-quality production.

10.1.3.1 Core process

The main processes that include the following steps:

- a) Providing and assembling moulds, positioning the reinforcement cage for reinforced concrete work and stressing the wires for prestressed elements;
- b) Installing concealed service conduits/pipes;
- c) Fixing inserts and tubes as necessary (for handling purposes);
- d) Pouring concrete into the moulds;
- e) Vibrating and finishing the concrete;
- f) Curing the concrete; and
- g) Demoulding the forms and stacking the precast products.

10.1.3.2 Auxiliary process

These processes are necessary for the successful completion of the core processes and include the following:

- a) Mixing and producing fresh concrete (carried out in a mixing station or batching plant);
- b) Prefabricating the reinforcement cage (performed in a steel yard or workshop);
- c) Manufacturing inserts and other finishing components to be incorporated into the main precast products; and
- d) Finishing the precast products.

10.1.3.3 Subsidiary process

Subsidiary processes shall include material handling, maintenance and utility services essential for uninterrupted production. These shall include:

- a) Proper storage of raw materials;
- b) Internal transport systems for fresh concrete and cages;
- c) Safe movement and stacking of precast elements;
- d) Periodic servicing of equipment and moulds; and
- e) Maintenance of curing and utility facilities (for example, steam).

10.1.4 The manufacturing of precast elements requires systematic planning of all the processes to achieve the following objectives:

- a) A cyclic working method to enhance speed and cost-efficiency in production;
- b) Mechanization of processes to boost productivity and improve quality;
- c) Optimal production levels that meet quality control standards while maintaining the desired construction speed;
- d) Improved working conditions for the workforce; and
- e) Minimization of weather-related disruptions to the manufacturing schedule.

10.1.5 The various stages of precasting, as outlined in [Table 12](#), are classified based on the equipment required for each stage. This classification facilitates the mechanization and rationalization of work throughout the process. Stages 6 and 7, as shown in [Table 12](#), represent the core processes in the manufacturing of precast concrete elements. For these stages, various technological processes are employed, tailored to the specific concrete product being produced. These methods have been proven to be rational, economical and time-saving. [Figure 40](#) provides a diagrammatic representation of the different stages involved in the plant process.

10.1.6 The accepted methods of manufacturing precast units can be broadly classified into two categories:

- a) The stationary method, where the moulds remain in a fixed position and the various processes are carried out in a cyclic order at the same location; and
- b) The circulation method, where the precast unit moves through different stages of production, with the processes being carried out in an assembly-line fashion.

The various accepted pre-casting methods are listed in [Table 13](#) along with the details regarding the elements that can be manufactured by these methods.

Table 12 Stages of Precasting of Concrete Products (Clause 10.1.5)			
Sl No.	Precasting Stage No.	Name of Process	Operations Involved
(1)	(2)	(3)	(4)
i)	1	Procurement and storage of construction materials	Unloading and transport of cement, coarse and fine aggregates and steel, and storing them in bins, silos or storage sheds
ii)	2	Testing of materials	Testing of all materials including steel
iii)	3	Design of concrete mix	Testing of raw materials, plotting of grading curves and trial of mixes in laboratory
iv)	4	Making of reinforcement cages	Unloading reinforcement bars from wagons or lorries and stacking them in the steel yard, cutting, bending, tying or welding the reinforcements and making in the form of a cage, which can be directly introduced into the mould
v)	5	Applying form release agent and laying of moulds in position	Moulds are cleaned, applied with form release agent, and assembled and placed at the right place
vi)	6	Placing of reinforcement cages, inserts and fixtures	The reinforcement cages are placed in the moulds with spacers, etc, as per data sheet prepared for the specific prefabricate units. Pre-tensioning of wires in the case of pre-stressing of concrete element

Table 12 (Concluded)

Sl No.	Precasting Stage No.	Name of Process	Operations Involved
(1)	(2)	(3)	(4)
vii)	7	Preparation of green concrete	Taking out aggregates and cement from bins, silos, etc, batching and mixing
viii)	8	Transport of green concrete	Transport of green concrete from the mixer to the moulds. In the case of precast method involving direct transfer of concrete from mixer to the mould or a concrete hopper attached to the mould this prefabrication stage is not necessary
ix)	9	Pouring and consolidation of concrete	Concrete is poured and vibrated to a good finish
x)	10	Curing of concrete and demoulding	Either natural curing with water or an accelerated curing using steam and other techniques. In the case of steam curing using trenches or autoclaves, this stage involves transport of moulds with the green concrete into the trench or autoclave and taking them out after the curing and demoulding elements. Cutting of protruding wires also falls in this stage. In certain cases, the moulds have to be partly removed and inserts have to be removed after the initial set. Final demoulding is done after concrete has gained designed strength, enough to resist handling stress, and the components are then allowed to be cured further. The minimum curing duration is as per Part C/Section 5 of this NBCS except in cases where the design strength of the precast element is achieved earlier. In that case, curing only up to achieving the design strength is sufficient, subject to appropriate approvals
xi)	11	Stacking of precast elements	Lifting of precast elements from the mould and transporting to the stacking yard for further transport by trailer or rail is part of this stage
xii)	12	Testing of finished components	Tests are carried out on the components individually and in combination to ensure the adequacy of their strength
xiii)	13	Miscellaneous	a) Generation of steam involving storing of coal or oil necessary for generation of steam and providing insulated steam pipe connection up to the various lines b) Repair of machines used in the production

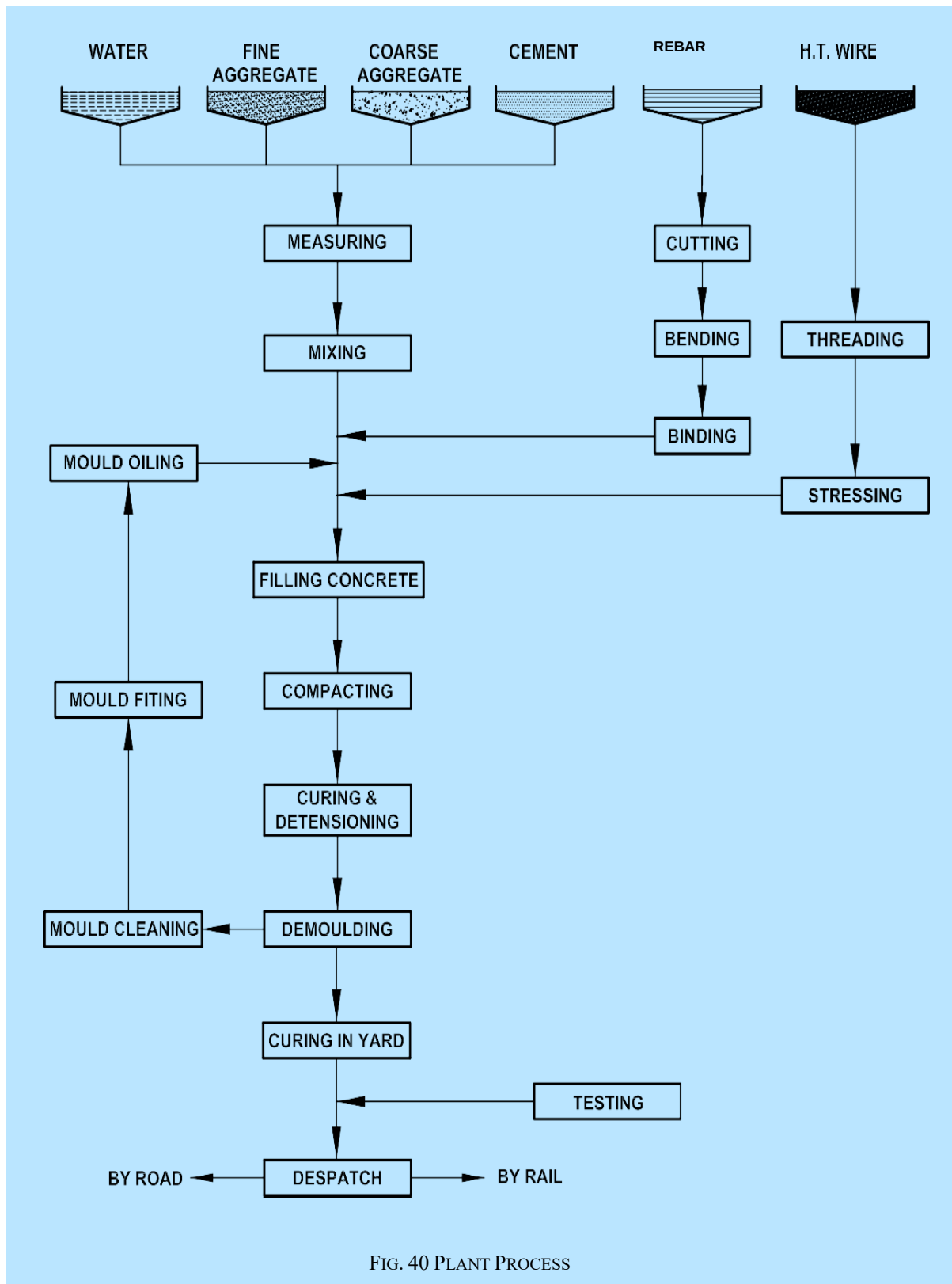


FIG. 40 PLANT PROCESS

Table 13 Precasting Methods

(Clauses [10.1.6](#) and [10.6.1](#))

Sl No.	Precasting Method	Where Used	Recommended Dimensions and Weights	Advantages and Remarks
(1)	(2)	(3)	(4)	(5)
i)	<i>Individual mould method</i> — Precasting method using mould which may be easily assembled out of bottom and sides, transportable, if necessary. This may be either in timber or in steel using needle or mould vibrators and capable of taking pre-stressing forces	Ribbed slabs, beams, girders, window panels, box type units and special elements, prestressed railway sleepers, parts of pre-stressed girders, etc	No limit in size and weight. It depends on the equipment used for demoulding, transporting and placing	Strengthening of the cross-section possible Openings are possible in two planes
ii)	<i>Battery mould method</i> — The shuttering panels may be adjusted into the form of a battery at the required distances equal to the thickness of the concrete member	Interior wall panels, shell elements, reinforced concrete battens, rafters, purlins, roof and floor slabs	Length ≤ 9 m Height ≤ 3 m Mass ≤ 5 t	Especially suitable for mass production of wall panels where shuttering cost largely reduced. Vapour or steam curing may be adopted by taking the steam pipes through the shuttering panels
iii)	<i>Stack method</i>	Floor and roof slab panels	Length: Any desired length Breadth: 1 to 4 m Mass ≤ 5 t	For casting identical reinforced or pre-stressed panels one over the other with separating media interposed in between
iv)	<i>Tilting mould method</i> — This method is capable of being skipped vertically using hydraulic jacks	Exterior wall panels where special finishes are required on one face or for sandwich panel	Length ≤ 6 m Breadth ≤ 4 m Mass ≤ 5 t	Suitable for manufacturing the external wall panels
v)	<i>Long line prestressing bed method</i>	Double tees, ribbed slabs, purlins, piles and beams	Length $\leq$ Any desired length Breadth ≤ 2 m Height ≤ 2 m Mass : Up to 10 t	Ideally suited for pretension members
vi)	<i>Extrusion method</i> — Long concrete mould with constant cross-section where concreting and vibration are done automatically just as in hollow cored slab casting	Roof slabs, foam concrete wall panels and beams cross-section	Length : Any desired length Breadth : < 2 m Height : < 3 m	May be used with advantage in the case of un-reinforced blocks, foam concrete panels

10.1.7 Preparation and Storage of Materials

Proper storage of materials is crucial in the precast industry, as errors in planning this aspect can significantly impact production economics. In a precast factory, where quality is paramount, the proper storage and preservation of building materials especially cement, coarse aggregates and fine aggregates are essential. Materials should be stored in accordance with accepted standard [C-7(21)]. Additionally, precast elements should follow a ‘first-in, first-out’ (FIFO) system to minimize or avoid issues related to shrinkage, creep and other material-related concerns.

10.2 Moulds

10.2.1 Moulds used for the manufacture of precast elements may be made from steel, timber, concrete, plastic, or a combination of these materials. When designing moulds for various elements, special attention should be given to facilitating easy demoulding and the assembly of parts. Additionally, the rigidity, strength, and watertightness of the mould are critical, especially considering the forces involved during the pouring of green concrete and the vibrating process. The deflection of any supporting member of the mould shall not exceed $L/350$ or 3 mm, whichever is less, where L represents the span of the member in millimetres).

10.2.2 Tolerances

The moulds shall be designed to accommodate the tolerances as per the casting tolerances given in [Table 14](#).

Table 14 Casting Tolerances for Precast Components (Clause 10.2.2)		
Sl No.	Product (See Key No.)	Product Tolerances
(1)	(2)	(3)
i)	<i>Length:</i>	
	1, 10, 12, 15, 16, 17	± 5 mm
	9, 11	± 10 mm
	2, 3, 4, 6, 7, 8	± 10 mm
	5	± 2 mm for length below and up to 500 mm ± 5 mm for length over 500 mm
	13, 14, 18, 19	± 15 mm
ii)	<i>Thickness/cross sectional dimensions:</i>	
	1	± 3 mm
	2, 8	± 3 mm or 0.1 percent, whichever is greater
	4, 5, 7, 10, 12, 15	± 3 mm, for thickness up to 300 mm ± 5 mm, for thickness greater than 300 mm
	3, 6, 9, 11, 16, 17, 18, 19	± 5 mm
	13, 14	+ 5 mm, - 3 mm
iii)	<i>Straightness/bow:</i>	
	2, 4, 5, 7, 8, 10, 15, 16, 18	1/360 of length, maximum of ± 10 mm
	1, 5	± 3 mm
iv)	<i>Squareness</i> — When considering the squareness of the corner, the longer of two adjacent sides being checked shall be taken as the base line:	
	<i>a) Skew across width of an end:</i>	
	15	± 3 mm per 300 mm length, with maximum of ± 5 mm
	7, 9	± 3 mm per 300 mm length, with maximum of ± 10 mm
	1, 2, 4, 5, 8, 10, 11	± 3 mm per 300 mm length, with maximum of ± 15 mm

Table 14 (Concluded)

Sl No.	Product (See Key No.)	Product Tolerances
(1)	(2)	(3)
	b) <i>Skew across thickness or depth or height of an end:</i>	
	12, 16	± 3 mm
	4, 7, 8, 16	± 3 mm per 300 mm length
	13, 14	Depth ≤ 600 mm, ± 5 mm Depth ≥ 600 mm ± 1.5 mm for 300 mm length with maximum of ± 15 mm
v)	<i>Twist</i> — Any corner shall not be more than the tolerance given below from the plane containing the other three corners:	
	2, 8	5 mm, Up to 600 mm in width and up to 6 m in length 10 mm, Over 600 mm in width and for any length
	4	± 1/1 500 of dimension or ± 5 mm, whichever is less
	1	± 3 mm
	7	± 1 mm
vi)	<i>Flatness</i> — The maximum deviation from 1.5 m straight edge placed in any position on a nominal plane surface shall not exceed:	
	2, 8	± 5 mm
	4	± 3 mm
	1	± 2 mm
	5	± 5 mm or maximum of 0.1 percent length
vii)	<i>Warping</i> — Deviation from plane in which the corners of the panel do not fall within the same plane:	
	2, 4, 5, 7, 8, 16, 18	± 1.5 mm per 300 mm length
	10, 15	± 5 mm
Key No.		
1	Channel unit	7 Precast planks
2	Ribbed slab unit/hollow slab	8 Ribbed/plain wall panel
3	Waffle unit	9 Column
4	Large panel prefabrication	10 Step unit
5	Cellular floor/roof slabs	11 Beam and spandrel
6	Prefabricated brick panel	12 Modular unit
		13 Single tee
		14 Double tee
		15 Stadium riser
		16 Cladding panel/facade
		17 H-Frame
		18 Foundation units
		19 Drain

10.2.3 Mould Wall Slopes

To facilitate the easy removal of precast elements from the mould, particularly when the mould has fixed sides and the elements need to be lifted out, the side slopes should ideally be 1 in 6 but shall not be less than 1 in 12. For slopes less than 1 in 6, the mould should have provisions of appropriate hydraulic jacks at the base of the mould to assist demoulding. A gentle slope helps prevent the elements from becoming stuck in the mould during demoulding.

10.3 Accelerated Curing

In precasting factories, using accelerated or artificial curing methods is economical, as it allows elements to be demoulded much earlier, enabling the early re-use of moulds. Any of the following methods shall be adopted:

10.3.1 Steam Curing

10.3.1.1 The steam curing of concrete products shall take place under tarpaulin in tents, under hoods, under chambers, in tunnels or in special autoclaves. The steam shall have a uniform flow throughout the length of the member. The precast elements shall be so stacked with sufficient clearance between each other and the bounding enclosure, to allow proper circulation of steam.

10.3.1.2 Before the concrete products are subjected to any accelerated method of curing, the cement to be used shall be tested for soundness, setting time and suitability for steam curing.

10.3.1.3 In the case of elements manufactured by accelerated curing methods, concrete admixtures to reduce the water content may be allowed to be used. The normal aeration agents used to increase the workability of concrete should not be allowed to be used. Use of calcium chloride-based admixtures should not be used for reinforced concrete elements so as to meet the chloride limits prescribed in Part C 'Structural design, Section 5 Structural Concrete' of this NBCS.

10.3.1.4 The inside face of the steam curing chamber, tunnel or hood shall have a damp-proof layer to maintain the humidity of steam. Moreover, a proper slope shall be given to the floor and the roof to allow the condensed water to be easily drained away. When steam is first let into the curing chambers, the air inside shall be allowed to go out through openings provided in the hoods or side walls which shall be closed soon after moist steam is seen jetting out.

10.3.1.5 It is preferable to let in steam from the top of the chamber through perforated pipelines to allow uniform entry of steam throughout the chamber.

10.3.1.6 The fresh concrete in the moulds should be allowed to get the initial set before allowing the concrete to come into contact with steam. The regular heating up of fresh concrete products from about 20 °C to 35 °C should start only after a waiting period ranging from 2 h to 5 h depending on the setting time of cement used. It may be further noted that steam can be let in earlier than this waiting period, provided the temperature of the concrete product does not rise beyond 35 °C within this waiting period.

10.3.1.7 The second stage in steam curing process is to heat up the concrete elements, moulds and the surroundings in the chamber in the following manners:

- a) In the low-pressure steam curing, the airspace around the member is heated up to a temperature of 75 °C to 80 °C at a gradual rate, usually not faster than 30 °C/h (this process takes around 1 h to 1 h 30 min depending upon outside temperature); and
- b) In the case of curing under high pressure steam in autoclaves, the temperature and pressure are gradually built up during a period of about 4 h.

10.3.1.8 The third stage of steam curing is to maintain the uniform temperature and pressure for a duration depending upon thickness of the section. This may vary from 3 h to 5½ h in the case of low-pressure steam curing and 4 h to 7 h in the case of high-pressure steam curing.

10.3.1.9 The fourth stage of steam curing is the gradual cooling down of concrete products and surroundings in the chamber and normalization of the pressure to bring it at par with outside air. The maximum cooling rate, which is dependent on the thickness of the member, should normally not exceed 30 °C/h.

10.3.1.10 In all these cases, the difference between the temperature of the concrete product and the outside temperature should not be more than 60 °C for concrete up to M30 and 75 °C for concretes of M45 or greater. In the case of light weight concrete, the difference in temperature should not be more than 60 °C for concretes less than M25. For concretes greater than M50, the temperature differences can go up to 75 °C.

10.3.2 Vapour Curing

Vapour curing shall consist of maintaining a high relative humidity around the concrete by means of application of moisture in the form of vapour or steam. The system and requirements shall be similar to those specified under [10.3.1](#). The temperature range to be maintained shall be as recommended by the manufacturer of the equipment, depending upon the product requirements.

10.3.3 Steam Injection during Mixing of Concrete

Low pressure saturated steam shall be injected into the mixer during the mixing of aggregates. This process shall raise the temperature of the concrete to about 60 °C. Concrete so produced, when placed in the moulds, shall develop high early strength.

10.3.4 Heated Air Method

The concrete elements are kept in contact with hot air with a relative humidity not less than 80 percent. This method is particularly suitable for light weight concrete products using porous coarse aggregates.

10.3.5 Hot Water Method

The concrete elements shall be immersed in a bath of hot water maintained at a temperature between 50 °C and 80 °C. The principles of this method shall generally be similar to those of steam curing.

10.3.6 CO₂ Curing

Carbon dioxide shall be injected into the concrete during mixing. This promotes faster hydration and is particularly effective in accelerating the hardening of precast concrete elements, thereby improving both efficiency and quality.

10.3.7 Concrete elements subjected to any of the accelerated curing methods specified in [10.3.2](#) to [10.3.6](#) shall be subjected thereafter to normal curing methods, so as to enable the development of the final strength.

10.3.8 Other Methods of Accelerated Hardening

Accelerated hardening of concrete may also be achieved by one or more of the following techniques:

- a) *Use of Construction Chemicals* — Suitable construction chemicals may be used for promoting early hardening;
- b) *Consolidation by Spinning* — This method shall generally be employed in centrifugal moulding of pipes and similar units. The spinning action removes excess water, effects consolidation, and permits earlier demoulding;
- c) *Pressed Concrete* — This method shall be suitable for the fabrication of small or large products at high rates of production. Presses in the range of 100 t to 200 t capacity shall be used to compress the wet concrete in rigid moulds, thereby expelling excess water. This results in early handling and produces dense, wear-resistant concrete;
- d) *Vacuum Treatment* — In this method, surplus air and water shall be removed from the freshly placed concrete (for example, slabs and similar elements). A suction of up to 70 percent of an atmosphere shall be applied for a period of 20 min to 30 min per cm thickness of the units; and
- e) *Consolidation by Shock* — This method shall be suitable for small concrete units placed in strong moulds and subjected to repeated dropping from a height. The number of shocks required to expel excess water and air may vary from 6 to 20 and the height of drop may be up to half the depth of the mould.

10.3.9 Stacking During Transport and Storage

Every precaution shall be taken against overstress or damage, by the provision of suitable packing at agreed points of support. In general, the final condition of the elements in the structure should be studied to finalize on the staking support. Even number of supports is preferred over the odd number of supports. Particular attention is directed to the inherent dangers of breakage and damage caused by supporting other than at two positions, and by the careless placing of packing (for example, not vertically one above the other). Ribs, corners and intricate projections from solid section should be adequately protected. Packing pieces shall not stain, disfigure or otherwise cause permanent mark on units or members. Stacking shall be arranged, or the precast units protected, to prevent the accumulation of trapped water or rubbish and, if necessary, to reduce the risk of efflorescence.

The following points shall be kept in view during stacking:

- a) Care should be taken to ensure that the flat elements are stacked as per their final position at the structure. For identification, top surfaces should be clearly marked;
- b) Stacking should be done on hard and suitable ground to avoid any sinking of support when elements are stacked;
- c) In case of horizontal stacking, packing materials shall be at specified locations and shall be exactly one over the other to avoid cantilever stress in panels;
- d) Components should be packed in a uniform way to avoid any undue projection of elements in the stack which normally is a source of accident;
- e) In general, vertical stacking should be limited to 6 components to 10 components. Minimum ground capacity shall be 5 t/m² for stacking of panels. Where the ground is only compacted, the maximum ground load at support should be restricted to 15 t/m². Where the ground is of PCC layer, the maximum ground load at support should be restricted to 30 t/m² subjected to capacity of the ground. In both cases, the ground should be levelled, and the support should be continuous along the width of the panels; and
- f) All wall panels shall be stacked, preferably symmetrically on both sides, against steel A-frames. The slope of the sides of the A-frame shall be between 75° and 80° to the horizontal and the bottom of the frame shall be provided with concrete footings to resist component of self-weight and horizontal wind pressure on the wall panel. A similar type of A-frame should be used for transporting wall panels.

10.4 Handling Arrangements

10.4.1 Lifting and handling positions shall be clearly defined particularly where these sections are critical. Where necessary special facilities, such as bolt holes or projecting loops, shall be provided in the units and full instructions supplied for handling. For bolts/hooks, bond strength shall be the criteria for embedded bolts and bearing strength for through bolts. For bond strength, pull out test of concrete shall be carried out. See [10.8](#) for additional design considerations.

10.4.2 For precast pre-stressed concrete members, the residual pre-stress at the age of operation of handling and erection shall be considered in conjunction with any stresses caused by the handling or erection of member. The compressive stress thus computed shall not exceed 50 percent of the cube strength of the concrete at the time of handling and erection. Tensile stresses up to a limit of 50 percent above those specified in Part C/Section 5 of this NBCS shall be permissible.

10.5 Identification and Marking

All precast units shall bear an indelible identification, location and orientation marks as and where necessary. The date of manufacture shall also be marked on the units.

The identification markings on the drawings shall be the same as that indicated in the manufacturer's literature and shall be shown in a tabular format on the setting schedule together with the length, type, size of the unit, and the sizes and arrangement of all reinforcement. Implementation of QR Code or similar stickers pasted on precast elements with above mentioned details are highly recommended.

10.6 Transport

Transport of precast elements inside the factory and to the site of erection is of considerable importance not only from the point of view of economics but also from the point of view of design and efficient management. Transport of precast elements shall be carried out with extreme care to avoid any impact and distress in elements and handled as far as possible in the same orientation as it is to be placed in the final position. See [10.8](#) for additional design considerations.

10.6.1 Transport Inside the Factory

Transport of precast elements moulded inside the factory depends on the method of production selected for the manufacture as given in [Table 13](#).

10.6.2 Transport from Stacking Yard Inside the Factory to the Site of Erection

10.6.2.1 Transport of precast concrete elements from the factory to the site of erection should be planned in such a way to be in conformity with the traffic rules and regulations as stipulated by the Authorities. The size of the elements is often restricted by the availability of suitable transport equipment, such as tractor-cum-trailers, to suit the load and dimensions of the member in addition to the opening dimensions under the bridge and load carrying capacity while transporting the elements over the bridge.

10.6.2.2 While transporting elements in various systems, that is, wagons, trailers, trucks, etc, care should be taken to avoid excessive cantilever actions and desired support are maintained. Transportation of prefabricated elements should be carried out with care. Sufficient care should be taken to also protect the steel inserts and reinforced bars of the precast elements during transportation. Precast elements should be secured with safety ties to ensure that vibrations to the elements in transit are minimum. Special care should be taken at location of sharp bends and on uneven or slushy roads to avoid undesirable stresses in elements.

10.6.2.3 Before loading the elements, care should be taken to ensure that the base packing for supporting the elements is located at specified positions only. Subsequent packing shall be kept strictly one over the other.

10.7 Erection

10.7.1 Tower cranes or other mobile cranes have varying lifting capacities at different radii, with the maximum weight at a radius closer to the centre and the minimum weight at a radius farther from the centre. To minimize the number of precast elements, the selected crane should be capable of handling the heaviest precast elements at its maximum radius. The floor plan should indicate the weight of panels at different crane radii, ensuring they align with the crane's lifting capacity at each radius.

In the erection of precast elements, all the following items of work are meant to be included:

- a) Slinging of the precast element;
- b) Tying up of erection ropes connecting to the erection hooks;
- c) Cleaning of the elements and the site of erection;
- d) Cleaning of the steel inserts before incorporation in the joints, lifting of the elements, setting them down into the correct envisaged position;
- e) Adjustment to get the stipulated level, line and plumb by means of levelling bolts and adjustable props;
- f) Welding of cleats;
- g) Changing of the erection tackles;
- h) Putting up and removing the necessary scaffolding or supports;
- j) Welding of the inserts, laying of reinforcements in joints and grouting the joints; and
- k) Finishing the joints to bring the whole work to be like finished product.

See [10.8](#) for additional design considerations.

10.7.2 Since the erection work in various construction jobs using prefabricated concrete elements differs from place to place depending on the site conditions, safety precautions in the work are of utmost importance. Hence, only those skilled foremen, trained workers and fitters who have been properly instructed about the safety precautions to be taken should be employed on the job.

10.7.3 Transport of people, workers or visitors, by using cranes shall be strictly prohibited on an erection site.

10.7.4 For Travelling Cranes

- a) The travel tracks shall be perfectly levelled both longitudinally and transversally;
- b) The gauge shall be constant and the rail tracks perfectly straight and with the same shape along the entire path;
- c) The rail tracks shall be placed on a solid base;

- d) Travelling stop buffers shall be placed at the ends of the rail tracks;
- e) For rail shape tolerances, the OEM's installation manual shall be referred; and
- f) The track of the crane should be checked daily to see that all fish plates and bolts connecting them to the sleepers are in place and in good condition.

10.7.5 The operation of all equipment used for handling and erection shall follow the operations manual provided by the manufacturer. All safety precautions shall be taken in the operations of handling and erection.

10.7.6 Erection Tolerances

As per [Table 15](#).

Table 15 Erection Tolerance for Precast Components		
<i>(Clause 10.7.6)</i>		
Sl No.	Product	Tolerances mm
(1)	(2)	(3)
i)	Level differences between support lines of floor or wall panels	± 5
ii)	Plumb lines of wall panels	± 5
iii)	Bearing for precast floor panels	± 5
iv)	Joint dimensions	± 5
v)	Maximum accumulated deviation	
	a) Till 10 storey building	1/1 250 of height or length or ± 20
	b) Greater than 20 storey building	1/1 250 of height or length or ± 50 (whichever is lesser)
vi)	Variation in plan location (any column or beam, any location)	± 10 for column ± 15 for beams
vii)	Variation in plan parallel to specified building lines	+ 2 mm per meter for any beam less than 6 m long or adjacent to columns paced 6 m apart or less 15 mm maximum for adjacent columns spaced more than 6 m

10.8 Design Considerations

Based on the design intent, production environment and aesthetic requirements of the prefabricated panels, the resultant stresses should be controlled to minimize cracking in precast and/or prestressed panels. Precast (non-prestressed) components are generally designed as 'cracked' unless cracking is an aesthetic concern. However, demoulding of precast elements shall not be allowed until the concrete achieves a strength of at least 1.25 times the (extreme fibre) stress which the concrete may be subjected at the time of demoulding. The same shall be applicable while transporting and erection. Reference may be made to [Table 7](#) for load multiplier to ascertain handling stress. The strength referred to shall be that of concrete using the same cement and aggregates and

admixture, if any, with the same proportions and cured under conditions of temperature and moisture similar to those existing on the work.

10.8.1 Structural design requirements of minimum spacing, skin reinforcement and crack width criteria as specified in Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS should be applicable.

10.8.2 Lifting hooks should be located such that while lifting the panel, minimum bending moments are produced.

10.8.3 Static load multipliers should be used to account for impact, and demoulding forces as per [Table 7](#).

10.8.4 The load multipliers are only needed for the flexural design and are not applicable for the lifting devices. Lifting hooks shall be designed as per [7.14.6](#).

10.8.4.1 For wall panels cast vertically on tilting moulds, battery moulds, and other vertically cast elements, the required concrete strength is not always dictated by flexural stresses. Instead, the minimum concrete strength during the initial lift may be governed by the capacity requirements of the lifting inserts. The capacity of the lifting inserts may be lower than their rated value due to factors such as inadequate embedment depth, insufficient edge distances, or low concrete strength at the time of lifting. Therefore, higher concrete strengths than those recommended in [Table 9](#) may be necessary to fully develop the capacity of certain lifting inserts and ensure the safe handling of the panels.

10.8.4.2 For prestressed elements, special care must be taken to ensure that lifting devices are anchored in compression zones, unless otherwise specified by the design.

10.8.4.3 Initial lifting should be done cautiously and gradually to overcome suction and friction, avoiding the application of sudden impact forces.

10.9 Equipment

10.9.1 The strategic equipment used in the precast concrete industry/construction may be classified into the following categories:

- a) *Production equipment* — Batching plant, concrete shuttle conveyors concrete vibrators, circulation tables, EOT, gantry, shear lines, stirrup making machines, weld mesh machines, robots;
- b) *Erection equipment* — Tower cranes, Mobile cranes, derricks, hoists, chain pulley blocks, etc;
- c) *Transport machinery* — Tractor-cum-trailers, dumpers, lorries, locomotives and rarely even helicopters;
- d) *Workshop machinery* — For making and repairing steel and timber moulds;
- e) Minor tools and tackles, such as lifting equipment, electric vehicle-dumpers, concrete buckets, etc; and
- f) Steam generation plant for accelerated curing.

Each of the above groups may further be classified into various categories of machines and further to various other types depending on the source of power and capacity. For use of renewable energy sources to achieve sustainability in precast construction [G-3](#) shall be referred.

10.9.2 Mechanization of the Construction and Erection Processes

The processes can be mechanized/automated, similar to other industries, to achieve the benefits of mass production of standardized components. This approach will enhance productivity, lower production costs over time and ensure consistent product quality.

10.10 Systems for Mixed and Composite Construction

The system of mixed and composite construction depends on the extent of the use of prefabricated components, their materials, sizes and the technique adopted for their manufacture and use in building.

10.10.1 Combinations of System Components for Mixed and Composite Construction

The following combinations may be used in mixed and composite construction:

- a) Structural steel work and timber roofs on precast frames;
- b) Precast half slab, solid slabs, hollow core slabs (floors) onto steel and concrete beams and masonry walls;
- c) Profiled metal decking on precast beams;
- d) Cast in situ slab on precast wall;
- e) Precast slab on cast in situ wall/3D printed wall;
- f) Precast frames onto cast in-situ foundations, retaining walls, etc;
- g) Precast frames stabilized by masonry walls, steel bracing, etc;
- h) Precast cladding in steel or cast in-situ frames and vice-versa;
- j) Glass curtain walling, stone cladding or metal sheeting onto precast concrete frames, etc; and
- k) Reinforced concrete and structural steel as composite columns and beams.

10.10.2 Precast concrete may be combined with cast in-situ concrete, often termed hybrid construction. Cast in-situ is mostly used to form homogenous connections between precast elements and provide a structural topping for horizontal diaphragm action. In other cases, it is used to form the foundations and sub-structure to the building.

10.10.3 Structural steel work is largely used in long span prestressed concrete floors (hollow core) supported on rolled and prefabricated steel beams and also as steel roof trusses supported on concrete columns.

10.10.4 Timber may be used as long span glue-laminated beams and rafters, with precast concrete. Precast floors may be used in timber frame construction. Similarly, timber frames with precast elements shall be used as a building system.

10.10.5 Brick and block masonry may be combined with precast concrete structures and floors. The most common combination is to use prestressed floors on load bearing walls.

NOTE — Any prefabricated floor system covered under other Indian Standards, for example, reinforced concrete plank system and speed floor system may be combined with load bearing masonry walls/insulating concrete form (ICF) walls/precast large reinforced concrete wall panels/precast or cast in-situ reinforced concrete column-beam frames provided overall structural design safety, stability and integrity for composite structure is ensured in accordance with this section and other safety standards, such as good practice [C-7(1)], 5 of Part C/Section 1 of this NBCS including fire safety and safety against progressive collapse as per provisions of respective good practices, such as [C-7(19)], Part C/Section 5 and Part C/Section 6A of this NBCS.

10.10.6 Design Considerations

The mixed and composite structures shall be analysed appropriately and the joints in them designed to take the forces of an equivalent discrete system. Resistance to horizontal loading shall be provided by placing beams, walls and/or bracings in two directions at right angles or otherwise. The individual components shall be designed, taking into consideration appropriate end conditions and loads at various stages of construction. The components of the structure shall be designed for loads in accordance with good practices [C-7(1)] and [C-7(20)] and 5 of Part C 'Structural Design, Section 1 Loads and Load Effects' of this NBCS. In addition, members shall be designed for handling, erection and impact loads that may be expected during handling and erection.

For mixed and composite construction, the following shall be considered:

- a) *Positions of stability cores, walls, bracing, etc* — In high rise buildings, the most popular method is a central core constructed several storeys ahead of the framework. The core could be built using either precast concrete panels, or cast in-situ walls, or steel cross bracing or precast concrete diagonal bracing;
- b) *Maturity of connections* — Cast-in-situ grouted joints need temporary propping until desired design strength is achieved unless combined mechanical connections are also used. The overall construction sequence should be planned ahead of time carefully as per the design requirements. The design of temporary bracing system using key components should be looked at critically;

- c) *Availability and/or positioning of equipment to transport and erect components* — The size and weight of the various components shall be organized to make optimum use of crane capacity, for example, the lightest units farthest from the operating zone;
- d) *Erection safety and speed of construction, with attention to cast in-situ concreting sequences* — This is particularly important where fixing gangs are unaccustomed to working with different materials; and
- e) *Tolerances for construction* — This is particularly important where different manufacturers are producing components in different materials.

Other design considerations and safety requirements against progressive collapse shall be in accordance with this NBCS.

10.11 Prefabricated Structural Units

For the design and construction of composite structures made up of prefabricated structural units and cast in-situ concrete, reference may be made to good practice [C-7(22)].

For design and construction of precast reinforced and prestressed concrete triangulated trusses, reference may be made to good practice [C-7(23)].

For design and construction of floors and roofs using various precast units, reference may be made to good practice [C-7(24)].

For construction of floors and roofs with joists and filler blocks, reference may be made to good practice [C-7(25)].

11 QUALITY ASSURANCE AND QUALITY CONTROL

The provisions for quality assurance as per good practice Part C/Section 5 of this NBCS shall apply. A documented quality assurance program covering various stages from precast unit/element production to erection shall be implemented and maintained. Common defects and remedies in precast concrete elements are presented in [Annex F](#). For achieving sustainability in precast concrete construction a reference may be made to [Annex G](#). Additionally, the flow chart given in [Fig. 41](#) shall be considered.

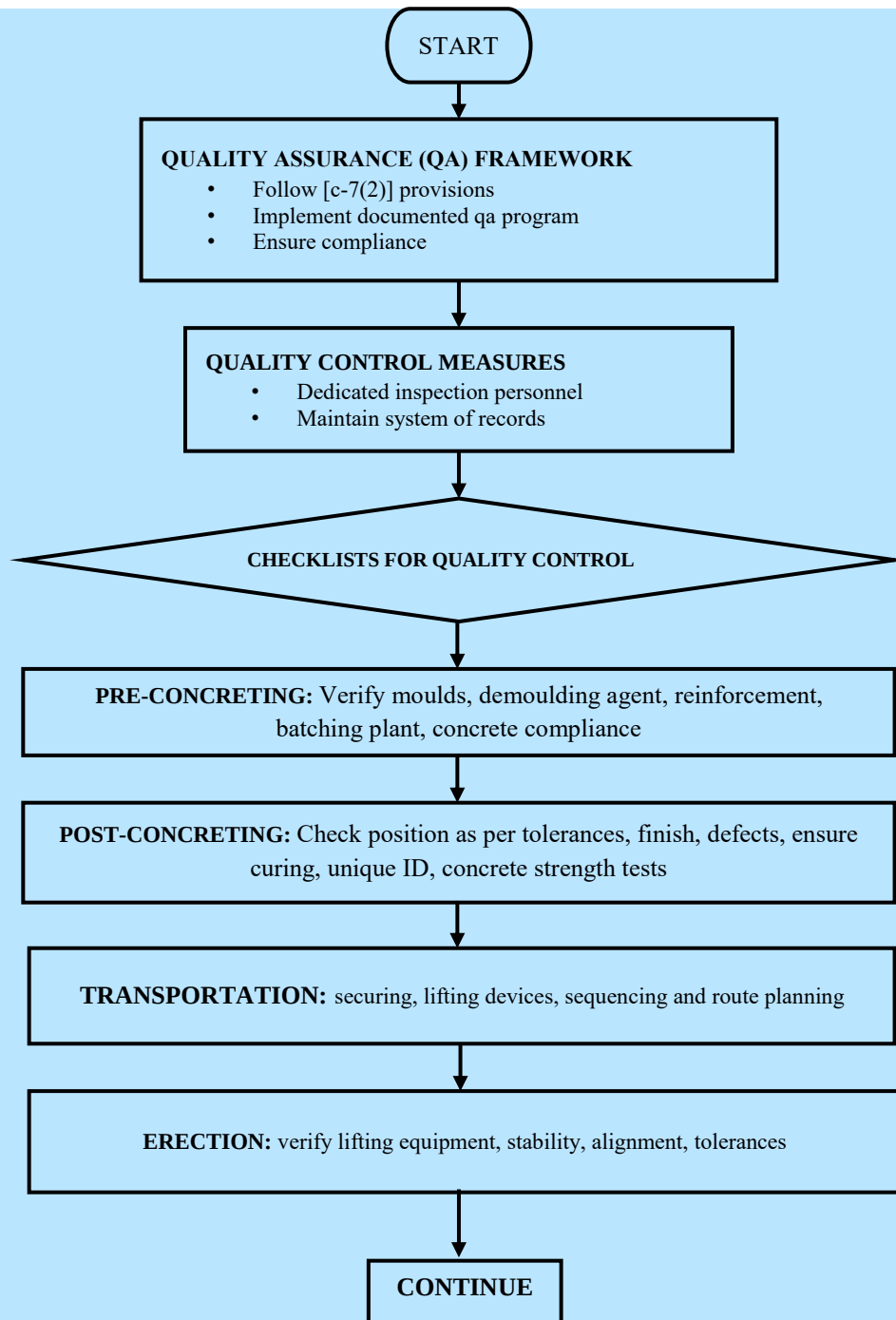
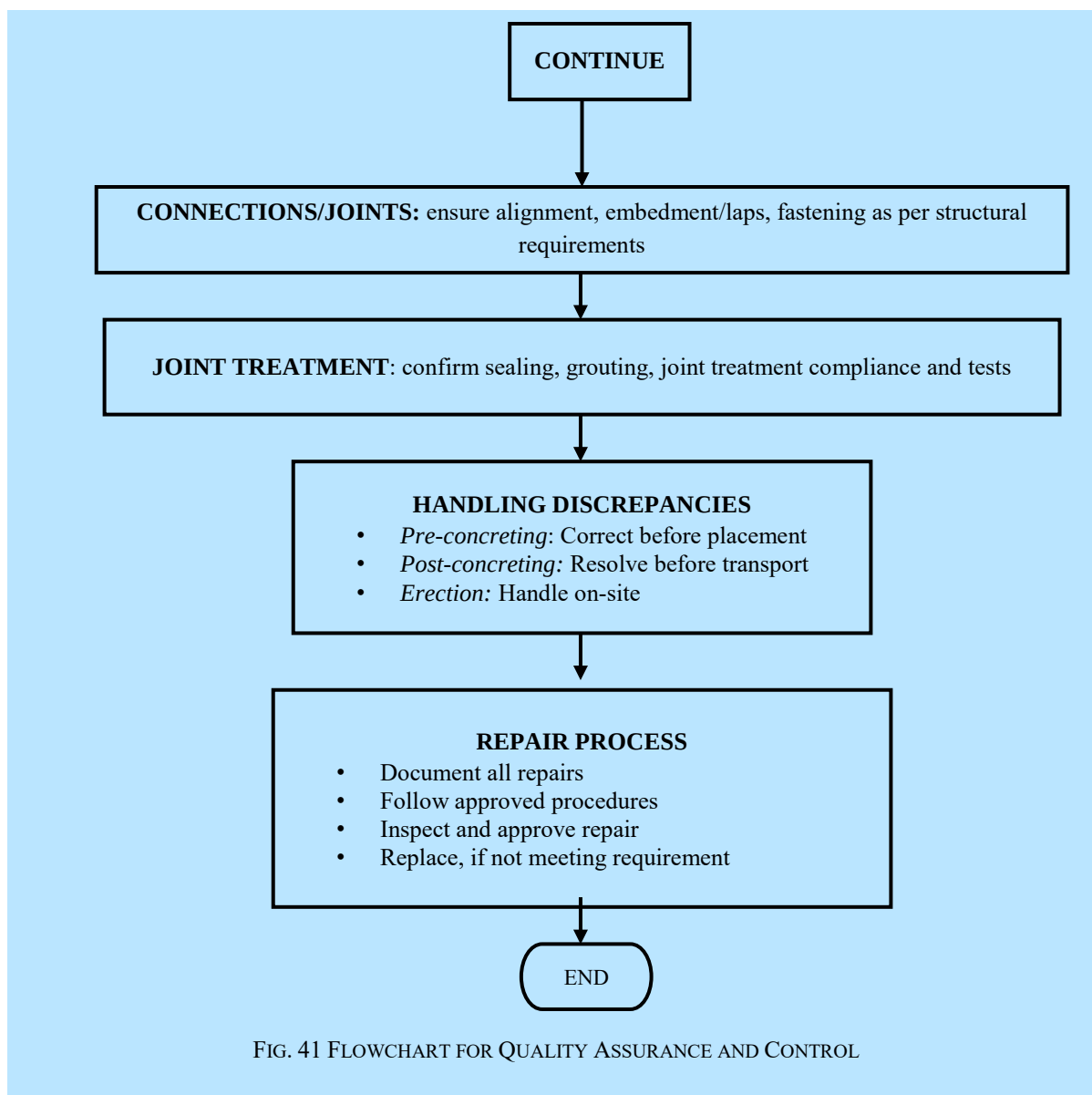


FIG. 41 FLOWCHART FOR QUALITY ASSURANCE AND CONTROL (CONTINUED)



12 TESTING

12.1 Sampling procedures

12.1.1 Lot

12.1.1.1 All the precast units of same size, manufactured from same material under similar conditions of production shall be grouped together to constitute a lot.

12.1.1.2 The number of units to be selected from each lot for dimensional requirements shall depend upon the size of the lot and shall be in accordance with col (3) and col (4) of [Table 16](#).

12.1.1.3 The units shall be selected from the lot at random. In order to ensure the randomness of the selection, reference may be made to accepted standard [C-7(26)].

Table 16 Sample Size and Rejection Number

(Clauses [12.1.1.2](#) and [12.1.2.1](#))

Sl No.	Lot Size	First Sample Size	Second Sample Size	First Rejection Number	Second Rejection Number
(1)	(2)	(3)	(4)	(5)	(6)
i)	Up to 100	5	5	2	2
ii)	101 to 300	8	8	2	2
iii)	301 to 500	13	13	2	2
iv)	501 and above	20	20	3	4

12.1.2 Number of Tests and Criteria for Conformity

12.1.2.1 All the units selected at random in accordance with [Table 16](#) shall be subjected to testing of dimensional tolerance requirements. A unit failing to satisfy any of the dimensional requirements shall be termed defective. The following shall be considered:

- If no defect is found in the sample, the lot shall be considered as conforming else shall be rejected, if the number of defectives is greater than or equal to the first rejection number;
- If the number of defectives is less than the first rejection number, the second lot shall be subjected to testing for dimensional tolerance requirements; and
- If the combined number of defectives is less than the second rejection number, the lot shall be considered as conforming to dimensional tolerance requirements; otherwise not.

12.1.2.2 The lot which has been found satisfactory with respect to dimensional tolerance requirements shall then be considered for load testing as per [12.4](#). One unit out of every 300 units or part thereof shall be selected for testing. The lot shall be considered as conforming to the strength requirement, if all units meet the requirement; otherwise not.

12.2 Testing on Individual Precast Units

12.2.1 The flexural components shall be load tested individually or along with the structural topping, depending on the requirements of the project or if any structural cracks (as per Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS) or damages are observed. The testing shall be carried out on the selected sample along with one undamaged unit or part for reference. If individual element is tested, then the loads to be placed should be prorated such that element experiences similar demand over capacity levels when loaded. In either case, the element should be loaded for 24 h at its full span with a total load of 1.25 times the sum of dead (including SIDL, etc) and imposed loads used in the design. At the end of this time, it should not show obvious signs of deterioration or excessive deflection. For large span elements, larger than 10 m span, with design stresses near cracking under the ultimate loads, it is not uncommon to see hairline cracks during the load test at maximum loads in sustained conditions. Except for the design criteria of un-cracked elements, if applicable, if the test elements in such cases perform at a satisfactory level with recovery as noted in [12.2.2](#) and [12.2.3](#) as applicable, the test should be considered passed. If such cracks are substantial, the test should be further evaluated by the engineer-in-charge.

12.2.2 The recovery of maximum deflection observed during testing after 24 h should not be less than 75 percent.

12.2.3 If prestressed, it should not show any visible cracks up to the working load, and the recovery of maximum deflection observed during testing after 3 h should not be less than 85 percent.

12.2.4 For inspection and testing of the compression members, provision of Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS shall apply.

12.3 Testing of Connecting Devices and Inserts

12.3.1 Grouted coupler or mechanical coupler shall be tested according to accepted standard [C-7(9)]. Where the grouted couplers are used for connections of lateral force resisting system (LFRS) transferring seismic and wind lateral loads, the grouted couplers should also be tested for the reverse cyclic load as per [Table 17](#), in an NABL accredited laboratory or institute of repute under the supervision of appropriate authority.

Table 17 Cyclic Tension and Compression Test for Grouted Coupler (Clause 12.3.1)				
Sl No.	Stage	Tension	Compression	Cycles
(1)	(2)	(3)	(4)	(5)
i)	I	$0.95 f_y$	$0.5 f_y$	20
ii)	II	$2\varepsilon_y$	$0.5 f_y$	4
iii)	III	$5\varepsilon_y$	$0.5 f_y$	4
iv)	IV	Load in tension till failure		

where

f_y = specified yield strength of the reinforcing bar; and

ε_y = strain of reinforcing bar at actual yield stress.

12.3.2 Any new connection assemblies such as, but not limited to, column shoe or shear connectors, that are used in connection of LFRS as noted in [12.3.1](#), shall be tested for monotonic as well as reverse cyclic load as per [12.3.1](#) in an NABL accredited laboratory or institutes of repute under the supervision of appropriate authority. The failure should only occur in the reinforcement with evidence of ductility without any damage or deterioration in the brittle components of the connection assembly.

12.3.3 Lifting inserts and devices shall be tested for pull-out and should not show any damage till design load (factor of safety times the working load). Factor of safety shall be according to [7.14](#).

12.4 Floor Load Testing of Structure or Part of Structure

12.4.1 Floor load testing on a part or entire floor shall be carried out before occupancy, depending on the requirements of the project or if any structural cracks or damages are observed or to ensure adequacy of strength or serviceability.

12.4.2 The structure should be loaded for 24 h with the full dead load and an imposed load equal to 1.25 times the specified imposed load used in design. The imposed load shall be removed, and the deflection shall be monitored. During the test, vertical struts should be placed appropriately to ensure safety.

NOTE — Dead load includes the self-weight of structural members plus weight of finishes and walls or partitions, if any, as considered in the design.

12.4.3 The recovery of maximum deflection shall be at least 75 percent otherwise the test shall be repeated after 72 h. If the recovery after second test is less than 80 percent the structure shall be deemed to be unacceptable. If unacceptable, the strengthening of this structure shall be carried out and 80 percent recovery of the structure shall be verified. The strengthening of the structure shall be verified by relevant authority.

12.4.4 If prestressed, the recovery of maximum deflection should be at least 85 percent, otherwise test shall be repeated after 72 h. If the recovery after second test is less than 85 percent the structure shall be deemed to be unacceptable.

12.4.5 If the maximum deflection in mm, shown during 24 h under load is less than $40 L^2/D$, where L is the effective span in m; and D is the overall depth of the section in mm, it is not necessary for the recovery to be measured and the recovery provisions of [12.4.3](#) and [12.4.4](#) shall not apply.

12.4.6 Testing of Connection/Joint and System

12.4.6.1 Testing of connection or system shall be performed for proprietary or unique connection or system.

12.4.6.2 Monotonic test shall be carried out for beam-column, walls-slab, wall-wall joints for quantifying the static behaviour.

12.4.6.3 Reverse cyclical load simulation test under intended loads of beam-column, wall-slab, wall-wall joints, shall be carried out for quantifying the energy dissipation and hysteresis behaviour.

12.4.6.4 System level testing shall be carried, if required by the contract or specification of the project.

12.4.6.5 Testing shall be carried out in reputed institutes with appropriate requirements.

12.4.6.6 Acceptance criteria for such test should be mutually established by the authority and the concerned institute where testing is being conducted.

12.5 Testing for Water Tightness

12.5.1 External Precast Wall/Facade Panels

Any of the testing procedures given in [12.5.1.1](#) and [12.5.1.2](#) may be followed for testing of external wall and facade panel joints for water tightness as per the engineer-in-charge.

12.5.1.1 Testing procedure 1

12.5.1.1.1 General

If it is deemed necessary to demonstrate that the precast facade joints are adequate to resist water penetration, a water penetration test on a full-scale panel including joint mock-ups may be undertaken.

12.5.1.1.2 Test procedures

Mock-up panels to be tested for water tightness should be subjected to both static and cyclic pressure in accordance with standard accepted procedures. Available international testing procedures are to be modified for local conditions as follows:

- a) Differential test pressure shall be 20 percent of the maximum inward design wind load but not less than 0.77 kPa; and
- b) The test shall be performed with a water flow rate of 3.4 litre/min/m² for 15 min.

12.5.1.1.3 Sampling for a water penetration test

Water tightness testing of vertical and horizontal joints in precast facade panels shall be carried out at the rate of 0.5 percent for each type of joint and combination of panels or at least one for each joint type and precast unit, whichever is greater.

12.5.1.1.4 Failure criterion

The facade units or joints are deemed to have failed, if signs of water seepage through the joints or through the facade unit, including signs of damp patches, are observed during the test or within the subsequent 2 h after the test.

12.5.1.1.5 Remedial action on failure

Upon failure of a water penetration test on a precast facade system the cause of failure should be identified and the system revised and re-tested until a satisfactory test result is achieved. Modifications shall be realistic in terms of job conditions and shall maintain standards of quality and durability.

12.5.1.2 Testing procedure 2

12.5.1.2.1 For external walls, a continuous jet of water shall be sprayed on the joint corresponding to 600 litre/h from a water hose having a nozzle velocity of 2 m/s and a cone scatter of approximately 60° held at a distance of 1.2 m from the wall surface under test. The duration of the test shall not be less than 2 h. The hose shall be placed centred on the horizontal or vertical joint width. For external walls, the following tests shall be conducted for each building block as per [Table 18](#).

Table 18 Frequency of Tests for Vertical and Horizontal Joints (Clause 12.5.1.2.1)		
SI No.	Building Height	Frequency of Tests
(1)	(2)	(3)
i)	Up to 10 storeys	5
ii)	Up to 20 storeys	10
iii)	Up to 30 storeys	15
iv)	Up to 40 storeys	20
v)	Up to 50 storeys	25

12.5.1.2.2 Twenty percent of these tests shall be carried out at the intersection of the vertical and horizontal joint.

12.5.1.2.3 These tests shall be carried out at any location as instructed by the engineer-in-charge.

12.5.1.2.4 Acceptance criterion

Water shall not seep into any part of the building during the test.

12.5.1.2.5 Remedial action on failure

For each test that fails, the tested joint shall be repaired as per the approved repair procedure as directed by the engineer-in-charge and the joint shall be retested for performance as per acceptance criterion. Two additional joints should be tested for each failed joint unless directed otherwise by the engineer-in-charge.

12.5.2 Floor or Roof Slabs, Toilet Units and Kitchen Units

12.5.2.1 A total of 5 percent of all horizontal floor slabs and the entire roof slab area, including bathroom/toilet floors, and kitchen floors shall be tested for water tightness by ponding of water for period of 4 h. This test shall be carried out before the application of floor finish or tiling. The depth of ponded water shall be taken as the maximum of the following:

- At least 25 mm; and
- 10 mm more than thickness of joint.

12.5.2.2 Acceptance criteria

Water shall not seep into any part of the building during the test and after 24 h. The rooms below the tested room shall not show any water leakage. Minor dampness or seepage without dripping water in limited area of less than approximately 50 cm² is considered acceptable at a few areas and should be repaired suitably to avoid seepage. In any other cases, the test should be repeated per [12.5.2.3](#).

12.5.2.3 Remedial action on failure

For each test which fails, the tested joint shall be repaired as per the approved repair procedure as directed by engineer-in-charge and the joint shall be retested for water tightness. Two additional joints/areas should be tested for each failed joint unless directed otherwise by the engineer-in-charge.

ANNEX A

(Clause 4)

SYMBOLS AND NOTATIONS

A_n	Area of tension reinforcement
A_{vf}	Area of shear friction reinforcement
A_1	Loaded area (bearing constant area)
A_2	Area of the lower base of pyramid, cone or tapered wedge
b_t	Bearing width
b_w	Average width of the stem for the full depth of the section
d_e	Distance from centre of load to the end of the ledge
f_{ck}	Characteristic strength of concrete
f_y	Yield strength of tension reinforcement
f_{ys}	Yield strength of shear reinforcement
h_l	Height of beam ledge
L	Anchorage length
l_p	Length of ledge projection
N_u	Factored axial force
s	Minimum spacing of concentrated loads
V_u	Factored shear force
V_{uR}	Design shear resistance
β	coefficient dependent on the bar type
τ_b	Bond stress
τ_{bu}	Design ultimate anchorage bond stress
φ_E	Effective bar size which, for a single bar is equal to the bar size and, for a group of bars

ANNEX B

([Foreword](#), [Clauses 6.5.1](#) and [G-6.2.2](#))

GUIDELINE FOR 3D PRINTED CONCRETE STRUCTURES

B-1 GENERAL

This annex provides an overview of the materials, methods and tests associated with concrete construction using 3D printing for various applications. The provisions contained herein are based on the current state of knowledge and practice in 3D printed concrete construction. In addition to these guidelines, reference shall be made to the applicable existing Indian standards and relevant research publications. Ongoing research at the national and international levels is expected to generate design-related data, which may subsequently be utilized for the purpose of standardization.

B-2 METHODOLOGY AND SCOPE

The following points are considered while defining the application and performance guidelines:

- a) *Printing technique* — This document considers only extrusion-based 3D printing technology, where layers are deposited one over the other to construct a structure;
- b) *Design freedom* — 3D printing allows for flexibility in the geometry and form. The shape, size and topology of any 3D printed concrete element for any application shall not be restricted. However, the performance requirements for 3D printing need to be satisfied;
- c) *Material* — The material shall be printable and buildable. As per the current state of knowledge of concrete for 3D printing, the limits on the usage of cementitious materials as given in good practice [C-7(6)] and Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS would not be applicable for 3D printed concrete. The assessment of 3D printability of any concrete mixture needs to be carried out as per the available specialist literature on the subject. The mechanical, durability and long-term properties of the printable material shall be determined, as applicable. In the absence of a well-established procedure for preparation of test specimens for such evaluations, reference shall be made to recognized specialized literature. The testing shall be conducted at accredited laboratories and approved by competent authority;
- d) *Performance criteria* — The performance criteria of any 3D printed components shall conform to the relevant Indian Standards for equivalent components. In the absence of any standards, an experiment-based approach shall be followed to satisfy the requirements for the specific element or component; and
- e) *Special applications* — In the absence of any relevant standard, the engineer-in-charge, authorised to define the qualification requirements in accordance with the applicable Indian Standards, shall take decisions based on sound engineering judgement and established practice. The testing shall be conducted at accredited laboratories and approved by competent authority.

B-3 APPLICATION — NON-LOAD BEARING

B-3.1 Falsework/Formwork

3D-printed concrete may be used as stay-in-place (permanent) or temporary formwork for new concrete construction. The structural integrity of the formwork shall be either certified by a practicing structural engineer before the application or the formwork shall satisfy deformation limits and structural requirement mentioned in good practice [C-7(26)]. The formwork shall be designed by testing. The stay in place formwork shall not be considered as a part of the structural member.

B-3.2 3D Printed Concrete Building Blocks

The blocks can be solid as well as hollow, following the definition in accepted standard [C-7(27)]. The blocks shall have any external dimension greater than the corresponding dimension of a brick as specified in accepted standard [C-7(28)]. The size and the mass shall be such that it can be handled by as single person. In case of mechanized placement of the blocks, larger units can be used. The height of the block shall not exceed either its

length or six times its width whichever is less. The faces of the block shall be manufactured based on the mutual agreement of the manufacturer and supplier/purchaser. Accepted standard [C-7(27)] shall be followed for considering the tests required to satisfy the performance criteria of the blocks.

B-3.3 Autoclaved 3D Printed Blocks

The autoclaved 3D printed solid blocks shall be tested based on the requirements mentioned in accepted standard [C-7(29)]. The blocks shall be of such dimensions and shape that they can be conveniently handled by a single person. The face of the block need not be plain, and the design shall be decided mutually by the manufacturer and supplier/purchaser.

B-3.4 Non-Load-Bearing 3D Printed Walls

3D printed concrete structures can be used as non-load bearing walls. The non-load-bearing walls can be divided into three types as mentioned as per Part C ‘Structural Design, Section 4 Masonry’ of this NBCS.

- a) *Panel walls* — An exterior non-load-bearing wall in framed construction, wholly supported at each story but subjected to lateral loads;
- b) *Curtain walls* — A non-load-bearing wall subject to lateral loads. It may be laterally supported by vertical or horizontal structural members, where necessary; and
- c) *Partition walls* — An interior non-load bearing wall, one storey or part storey in height.

Such walls can be assessed for the performance requirements in the given application, and if necessary, testing of individual elements with necessary connection details can be done to ascertain lateral load resistance and intended performance.

B-3.5 Precast 3D Printed Panels

Precast 3D printed panels can be used for external or internal non-load-bearing walls. The panels shall be connected considering the guidelines provided in 8. The printing and erection tolerance shall follow the permissible limits mentioned in 10.7. The designing of the wall panels (including reinforcement) shall be provided by a certified structural engineer using this NBCS or following the concept of design by testing (in the absence of standardized design methods).

B-3.6 External Facing and Veneers

3D printed components can be used for external facing following the guidelines mentioned in good practice [C-7(30)]. Precast-facing concrete blocks shall generally conform to the requirements of accepted standard [C-7(27)] and, in addition, shall have special treatment regarding durability, colour and surface textures for the exposed facing. The face finish of slabs shall also be adequately waterproofed. The joint details, fixing technique and method of attachment shall be provided by a certified structural engineer following the concept of design by testing, in the absence of standardized design methods.

B-3.7 Acoustic

The acoustic treatment of non-industrial buildings, industrial establishments, and assembly spaces such as cinema halls or conference rooms is achieved through the design flexibility offered by 3D printing. The incorporation of air cavities between the inner and outer shells of walls contributes to the reduction of sound transmission between rooms. Panel and partition walls may be printed with a suitable spacing between face shells as suggested in good practice [C-7(31)]. The semi-discontinuous and discontinuous construction mentioned in the good practice [C-7(31)] can be followed for non-industrial buildings. For industrial establishments and cinema halls or conference rooms, the reverberation time governs the acoustic experience. Good practices [C-7(32)] shall be followed to determine the reverberation time required for each application. The face of the walls and ceilings can be designed to govern the reverberation time while satisfying the requirements in the standards mentioned.

B-4 APPLICATION — LOAD BEARING

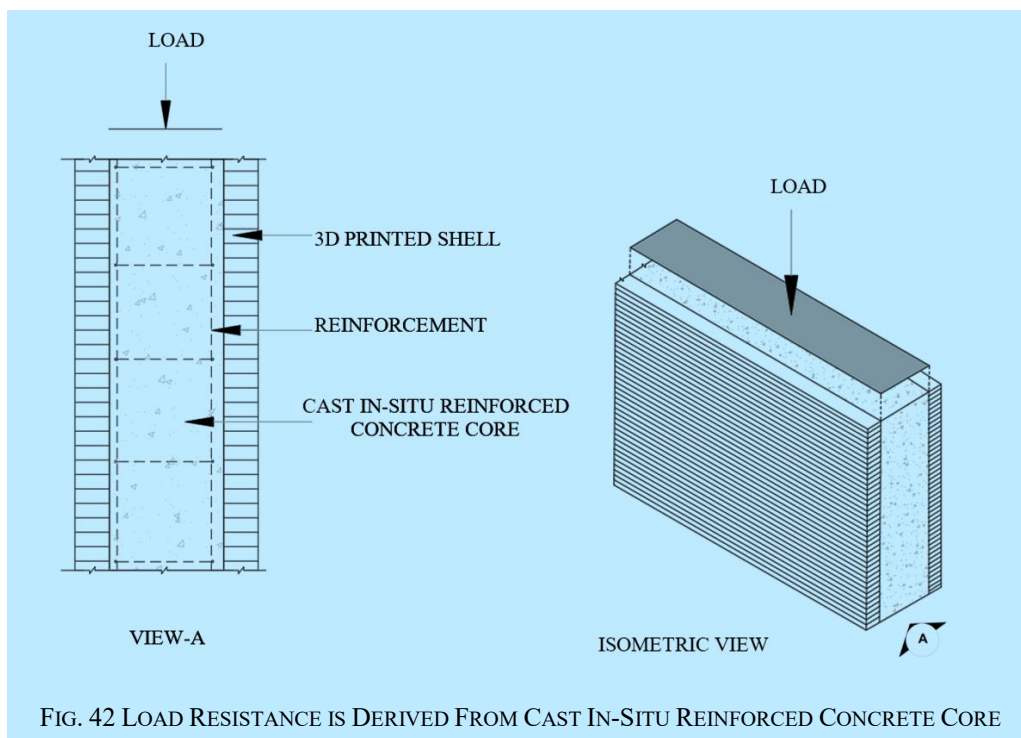
B-4.1 Design Philosophy

3D printed concrete structure shall satisfy the strength as well as serviceability requirements. The scope of this NBCS is limited to the load-bearing members in a building. Other structural applications are not considered for the purpose of this NBCS. Additionally, the durability and long-term behaviour of the printed structure shall be considered during design.

The load-bearing walls consist of a 3D printed shell and cast in-situ reinforced concrete core. In these cases, the 3D printed shell is considered to act integrally with the core. Depending on the application and the loading, the requirement of the reinforced concrete core may be done away with, in which case, the shell itself is capable of resisting both in-plane and lateral loads.

The design of load bearing walls shall be dependent on the load resistance mechanism. For the following two applications, the design using available information, and based on established principles of analysis and deriving load resistance, as commonly used in established codes of practice may be followed:

- a) If the load resistance is derived from cast in-situ reinforced concrete core, (see [Fig. 42](#)); and



- b) If both 3D printed shell and cast in situ reinforced core are made using the same concrete mix, resulting in monolithic behaviour (see [Fig. 43](#)).

Where load resistance is derived using composite action between printed shell and cast in-situ core, design by testing the individual components may be followed.

All attempts to make the printed object behave in a ductile manner should be made to the extent possible, irrespective of the nature of usage.

NOTE — Specialized literature may be referred to while designing the test methods for the evaluation of structural walls. Alternatively, load-bearing 3D printed structures can be constructed with the approval of a certified external authority.

The use of load bearing walls is limited to gravity load. The use for load resistance subjected to earthquake loading as per good practice [C-7(20)] and 5 of Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS and the requirements of ductile detailing as per good practice [C-7(1)] are outside the scope of this annex. Design by testing approach may be followed for any specific application. Additionally, the foundation shall comply all the requirements of good practice [C-7(33)] and to any other relevant Indian standards.

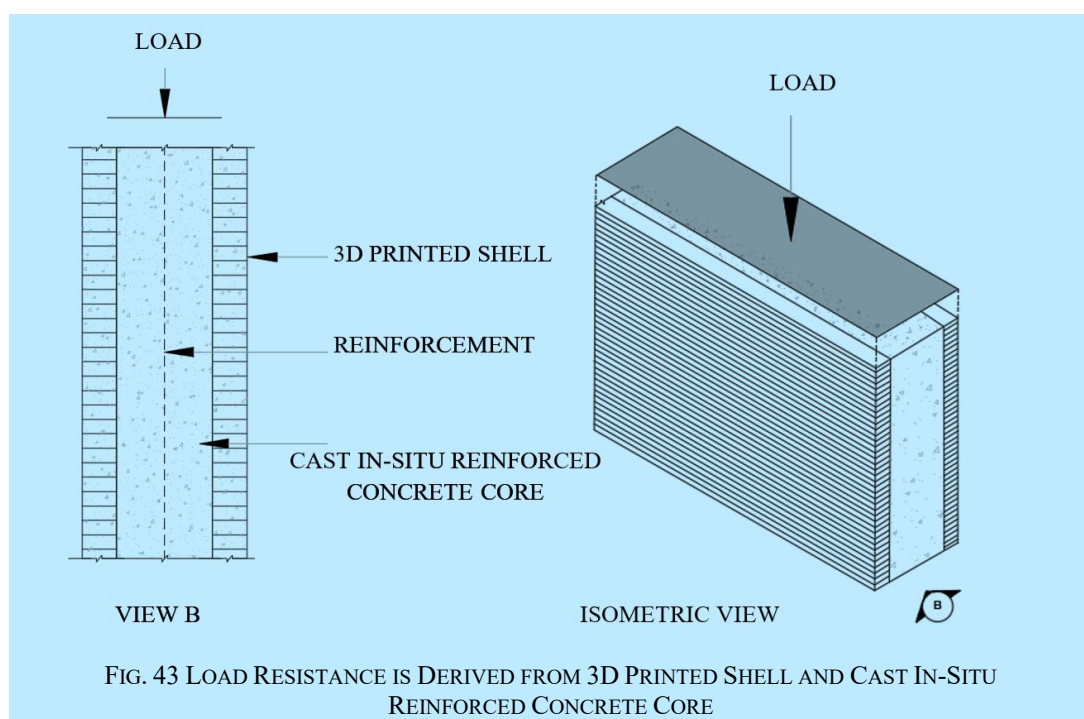


FIG. 43 LOAD RESISTANCE IS DERIVED FROM 3D PRINTED SHELL AND CAST IN-SITU REINFORCED CONCRETE CORE

B-4.2 Loads

The load criteria shall be as per Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS. The lateral and earthquake load shall be taken special care of while designing the load bearing structures. The components of the structures shall be tested at full scale individual element level with necessary boundary conditions.

B-4.3 On-Site and Off-Site Construction

In case of an on-site construction, the printer is expected to be stationed at the location and construction to be carried out in-situ. For off-site construction, panels/sub-parts of the structure are printed in a factory and assembled at the site. In such a case, particular attention is to be paid to the design of the connections, as per the prevailing standards and codes.

B-4.4 Load Bearing Wall

3D printed load bearing walls can be used for buildings up to a height where the effect of lateral load is not severe, which may be restricted to building of height of 12 m. For buildings beyond 12 m height, the structural design and construction methodology shall be approved by the competent authority. Special care shall be taken while designing the connection between wall to slab, wall to beam and wall to footing. The minimum effective thickness of the load bearing wall shall be 100 mm in accordance with the guidelines of Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS. The 3D printed load bearing wall can be considered as:

- a) 3D printed perimeter (with or without printed internal rib filaments) and cast in situ concrete core. Depending on the application and extent of loading, these walls can be fully solid (that is with the core completely filled with cast in situ reinforced concrete), hollow (that is without any internal filling) or partially filled (with reinforced concrete in some of the cavities); or
- b) Completely printed wall with no hollow space.

The cover depth of 3D printed wall shall consider exposure condition and fire resistance as per Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS. The cover depth should include the thickness of the 3D printed perimeter filament and the cover (if any) provided by the core concrete. The minimum reinforcement criteria laid down in Part C ‘Structural Design, Section 5 Structural Concrete’ 5 of this NBCS shall be followed. Horizontal reinforcement in the form of high yield strength deformed (HYSD) rebar or welded wire mesh or textile mesh shall be provided to connect the outer leaves to the inner core, when the design assumes a composite action of the 3D printed perimeter and the inner core. In other cases, the printed leaves shall be considered as stay-in-place formwork and shall not be taken as a part of the load-bearing wall. However, even in such cases, efforts

should be made to ensure that spalling or collapse of the outer periphery does not occur in extreme events such as earthquakes.

The walls shall satisfy the strength and serviceability criteria of the load bearing walls as per Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS. If required, the design shall be on basis of testing of individual elements with necessary boundary conditions or connection details. For testing guidelines, specialist literature and publication may be referred if necessary.

NOTE — Specialist literature/publication may include those published by International Code Council, etc.

B-4.5 Load Bearing Column

Column shall be designed using as per Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS. If required, test results obtained by full scale testing of individual elements with necessary boundary conditions or connection details shall be used. Buckling load criteria shall be satisfied considering the end-restraint conditions by testing. A certified structural engineer shall design the column as per logical reasoning based on experimental evidence.

B-4.6 Shear Walls

The shear walls shall be solid walls designed as per the requirements of good practice [C-7(2)]. The design guidelines similar to that of load-bearing 3D printed walls shall apply. The performance of 3D printed shear walls under lateral and earthquake load shall be tested. The design shall be on basis of experimental outcomes by a certified structural engineer.

B-4.7 Load-Bearing Beams and Slabs

The 3D printed beam and slab shall satisfy the load and serviceability requirements mentioned in Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS. The flexural capacity of the printed beam shall be tested using 3-point and 4-point bending test at full-length-scale element level. The characteristic strength of the printable mix shall not be used to design the flexure member. The results from full-length-scale element level tests shall be used. A certified structural engineer shall design the flexural members as per the reports from the testing.

B-4.8 Foundation

A 3D printed foundation shall be designed following the design by testing method. The foundation can be printed in-situ or assembled at site like precast foundation after printing at the factory. The foundation shall be designed by a certified structural engineer while considering inputs from geotechnical engineers about the soil bearing characteristics. Durability requirements, as mentioned in good practice [C-7(2)], for the foundation shall be strictly satisfied.

If the outer perimeter of the foundation is only printed and the core is cast as conventional foundation, then the printed part shall not be taken into the load calculation and shall be treated as a formwork.

B-4.9 Retaining Wall and Basement

Retaining wall and basement shall be solid for load-bearing applications. Reinforcement strategy shall follow the guidelines mentioned for 3D printed load bearing walls.

B-4.10 Concrete-Reinforcement Bond Strength and Development Length

These are to be determined for each case and the structural engineer shall take into account the same while designing the reinforcement. The effect of layer time gap and moisture loss shall also be considered while designing the reinforcement. The bond between the rebar and the printed concrete shall be measured before designing the reinforcement.

B-4.11 Connections

Connections between the elements, including, but not limited to between two printed elements; printed non-structural element and cast in-situ structural reinforced concrete element; printed non-structural element and

precast structural element; printed structural element and precast structural element; follow the requirements given in Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS. If required, the connections shall be designed by testing. However, the connection between cast in-situ reinforced core and other structural elements shall follow standard design practice.

B-5 OTHER APPLICATIONS

B-5.1 Boundary Walls

The 3D-printed boundary wall shall not be a retaining wall or any load-bearing structure.

B-5.2 Design Elements for Outdoor Spaces

These elements shall comprise, but are not limited to, the following:

- a) Pavement and other pedestrian movement spaces such as footpaths, plaza assembly spaces, kerb to footpath, steps and ramps;
- b) Parking and vehicular movement corridors such as road dividers, road marking, etc;
- c) Traffic management units such as gate/access control, barriers, traffic separators, etc;
- d) Outdoor public conveniences such as seating, drinking fountains, toilet/washrooms, etc;
- e) Shelter and kiosks such as bus shelters, police and telephone booth, food stall, information desk, etc;
- f) Outdoor illumination such as street light pedestals, etc;
- g) Tree guards, tree grates and planters;
- h) Litter bins, and spittoons; and
- j) Service utilities relating to water supply network, storm water network, sewerage system, irrigation network, etc.

These structures shall be designed by a certified structural engineer following the reports from experimental evaluation.

B-6 CURING

The printed structures shall be cured following the provisions mentioned in Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS. However, curing shall be initiated at the early age to avoid plastic shrinkage cracks. Water spraying in the form of a fine mist must be started even before the initial set. At any given instance, the surface shall not be dry. The curing shall be done until the designed strength is reached for the 3D printed walls and slabs. Curing compounds can be used to reduce the number of days of water curing. The engineer-in-charge shall take the decision regarding the curing requirements based on experimental evidence.

B-7 QUALITY ASSURANCE AND QUALITY CONTROL

The following properties shall be assessed for determining the quality of the 3D printable concrete mix. The sampling shall be done as per Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS:

- a) The mechanical properties including cube compressive strength, elastic modulus, split tensile strength, and flexural strength of the concrete shall be tested following the method in accepted standard [C-7(34)]. The properties of water permeability, drying shrinkage, and rapid chloride permeability shall also be tested following the relevant Indian standards. The tests shall be performed on the specimens sawn/cored from printed samples. The anisotropy of the structure shall be tested along X, Y and Z coordinates. The test shall be conducted on the cast specimen also with the concrete used for printing the structure; and
- b) Additionally, the bond strength of the layer shall be tested by extracting beam specimens of 40 mm × 40 mm × 160 mm from the printed structure, or a special test element. 3-point and 4-point bending test shall be performed on the extracted specimens with a minimum number of 5 specimens for each sample.

ANNEX C

(Foreword and Clause 6.5.2)

PREFABRICATED PREFINISHED VOLUMETRIC CONSTRUCTION (PPVC)

C-1 GENERAL

This annex provides a comprehensive guideline on prefabricated prefinished volumetric construction (PPVC) methods, materials, design standards and performance criteria for use by building professionals and other stakeholders involved in modular construction. This annex aims to facilitate the understanding and practical application of PPVC across various types of construction projects, including residential, commercial and industrial buildings. By outlining the key requirements and standards, this annex supports efficient and effective PPVC project planning, production, and assembly of PPVC to ensure high-quality, durable and sustainable modular structures.

This annex provides insights into essential aspects of prefabricated prefinished volumetric construction (PPVC), including design principles, manufacturing and installation practices, quality assurance, transportation and site assembly. It also highlights critical considerations for structural integrity, fire safety, environmental sustainability and inspection protocols, thereby promoting the adoption of best practices and supporting ongoing efforts to advance safe and innovative construction methods.

C-2 METHODOLOGY

The following points are considered while defining the application and performance guidelines.

C-2.1 Design

Design of PPVC modules considers both architectural and structural planning, taking into account modularization, alignment, and standardization of dimensions. It ensures that the modules comply with specifications for efficient on-site assembly, structural integrity and adherence to applicable building standards and design requirements.

C-2.2 Off-Site Manufacturing

The modules should be constructed in a controlled factory environment. This allows precise manufacturing with minimal errors and faster production times. Components, including concrete and steel modules, are cast, cured and finished with necessary fixtures, plumbing, and electrical setups before transportation to the construction site.

C-2.3 Transportation

Once manufactured, modules are carefully transported to the construction site. The transportation plan ensures that modules remain damaged and conform to transportation regulations, accounting for dimensions, weight and route restrictions.

C-2.4 On-Site Assembly

This stage involves the installation and connection of PPVC modules at the construction site. Modules are hoisted and secured in position, followed by finishing operations, including joint sealing and final inspections, to ensure that the assembled structure complies with all applicable quality and safety standards.

C-2.5 Modular Dimensions

PPVC modules should be designed to fit within standard transportation limits to ensure easy delivery to construction sites. Common dimensions range from 3 m to 12 m in length, 2.5 m to 3.5 m in width, and 2.5 m to 4.5 m in height. Specific adaptations shall account for the maximum width (2.5 m), height (4.5 m including truck height), and length (up to 18 m) based on transportation guideline.

C-2.6 Transportability and Site Constraints

The size and weight of PPVC modules are influenced by on-site access, route clearance and crane requirements. The structure shall be modularized to ensure efficient on-site alignment and joining of modules, while minimizing the need for additional handling at the construction site.

C-2.7 Materials

- a) *Concrete and steel modules* — PPVC modules are typically constructed from high-performance materials like reinforced concrete (RC) or steel. Reinforced concrete offers durability and structural integrity, whereas steel provides a lighter alternative, especially useful for high-rise buildings; and
- b) *Enhanced material specifications* — Materials are chosen for performance under factory conditions and transportation stresses. For instance, high-strength/performance concrete (HPC) and steel conforming to relevant Indian Standard (such as Fe 500 or higher for reinforcement) are preferred. The use of sustainable materials, such as fly ash and other admixtures, is also encouraged to improve durability and reduce environmental impact.

C-2.8 Connections

Connections should be strong enough to secure modules under various stresses, including lifting, transportation, and environmental forces. Common PPVC connection methods include grouted couplers, wire loop boxes, dowel systems, and mechanical fasteners. These ensure that modules are securely integrated without compromising flexibility needed for seismic or wind forces.

Vertical joints (for stacking modules) and horizontal connections (for lateral stability) are essential for PPVC structures. These connections often involve interlocking systems or bolted end plates that provide stability while allowing for some movement to accommodate load and environmental stresses.

Connections should be designed to follow the Indian Standards for seismic and wind loads, such as good practice [C-7(35)] and 5 of Part C/Section 1 of this NBCS, to ensure safety and resilience under local conditions. Proper connection systems are necessary to transfer loads between modules and secure them as a unified structure.

C-2.9 Alignment

The alignment of prefabricated prefinished volumetric construction (PPVC) modules is critical to ensure that the final building is structurally sound and aesthetically pleasing. There are several key considerations to keep in mind when aligning PPVC modules:

- a) The foundation on which the PPVC modules are placed should be level and square. If the foundation is not level, it can cause the modules to shift and potentially compromise the structural integrity of the building. Unless it is designed as a paper joint, a minimum 20 mm joint thickness shall be detailed between the top of the foundation and the PPVC module to account for site tolerances. The joint shall be later grouted with high strength grout as per design specification for structural integrity as well as continuity for functional performance;
- b) Each PPVC module should be precisely aligned with the adjacent modules to ensure that the building is straight and true. This is typically achieved using alignment pins or other methods to ensure that the modules are aligned both vertically and horizontally. The erection tolerances shall be followed as per [10.7.6](#);
- c) Ensuring that the PPVC modules are level vertically is also critical to the structural integrity of the building. Laser levels and other tools can be used to ensure that the modules are plumb and aligned with the adjacent modules; and
- d) The facade of the building, which includes windows, doors, and exterior finishes, should also be precisely aligned with the PPVC modules to ensure that the final building looks aesthetically pleasing.

C-2.10 Structural Design and Load Compliance

PPVC modules should comply with load standards as outlined in Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS, for addressing dead loads, live loads, wind loads and other special loads requirements.

Modules should also meet seismic design requirements under 5 of Part C/Section 1 of this NBCS to ensure stability during earthquakes.

Structural elements for concrete and steel structures shall follow Part C ‘Structural Design, Section 5 Structural Concrete’ and Part C ‘Structural Design, Section 6A Steel’ of this NBCS, ensuring that all modules meet load-bearing and durability criteria. The Limit State Method is typically applied for PPVC modules, which includes safety factors to manage risks associated with live loads, dead loads and extreme environmental forces.

C-2.11 Material Standards

- a) *Concrete and reinforcement compliance* — For concrete quality, PPVC construction shall follow Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS, including high-performance concrete or self-compacting concrete for durability. Reinforcements use deformed steel bars as per accepted standard [C-7(36)] for high-strength requirements, ensuring modules can withstand handling, transportation and on-site stresses; and
- b) *Admixtures and aggregates* — Chemical and mineral admixtures should comply with accepted standards [C-7(5)] and Part C ‘Structural Design, Section 5 Structural Concrete’ of this NBCS, which specify performance standards for durability and workability. Aggregates are selected as per accepted standard [C-7(4)], ensuring strength and proper gradation.

C-2.12 Fire Safety and Compartmentation

For fire resistance, PPVC structures shall comply with the requirements specified in Part F ‘Fire and Life Safety’ of this NBCS, including provisions for compartmentation. Each PPVC unit shall be provided with fire-rated walls, floors and doors, as required, to restrict the spread of fire and smoke and to ensure adequate means of egress and safety of occupants.

All materials used shall meet the relevant requirements of Part F ‘Fire and Life Safety’ of this NBCS for fire safety. Structural elements shall be designed to withstand high temperatures and achieve the required fire resistance in accordance with the relevant provisions of Part C/Section 5 of this NBCS.

C-2.13 MEP (Mechanical, Electrical, Plumbing) Standards

- a) Electrical installations in PPVC buildings should meet the accepted standard [C-7(30)] and Part D ‘Building Services/Section 2 Electrical and Allied Installations’ of this NBCS, ensuring safe and reliable connections between modules. Additionally, accepted standard [C-7(38)] and Part D/Section 2 of this NBCS should be used for lightning protection systems to prevent electrical hazards; and
- b) Plumbing systems follow good practice [C-7(39)] and Part E ‘Plumbing Services, Section 1 Water Supply and Section 2 Drainage and Sanitation’ for water supply and drainage, ensuring leak-proof, pressure-resistant pipes that meet durability and sanitation requirements in modular installations.

CI-2.14 Construction Tolerances

Construction tolerance limits in prefabricated prefinished volumetric construction (PPVC) refer to the allowable deviation from the intended design and specified dimensions. These limits are important to ensure that the final building is structurally sound and aesthetically pleasing. The construction tolerance limit in PPVC can vary depending on the specific materials and construction techniques used, as well as the local building standards and regulations. Permissible tolerances from the specified dimension of a PPVC module should conform to casting tolerances provided in [10.7.5](#).

C-2.15 Performance Criteria

The performance criteria of any PPVC components should conform to the relevant Indian Standard for equivalent components. In the absence of any Indian Standard, the safety and durability of the specific element or component should be demonstrated through testing.

C-2.16 Special Applications

In the absence of any standards, the engineer-in-charge should make decisions based on logical reasoning and practical evidence. The test program should be conducted at accredited/approved laboratories and approved by statutory bodies.

C-3 APPLICATIONS

C-3.1 Residential and Commercial Buildings

PPVC modules for residential and commercial buildings shall be permitted for use in multi-storey structures and may be designed as either load-bearing or non-load-bearing units. Load-bearing modules shall be designed to resist vertical loads and shall satisfy requirements of stability and safety for multi-level configurations. Non-load-bearing modules shall be used as partitions or infill components. The provisions of Part F 'Fire and Life Safety' of this NBCS shall apply for fire safety requirements, while Part C 'Structural Design, Section 4 Masonry' of this NBCS shall govern structural requirements, sound insulation and seismic load-bearing capacity. The design shall ensure adequate connections and overall structural integrity of the system.

C-3.2 Educational and Healthcare Facilities

Modules are pre-finished to meet spatial needs, such as classrooms and patient areas, integrating critical systems like mechanical, electrical and plumbing (MEP) as per good practice [C-7(40)] and Part E 'Plumbing Services, Section 1 Water Supply and Section 2 Drainage and Sanitation' for water supply and drainage.

C-3.3 Industrial Buildings

PPVC in industrial buildings such as warehouses and factories requires modules engineered to meet the load-bearing and lateral resistance standards outlined in Part C 'Structural Design, Section 6A Steel' of this NBCS for structural steel and Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS for reinforced concrete. The modular units allow for open, high-capacity spaces that can bear substantial loads, with compliance to Part C 'Structural Design, Section 1 Load and Load Effects' of this NBCS for design loads, wind, snow loads and for seismic considerations. Structural configurations shall also align with fire safety standards as per good practice [C-7(19)] to facilitate efficient integration of services. This approach enables high-speed assembly, structural integrity and durability for industrial purposes while meeting all requirements.

C-3.4 Design Philosophy

PPVC structures should be designed to meet strength and serviceability requirements, ensuring structural integrity under the anticipated loads. This focuses on load-bearing applications within PPVC, excluding non-structural or purely aesthetic uses. Additionally, the design should account for durability and long-term performance, particularly as modules are prefabricated off-site and subjected to transportation and on-site assembly stresses.

Load-bearing PPVC modules typically consist of a reinforced concrete or steel core surrounded by a prefinished volumetric structure. The core provides essential stability, especially under vertical and lateral loads. In certain applications, the volumetric shell itself may be designed to carry both in-plane and lateral loads, with the core providing additional structural redundancy.

The design approach for load-bearing PPVC modules depends on the load resistance mechanism employed, as outlined below:

- a) *Core-based load resistance* — Where the primary load resistance is provided by a reinforced concrete or steel core, which may be cast in-situ or prefabricated; and
- b) *Monolithic module design* — Where the PPVC module's core and shell use a compatible concrete mix or reinforcement design to behave as a single monolithic unit under load.

For cases where the load resistance is derived from the combined action of the volumetric shell and the core, individual component testing is recommended to validate the composite action. Efforts to promote ductile behaviour are essential, ensuring that load-bearing PPVC elements respond safely under stress.

NOTE — Testing and certification can be carried out as per specialised literature (reference may be made to applicable standards/Codes of the International Code Council (ICC) for structural evaluation methods), or structures may be certified through recognized government bodies in India. The scope of this section is limited to gravity loads. For resistance to earthquake-induced forces and ductile detailing as per Part C/Section 1 and Part C/Section 5 of this NBCS, additional testing and validation may be required.

C-3.5 Loads

Load criteria for PPVC modules should conform to good practices, with particular attention to lateral and seismic loads. Structural components should be validated through full-scale testing of individual elements, ensuring that boundary conditions simulate actual applications. Testing should confirm that PPVC modules meet all load-bearing and serviceability requirements, as specified accepted standards, for safe deployment in permanent structures.

C-3.6 Foundations

A PPVC foundation should be designed following a design-by-testing approach to confirm compliance with all load-bearing and durability standards. The foundation may be cast in-situ or prefabricated off-site as a precast unit for on-site assembly. Design shall be overseen by a certified structural engineer, with essential geotechnical input to evaluate soil bearing capacity, ensuring that the foundation is appropriate for the specific site conditions. Durability requirements per ‘Structural Design, Section 5 Structural Concrete’ of this NBCS should be strictly observed to maintain long-term performance and stability under applied loads.

To ensure a secure connection between the PPVC modules and the foundation, appropriate dowel bars or reinforcement should be incorporated into the foundation design. These reinforcement provisions should align with the load transfer requirements and provide structural continuity between the foundation and the modular components. Proper anchorage and placement of dowels will prevent differential movement and provide stability for the assembled structure.

C-4 CURING

As PPVC modules are manufactured in an off-site, controlled environment, advanced curing methods may be implemented to expedite the process and ensure consistent quality. In addition to following ‘Structural Design, Section 5 Structural Concrete’ of this NBCS, rapid curing techniques, such as steam curing, warm water curing, or the use of curing compounds, can be employed to accelerate strength development. These methods enable the modules to achieve the desired performance levels within a shorter time frame, which is critical for modular construction timelines.

The curing approach should ensure that modules reach their designed strength before transportation to the construction site. The decision on curing methods and duration should be based on testing and performance requirements, as determined by the supervising engineer, to guarantee durability and structural integrity.

C-5 QUALITY ASSURANCE AND QUALITY CONTROL

Compliance with good practices includes ongoing quality checks at both the manufacturing and construction stages. Regular inspections, as per relevant provisions this NBCS and relevant Indian Standards, to verify structural integrity, alignment and safety of each module, especially focusing on connections and material durability over time.

C-6 MAINTENANCE AND INSPECTION

Regular inspection and maintenance protocols should be set to identify wear or structural concerns in connections and external finishes, ensuring longevity and safety as outlined in relevant provision of this NBCS.

ANNEX D

([Foreword](#) and [Clause 6.6](#))

GUIDELINES FOR DESIGN FOR MANUFACTURING AND ASSEMBLY (DFMA)

D-1 The concept of design for manufacturing and assembly (DFMA) emphasizes the use of precast and prefabricated building components. The underlying principles of DFMA call for more efficient building design and construction detailing, especially in the area of precast concrete connection designs, so as to minimize the amount of wet works on-site.

To develop an advanced pre-cast concrete system (APCS), it is important for the various building disciplines to first understand the functional considerations, building aesthetics, and design approaches. This understanding will enable the conceptualization and development of the APCS design, ensuring it meets the project specifications. It is essential that the APCS design is coordinated with the manufacturing specification for components, construction planning and site work to ensure seamless integration and efficient construction processes.

D-2 DESIGN BASIS FOR ADVANCED PRECAST CONCRETE

The following are key principles of efficient construction using APCS systems:

- a) *Modular architecture layouts* — Designing buildings in modules to allow for greater use of precast and prefabricated components, reducing on-site construction time and costs;
- b) *Precast repetition* — Using standard precast elements repeatedly throughout the building, simplifying the construction process and reducing the need for customization;
- c) *Efficient structural design* detailing (standardize design and detailing) — Designing precast connections that can be easily assembled on-site, reducing the need for wet works and temporary supports;
- d) *Quick assembly detailing* — Designing connections and components that can be quickly and easily assembled on-site, reducing construction time and costs; and
- e) *Limited use of temporary supports* — Designing precast components that can be assembled without the need for temporary supports, further reducing construction time and costs.

D-3 FACADE STANDARDIZATION

Architectural design is a process to create a built environment with functionality for human comfort and sustainability through art and engineering design. In an effective precast concrete construction, the external facade design is generally achieved through optimal standardization in the following:

- a) *Shapes* — FRP/rubber-lined moulds can be used to create various shapes for precast concrete components;
- b) Textures such as sandblasted or acid-etched finishes can also be added to enhance the visual appearance of the precast components;
- c) Different colours can be achieved using pigments or concrete colours, allowing for a wide range of design options;
- d) *Repetition* — Standard precast elements can be repeated throughout a building's facade, making construction faster and more efficient;
- e) *Modular design* — It allows for the precast components to be easily assembled on-site, reducing the need for wet works and temporary supports;
- f) *Small complex facade panels* — Glass reinforced concrete (GRC) can be used instead of concrete for small complex facade panels;
- g) *Complex facade as a separate add-on* — Complex facade elements can be designed and manufactured separately from the building structure, allowing for easier handling and installation on-site;
- h) *Layouts rotated over mirrored* — Rotating layouts over mirrored elements can make use of the same assembly for construction; and

- j) *Checklists* — Checklists can be used to ensure good precast arch design, covering aspects such as structural integrity, aesthetic appeal and ease of construction.

D-4 STRUCTURAL DESIGN

Structural design has a direct impact on the ease of precast concrete manufacturing and site erection works. The key consideration for structural system design solutions are:

- a) *Simple design and detailing* — Precast concrete components should be designed with simple connections that can be easily assembled on-site, reducing construction time and costs;
- b) *Commercial - long spans* — Precast concrete components can be used for long-span commercial buildings, such as warehouses or shopping centres;
- c) *High rise residential* — In high-rise residential buildings, precast concrete wall panels can be used in conjunction with precast concrete floor slabs;
- d) *Low/mid rise residential* — In low to mid-rise residential buildings, precast concrete wall panels can be used with precast concrete floor slabs and beam-column combinations; and
- e) *Design to cater for temporary loads during construction* — The structural design should cater for temporary loads during the construction phase, such as the weight of equipment and construction workers.

D-5 TYPES OF STRUCTURAL SYSTEMS

All semi or full precast concrete structural systems need clear consideration for stability. It may take one of the following forms:

- a) *Skeletal frames* — Precast concrete components can be used to create skeletal frames, with columns and beams connected with precast framed connections;
- b) *Load bearing walls* — Precast concrete wall panels can be used as load-bearing walls, supporting the weight of the floors and roof;
- c) *Load bearing facade* — In a load-bearing facade system, the precast facade concrete panels bear the weight of the building and transfer the loads to the foundation independently to the core of the structure;
- d) *3-D boxes* — Precast concrete components can be used to create 3-D boxes, with precast walls, floors, and roofs forming the 3-D building unit;
- e) *Mixed systems* — A combination of precast concrete components can be used to create mixed systems, such as precast concrete walls with cast-in-place concrete floors; and
- f) *Checklists for good precast structural design* — Checklists can be used to ensure good precast structural design, covering aspects such as strength and durability, fire resistance, seismic design and ease of construction.

D-6 CONNECTIONS

Connection in precast concrete construction must be designed to take the forces derived in the structural modelling. To achieve work efficiency, the design approach should consider the following aspects:

- a) *Standardized connections* — Standardized precast concrete connections are designed to be easily designed, manufactured and assembled on-site;
- b) *Ease of erection* — Precast concrete connections should be designed with ease of erection in mind, to ensure that they can be quickly and safely assembled on-site;
- c) *Simple and minimalistic* — Simple and minimalistic connections are preferred in precast concrete construction, as they are more efficient and cost-effective;
- d) *Checklists for good connection systems* — Checklists can be used to ensure good precast concrete connection systems, covering aspects such as durability, strength, and ease of assembly; and
- e) *Innovative connections* — Innovative precast concrete connections can be used to create unique and complex structures, such as hybrid steel and precast systems or lotus root precast systems. These connections can offer advantages such as improved structural performance, and reduced construction time.

D-7 DESIGN OF CONNECTIONS

Design methods for joints in precast construction have been covered by various international technical journals and published academic research papers. They briefly cover the connections listed below:

- a) *Corbel-beam end joint* — A hidden mechanical connector can be used for the corbel-beam end joint, similar to hanger connections;
- b) *Column/wall base mechanical connections* — Column shoes and wall shoes with anchor bolts can be used for mechanical connections at the base of columns and walls;
- c) *Wall to wall connections* — Wire loops can be used for wall to wall connections;
- d) *Beam to beam* — Steel hangers can be used for connecting precast concrete beams; and
- e) *Rigid beam-column* — Top steel couplers and bottom steel welded connections can be used for rigid beam-column connections, ensuring that the connection is strong and can resist bending and shear forces.

D-8 CONNECTION DESIGN CONCEPT

This can cover the following:

- a) Load paths — Diaphragm, slab to wall/column paths, slab-beam shear connection;
- b) Frame behaviours;
- c) Accidental loads, progressive collapse;
- d) Tie design forces;
- e) Tie continuity — Laps, tie with enclosed links; and
- f) Tie anchorages with other ties.

D-9 DESIGN EXAMPLES

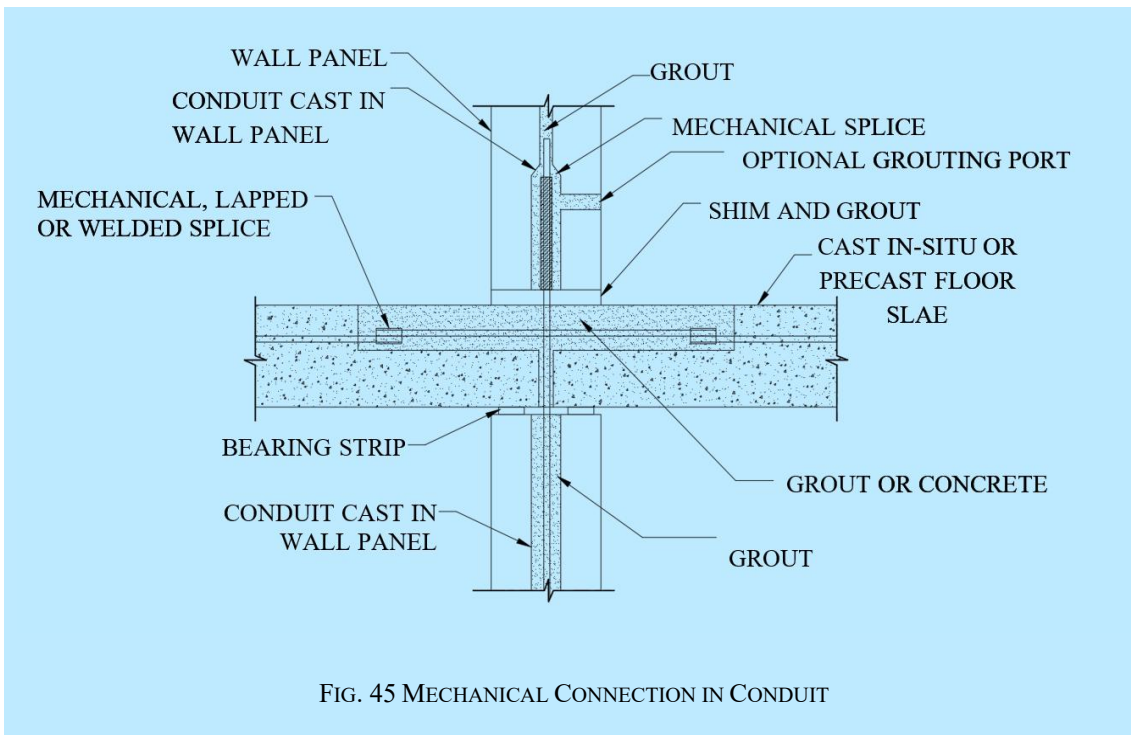
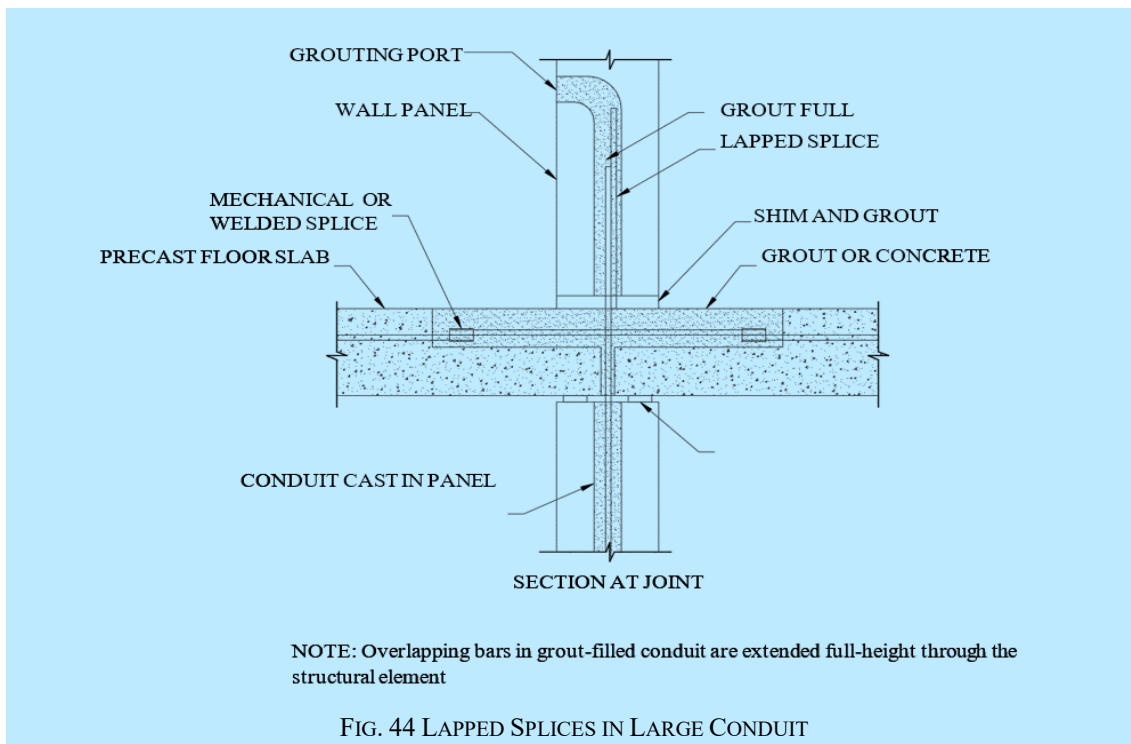
- a) Beam half joint (dap connection);
- b) Beam half joint (hanger/beam shoe connection);
- c) Corbel (conventional);
- d) Peikko corbel;
- e) Column base — Splice sleeve/grouted couplers;
- f) Column base — Shoe connection;
- g) Wall base — Splice sleeve/grouted couplers;
- h) Wall base — Shoe connection;
- j) Wall — Wall joint — Shear keys only;
- k) Wall — Wall joint — With loop reinforcement;
- m) Beam steel hangers — Generic;
- n) Beam steel hangers — Peikko;
- p) Proprietary mechanical connections;
- q) Column to base and column to column;
- r) Beam to column — Moment connection;
- s) Beam to column — Simply supported;
- t) Beam to beam;
- u) Wall to wall — Vertical and horizontal connections; and
- v) Landing slabs — TSS.

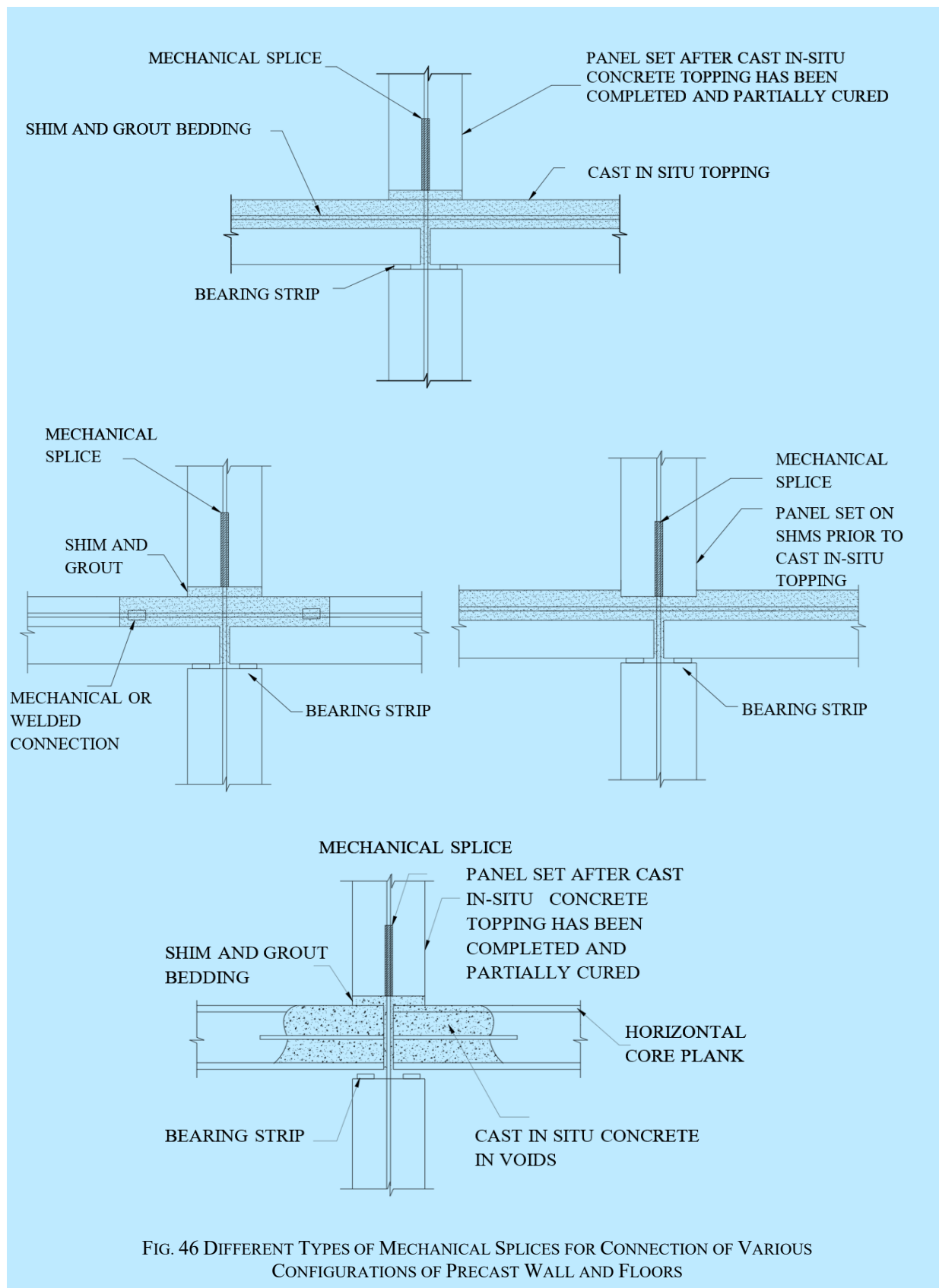
ANNEX E

([Foreword](#) and [Clause 8.3](#))

COMMON PRECAST CONNECTIONS

E-1 Typical figures of some common precast joint connections are given in [Fig. 44](#) to [Fig. 56](#).





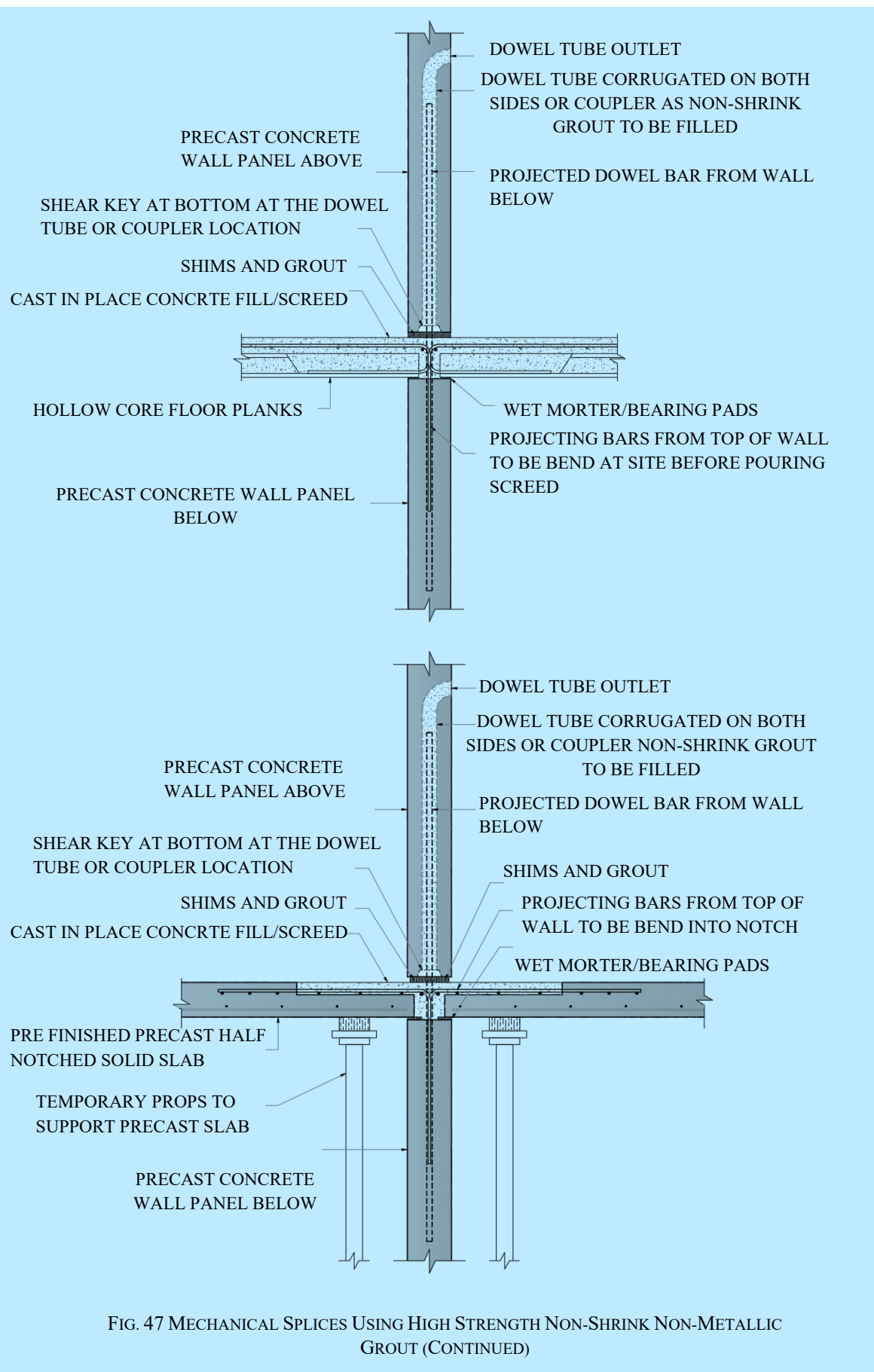


FIG. 47 MECHANICAL SPLICES USING HIGH STRENGTH NON-SHRINK NON-METALLIC GROUT (CONTINUED)

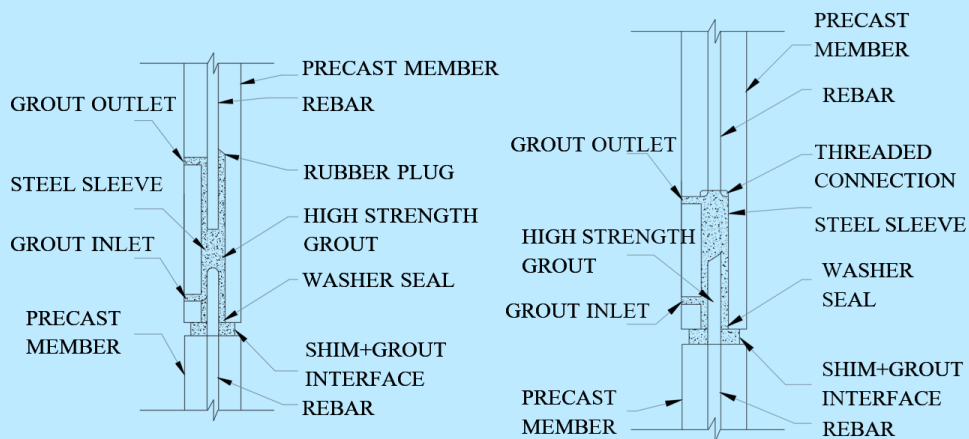
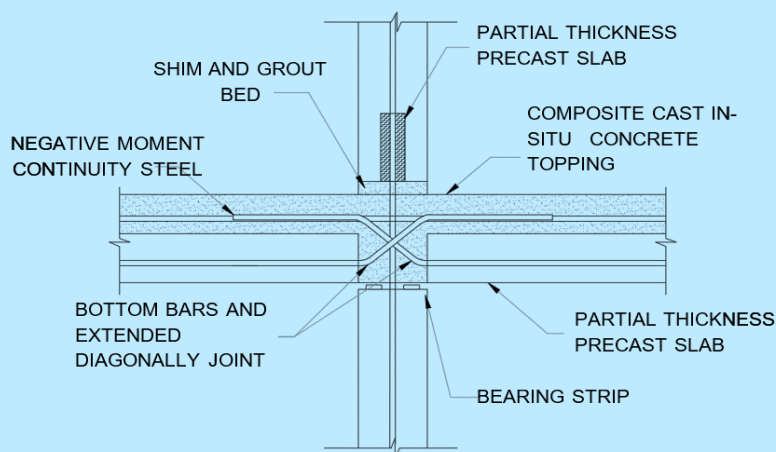
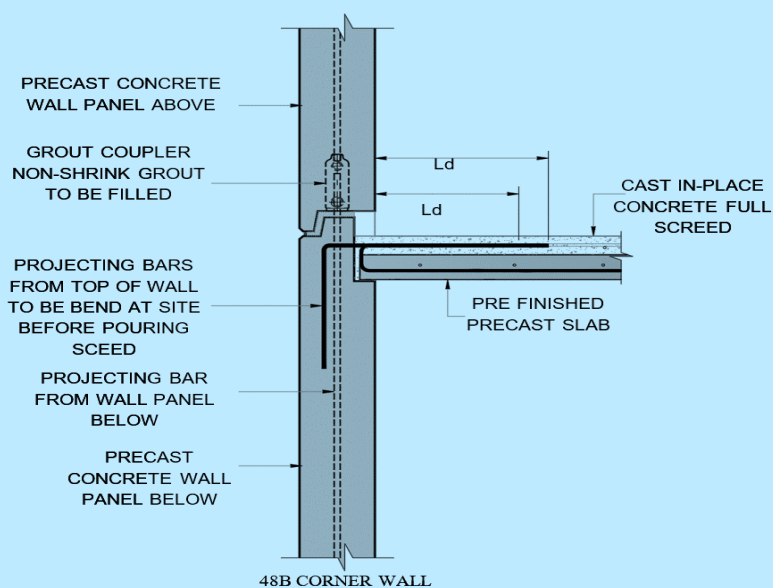


FIG. 47 MECHANICAL SPLICES USING HIGH STRENGTH NON-SHRINK NON-METALLIC GROUT



48A CENTRAL WALL



48B CORNER WALL

FIG. 48 FLOOR SLAB TO WALL DETAILS

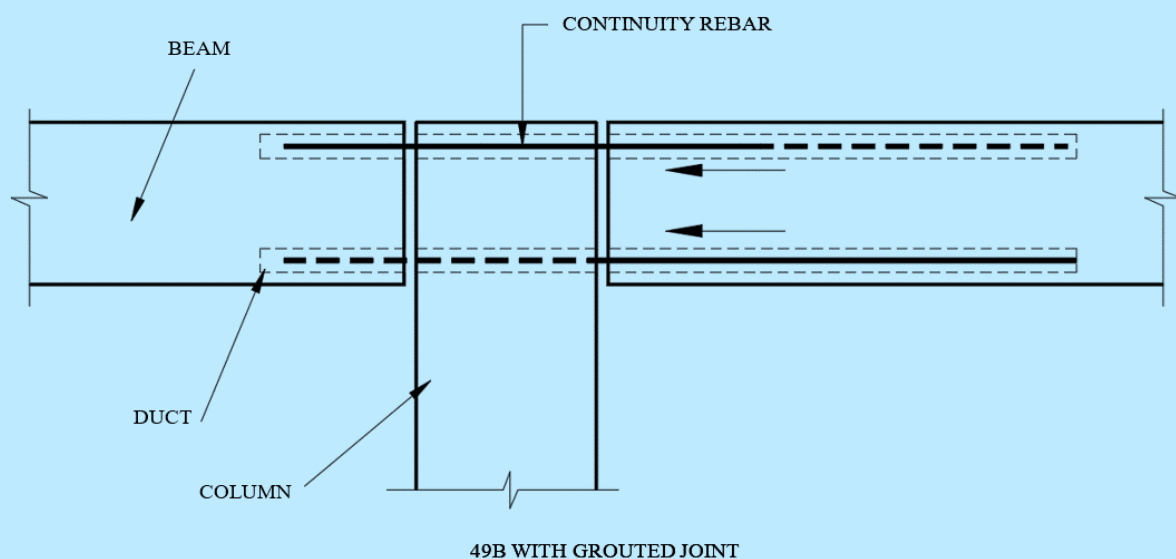
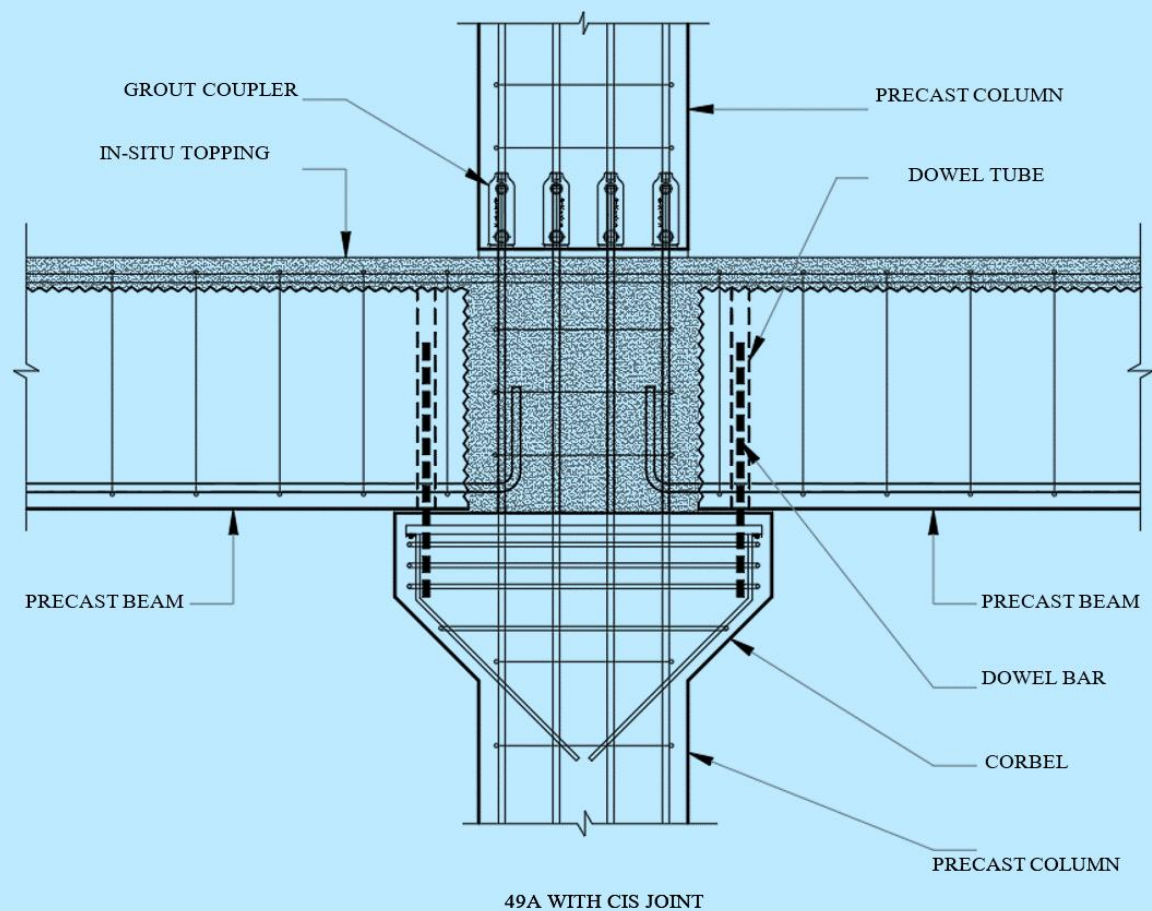
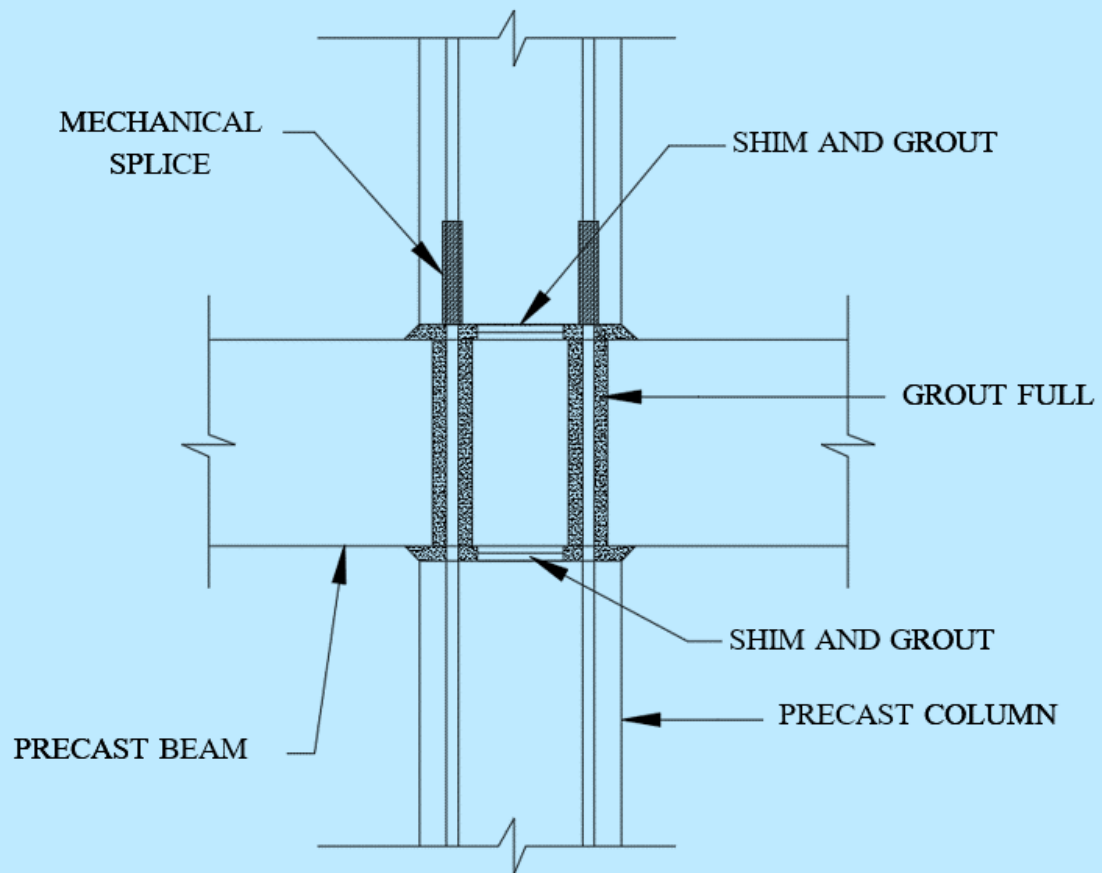


FIG. 49 BEAM AND COLUMN



NOTE: Conduit diameter should be two to four times the bar diameter for tolerance in field erection

FIG. 50 COLUMN TO COLUMN THROUGH CONDUIT IN BEAM

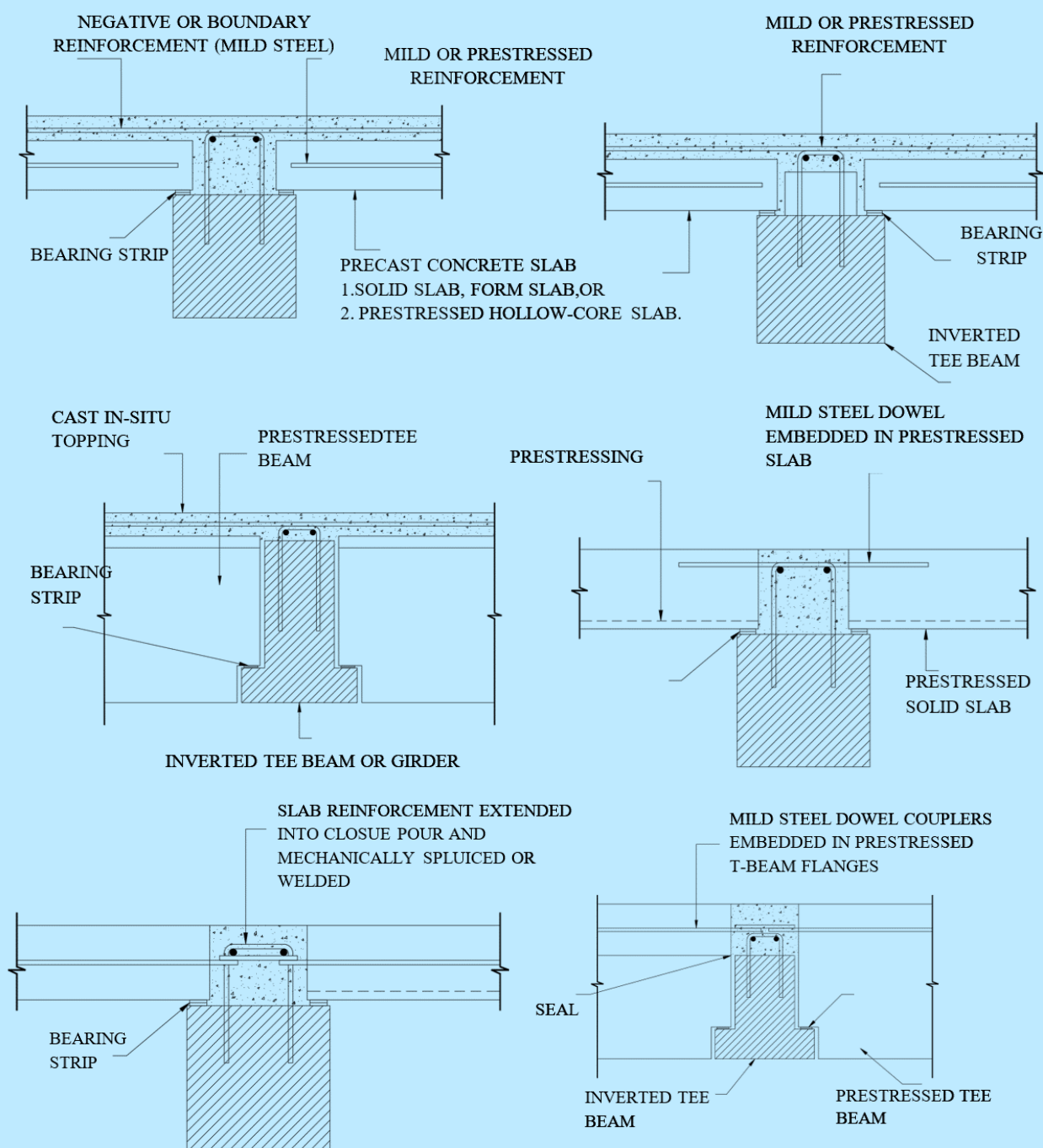
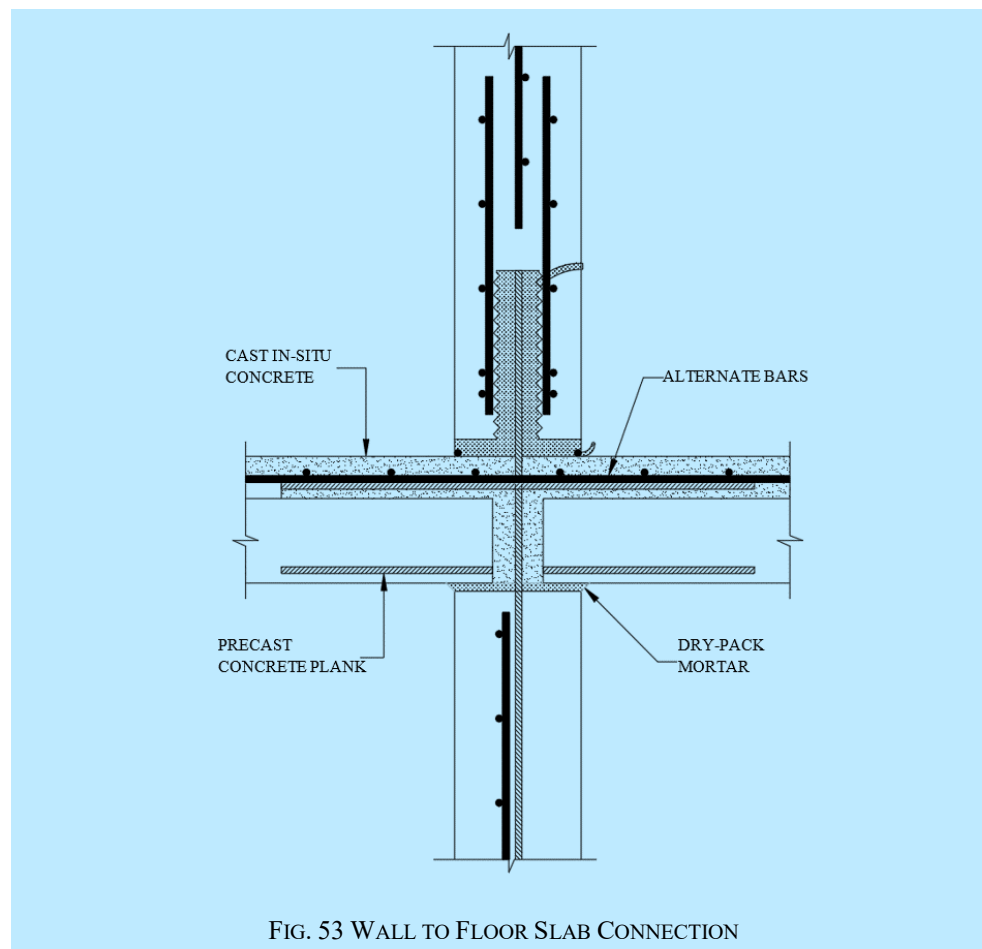
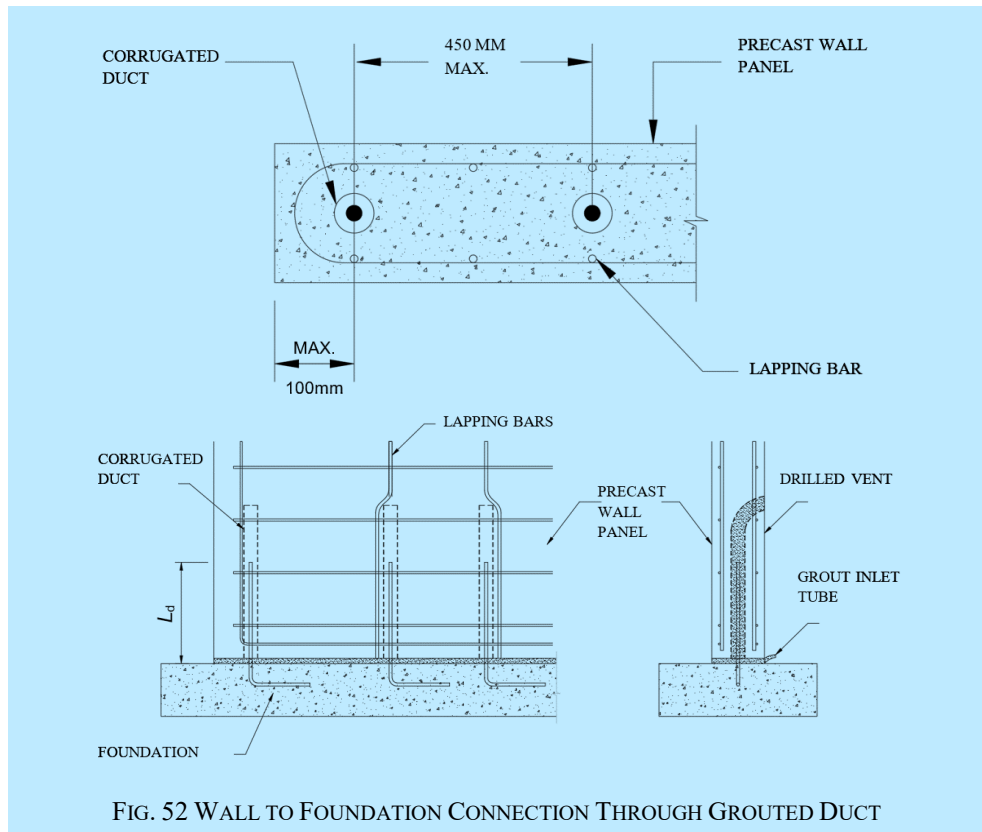


FIG. 51 TYPICAL CONNECTIONS OF PRECAST FLOOR SLAB ELEMENTS



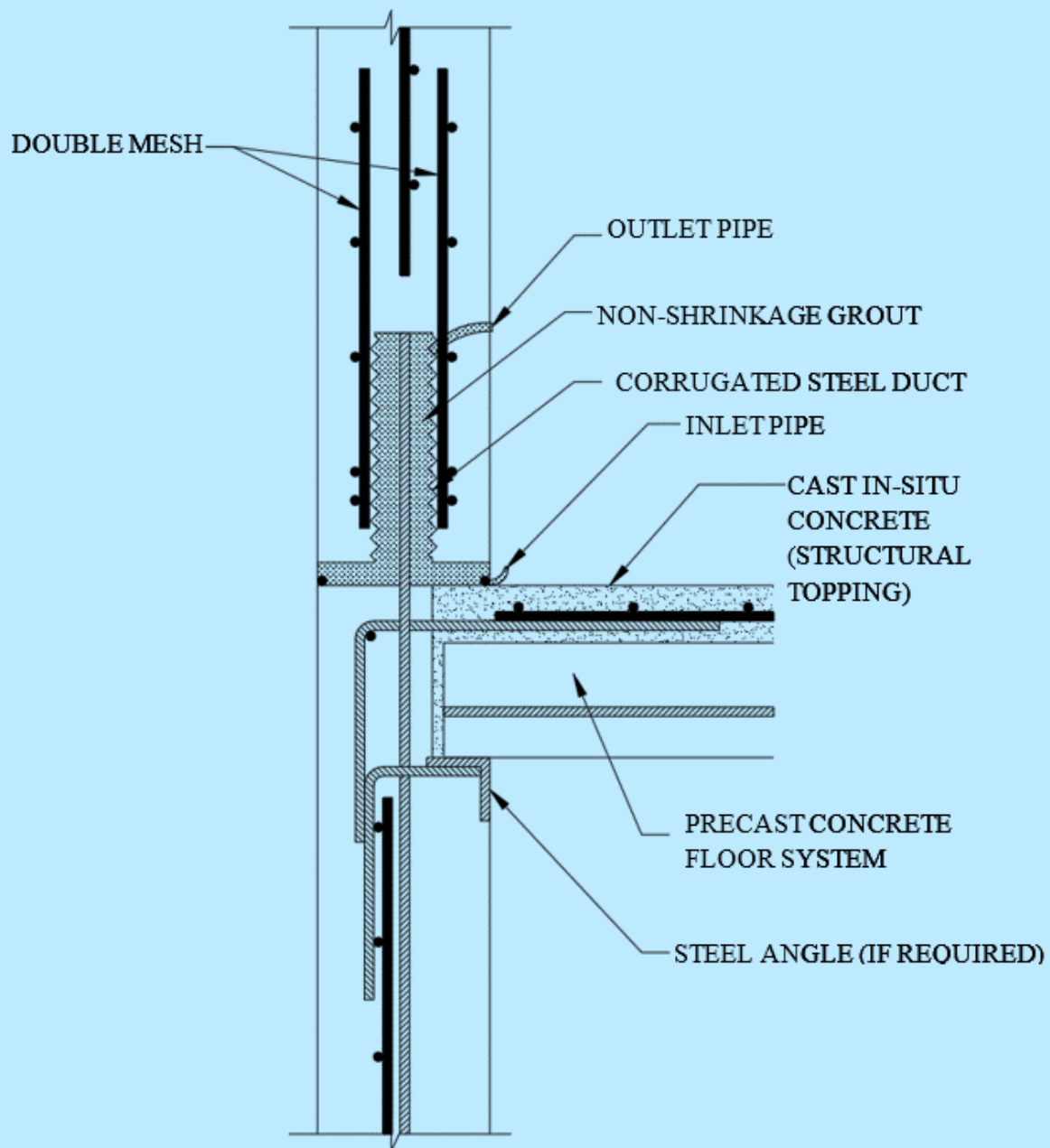


FIG. 54 EXTERIOR WALL TO FLOOR SLAB CONNECTION

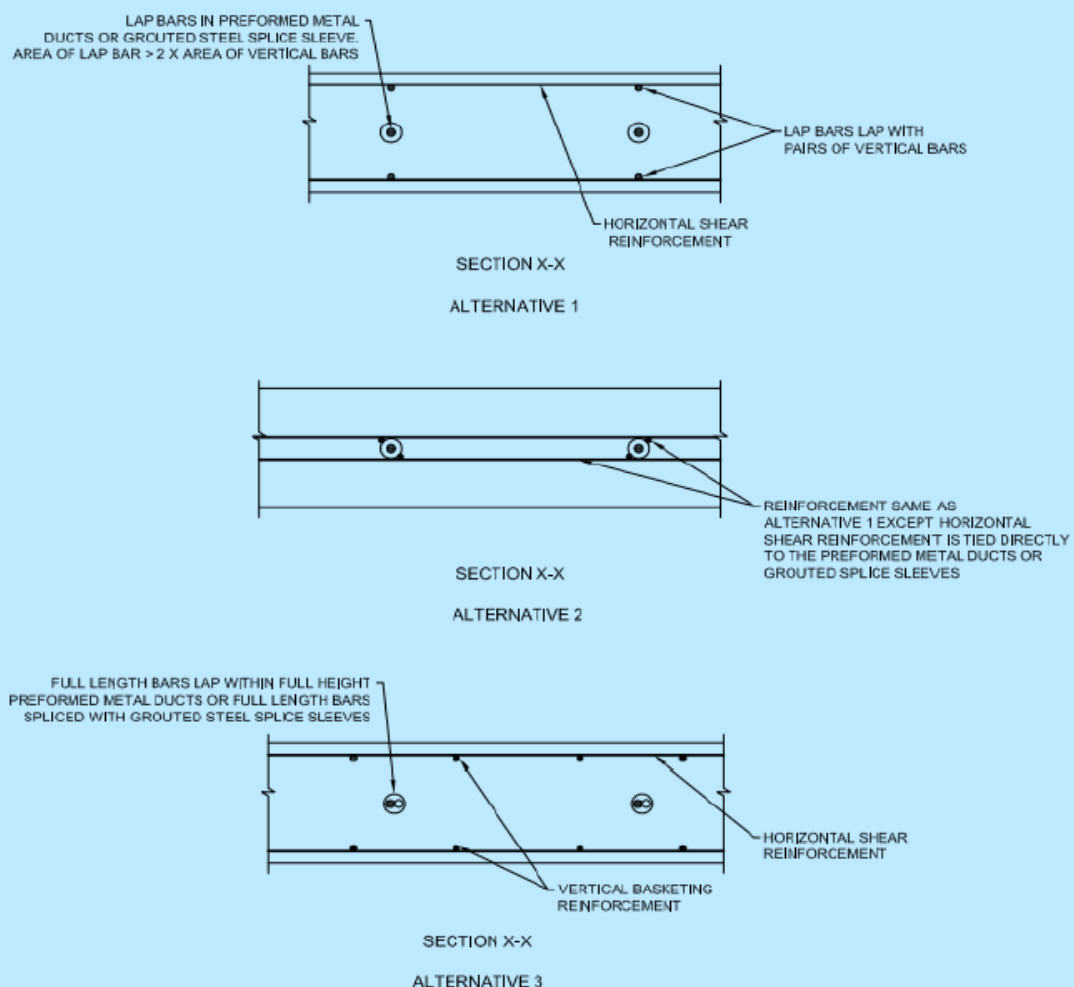
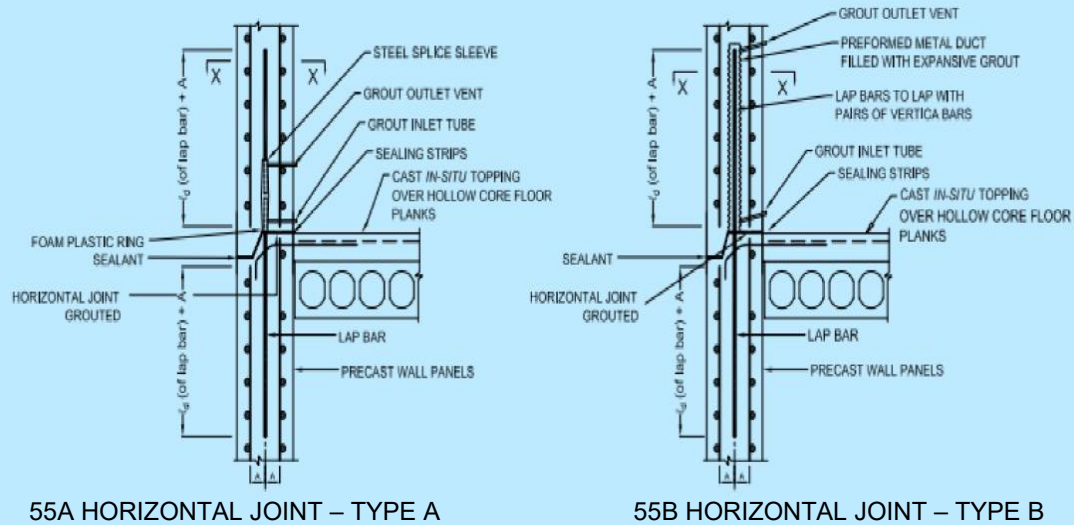


FIG. 55 MONOLITHIC PRECAST WALL CONSTRUCTION

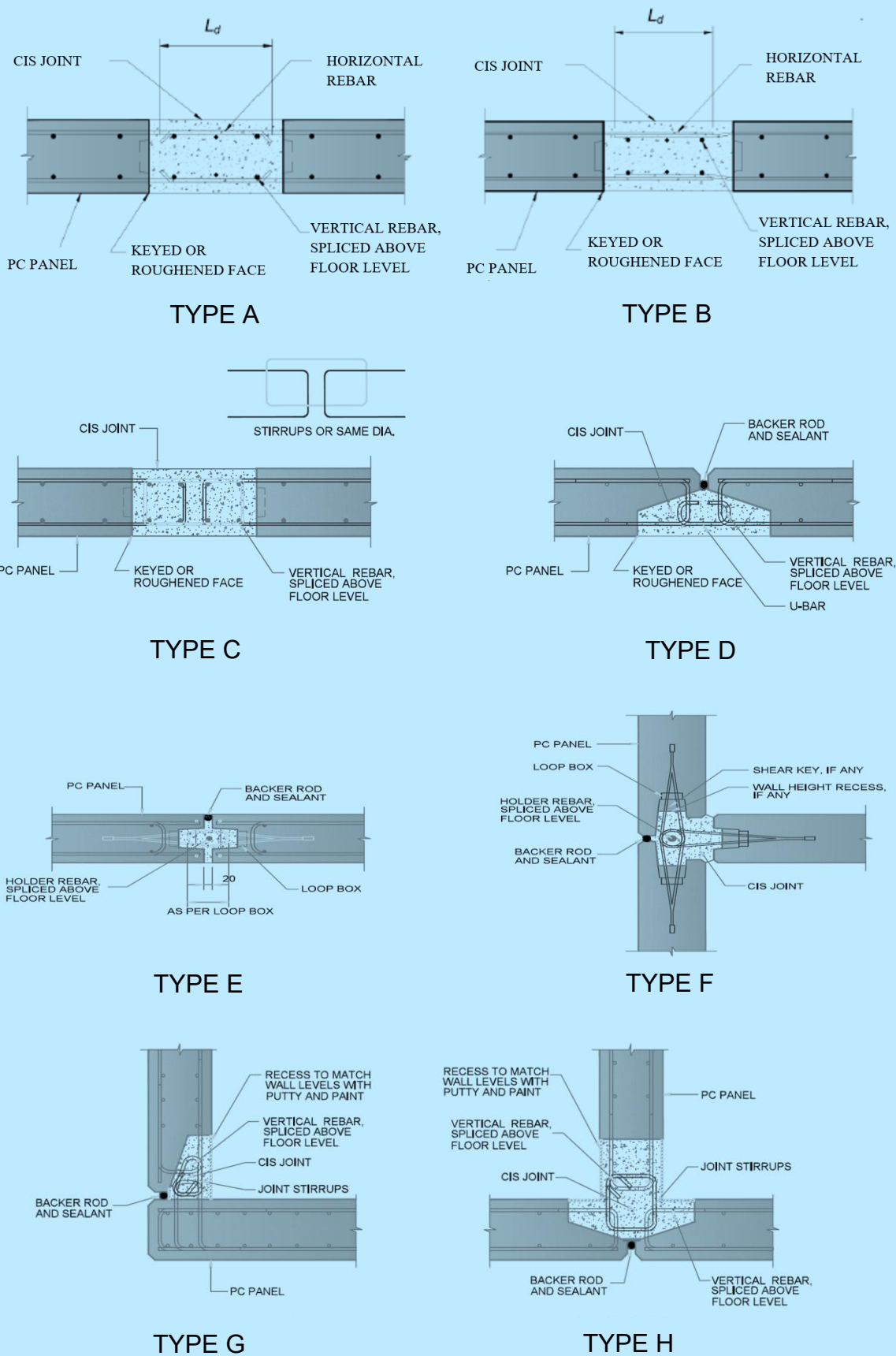
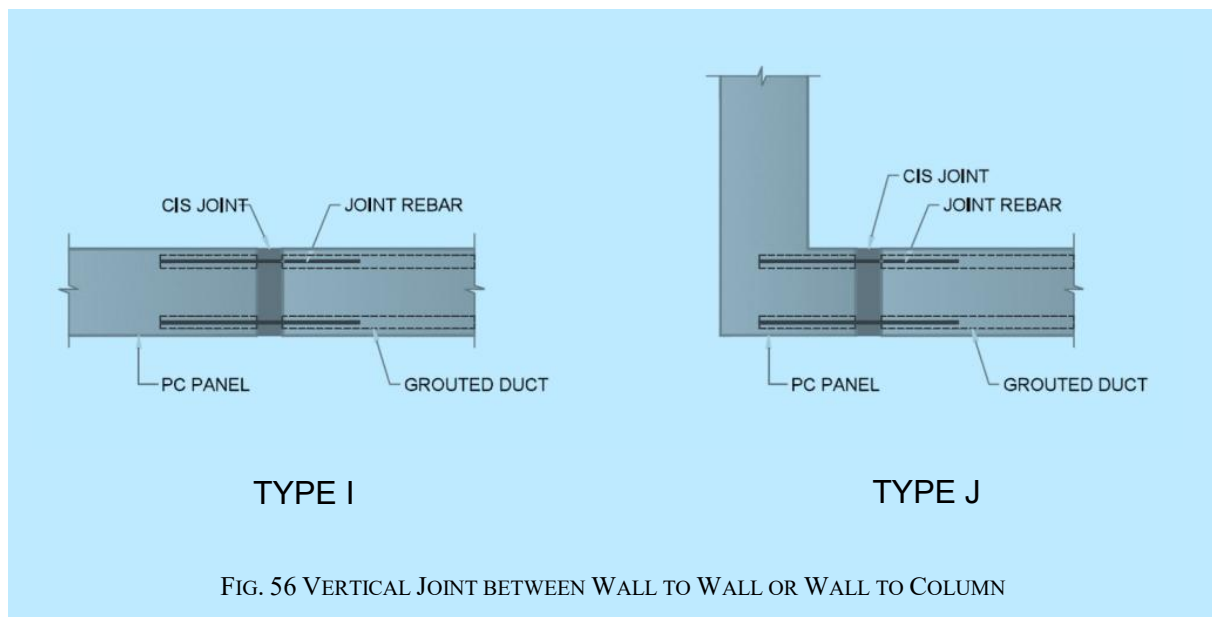


FIG. 56 VERTICAL JOINT BETWEEN WALL TO WALL OR WALL TO COLUMN (CONTINUED)



ANNEX F

(Clause 11)

COMMON DEFECTS AND REMEDIES

F-1 As defects in precast concrete elements result in direct and indirect cost in terms of rectification and construction time, it is worthwhile to ensure that they are produced and handled in a way to avoid/reduce such incidences. [Table 19](#) illustrates some of these common defects, their causes and preventive and remedial measures.

Table 19 List of Common Defects and Recommended Remedial Measures

(Clause F-1)

Sl No.	Common Defects	Possible Causes	Recommendations	Remedial Measures
(1)	(2)	(3)	(4)	(5)
i)	<i>Dimensional deviation:</i>			
	a) Variation in the dimension of precast elements would affect the joint alignment between these elements when erected; and	a) Mould forms may not be sufficiently rigid to maintain specified tolerances during concrete placement;	a) Regular check on the dimensions and rigidity of mould forms before casting operations. As a general guide the recommended thickness for steel mould are:	a) For minor deviation, corrective measures such as surface grinding, trimming/ hacking and skim coat application can be appropriately used to remedy the situation; and
	b) Precast slab element may warp due to insufficient concrete strength at lifting or improper storage condition.	b) Precast elements may not have gained sufficient concrete strength when demoulded;	1) 4.5 mm — Up to 50 castings;	b) Precast elements that are not within acceptable tolerance limits and have significant effects on the structural integrity or architectural performance should not be used.
		c) Top surface finish of precast elements may not be properly levelled and trowelled during production which result in differential thickness; and	2) 6 mm — Up to 100 castings; and 3) 9 mm — Up to 200 castings.	
		d) Precast elements (especially slender wall or slab panels) may be subjected to undue stress and deformation when they are not properly supported during storage.	Mould forms conditions will deteriorate with time and usage. They should be repaired, stiffened or replaced when needed. b) Cube tests should be conducted to ascertain the concrete strength of elements before demoulding;	

Table 19 (*Continued*)

Sl No. (1)	Common Defects (2)	Possible Causes (3)	Recommendations (4)	Remedial Measures (5)
ii)	Cracks	a) Precast elements may not have gained sufficient concrete strength before demoulding; b) Cracks may have occurred during initial lifting due to friction between the elements and the casting mould forms; c) Thickness of the precast elements may be too thin (70 mm or less) for safe demoulding and handling; and d) Cracks may have occurred during erection due to lack of planning and provisions for the precast panel geometry, crane rigging configuration and location of openings.	c) Spreading and levelling of concrete placement using appropriate tools such as structural screeder; and d) Precast elements should be properly stored and stacked at designated points using suitable support spacers and frame rack system. a) Proper curing method, curing time and temperature should be maintained; b) Cube tests should be conducted to ascertain the concrete strength of elements before demoulding; c) Appropriate form release agents should be used and uniformly applied onto the mould surface to minimize friction; d) Sectional thickness of precast elements should be increased to accommodate demoulding and handling stresses; e) Proper handling techniques should be adopted; f) Sufficient lifting points should be used to minimize over-stressing on certain areas;	a) All cracks should be examined by a qualified engineer to determine if they present a structural problem; b) Depending on the locations and seriousness of the cracks, different repair methods can be used to make good the affected precast elements; and c) Hairline cracks (not more than design crack width) can be repaired by cutting V-groove of specified minimum depth along the crack-lines, followed by patching subject to location and length of the crack in relation to length of units and not having adverse effect on the structural strength of the unit. For surface cracks (more than design crack width) or through cracks, epoxy injection method should be used

Table 19 (Continued)

Sl No. (1)	Common Defects (2)	Possible Causes (3)	Recommendations (4)	Remedial Measures (5)
iii)	Chip-off and damages	a) Chip-off at panel edges are usually caused by the hard bearing at support or excessive force exerted on the elements when handling; and 4) Improper storage method.	g) Additional reinforcing bars should be placed around the opening and odd corners; and h) Temporary stiffeners for openings should be provided during erection. a) Precautions should be taken to avoid damaging the elements during placement on vehicle, travel to site and during the unloading operation. Bearing pads should be used to cushion the contact areas from damage; b) The storage area should be relatively flat and dry; c) Precast elements should be properly stored and stacked at designated points using suitable support spacers and frame rack system; d) The casting slab or mould form should be thoroughly cleansed, levelled to achieve a smooth surface; e) The coverage of form release agents should be adequate and applied uniformly onto the mould surface;	to ensure that the cracks are completely bound and filled with epoxy. a) Remove all loose concrete and wash off any dust or dirt in the affected area; b) Apply bonding agent to affected concrete surface; c) Welded wire mesh can be included to provide support for the patch mix concrete or grout; d) Patch mix composition or grout should be consistent with the strength requirements of the adjacent concrete; e) Formwork should be placed whenever it is necessary to contain the patch mix or grout; and f) The affected area should be protected from any disturbance during the curing period.

Table 19 (Continued)

Sl No. (1)	Common Defects (2)	Possible Causes (3)	Recommendations (4)	Remedial Measures (5)
iv)	Honeycomb and excessive pinholes	a) Poor concrete compaction due to ineffective vibration or rebars congestion; and b) Grout leakage along the perimeter of side forms due to: 1) Loose or missing bolts, fixing pins; 2) Damaged rubber gasket seal; 3) Mould part, if defective.	f) Spacers of the correct sizes should be used and well secured to maintain the concrete cover required during casting; g) Lifting inserts and concrete strength of elements should be of adequate capacity for the intended lift. Proprietary products for safe and efficient handling may be used for this purpose; and h) Lifting inserts or hooks should be inspected for proper location. They should be fastened to specified depth prior to concrete placement. a) Proper compaction methods should be adopted and carried out; b) Concrete mix design and workability should be reviewed and adjusted when needed; c) Appropriate concrete vibrator such as clamp-on form vibrator may be used to attain better compaction; d) Rebar congestion can be alleviated by having larger size (that is, lesser number) rebars or by increasing the sectional	a) Remove all loose concrete and wash off any dust or dirt in the affected area; b) Apply bonding agent to affected concrete surface; c) Welded wire mesh may be included to provide support for the patch mix concrete or grout; d) Patch mix composition or grout should be consistent with the strength requirements of the adjacent concrete;

Table 19 (Continued)

SI No. (1)	Common Defects (2)	Possible Causes (3)	Recommendations (4)	Remedial Measures (5)
			dimensions of the elements where possible; e) Mechanical couplers or sleeves can be used to simplify the reinforcement layout and to minimize rebar congestion; and f) Defective mould forms and accessories should be repaired or replaced to prevent grout leakage during concrete.	e) Formwork is to be put up where it is necessary to contain the patch mix or grout; and f) The affected area should be protected from any disturbance during the curing period.
v)	<i>Missing or wrong details</i> — such as cast-in items, architectural nib and groove details, lifting hooks, reinforcement/starter bars/block-out/corrugated pipes.	a) Items may not have been included in the shop drawing; and b) Quality checks may not be properly in place.	a) All items should be reflected in the shop drawings for production. Any changes should be made known to the production team; and b) The use of checklists during inspection can help to ensure that all items specified in the drawing are included before casting.	a) Certain details such as missing starter bars/reinforcement and lifting hooks can be replaced by welding additional reinforcement bars after hacking off the concrete at the affected area; and b) Other items such as cast-in items, groove and block-out can be provided by chasing or chiselling out the face of the precast panels.
vi)	Strand slippage which exceed allowable design values (applicable to pre-stressed elements only) (Slippages of pre-stressing strands can be detected by visual inspection).	a) Insufficient bond strength between concrete and the pre-stressing strands; and b) Poor compaction of concrete around pre-stressed strands.	a) Required concrete strength of the precast elements should be attained and verified by cube test results before de-tensioning of strands; b) Proper compaction method should be adopted and carried out during casting;	a) It is not possible to rectify the elements from strand slippage; and b) Design verification should be carried out to ascertain the reduced capacity of the elements due to slippage, if adopted.

Table 19 (Concluded)

SI No. (1)	Common Defects (2)	Possible Causes (3)	Recommendations (4)	Remedial Measures (5)
vi)	Alignment	a) Inaccurate setting out and positioning of precast elements during erection; and b) Deviation in the dimensions of the precast element.	c) Concrete mix design and workability should be reviewed and adjusted when needed; and b) Appropriate concrete vibrator to attain better compaction may be provided. a) Appropriate surveying and levelling equipment should be used to achieve better alignment; b) Required alignment and level should be confirmed before permanent jointing; and c) Critical dimensions of precast concrete elements should be verified before installation.	a) Minor adjustments of the element/panel alignment can be done during installation. However the effects on the final alignment and deviation of the building should be evaluated; b) For minor deviations, corrective measures such as surface grinding, trimming/ hacking and skim coat application can be appropriately used to rectify the precast elements before installation; and c) Precast elements that are not within acceptable tolerance limits and have significant effects on the structural integrity or architectural performance should not be used.

ANNEX G

([Foreword](#), [Clauses 5, 7.1, 10.1.2, 10.9.1 and 11](#))

SUSTAINABILITY IN PRECAST CONCRETE CONSTRUCTION

G-1 GENERAL

This annex presents sustainable practices in precast concrete construction, emphasizing methods for reducing environmental impacts, promoting resource conservation and ensuring long-term viability. The following key sustainability factors can enable the adoption of greener and efficient solutions provided the challenges involved are appropriately addressed.

G-2 USE OF RECYCLED AGGREGATES IN PRECAST CONCRETE

G-2.1 Environmental Benefits

G-2.1.1 Conservation of Natural Resources

By reusing construction and demolition debris as recycled aggregates, the need for extracting virgin raw materials like sand and granite is significantly reduced, preserving ecosystems and minimizing environmental disruption.

G-2.1.2 Reduction in Energy Use

Recycled aggregates require less energy to process compared to natural aggregates, which leads to a lower carbon footprint during production. This also reduces the energy demand associated with quarrying, transportation and processing.

G-2.1.3 Circular Economy Impact

Incorporating recycled aggregates into precast concrete non-structural components—such as paving slabs, kerbs, and drainage of systems, etc., supports a circular economy, minimizing waste sent to landfills and promoting the reuse of construction materials.

G-2.2 Challenges and Innovations

G-2.2.1 Ensuring Consistent Quality

Recycled aggregates may exhibit variability in size and composition, potentially impacting concrete strength and durability. Innovations in sorting, crushing and refining technologies can improve the consistency and reliability of recycled aggregates.

G-2.2.2 Efficient Use of Water

In general, excessive water is required to be used while using recycled aggregate and can have negative environmental impacts. By adopting optimized mix designs and water-efficient technologies, the water demand for concrete production can be significantly reduced.

G-2.2.3 Advancement in Sustainable Mixes

New research into mix designs is focusing on achieving better integration of recycled aggregates without compromising the material's strength or durability. This can lead to a more sustainable, high-performance concrete product.

G-3 ADOPTION OF RENEWABLE ENERGY SOURCES

G-3.1 Environmental Benefits

G-3.1.1 Decarbonizing Operations

Shifting precast concrete manufacturing facilities to renewable energy sources, such as solar, wind and

hydropower, significantly reduces reliance on fossil fuels, helping to lower overall greenhouse gas emissions and mitigate climate change impacts.

G-3.1.2 Electric and Hybrid Machinery

By replacing conventional diesel-powered equipment with electric and hybrid alternatives, the precast concrete industry can reduce air pollution and carbon emissions associated with construction machinery.

G-3.2 Challenges and Innovations

G-3.2.1 High Initial Costs

The transition to renewable energy systems and low-emission machinery involves substantial initial investment. However, long-term operational savings, combined with available financial incentives for green technologies, can help offset the upfront costs.

G-3.2.2 Energy Storage and Smart Grids

Installing energy storage solutions, such as batteries, in conjunction with on-site renewable energy systems can help mitigate energy supply issues during periods of high demand or when renewable energy generation is low, further reducing dependence on the traditional energy grid.

G-4 REDUCING SITE POLLUTION AND WASTE

G-4.1 Environmental Benefits

G-4.1.1 Waste Minimization

Precast construction reduces on-site waste generation because most components are manufactured off-site in controlled factory settings. This reduces generation of waste and their dispersal at landfills and minimizes the potential for soil and water contamination and helps in achieving undisturbed site.

G-4.1.2 Reduced Emissions and Noise Pollution

Precast methods reduce the need for heavy on-site machinery and long construction timelines. As a result, there is less air pollution, noise pollution, and disturbance to nearby ecosystems or communities.

G-4.2 Challenges and Innovations

G-4.2.1 Reducing Transportation Emissions

Transporting precast components to construction sites can contribute to emissions, especially when distances are long. This can be minimized by optimizing transportation logistics, using hybrid or electric vehicles and focussing on the use of modular components that reduce transportation needs.

G-4.2.2 On-Site Pollution Prevention

During installation, on-site pollution, such as dust or runoff, can still occur. Simple practices like water misting to suppress dust and covering concrete surfaces to prevent runoff can significantly reduce the environmental impact of installation activities.

G-5 USE OF SUPPLEMENTARY CEMENTITIOUS MATERIALS (SCMs) AND OPTIMIZED CONCRETE MIXES

G-5.1 Environmental Benefits

G-5.1.1 Low-Carbon Alternatives to Cement

Replacing traditional Portland cement with supplementary cementitious materials (SCMs) such as fly ash, slag,

or natural pozzolans can drastically reduce the carbon footprint of concrete production by minimizing the need for energy-intensive cement manufacturing.

G-5.1.2 Efficient Resource Use

Advanced mix designs, including high-performance concrete (HPC) and ultra-high-performance concrete (UHPC), allow for the reduction of material consumption without sacrificing strength or durability. These innovations help to optimize the use of raw materials, ensuring sustainability in production.

G-5.2 Challenges and Innovations

G-5.2.1 Durability and Performance

While reducing cement content or using alternative binders can lower the carbon footprint, it is essential to ensure the long-term durability and performance of the concrete.

G-5.2.2 Geopolymer Concrete

Geopolymer concrete, made from industrial by-products has a lower carbon footprint compared to traditional cement-based concrete and can be used in precast applications without compromising strength and durability.

G-6 SUSTAINABLE DESIGN AND PRODUCTION OPTIMIZATION

G-6.1 Environmental Benefits

G-6.1.1 Material Efficiency through Design

Precast elements can be optimised for material efficiency by carrying out optimum design. Building information modelling (BIM) allows for detailed digital models that reduce waste during both the design and construction phases.

G-6.1.2 Modular and Prefabricated Systems

The use of modular designs with standardization in precast construction can reduce material waste, reducing production time, reuse of mould and optimize the process of precasting.

G-6.2 Challenges and Innovations

G-6.2.1 Customization versus Standardization

While standardization helps reduce waste and increase efficiency, some projects require customised solutions. The challenge lies in finding a balance between benefits of standardized components and the need for bespoke, project-specific designs.

G-6.2.2 3D Printing and Additive Manufacturing

Emerging technologies like 3D printing could revolutionise precast construction (refer [Annex B](#)). By using digital fabrication techniques, custom shapes and optimized designs can be produced efficiently, with less material wastage.

G-7 TRANSPARENCY, CERTIFICATION AND ETHICAL PRACTICES

G-7.1 Sustainability Governance

G-7.1.1 Environmental Certifications

Precast manufacturers and construction companies can pursue certifications such as environmental management systems, to demonstrate their commitment towards achieving sustainability. These certifications set a global benchmark for eco-friendly practices in construction.

G-7.1.2 Ethical Sourcing and Labour Practices

Sustainable construction practices require ethical sourcing of materials, ensuring that raw materials are extracted responsibly and workers are treated fairly. This also includes adherence to applicable labour and governance standards/practices.

G-7.2 Challenges and Innovations

G-7.2.1 Data Transparency

As consumer and regulatory demand for sustainability data increases, companies are under pressure to disclose their environmental impacts. Technologies like block-chain offer innovative solutions for ensuring transparency and accountability in reporting environmental performance.

G-7.2.2 Community and Stakeholder Engagement

Actively engaging with local communities, government bodies, and customers to ensure that sustainability goals align with the social expectations and needs of stakeholders.

ANNEX H

(Clause [7.9.4.8](#))

DESIGN AND DETAILING GUIDELINES FOR PRECAST STRUCTURES WITH HOLLOW CORE SLABS

H-1 GENERAL

This annex presents the design guidelines and detailing parameters essential for structures using hollow core slabs as the flooring system. This annexure ensures that all parameters pertaining to structural stability and end user safety of hollow core slab systems are attended to.

H-2 DESIGN GUIDELINES

H-2.1 Hollow core slabs are to be designed as a simply supported slab system. The use of hollow core slabs is not permitted as a continuous flooring system.

H-2.2 Hollow core slabs are to be mandatorily designed using structural topping as mentioned in [6.3.6.1](#). This topping shall be considered acting composite with the hollow core slab provided conditions mentioned in [6.3.6.3](#) are met.

H-2.3 For flooring systems consisting of hollow core slabs, only the structural topping shall be considered as part of the diaphragm as noted in [6.3.6.3](#). No part of the hollow core slab shall be considered as part of diaphragm or considered participating in lateral load transfer. Structural topping design to follow guidelines as prescribed in [7.10.8](#).

H-2.4 The design engineer is to assess the torsional rotation of the hollow core slab caused by the differential settlement of the slab supports. Special attention is to be undertaken by the structural engineer since hollow core slabs have limited torsional resistance capacity.

H-3 DETAILING PARAMETERS

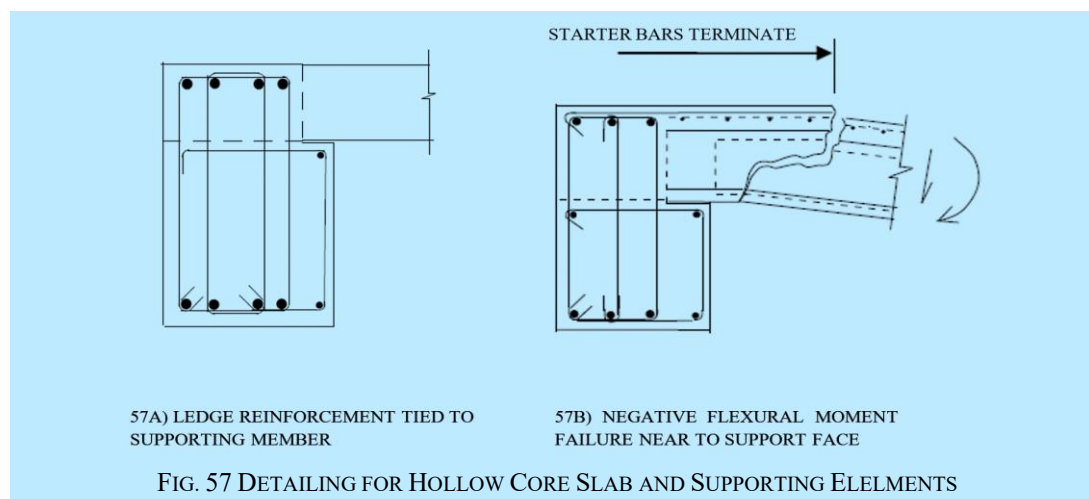
H-3.1 Low friction bearing shims or strips are to be placed under the hollow core slabs to mitigate the formation of flexural crack close to the support in the hollow core element and to mitigate spalling of support ledge concrete.

H-3.2 Concrete ledges to be used as bearing support for hollow core slabs must have the ledge reinforcement tied with the primary supporting beam or wall as shown in [Fig. 57A](#).

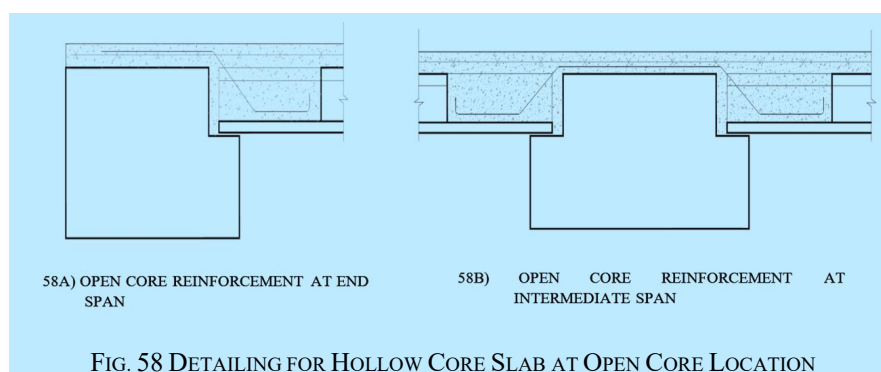
H-3.3 Hollow core slabs to have minimum bearing of 80 mm to mitigate loss of support due to spalling and relative movement of the slabs.

H-3.4 Nominal starter bars must extend for a length of 600 mm from the face of the supporting members and tied to the structural topping reinforcement to mitigate failure due to negative moments close to the support during seismic events as shown in [Fig. 57B](#). Starter bar reinforcement diameter may be 8 mm for slabs up to 300 mm depth and 10 mm for slabs over 300 mm depth at a spacing not more than 250 mm.

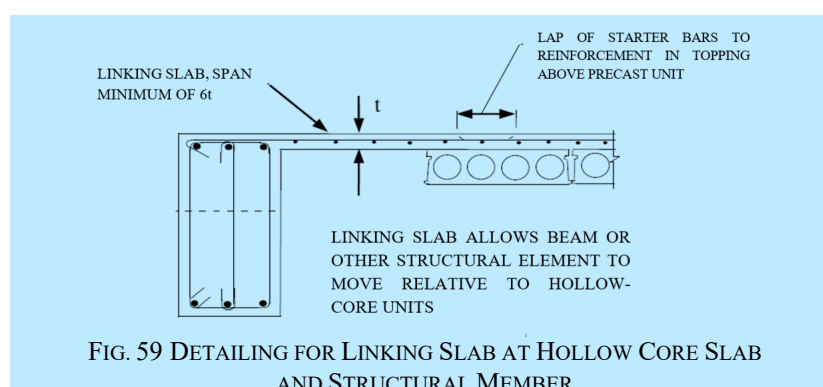
H-3.5 Structural topping to be preferably reinforced with reinforcement with elongation more than 14.5 percent to mitigate brittle failure of topping. Welded mesh may be permitted as topping reinforcement provided the mesh is fabricated from reinforcement having elongation more than 14.5 percent.



H-3.6 A minimum of 2 cores to be opened for a length of 600 mm at both span ends of the hollow core slab, they are to be reinforced with 8 mm reinforcement for slabs up to 300 mm depth and 10 mm for slabs over 300 mm depth and filled simultaneously with the structural topping concrete. [Figure 58](#) elaborates on the detailing that may be adopted at open core locations.



H-3.7 Linking slabs to be provided to connect structural members to non-bearing ends of hollow core slabs. The length of the linking slab to be a minimum of 6 times the structural topping thickness as described in [Fig. 59](#). The depth of the linking slab will be as per structural or load requirements.



LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used as a guide in conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Section.

	<i>IS No.</i>	<i>Title</i>
(1)	IS 13920 : 2016	Ductile design and detailing of reinforced concrete structures subjected to seismic forces — Code of practice (<i>first revision</i>)
(2)	IS 456 : 2000	Plain and reinforced concrete — Code of practice (<i>fourth revision</i>)
(3)	IS 1343 : 2012	Prestressed concrete — Code of practice (<i>second revision</i>)
(4)	IS 383 : 2016	Coarse and fine aggregate for concrete — Specification (<i>third revision</i>)
	IS 456 : 2000	Plain and reinforced concrete — Code of practice (<i>fourth revision</i>)
(5)	IS 9103 : 1999	Concrete admixtures — Specification (<i>first revision</i>)
	IS 456 : 2000	Plain and reinforced concrete — Code of practice (<i>fourth revision</i>)
(6)	IS 10262 : 2019	Concrete mix proportioning — Guidelines (<i>second revision</i>)
(7)	IS 1199 : 1959	Methods of sampling and analysis of concrete
(8)	IS 10262 : 2019	Concrete mix proportioning — Guidelines (<i>second revision</i>)
	IS 1199 : 1959	Methods of sampling and analysis of concrete
(9)	IS 9417 : 2018	Welding of high strength steel bars for reinforced concrete construction — Recommendations (<i>second revision</i>)
(10)	IS 16172 : 2023	Reinforcement couplers for mechanical splices of steel bars in concrete — Specification (<i>first revision</i>)
(11)	IS 18256 : 2023	Solid round glass fibre reinforced polymer (GFRP) bars for concrete Reinforcement — Specification
(12)	IS 1566 : 1982	Specification for hard-drawn steel wire fabric for concrete reinforcement (<i>second revision</i>)
(13)	IS 14343 : 1996	Choice of grouting, materials for alluvial grouting — Guidelines
(14)	IS 4671 : 2018	Expanded polystyrene for thermal insulation purposes — Specification (<i>second revision</i>)
(15)	IS 8183 : 2024	Bonded mineral wool — Specification (<i>second revision</i>)
(16)	IS 6598 : 2018	Cellular concrete for thermal insulation — Specification (<i>first revision</i>)
(17)	IS 10297 : 1982	Code of practice for design and construction of floors and roofs using precast reinforced/prestressed concrete ribbed or cored slab units
(18)	IS 16700 : 2023	Criteria for structural safety of tall concrete buildings (<i>first revision</i>)

	<i>IS No.</i>	<i>Title</i>
(19)	IS 1641 : 2013	Fire safety of buildings (general): General principles of fire grading and classification — Code of practice (<i>second revision</i>)
	IS 1642 : 2013	Fire safety of buildings (general): Details of construction — Code of practice (<i>second revision</i>)
(20)	IS 4326 : 2013	Earthquake resistant design and construction of buildings — Code of practice (<i>third revision</i>)
(21)	IS 4082 : 1996	Stacking and storage of construction materials and components at site — Recommendations (<i>second revision</i>)
(22)	IS 11384 : 2022	Composite construction in structural steel and concrete — Code of Practice (<i>first revision</i>)
(23)	IS 3201 : 1988	Criteria for design and construction of precast concrete trusses and purlins (<i>first revision</i>)
(24)	IS 6332 : 1984	Code of practice for construction of floor and roofs using precast doubly-curved shell units
	IS 10297 : 1982	Code of practice for design and construction of floors and roofs using precast reinforced/prestressed concrete ribbed or cored slab units
	IS 10505 : 1983	Code of practice for construction of floors and roofs using precast concrete waffle units
	IS 13994 : 1994	Design and construction of floor and roof with precast reinforced concrete planks and joists — Code of practice
	IS 14142 : 1994	Design and construction of floors and roofs with prefabricated brick panel — Code of practice
	IS 14215 : 1994	Design and construction of floors and roofs with precast reinforced concrete channel units — Code of practice
	IS 14242 : 1995	Design and construction of roofs using precast reinforced concrete L-Panels — Code of practice
(25)	IS 6061	Code of practice for construction of floor and roof with joists and filler blocks:
	(Part 1) : 1971	With hollow concrete filler blocks
	(Part 2) : 1981	With hollow clay filler blocks (<i>first revision</i>)
	(Part 3) : 1981	With precast hollow clay block joists and hollow clay filler blocks
	(Part 4) : 1981	With precast hollow clay block slab panels
(26)	IS 14687 : 1999	Falsework for concrete structures — Guidelines
(27)	IS 2185 (Part 1) : 2005	Concrete masonry units — Specification: Part 1 Hollow and solid concrete blocks (<i>third revision</i>)
(28)	IS 3952 : 2013	Burnt clay hollow bricks and blocks for walls and partitions — Specification (<i>third revision</i>)
(29)	IS 2185 (Part 3) : 1984	Specification for concrete masonry units: Part 3 Autoclaved cellular (Aerated) concrete blocks (<i>first revision</i>)

	<i>IS No.</i>	<i>Title</i>
(30)	IS 4101 (Part 2) : 1967	Code of practice for external facings and veneers: Part 2 Cement concrete facing
(31)	IS 1950 : 1962	Code of practice for sound insulation of non-industrial buildings
(32)	IS 3483 : 1965	Code of practice for noise reduction in industrial buildings
	IS 2526 : 1963	Code of practice for acoustical design of auditoriums and conference halls
(33)	IS 1904 : 2021	General requirements for design and construction of foundations in soils — Code of practice (<i>fourth revision</i>)
(34)	IS 516 (Part 1/Sec 1) : 2021	Hardened concrete — Methods of test: Part 1 Testing of strength of hardened concrete, Section 1 Compressive, flexural and split tensile strength (<i>first revision</i>)
(35)	IS 3414 : 1968	Code of practice for design and installation of joints in buildings
(36)	IS 1786 : 2008	High strength deformed steel bars and wires for concrete reinforcement — Specification (<i>fourth revision</i>)
(37)	SP 30 : 2023	National Electrical Code of India 2023 (<i>second revision</i>)
	IS 732 : 2019	Code of practice for electrical wiring installations (<i>fourth revision</i>)
(38)	IS/IEC 62305-1 : 2010	Protection against lightning: Part 1 General principles
(39)	IS 2065 : 1983	Code of practice for water supply in buildings (<i>second revision</i>)
	IS 1742 : 1983	Code of practice for building drainage (<i>second revision</i>)
(40)	IS 1742 : 1983	Code of practice for building drainage (<i>second revision</i>)
	IS 12183 (Part 1) : 1987	Code of practice for plumbing in multi-storeyed buildings: Part 1 Water supply

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 8 GLASS AND GLAZING

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Section 8) deals with the use of glass in buildings. Glazing has become an important item in building construction. Glass, the primary glazing element has to be selected to cater to several requirements and the glazing has to be designed to meet various engineering requirements. Fixing of glass, a specialized operation, when properly done, will avoid the hazards of broken glass. Growing trend in resorting to glazed windows/doors in buildings and structures has considerably increased the importance of glazing and the need for proper workmanship. Considering all these, this Section was introduced in the third revision of the Code in 2016 as part of the National Building Code of India.

This Section provides guidance in the selection of appropriate glazing for various types for building construction; guidance on the energy and light aspects while choosing glazing; guidance on determining the appropriate thickness of glass used in building glazing and guidance with respect to human impact safety while using glazing in buildings.

Use of organic coated films including safety films on glass shall not classify the glass as safety glass as per this NBCS, and hence users should exercise caution in choosing their options.

Information on installation of glass mirrors and glass railings are given in the informative [Annex H](#) and [Annex J](#), respectively.

A conformity assessment sheet for evaluating the compliance of glass and glazing systems with the requirements of this Section has been included in informative [Annex K](#).

This Section was introduced in the Code in 2016 as part of the National Building Code of India. As a result of experience gained in implementation of 2016 version of this Section, feedback received as well as in view of formulation of new standards in the field of glass and glazing and revision of some existing standards, a need to revise this Section was felt. This revision was, therefore, prepared to take care of these aspects.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

The significant changes incorporated in this revision include:

- a) Definitions of various existing terms have been reviewed and updated, wherever required;
- b) Provisions related to the requirements for laminated safety glass, toughened (tempered) safety glass and heat strengthened glass in [4](#) have been updated;
- c) Clause [5](#) on energy and light has been reviewed and updated, and new provisions on energy design considerations and window design guidelines for non-conditioned/mixed mode buildings have been included;
- d) The types of glass permitted for fire-resistant glazing within the scope of the Section in [6](#) have been updated;
- e) Inclusion of provisions for determining the fire resistance performance of perimeter fire barrier joint systems, thereby enhancing fire safety measures;
- f) New provisions addressing anti-fallout features, window restrictors for child safety, and dynamic wind restraint mechanisms, have been included in [7.2](#), [7.3](#) and [7.4](#) respectively;
- g) A new clause [9](#) on security glass has been introduced, covering glass for security applications such as bullet resistance, burglary resistance, and blast resistance;
- j) Incorporation of provisions for assessing curtain walls and storefront systems under inter-storey drift conditions in [Annex A](#);
- h) Provisions related to railings with glass infill panels and free-standing glass protective railings have been modified in [Annex J](#);

- k) Addition of a conformity assessment sheet to evaluate the compliance of glass and glazing systems with the requirements outlined in this Section in Annex K; and
- m) References to all the concerned Indian Standards have been updated.

The information contained in this Section is largely based on the following Indian Standards:

<i>IS No.</i>	<i>Title</i>
IS 16231	Use of glass in buildings — Code of practice:
(Part 1) : 2019	General methodology for selection (<i>first revision</i>)
(Part 2) : 2019	Energy and light (<i>first revision</i>)
(Part 3) : 2019	Fire and loading (<i>first revision</i>)
(Part 4) : 2019	Safety related to human impact (<i>first revision</i>)

Assistance has also been derived from the following publications in the formulation of this Section:

ISO 9845-1 : 2022	Solar energy — Reference solar spectral irradiance at the ground at different receiving conditions — Part 1: Direct normal and hemispherical solar irradiance for air mass 1,5
ISO 10291 : 1994	Glass in building — Determination of steady-state U values (thermal transmittance) of multiple glazing — Guarded hot plate method
ISO 10292 : 1994	Glass in building — Calculation of steady-state U values (thermal transmittance) of multiple glazing
ISO 10293 : 1997	Glass in building — Determination of steady-state U values (thermal transmittance) of multiple glazing — Heat flow metre method

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with in the rounded off value should be the same as that of the specified value in this Section.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to “**accepted standard**” in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/ Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Accepted standards [C-8(1)] refers to the Indian Standards given at serial number (1) of the list of standards given at the end of this Part C/Section 8, that is, IS 14900 : 2018 ‘Specification for transparent float glass (*first revision*)’ and IS 2553 (Part 1) : 2018 ‘Specification for safety glass: Part 1 Architectural, building and general uses’ (*fourth revision*).

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 8 GLASS AND GLAZING

1 SCOPE

1.1 This NBCS (Part C/Section 8) covers,

- a) selection and application of glass in buildings, different types of glass, their requirements and associated glazing materials;
- b) guiding provision for glazing in buildings with respect to their effect on energy, visual (light) and solar environments in the building;
- c) selection of glass in buildings, subject to wind loading, seismic loading and special considerations for fire rated glass and related materials;
- d) provisions for the selection and manifestation of glass in buildings, subject to safety with respect to human impact of the occupants;
- e) provisions relating to glazing systems such as selection, design, fabrication, installation, testing and maintenance; and
- f) requirement and selection criteria for security glass in buildings.

1.2 This Section does not cover patent glazing.

NOTE — The provision of this Section pertains to glass and installation thereof. However, parties, as per their mutual agreement, may appropriately utilize the relevant provisions for installation of alternative sheeting materials used for similar purposes such as plastic glazing sheets/polycarbonate sheets taking also into account requirements of safety issues as applicable with the specific material.

1.3 The provisions in respect of sustainable use of glass in buildings are covered in relevant guidelines/best practices (Handbook on ‘Sustainability in Buildings’).

2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply, and for other terms those given in accepted standards [C-8(1)] shall apply.

2.1 Annealed Glass (*see Glass*) — Also known as ‘normal’ glass, which has not been subjected to toughening, lamination and heat strengthening. It is the product obtained from the process of slow and steady cooling of hot glass in order to relieve internal stresses or thermal induced stresses.

NOTE — This term is most commonly used for flat glass.

2.2 Annealed Fire Resistant Glass — Type of glass which is made out of annealed glass combination and is tested for the required duration and satisfies the intended criteria of fire resistance. These types of glasses may not possess the best of the impact resistance characteristics and may be used as vision panels.

2.3 As-Cut Finished Sizes — Finished panes of flat glass cut from stock/standard sizes. They may be subject to further processing, such as edge working, drilling, face decoration, etc.

2.4 Aspect Ratio — The ratio of a longer side of glass pane to its shorter side.

2.5 Balustrade — A low barrier forming a parapet to a stair, ramp, balcony, raised level or a change in level.

2.6 Beads or Glazing Beads — A strip of wood, metal or other suitable material attached to the rebate to retain the glass in position in the frame.

2.7 Bite — The minimum distance by which the inner edge of a frame (or a stop) overlaps the edge of the glass panel.

2.7.1 Glazing Bite — The dimension by which the framing system overlaps the edge of the glazing infill. Minimum glazing bite shall be 10 mm for non-structural sealant applications.

2.8 Block (Setting Block) — A small piece of wood, plastic, rubber or other suitable material used between the edges of the glass (generally the bottom edge only) to centralize the glass in the frame (frequently called a setting block).

2.9 Chair Rail — A fixed rigid bar that provides protection from human impact.

2.10 Clear Glass — A transparent glass with neutral/near colourless appearance.

2.11 Coefficient of Linear Expansion — It is expressed as the strain (change in length per unit length) per unit variation of temperature of a material.

2.12 Combustible Material — The material which when burnt adds heat to a fire when tested for combustibility in accordance with accepted standard [C-8(2)].

2.13 Corridor — A common passage or circulation space including a common hall.

2.14 Coupled Glazing (also known as Secondary Glazing) — Two panes of glazing spaced apart in an opening, either in the frame or glazed separately, to form an unsealed cavity.

2.15 Curtain Wall — Non-load bearing structure/partition of glass attached to the building frame, usually on the exterior face. Curtain wall vertical framing members run past the face of floor slabs, and provision for anchorage is typically made at vertical framing members only.

NOTE — In contrast to combination assemblies and composite units, curtain wall systems often need to meet additional performance requirements for inter-storey differential movement, seismic drift (*see Annex A*), dynamic water infiltration, etc. Operating vents and entrance doors are provided as separate inserts.

2.16 Curtain Wall Assembly — A fire rated or non-fire rated, non-load bearing exterior wall assembly secured to the structural members of the building.

2.17 Distance Piece — A small piece of wood or other suitable material used to locate the glass between the bead and the back of the rebate to prevent lateral movement.

2.18 Door or Shutter Assembly Door-Set — A pedestrian door-set or industrial type door-set including any frame or guide, door leaf or leaves, rolling or folding curtain, etc, which is provided to give a fire resisting capability when used for the closing of permanent openings in fire resisting separating elements, which includes any side panel(s), vision panel(s), flush over panel(s) and/or transom panel(s) together with the building hardware and any seals (whether provided for the purpose of fire resistance or smoke control or for other purposes such as draught or acoustics) which form the assembly.

2.19 Double Glazed Fire Resistant Glass — A fire resistant glass that is used in a double glazed unit.

NOTE — These shall be tested as a complete Double Glazed Unit (DGU) or other multiple glazed units as the case may be. Double glazed fire resistant glass can be a combination of intumescent laminated fire rated or tempered fire resistant glass. Fire resistant glass cannot be combined with another glass and used as a DGU unless the combined double glazed or multiple glazed units are also tested.

2.20 Double Glazing — Glazing formed by an assembly of two glass panes separated by a spacer and the gap may be filled by vacuum, air or inert gases and hermetically sealed along the periphery to improve the thermal insulation.

2.21 Edge Deterioration — The discolouration of the reflective coating at the edge of the silvered glass.

2.22 Edge Faults — The faults that affect the as-cut edge of the glass. They may include entrant/emergent faults, shelling, corners on/off and vents.

2.23 Edge Polished — It is usually applied to flat glass, the edges of which have been polished after cutting.

2.24 Edging — Grinding the edge of flat glass, to a desired shape or size.

2.25 Exposed Edge — A glass panes' edge that is not covered.

2.26 Facade — It is the front or face of the building which is part of framed or frameless system.

2.27 Faceted Glazing — It is a type of glazing in which flat panes of glass are installed vertically at an angle to each other to form a faceted surface.

2.28 Fenestration — All area (including frame) in the building envelope that let in light, including window, plastic panels, clerestories, skylight, fanlight and glass doors that are generally more than one-half the floor height, and glass block walls.

2.29 Fin — A piece of glass positioned and fastened to provide lateral support.

2.30 Fire Separation — The shortest distance, in metre, measured from the external wall of the building concerned to the external wall of any other building on the site, or from other site, or from the opposite side of street or other public space for the purpose of preventing the spread of fire.

2.31 Float Glass — Flat, transparent, clear or tinted soda-lime silicate glass having parallel and polished surfaces obtained by continuous casting and floatation on a metal bath.

NOTE — Flat glass is a general term covering sheet glass, float glass and various forms of rolled and plate glass in which shape of the glass is flat and commonly used for windows, glass doors, transparent walls and other architectural applications.

2.32 Flush Over Panel — A panel which is incorporated within a door-set or openable window and fitted above the leaf or leaves within the frame head and the jambs and with no transom.

2.33 Frame — An element made of timber/metal/aluminium or other durable material/combinations of materials, such as glass fins and structural sealant, supporting the full length of a glazed panel edge.

2.34 Frameless Glazing — An unframed glazing which maintains the integrity through pointed support structures.

2.35 Front Putty — The compound forming a triangular fillet between the surface of the glass and the front edge of the rebate.

2.36 Fully Framed Glazing — The glazing whose panels have all its four edges framed.

2.37 Gap — A clearance between two adjacent surfaces and/or edges, for example between the edge of the leaf and the reveal of the frame or between the face of the leaf and the frame stop.

2.38 Glass — An inorganic product of fusion which has cooled to a rigid condition without crystallizing. It is typically hard and brittle, and has a conchoidal fracture. It may be colourless or tinted and transparent to opaque. The term glass refers to monolithic glass unless specified otherwise.

2.39 Glass Appearance Faults — Faults such as spot and/or linear and/or enlarged area faults which alter the visual quality of glass.

2.40 Glazing — Act of securing of glass or other glazing material into a building in prepared openings in windows, door panels, partitions, etc.

2.41 Guard Rail — A permanent physical barricade used to prevent human impact on glass/glazing.

2.42 Hairline Scratch — Very fine scratch that can hardly be seen with naked eye and is associated with glass cleaning techniques.

2.43 Halo — It is the distortion zone around a spot fault.

2.44 Heat Soaking — It is done on toughened/tempered glass by reheating to a temperature of 290 °C and keeping it at this temperature for 8 h and cooling it gradually. The glass can break spontaneously and without provocation due to possible impurity of nickel sulphide in basic glass used for toughening/tempering. This risk of spontaneous breakage can be minimised by heat soaking process by forcing such glasses to break during the test itself.

2.45 Heat Strengthened Glass — A single piece of glass within which a permanent surface compressive stress, additionally to the basic mechanical strength, has been induced by a controlled heating and cooling process in order to give it increased resistance to mechanical and thermal stress and prescribed fracture characteristics {see 6.1.5 of accepted standard [C-8(3)]}.

2.46 Hermetic Seal — A complete seal (especially against the escape or entry of air which is impervious to outside interference or influence) done on insulating glass units.

2.47 High Activity Area — An area where multiple and major human activity takes place.

2.48 High Risk Area — An area prone to human injury and casualty.

2.49 Infill Balustrades — The balustrades in which the supported glass resists an infill pressure and/or point load applied to the glass panel.

2.50 Insulating Glass Unit (IGU) — An assembly consisting of at least two panes of glass, separated by one or more spaces, hermetically sealed along the periphery, mechanically stable and durable.

2.51 Interlayer — Layer or material acting as an adhesive and separator between plies of glass and/or plastic glazing sheets.

NOTE — It can be designed to give additional performance to the finished product, for example impact resistance, fire resistance, solar control and acoustic insulation.

2.52 Internal Partition — An interior dividing wall or such portion of an interior dividing wall that is not a door, side panel, shop front or atrium wall.

2.53 Light Transmission — It is the measure of light passing through a transparent or a translucent material.

2.54 Light Transmittance — It is the measure of percent of visible light transmitted through glass pane which depends on type of body substrate and coating done on glass.

2.55 Linear Defects — They are the scratches and extended spot faults on the glass.

2.56 Manifestation — Any technique for enhancing a person's awareness of the presence of transparent glass.

2.57 Maximum Thickness — The thickness of a pane of glass at the maximum thickness tolerance.

2.58 Minimum Thickness — The thickness of a pane of glass at the minimum thickness tolerance.

2.59 Modulus of Elasticity (Young's Modulus) — This modulus expresses the tensile force that theoretically has to be applied to a glass specimen to stretch it by an amount equal to its original length. It is expressed as a force per unit area (for glass, $E = 0.7 \times 10^5 \text{ N/mm}^2 = 70 \text{ GPa}$).

2.60 Monolithic Glass — A single sheet of flat glass which could be either annealed, toughened or heat strengthened.

2.61 Mullion — Vertical framing member supporting the edge or surface of the glass. It is responsible for transferring loads to the sub-structure.

2.62 Nominal Thickness — A numeric designation that indicates the approximate thickness of glass.

2.63 Non-Combustible Material — A material not liable to burn itself or to add heat to a fire when tested for combustibility in accordance with good practice [C-8(2)].

2.64 Organic Coated Glass — A sheet of glass coated on one or both sides with an applied organic coating or sheeting.

2.65 Pane — A single piece of glass or plastic glazing sheet material in a finished size ready for glazing.

2.66 Panel — An assembly containing one or more panes of glass.

2.67 Partition — It means an interior non-load bearing divider, one storey in height or part.

2.68 Partly Framed or Unframed Glazing — The panels that have one or more edges unframed.

2.69 Passive Solar Gain — The solar radiation in the form of energy and light which is transmitted through the glazing into the building and which can be utilized as a source of energy to reduce the need for artificial lighting.

2.70 Patterned Glass — A rolled glass having a distinct pattern on one or both surfaces.

2.71 Perimeter Fire Barrier — The perimeter joint protection that provides resistance to prevent the passage of fire and/or smoke from floor to floor within the building at the opening between the exterior wall assembly and the floor assembly (*see also* [6.4.3.7](#)).

2.72 Protective Coating(s) Faults — The faults where the metallic layer is exposed. They can be scratches or loss of adhesion of the protective coating(s).

2.73 Rebate — The part of a surround; the cross-section of which forms an angle into which the edge of the glass is received.

2.74 Reflective Coated Glass — It is a glass with reflective coating which uses the principle of increasing the direct reflection to maximize solar energy attenuation. In comparison with clear glass, its absorption of solar energy is also increased.

NOTE — In comparison with float glass surface, these reflecting coatings (due to their composition) exhibit lower level of emissivity which improves their *U*-value.

2.75 Reflective Silver Coating Faults — It is the fault on reflective silver layer which shall alter the appearance of the silvered glass. They consist of scratches, stains, colour spots and edge deteriorations.

2.76 Residual Protection — It is the protection provided to avoid the human impact on glass. It is provided on the side of glass where there are chances of human impact. It can be achieved by providing a sill structure or a grill inside.

2.77 Safety Glass

2.77.1 Laminated Safety Glass — Two or more pieces of glass held together by an interleaving layer or layers of plastic sheet material. The laminated safety glass will crack and break under sufficient impact, but the pieces of glass tend to adhere to the plastic and do not fly.

NOTE — The railing and overhead glazing shall consist exclusively of laminated safety glass.

2.77.2 Toughened (Tempered) Safety Glass — A single piece of specially heat treated glass, with a stress pattern such that the piece when fractured reduces to numerous granular fragments, with no large jagged edges.

2.78 Shading Coefficient — The ratio of the rate of solar heat gain through a specific unit assembly of glass to the solar heat gain through a single light of 3 mm clear glass in the same environment.

2.79 Sheet Glass — Glass made in large sheet directly from furnace or by making a cylinder and then flattening it.

NOTE — These transparent glass have a glossy, fire-finished, apparently plane and smooth surfaces, but having a characteristic waviness of surface.

2.80 Skylight — A fenestration surface having a slope of less than 60° from the horizontal plane. Other fenestration, even if mounted on roof of a building is considered as either a vertical glazing or sloped glazing depending upon the angle of the glazing.

2.81 Sloped Overhead Glazing — A glazing that is inclined at less than 75° to the horizontal and located, wholly or partially, directly above an area that may be used by people.

2.82 Solar Energy Absorption — The percentage of the solar spectrum energy (ultraviolet, visible, and near-infrared) from 300 nm to 2 500 nm that is absorbed by a glass product.

2.83 Solar Energy Transmittance (Direct) — The percentage of energy in the solar spectrum, ultraviolet, visible, and near-infrared energy, 300 nm to 2 500 nm, that is directly transmitted through the glass.

2.84 Solar Heat Gain Coefficient (SHGC) — The SHGC is the fraction of incident solar radiation admitted through a fenestration, both directly transmitted, and absorbed and subsequently released inward through conduction, convection and radiation.

2.85 Spandrel — A non-vision portion of the exterior wall of a building.

2.86 Spot Faults — The nuclei (solid or gaseous inclusions), deposits, crush marks, etc, in the glass. In certain instances, spot faults are accompanied by a distortion zone called 'halo'. The nucleus of the spot fault is measurable.

2.87 Stain — An alteration of the reflective coating characterized by a more or less brownish, yellowish or greyish colouration of zones which can sometimes cover the whole reflective surface.

2.88 Stock/Standard Sizes — The panes of flat glass such as coated, mirror, enamelled glass, etc supplied with as-cut edges which are intended for further processing. These are generally corresponded with manufacturer's production size.

2.89 Tempered Fire Resistant Glass — A single piece of specially thermally-treated or chemically treated glass, tested for fire resistance for the required duration and having such a stress pattern that the piece when fractured, reduces to numerous granular fragments, with no large jagged edges.

2.90 Test Assembly — The complete assembly of the test specimen together with the test apparatus.

2.91 Thermal Transmittance (U) — Thermal transmission through unit area of the given building unit divided by the temperature difference between the air or other fluid on either side of the building unit in steady state conditions. It is also called as U -value. Its unit is W/m^2K .

NOTES

1 Thermal transmittance differs from thermal conductance insofar as temperatures are measured on the two surfaces of a building unit in the latter case and in the surrounding air (or other fluid) of the material on the two sides, in the former case. Thermal conductance is a characteristic of the building unit whereas thermal transmittance depends on conductance and surface coefficients of the building unit under the conditions of use.

2 In the case of glazing, thermal transmission is taken through the central part of the glazing, that is, without edge effects.

2.92 Through Component/Connection — The internal spacer or fixing that either penetrates through the door-set or openable window from one face to another or directly connects the faces of one to the other.

2.93 Tinted Glass — A normal flat glass to which colourants (normally metal oxides) are added during manufacturing process to achieve tinting and solar radiation absorption properties. It is also referred to as body tinted glass {see accepted standard [C-8(4)]}.

2.94 Transom — A horizontal cross bar connecting two vertical members in a glazing frame.

2.95 Unframed Glazing — A glazing with one or more edges unframed.

2.96 UV Transmittance — The percentage of energy in the ultraviolet (UV) spectrum from 300 nm to 380 nm that is directly transmitted through the glass.

2.97 Vertical Fenestration — All fenestration other than skylights. Trombe wall assemblies, where glazing is installed within 300 mm of a mass wall are considered walls and not fenestration.

2.98 Wired Glass — It is a type of glass into which a wire mesh is embedded during production. Wired glass has an impact resistance similar to that of normal glass, but in case of breakage, the mesh retains the pieces of glass.

3 APPLICATION

The type of glass chosen shall satisfy the following criteria:

- a) For a given application, the type and thickness of glass selected shall be in accordance with the requirements provided herein and the thickness calculation as subjected to wind load shall also satisfy requirements for human impact safety;
- b) Installation of glass to the building is done in accordance with the various provisions available in this Section and in the absence of any specific provisions the same shall be as per manufacturer's instructions;
- c) Optical, solar, thermal performance of glazing (clear/tinted/coated reflective glass) shall have the characteristic performances, such as:
 - 1) spectral transmittance, external and internal spectral reflectance;
 - 2) light transmittance, external and internal light reflectance;
 - 3) solar direct transmittance and reflectance;
 - 4) UV transmittance; and
 - 5) better thermal transmittance (U -value).

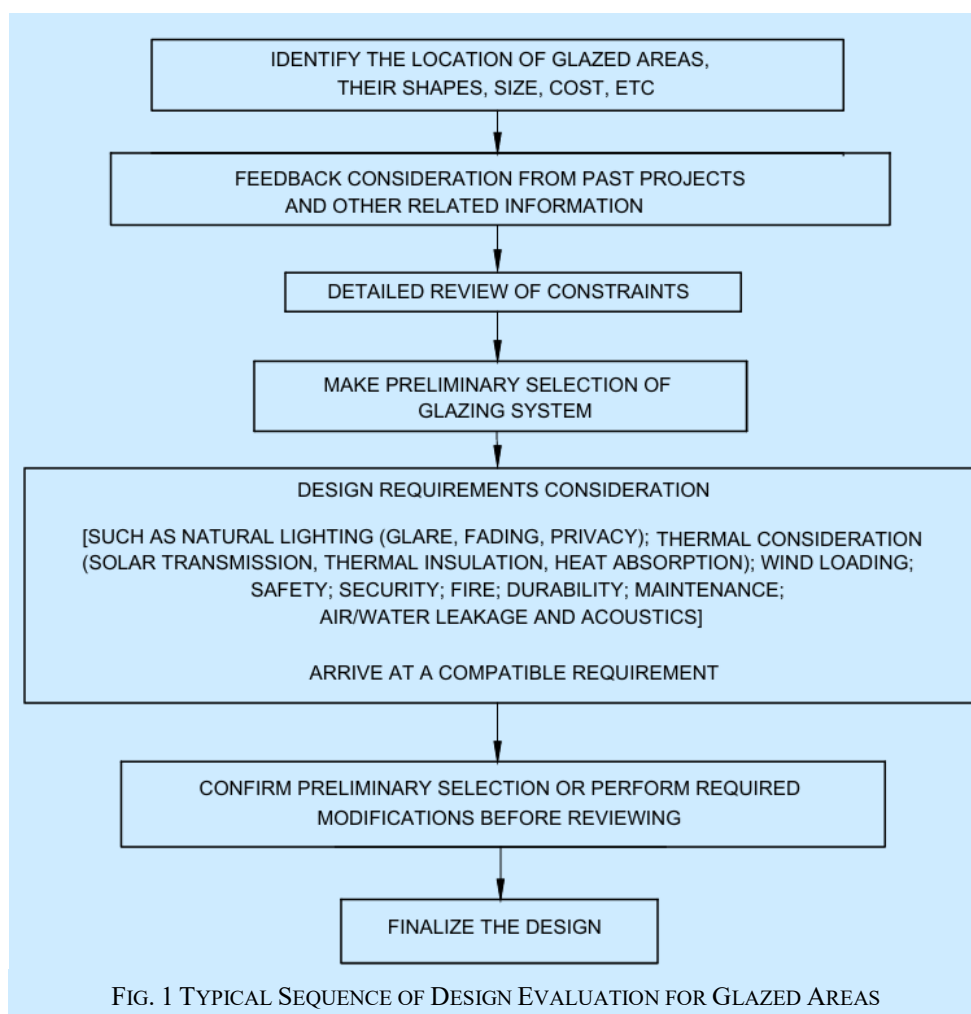
4 GENERAL METHODOLOGY FOR SELECTION

4.1 Selection

4.1.1 Right from the initial project evaluation stage, a suitable methodology that incorporates the implications of use of glass materials and their influence on performance of the building shall have to be looked into. A typical flowchart outlining such a methodology is provided (*see* [Fig. 1](#)).

It is assumed that,

- a) the location of area to be glazed and its preliminary size/shape have been decided as part of normal building design process,
- b) the designer has gained information on glass/and plastic glazing sheet materials used in similar circumstances, and
- c) the designer is cognizant of implications of any innovation in design.



4.1.2 Design requirement including aesthetic considerations and client specified requirements such as security and maintenance are to be considered.

4.1.3 Effect of design on cost such as initial cost of glazing (specification of glass/plastic sheeting material; glazing method; access for initial glazing; work schedule; protection during construction); effect of glass/plastic sheeting material on capital/running cost related to building heating/cooling, lighting and ventilation; and maintenance cost of glazed areas (access for cleaning and reglazing) should be considered.

4.1.4 Selection of appropriate glass may be done considering the cost benefit analysis in respect of the benefit accrued against the alternatives available. Also, the comparative cost of glazing materials should be considered together with their life expectancy and probability of need for maintenance. In the context of selection of glass, the provisions given in the relevant guidelines/best practices (Handbook on ‘Sustainability in Buildings’) may also be considered in addition to the cost benefit analysis on the use of glass *vis-a-vis* use of other alternatives.

4.2 Types of Glass

4.2.1 The soda lime silicate glass is classified based on the type of internal stress which determines the strength of the glazing towards breakage, design and installation procedure and based on application. [Table 1](#) gives the typical properties of various types of glass.

4.2.2 Normal (Annealed) Glass

Normal (Annealed) glass shall be as per accepted standard [C-8(5)] for flat transparent sheet glass or as per accepted standard [C-8(6)] for transparent float glass.

4.2.3 Laminated Safety Glass

Two or more pieces of glass held together by an interleaving layer or layers of plastic sheet material. The laminated safety glass will crack and break under sufficient impact, but the pieces of glass tend to adhere to the plastic and do not fly. For requirements of laminated safety glass, accepted standard [C-8(7)] shall be referred.

Table 1 Typical Properties of Various Types of Glass

(Clause 4.2.1)

Sl No.	Properties	Annealed Glass	Float Glass	Annealed Laminated Glass	Tempered or Toughened Glass
(1)	(2)	(3)	(4)	(5)	(6)
i)	Modulus of rupture	40 N/mm ²	40 N/mm ²	40 N/mm ²	120 N/mm ² to 200 N/mm ²
ii)	Thickness	As per accepted standard [C-8(5)]	As per accepted standard [C-8(6)]	6.38 mm to 19 mm	3 mm to 19 mm
iii)	Density (approximate)	← 2 400 kg/m ³ to 2 550 kg/m ³ →			
iv)	Modulus of elasticity	← 70 GPa →			
v)	Coefficient of linear expansion	← 9 × 10 ⁻⁶ m/m K →			
vi)	Compressive strength	← 1 000 N/mm ² →			
NOTE — Conventionally, the following types of glass exist:					
a) Annealed (clear) glass;					
b) Extra clear glass;					
c) Body tinted glass;					
d) Coated glass (including Lacquered glass); and					
		1) Online coated glass; and			
		2) Offline coated glass.			
e) Processed glass		Each of the glasses (a) to (e) above can be subjected to further processing to produce the following types:			
		1) Heat strengthened;			
		2) Toughened or tempered;			
		3) Double glazed unit;			
		4) Laminated glass (single laminated);			
		5) Laminated double glass unit;			
		6) Ceramic/screen printed fritted glass (patterned glass); and			
		7) Fire safety glass.			

4.2.4 Toughened (Tempered) Safety Glass

A single piece of specially heat treated glass, with a stress pattern such that the piece when fractured reduces to numerous granular fragments, with no large jagged edges. For requirements of toughened safety glass, accepted standard [C-8(7)] shall be referred.

4.2.5 Heat Strengthened Glass

A single piece of glass within which a permanent surface compressive stress, additionally to the basic mechanical strength, has been induced by a controlled heating and cooling process in order to give it increased resistance to mechanical and thermal stress and prescribed fracture characteristics {see 6.1.5 of accepted standard [C-8(3)]}. Heat strengthened glass shall have a surface compression between 31 MPa and 52 MPa (4 500 and 7 500 psi) when tested as prescribed in 6.5 of accepted standard [C-8(3)]. For requirements of heat strengthened glass, accepted standard [C-8(8)] shall be referred.

NOTE — Heat strengthened glass is not a safety glass.

4.2.6 Coated Glass

A metallic coating is applied to one side of the glass in order to significantly increase the amount of reflection by the glass of both the visible and infrared (light and heat) ranges of the electromagnetic spectrum. This metallic coating can be applied to clear or body tinted glass, without adversely affecting the transparency of the glass. The reflective glass imparts an enhanced appearance to the exterior of buildings due to the coating of metal oxides on the glass. It is widely applied as an aesthetic and energy efficient product in buildings for its highly reflective surface and wide palette of colours. It reduces heat gain and glare from the exterior and allows optimum visible light transmission to the interior. If designed/used properly, reflective glass can aid in reducing the air conditioning load of buildings.

NOTE — Highly reflective glass should be avoided, especially in internal environments. The reflections from these surfaces can be particularly confusing for persons with vision impairment.

4.2.7 Insulating Glazing Unit

Insulating glazing units are two or more panels of glass spaced apart and hermetically sealed to form a single unit with air or inert gas filled space between the panels. The spacer is made of aluminium that is either mill-finished or painted/anodized in different colours or finishes, or an equivalent alternative material. Desiccants are filled inside the spacer framework to prevent condensation inside the unit. The primary sealant acts as a barrier to moisture and vapour penetration. The secondary sealant provides structural integrity to the unit. For requirements of insulating glazing unit, accepted standard [C-8(9)] shall be referred.

NOTE — Moisture penetration test and argon/krypton gas level test are typically carried out to determine the characteristic of IGUs.

4.2.8 Rolled Glass — Patterned, Extra Clear Patterned, Wired and Wired-Patterned Glass

The general requirements including thickness, dimensional tolerance and tests for wired glass as specified in accepted standard [C-8(10)] are valid.

4.2.9 Glass Fins

When glass fins are used as a decorative member, it can be either toughened (tempered) or laminated glass.

When glass fins are used as a structural member, it shall be toughened and laminated. The depth and thickness may vary as per elevation and wind load requirements. Design should also accommodate movements and verified for lateral torsional buckling. It is recommended to use finite element analysis to determine the appropriate glass fin sizes.

4.3 Application

Glass can be designed/suited to use in building occupancies such as in residential, assembly and mercantile in elements such as facades, windows, doors, partitions, display units, shop fronts, greenhouses (where plants are grown) and atrium. Different types of glass are selected based on requirements. For safety against accidental breakage and strength requirement (mechanical and thermal), toughened glass or laminated glass may be considered over annealed glass. For increased high security requirements, acoustic insulation, sloped glazing and skylights, laminated glass may be considered.

4.4 Associated Glazing Materials

4.4.1 Structural Sealant

The structural sealant could be of silicone type with requirements meeting the design criteria. Structural sealant shall be shelf-stable, natural-cured, elastomeric adhesive exhibiting the desired adhesion to building substrates. The structural sealant may require cleaning and/or priming to get good adhesion for structural glazing application. For laminated glass applications where glass edges are sealed with a sealant, compatibility shall be ensured with the interlayer to avoid delamination.

Structural sealant shall be as per accepted standard [C-8(11)].

4.4.2 Gaskets

Gasket is a mechanical seal that fills the space between two mating surfaces, to prevent leakage from or into the joined objects and will be able to deform while under compression and tightly fills the space it is designed for, including any slight irregularities. Gaskets shall be as per 5.10 of accepted standard [C-8(12)].

4.4.3 Preformed Tape

Typical requirements of preformed butyl and foam tapes shall be as given in [Table 2](#).

4.4.4 Setting Blocks, Location Blocks and Distance Pieces

Setting blocks, location blocks, and distance pieces shall be designed in accordance with [6](#) which are:

- a) resilient, load-bearing, non-absorbent and rot-proof materials; and
- b) compatible with all other materials that may come in contact with them.

4.5 General Requirements

4.5.1 Glazing and associated materials used for the proper installation shall satisfy the minimum dimensional and framing requirements in accordance with the procedures given in this Section. The dimensional and thickness tolerance of the glass used in exterior and interior is determined from the provisions given hereunder.

4.5.2 Dimensional Requirement

4.5.2.1 The dimensions for edge clearance, edge cover, and front and back clearance, as defined in [Fig. 2](#), shall be not less than the values given in [Table 3](#).

NOTE — For glazing systems where the framing member is aluminium or steel and gaskets are used to capture the glazing, minimum glazing bite of 12.7 mm is recommended for architectural glazing.

4.5.2.2 Front putty width

The front putty width shall be not less than 10 mm for panels up to 1 m², and not less than 12 mm for panels between 1 m² and 2 m².

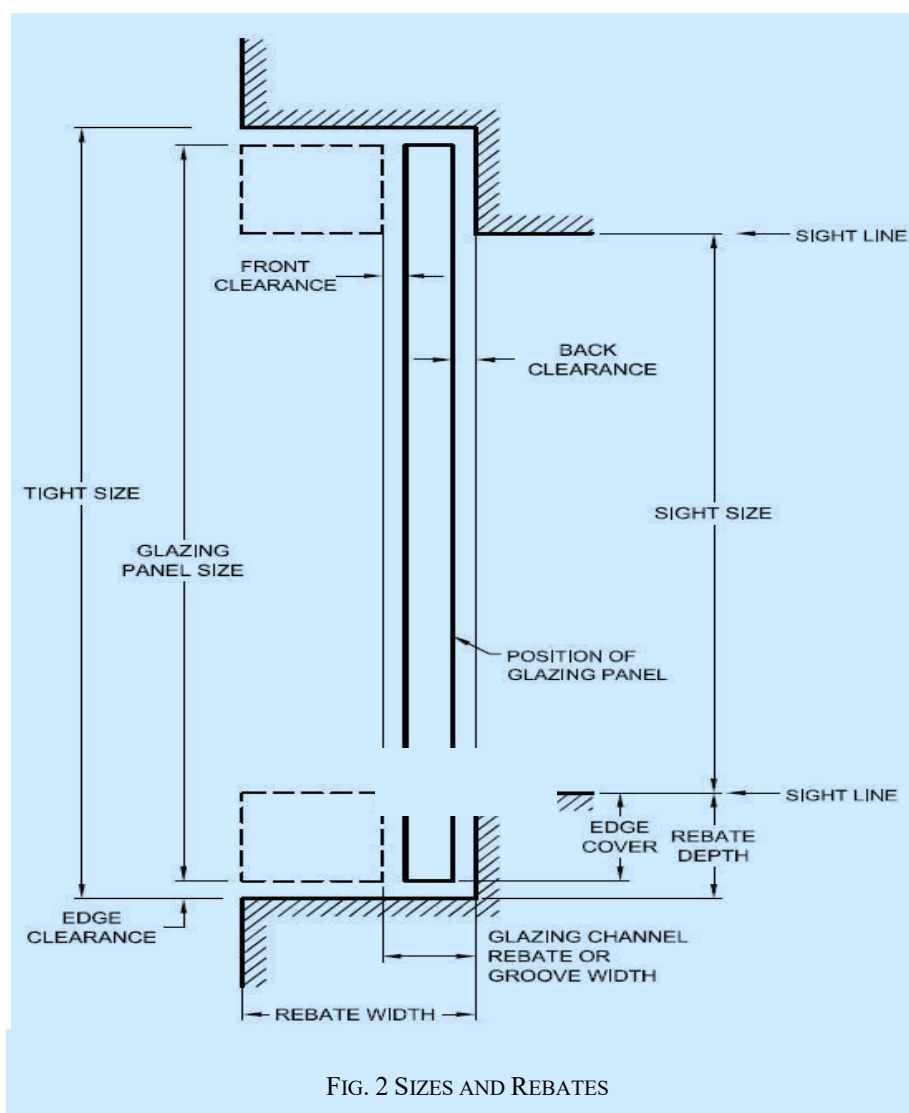
Table 2 Typical Properties of Preformed Tapes

(Clause [4.4.3](#))

Sl No.	Properties	Requirement
(1)	(2)	(3)
i)	Low temperature flex	No cracks at – 23 °C
ii)	Weight loss	2 percent, <i>Max</i>
iii)	Vehicle migration	1 paper stained maximum and this stain shall be no more than 3.2 mm from edge of sample maximum

Table 2 (Concluded)

Table 2 Typical Properties of Preformed Tapes		
(Clause 4.4.3)		
Sl No.	Properties	Requirement
(1)	(2)	(3)
iv)	Backing removal	No transfer of tape compound to the paper
v)	Yield strength	41.4 kPa, <i>Min</i>
vi)	Compression/Recovery	Compression index, 1.22 N/mm ² , <i>Max</i>
vii)	Water absorption	Weight gain maximum after boiling, 40 kg/m ³
viii)	Flow test	Loss of height, 60 percent, <i>Max</i>



4.5.2.3 Dimensions of rebates and grooves (width and depth)

Dimensions of rebates and grooves shall accommodate the requirements of [Table 3](#) and allow for the appropriate setting and location blocks ([see 6](#)).

4.5.2.4 Glass dimensional tolerance

The tolerance for dimensions shall be as per those in the respective Indian Standards.

4.6 Frame Requirements

4.6.1 When completely assembled and glazed, the secondary design action effects due to deflection of the frame member supporting the edge of the glass shall be allowed for.

4.6.2 Deflection Limits

The frame deflection limits under serviceability loading shall satisfy the conditions for four sided frames as given in [6.3.6.2\(c\)](#).

NOTE — In framing members, where reinforcement/fixing brackets are required to meet the deflection/stress criteria, it may be of aluminium or steel. When steel is used, proper dielectric galvanic separation shall be used. The material chosen for reinforcement shall be in close proximity in the galvanic series, to ensure dielectric separation.

4.6.3 Panels Glazed into the Building Structure

A panel glazed directly into a building structure by means of appropriate beads or stops shall be considered to be framed, provided the assembly complies with the deflection requirements of [4.6.2](#).

4.6.4 Mixed Framing

Glass supported along the top and bottom edges by one means and along the vertical edges by another means shall be considered to be framed provided each frame member of the assembly complies with the deflection requirements of [4.6.2](#).

4.7 Glass Dimensions

4.7.1 The actual thickness for each pane of glass shall be measured using screw calliper as per [4.3](#) of accepted standard [C-8(6)].

The actual thickness of glass, rounded to the nearest 0.1 mm shall not vary from the nominal thickness by more than the tolerances specified in respective Indian Standard or the values provided in this Section (where Indian Standard specifications are not available).

The minimum thickness allowed in insulated glazing unit (IGU) should be the sum of minimum thickness allowed of two panes as per accepted standard [C-8(6)] plus thickness of air-gap.

As per the accepted standard [C-8(7)], the nominal thickness of laminated safety glass shall be the sum of the nominal thickness of constituent panes of glass and interlayer material. The limit deviation of the interlayer shall not be taken into account if the total interlayer thickness is < 2 mm.

If the total interlayer thickness is > 2 mm, a limit deviation of ± 0.2 mm shall apply. For plastic glazing sheet material, the limit of deviation on thickness shall be the same as applicable to a float glass of the same nominal thickness.

The limit deviations on thickness of laminated glass shall not exceed the sum of the limit deviations of the constituent glass panes as specified in accepted standard [C-8(6)].

Table 3 Minimum Dimensions for Glazing Material

(Clauses 4.5.2.1 and 4.5.2.3)

All dimensions in millimetres.

Sl No.	Nominal Thickness	Front and Back Clearance			Edge Clearance	Edge Cover	Rebate Depth
		Type A (See Note 1)	Type B (See Note 2)	Type C (See Note 3)			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	3 (for pane area up to 0.1 m ²)	2	—	—	2	4	6
ii)	3 (for pane area > 0.1 m ²)	2	—	—	3	6	9
iii)	3	—	2	1	3	6	9
iv)	4	2	—	—	2	6	8
v)	4	—	2	1	3	6	9
vi)	5	2	2	2	4	6	10
vii)	6	2	2	2	4	6	10
viii)	8	—	3	2	5	8	13
ix)	10	—	3	2	5	8	13
x)	12	—	3	2	6	9	15
xi)	15	—	5	4	8	10	18
xii)	19	—	5	4	10	12	22
xiii)	25	—	5	4	10	15	15

NOTES

1 Type A applies to linseed oil and metal casement putties.

2 Type B applies to non-setting glazing compounds, plastic glazing compounds, two-part rubberizing compounds, sealants and preformed strip materials.

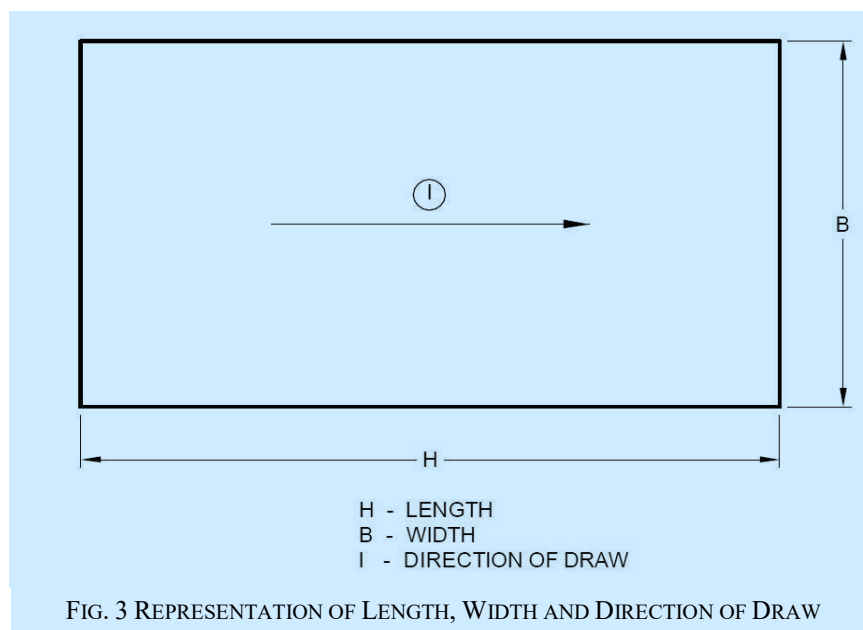
3 Type C applies to gaskets made from extruded materials such as butyl strip, PVC, neoprene and sanoprene held in position by pressure upon the glass.

4 For non-standard glass thicknesses the nearest values of nominal thickness, shall be used.

5 Timber and PVC frames may not require the specified front and back clearances provided the waterproofing performance requirements are met.

6 The dimensions are the minimum necessary for the structural integrity of glass only but do not apply to insulating glass units.

4.7.2 Length and width are defined with reference to the direction of draw of the flat glass ribbon (see Fig. 3).



4.7.3 Jumbo Sizes

Glass is available commonly in the following sizes, nominal length (H) 4 500 mm to 6 000 mm and nominal width, (B) 3 660 mm or 3 210 mm.

NOTE — The usual width is 3 660 mm or 3 210 mm. Other sizes may be mutually agreed to between the manufacturer and the user based on their requirement.

4.7.4 Stock Size

Glass is available commonly in the following sizes, nominal length (H) 2 250 mm or 2 440 mm and nominal width, (B) 3 660 mm or 3 210 mm.

NOTE — Maximum glass size available from manufacturer are one consideration, the processing equipment limitation, capabilities of contract fabricator to install unit, availability of specialized transport and handling equipment to deliver unit should be checked in addition.

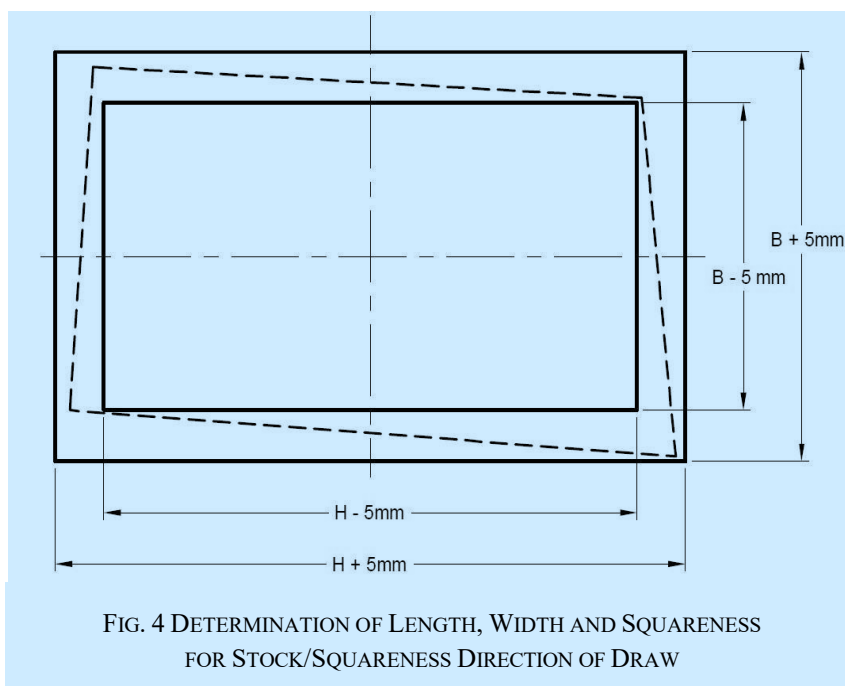
4.7.5 Split Sizes

Glass is delivered in equal or unequal split sizes from the jumbo or stock size.

4.7.6 Squareness

For given the nominal dimensions for length (H) and width (B), the pane shall not be larger than a prescribed rectangle resulting from the nominal dimensions increased by the permissible plus tolerance or smaller than the prescribed rectangle reduced by the permissible minus tolerance.

The sides of the prescribed rectangles (*see Fig. 4*) shall be parallel to one another and these rectangles shall have a common centre. For stock/standard sizes the tolerances on nominal dimensions length (H) and width (B), are ± 5 mm.



4.7.7 As-Cut Finished Size

The nominal dimension for width and length being given, the finished pane or as-cut finished size shall not be greater than the prescribed rectangle from the nominal dimension increased by the tolerance ' t ' or smaller than the prescribed rectangle reduced by the tolerance ' t '. The sides of the prescribed rectangles are parallel to one another and these shall have a common centre in accordance with Fig. 5. The limits of the squareness are also mentioned. Tolerances shall be as given in respective accepted standards. An indicative list of such accepted standards is provided in Table 4.

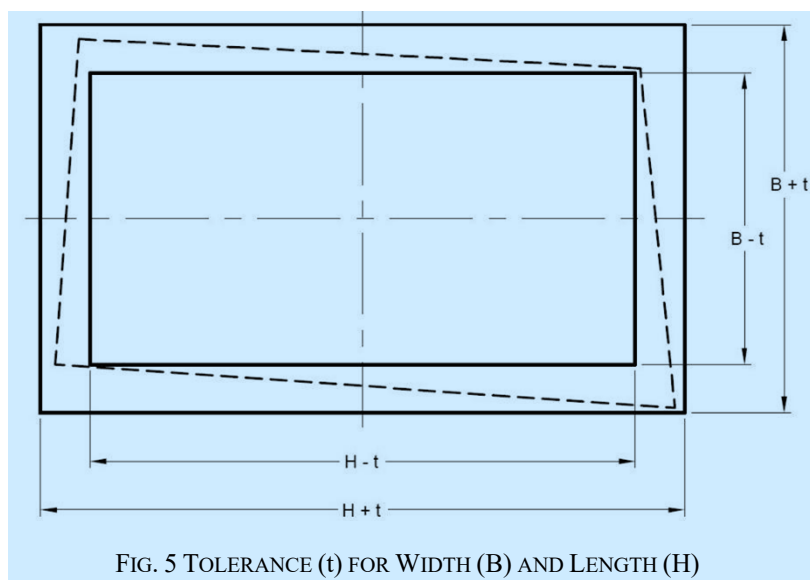


Table 4 Indicative List of Accepted Standards for Different Types of Glass

(Clause [4.7.7](#))

SI No.	Type of Glass	Accepted Standard
(1)	(2)	(3)
i)	Flat transparent sheet glass	[C-8(5)]
ii)	Transparent float glass	[C-8(6)]
iii)	Rolled glass — Patterned, Extra Clear patterned, Wired and Wired patterned glass	[C-8(10)]
iv)	Heat strengthened glass	[C-8(8)]
v)	Insulating glazing unit	[C-8(9)]
vi)	Safety glass for architectural, building and general uses	[C-8(7)]
vii)	Tinted float glass	[C-8(4)]

5 ENERGY AND LIGHT

5.1 General

Energy and light performance of glazing system is defined in terms of U-value, solar heat gain coefficient, visible light transmission, internal and external reflection. The overall performance is determined by the efficiency of glass, frame, shading systems and the air tightness of the system. The performance requirement for glazing system shall be suitably selected. Minimum performance requirement is mentioned in the [Annex B](#).

Specific methods of calculating light and energy factors for glass in buildings are covered here and their characteristic data can serve as a basis for light, heating and ventilation calculations of rooms and can permit comparison between different types of glazing. It is applicable to all flat glass material.

5.2 Thermal Energy Transmission

5.2.1 General

Glazed areas in buildings should be designed considering overall energy balance, the effects on thermal comfort of occupants and the cooling and heating energy requirements. Solar energy gain is due to transmission of solar radiant energy through the glass into the building. Solar energy is all at relatively short wavelengths and is controlled by using the glass to absorb or reflect the energy. Energy loss is due to the transfer of energy by conduction, convection and long wavelength radiation. The solar energy gain and loss can be regulated by several factors, for example, incorporating air cavities, including low emissivity glass and use of shading systems. Solar energy gain through glazing systems can be advantageous in cold climates for improving thermal comfort and reducing the heating energy requirements.

5.2.2 Thermal Comfort

Energy transmission through glazing can significantly influence occupant comfort by raising or lowering room air temperatures. For example, hot sunny weather can lead to excessive energy gains and energy losses can occur through the glazing during cold weather or at night. Thermal comfort can also be influenced by direct solar radiation through the glazing and/or by radiation exchange between the glazing and the occupants. Appropriate choice of glazing and opaque wall systems is essential to avoid radiant asymmetry related thermal discomfort.

5.2.3 Solar Energy Gain

The solar energy transmitted through the glass from the outside environment is the part of solar radiation which is transmitted directly through the glass and the part which is re-emitted from the glass after being absorbed. In warm and hot climates, this heat energy which is transmitted through the glass has to be limited in-order to achieve

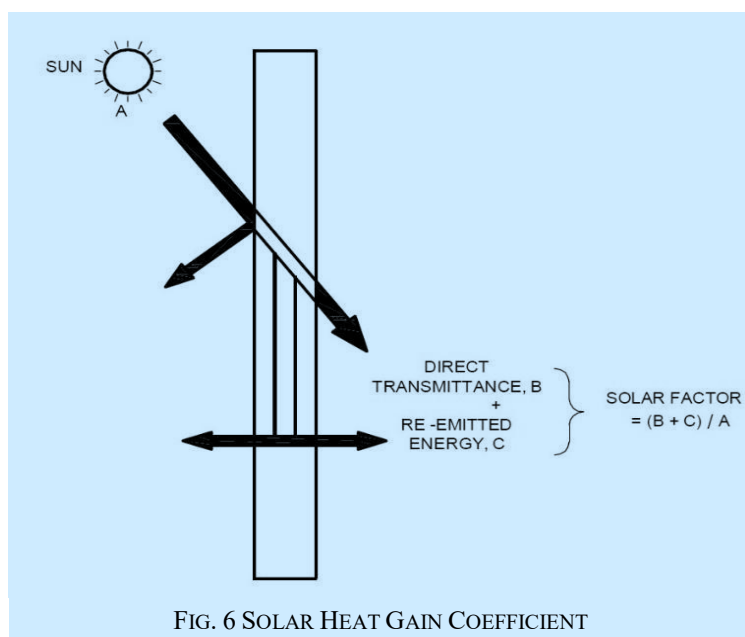
thermal comfort for the occupants and also to reduce the energy demands of the building. Factors that can influence the amount of solar gain through the glazing include:

- a) orientation;
- b) glazed area;
- c) shading devices, both internal and external; and
- d) properties of the glazing.

5.2.4 Total Solar Energy Transmittance

Total solar energy transmittance is the proportion of solar radiation at normal incidence transferred through the glazing. It is composed of the direct transmittance (short wave component) and the part of the solar absorptance dissipated inwards by long wave radiation and convection (long wave component). The total solar energy transmission properties of solar control glazing can be described by their shading coefficients and solar heat gain coefficient (SHGC)/solar factor (SF). The shading coefficient is derived by comparing the properties of the solar control glazing with a clear float glass having a total solar energy transmittance of 0.87 (that is clear glass between 3 mm and 4 mm thick).

Solar heat gain coefficient (SHGC)/solar factor (SF) is the measure of amount of heat transfer from outside to inside by direct transmission (short wave radiation) and internal remitted radiation (long wave radiation). The SHGC (see Fig. 6) or SF depends on the tint of the substrate and the type of coating that is done on the base glass. The amount of solar energy gained through glass is between 0 and 1. Multiplying the SHGC or SF by 100 gives the percentage of solar energy allowed into the building.



The total solar energy transmittance g is the sum of the solar direct transmittance τ_e and the secondary heat transfer factor q_i towards the inside, the latter resulting from heat transfer by convection and long-wave IR-radiation of that part of the incident solar radiation which has been absorbed by the glazing and is expressed as:

$$g = \tau_e + q_i$$

5.2.5 Division of Incident Solar Radiation Flux

The incident solar radiant flux per unit area ϕ_e is divided into the following three parts (see Fig. 7):

- a) transmitted part, $\tau_e \phi_e$;
- b) reflected part, $\rho_e \phi_e$; and
- c) absorbed part, $\alpha_e \phi_e$.

where

τ_e = solar direct transmittance;
 ρ_e = solar direct reflectance; and
 α_e = solar direct absorptances.

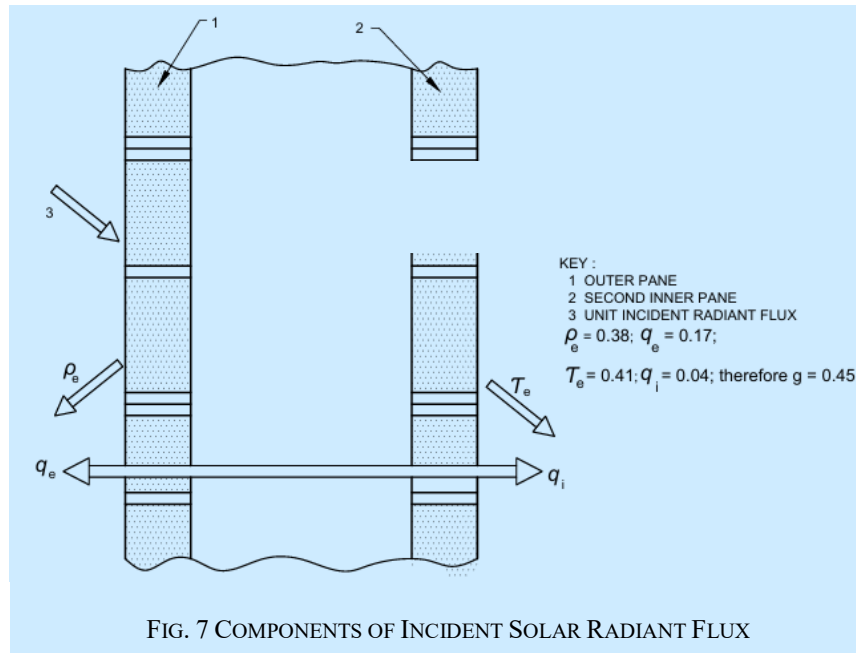


FIG. 7 COMPONENTS OF INCIDENT SOLAR RADIANT FLUX

The relationship between the three characteristics is:

$$\tau_e + \rho_e + \alpha_e = 1$$

The absorbed part $\alpha_e \phi_e$ is subsequently divided into two parts $q_i \phi_e$ and $q_e \phi_e$, which are energy transferred to the inside and outside, respectively:

$$\alpha_e = q_i + q_e$$

where

q_i = secondary heat transfer factor of the glazing towards the inside; and

q_e = secondary heat transfer factor of the glazing towards the outside.

5.2.6 Solar Direct Transmittance

The solar direct transmittance τ_e of glazing shall be calculated using the following formula:

$$\tau_e = \frac{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} \tau(\lambda) S_{\lambda} \Delta\lambda}{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} S_{\lambda} \Delta\lambda}$$

where

S_{λ} = relative spectral distribution of the solar radiation;

$\tau(\lambda)$ = spectral transmittance of the glazing; and

$\Delta\lambda$ = integration procedure are the same as in [5.4.5.4](#) except that the data points shall be chosen at the wavelengths given in [Table 5](#).

5.2.7 Solar Direct Reflectance

The solar direct reflectance ρ_e of the glazing shall be calculated using the following formula:

$$\rho_e = \frac{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} \rho_o(\lambda) S_{\lambda} \Delta\lambda}{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} S_{\lambda} \Delta\lambda}$$

where

- S_{λ} = relative spectral distribution of the solar radiation (see [5.2.6](#));
- $\rho_o(\lambda)$ = spectral external reflectance of the glazing; and
- $\Delta\lambda$ = integration procedure are the same as in light transmittance (see [5.4.5.4](#)) except that the data points shall be chosen at the wavelengths given in [Table 5](#).

In the case of multiple glazing, the spectral external reflectance $\rho_o(\lambda)$ is calculated in accordance with [5.5.1](#).

Table 5 Normalized Relative Spectral Distribution of Global Solar Radiation
(Clauses [5.2.6](#), [5.2.7](#), [5.2.9.3](#) and [5.2.9.4](#))

SI No.	λ nm	$S_{\lambda}\Delta\lambda$
(1)	(2)	(3)
i)	300	0
ii)	305	0.000 057
iii)	310	0.000 236
iv)	315	0.000 554
v)	320	0.000 916
vi)	325	0.001 309
vii)	330	0.001 914
viii)	335	0.002 018
ix)	340	0.002 189
x)	345	0.000 260
xi)	350	0.002 445
xii)	355	0.002 555
xiii)	360	0.002 683
xiv)	365	0.003 020
xv)	370	0.003 359
xvi)	375	0.003 509
xvii)	380	0.003 600
xviii)	385	0.003 529
xix)	390	0.003 551
xx)	395	0.004 294
xxi)	400	0.007 812
xxii)	410	0.011 638
xxiii)	420	0.011 877
xxiv)	430	0.011 347
xxv)	440	0.013 245
xxvi)	450	0.015 343
xxvii)	460	0.016 166
xxviii)	470	0.016 178
xxix)	480	0.016 402
xxx)	490	0.015 794

Table 5 (Concluded)

Sl No.	λ nm	$S_{\lambda}\Delta\lambda$
(1)	(2)	(3)
xxxix)	500	0.015 801
xxxix)	510	0.015 973
xxxix)	520	0.015 357
xxxix)	530	0.015 857
xxxix)	540	0.015 827
xxxix)	550	0.015 844
xxxix)	560	0.015 590
xxxix)	570	0.015 255
xxxix)	580	0.014 745
xl)	590	0.014 330
xli)	600	0.014 663
xlii)	610	0.015 030
xliii)	620	0.014 859
xliv)	630	0.014 622
xliv)	640	0.014 526
xlvi)	650	0.014 445
xlvi)	660	0.014 313
xlvi)	670	0.014 023
xlvi)	680	0.012 838
li)	690	0.011 788
li)	700	0.012 453
lii)	710	0.012 798
liii)	720	0.010 589
liv)	730	0.011 233
lv)	740	0.012 175
lvi)	750	0.012 181
lvii)	760	0.009 515
lviii)	770	0.010 479
lix)	780	0.011 381
lx)	790	0.011 262
lxi)	800	0.026 718
lxii)	850	0.048 240
lxiii)	900	0.040 297
lxiv)	950	0.021 384
lxv)	1 000	0.036 097

Table 5 (Concluded)

SI No. (1)	λ nm (2)	$S_{\lambda}\Delta\lambda$ (3)
lxvi)	1 050	0.034 110
lxvii)	1 100	0.018 861
lxviii)	1 150	0.013 228
lxix)	1 200	0.022 551
lxx)	1 250	0.023 376
lxxi)	1 300	0.017 756
lxxii)	1 350	0.003 743
lxxiii)	1 400	0.000 741
lxxiv)	1 450	0.003 792
lxxv)	1 500	0.009 693
lxxvi)	1 550	0.013 693
lxxvii)	1 600	0.012 203
lxxviii)	1 650	0.010 615
lxxix)	1 700	0.007 255
lxxx)	1 750	0.007 183
lxxxi)	1 800	0.002 157
lxxxii)	1 850	0.000 395
lxxxiii)	1 900	0.000 082
lxxxiv)	1 950	0.001 087
lxxxv)	2 000	0.003 024
lxxxvi)	2 050	0.003 988
lxxxvii)	2 100	0.004 229
lxxxviii)	2 150	0.004 142
lxxxix)	2 200	0.003 690
xc)	2 250	0.003 592
xc i)	2 300	0.003 436
xc ii)	2 350	0.003 163
xc iii)	2 400	0.002 233
xc iv)	2 450	0.001 202
xc v)	2 500	0.000 475

NOTE — The values in this table are calculated according to the trapezoidal rule.

5.2.8 Solar Direct Absorptance

The solar direct absorptance α_e shall be calculated from the following formula:

$$\tau_e + \rho_e + \alpha_e = 1$$

5.2.9 Secondary Heat Transfer Factor Towards the Inside

5.2.9.1 Boundary conditions

For the calculation of the secondary heat transfer factor towards the inside, q_i , the heat transfer coefficients of the glazing towards the outside, h_e , and towards the inside, h_i , are needed. These values mainly depend on the position of the glazing, wind velocity, inside and outside temperatures and, furthermore, on the temperature of the two external glazing surfaces.

As the purpose of this Section is to provide basic information on the performance of glazing, the following conventional conditions have been stated for simplicity:

- Position of the glazing: vertical;
- Outside surface: wind velocity approximately 4 m/s; corrected emissivity 0.837;
- Inside surface: natural convection; emissivity optional; and
- Air spaces are unventilated.

Under these conventional, average conditions, standard values for h_e and h_i are obtained:

$$h_e = 23 \text{ W/(m}^2\text{K)}$$

$$h_i = \left(3.6 + \frac{4.4 \varepsilon_i}{0.837} \right) \text{ W/(m}^2\text{.K)}$$

where

ε_i = corrected emissivity of the inside surface [for soda lime glass, $\varepsilon_i = 0.837$ and $h_i = 8 \text{ W/(m}^2\text{K)}$].

If other boundary conditions are used to meet special requirements they shall be stated in the test report. Values for ε_i lower than 0.837 (due to surface coatings with higher reflectance in the far infrared) should only be taken into account if condensation on the coated surface can be excluded.

5.2.9.2 Single glazing

The secondary heat transfer factor towards the inside, q_i , of single glazing shall be calculated using the following formula:

$$q_i = \alpha_e \frac{h_i}{h_e + h_i}$$

where

α_e = solar direct absorptance obtained from the in accordance with [5.2.5](#), and

h_e, h_i = heat transfer coefficients towards the outside and inside, respectively, in accordance with [5.2.9.1](#).

5.2.9.3 Double glazing

The secondary heat transfer factor towards the inside, q_i , of double glazing shall be calculated using the following formula:

$$q_i = \frac{\left(\frac{\alpha_{e1} + \alpha_{e2}}{h_e} + \frac{\alpha_{e2}}{\Lambda} \right)}{\left(\frac{1}{h_i} + \frac{1}{h_e} + \frac{1}{\Lambda} \right)}$$

where

α_{e1} = solar direct absorptance of the outer (first) pane within the double glazing;

- α_{e2} = solar direct absorptance of the second pane within the double glazing;
 A = thermal conductance between the outer surface and the innermost surface of the double glazing (see Fig. 8), in watts per square metre per Kelvin ($\text{W/m}^2\cdot\text{K}$); and
 h_e, h_i = heat transfer coefficients towards the outside and the inside, respectively in accordance with 5.2.9.1.

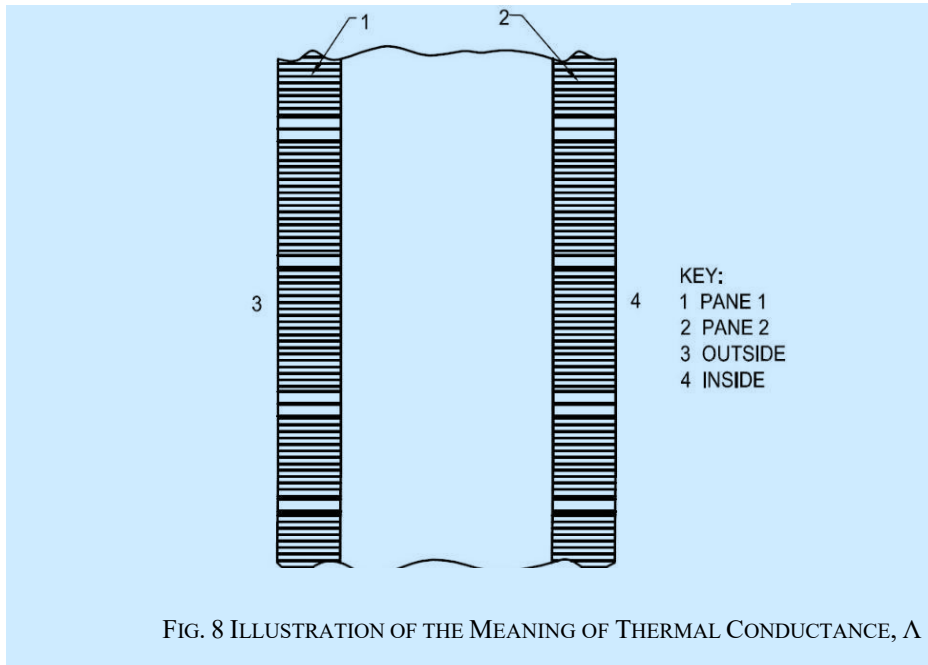


FIG. 8 ILLUSTRATION OF THE MEANING OF THERMAL CONDUCTANCE, A

Characteristics α_{e1} and α_{e2} are calculated as follows:

$$\alpha_{e1} = \frac{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} \left\{ \alpha_1(\lambda) + \frac{\alpha'_1(\lambda)\tau_1(\lambda)\rho_2(\lambda)}{1 - \rho'_1(\lambda)\rho_2(\lambda)} \right\} S_\lambda \Delta\lambda}{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} S_\lambda \Delta\lambda}$$

$$\alpha_{e2} = \frac{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} \left\{ \frac{\alpha_2(\lambda)\tau_1(\lambda)}{1 - \rho'_1(\lambda)\rho_2(\lambda)} \right\} S_\lambda \Delta\lambda}{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} S_\lambda \Delta\lambda}$$

where

$\tau_1(\lambda), \tau_2(\lambda), \rho_1(\lambda), \rho'_1(\lambda), \rho_2(\lambda)$ are as defined in light transmittance (see 5.4.5.4)

$\alpha_1(\lambda)$ = spectral direct absorptance of the outer pane, measured in the direction of the incident radiation, given by the relationship:

$$\alpha_1(\lambda) = 1 - \tau_1(\lambda) - \rho_1(\lambda)$$

$\alpha'_1(\lambda)$ = spectral direct absorptance of the outer pane, measured in the opposite direction to the incident radiation, given by the relationship:

$$\alpha'_1(\lambda) = 1 - \tau_1(\lambda) - \rho'_1(\lambda)$$

$\alpha_2(\lambda)$ = spectral direct absorptance of the second pane, measured in the direction of the incident radiation, given by the relationship:

$$\alpha_2(\lambda) = 1 - \tau_2(\lambda) - \rho_2(\lambda)$$

$\Delta\lambda$ and the integration procedure are the same as in 5.4.5.4 except that the data points shall be chosen at the wavelengths given in Table 5.

The thermal conductance A shall be determined for a temperature difference of $\Delta T = 15^\circ\text{C}$ across the sample and a mean temperature of the sample of 10°C , or by measuring methods using the standard guarded hot plate method, or the standard heat flow metre method.

If another temperature difference ΔT across the sample and/or another mean temperature of the sample is used for the determination of the thermal conductance A to meet special requirements, this shall be stated in the test report.

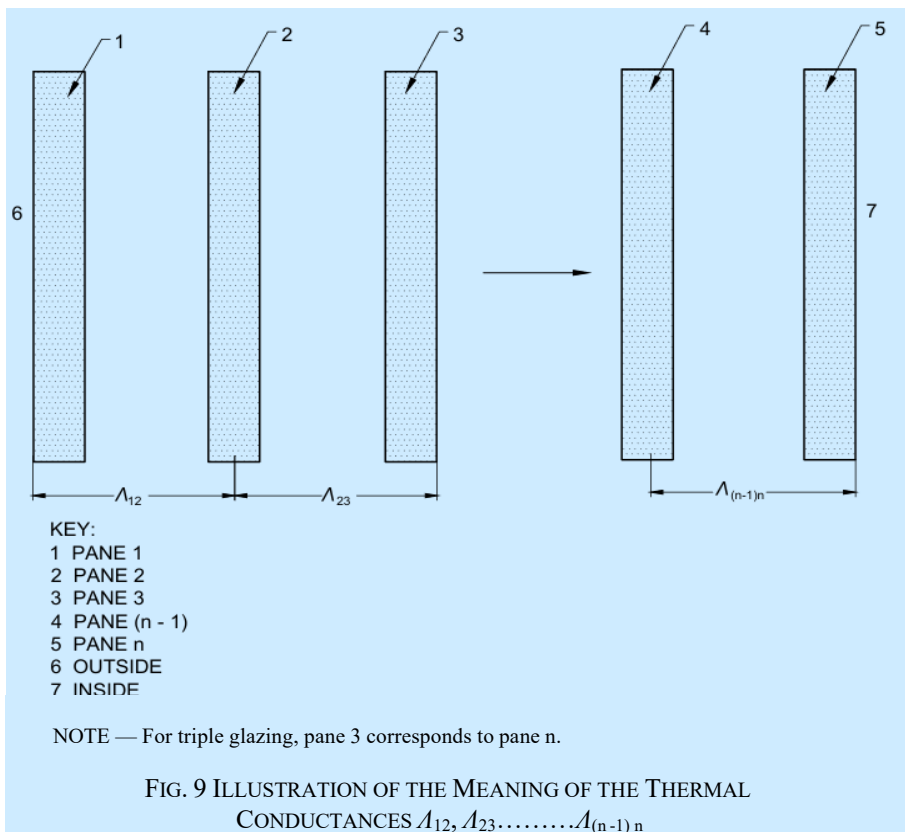
5.2.9.4 Multiple glazing with $n > 2$ components

The secondary heat transfer factor towards the inside, q_i , of a multiple glazing with more than two components shall be calculated using the following formula:

$$q_i = \frac{\frac{\alpha_{e1} + \alpha_{e2} + \alpha_{e3} + \dots + \alpha_{en}}{h_e} + \frac{\alpha_{e2} + \alpha_{e3} + \dots + \alpha_{en}}{A_{12}} + \frac{\alpha_{e3} + \dots + \alpha_{en}}{A_{23}} + \dots + \frac{\alpha_{en}}{h_e}}{\frac{1}{h_i} + \frac{1}{h_e} + \frac{1}{A_{12}} + \frac{1}{A_{23}} + \dots + \frac{1}{A_{(n-1)n}}}$$

where

- α_{e1} = solar direct absorptance of the outer (first) pane within the n -fold glazing;
- α_{e2} = solar direct absorptance of the second pane within the n -fold glazing;
- α_{en} = solar direct absorptance of the n^{th} (inner) pane of the n -fold glazing;
- h_e, h_i = heat transfer coefficients towards the outside and towards the inside, respectively in accordance with [5.2.9.1](#);
- A_{12} = thermal conductance between the outer surface of the outer (first) pane and the centre of the second pane (see [Fig. 9](#));
- $A_{(n-1)n}$ = thermal conductance between the centre of the $(n-1)^{\text{th}}$ pane and the outer surface of the n^{th} (inner) pane (see [Fig. 9](#)); and
- A_{23} = thermal conductance between the centre of the second pane and the centre of the third pane (see [Fig. 9](#));



The thermal conductances $A_{12}, A_{23}, \dots, A_{(n-1)n}$ shall be determined by iteration technique. The calculation of the direct solar absorptances $\alpha_{e1}, \alpha_{e2}, \dots, \alpha_{en}$ shall be performed using the methods given in [5.2.9.3](#).

As an example for the calculation of the direct solar absorptances the following procedure is given which consist of the following $(n-1)$ steps for a glazing consisting of n components:

- First step — calculate the spectral characteristics of a unit consisting of $(n-1)$ components 2,3,..., n according to what has been prescribed in [5.4.5.4](#) and [6.1](#). Then combine this unit with the first (outer) pane as a double glazing;
- Second step — calculate the spectral characteristics of a unit consisting of the $(n-2)$ components 3,..., n and furthermore, those of a double glazing consisting of pane 1 and pane 2. These units are then combined as a double glazing. The sum $\alpha_{e1} + \alpha_{e2}$ is then obtained. This procedure is continued up to the last $(n-1)^{\text{th}}$ step; and
- $(n-1)^{\text{th}}$ step — combine the $(n-1)$ panes 1,2,..., $(n-1)$ and determine the spectral characteristics of this unit. This unit is then combined with the n^{th} (inner) pane as a double glazing. From the second equation in [5.2.9.3](#), the sum $\alpha_{e1}, \alpha_{e2}, \dots, \alpha_{e(n-1)}$ is obtained, that is with the known values $\alpha_{e1}, \alpha_{e2}, \dots, \alpha_{e(n-2)}$ from the previous steps $\alpha_{e(n-1)}$ is determined. α_{en} is obtained according to the expression of α_{e2} in [5.2.9.3](#).

In the case of triple glazing for the solar absorptances α_{e1}, α_{e2} and α_{e3} as a function of the spectral characteristics of the individual components of the unit, the following formulae are obtained:

$$\alpha_{e1} = \frac{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} \left\{ \alpha_1(\lambda) + \frac{\tau_1(\lambda)\alpha'_1(\lambda)\rho_2(\lambda) [1 - \rho'_2(\lambda)\rho_3(\lambda)] + \tau_1(\lambda)\tau_2^2(\lambda)\alpha'_1(\lambda)\rho_3(\lambda)}{[1 - \rho'_1(\lambda)\rho_2(\lambda)] \cdot [1 - \rho'_2(\lambda)\rho_3(\lambda)] - \tau_2^2(\lambda)\rho'_1(\lambda)\rho_3(\lambda)} \right\} S_{\lambda} \Delta\lambda}{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} S_{\lambda} \Delta\lambda}$$

$$\alpha_{e2} = \frac{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} \left\{ \frac{\tau_1(\lambda)\alpha_2(\lambda) [1 - \rho'_2(\lambda)\rho_3(\lambda)] + \tau_1(\lambda)\tau_2(\lambda)\alpha'_2(\lambda)\rho_3(\lambda)}{[1 - \rho'_1(\lambda)\rho_2(\lambda)] \cdot [1 - \rho'_2(\lambda)\rho_3(\lambda)] - \tau_2^2(\lambda)\rho'_1(\lambda)\rho_3(\lambda)} \right\} S_{\lambda} \Delta\lambda}{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} S_{\lambda} \Delta\lambda}$$

$$\alpha_{e3} = \frac{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} \left\{ \frac{\tau_1(\lambda)\tau_2(\lambda)\alpha_3(\lambda)}{[1 - \rho'_1(\lambda)\rho_2(\lambda)] \cdot [1 - \rho'_2(\lambda)\rho_3(\lambda)] - \tau_2^2(\lambda)\rho'_1(\lambda)\rho_3(\lambda)} \right\} S_{\lambda} \Delta\lambda}{\sum_{\lambda=300 \text{ nm}}^{2500 \text{ nm}} S_{\lambda} \Delta\lambda}$$

where

$\tau_1(\lambda), \tau_2(\lambda), \tau_3(\lambda), \rho_1(\lambda), \rho'_1(\lambda), \rho_2(\lambda), \rho'_2(\lambda), \rho_3(\lambda)$ are as defined in light transmittance;

$\alpha_1(\lambda), \alpha'_1(\lambda)$ and $\alpha_2(\lambda)$ are as defined in [5.2.9.3](#) for double glazing;

$\alpha'_2(\lambda)$ = spectral direct absorptance of the second pane, measured in the opposite direction to the incident radiation, given by the relationship:

$$\alpha'_2(\lambda) = 1 - \tau_2(\lambda) - \rho'_2(\lambda)$$

$\alpha_3(\lambda)$ = spectral direct absorptance of the third pane, measured in the direction of the incident radiation, given by the relationship:

$$\alpha_3(\lambda) = 1 - \tau_3(\lambda) - \rho_3(\lambda)$$

$\Delta\lambda$ and the integration procedure are the same as in [5.4.5.4](#) except that the data points shall be chosen at the wavelengths given in [Table 5](#).

For a glazing with more than three components, the formulae for the solar absorptances $\alpha_{e1}, \alpha_{e2}, \dots, \alpha_{en}$ as a function of the individual components.

5.2.9.5 Mean radiant temperature (MRT)

Besides solar energy transmittance, heat gain and losses occur by radiation exchange between the glazing which influences thermal comfort of the occupants. Mean radiant temperature (MRT), which is defined as the uniform temperature of an imaginary enclosure in which radiant heat transfer from the human body is equal to the radiant heat transfer in the actual non-uniform enclosure, is used as a metric in thermal comfort assessment. MRT can be calculated by various methods, among which measuring with globe thermometer is the most common and is given by:

$$MRT = \sqrt[4]{T_g^4 + \frac{h_{cg}}{\varepsilon_g \sigma} (T_g - T_a)^4}$$

where

- T_a = air temperature, in Kelvin;
- T_g = black globe temperature, in Kelvin;
- ε_g = emissivity of black globe;
- σ = Stefane-Boltzmann constant; and
- h_{cg} = convective heat transfer coefficient of the black globe.

5.2.10 Thermal Transmittance due to Temperature Difference

The other part to be looked at is the transfer of heat from outside to inside environment through the glass, due to the temperature difference. This is known as U-value, which is the rate of heat flow through one square metre of glazing when there is a temperature difference of 1 °C. The lower the U-value, the lesser is the heat transfer.

5.2.11 Total Solar Energy Transmission

The total solar energy transmitted into the room per unit area of glazing φ_{ei} is given by the relationship:

$$\varphi_{ei} = \varphi_e g$$

where

- φ_e = incident solar radiation flux per unit area;
- g = total solar energy transmittance of the glazing; and

The values of φ_e can be obtained from appropriate tables in meteorological literature.

NOTE — The SHGC/SF requirement for the glazing should be selected based on the prescriptive requirements are mentioned in [Annex B](#).

5.2.11.1 Additional heat transfer

If the room temperature T_i differs from the outside temperature T_o , an additional heat transfer occurs in addition to φ_{ei} . This additional heat flow q_z can be calculated as follows:

$$q_z = U (T_o - T_i)$$

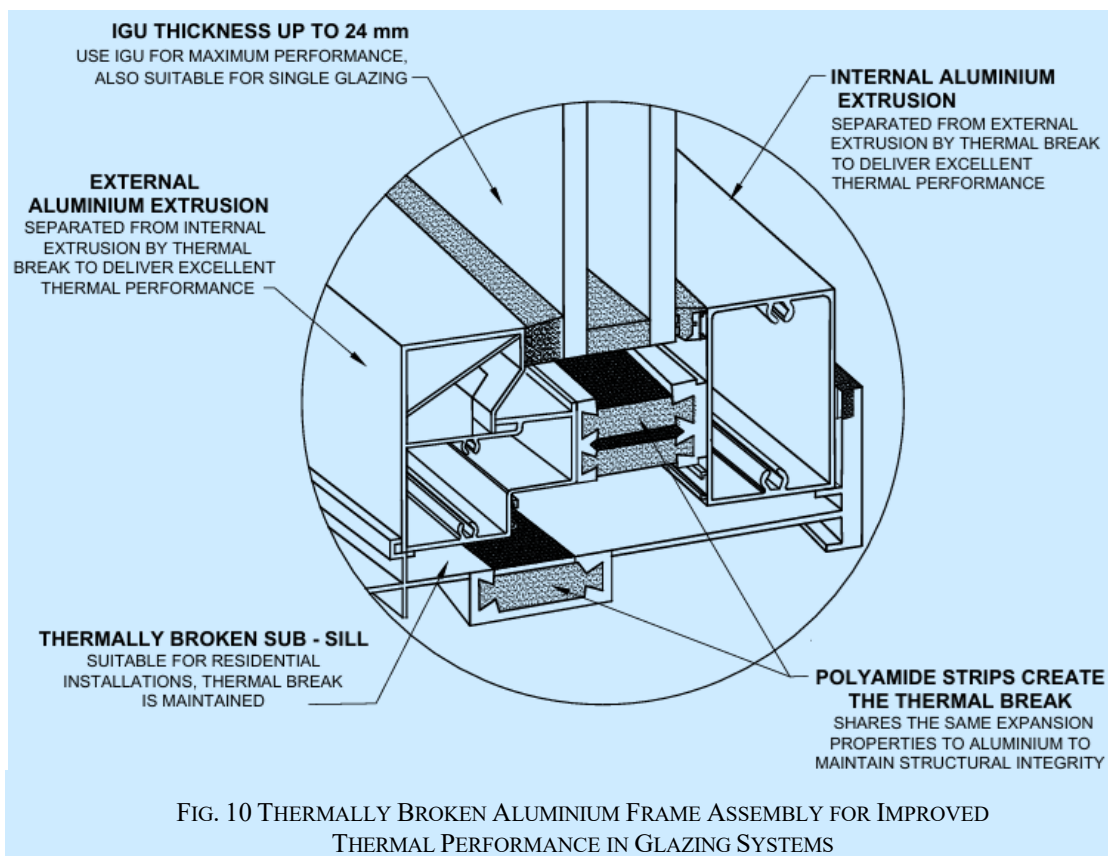
where U is the U-value (thermal transmittance) of glazing.

5.2.12 Frames

These can contribute to a significant amount of energy gain and energy loss through the glazing system. For improved efficiency, some frames are designed with internal thermal breaks that reduce energy flow through the

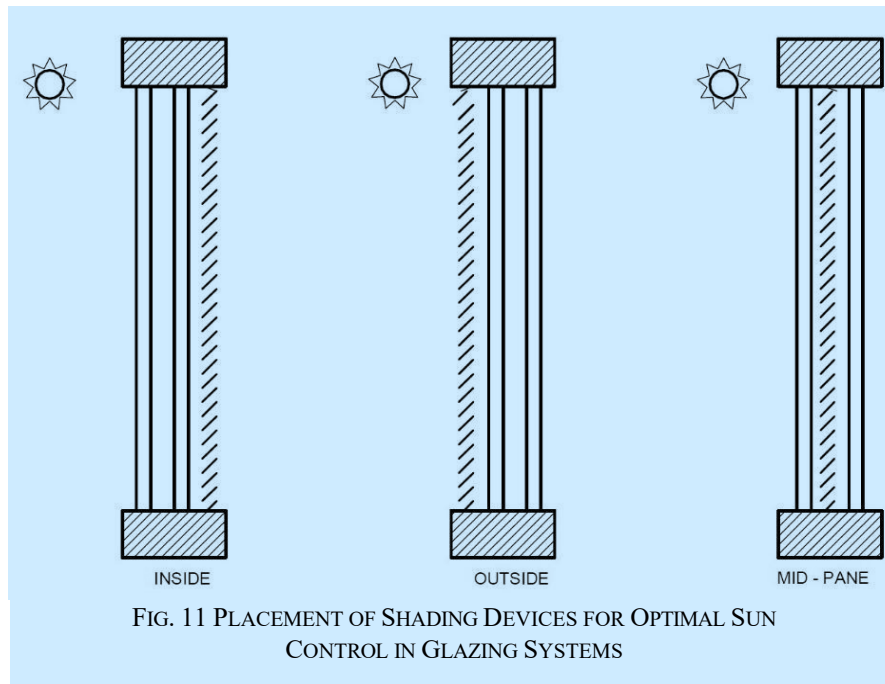
frame (see Fig. 10). The sealing between the openings and the window frames and between the opening sashes of windows and fixed portions are important areas to be addressed as major heat loss happens from this area due to poor design, selection and application of sealant material. Adequate provision shall also be made to ensure that the frames are not susceptible to water ingress in case of driving rain.

NOTE — As the evaluation of structural property differs from a monolithic profile, for evaluation of the structural property of thermally broken profile, the method described in Annex G of accepted standard [C-8(12)] shall be followed.



5.2.13 Shading Systems

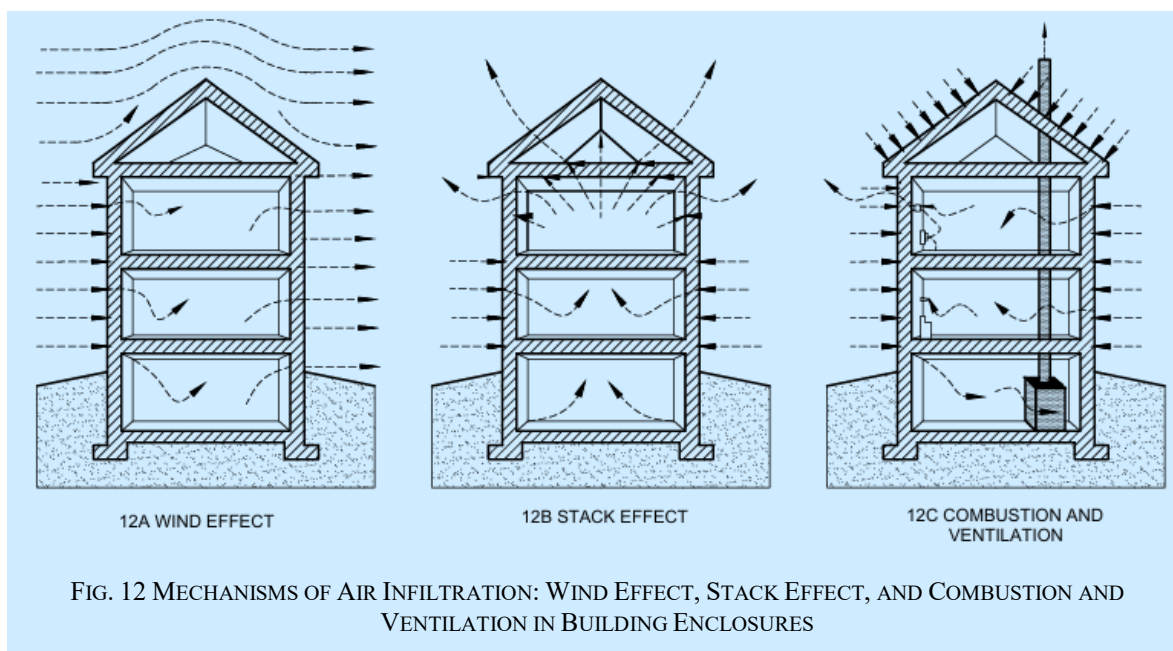
These regulate the solar radiation incident on the glazing thereby reducing the effective solar heat gain coefficient of the glazing system. Besides reduction in direct solar gains inside the building, shading systems reduce the radiant heat gain from the glazing surface. Shading system design requires due consideration for the geographic location of the building, orientation of the facade, location of the system, and solar geometry. External shading systems are effective in solar control compared to internal shading systems (see Fig. 11). Shading systems may impact the daylight availability and the views available for occupants. Kinetic systems, either automated or manually operated, can be used to vary shading system positions dynamically based on solar position, daylight requirements and occupant preferences.



5.2.14 Air Infiltration

Inadequate control of airflow through the building envelope is often a primary factor contributing to premature building envelope failures. Air infiltration may be caused by wind pressure, stack pressure or fan pressure (see Fig. 12). Wind creates a positive pressure on the windward face and negative pressure on the leeward face, causing infiltration and exfiltration. Besides the positive or negative pressure created by the heating, ventilation and air conditioning system may lead to exfiltration or infiltration, respectively.

The stack effect is when warm air moves upward in a building because it is lighter than cold air and escapes out of the upper levels of the building, through penetrations and cracks in the building envelope or other openings. The rising warm air reduces the pressure in the base of the building, forcing cold air to infiltrate through open doors, windows, or other openings. The stack effect basically causes air infiltration on the lower portion of a building and exfiltration on the upper part. Mechanical equipment such as fans and blowers cause the movement of air within buildings and through enclosures, which can generate pressure differences. If more air is exhausted from a building than is supplied, a net negative pressure is generated, which can induce unwanted airflow through the building envelope.



5.2.14.1 Considerations for controlling air infiltration

- a) Ensuring the continuity of each component serving its role in resisting infiltration, such as a window assembly or a curtain, and ensuring they are interconnected to prevent air leakage at the joints between materials, components, assemblies, and systems; and
- b) Effective structural support shall be ensured so that all components can resist the positive or negative structural loads imposed on them by wind, stack effect, and HVAC fan pressures without rupture, displacement or undue deflection, thereby resisting air infiltration through them. Design considerations shall ensure that these loads are safely transferred to the structure and provision of adequate resistance to these pressures by use of fasteners, tapes, adhesives, etc.

Inadequate sealing of the glazing system may lead to energy gains and losses, thereby affecting thermal comfort and increasing the cooling and heating energy demand. If moisture-laden air infiltrates through the glazing system, the moisture may, under certain environmental conditions condense, especially in cold climates.

5.3 Methods for Improving Thermal Insulation

Glass readily conducts energy and so is a poor insulator. To improve resistance to energy transfer, insulating glass units, coupled glazing or high performance coatings can be used. Increasing the thickness of the glass makes little difference to the *U*-value of the glazing.

5.3.1 Use of Low Emissivity Coatings

Low emissivity (low-E) coatings have surface emissivity of less than 0.2. The use of such a solar control low-E coating on glass improves the thermal insulation. They are most efficient when used on the cavity surfaces of insulating glass units.

5.3.2 Solar Control Glass

Solar control glass can be manufactured in several forms. The function of solar control glass is to reduce the total solar energy transmittance, which usually leads to a decrease in the transmission of the visible part of the solar spectrum. However, some tints and coatings are able to attenuate preferentially non-visible solar radiation, leaving the transmission of the greater proportion of the visible radiation largely unchanged.

The application of ceramic frit-fired into the surface of the glass can also be used to modify the energy and light transmission of the glass. The effect can be varied across a pane depending on the screen print pattern. Manufacturers should be consulted for specific details.

Solar control glass can be used in single glazed or double glazed unit depending upon the site requirements. The relationship between light transmission and total solar energy transmission is expressed as the light/energy ratio.

There is no optimum light/energy ratio. The selection of an appropriate product depends on the requirements of the building. Solar gain may be relatively high for making use of passive solar gains, or low to reduce air conditioning loads. Light transmission will depend on the extent to which daylight is used to obviate artificial lighting.

Some solar control glasses can be toughened or heat strengthened and gives a means of raising the design stress and enhances safety of thermal fracture. The following are the common varieties.

NOTE — Some manufacturers use this as a descriptive code, quoting a light transmission figure followed by a total solar energy transmission figure.

- a) *Body tinted glass (for increased absorption)* — Solar control properties and colour vary with the thickness of the glass. When used in insulating glass units, body tinted glass should be positioned as the outer pane because the energy due to the absorbed radiation is more easily dissipated to the outside.
- b) *Reflective coated glass* — It uses the principle of increasing the direct reflection to maximize solar energy attenuation. In comparison with clear glass, its absorption of solar energy is also increased. An informative annex on test requirements is given at [Annex C](#).

The advantages of such glass types are:

- 1) Greater performance range than body tinted glass;
- 2) Higher performances (greater solar energy attenuation);
- 3) Light/energy ratios nearer to the theoretical limit; and
- 4) A range of colour appearances in transmission and reflection.

The coatings may be placed on to body-tinted glass to extend the range of performances.

NOTES

1 Compared with a float glass surface, these reflective coatings (owing to their composition) exhibit lower levels of emissivity which improves their U -value.

2 Performance data in comparison to clear glass can be had from manufacturers.

- c) *Laminated glass* — It is commonly manufactured with clear glass and clear interlayers, but solar control properties can be incorporated into laminated glass by including either solar control glass or tinted interlayers or both.

Laminated glass with clear interlayers and solar control glass exhibit similar properties to the solar control glass from which it is made.

Laminated glass with a tinted interlayer acts in a similar manner to body tinted glass, by absorbing the solar radiation, but with a different range of colours and performances.

NOTE — Performance data in comparison to clear glass can be had from manufacturers. Performances of a typical range of laminated glass products with clear glass and tinted interlayers can be had from manufacturers.

- d) *Insulating glass units* — These are used primarily for improved thermal transmittance (U -value) (see [5.2.10](#)), but can also improve the total solar energy transmittance. This improvement is a result of the incorporation of a second pane of glass together with the hermetically sealed air space. The second glass pane can be of any glass type. Outer pane can be solar control low emissivity glass (surface 2) and inner pane can be clear glass (see [Fig. 13A](#)). The other option is outer pane can be solar control glass (surface 2) and inner pane with low emissivity coating glass (surface 3) (see [Fig. 13B](#)).

NOTES

1 Soft coat low emissivity glass can be used as the hermetically sealed airspace protects the coating.

2 Performances of a typical range of insulating glass units with clear float inner pane or with hard/soft coat low emissivity glass inner pane can be had from manufacturers.

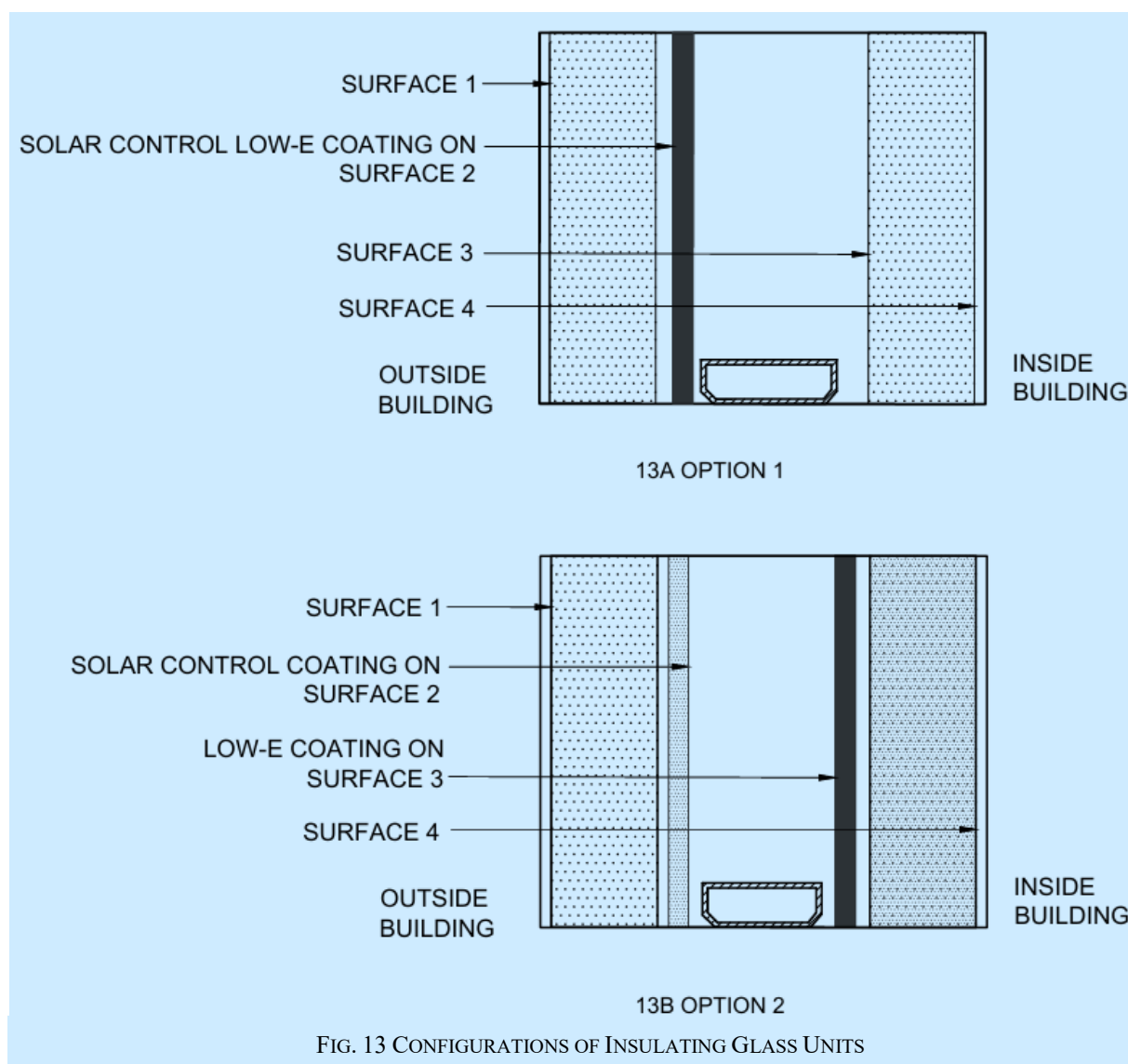


FIG. 13 CONFIGURATIONS OF INSULATING GLASS UNITS

5.3.3 Reducing Thermal Transmittance of Glazing System

5.3.3.1 Increasing the width of the air space

Enhanced thermal insulation can be achieved by increasing the width of the airspace. However, there is a convection of the gas in the cavity.

5.3.3.2 Using gases of lower thermal conductivity

Replacing the air in the cavity with, for example, argon/krypton, can improve the thermal insulation.

5.3.3.3 Inhibiting convection within the air space

Filling the cavity with cellulose material reduces convection and makes the cavity a more efficient insulator. However, this usually results in loss of vision, since the materials are, at best, translucent.

5.3.3.4 Evacuation of the air space

In theory, a vacuum shall eliminate energy transfer by conduction and convection. However, a vacuum puts high demands on the glass from the external air pressure.

NOTE — Evacuation of the air space is currently considered to be technically impractical.

5.3.3.5 Evaluation of type of spacer used in double glazed unit

Thermal performance of the glazing system for conduction can be improved by changing the material of the spacer.

5.3.3.6 Typical U-values of glass products

[Table 6](#), [Table 7](#) and [Table 8](#) give typical U-values.

Table 6 Thermal Transmittance (U-Value) of Glass Products: Single Glazing (Clause 5.3.3.6)		
Sl No.	Glass Thickness mm	U-Value W/m ² K
(1)	(2)	(3)
i)	4	5.8
ii)	6	5.7
iii)	10	5.6
iv)	12	5.5

Table 7 Thermal Transmittance (U-Value) of Glass Products: Insulating Glass Units (Clause 5.3.3.6)				
Sl No.	Clear glass Thickness and type mm	Cavity Width mm	U-Value W/m ² K	
			Air	Argon
(1)	(2)	(3)	(4)	(5)
i)	4+4	6	3.3	3.1
		12	2.9	2.7
		16	2.7	2.6
		20	2.8	2.6
ii)	4+4 Low E ($\epsilon_d = 0.15$)	6	2.7	2.3
		12	1.9	1.6
		16	1.7	1.5
		20	1.7	1.5
iii)	4 + 4 Low E ($\epsilon_d = 0.04$)	6	2.5	2.0
		12	1.6	1.3
		16	1.4	1.2
		20	1.4	1.2
iv)	4 + 4 + 4	6	2.4	2.1
		12	1.9	1.8
		16	1.8	1.7
		20	1.8	1.7
v)	4 Low E ($\epsilon_d = 0.15$) + 4 + 4 Low E ($\epsilon_d = 0.15$)	6	1.7	1.4
		12	1.7	1.0
		16	1.0	0.8
		20	0.9	0.8
vi)	4 Low E ($\epsilon_d = 0.04$) + 4 + 4 Low E ($\epsilon_d = 0.04$)	6	1.6	1.2
		12	1.0	0.7
		16	0.8	0.6
		20	0.7	0.6

Table 7 (Concluded)

Table 7 Thermal Transmittance (<i>U</i>-Value) of Glass Products: Insulating Glass Units (Clause 5.3.3.6)				
SI No.	Clear glass Thickness and type mm	Cavity Width mm	U -Value W/m^2K Air Argon	
(1)	(2)	(3)	(4)	(5)
NOTES 1 U -values for argon gas-filled cavity based on 90 percent argon/10 percent air. 2 ϵ_d is the declared (normal) emissivity.				

Table 8 Thermal Transmittance (<i>U</i>-Value) of Glass Products: Coupled Glazing (Clause 5.3.3.6)			
SI No.	Clear Glass Thickness and Type mm	Separation mm	U -Value W/m^2K
(1)	(2)	(3)	(4)
i)	4 + 4	25	2.8
		75	2.8
ii)	4 + 4 Low E	25	1.7
		75	1.9
NOTE — ϵ_d is the declared (normal) emissivity.			

The U -value requirement for the glazing system should be selected in accordance with the [Table 9](#) and [Annex B](#).

Table 9 Thermal Performance of Different Glass Shading Devices

(Clause 5.3.3.6)

Sl No.	Name of the Shading Device	Transmittance, U-Value	Shade Factor
(1)	(2)	(3)	(4)
i)	Plain glass sheet (3.0 mm thick)	5.23	1.00
ii)	Plain glass + Wire mesh outside	5.00	0.65
iii)	Painted glass:		
	a) White paint	5.22	0.35
	b) Yellow paint	5.22	0.37
	c) Green paint	5.22	0.40
iv)	Heat absorbing glass	4.65	0.45
v)	Plain glass sheet + Venetian blind inside:	3.14	
	a) Light colour		0.35
	b) Dark colour		0.40
vi)	Plain glass sheet which is,	5.23	
	a) 100 percent shaded		0.14
	b) 75 percent shaded		0.34
	c) 60 percent shaded		0.56
NOTE — Indicative values, actual value of the system to be measured using either approved simulation software or testing at laboratory.			

5.3.4 Thermal Safety of Glass

Thermal safety of glass should be assessed considering the amount of radiation incident on the surface and the thermal capabilities of the glass. For example, the solar radiation intensity on the glass surface should be determined along with the air temperature range applicable to the location of the building. These measurements, together with the energy transfer coefficients and the glass absorption allow determination of the appropriate basic temperature difference between the central area of the glass and its edge. The difference is related to the thermal stress and then modified for the type of glazing system, taking account of extraneous effects resulting from curtains, blinds, back-up walls, proximity to heaters, etc to derive a stress in actual service conditions.

NOTES

1 High air temperatures, low rates of air movement, and the insulation provided by curtains, blinds, back-up walls and multiple glazing tend to reduce the loss of energy and uphold the centre temperature. Low temperatures at the edges are maintained by conduction from the glass through the frame to a cold building structure with a large thermal capacity.

2 Advice may be sought from the manufacturer of glass as to the methods for assessing the thermal safety of the glass.

The resultant service stress should then be compared with the design stress for the glass. If on comparison, the service stress is less than or equal to the design stress, the glass and glazing system may be accepted as thermally safe provided, that the edges of the glass are of adequate quality.

NOTE — Where the application of a solar control film is being considered to existing glazing, advice should be sought from the manufacturer on the effect of any additional thermal stress likely to be induced in the glass.

The normal mode of thermal breakage of glass is by the action of tensile stress located in and parallel to an edge, and so the breaking stress of the glass is mainly dependent on the extent and position of flaws in the edges. The condition of the glass edge is therefore extremely important.

Solar control glasses should not be nipped to size and any panes with shelled or vented edges should not be accepted for glazing in orientations subject to direct sunlight. Although a wheel-cut edge is the most satisfactory, laminated glasses with worked edges may be used.

Where clean-cut edges are not permitted, arises should be created by a wet process, working parallel to the edge and not across the thickness, and the design implications of such an action should be examined.

Where solar control glasses are to be used in sliding doors and windows there is always the possibility that, when opened during sunny periods, the overlapping will function as double glazing with little ventilation in the air space, and it is this condition that should be assumed in assessing the thermal safety of glass.

Thermal safety assessment is based on the behaviour of glass in good condition and properly glazed. Even if the glass is shown to be thermally safe, this depends on close adherence to the recommend glazing procedures. All necessary precautions should be taken to see that only glass with edges of an acceptable condition is used. The glass should be stored and handled so that no contact with hard bodies can damage the edges and each pane or insulating glass unit should be carefully examined immediately before glazing.

5.3.5 UV-Transmittance

The UV-transmittance of glazing is the fraction of the incident solar radiation transmitted by the glazing in the 300 nm to 380 nm range (UV-B range from 300 nm to 315 nm and UV-A range from 315 nm to 380 nm). The UV-transmittance τ_{UV} is calculated as follows:

$$\tau_{UV} = \frac{\sum_{\lambda=300 \text{ nm}}^{380 \text{ nm}} \tau(\lambda) S_{\lambda} \Delta\lambda}{\sum_{\lambda=300 \text{ nm}}^{380 \text{ nm}} S_{\lambda} \Delta\lambda}$$

where

S_{λ} = relative spectral distribution of UV radiation;

$\tau(\lambda)$ = spectral transmittance of the glazing; and

$\Delta\lambda$ and the integration procedure are the same as [5.4.5.4](#) except that the data points shall be chosen at the wavelength given in [Table 10](#).

This average extends over the defined UV-portion of the solar spectrum. It may not be correlated with solar radiation damage of materials and skin.

Table 10 Normalized Relative Spectral Distribution of Global Solar Radiation

(Clause [5.3.5](#))

Sl No.	λ nm	$S_{\lambda}\Delta\lambda$
(1)	(2)	(3)
i)	300	0
ii)	305	0.001 859
iii)	310	0.007 665
iv)	315	0.017 961
v)	320	0.029 732
vi)	325	0.042 466
vii)	330	0.062 108
viii)	335	0.065 462
ix)	340	0.071 020
x)	345	0.073 326
xi)	350	0.079 330

Table 10 (Concluded)

Table 10 Normalized Relative Spectral Distribution of Global Solar Radiation <i>(Clause 5.3.5)</i>		
SI No.	λ nm	$S_{\lambda}\Delta\lambda$
(1)	(2)	(3)
xii)	355	0.082 894
xviii)	360	0.087 039
xix)	365	0.097 963
xx)	370	0.108 987
xxi)	375	0.113 837
xxii)	380	0.058 351
NOTE — The values in this table are calculated according to the trapezoidal rule.		

5.4 Daylight Transmission

Good glazing system design can, by reducing reliance on artificial lighting, can be one of the largest single means of saving energy and should be carefully considered.

Account should be taken of shape of the room, window orientation, occupancy patterns and task, together with the relationship of windows to surrounding buildings and other obstruction. As with task lighting, possible solar overheating might result and similar precautions should be taken.

5.4.1 Light Transmission Properties of Window

Light transmission is defined as the fraction of visible light at normal incidence transmitted through the glazing. For properties of plastic glazing sheet materials, manufacturers may be consulted.

Dirt on glazing reduces the light transmission, often by an appreciable extent before becoming noticeable. To ensure day lighting levels are adequate, an allowance for the reduced light transmission should be made in day lighting calculation by introducing a 'dirt factor' between 0.7 and 1.0. Regular cleaning of glass and plastic glazing sheet materials is therefore important.

5.4.2 Glare

5.4.2.1 General

Glare results from excessive contrast of illumination, or from an excess of illumination in the field of view. Reaction to it is subjective. When correctly designed, natural lighting should not result in glare problem.

NOTE — Contrasts in excess of 10 : 1 in illumination in different parts of the field of view might give rise to glare in some form.

5.4.2.2 Disability glare

Disability glare is glare which impairs the visibility of objects without necessarily causing discomfort.

During a sunny day, even a light transmission of the glazing resulting in 10 000 lux or lower, is likely to cause disability glare.

Glare can be reduced by some form of mechanical shading, for example a canopy, an overhanging floor, a balcony or a louvre system. Alternatively, internal screening can be provided by louvres or blinds.

NOTE — Any fixed shading system reduces the amount of natural light entering the building throughout the year, irrespective of whether there is a glare problem at any particular time.

It may also be possible to re-orientate the glazing to avoid entry of direct solar radiation. Alternatively, the interior layout can be suitably designed to eliminate glare.

5.4.2.3 Discomfort glare

Glazing products with light transmission lower than 50 percent can ameliorate discomfort glare. These products decrease the sky luminance components but permanently reduce the admission of daylight. Alternatively, shading devices, internal or external, movable or fixed, may be used.

Other methods of reducing the problems of glare should be considered, including,

- a) installation of windows in more than one wall to raise the general background illumination and, in so doing, to reduce the contrast between a window and its surrounding surfaces;
- b) use of light coloured matt finishes for the window frames and the surrounding surfaces;
- c) splayed reveals, to assist in reducing the contrast between the window and its surroundings;
- d) use of slender glazing bars and transoms of high reflectance; and
- e) lowering window sills to allow increased illumination to enter, which increases the adaptation level and reduces the likelihood of discomfort glare.

5.4.3 Diffusion and Obscuration

The nature of some glazing products (for example, patterned or acid etched glass) can cause the direct incident solar beam to be scattered diffusely. Hence the window might assume an uncomfortable high brightness and become a discomfort glare source in its own right. Diffusing glazing used within the normal field of view should be used with caution.

5.4.4 Fading

Most materials can fade when subjected to either daylight (particularly direct sunlight) or artificial light. Fading is a complex phenomenon involving many chemical reactions, initiated or accelerated by light of different wavelengths. Generally, the better quality dyes and pigments fade relatively slowly and react only to the shorter wavelengths (ultraviolet and the blue end of the visible spectrum). Other materials can fade quickly and might do so under light of much longer wavelengths. It is the combination of wavelength, available light and transmission which determines glass selection to minimise fading, not simply the UV transmission.

5.4.5 Determination of Characteristic Parameters

5.4.5.1 General

The characteristic parameters are determined for quasi-parallel, almost normal radiation incidence.

5.4.5.2 Test setup

The samples shall be irradiated by a beam whose axis is at an angle not exceeding 10° from the normal to the surface. The angle between the axis and any ray of the illuminating beam shall not exceed 5° .

The characteristic parameters are as follows:

- a) spectral transmittance $\tau(\lambda)$, the spectral external reflectance $\rho_o(\lambda)$ and the spectral internal reflectance $\rho_i(\lambda)$ in the wavelength range of 300 nm to 2 500 nm;
- b) light transmittance τ_V , the external light reflectance $\rho_{v,0}$ and the internal light reflectance $\rho_{v,i}$ for illuminant D65;
- c) solar direct transmittance τ_e and the solar direct reflectance ρ_e ;
- d) total solar energy transmittance (solar heat gain coefficient SHGC) g ; and
- e) UV-transmittance τ_{UV} ;

When calculating the characteristic parameters of multiple glazing, the spectral data of each glass component is to be used instead of integrating data of the complete unit.

5.4.5.3 Performance requirements of optical measurements

Optical measurements in transmission and reflection require special care to achieve accuracy in transmittance and reflectance of about ± 0.01 .

The wavelength calibration and the photometric linearity of commercial spectrophotometers shall be checked periodically using reference materials obtained from metrological laboratories.

The wavelength calibration shall be performed by measuring glass plates which feature relatively sharp absorption bands at specified wavelengths; the photometric linearity shall be checked using grey filters with a certified transmittance.

For reflectance measurements, reference materials having reflection behaviour (that is, reflectance level and ratio of diffuse and direct reflectance).

Thick samples (that is laminated glass or insulating units) can modify the optical path of the instrument's beam as compared to the path in air and therefore the sample beam hits an area of the detector having a different responsivity.

A similar source of inaccuracy occurs in case of samples with significant wedge angles which deflect the transmitted (and reflected) beams. It is recommended to check the reproducibility by repeating the measurement after rotating the sample.

Additionally, in the case of reflectance measurements, glass sheets cause a lateral shear of the beam reflected by the second surface, causing reflectance losses (whose extent is particularly evident in the case of thick and/or wedged samples). This source of inaccuracy shall be taken into account particularly in the case of reflectance measurements through the uncoated side. In order to quantify and correct systematic errors, it is recommended to use calibrated reflectance standards with a thickness similar to the unknown sample.

In the case of diffusing samples (or samples with a non-negligible diffusing component or wedged samples), transmittance and reflectance measurements shall be performed using integrating spheres whose openings are sufficiently large to collect the entire diffusely transmitted or reflected beam. The sphere diameter shall be adequate and the internal surface adequately coated with a highly diffusing reflectance material, so that the internal area can provide the necessary multiple reflections. Reference materials with characteristics similar to the unknown sample as specified above shall be used. If the transmittance or reflectance curve recorded by the spectrometer exhibits a high level of noise for some wavelengths, the values to be considered for those wavelengths should be obtained after a smoothing of the noise.

5.4.5.4 Light transmittance

Light transmittance is the measure of the amount of light passing through the glazing. The visible light lies between the wavelength 380 nm and 780 nm. The light transmission of the glazing depends on the substrate type and coating layer done on the base glass.

The light transmittance, τ_v of glazing shall be calculated using the following formula:

$$\tau_v = \frac{\sum_{\lambda=380 \text{ nm}}^{780 \text{ nm}} \tau(\lambda) D_\lambda V(\lambda) \Delta\lambda}{\sum_{\lambda=380 \text{ nm}}^{780 \text{ nm}} D_\lambda V(\lambda) \Delta\lambda}$$

where

D_λ = relative spectral distribution of illuminant D65;

$\tau(\lambda)$ = the spectral transmittance of glazing;

$V(\lambda)$ = spectral luminous efficiency for photopic vision defining the standard observer for photometry; and

$\Delta\lambda$ = wavelength interval.

[Table 11](#) indicates the values for $D_\lambda V(\lambda) \Delta\lambda$ for wavelength intervals of 10 nm. The table has been drawn up in such a way that $\sum D_\lambda V(\lambda) \Delta\lambda = 1$.

In the case of multiple glazing, the spectral transmittance $\tau(\lambda)$ shall be obtained by calculation from the spectral characteristics of the individual components. Alternatively, measurements on non-diffusing multiple units may be performed using an integrating sphere. This may be achieved after reducing the interspaces under conditions that allow the collection of the whole transmitted beam.

The calculation of the spectral transmittance $\tau(\lambda)$ shall be performed using methods such as algebraic manipulation. Any algorithm that can be shown to yield consistently the correct solution is acceptable.

For the calculation of $\tau(\lambda)$ as well as for the calculation of spectral reflectance the following symbols for the spectral transmittance and spectral reflectance of the individual components are used:

$\tau_1(\lambda)$	=	spectral transmittance of the outer (first) pane;
$\tau_2(\lambda)$	=	spectral transmittance of the second pane;
$\tau_n(\lambda)$	=	the spectral transmittance of the n^{th} (inner) pane (for example, for triple glazing $n=3$);
$\rho_1(\lambda)$	=	spectral reflectance of the outer (first) pane measured in the direction of incident radiation;
$\rho'_1(\lambda)$	=	spectral reflectance of the outer (first) pane measured in the opposite direction of incident radiation;
$\rho_2(\lambda)$	=	spectral reflectance of the second pane measured in the direction of incident radiation;
$\rho'_2(\lambda)$	=	spectral reflectance of the second pane measured in the opposite direction of incident radiation;
$\rho_n(\lambda)$	=	spectral reflectance of the n^{th} (inner) pane measured in the direction of incident radiation;
$\rho'_n(\lambda)$	=	spectral reflectance of the n^{th} (inner) pane measured in the opposite direction of incident radiation.

For the spectral transmittance $\tau(\lambda)$ as a function of the spectral characteristics of the individual components of the unit, the following formulae are obtained:

- a) For double glazing; and
- b) For triple glazing.

For multiple glazing with more than three components, relationships similar to above expressions are found to calculate $\tau(\lambda)$ of such glazing from the spectral characteristics of the individual components.

As an example for calculating $\tau(\lambda)$ according to the procedures of this Section, a glazing composed of five components may be treated as follows:

- 1) First consider the first three components as triple glazing and calculate the spectral characteristics of this combination;
- 2) Next, run the same procedure for the next two components as double glazing; and
- 3) Then calculate $\tau(\lambda)$ for the five component glazing, considering it as double glazing consisting of the preceding triple and double glazing.

The light transmission requirement for the glazing should be selected in accordance with relevant Indian Standards.

Table 11 Normalized Relative Spectral Distribution $D_{\lambda}V(\lambda)\Delta\lambda$

(Clause [5.4.5.4](#))

SI No.	λ nm	$D_{\lambda}V(\lambda)\Delta\lambda \times 10^2$
(1)	(2)	(3)
i)	380	0
ii)	390	0.000 5
iii)	400	0.003 0
iv)	410	0.0103
v)	420	0.035 2
vi)	430	0.094 8
vii)	440	0.227 4
viii)	450	0.419 2
ix)	460	0.666 3
x)	470	0.985 0
xi)	480	1.518 9
xii)	490	2.133 6
xiii)	500	3.349 1
xiv)	510	5.139 3
xv)	520	7.052 3
xvi)	530	8.799 0
xvii)	540	9.442 7
xviii)	550	9.807 7
xix)	560	9.430 6
xx)	570	8.689 1
xxi)	580	7.899 4
xxii)	590	6.330 6
xxiii)	600	5.354 2
xxiv)	610	4.249 1
xxv)	620	3.150 2
xxvi)	630	2.081 2
xxvii)	640	1.381 0
xxviii)	650	0.807 0
xxix)	660	0.461 2
xxx)	670	0.248 5
xxxi)	680	0.125 5
xxxii)	690	0.053 6
xxxiii)	700	0.027 6
xxxiv)	710	0.014 6

Table 11 (Concluded)

Sl No.	λ nm	$D_{\lambda}V(\lambda)\Delta\lambda \times 10^2$
(1)	(2)	(3)
xxxv)	720	0.005 7
xxxvi)	730	0.003 5
xxxvii)	740	0.002 1
xxxviii)	750	0.000 8
xxxix)	760	0.000 1
xxxx)	770	0.000 0
xxxxi)	780	0.000 0

NOTE — Normalized relative spectral distribution D_{λ} of illuminant D65 multiplied by the spectral luminous efficiency $V(\lambda)$ and by the wavelength interval $\Delta\lambda$. The values in this table are calculated according to the trapezoidal rule.

5.5 Light Reflectance

5.5.1 External Light Reflectance of Glazing

External light reflectance corresponds to the amount of sunlight reflected by the external glass surface. The amount of light reflection depends on the type of base glass (substrate tint) and the position of coating on the glass surface.

The external light reflectance of glazing $\rho_{v,o}$ shall be calculated using the following formula:

$$\rho_{v,o} = \frac{\sum_{\lambda=380 \text{ nm}}^{780 \text{ nm}} \rho_o(\lambda) D_{\lambda} V(\lambda) \Delta\lambda}{\sum_{\lambda=380 \text{ nm}}^{780 \text{ nm}} D_{\lambda} V(\lambda) \Delta\lambda}$$

where

$\rho_o(\lambda)$ = spectral external reflectance of glazing, and D_{λ} , $V(\lambda)$, $\Delta\lambda$ and the integration procedure are defined in [5.4.5.4](#).

For multiple glazing, the calculation of the spectral external reflectance $\rho_o(\lambda)$ shall be performed using the same methods as given in [5.4.5.4](#) for the calculation of the spectral transmittance $\tau(\lambda)$.

For the spectral external reflectance $\rho_o(\lambda)$ as a function of the spectral characteristics of the individual components of the unit, the following formulae are applied:

a) For double glazing:

$$\rho_o(\lambda) = \rho_1(\lambda) + \frac{\tau_1^2(\lambda) \rho_2(\lambda)}{1 - \rho_1'(\lambda) \rho_2(\lambda)}$$

b) For triple glazing:

$$\rho_o(\lambda) = \rho_1(\lambda) + \frac{\tau_1^2(\lambda) \rho_2(\lambda) [1 - \rho_2'(\lambda) \rho_3(\lambda)] + \tau_1^2(\lambda) \tau_2^2(\lambda) \rho_3(\lambda)}{[1 - \rho_1'(\lambda) \rho_2(\lambda)] \cdot [1 - \rho_2'(\lambda) \rho_3(\lambda)] - \tau_2^2(\lambda) \rho_1'(\lambda) \rho_3(\lambda)}$$

For multiple glazing with more than three components, relationships similar to above equations are found to calculate $\rho_o(\lambda)$ of such glazing from the spectral characteristics of the individual components.

As an example for calculating $\rho_o(\lambda)$, a glazing composed of five components may be treated in the same way as described in [5.4.5.4](#).

5.5.2 Internal Light Reflectance of Glazing

Internal light reflectance corresponds to the amount of light reflected by the inner glass surface. The amount of light reflection depends on the type of base glass (substrate tint) and the position of coating on the glass surface.

The internal light reflectance of glazing $\rho_{v,i}$ shall be calculated using the following formula:

$$\rho_{v,i} = \frac{\sum_{\lambda=380 \text{ nm}}^{780 \text{ nm}} \rho_i(\lambda) D_\lambda V(\lambda) \Delta\lambda}{\sum_{\lambda=380 \text{ nm}}^{780 \text{ nm}} D_\lambda V(\lambda) \Delta\lambda}$$

where

$\rho_i(\lambda)$ = spectral internal reflectance of glazing; and

D_λ , $V(\lambda)$, $\Delta\lambda$ and the integration procedure are as defined in light transmittance in [5.4.5.4](#).

For multiple glazing, the calculation of the spectral internal reflectance $\rho_i(\lambda)$ shall be performed using the same methods as given in [5.4.5.4](#) for the calculation of the spectral transmittance $\tau(\lambda)$.

For the spectral internal reflectance $\rho_i(\lambda)$ as a function of the spectral characteristics of the individual components of the unit, the following formulae are applied.

b) For double glazing:

$$\rho_i(\lambda) = \rho'_2(\lambda) + \frac{\tau_2^2(\lambda) \rho'_1(\lambda)}{1 - \rho'_1(\lambda) \rho_2(\lambda)}$$

c) For triple glazing:

$$\rho_i(\lambda) = \rho'_3(\lambda) + \frac{\tau_3^2(\lambda) \rho'_2(\lambda) [1 - \rho_2(\lambda) \rho'_1(\lambda)] + \tau_3^2(\lambda) \tau_2^2(\lambda) \rho'_1(\lambda)}{[1 - \rho_3(\lambda) \rho'_2(\lambda)] \cdot [1 - \rho_2(\lambda) \rho'_1(\lambda)] - \tau_2^2(\lambda) \rho_3(\lambda) \rho'_1(\lambda)}$$

For multiple glazing with more than three components, relationships similar to above expressions are found to calculate $\rho_i(\lambda)$ of such glazing from the spectral characteristics of the individual components.

5.6 Energy Design Considerations

5.6.1 General

Glass and glazing system design shall enhance daylighting into the building, improve the views to outdoor environments for the occupants, minimize the energy gains in hot climates, energy losses in cold climates and facilitate air movement in naturally ventilated spaces.

Fenestration, glazing and shading systems shall be designed by considering the climate characteristics, solar geometry, design sky conditions, orientation of the facade and building use. Effective design of the glazing system can minimize energy gains during summer and energy losses during winter. Height of the vision panel and the lintel height can be effectively adjusted to optimize the daylight penetration into the building.

5.6.2 Design factors

The following factors/components of a glazing system shall be considered during design for achieving energy efficiency:

- Light transmittance and glare,
- Thermal transmittance (solar heat gain co-efficient and U-value),
- Fenestration size and placement,
- Thermal properties of the frames,
- Shading devices, and
- Air infiltration.

5.6.3 Analysis for Glazing System Selection

5.6.3.1 Building orientation and solar geometry analysis

Optimal building orientation is a critical aspect for climate responsive design. Knowledge of the solar geometry at a site is essential for designing building facades to regulate energy and daylight transmission as well as reduce glare and overheating inside the building. Choice of glass and the shading system can considerably vary based on orientation of the facade and the solar geometry. A well oriented building maximises daylighting through its facades, thereby reducing the need for artificial lighting. Such building regulates the energy gains and losses through the facade, thereby minimizing thermal discomfort and cooling and heating energy demand.

5.6.3.2 Solar insolation and shading analysis

Design of glazing system is driven by the insolation levels on the facade, the desirable energy gains/losses and daylight requirements for a given building. Solar insolation varies with geographic location, orientation of the facade, time of the day and season. Direct solar incident radiation is the energy transmitted from the sun, which lies between the wavelength range of 250 nm and 2 500 nm. Solar radiation typically consists of 3 percent ultraviolet light, 42 percent visible light, and 55 percent infrared light. Infrared radiation is the major heat carrying radiation, followed by visible light. Analysing solar radiation helps to assess its impact on the building's elevations, allowing for a better understanding of how this influences the building's performance. Solar insolation and shading analysis are employed for determining optimum SHGC and U-value of glass, U-value of the frame and the design of shading system necessary for minimizing the energy gains and losses through the facade.

5.6.3.3 Daylight analysis

Daylight analysis is used to assess the penetration of natural light on the building floor plan or at the working plane level. A good lighting strategy involves optimizing the glazed area, increasing the daylight autonomy and thereby reducing the dependency on artificial lighting. Using a glass with optimal light transmission can go a long way in curtailing the dependence on artificial lighting. Additionally, the presence of natural light alleviates the claustrophobic feeling often associated with buildings that have limited glazing and outdoor views. Daylight calculations are conducted for overcast conditions to represent the worst-case scenario for outdoor lighting levels. Detailed methods for daylight analysis are provided in good practice [C-8(13)].

Any glass with enhanced energy performance will inevitably reduce the amount of daylight transmitted, even though it remains transparent for outward vision. The higher the demand for energy performance, the greater the compromise on daylight. Therefore, the selection of glass should begin by assessing the daylight requirements and deciding the optimum visible light transmittance (VLT). This approach helps narrow down the options, allowing for the selection of glass that provides the best performance for the specified VLT while also considering cost and aesthetics.

Floor plate geometry plays an important role as the daylight distribution will depend on the depth of the plate from the glazed area. Planforms incorporating courtyard and atrium have a great advantage of harnessing maximum daylight.

5.6.4 Integrated Design Approach

Glass and glazing system design shall duly consider the trade-offs in enhancing daylight availability, maximizing or minimizing the heat gains/losses, reduction in glare and improving views for the occupants. [Table 12](#) provides the relation between glazing system parameters and different performance attributes.

Table 12 Relation Between Glazing System Parameters and Different Performance Attributes
(Clause 5.6.4)

Sl No.	Performance Attributes	Glazing System Parameters				
		VLT	U-Value	SHGC	Shading system	Fenestration size
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Daylight	$\propto$	—	ID	$\frac{1}{\propto}$	$\propto$
ii)	View	$\propto$	—	ID	$\frac{1}{\propto}$	$\propto$
iii)	Heat transfer	—	$\propto$	$\propto$	$\frac{1}{\propto}$	$\propto$
iv)	Ventilation	—	—	—	$\frac{1}{\propto}$	$\propto$
$\propto$: directly proportional; $\frac{1}{\propto}$: Inversely proportional; ID: Indirect relationship						
NOTE — The effects may vary depending on the geographic location, orientation of the facade and on-site and off-site features. The factors also have inter-dependencies.						

5.7 Window Design Guideline for Non-Conditioned/Mixed Mode Buildings

In non-conditioned spaces, windows typically use single-glazed units, as they are often opened to allow ventilation. In such buildings, shading devices play a critical role in the thermal performance of windows. Considerations for selecting type of window and the necessary design considerations for energy efficiency and natural ventilation may be made in accordance with the relevant guidelines/best practices (Handbook on ‘Sustainability in Buildings’).

NOTE — Air infiltration Class B in accordance with accepted standard [C-8(14)] is adequate for non-conditioned buildings.

6 STRUCTURAL PARAMETERS AND FIRE SAFETY

6.1 Determination of Appropriate Glass Thickness

6.1.1 General

This Section gives provisions to determine the minimum glass thickness to be used to resist the ultimate limit state design wind pressures. However, the location of the glass within the building require additional considerations, such as safety with respect to human which may impose either a minimum glass thickness above the basic calculation of this part or a provision for a specific glass type or both.

In addition to the provisions herein, the design requirements such as wind loads determined in accordance with good practice [C-8(15)] and the serviceability deflection requirement limit given in this Section shall have to be complied.

The provisions under 6 are applicable provided the following parameters are satisfied:

- Ultimate limit state wind pressure shall not be greater than 10 kPa;

- b) For laminated glass, the two sheets may be of equal or unequal thickness and the interlayer material is either polyvinyl butyral or an equivalent type of interlayer with a modulus of elasticity of about 24 MPa and a Poisson's ratio of 0.50 at 20 °C;
- c) Design flexural tensile strength of glass depends upon the design strength obtained after applying a factor of safety of 2.5. The minimum design strength of normal glass for thickness up to 6 mm is 16.7 N/mm² and for thickness above 6 mm is 15.2 N/mm².

6.1.2 Design Considerations

The following design considerations shall be taken into account in determination of appropriate thickness of glass:

- a) Maximum area of glass panel is restricted to 15 m²;
- b) Maximum span of glass panel is restricted to 4 m;
- c) Aspect ratio of the glass panel should be greater than 1.5. If it is less than 1.5, next higher available thickness should be selected in order to reduce the stress level;
- d) Factor of safety applied is 2.5 considering the variability in strength of glass;
- e) Applicable to normal, reflective, laminated, tempered and insulating glass;
- f) Applicable to rectangular panels properly secured;
- g) Design minimum thickness of laminated glass will be the maximum value of thickness in accordance with Table 2 of accepted standard [C-8(6)]; and
- h) The parameters listed in (a) to (g) are valid for calculating glass thickness using the empirical relationship in 6.1.3. For glass sizes larger than those mentioned, any recognized method of analysis may be used for design, in consultation with the manufacturer and the expert in the field.

6.1.3 Empirical Relationship

Empirical relation between the wind pressure, area of the glass panel and the required glass thickness can be used:

$$P_{\text{net}} \cdot A = 200T^k \text{ (for } T \leq 6 \text{ mm)}$$

$$P_{\text{net}} \cdot A = 200T^k + 1\,900 \text{ (for } T > 6 \text{ mm)}$$

where

P_{net} = Net design wind pressure as per 6.1.4, in N/m²;

A = Area of glass panel, in m²;

T = Standard nominal thickness (SNT) of glass from Table 13, in mm; and

k = Constant from the Table 13.

Table 13 Value of k for the Corresponding Standard Nominal Thickness of Float Glass (Clause 6.1.3)										
T , mm	3	4	5	6	8	10	12	15	19	25
k	1.683	1.732	1.753	1.765	1.57	1.578	1.583	1.579	1.569	1.569

6.1.3.1 Glass supported on two opposite sides

- a) Normal and laminated glass panels supported on two opposite sides can be designed using following empirical relations:
 - 1) For $T \leq 6$ mm :

$$b = \frac{4.39 T}{\sqrt{\left(\frac{P_{net}}{P_f}\right)}}$$

2) For $T > 6$ mm :

$$b = \frac{4.22 T}{\sqrt{\left(\frac{P_{net}}{P_f}\right)}}$$

b) Tempered/toughened panel supported on two opposite sides can be designed using following empirical relations:

1) For $T \leq 6$ mm :

$$b = \frac{3.2688 T}{\sqrt{\left(\frac{P_{net}}{P_f}\right)}}$$

2) For $T > 6$ mm :

$$b = \frac{2.9069 T}{\sqrt{\left(\frac{P_{net}}{P_f}\right)}}$$

where

b = span, in m; and

P_f = strength factors as in [Table 14](#).

For insulated glass, thickness of only one glass pane shall be considered. If the glass panes are of different thickness, the minimum of the two thicknesses shall be considered.

For laminated glass, the thickness of PVB has not been considered. The value to be used is:

T = Standard nominal thickness —Thickness of PVB.

6.1.3.2 For non-linear analysis, specialist literature may be consulted for various shapes, sizes, and types of support conditions of glass.

6.1.4 Determination of Design Wind Pressure

Net design wind pressure (P_{net}) is an important parameter governing the thickness of glass to be used in the window panels. It depends on factors as given below:

- Location of the building (wind zone),
- Construction patterns around buildings (terrain category),
- Topography of site,
- Building plan and height, etc.

Net design wind pressure (P_{net}), may be defined using the following equation:

$$P_{net} = p_z \cdot C_p$$

where

C_p = net pressure coefficient as per good practice [C-8(15)]; and

p_z = design wind pressure as per good practice [C-8(15)], in N/m^2 .
 $= 0.6 V_z^2$

$$V_z = V_b.k_1.k_2.k_3.k_4$$

where

V_b = basic wind speed based on location {as per Fig. 1 of good practice [C-8(15)]};

k_1 = risk co-efficient factor;

k_2 = terrain factor;

k_3 = topography factor;

k_4 = importance factor for the cyclonic region; and

V_z = design wind speed, in m/s.

6.1.4.1 Laminated/tempered/insulating glass

To determine the thickness of laminated/ tempered/ insulating glass, the design wind pressure P_{net} is modified by dividing it with the strength or pressure factor P_f , dependent on the type of glass. The values of the P_f are given in [Table 14](#).

Using the modified values of P_{net} , the thickness of other glass types can be obtained in accordance with [6.1.3](#).

Table 14 Strength Factor, P_f [Clauses 6.1.3.1(b) , 6.1.4.1 , 6.3.5 and E-1.1.1]		
Sl No. (1)	Glass Type (2)	P_f (3)
i)	Normal (Annealed)	1.00
ii)	Laminated	0.80
iii)	Tempered	2.50
iv)	Insulated	1.50
v)	Heat strengthened	1.60
vi)	Wired	0.50

6.1.5 Aspect Ratio (AR_{max})

The design of the thickness using empirical relation in accordance with [6.1.3](#) shall be valid up to a limiting aspect ratio AR_{max} . The value of AR_{max} for different thickness (T) of glass is given in [Table 15](#).

Table 15 Thickness and Corresponding AR_{max} Values (Clause 6.1.5 and E-1.1.1)										
T , in mm	3	4	5	6	8	10	12	15	19	25
AR_{max}	7.3	6.8	6.5	6.3	5.9	4.9	4.3	3.8	3.3	2.9

6.1.6 Seismic Design

6.1.6.1 Seismic load (EQ)

The design for seismic forces shall be performed in steps to establish upper and lower bound force and movement response parameters. In addition, the response of the structure sensitive to certain parameters such as the damping and the stiffness of the support points may be evaluated.

The principal design criteria for the seismic design are,

- a) Structure will remain elastic during the design seismic event.
- b) ‘Hangers’ will stiffen the cantilever ‘tree’ structure.
- c) ‘Hangers’ will not go slack.

6.1.6.2 Architectural elements and utilities

Depending on response sensitivity, architectural elements and utilities can be classified as deformation sensitive, acceleration sensitive, or both deformation and acceleration sensitive. [Table 16](#) classifies non-structural elements according to their response sensitivity.

Table 16 Response Sensitivity of Architectural Component <i>(Clause 6.1.6.2)</i>			
SI No.	Component	Sensitivity	
		Acceleration Sensitive	Deformation Sensitive
(1)	(2)	(3)	(4)
i)	Exterior skin:		
	a) Adhered veneer	S	P
	b) Anchored veneer	S	P
	c) Glass blocks	S	P
	d) Prefabricated panel	S	P
	e) Glazing system	S	P
ii)	Partitions:		
	a) Heavy	S	P
	b) Light	S	P
iii)	Interior veneers:		
	a) Stone, including marble	S	P
	b) Ceramic tile	S	P
iv)	Parapets and appendages	P	—
v)	Canopies and marquees	P	—
vi)	Stairs	P	S
NOTES			
1 P = Primary response			
2 S = Secondary response			

6.1.6.3 Design seismic force

Design seismic force on the architectural elements and utilities shall be calculated as per Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS.

6.1.6.4 Seismic relative displacement

- a) For two connection points on the same structure A , one at a height h_x , and other at a height h_y , seismic relative displacement shall be determined as:

$$D_p = \delta_{xA} - \delta_{yA}$$

$$D_p \text{ is not required to be taken as greater than } R(h_x - h_y) \frac{\Delta_{aA}}{h_{sx}}$$

where

δ_{xA} = deflection at building level x of structure A due to design seismic load determined by elastic analysis, and multiplied by response reduction factor R of the building as per Table 9 of good practice [C-8(16)];

δ_{yA} = deflection at building level y of structure A due to design seismic load determined by elastic analysis, and multiplied by response reduction factor R of the building as per Table 9 of good practice [C-8(16)];

h_x = height of level x to which upper connection point is attached;

h_y = height of level y to which lower connection point is attached;

Δ_{aA} = allowable storey drift for structure A calculated as per 7.11.1.1 of good practice [C-8(16)]; and

h_{sx} = storey height below level x .

- b) For two connection points on separate structures A and B , or separate structural systems, one at height, h_x and the other at a height, h_y , D_p shall be determined as:

$$D_p = |\delta_{xA}| + |\delta_{yA}|$$

D_p is not required to be taken as greater than

$$R \left(h_x \frac{\Delta_{aA}}{h_{sx}} + h_y \frac{\Delta_{aB}}{h_{sy}} \right)$$

where

δ_{yB} = deflection at building level y of structure B due to design seismic load determined by elastic analysis, and multiplied by response reduction factor, R , of the building as per Table 9 of good practice [C-8(16)]; and

Δ_{aB} = allowable storey drift for structure B calculated as per 7.11.1.1 of good practice [C-8(16)].

6.1.7 Imposed Loads

The uniform distributed load and point load is to be considered based on application and is to be selected in accordance with good practice [C-8(17)]. Appropriate uniform distributed load (UDL) and concentrated load are selected in accordance to good practice [C-8(17)] based on nature of building (for example, residential, institutional, industrial building, etc) and nature of activity under each building type.

6.1.8 Load Contribution

6.1.8.1 Wind load

The wind load is the effect of wind pressure which is dependent on the location, building height, topography and ground roughness factor. The wind speed is taken from good practice [C-8(15)] for the geographic location.

6.1.8.2 Dead load

In case of inclined sloped glazing, the dead weight of the glazing should be considered. In case of component of sloped glazing, the following corresponding component of the total glazing may be used:

- Point load shall be applied as a uniformly distributed load over a circular area of 0.01 m²;
- For a glazed panel supported on all edges, the point loading shall be applied at the centre;
- For free glazing edges, the point load shall be applied adjacent to the centre of the free edge; and
- Ultimate limit state design wind pressure shall not be greater than 1.2 kPa. For greater design wind pressures, the glass shall be separately designed to withstand wind loading in accordance with [6.1.4](#).

6.1.9 Reference Chart Relating Wind Load, Maximum Allowable Area for a Thickness of Glass with Respect to Terrain Category

[Table 17](#) to [Table 25](#) provides relation between design wind pressure and maximum allowable area for a thickness of glass with respect to specific terrain category. User may select the type of glass depending on the support conditions.

Specialist literature may be consulted for use of non-linear analysis.

NOTE — Specialist literature shall mean any national/international publication, codes or peer reviewed research papers on the topic.

The deflection at the centre of glass pane can be calculated using the formula in accordance to [Annex D](#).

Typical model calculation to calculate thickness and determine wind load and deflection at the centre of the glazing is given at [Annex E](#).

Table 17 Maximum Areas for Annealed Glass Fixed on All Four Sides, m²

(Clauses [6.1.9](#) and [6.3.6.1](#))

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	500	2.13	3.86	6.03	8.63	13.06	15.00	15.00	15.00	15.00
ii)	550	1.93	3.51	5.48	7.85	11.88	15.00	15.00	15.00	15.00
iii)	600	1.77	3.21	5.02	7.19	10.89	14.61	15.00	15.00	15.00
iv)	650	1.64	2.97	4.64	6.64	10.05	13.48	15.00	15.00	15.00
v)	700	1.52	2.75	4.31	6.17	9.33	12.52	15.00	15.00	15.00
vi)	750	1.42	2.57	4.02	5.76	8.71	11.69	14.75	15.00	15.00
vii)	800	1.33	2.41	3.77	5.40	8.16	10.96	13.83	15.00	15.00
viii)	850	1.25	2.27	3.55	5.08	7.68	10.31	13.01	15.00	15.00
ix)	900	1.18	2.14	3.35	4.80	7.26	9.74	12.29	15.00	15.00
x)	950	1.12	2.03	3.17	4.54	6.88	9.23	11.64	15.00	15.00
xi)	1 000	1.06	1.93	3.01	4.32	6.53	8.76	11.06	15.00	15.00
xii)	1 050	1.01	1.84	2.87	4.11	6.22	8.35	10.53	14.38	15.00
xiii)	1 100	0.97	1.75	2.74	3.92	5.94	7.97	10.06	13.73	15.00
xiv)	1 150	0.93	1.68	2.62	3.75	5.68	7.62	9.62	13.13	15.00

Table 17 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Glass, <i>T</i>								
		mm								
(1)	(2)	3	4	5	6	8	10	12	15	19
(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)		
xv)	1 200	0.89	1.61	2.51	3.60	5.44	7.30	9.22	12.58	15.00
xvi)	1 250	0.85	1.54	2.41	3.45	5.23	7.01	8.85	12.08	15.00
xvii)	1 300	0.82	1.48	2.32	3.32	5.02	6.74	8.51	11.61	15.00
xviii)	1 350	0.79	1.43	2.23	3.20	4.84	6.49	8.19	11.18	14.98
xix)	1 400	0.76	1.38	2.15	3.08	4.67	6.26	7.90	10.78	14.44
xx)	1 450	0.73	1.33	2.08	2.98	4.50	6.04	7.63	10.41	13.95
xxi)	1 500	0.71	1.29	2.01	2.88	4.35	5.84	7.37	10.07	13.48
xxii)	1 550	0.69	1.24	1.94	2.78	4.21	5.65	7.14	9.74	13.05
xxiii)	1 600	0.67	1.21	1.88	2.70	4.08	5.48	6.91	9.44	12.64
xxiv)	1 650	0.64	1.17	1.83	2.62	3.96	5.31	6.70	9.15	12.25
xxv)	1 700	0.63	1.13	1.77	2.54	3.84	5.16	6.51	8.88	11.89
xxvi)	1 750	0.61	1.10	1.72	2.47	3.73	5.01	6.32	8.63	11.55
xxvii)	1 800	0.59	1.07	1.67	2.40	3.63	4.87	6.14	8.39	11.23
xxviii)	1 850	0.58	1.04	1.63	2.33	3.53	4.74	5.98	8.16	10.93
xxix)	1 900	0.56	1.01	1.59	2.27	3.44	4.61	5.82	7.95	10.64
xxx)	1 950	0.55	0.99	1.55	2.21	3.35	4.49	5.67	7.74	10.37
xxxi)	2 000	0.53	0.96	1.51	2.16	3.27	4.38	5.53	7.55	10.11
xxxii)	2 050	0.52	0.94	1.47	2.11	3.19	4.28	5.40	7.36	9.86
xxxiii)	2 100	0.51	0.92	1.44	2.06	3.11	4.17	5.27	7.19	9.63
xxxiv)	2 150	0.49	0.90	1.40	2.01	3.04	4.08	5.14	7.02	9.40
xxxv)	2 200	0.48	0.88	1.37	1.96	2.97	3.98	5.03	6.86	9.19
xxxvi)	2 250	0.47	0.86	1.34	1.92	2.90	3.90	4.92	6.71	8.99
xxxvii)	2 300	0.46	0.84	1.31	1.88	2.84	3.81	4.81	6.56	8.79
xxxviii)	2 350	0.45	0.82	1.28	1.84	2.78	3.73	4.71	6.42	8.60
xxxix)	2 400	0.44	0.80	1.26	1.80	2.72	3.65	4.61	6.29	8.43
xl)	2 450	0.43	0.79	1.23	1.76	2.67	3.58	4.51	6.16	8.25
xli)	2 500	0.43	0.77	1.21	1.73	2.61	3.51	4.42	6.04	8.09
xlii)	2 550	0.42	0.76	1.18	1.69	2.56	3.44	4.34	5.92	7.93
xliii)	2 600	0.41	0.74	1.16	1.66	2.51	3.37	4.25	5.81	7.78
xliv)	2 650	0.40	0.73	1.14	1.63	2.46	3.31	4.17	5.70	7.63
xl v)	2 700	0.39	0.71	1.12	1.60	2.42	3.25	4.10	5.59	7.49
xlvi)	2 750	0.39	0.70	1.10	1.57	2.38	3.19	4.02	5.49	7.35
xl vii)	2 800	0.38	0.69	1.08	1.54	2.33	3.13	3.95	5.39	7.22

Table 17 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Glass, <i>T</i>								
		mm								
(1)	(2)	3	4	5	6	8	10	12	15	19
(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)		
xlvi)	2 850	0.37	0.68	1.06	1.51	2.29	3.08	3.88	5.30	7.09
xlix)	2 900	0.37	0.66	1.04	1.49	2.25	3.02	3.81	5.21	6.97
l)	2 950	0.36	0.65	1.02	1.46	2.21	2.97	3.75	5.12	6.85
li)	3 000	0.35	0.64	1.00	1.44	2.18	2.92	3.69	5.03	6.74
lii)	3 050	0.35	0.63	0.99	1.42	2.14	2.87	3.63	4.95	6.63
liii)	3 100	0.34	0.62	0.97	1.39	2.11	2.83	3.57	4.87	6.52
liv)	3 150	0.34	0.61	0.96	1.37	2.07	2.78	3.51	4.79	6.42
lv)	3 200	0.33	0.60	0.94	1.35	2.04	2.74	3.46	4.72	6.32
lvi)	3 250	0.33	0.59	0.93	1.33	2.01	2.70	3.40	4.65	6.22
lvii)	3 300	0.32	0.58	0.91	1.31	1.98	2.66	3.35	4.58	6.13
lviii)	3 350	0.32	0.58	0.90	1.29	1.95	2.62	3.30	4.51	6.04
lix)	3 400	0.31	0.57	0.89	1.27	1.92	2.58	3.25	4.44	5.95
lx)	3 450	0.31	0.56	0.87	1.25	1.89	2.54	3.21	4.38	5.86
lxi)	3 500	0.30	0.55	0.86	1.23	1.87	2.50	3.16	4.31	5.78
lxii)	3 550	0.30	0.54	0.85	1.22	1.84	2.47	3.12	4.25	5.70
lxiii)	3 600	0.30	0.54	0.84	1.20	1.81	2.43	3.07	4.19	5.62
lxiv)	3 650	0.29	0.53	0.83	1.18	1.79	2.40	3.03	4.14	5.54
lxv)	3 700	0.29	0.52	0.81	1.17	1.77	2.37	2.99	4.08	5.47
lxvi)	3 750	0.28	0.51	0.80	1.15	1.74	2.34	2.95	4.03	5.39
lxvii)	3 800	0.28	0.51	0.79	1.14	1.72	2.31	2.91	3.97	5.32
lxviii)	3 850	0.28	0.50	0.78	1.12	1.70	2.28	2.87	3.92	5.25
lxix)	3 900	0.27	0.49	0.77	1.11	1.67	2.25	2.84	3.87	5.18
lxx)	3 950	0.27	0.49	0.76	1.09	1.65	2.22	2.80	3.82	5.12
lxxi)	4 000	0.27	0.48	0.75	1.08	1.63	2.19	2.77	3.77	5.06
lxxii)	4 050	0.26	0.48	0.74	1.07	1.61	2.16	2.73	3.73	4.99
lxxiii)	4 100	0.26	0.47	0.74	1.05	1.59	2.14	2.70	3.68	4.93
lxxiv)	4 150	0.26	0.46	0.73	1.04	1.57	2.11	2.67	3.64	4.87
lxxv)	4 200	0.25	0.46	0.72	1.03	1.56	2.09	2.63	3.59	4.81
lxxvi)	4 250	0.25	0.45	0.71	1.02	1.54	2.06	2.60	3.55	4.76
lxxvii)	4 300	0.25	0.45	0.70	1.00	1.52	2.04	2.57	3.51	4.70
lxxviii)	4 350	0.24	0.44	0.69	0.99	1.50	2.01	2.54	3.47	4.65
lxxix)	4 400	0.24	0.44	0.69	0.98	1.48	1.99	2.51	3.43	4.60
lxxx)	4 450	0.24	0.43	0.68	0.97	1.47	1.97	2.49	3.39	4.54

Table 17 (Concluded)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
lxxxix)	4 500	0.24	0.43	0.67	0.96	1.45	1.95	2.46	3.36	4.49
lxxxii)	4 550	0.23	0.42	0.66	0.95	1.44	1.93	2.43	3.32	4.44
lxxxiii)	4 600	0.23	0.42	0.66	0.94	1.42	1.91	2.40	3.28	4.40
lxxxiv)	4 650	0.23	0.41	0.65	0.93	1.40	1.88	2.38	3.25	4.35
lxxxv)	4 700	0.23	0.41	0.64	0.92	1.39	1.86	2.35	3.21	4.30
lxxxvi)	4 750	0.22	0.41	0.63	0.91	1.38	1.85	2.33	3.18	4.26
lxxxvii)	4 800	0.22	0.40	0.63	0.90	1.36	1.83	2.30	3.15	4.21
lxxxviii)	4 850	0.22	0.40	0.62	0.89	1.35	1.81	2.28	3.11	4.17
lxxxix)	4 900	0.22	0.39	0.62	0.88	1.33	1.79	2.26	3.08	4.13
xc)	4 950	0.21	0.39	0.61	0.87	1.32	1.77	2.23	3.05	4.08
xc)	5 000	0.21	0.39	0.60	0.86	1.31	1.75	2.21	3.02	4.04
xcii)	Maximum aspect ratio	7.3	6.8	6.5	6.3	5.9	4.9	4.3	3.8	3.3

Table 18 Maximum Areas for Laminated Glass Fixed on All Four Sides, m²

(Clauses [6.1.9](#), [6.3.5](#) and [6.3.6.1](#))

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Laminated Glass, <i>T</i>					
		mm					
		5.38	6.38	8.38	10.38	12.38	16.38
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	500	4.47	6.28	10.45	14.02	15.00	15.00
ii)	550	4.06	5.71	9.50	12.75	15.00	15.00
iii)	600	3.72	5.23	8.71	11.69	15.00	15.00
iv)	650	3.44	4.83	8.04	10.79	13.93	15.00
v)	700	3.19	4.48	7.46	10.02	12.94	15.00
vi)	750	2.98	4.19	6.97	9.35	12.08	15.00
vii)	800	2.79	3.92	6.53	8.76	11.32	15.00
viii)	850	2.63	3.69	6.15	8.25	10.66	15.00
ix)	900	2.48	3.49	5.81	7.79	10.06	14.21

Table 18 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Laminated Glass, <i>T</i>					
		mm					
		5.38	6.38	8.38	10.38	12.38	16.38
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
x)	950	2.35	3.30	5.50	7.38	9.53	13.46
xi)	1 000	2.23	3.14	5.23	7.01	9.06	12.79
xii)	1 050	2.13	2.99	4.98	6.68	8.63	12.18
xiii)	1 100	2.03	2.85	4.75	6.37	8.23	11.63
xiv)	1 150	1.94	2.73	4.54	6.10	7.88	11.12
xv)	1 200	1.86	2.62	4.35	5.84	7.55	10.66
xvi)	1 250	1.79	2.51	4.18	5.61	7.25	10.23
xvii)	1 300	1.72	2.41	4.02	5.39	6.97	9.84
xviii)	1 350	1.66	2.33	3.87	5.19	6.71	9.47
xix)	1 400	1.60	2.24	3.73	5.01	6.47	9.14
xx)	1 450	1.54	2.16	3.60	4.84	6.25	8.82
xxi)	1 500	1.49	2.09	3.48	4.67	6.04	8.53
xxii)	1 550	1.44	2.03	3.37	4.52	5.84	8.25
xxiii)	1 600	1.40	1.96	3.27	4.38	5.66	7.99
xxiv)	1 650	1.35	1.90	3.17	4.25	5.49	7.75
xxv)	1 700	1.31	1.85	3.07	4.12	5.33	7.52
xxvi)	1 750	1.28	1.79	2.99	4.01	5.18	7.31
xxvii)	1 800	1.24	1.74	2.90	3.90	5.03	7.11
xxviii)	1 850	1.21	1.70	2.82	3.79	4.90	6.91
xxix)	1 900	1.18	1.65	2.75	3.69	4.77	6.73
xxx)	1 950	1.15	1.61	2.68	3.60	4.64	6.56
xxxi)	2 000	1.12	1.57	2.61	3.51	4.53	6.40
xxxii)	2 050	1.09	1.53	2.55	3.42	4.42	6.24
xxxiii)	2 100	1.06	1.49	2.49	3.34	4.31	6.09
xxxiv)	2 150	1.04	1.46	2.43	3.26	4.21	5.95
xxxv)	2 200	1.02	1.43	2.38	3.19	4.12	5.81
xxxvi)	2 250	0.99	1.40	2.32	3.12	4.03	5.68
xxxvii)	2 300	0.97	1.36	2.27	3.05	3.94	5.56

Table 18 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Laminated Glass, <i>T</i>					
		mm					
		5.38	6.38	8.38	10.38	12.38	16.38
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
xxxviii)	2 350	0.95	1.34	2.22	2.98	3.85	5.44
xxxix)	2 400	0.93	1.31	2.18	2.92	3.77	5.33
xl)	2 450	0.91	1.28	2.13	2.86	3.70	5.22
xli)	2 500	0.89	1.26	2.09	2.80	3.62	5.12
xl ii)	2 550	0.88	1.23	2.05	2.75	3.55	5.02
xl iii)	2 600	0.86	1.21	2.01	2.70	3.48	4.92
xl iv)	2 650	0.84	1.18	1.97	2.65	3.42	4.83
xl v)	2 700	0.83	1.16	1.94	2.60	3.35	4.74
xl vi)	2 750	0.81	1.14	1.90	2.55	3.29	4.65
xl vii)	2 800	0.80	1.12	1.87	2.50	3.23	4.57
xl viii)	2 850	0.78	1.10	1.83	2.46	3.18	4.49
xl ix)	2 900	0.77	1.08	1.80	2.42	3.12	4.41
l)	2 950	0.76	1.06	1.77	2.38	3.07	4.34
li)	3 000	0.74	1.05	1.74	2.34	3.02	4.26
li i)	3 050	0.73	1.03	1.71	2.30	2.97	4.19
li ii)	3 100	0.72	1.01	1.69	2.26	2.92	4.13
li v)	3 150	0.71	1.00	1.66	2.23	2.88	4.06
li v)	3 200	0.70	0.98	1.63	2.19	2.83	4.00
li vi)	3 250	0.69	0.97	1.61	2.16	2.79	3.94
li vii)	3 300	0.68	0.95	1.58	2.12	2.74	3.88
li viii)	3 350	0.67	0.94	1.56	2.09	2.70	3.82
li x)	3 400	0.66	0.92	1.54	2.06	2.66	3.76
li x)	3 450	0.65	0.91	1.51	2.03	2.63	3.71
li xi)	3 500	0.64	0.90	1.49	2.00	2.59	3.65
li xii)	3 550	0.63	0.88	1.47	1.98	2.55	3.60
li xiii)	3 600	0.62	0.87	1.45	1.95	2.52	3.55
li xiv)	3 650	0.61	0.86	1.43	1.92	2.48	3.50
li xv)	3 700	0.60	0.85	1.41	1.90	2.45	3.46

Table 18 (Concluded)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Laminated Glass, <i>T</i>					
		mm					
		5.38	6.38	8.38	10.38	12.38	16.38
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
lxvi)	3 750	0.60	0.84	1.39	1.87	2.42	3.41
lxvii)	3 800	0.59	0.83	1.38	1.85	2.38	3.37
lxviii)	3 850	0.58	0.82	1.36	1.82	2.35	3.32
lxix)	3 900	0.57	0.80	1.34	1.80	2.32	3.28
lxx)	3 950	0.57	0.79	1.32	1.78	2.29	3.24
lxxi)	4 000	0.56	0.78	1.31	1.75	2.26	3.20
lxxii)	4 050	0.55	0.78	1.29	1.73	2.24	3.16
lxxiii)	4 100	0.55	0.77	1.27	1.71	2.21	3.12
lxxiv)	4 150	0.54	0.76	1.26	1.69	2.18	3.08
lxxv)	4 200	0.53	0.75	1.24	1.67	2.16	3.05
lxxvi)	4 250	0.53	0.74	1.23	1.65	2.13	3.01
lxxvii)	4 300	0.52	0.73	1.22	1.63	2.11	2.97
lxxviii)	4 350	0.51	0.72	1.20	1.61	2.08	2.94
lxxix)	4 400	0.51	0.71	1.19	1.59	2.06	2.91
lxxx)	4 450	0.50	0.71	1.17	1.58	2.04	2.87
lxxxi)	4 500	0.50	0.70	1.16	1.56	2.01	2.84
lxxxii)	4 550	0.49	0.69	1.15	1.54	1.99	2.81
lxxxiii)	4 600	0.49	0.68	1.14	1.52	1.97	2.78
lxxxiv)	4 650	0.48	0.68	1.12	1.51	1.95	2.75
lxxxv)	4 700	0.48	0.67	1.11	1.49	1.93	2.72
lxxxvi)	4 750	0.47	0.66	1.10	1.48	1.91	2.69
lxxxvii)	4 800	0.47	0.65	1.09	1.46	1.89	2.66
lxxxviii)	4 850	0.46	0.65	1.08	1.45	1.87	2.64
lxxxix)	4 900	0.46	0.64	1.07	1.43	1.85	2.61
xc)	4 950	0.45	0.63	1.06	1.42	1.83	2.58
xc i)	5 000	0.45	0.63	1.05	1.40	1.81	2.56
xcii)	Maximum aspect ratio	6.5	6.2	5.7	4.8	4.3	3.6

Table 19 Maximum Areas for Tempered Glass Fixed on All Four Sides, m²

(Clauses [6.1.9](#) and [6.3.6.1](#))

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Tempered Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	500	5.32	9.64	15.00	15.00	15.00	15.00	15.00	15.00	15.00
ii)	550	4.84	8.76	13.70	15.00	15.00	15.00	15.00	15.00	15.00
iii)	600	4.43	8.03	12.56	15.00	15.00	15.00	15.00	15.00	15.00
iv)	650	4.09	7.42	11.59	15.00	15.00	15.00	15.00	15.00	15.00
v)	700	3.80	6.89	10.77	15.00	15.00	15.00	15.00	15.00	15.00
vi)	750	3.55	6.43	10.05	14.39	15.00	15.00	15.00	15.00	15.00
vii)	800	3.33	6.03	9.42	13.49	15.00	15.00	15.00	15.00	15.00
viii)	850	3.13	5.67	8.87	12.70	15.00	15.00	15.00	15.00	15.00
ix)	900	2.96	5.36	8.37	11.99	15.00	15.00	15.00	15.00	15.00
x)	950	2.80	5.07	7.93	11.36	15.00	15.00	15.00	15.00	15.00
xi)	1 000	2.66	4.82	7.54	10.79	15.00	15.00	15.00	15.00	15.00
xii)	1 050	2.53	4.59	7.18	10.28	15.00	15.00	15.00	15.00	15.00
xiii)	1 100	2.42	4.38	6.85	9.81	14.84	15.00	15.00	15.00	15.00
xiv)	1 150	2.31	4.19	6.55	9.38	14.20	15.00	15.00	15.00	15.00
xv)	1 200	2.22	4.02	6.28	8.99	13.61	15.00	15.00	15.00	15.00
xvi)	1 250	2.13	3.86	6.03	8.63	13.06	15.00	15.00	15.00	15.00
xvii)	1 300	2.05	3.71	5.80	8.30	12.56	15.00	15.00	15.00	15.00
xviii)	1 350	1.97	3.57	5.58	7.99	12.10	15.00	15.00	15.00	15.00
xix)	1 400	1.90	3.44	5.38	7.71	11.66	15.00	15.00	15.00	15.00
xx)	1 450	1.83	3.32	5.20	7.44	11.26	15.00	15.00	15.00	15.00
xxi)	1 500	1.77	3.21	5.02	7.19	10.89	14.61	15.00	15.00	15.00
xxii)	1 550	1.72	3.11	4.86	6.96	10.53	14.14	15.00	15.00	15.00
xxiii)	1 600	1.66	3.01	4.71	6.74	10.21	13.69	15.00	15.00	15.00
xxiv)	1 650	1.61	2.92	4.57	6.54	9.90	13.28	15.00	15.00	15.00
xxv)	1 700	1.56	2.84	4.43	6.35	9.61	12.89	15.00	15.00	15.00
xxvi)	1 750	1.52	2.75	4.31	6.17	9.33	12.52	15.00	15.00	15.00
xxvii)	1 800	1.48	2.68	4.19	6.00	9.07	12.17	15.00	15.00	15.00
xxviii)	1 850	1.44	2.61	4.07	5.83	8.83	11.84	14.95	15.00	15.00

Table 19 (Continued)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Tempered Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
xxix)	1 900	1.40	2.54	3.97	5.68	8.59	11.53	14.55	15.00	15.00
xxx)	1 950	1.36	2.47	3.86	5.53	8.37	11.24	14.18	15.00	15.00
xxxi)	2 000	1.33	2.41	3.77	5.40	8.16	10.96	13.83	15.00	15.00
xxxii)	2 050	1.30	2.35	3.68	5.26	7.97	10.69	13.49	15.00	15.00
xxxiii)	2 100	1.27	2.30	3.59	5.14	7.78	10.43	13.17	15.00	15.00
xxxiv)	2 150	1.24	2.24	3.51	5.02	7.59	10.19	12.86	15.00	15.00
xxxv)	2 200	1.21	2.19	3.43	4.91	7.42	9.96	12.57	15.00	15.00
xxxvi)	2 250	1.18	2.14	3.35	4.80	7.26	9.74	12.29	15.00	15.00
xxxvii)	2 300	1.16	2.10	3.28	4.69	7.10	9.53	12.02	15.00	15.00
xxxviii)	2 350	1.13	2.05	3.21	4.59	6.95	9.32	11.77	15.00	15.00
xxxix)	2 400	1.11	2.01	3.14	4.50	6.80	9.13	11.52	15.00	15.00
xl)	2 450	1.09	1.97	3.08	4.40	6.66	8.94	11.29	15.00	15.00
xli)	2 500	1.06	1.93	3.01	4.32	6.53	8.76	11.06	15.00	15.00
xl ii)	2 550	1.04	1.89	2.96	4.23	6.40	8.59	10.84	14.80	15.00
xl iii)	2 600	1.02	1.85	2.90	4.15	6.28	8.43	10.64	14.52	15.00
xl iv)	2 650	1.00	1.82	2.84	4.07	6.16	8.27	10.43	14.24	15.00
xl v)	2 700	0.99	1.79	2.79	4.00	6.05	8.12	10.24	13.98	15.00
xl vi)	2 750	0.97	1.75	2.74	3.92	5.94	7.97	10.06	13.73	15.00
xl vii)	2 800	0.95	1.72	2.69	3.85	5.83	7.83	9.88	13.48	15.00
xl viii)	2 850	0.93	1.69	2.64	3.79	5.73	7.69	9.70	13.24	15.00
xl ix)	2 900	0.92	1.66	2.60	3.72	5.63	7.56	9.54	13.02	15.00
l)	2 950	0.90	1.63	2.55	3.66	5.54	7.43	9.37	12.79	15.00
li)	3 000	0.89	1.61	2.51	3.60	5.44	7.30	9.22	12.58	15.00
li i)	3 050	0.87	1.58	2.47	3.54	5.35	7.18	9.07	12.38	15.00
li ii)	3 100	0.86	1.56	2.43	3.48	5.27	7.07	8.92	12.18	15.00
li v)	3 150	0.84	1.53	2.39	3.43	5.18	6.96	8.78	11.98	15.00
li v)	3 200	0.83	1.51	2.36	3.37	5.10	6.85	8.64	11.79	15.00
li vi)	3 250	0.82	1.48	2.32	3.32	5.02	6.74	8.51	11.61	15.00
li vii)	3 300	0.81	1.46	2.28	3.27	4.95	6.64	8.38	11.44	15.00
li viii)	3 350	0.79	1.44	2.25	3.22	4.87	6.54	8.25	11.27	15.00

Table 19 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Tempered Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
lix)	3 400	0.78	1.42	2.22	3.17	4.80	6.44	8.13	11.10	14.87
lx)	3 450	0.77	1.40	2.18	3.13	4.73	6.35	8.02	10.94	14.65
lxi)	3 500	0.76	1.38	2.15	3.08	4.67	6.26	7.90	10.78	14.44
lxii)	3 550	0.75	1.36	2.12	3.04	4.60	6.17	7.79	10.63	14.24
lxiii)	3 600	0.74	1.34	2.09	3.00	4.54	6.09	7.68	10.48	14.04
lxiv)	3 650	0.73	1.32	2.06	2.96	4.47	6.00	7.58	10.34	13.85
lxv)	3 700	0.72	1.30	2.04	2.92	4.41	5.92	7.47	10.20	13.66
lxvi)	3 750	0.71	1.29	2.01	2.88	4.35	5.84	7.37	10.07	13.48
lxvii)	3 800	0.70	1.27	1.98	2.84	4.30	5.77	7.28	9.93	13.30
lxviii)	3 850	0.69	1.25	1.96	2.80	4.24	5.69	7.18	9.80	13.13
lxix)	3 900	0.68	1.24	1.93	2.77	4.19	5.62	7.09	9.68	12.96
lxx)	3 950	0.67	1.22	1.91	2.73	4.13	5.55	7.00	9.56	12.80
lxxi)	4 000	0.67	1.21	1.88	2.70	4.08	5.48	6.91	9.44	12.64
lxxii)	4 050	0.66	1.19	1.86	2.66	4.03	5.41	6.83	9.32	12.48
lxxiii)	4 100	0.65	1.18	1.84	2.63	3.98	5.34	6.74	9.21	12.33
lxxiv)	4 150	0.64	1.16	1.82	2.60	3.93	5.28	6.66	9.09	12.18
lxxv)	4 200	0.63	1.15	1.79	2.57	3.89	5.22	6.58	8.99	12.04
lxxvi)	4 250	0.63	1.13	1.77	2.54	3.84	5.16	6.51	8.88	11.89
lxxvii)	4 300	0.62	1.12	1.75	2.51	3.80	5.10	6.43	8.78	11.76
lxxviii)	4 350	0.61	1.11	1.73	2.48	3.75	5.04	6.36	8.68	11.62
lxxix)	4 400	0.60	1.10	1.71	2.45	3.71	4.98	6.28	8.58	11.49
lxxx)	4 450	0.60	1.08	1.69	2.43	3.67	4.92	6.21	8.48	11.36
lxxxi)	4 500	0.59	1.07	1.67	2.40	3.63	4.87	6.14	8.39	11.23
lxxxii)	4 550	0.58	1.06	1.66	2.37	3.59	4.82	6.08	8.30	11.11
lxxxiii)	4 600	0.58	1.05	1.64	2.35	3.55	4.76	6.01	8.21	10.99
lxxxiv)	4 650	0.57	1.04	1.62	2.32	3.51	4.71	5.95	8.12	10.87
lxxxv)	4 700	0.57	1.03	1.60	2.30	3.47	4.66	5.88	8.03	10.76
lxxxvi)	4 750	0.56	1.01	1.59	2.27	3.44	4.61	5.82	7.95	10.64
lxxxvii)	4 800	0.55	1.00	1.57	2.25	3.40	4.56	5.76	7.86	10.53
lxxxviii)	4 850	0.55	0.99	1.55	2.23	3.37	4.52	5.70	7.78	10.42

Table 19 (Concluded)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Tempered Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
lxxxix)	4 900	0.54	0.98	1.54	2.20	3.33	4.47	5.64	7.70	10.32
xc)	4 950	0.54	0.97	1.52	2.18	3.30	4.43	5.59	7.63	10.21
xc i)	5 000	0.53	0.96	1.51	2.16	3.27	4.38	5.53	7.55	10.11
xc ii)	Maximum aspect ratio	7.3	6.8	6.5	6.3	5.9	4.9	4.3	3.8	3.3

Table 20 Maximum Areas for Insulating Glass Fixed on All Four Sides, m²

(Clauses [6.1.9](#) and [6.3.6.1](#))

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Glass, <i>T</i>						
		mm						
		3 + 3	4 + 4	5 + 5	6 + 6	8 + 8	10 + 10	12 + 12
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	500	3.19	5.78	9.04	12.95	15.00	15.00	15.00
ii)	550	2.90	5.26	8.22	11.77	15.00	15.00	15.00
iii)	600	2.66	4.82	7.54	10.79	15.00	15.00	15.00
iv)	650	2.46	4.45	6.96	9.96	15.00	15.00	15.00
v)	700	2.28	4.13	6.46	9.25	14.00	15.00	15.00
vi)	750	2.13	3.86	6.03	8.63	13.06	15.00	15.00
vii)	800	2.00	3.62	5.65	8.09	12.25	15.00	15.00
viii)	850	1.88	3.40	5.32	7.62	11.53	15.00	15.00
ix)	900	1.77	3.21	5.02	7.19	10.89	14.61	15.00
x)	950	1.68	3.04	4.76	6.82	10.31	13.84	15.00
xi)	1 000	1.60	2.89	4.52	6.48	9.80	13.15	15.00
xii)	1 050	1.52	2.75	4.31	6.17	9.33	12.52	15.00
xiii)	1 100	1.45	2.63	4.11	5.89	8.91	11.95	15.00
xiv)	1 150	1.39	2.52	3.93	5.63	8.52	11.43	14.43
xv)	1 200	1.33	2.41	3.77	5.40	8.16	10.96	13.83
xvi)	1 250	1.28	2.31	3.62	5.18	7.84	10.52	13.27
xvii)	1 300	1.23	2.22	3.48	4.98	7.54	10.11	12.76
xviii)	1 350	1.18	2.14	3.35	4.80	7.26	9.74	12.29

Table 20 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Glass, <i>T</i>						
		mm						
		3 + 3	4 + 4	5 + 5	6 + 6	8 + 8	10 + 10	12 + 12
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xix)	1 400	1.14	2.07	3.23	4.63	7.00	9.39	11.85
xx)	1 450	1.10	1.99	3.12	4.47	6.76	9.07	11.44
xxi)	1 500	1.06	1.93	3.01	4.32	6.53	8.76	11.06
xxii)	1 550	1.03	1.87	2.92	4.18	6.32	8.48	10.70
xxiii)	1 600	1.00	1.81	2.83	4.05	6.12	8.22	10.37
xxiv)	1 650	0.97	1.75	2.74	3.92	5.94	7.97	10.06
xxv)	1 700	0.94	1.70	2.66	3.81	5.76	7.73	9.76
xxvi)	1 750	0.91	1.65	2.58	3.70	5.60	7.51	9.48
xxvii)	1 800	0.89	1.61	2.51	3.60	5.44	7.30	9.22
xxviii)	1 850	0.86	1.56	2.44	3.50	5.30	7.11	8.97
xxix)	1 900	0.84	1.52	2.38	3.41	5.16	6.92	8.73
xxx)	1 950	0.82	1.48	2.32	3.32	5.02	6.74	8.51
xxxi)	2 000	0.80	1.45	2.26	3.24	4.90	6.57	8.30
xxxii)	2 050	0.78	1.41	2.21	3.16	4.78	6.41	8.09
xxxiii)	2 100	0.76	1.38	2.15	3.08	4.67	6.26	7.90
xxxiv)	2 150	0.74	1.35	2.10	3.01	4.56	6.11	7.72
xxxv)	2 200	0.73	1.31	2.06	2.94	4.45	5.98	7.54
xxxvi)	2 250	0.71	1.29	2.01	2.88	4.35	5.84	7.37
xxxvii)	2 300	0.69	1.26	1.97	2.82	4.26	5.72	7.21
xxxviii)	2 350	0.68	1.23	1.92	2.76	4.17	5.59	7.06
xxxix)	2 400	0.67	1.21	1.88	2.70	4.08	5.48	6.91
xl)	2 450	0.65	1.18	1.85	2.64	4.00	5.37	6.77
xli)	2 500	0.64	1.16	1.81	2.59	3.92	5.26	6.64
xlii)	2 550	0.63	1.13	1.77	2.54	3.84	5.16	6.51
xliii)	2 600	0.61	1.11	1.74	2.49	3.77	5.06	6.38
xliv)	2 650	0.60	1.09	1.71	2.44	3.70	4.96	6.26
xlv)	2 700	0.59	1.07	1.67	2.40	3.63	4.87	6.14
xlvi)	2 750	0.58	1.05	1.64	2.35	3.56	4.78	6.03
xlvii)	2 800	0.57	1.03	1.61	2.31	3.50	4.70	5.93
xlviii)	2 850	0.56	1.01	1.59	2.27	3.44	4.61	5.82

Table 20 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Glass, <i>T</i>						
		mm						
		3 + 3	4 + 4	5 + 5	6 + 6	8 + 8	10 + 10	12 + 12
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xlix)	2 900	0.55	1.00	1.56	2.23	3.38	4.53	5.72
l)	2 950	0.54	0.98	1.53	2.19	3.32	4.46	5.62
li)	3 000	0.53	0.96	1.51	2.16	3.27	4.38	5.53
lii)	3 050	0.52	0.95	1.48	2.12	3.21	4.31	5.44
liii)	3 100	0.51	0.93	1.46	2.09	3.16	4.24	5.35
liv)	3 150	0.51	0.92	1.44	2.06	3.11	4.17	5.27
lv)	3 200	0.50	0.90	1.41	2.02	3.06	4.11	5.18
lvi)	3 250	0.49	0.89	1.39	1.99	3.01	4.05	5.11
lvii)	3 300	0.48	0.88	1.37	1.96	2.97	3.98	5.03
lviii)	3 350	0.48	0.86	1.35	1.93	2.92	3.92	4.95
lix)	3 400	0.47	0.85	1.33	1.90	2.88	3.87	4.88
lx)	3 450	0.46	0.84	1.31	1.88	2.84	3.81	4.81
lxi)	3 500	0.46	0.83	1.29	1.85	2.80	3.76	4.74
lxii)	3 550	0.45	0.81	1.27	1.82	2.76	3.70	4.67
lxiii)	3 600	0.44	0.80	1.26	1.80	2.72	3.65	4.61
lxiv)	3 650	0.44	0.79	1.24	1.77	2.68	3.60	4.55
lxv)	3 700	0.43	0.78	1.22	1.75	2.65	3.55	4.48
lxvi)	3 750	0.43	0.77	1.21	1.73	2.61	3.51	4.42
lxvii)	3 800	0.42	0.76	1.19	1.70	2.58	3.46	4.37
lxviii)	3 850	0.41	0.75	1.17	1.68	2.54	3.41	4.31
lxix)	3 900	0.41	0.74	1.16	1.66	2.51	3.37	4.25
lxx)	3 950	0.40	0.73	1.14	1.64	2.48	3.33	4.20
lxxi)	4 000	0.40	0.72	1.13	1.62	2.45	3.29	4.15
lxxii)	4 050	0.39	0.71	1.12	1.60	2.42	3.25	4.10
lxxiii)	4 100	0.39	0.71	1.10	1.58	2.39	3.21	4.05
lxxiv)	4 150	0.38	0.70	1.09	1.56	2.36	3.17	4.00
lxxv)	4 200	0.38	0.69	1.08	1.54	2.33	3.13	3.95
lxxvi)	4 250	0.38	0.68	1.06	1.52	2.31	3.09	3.90
lxxvii)	4 300	0.37	0.67	1.05	1.51	2.28	3.06	3.86
lxxviii)	4 350	0.37	0.66	1.04	1.49	2.25	3.02	3.81

Table 20 (Concluded)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Glass, <i>T</i>						
		mm						
		3 + 3	4 + 4	5 + 5	6 + 6	8 + 8	10 + 10	12 + 12
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
lxxix)	4 400	0.36	0.66	1.03	1.47	2.23	2.99	3.77
lxxx)	4 450	0.36	0.65	1.02	1.46	2.20	2.95	3.73
lxxxi)	4 500	0.35	0.64	1.00	1.44	2.18	2.92	3.69
lxxxii)	4 550	0.35	0.64	0.99	1.42	2.15	2.89	3.65
lxxxiii)	4 600	0.35	0.63	0.98	1.41	2.13	2.86	3.61
lxxxiv)	4 650	0.34	0.62	0.97	1.39	2.11	2.83	3.57
lxxxv)	4 700	0.34	0.62	0.96	1.38	2.08	2.80	3.53
lxxxvi)	4 750	0.34	0.61	0.95	1.36	2.06	2.77	3.49
lxxxvii)	4 800	0.33	0.60	0.94	1.35	2.04	2.74	3.46
lxxxviii)	4 850	0.33	0.60	0.93	1.34	2.02	2.71	3.42
lxxxix)	4 900	0.33	0.59	0.92	1.32	2.00	2.68	3.39
xc)	4 950	0.32	0.58	0.91	1.31	1.98	2.66	3.35
xc i)	5 000	0.32	0.58	0.90	1.30	1.96	2.63	3.32
xc ii)	Maximum aspect ratio	7.3	6.8	6.5	6.3	5.9	4.9	4.3

Table 21 Maximum Area for Heat Strengthened Glass Fixed on Four Sides, m²

(Clauses [6.1.9](#), [6.3.5](#) and [6.3.6.1](#))

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Heat Strengthened Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	500	3.41	6.17	9.65	13.81	15.00	15.00	15.00	15.00	15.00
ii)	550	3.10	5.61	8.77	12.56	15.00	15.00	15.00	15.00	15.00
iii)	600	2.84	5.14	8.04	11.51	15.00	15.00	15.00	15.00	15.00
iv)	650	2.62	4.75	7.42	10.63	15.00	15.00	15.00	15.00	15.00
v)	700	2.43	4.41	6.89	9.87	14.93	15.00	15.00	15.00	15.00
vi)	750	2.27	4.11	6.43	9.21	13.93	15.00	15.00	15.00	15.00
vii)	800	2.13	3.86	6.03	8.63	13.06	15.00	15.00	15.00	15.00

Table 21 (Continued)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Heat Strengthened Glass, <i>T</i>								
		mm								
(1)	(2)	3	4	5	6	8	10	12	15	19
(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
viii)	850	2.00	3.63	5.67	8.13	12.29	15.00	15.00	15.00	15.00
ix)	900	1.89	3.43	5.36	7.67	11.61	15.00	15.00	15.00	15.00
x)	950	1.79	3.25	5.08	7.27	11.00	14.76	15.00	15.00	15.00
xi)	1 000	1.70	3.09	4.82	6.91	10.45	14.02	15.00	15.00	15.00
xii)	1 050	1.62	2.94	4.59	6.58	9.95	13.36	15.00	15.00	15.00
xiii)	1 100	1.55	2.80	4.38	6.28	9.50	12.75	15.00	15.00	15.00
xiv)	1 150	1.48	2.68	4.19	6.01	9.09	12.19	15.00	15.00	15.00
xv)	1 200	1.42	2.57	4.02	5.76	8.71	11.69	14.75	15.00	15.00
xvi)	1 250	1.36	2.47	3.86	5.53	8.36	11.22	14.16	15.00	15.00
xvii)	1 300	1.31	2.37	3.71	5.31	8.04	10.79	13.61	15.00	15.00
xviii)	1 350	1.26	2.29	3.57	5.12	7.74	10.39	13.11	15.00	15.00
xix)	1 400	1.22	2.20	3.45	4.93	7.46	10.02	12.64	15.00	15.00
xx)	1 450	1.17	2.13	3.33	4.76	7.21	9.67	12.21	15.00	15.00
xxi)	1 500	1.14	2.06	3.22	4.60	6.97	9.35	11.80	15.00	15.00
xxii)	1 550	1.10	1.99	3.11	4.46	6.74	9.05	11.42	15.00	15.00
xxiii)	1 600	1.06	1.93	3.01	4.32	6.53	8.76	11.06	15.00	15.00
xxiv)	1 650	1.03	1.87	2.92	4.19	6.33	8.50	10.73	14.64	15.00
xxv)	1 700	1.00	1.81	2.84	4.06	6.15	8.25	10.41	14.21	15.00
xxvi)	1 750	0.97	1.76	2.76	3.95	5.97	8.01	10.11	13.80	15.00
xxvii)	1 800	0.95	1.71	2.68	3.84	5.81	7.79	9.83	13.42	15.00
xxviii)	1 850	0.92	1.67	2.61	3.73	5.65	7.58	9.57	13.06	15.00
xxix)	1 900	0.90	1.62	2.54	3.64	5.50	7.38	9.31	12.71	15.00
xxx)	1 950	0.87	1.58	2.47	3.54	5.36	7.19	9.08	12.39	15.00
xxxi)	2 000	0.85	1.54	2.41	3.45	5.23	7.01	8.85	12.08	15.00
xxxii)	2 050	0.83	1.50	2.35	3.37	5.10	6.84	8.63	11.78	15.00
xxxiii)	2 100	0.81	1.47	2.30	3.29	4.98	6.68	8.43	11.50	15.00
xxxiv)	2 150	0.79	1.43	2.24	3.21	4.86	6.52	8.23	11.24	15.00
xxxv)	2 200	0.77	1.40	2.19	3.14	4.75	6.37	8.04	10.98	14.71
xxxvi)	2 250	0.76	1.37	2.14	3.07	4.64	6.23	7.87	10.74	14.38
xxxvii)	2 300	0.74	1.34	2.10	3.00	4.54	6.10	7.69	10.50	14.07

Table 21 (Continued)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Heat Strengthened Glass, <i>T</i>								
		mm								
(1)	(2)	3	4	5	6	8	10	12	15	19
(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)		
xxxviii)	2 350	0.72	1.31	2.05	2.94	4.45	5.97	7.53	10.28	13.77
xxxix)	2 400	0.71	1.29	2.01	2.88	4.35	5.84	7.37	10.07	13.48
xl)	2 450	0.69	1.26	1.97	2.82	4.27	5.72	7.22	9.86	13.21
xli)	2 500	0.68	1.23	1.93	2.76	4.18	5.61	7.08	9.66	12.94
xl ii)	2 550	0.67	1.21	1.89	2.71	4.10	5.50	6.94	9.47	12.69
xl iii)	2 600	0.65	1.19	1.86	2.66	4.02	5.39	6.81	9.29	12.44
xl iv)	2 650	0.64	1.16	1.82	2.61	3.94	5.29	6.68	9.12	12.21
xl v)	2 700	0.63	1.14	1.79	2.56	3.87	5.19	6.55	8.95	11.98
xl vi)	2 750	0.62	1.12	1.75	2.51	3.80	5.10	6.44	8.78	11.76
xl vii)	2 800	0.61	1.10	1.72	2.47	3.73	5.01	6.32	8.63	11.55
xl viii)	2 850	0.60	1.08	1.69	2.42	3.67	4.92	6.21	8.48	11.35
xl ix)	2 900	0.59	1.06	1.66	2.38	3.60	4.84	6.10	8.33	11.16
l)	2 950	0.58	1.05	1.63	2.34	3.54	4.75	6.00	8.19	10.97
li)	3 000	0.57	1.03	1.61	2.30	3.48	4.67	5.90	8.05	10.78
li i)	3 050	0.56	1.01	1.58	2.26	3.43	4.60	5.80	7.92	10.61
li ii)	3 100	0.55	1.00	1.56	2.23	3.37	4.52	5.71	7.79	10.44
li v)	3 150	0.54	0.98	1.53	2.19	3.32	4.45	5.62	7.67	10.27
li v)	3 200	0.53	0.96	1.51	2.16	3.27	4.38	5.53	7.55	10.11
li vi)	3 250	0.52	0.95	1.48	2.13	3.22	4.31	5.45	7.43	9.95
li vii)	3 300	0.52	0.93	1.46	2.09	3.17	4.25	5.36	7.32	9.80
li viii)	3 350	0.51	0.92	1.44	2.06	3.12	4.19	5.28	7.21	9.66
li x)	3 400	0.50	0.91	1.42	2.03	3.07	4.12	5.21	7.10	9.52
li x)	3 450	0.49	0.89	1.40	2.00	3.03	4.06	5.13	7.00	9.38
li xi)	3 500	0.49	0.88	1.38	1.97	2.99	4.01	5.06	6.90	9.24
li xi i)	3 550	0.48	0.87	1.36	1.95	2.94	3.95	4.99	6.80	9.11
li xi ii)	3 600	0.47	0.86	1.34	1.92	2.90	3.90	4.92	6.71	8.99
li xi iv)	3 650	0.47	0.85	1.32	1.89	2.86	3.84	4.85	6.62	8.86
li xi v)	3 700	0.46	0.83	1.30	1.87	2.82	3.79	4.78	6.53	8.74
li xi vi)	3 750	0.45	0.82	1.29	1.84	2.79	3.74	4.72	6.44	8.63
li xi vii)	3 800	0.45	0.81	1.27	1.82	2.75	3.69	4.66	6.36	8.51

Table 21 (Concluded)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Heat Strengthened Glass, <i>T</i>								
		mm								
(1)	(2)	3	4	5	6	8	10	12	15	19
(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)		
lxviii)	3 850	0.44	0.80	1.25	1.79	2.71	3.64	4.60	6.27	8.40
lxix)	3 900	0.44	0.79	1.24	1.77	2.68	3.60	4.54	6.19	8.30
lxx)	3 950	0.43	0.78	1.22	1.75	2.65	3.55	4.48	6.12	8.19
lxxi)	4 000	0.43	0.77	1.21	1.73	2.61	3.51	4.42	6.04	8.09
lxxii)	4 050	0.42	0.76	1.19	1.71	2.58	3.46	4.37	5.96	7.99
lxxiii)	4 100	0.42	0.75	1.18	1.68	2.55	3.42	4.32	5.89	7.89
lxxiv)	4 150	0.41	0.74	1.16	1.66	2.52	3.38	4.26	5.82	7.80
lxxv)	4 200	0.41	0.73	1.15	1.64	2.49	3.34	4.21	5.75	7.70
lxxvi)	4 250	0.40	0.73	1.13	1.63	2.46	3.30	4.16	5.68	7.61
lxxvii)	4 300	0.40	0.72	1.12	1.61	2.43	3.26	4.12	5.62	7.52
lxxviii)	4 350	0.39	0.71	1.11	1.59	2.40	3.22	4.07	5.55	7.44
lxxix)	4 400	0.39	0.70	1.10	1.57	2.38	3.19	4.02	5.49	7.35
lxxx)	4 450	0.38	0.69	1.08	1.55	2.35	3.15	3.98	5.43	7.27
lxxxii)	4 500	0.38	0.69	1.07	1.53	2.32	3.12	3.93	5.37	7.19
lxxxiii)	4 550	0.37	0.68	1.06	1.52	2.30	3.08	3.89	5.31	7.11
lxxxiv)	4 600	0.37	0.67	1.05	1.50	2.27	3.05	3.85	5.25	7.03
lxxxv)	4 650	0.37	0.66	1.04	1.49	2.25	3.02	3.81	5.19	6.96
lxxxvi)	4 700	0.36	0.66	1.03	1.47	2.22	2.98	3.77	5.14	6.88
lxxxvii)	4 750	0.36	0.65	1.02	1.45	2.20	2.95	3.73	5.09	6.81
lxxxviii)	4 800	0.35	0.64	1.00	1.44	2.18	2.92	3.69	5.03	6.74
lxxxix)	4 850	0.35	0.64	0.99	1.42	2.15	2.89	3.65	4.98	6.67
lxxxix)	4 900	0.35	0.63	0.98	1.41	2.13	2.86	3.61	4.93	6.60
xc)	4 950	0.34	0.62	0.97	1.40	2.11	2.83	3.58	4.88	6.54
xcii)	5 000	0.34	0.62	0.96	1.38	2.09	2.80	3.54	4.83	6.47
xcii)	Maximum aspect ratio	7.3	6.8	6.5	6.3	5.9	4.9	4.3	3.8	3.3

Table 22 Maximum Span for Annealed Glass Fixed on Two Opposite Sides, m

(Clauses [6.1.9](#) and [6.3.6.1](#))

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Annealed Glass, <i>T</i>								
		mm								
(1)	(2)	3	4	5	6	8	10	12	15	19
(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)		
i)	500	0.53	0.73	0.92	1.12	1.40	1.77	2.11	2.68	3.36
ii)	550	0.51	0.69	0.88	1.07	1.33	1.69	2.02	2.56	3.20
iii)	600	0.48	0.66	0.84	1.02	1.27	1.62	1.93	2.45	3.07
iv)	650	0.46	0.64	0.81	0.98	1.22	1.56	1.85	2.35	2.95
v)	700	0.45	0.61	0.78	0.95	1.18	1.50	1.79	2.26	2.84
vi)	750	0.43	0.59	0.75	0.91	1.14	1.45	1.73	2.19	2.74
vii)	800	0.42	0.57	0.73	0.88	1.10	1.40	1.67	2.12	2.66
viii)	850	0.41	0.56	0.71	0.86	1.07	1.36	1.62	2.06	2.58
ix)	900	0.40	0.54	0.69	0.83	1.04	1.32	1.58	2.00	2.50
x)	950	0.38	0.53	0.67	0.81	1.01	1.29	1.53	1.94	2.44
xi)	1 000	0.37	0.51	0.65	0.79	0.99	1.25	1.49	1.89	2.38
xii)	1 050	0.37	0.50	0.64	0.77	0.96	1.22	1.46	1.85	2.32
xiii)	1 100	0.36	0.49	0.62	0.75	0.94	1.20	1.43	1.81	2.26
xiv)	1 150	0.35	0.48	0.61	0.74	0.92	1.17	1.39	1.77	2.22
xv)	1 200	0.34	0.47	0.60	0.72	0.90	1.15	1.36	1.73	2.17
xvi)	1 250	0.34	0.46	0.58	0.71	0.88	1.12	1.34	1.69	2.12
xvii)	1 300	0.33	0.45	0.57	0.69	0.87	1.10	1.31	1.66	2.08
xviii)	1 350	0.32	0.44	0.56	0.68	0.85	1.08	1.29	1.63	2.04
xix)	1 400	0.32	0.43	0.55	0.67	0.83	1.06	1.26	1.60	2.01
xx)	1 450	0.31	0.43	0.54	0.66	0.82	1.04	1.24	1.57	1.97
xxi)	1 500	0.31	0.42	0.53	0.65	0.81	1.02	1.22	1.55	1.94
xxii)	1 550	0.30	0.41	0.52	0.64	0.79	1.01	1.20	1.52	1.91
xxiii)	1 600	0.30	0.41	0.52	0.63	0.78	0.99	1.18	1.50	1.88
xxiv)	1 650	0.29	0.40	0.51	0.62	0.77	0.98	1.16	1.48	1.85
xxv)	1 700	0.29	0.39	0.50	0.61	0.76	0.96	1.15	1.45	1.82
xxvi)	1 750	0.28	0.39	0.49	0.60	0.75	0.95	1.13	1.43	1.80
xxvii)	1 800	0.28	0.38	0.49	0.59	0.74	0.93	1.11	1.41	1.77
xxviii)	1 850	0.28	0.38	0.48	0.58	0.73	0.92	1.10	1.39	1.75
xxix)	1 900	0.27	0.37	0.47	0.57	0.72	0.91	1.08	1.37	1.72
xxx)	1 950	0.27	0.37	0.47	0.57	0.71	0.90	1.07	1.36	1.70

Table 22 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Annealed Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
xxxix)	2 000	0.27	0.36	0.46	0.56	0.70	0.89	1.06	1.34	1.68
xxxix)	2 050	0.26	0.36	0.46	0.55	0.69	0.88	1.04	1.32	1.66
xxxix)	2 100	0.26	0.35	0.45	0.55	0.68	0.87	1.03	1.31	1.64
xxxix)	2 150	0.26	0.35	0.44	0.54	0.67	0.86	1.02	1.29	1.62
xxxix)	2 200	0.25	0.35	0.44	0.53	0.67	0.85	1.01	1.28	1.60
xxxix)	2 250	0.25	0.34	0.43	0.53	0.66	0.84	1.00	1.26	1.58
xxxix)	2 300	0.25	0.34	0.43	0.52	0.65	0.83	0.99	1.25	1.57
xxxix)	2 350	0.24	0.34	0.43	0.52	0.64	0.82	0.97	1.24	1.55
xxxix)	2 400	0.24	0.33	0.42	0.51	0.64	0.81	0.96	1.22	1.53
xl)	2 450	0.24	0.33	0.42	0.51	0.63	0.80	0.95	1.21	1.52
xli)	2 500	0.24	0.32	0.41	0.50	0.62	0.79	0.95	1.20	1.50
xlii)	2 550	0.23	0.32	0.41	0.50	0.62	0.79	0.94	1.19	1.49
xliii)	2 600	0.23	0.32	0.40	0.49	0.61	0.78	0.93	1.18	1.47
xliv)	2 650	0.23	0.32	0.40	0.49	0.61	0.77	0.92	1.16	1.46
xlvi)	2 700	0.23	0.31	0.40	0.48	0.60	0.76	0.91	1.15	1.45
xlvi)	2 750	0.23	0.31	0.39	0.48	0.60	0.76	0.90	1.14	1.43
xlvi)	2 800	0.22	0.31	0.39	0.47	0.59	0.75	0.89	1.13	1.42
xlvi)	2 850	0.22	0.30	0.39	0.47	0.58	0.74	0.89	1.12	1.41
xlvi)	2 900	0.22	0.30	0.38	0.46	0.58	0.74	0.88	1.11	1.39
l)	2 950	0.22	0.30	0.38	0.46	0.57	0.73	0.87	1.10	1.38
li)	3 000	0.22	0.30	0.38	0.46	0.57	0.72	0.86	1.09	1.37
lii)	3 050	0.21	0.29	0.37	0.45	0.57	0.72	0.86	1.09	1.36
liii)	3 100	0.21	0.29	0.37	0.45	0.56	0.71	0.85	1.08	1.35
liv)	3 150	0.21	0.29	0.37	0.45	0.56	0.71	0.84	1.07	1.34
lv)	3 200	0.21	0.29	0.36	0.44	0.55	0.70	0.84	1.06	1.33
lvi)	3 250	0.21	0.28	0.36	0.44	0.55	0.70	0.83	1.05	1.32
lvii)	3 300	0.21	0.28	0.36	0.44	0.54	0.69	0.82	1.04	1.31
lviii)	3 350	0.20	0.28	0.36	0.43	0.54	0.69	0.82	1.04	1.30
lix)	3 400	0.20	0.28	0.35	0.43	0.54	0.68	0.81	1.03	1.29
lx)	3 450	0.20	0.28	0.35	0.43	0.53	0.68	0.80	1.02	1.28

Table 22 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Annealed Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
lxi)	3 500	0.20	0.27	0.35	0.42	0.53	0.67	0.80	1.01	1.27
lxii)	3 550	0.20	0.27	0.35	0.42	0.52	0.67	0.79	1.01	1.26
lxiii)	3 600	0.20	0.27	0.34	0.42	0.52	0.66	0.79	1.00	1.25
lxiv)	3 650	0.20	0.27	0.34	0.41	0.52	0.66	0.78	0.99	1.24
lxv)	3 700	0.19	0.27	0.34	0.41	0.51	0.65	0.78	0.99	1.23
lxvi)	3 750	0.19	0.27	0.34	0.41	0.51	0.65	0.77	0.98	1.23
lxvii)	3 800	0.19	0.26	0.33	0.41	0.51	0.64	0.77	0.97	1.22
lxviii)	3 850	0.19	0.26	0.33	0.40	0.50	0.64	0.76	0.97	1.21
lxix)	3 900	0.19	0.26	0.33	0.40	0.50	0.64	0.76	0.96	1.20
lxx)	3 950	0.19	0.26	0.33	0.40	0.50	0.63	0.75	0.95	1.20
lxxi)	4 000	0.19	0.26	0.33	0.40	0.49	0.63	0.75	0.95	1.19
lxxii)	4 050	0.19	0.26	0.32	0.39	0.49	0.62	0.74	0.94	1.18
lxxiii)	4 100	0.19	0.25	0.32	0.39	0.49	0.62	0.74	0.94	1.17
lxxiv)	4 150	0.18	0.25	0.32	0.39	0.48	0.62	0.73	0.93	1.17
lxxv)	4 200	0.18	0.25	0.32	0.39	0.48	0.61	0.73	0.92	1.16
lxxvi)	4 250	0.18	0.25	0.32	0.38	0.48	0.61	0.72	0.92	1.15
lxxvii)	4 300	0.18	0.25	0.31	0.38	0.48	0.60	0.72	0.91	1.15
lxxviii)	4 350	0.18	0.25	0.31	0.38	0.47	0.60	0.72	0.91	1.14
lxxix)	4 400	0.18	0.24	0.31	0.38	0.47	0.60	0.71	0.90	1.13
lxxx)	4 450	0.18	0.24	0.31	0.38	0.47	0.59	0.71	0.90	1.13
lxxxi)	4 500	0.18	0.24	0.31	0.37	0.47	0.59	0.70	0.89	1.12
lxxxii)	4 550	0.18	0.24	0.31	0.37	0.46	0.59	0.70	0.89	1.11
lxxxiii)	4 600	0.17	0.24	0.30	0.37	0.46	0.58	0.70	0.88	1.11
lxxxiv)	4 650	0.17	0.24	0.30	0.37	0.46	0.58	0.69	0.88	1.10
lxxxv)	4 700	0.17	0.24	0.30	0.36	0.46	0.58	0.69	0.87	1.10
lxxxvi)	4 750	0.17	0.24	0.30	0.36	0.45	0.58	0.69	0.87	1.09
lxxxvii)	4 800	0.17	0.23	0.30	0.36	0.45	0.57	0.68	0.86	1.08
lxxxviii)	4 850	0.17	0.23	0.30	0.36	0.45	0.57	0.68	0.86	1.08
lxxxix)	4 900	0.17	0.23	0.29	0.36	0.45	0.57	0.68	0.86	1.07
xc)	4 950	0.17	0.23	0.29	0.36	0.44	0.56	0.67	0.85	1.07

Table 22 (Concluded)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Annealed Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
xci)	5 000	0.17	0.23	0.29	0.35	0.44	0.56	0.67	0.85	1.06

Table 23 Maximum Span for Laminated Glass Fixed on Two Opposite Sides, m

(Clauses [6.1.9](#), [6.3.5](#) and [6.3.6.1](#))

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Laminated Glass, <i>T</i>					
		mm					
		5.38	6.38	8.38	10.38	12.38	16.38
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	500	0.79	0.95	1.25	1.59	1.92	2.50
ii)	550	0.75	0.90	1.19	1.51	1.83	2.38
iii)	600	0.72	0.87	1.14	1.45	1.76	2.28
iv)	650	0.69	0.83	1.10	1.39	1.69	2.19
v)	700	0.67	0.80	1.06	1.34	1.63	2.11
vi)	750	0.65	0.77	1.02	1.30	1.57	2.04
vii)	800	0.62	0.75	0.99	1.25	1.52	1.98
viii)	850	0.61	0.73	0.96	1.22	1.48	1.92
ix)	900	0.59	0.71	0.93	1.18	1.43	1.86
x)	950	0.57	0.69	0.91	1.15	1.40	1.81
xi)	1 000	0.56	0.67	0.88	1.12	1.36	1.77
xii)	1 050	0.55	0.65	0.86	1.09	1.33	1.72
xiii)	1 100	0.53	0.64	0.84	1.07	1.30	1.68
xiv)	1 150	0.52	0.63	0.82	1.05	1.27	1.65
xv)	1 200	0.51	0.61	0.81	1.02	1.24	1.61
xvi)	1 250	0.50	0.60	0.79	1.00	1.22	1.58
xvii)	1 300	0.49	0.59	0.77	0.98	1.19	1.55
xviii)	1 350	0.48	0.58	0.76	0.97	1.17	1.52
xix)	1 400	0.47	0.57	0.75	0.95	1.15	1.49
xx)	1 450	0.46	0.56	0.73	0.93	1.13	1.47
xxi)	1 500	0.46	0.55	0.72	0.92	1.11	1.44

Table 23 (Continued)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Laminated Glass, <i>T</i>					
		mm					
		5.38	6.38	8.38	10.38	12.38	16.38
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
xxii)	1 550	0.45	0.54	0.71	0.90	1.09	1.42
xxiii)	1 600	0.44	0.53	0.70	0.89	1.08	1.40
xxiv)	1 650	0.43	0.52	0.69	0.87	1.06	1.38
xxv)	1 700	0.43	0.51	0.68	0.86	1.04	1.35
xxvi)	1 750	0.42	0.51	0.67	0.85	1.03	1.34
xxvii)	1 800	0.42	0.50	0.66	0.84	1.01	1.32
xxviii)	1 850	0.41	0.49	0.65	0.82	1.00	1.30
xxix)	1 900	0.41	0.49	0.64	0.81	0.99	1.28
xxx)	1 950	0.40	0.48	0.63	0.80	0.97	1.27
xxxi)	2 000	0.40	0.47	0.62	0.79	0.96	1.25
xxxii)	2 050	0.39	0.47	0.62	0.78	0.95	1.23
xxxiii)	2 100	0.39	0.46	0.61	0.77	0.94	1.22
xxxiv)	2 150	0.38	0.46	0.60	0.77	0.93	1.20
xxxv)	2 200	0.38	0.45	0.60	0.76	0.92	1.19
xxxvi)	2 250	0.37	0.45	0.59	0.75	0.91	1.18
xxxvii)	2 300	0.37	0.44	0.58	0.74	0.90	1.16
xxxviii)	2 350	0.36	0.44	0.58	0.73	0.89	1.15
xxxix)	2 400	0.36	0.43	0.57	0.72	0.88	1.14
xl)	2 450	0.36	0.43	0.56	0.72	0.87	1.13
xli)	2 500	0.35	0.42	0.56	0.71	0.86	1.12
xl ii)	2 550	0.35	0.42	0.55	0.70	0.85	1.11
xl iii)	2 600	0.35	0.42	0.55	0.70	0.84	1.10
xl iv)	2 650	0.34	0.41	0.54	0.69	0.84	1.09
xl v)	2 700	0.34	0.41	0.54	0.68	0.83	1.08
xl vi)	2 750	0.34	0.40	0.53	0.68	0.82	1.07
xl vii)	2 800	0.33	0.40	0.53	0.67	0.81	1.06
xl viii)	2 850	0.33	0.40	0.52	0.66	0.81	1.05
xl ix)	2 900	0.33	0.39	0.52	0.66	0.80	1.04
l)	2 950	0.33	0.39	0.51	0.65	0.79	1.03

Table 23 (Continued)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Laminated Glass, <i>T</i>					
		mm					
		5.38	6.38	8.38	10.38	12.38	16.38
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
li)	3 000	0.32	0.39	0.51	0.65	0.79	1.02
lii)	3 050	0.32	0.38	0.51	0.64	0.78	1.01
liii)	3 100	0.32	0.38	0.50	0.64	0.77	1.00
liv)	3 150	0.31	0.38	0.50	0.63	0.77	1.00
lv)	3 200	0.31	0.37	0.49	0.63	0.76	0.99
lvi)	3 250	0.31	0.37	0.49	0.62	0.75	0.98
lvii)	3 300	0.31	0.37	0.49	0.62	0.75	0.97
lviii)	3 350	0.31	0.37	0.48	0.61	0.74	0.97
lix)	3 400	0.30	0.36	0.48	0.61	0.74	0.96
lx)	3 450	0.30	0.36	0.48	0.60	0.73	0.95
lxi)	3 500	0.30	0.36	0.47	0.60	0.73	0.94
lxii)	3 550	0.30	0.36	0.47	0.60	0.72	0.94
lxiii)	3 600	0.29	0.35	0.47	0.59	0.72	0.93
lxiv)	3 650	0.29	0.35	0.46	0.59	0.71	0.92
lxv)	3 700	0.29	0.35	0.46	0.58	0.71	0.92
lxvi)	3 750	0.29	0.35	0.46	0.58	0.70	0.91
lxvii)	3 800	0.29	0.34	0.45	0.58	0.70	0.91
lxviii)	3 850	0.28	0.34	0.45	0.57	0.69	0.90
lxix)	3 900	0.28	0.34	0.45	0.57	0.69	0.89
lxx)	3 950	0.28	0.34	0.44	0.56	0.68	0.89
lxxi)	4 000	0.28	0.34	0.44	0.56	0.68	0.88
lxxii)	4 050	0.28	0.33	0.44	0.56	0.68	0.88
lxxiii)	4 100	0.28	0.33	0.44	0.55	0.67	0.87
lxxiv)	4 150	0.27	0.33	0.43	0.55	0.67	0.87
lxxv)	4 200	0.27	0.33	0.43	0.55	0.66	0.86
lxxvi)	4 250	0.27	0.33	0.43	0.54	0.66	0.86
lxxvii)	4 300	0.27	0.32	0.43	0.54	0.66	0.85
lxxviii)	4 350	0.27	0.32	0.42	0.54	0.65	0.85
lxxix)	4 400	0.27	0.32	0.42	0.53	0.65	0.84

Table 23 (Concluded)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Laminated Glass, <i>T</i>					
		mm					
		5.38	6.38	8.38	10.38	12.38	16.38
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
lxxx)	4 450	0.26	0.32	0.42	0.53	0.65	0.84
lxxx i)	4 500	0.26	0.32	0.42	0.53	0.64	0.83
lxxx ii)	4 550	0.26	0.31	0.41	0.53	0.64	0.83
lxxx iii)	4 600	0.26	0.31	0.41	0.52	0.63	0.82
lxxx iv)	4 650	0.26	0.31	0.41	0.52	0.63	0.82
lxxx v)	4 700	0.26	0.31	0.41	0.52	0.63	0.81
lxxx vi)	4 750	0.26	0.31	0.41	0.51	0.62	0.81
lxxx vii)	4 800	0.26	0.31	0.40	0.51	0.62	0.81
lxxx viii)	4 850	0.25	0.30	0.40	0.51	0.62	0.80
lxxx ix)	4 900	0.25	0.30	0.40	0.51	0.61	0.80
xc)	4 950	0.25	0.30	0.40	0.50	0.61	0.79
xc i)	5 000	0.25	0.30	0.40	0.50	0.61	0.79

Table 24 Maximum Span for Tempered Glass Fixed on Two Opposite Sides, m

(Clauses [6.1.9](#) and [6.3.6.1](#))

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Tempered Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	500	0.62	0.86	1.09	1.32	1.52	1.93	2.30	2.92	3.66
ii)	550	0.60	0.82	1.04	1.26	1.45	1.84	2.20	2.78	3.49
iii)	600	0.57	0.78	0.99	1.20	1.39	1.76	2.10	2.66	3.34
iv)	650	0.55	0.75	0.95	1.16	1.33	1.69	2.02	2.56	3.21
v)	700	0.53	0.72	0.92	1.11	1.29	1.63	1.95	2.47	3.09
vi)	750	0.51	0.70	0.89	1.08	1.24	1.58	1.88	2.38	2.99
vii)	800	0.49	0.68	0.86	1.04	1.20	1.53	1.82	2.31	2.89
viii)	850	0.48	0.66	0.83	1.01	1.17	1.48	1.77	2.24	2.81
ix)	900	0.47	0.64	0.81	0.98	1.13	1.44	1.72	2.18	2.73

Table 24 (Continued)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Tempered Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
x)	950	0.45	0.62	0.79	0.96	1.10	1.40	1.67	2.12	2.65
xi)	1 000	0.44	0.60	0.77	0.93	1.08	1.37	1.63	2.06	2.59
xii)	1 050	0.43	0.59	0.75	0.91	1.05	1.33	1.59	2.01	2.52
xiii)	1 100	0.42	0.58	0.73	0.89	1.03	1.30	1.55	1.97	2.47
xiv)	1 150	0.41	0.56	0.72	0.87	1.00	1.27	1.52	1.92	2.41
xv)	1 200	0.40	0.55	0.70	0.85	0.98	1.25	1.49	1.88	2.36
xvi)	1 250	0.39	0.54	0.69	0.83	0.96	1.22	1.46	1.85	2.31
xvii)	1 300	0.39	0.53	0.67	0.82	0.94	1.20	1.43	1.81	2.27
xviii)	1 350	0.38	0.52	0.66	0.80	0.93	1.18	1.40	1.78	2.23
xix)	1 400	0.37	0.51	0.65	0.79	0.91	1.15	1.38	1.74	2.19
xx)	1 450	0.37	0.50	0.64	0.77	0.89	1.13	1.35	1.71	2.15
xxi)	1 500	0.36	0.49	0.63	0.76	0.88	1.12	1.33	1.69	2.11
xxii)	1 550	0.35	0.49	0.62	0.75	0.86	1.10	1.31	1.66	2.08
xxiii)	1 600	0.35	0.48	0.61	0.74	0.85	1.08	1.29	1.63	2.05
xxiv)	1 650	0.34	0.47	0.60	0.73	0.84	1.06	1.27	1.61	2.01
xxv)	1 700	0.34	0.46	0.59	0.71	0.82	1.05	1.25	1.58	1.98
xxvi)	1 750	0.33	0.46	0.58	0.70	0.81	1.03	1.23	1.56	1.96
xxvii)	1 800	0.33	0.45	0.57	0.69	0.80	1.02	1.21	1.54	1.93
xxviii)	1 850	0.32	0.44	0.56	0.68	0.79	1.00	1.20	1.52	1.90
xxix)	1 900	0.32	0.44	0.56	0.68	0.78	0.99	1.18	1.50	1.88
xxx)	1 950	0.32	0.43	0.55	0.67	0.77	0.98	1.17	1.48	1.85
xxxi)	2 000	0.31	0.43	0.54	0.66	0.76	0.97	1.15	1.46	1.83
xxxii)	2 050	0.31	0.42	0.54	0.65	0.75	0.95	1.14	1.44	1.81
xxxiii)	2 100	0.30	0.42	0.53	0.64	0.74	0.94	1.12	1.42	1.79
xxxiv)	2 150	0.30	0.41	0.52	0.64	0.73	0.93	1.11	1.41	1.76
xxxv)	2 200	0.30	0.41	0.52	0.63	0.73	0.92	1.10	1.39	1.74
xxxvi)	2 250	0.29	0.40	0.51	0.62	0.72	0.91	1.09	1.38	1.72
xxxvii)	2 300	0.29	0.40	0.51	0.61	0.71	0.90	1.07	1.36	1.71
xxxviii)	2 350	0.29	0.39	0.50	0.61	0.70	0.89	1.06	1.35	1.69
xxxix)	2 400	0.28	0.39	0.50	0.60	0.69	0.88	1.05	1.33	1.67

Table 24 (Continued)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Tempered Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
xl)	2 450	0.28	0.39	0.49	0.60	0.69	0.87	1.04	1.32	1.65
xli)	2 500	0.28	0.38	0.49	0.59	0.68	0.86	1.03	1.31	1.64
xl ii)	2 550	0.28	0.38	0.48	0.58	0.67	0.86	1.02	1.29	1.62
xl iii)	2 600	0.27	0.38	0.48	0.58	0.67	0.85	1.01	1.28	1.60
xl iv)	2 650	0.27	0.37	0.47	0.57	0.66	0.84	1.00	1.27	1.59
xl v)	2 700	0.27	0.37	0.47	0.57	0.65	0.83	0.99	1.26	1.57
xl vi)	2 750	0.27	0.36	0.46	0.56	0.65	0.82	0.98	1.24	1.56
xl vii)	2 800	0.26	0.36	0.46	0.56	0.64	0.82	0.97	1.23	1.55
xl viii)	2 850	0.26	0.36	0.46	0.55	0.64	0.81	0.96	1.22	1.53
xl ix)	2 900	0.26	0.36	0.45	0.55	0.63	0.80	0.96	1.21	1.52
l)	2 950	0.26	0.35	0.45	0.54	0.63	0.80	0.95	1.20	1.51
li)	3 000	0.25	0.35	0.44	0.54	0.62	0.79	0.94	1.19	1.49
li i)	3 050	0.25	0.35	0.44	0.53	0.62	0.78	0.93	1.18	1.48
li ii)	3 100	0.25	0.34	0.44	0.53	0.61	0.78	0.92	1.17	1.47
li v)	3 150	0.25	0.34	0.43	0.52	0.61	0.77	0.92	1.16	1.46
li v)	3 200	0.25	0.34	0.43	0.52	0.60	0.76	0.91	1.15	1.45
li vi)	3 250	0.24	0.34	0.43	0.52	0.60	0.76	0.90	1.14	1.44
li vii)	3 300	0.24	0.33	0.42	0.51	0.59	0.75	0.90	1.14	1.42
li viii)	3 350	0.24	0.33	0.42	0.51	0.59	0.75	0.89	1.13	1.41
li x)	3 400	0.24	0.33	0.42	0.51	0.58	0.74	0.88	1.12	1.40
li x)	3 450	0.24	0.33	0.41	0.50	0.58	0.74	0.88	1.11	1.39
li xi)	3 500	0.24	0.32	0.41	0.50	0.57	0.73	0.87	1.10	1.38
li xii)	3 550	0.23	0.32	0.41	0.49	0.57	0.73	0.86	1.10	1.37
li xiii)	3 600	0.23	0.32	0.40	0.49	0.57	0.72	0.86	1.09	1.36
li xiv)	3 650	0.23	0.32	0.40	0.49	0.56	0.72	0.85	1.08	1.35
li xv)	3 700	0.23	0.31	0.40	0.48	0.56	0.71	0.85	1.07	1.34
li xvi)	3 750	0.23	0.31	0.40	0.48	0.56	0.71	0.84	1.07	1.34
li xvii)	3 800	0.23	0.31	0.39	0.48	0.55	0.70	0.84	1.06	1.33
li xviii)	3 850	0.22	0.31	0.39	0.47	0.55	0.70	0.83	1.05	1.32
li xix)	3 900	0.22	0.31	0.39	0.47	0.54	0.69	0.82	1.05	1.31

Table 24 (Concluded)

SI No.	Design Wind Pressure N/m ²	Nominal Thickness of Tempered Glass, <i>T</i>								
		mm								
(1)	(2)	3	4	5	6	8	10	12	15	19
(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)		
lxx)	3 950	0.22	0.30	0.39	0.47	0.54	0.69	0.82	1.04	1.30
lxxi)	4 000	0.22	0.30	0.38	0.47	0.54	0.68	0.81	1.03	1.29
lxxii)	4 050	0.22	0.30	0.38	0.46	0.53	0.68	0.81	1.03	1.29
lxxiii)	4 100	0.22	0.30	0.38	0.46	0.53	0.67	0.80	1.02	1.28
lxxiv)	4 150	0.22	0.30	0.38	0.46	0.53	0.67	0.80	1.01	1.27
lxxv)	4 200	0.22	0.30	0.37	0.45	0.52	0.67	0.79	1.01	1.26
lxxvi)	4 250	0.21	0.29	0.37	0.45	0.52	0.66	0.79	1.00	1.25
lxxvii)	4 300	0.21	0.29	0.37	0.45	0.52	0.66	0.79	1.00	1.25
lxxviii)	4 350	0.21	0.29	0.37	0.45	0.52	0.66	0.78	0.99	1.24
lxxix)	4 400	0.21	0.29	0.37	0.44	0.51	0.65	0.78	0.98	1.23
lxxx)	4 450	0.21	0.29	0.36	0.44	0.51	0.65	0.77	0.98	1.23
lxxxi)	4 500	0.21	0.29	0.36	0.44	0.51	0.64	0.77	0.97	1.22
lxxxii)	4 550	0.21	0.28	0.36	0.44	0.50	0.64	0.76	0.97	1.21
lxxxiii)	4 600	0.21	0.28	0.36	0.43	0.50	0.64	0.76	0.96	1.21
lxxxiv)	4 650	0.20	0.28	0.36	0.43	0.50	0.63	0.75	0.96	1.20
lxxxv)	4 700	0.20	0.28	0.35	0.43	0.50	0.63	0.75	0.95	1.19
lxxxvi)	4 750	0.20	0.28	0.35	0.43	0.49	0.63	0.75	0.95	1.19
lxxxvii)	4 800	0.20	0.28	0.35	0.43	0.49	0.62	0.74	0.94	1.18
lxxxviii)	4 850	0.20	0.27	0.35	0.42	0.49	0.62	0.74	0.94	1.17
lxxxix)	4 900	0.20	0.27	0.35	0.42	0.49	0.62	0.74	0.93	1.17
xc)	4 950	0.20	0.27	0.35	0.42	0.48	0.61	0.73	0.93	1.16
xc)	5 000	0.20	0.27	0.34	0.42	0.48	0.61	0.73	0.92	1.16

Table 25 Maximum Span for Heat Strengthened Glass Fixed on Two Opposite Sides, m
(Clauses [6.1.9](#), [6.3.5](#) and [6.3.6.1](#))

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Heat Strengthened Glass, <i>T</i> mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	500	0.50	0.68	0.87	1.05	1.22	1.55	1.84	2.34	2.93
ii)	550	0.48	0.65	0.83	1.00	1.16	1.47	1.76	2.23	2.79
iii)	600	0.46	0.62	0.79	0.96	1.11	1.41	1.68	2.13	2.67
iv)	650	0.44	0.60	0.76	0.92	1.07	1.36	1.62	2.05	2.57
v)	700	0.42	0.58	0.73	0.89	1.03	1.31	1.56	1.97	2.47
vi)	750	0.41	0.56	0.71	0.86	0.99	1.26	1.50	1.91	2.39
vii)	800	0.39	0.54	0.69	0.83	0.96	1.22	1.46	1.85	2.31
viii)	850	0.38	0.52	0.67	0.81	0.93	1.19	1.41	1.79	2.24
ix)	900	0.37	0.51	0.65	0.79	0.91	1.15	1.37	1.74	2.18
x)	950	0.36	0.50	0.63	0.76	0.88	1.12	1.34	1.69	2.12
xi)	1 000	0.35	0.48	0.61	0.75	0.86	1.09	1.30	1.65	2.07
xii)	1 050	0.34	0.47	0.60	0.73	0.84	1.07	1.27	1.61	2.02
xiii)	1 100	0.34	0.46	0.59	0.71	0.82	1.04	1.24	1.57	1.97
xiv)	1 150	0.33	0.45	0.57	0.69	0.80	1.02	1.21	1.54	1.93
xv)	1 200	0.32	0.44	0.56	0.68	0.79	1.00	1.19	1.51	1.89
xvi)	1 250	0.32	0.43	0.55	0.67	0.77	0.98	1.16	1.48	1.85
xvii)	1 300	0.31	0.42	0.54	0.65	0.75	0.96	1.14	1.45	1.82
xviii)	1 350	0.30	0.42	0.53	0.64	0.74	0.94	1.12	1.42	1.78
xix)	1 400	0.30	0.41	0.52	0.63	0.73	0.92	1.10	1.40	1.75
xx)	1 450	0.29	0.40	0.51	0.62	0.71	0.91	1.08	1.37	1.72
xxi)	1 500	0.29	0.40	0.50	0.61	0.70	0.89	1.06	1.35	1.69
xxii)	1 550	0.28	0.39	0.49	0.60	0.69	0.88	1.05	1.33	1.66
xxiii)	1 600	0.28	0.38	0.49	0.59	0.68	0.86	1.03	1.31	1.64
xxiv)	1 650	0.27	0.38	0.48	0.58	0.67	0.85	1.01	1.29	1.61
xxv)	1 700	0.27	0.37	0.47	0.57	0.66	0.84	1.00	1.27	1.59
xxvi)	1 750	0.27	0.37	0.46	0.56	0.65	0.83	0.98	1.25	1.56
xxvii)	1 800	0.26	0.36	0.46	0.56	0.64	0.81	0.97	1.23	1.54
xxviii)	1 850	0.26	0.36	0.45	0.55	0.63	0.80	0.96	1.21	1.52
xxix)	1 900	0.26	0.35	0.45	0.54	0.62	0.79	0.94	1.20	1.50
xxx)	1 950	0.25	0.35	0.44	0.53	0.62	0.78	0.93	1.18	1.48

Table 25 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Heat Strengthened Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
xxxi)	2 000	0.25	0.34	0.43	0.53	0.61	0.77	0.92	1.17	1.46
xxxii)	2 050	0.25	0.34	0.43	0.52	0.60	0.76	0.91	1.15	1.45
xxxiii)	2 100	0.24	0.33	0.42	0.51	0.59	0.75	0.90	1.14	1.43
xxxiv)	2 150	0.24	0.33	0.42	0.51	0.59	0.75	0.89	1.13	1.41
xxxv)	2 200	0.24	0.33	0.41	0.50	0.58	0.74	0.88	1.11	1.40
xxxvi)	2 250	0.24	0.32	0.41	0.50	0.57	0.73	0.87	1.10	1.38
xxxvii)	2 300	0.23	0.32	0.41	0.49	0.57	0.72	0.86	1.09	1.36
xxxviii)	2 350	0.23	0.32	0.40	0.49	0.56	0.71	0.85	1.08	1.35
xxxix)	2 400	0.23	0.31	0.40	0.48	0.56	0.71	0.84	1.07	1.34
xl)	2 450	0.23	0.31	0.39	0.48	0.55	0.70	0.83	1.05	1.32
xli)	2 500	0.22	0.31	0.39	0.47	0.54	0.69	0.82	1.04	1.31
xlii)	2 550	0.22	0.30	0.38	0.47	0.54	0.68	0.82	1.03	1.30
xliii)	2 600	0.22	0.30	0.38	0.46	0.53	0.68	0.81	1.02	1.28
xliv)	2 650	0.22	0.30	0.38	0.46	0.53	0.67	0.80	1.01	1.27
xlvi)	2 700	0.21	0.29	0.37	0.45	0.52	0.67	0.79	1.00	1.26
xlvi)	2 750	0.21	0.29	0.37	0.45	0.52	0.66	0.79	1.00	1.25
xlvi)	2 800	0.21	0.29	0.37	0.45	0.51	0.65	0.78	0.99	1.24
xlvi)	2 850	0.21	0.29	0.36	0.44	0.51	0.65	0.77	0.98	1.23
xlvi)	2 900	0.21	0.28	0.36	0.44	0.51	0.64	0.76	0.97	1.22
l)	2 950	0.21	0.28	0.36	0.43	0.50	0.64	0.76	0.96	1.21
li)	3 000	0.20	0.28	0.35	0.43	0.50	0.63	0.75	0.95	1.19
lii)	3 050	0.20	0.28	0.35	0.43	0.49	0.63	0.75	0.95	1.19
liii)	3 100	0.20	0.27	0.35	0.42	0.49	0.62	0.74	0.94	1.18
liv)	3 150	0.20	0.27	0.35	0.42	0.48	0.62	0.73	0.93	1.17
lv)	3 200	0.20	0.27	0.34	0.42	0.48	0.61	0.73	0.92	1.16
lvi)	3 250	0.20	0.27	0.34	0.41	0.48	0.61	0.72	0.92	1.15
lvii)	3 300	0.19	0.27	0.34	0.41	0.47	0.60	0.72	0.91	1.14
lviii)	3 350	0.19	0.26	0.34	0.41	0.47	0.60	0.71	0.90	1.13
lix)	3 400	0.19	0.26	0.33	0.40	0.47	0.59	0.71	0.90	1.12
lx)	3 450	0.19	0.26	0.33	0.40	0.46	0.59	0.70	0.89	1.11

Table 25 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Heat Strengthened Glass, <i>T</i>								
		mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
lxi)	3 500	0.19	0.26	0.33	0.40	0.46	0.58	0.70	0.88	1.11
lxii)	3 550	0.19	0.26	0.33	0.40	0.46	0.58	0.69	0.88	1.10
lxiii)	3 600	0.19	0.25	0.32	0.39	0.45	0.58	0.69	0.87	1.09
lxiv)	3 650	0.18	0.25	0.32	0.39	0.45	0.57	0.68	0.86	1.08
lxv)	3 700	0.18	0.25	0.32	0.39	0.45	0.57	0.68	0.86	1.08
lxvi)	3 750	0.18	0.25	0.32	0.38	0.44	0.56	0.67	0.85	1.07
lxvii)	3 800	0.18	0.25	0.32	0.38	0.44	0.56	0.67	0.85	1.06
lxviii)	3 850	0.18	0.25	0.31	0.38	0.44	0.56	0.66	0.84	1.05
lxix)	3 900	0.18	0.24	0.31	0.38	0.44	0.55	0.66	0.84	1.05
lxx)	3 950	0.18	0.24	0.31	0.37	0.43	0.55	0.66	0.83	1.04
lxxi)	4 000	0.18	0.24	0.31	0.37	0.43	0.55	0.65	0.83	1.03
lxxii)	4 050	0.18	0.24	0.31	0.37	0.43	0.54	0.65	0.82	1.03
lxxiii)	4 100	0.17	0.24	0.30	0.37	0.42	0.54	0.64	0.82	1.02
lxxiv)	4 150	0.17	0.24	0.30	0.37	0.42	0.54	0.64	0.81	1.02
lxxv)	4 200	0.17	0.24	0.30	0.36	0.42	0.53	0.64	0.81	1.01
lxxvi)	4 250	0.17	0.23	0.30	0.36	0.42	0.53	0.63	0.80	1.00
lxxvii)	4 300	0.17	0.23	0.30	0.36	0.41	0.53	0.63	0.80	1.00
lxxviii)	4 350	0.17	0.23	0.29	0.36	0.41	0.52	0.62	0.79	0.99
lxxix)	4 400	0.17	0.23	0.29	0.36	0.41	0.52	0.62	0.79	0.99
lxxx)	4 450	0.17	0.23	0.29	0.35	0.41	0.52	0.62	0.78	0.98
lxxxi)	4 500	0.17	0.23	0.29	0.35	0.41	0.52	0.61	0.78	0.98
lxxxii)	4 550	0.17	0.23	0.29	0.35	0.40	0.51	0.61	0.77	0.97
lxxxiii)	4 600	0.16	0.23	0.29	0.35	0.40	0.51	0.61	0.77	0.97
lxxxiv)	4 650	0.16	0.22	0.28	0.35	0.40	0.51	0.60	0.77	0.96
lxxxv)	4 700	0.16	0.22	0.28	0.34	0.40	0.50	0.60	0.76	0.95
lxxxvi)	4 750	0.16	0.22	0.28	0.34	0.39	0.50	0.60	0.76	0.95
lxxxvii)	4 800	0.16	0.22	0.28	0.34	0.39	0.50	0.59	0.75	0.94
lxxxviii)	4 850	0.16	0.22	0.28	0.34	0.39	0.50	0.59	0.75	0.94
lxxxix)	4 900	0.16	0.22	0.28	0.34	0.39	0.49	0.59	0.75	0.94
xc)	4 950	0.16	0.22	0.28	0.33	0.39	0.49	0.59	0.74	0.93

Table 25 (Continued)

Sl No.	Design Wind Pressure N/m ²	Nominal Thickness of Heat Strengthened Glass, <i>T</i> mm								
		3	4	5	6	8	10	12	15	19
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
xcī)	5 000	0.16	0.22	0.27	0.33	0.38	0.49	0.58	0.74	0.93

6.2 Installation

6.2.1 Basic installation requirements and materials used for glass are given hereunder:

NOTES

- 1 The provisions under this item do not include the use of other methods or systems for glazing, provided the alternate method or system can be demonstrated to satisfy the requirements for correctly supporting the glass within the frame, or glazing system.
- 2 Patent and other proprietary systems are not described/ covered here.
- 3 Installation technique for frameless glazing system is not covered here.

6.2.2 Site Working and Glass Machining Operation

6.2.2.1 Heat-strengthened and toughened glass shall not be cut or worked after heat treatment. All necessary cutting, drilling, notching and edge-working shall be carried out to correct dimensions prior to value addition to the glass, such as toughening/tempering/heat strengthening. The edges and surfaces of all glass types shall not be damaged during fixing.

6.2.3 Dimensional Requirement

The dimensions for edge clearance, edge cover, front and back clearance, rebate depth is selected in accordance with the requirements in [4.5.2](#).

6.2.4 Glazing Materials

6.2.4.1 Compatibility of materials

A glazing material shall be used only when compatible with contiguous materials, including the rebate surface finish, setting or location blocks, distance pieces and glass type.

6.2.4.2 Application of materials

The application of glazing materials shall be in accordance with relevant/available Indian Standard or manufacturer's recommendations.

6.2.4.3 Life expectancy of materials

A glazing material shall only be used where its life expectancy (durability) has been established. The manufacturers' advice/guarantee should be sought for information regarding life expectancy.

6.2.5 Setting Blocks

The number and location of setting blocks shall be as shown in [Fig. 14](#) and [Fig 16](#). Generally, setting blocks shall be,

- a) positioned at quarter points or not less than 30 mm from the corner, whichever is less;
- b) the minimum width of each setting block shall be not less than the glass thickness; and

- c) if a glazing bridge is used then the thickness of the glazing bridge can be added to the thickness of the setting block to achieve the minimum gap of 6 mm between the edge of glass and the framing member.

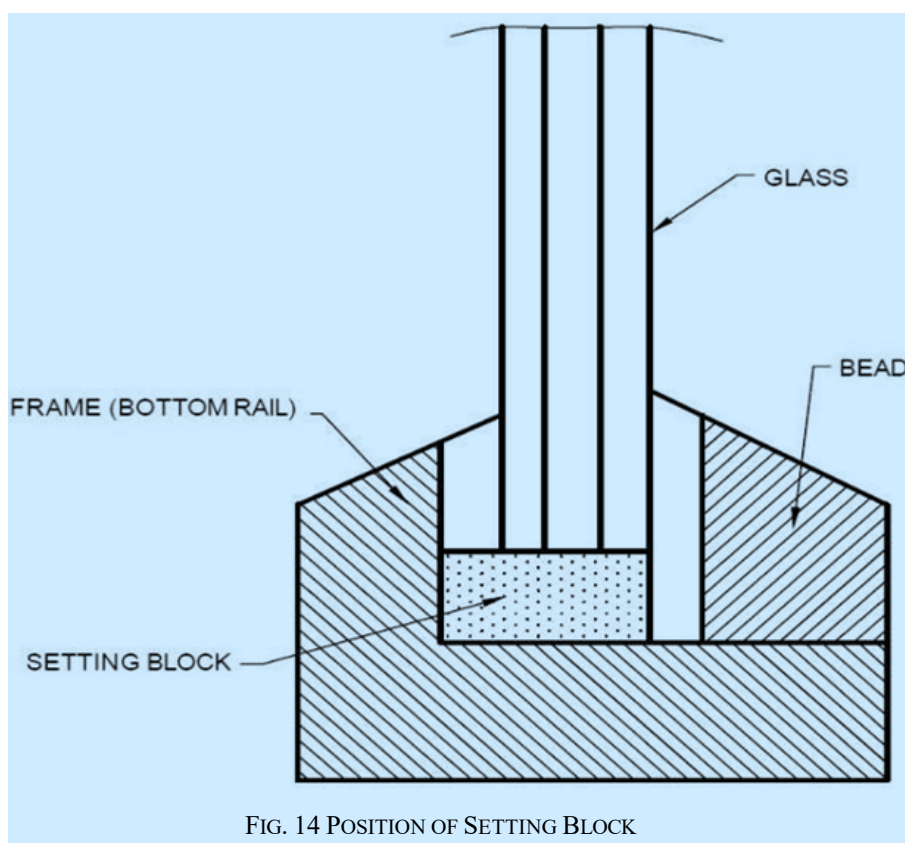
Setting blocks shall be located to equally support all panes of glass, and shall be fixed to prevent displacement during installation and service. The minimum length of each setting block (or two blocks side by side) shall be 25 mm in length for every square metre of glass area, with a minimum length of 50 mm. When wood is used as the material for setting blocks, only the seasoned ones should be used as green wood may shrink laterally (in the process of attaining equilibrium moisture content over a period of time) and result in loosened installations.

For example, for a 3.0 m² glass area, 3.0 mm × 25 mm = 75 mm long, that is, 75 mm is the length of each setting block.

Setting blocks shall be of resilient, load-bearing, non-absorbent, rot-proof, and material that is compatible with all other glazing materials conforming to available standards (if any) that may come into contact with the blocks.

NOTES

- 1 Setting blocks are used between the bottom edge of the unit and the frame to centralize and equally support both panes of glass.
- 2 Setting block width and location should not restrict water drainage.
- 3 Extruded rubber material with 80 to 90 shore A hardness is recommended.
- 4 Shaped setting blocks will be required for a glazing platform.



6.2.6 Location Blocks

The number and position of location blocks shall be as shown in [Fig. 15](#) and [Fig 16](#).

Location blocks shall be,

- a) of minimum of 25 mm in length;
- b) at least as wide as the glass thickness;

- c) positively located to prevent displacement in service; and
- d) sufficiently resilient to accommodate movement within the frame, without imposing stress on the glass, and of resilient, non-absorbent material.

NOTES

1 Location blocks are used between the edges of the glass and the frame to prevent movement of the glass within the frame by thermal expansion or when the window or door is opened or closed. They are required to prevent the weight of the glass from causing the frame to become out of square.

2 Extruded rubber material with 55 to 65 shore A hardness is recommended.

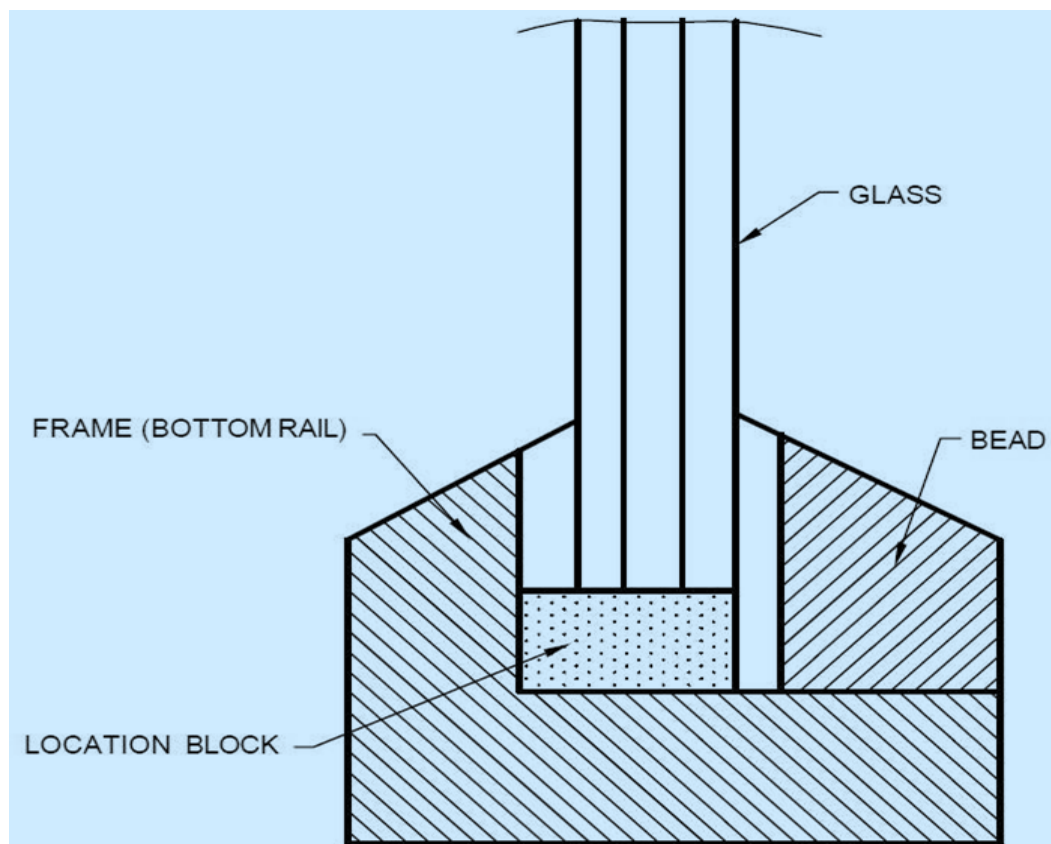
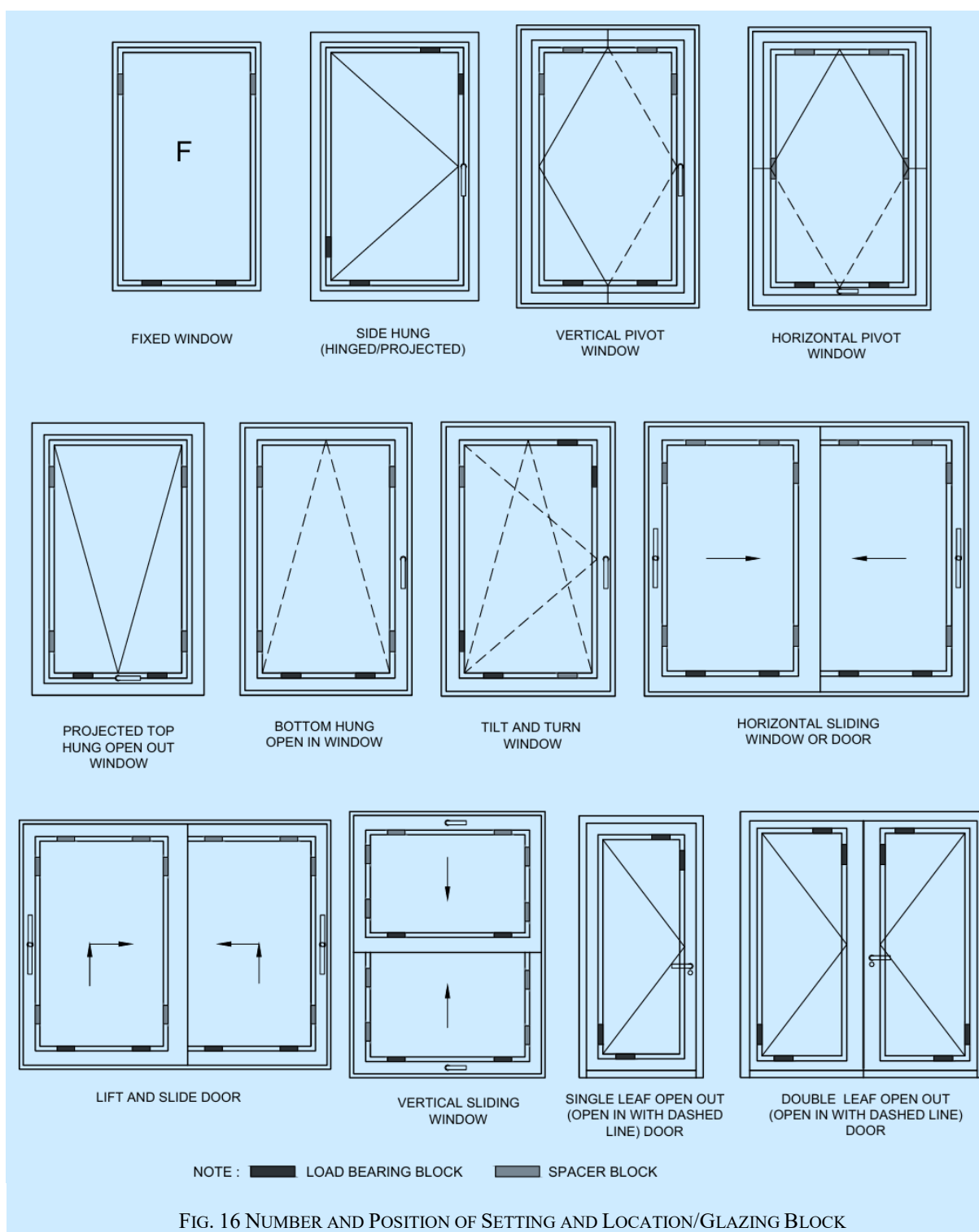


FIG. 15 POSITION OF LOCATION BLOCK



6.2.7 Distance Pieces

Distance pieces, as shown in [Fig. 17](#), used where required, shall be,

- of resilient, non-absorbent material;
- 25 mm long and of a height to suit the depth of the rebate and the method of glazing; and
- spaced opposite each other, approximately 50 mm from each corner at intervals of not more than 300 mm.

The thickness shall be equal to the front and back clearance, to retain the glass firmly in the frame.

NOTES

- 1 Distance pieces are required to prevent displacement of glazing compounds or sealant by external loading, such as wind pressure.
- 2 Extruded rubber material with 55 to 65 shore A hardness is required.

6.2.8 Rebates and Grooves for Installation

Rebates, grooves and beads shall be cleaned and free from grease, moisture and other contaminants. All sealant surfaces shall be primed or sealed in accordance with the recommendations of the manufacturer of the glazing material.

The number and location of setting blocks and location blocks shall be as shown in [Fig. 14](#) to [Fig. 16](#), wherever necessary to maintain the requisite edge clearance. Each block shall support the full thickness of the glass.

6.2.9 Glazing Beads

Where used, glazing beads shall be capable of restraining the glass under all design forces.

6.2.10 Structural Sealants

Structural silicone shall be installed such that a full adhesive bond to the substrate is achieved. Application of structural silicone may require prior cleaning and or priming of the substrate.

Applied structural silicones shall not be installed adjacent to other materials that may be chemically incompatible with the structural sealant and cause a loss of adhesion or adverse chemical changes within the structural silicone that could lead to a loss of strength in the structural silicone. For guidance on the installation of structural silicone, reference to available standards/literatures may be made.

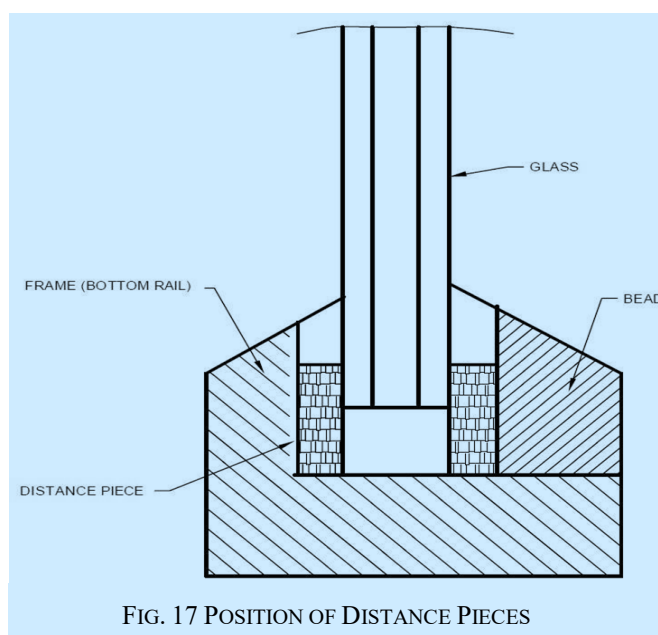


FIG. 17 POSITION OF DISTANCE PIECES

6.3 Design Criteria

6.3.1 Frame Support Condition

6.3.1.1 Maximum area of glass panels subjected to wind loading

The maximum span for a given standard nominal thickness of ordinary annealed, laminated, heat-strengthened and toughened glass for a given panel size shall be determined in accordance with [6.3.2](#), [6.3.3](#) and [6.3.4](#) as applicable for the relevant support conditions. For heat strengthened laminated and toughened laminated glass, maximum allowable span shall be determined in accordance with [6.3.5](#). For 3 mm monolithic annealed glass, the maximum area shall not exceed 0.85 m².

The minimum thickness of the nominal glass size shall be as per the accepted practice in accordance with Indian Standard specifications.

For laminated glass composites, the combined minimum thickness of the glass sheets may be used excluding the interlayer thickness. Alternatively, reference to specialist literature may be drawn for the same purpose.

6.3.2 Rectangular Glass Supported on All Four Edges

For rectangular glass supported on all four edges, the maximum area for the design wind pressure shall be determined using the corresponding tables provided for each type of glass. Linear interpolation may be used for any value in between those given therein while maintaining the aspect ratio.

6.3.3 Rectangles of Glass Supported on Two Opposite Edges

For rectangles of glass supported on two opposite edges, the maximum glass area for the design wind pressure shall be determined from [Table 22](#) to [Table 25](#) for ordinary annealed, laminated, toughened and heat-strengthened glass, respectively.

6.3.4 Rectangles of Glass Supported on Three Edges

For rectangles of glass supported on three edges, the maximum glass area shall be determined as for two-edge support spanning along the unsupported edge. Alternatively, basic engineering principles shall be adopted in determining the glass thickness for all applications where only three edges of the panel are supported.

NOTE — Butt glazing of adjacent panels in the same plane should not be considered as a support.

6.3.5 Heat-Strengthened Laminated and Toughened Laminated Glass

For heat-strengthened laminated and toughened laminated glass, the maximum span for a given standard nominal thickness for a given panel size shall be determined from [Table 21](#) and [Table 25](#), and [Table 19](#) and [Table 24](#), respectively with the design wind pressure being divided by the appropriate glass type factor, P_f (see [Table 14](#)), as applicable for the relevant support conditions.

6.3.6 Serviceability Checks

6.3.6.1 Glass complying to [Table 17](#) to [Table 25](#) are deemed to meet, the serviceability deflection limits specified. In the absence of any other specified values, the following values may be used for guidance.

6.3.6.2 Maximum allowable deflection of framing members

a) *For mullions:*

- 1) Single height glazing — Span/175 or 19 mm, whichever is less.
- 2) Double height glazing — For spans up to 4 110 mm, same as single height glazing; and for spans above 4 110 mm, the same shall be $(\text{Span}/240) + 6.35$ mm.

b) *For transoms:*

- 1) Span/500 or 3 mm, whichever is the least for dead load.
- 2) Span/175 or 19 mm, whichever is the least for wind load.

c) *Deflection at the centre of the glass:*

- 1) Monolithic glass — Shortest span/60 or 19 mm, whichever is the least.
- 2) Double glazed unit — Shortest span/90 or 19 mm, whichever is the least.

d) *Deflection at edge of the glass:*

- 1) Shall be limited to 15 mm.

6.3.6.2.1 Dielectric separation of dissimilar metals

When the framing member for the glazing is aluminium and reinforcement is required to meet the deflection criteria, the reinforcement may be of aluminium or galvanized steel (red oxide coating being not recommended) with an insulating material between them at contact points. It is recommended to choose materials as close to each other in the galvanic series as possible.

6.3.7 Ultimate Stresses

The ultimate stresses for various types of glass are given in [Table 26](#).

Table 26 Ultimate Limit State Design Stresses for Glass Subjected to Wind Loading <i>(Clause 6.3.7)</i>				
Sl No.	Glass Type	Standard Nominal Thickness 'T' mm	Ultimate Limit State Design Stress at Given Location	
			Away from Edge MPa	At Edge MPa
(1)	(2)	(3)	(4)	(5)
i)	Annealed	3	41.00	32.8
		4	38.99	31.19
		5	37.45	29.96
		6	36.2	28.96
		8	34.33	27.46
		10	32.8	26.24
		12	31.57	25.25
		15	30.15	24.12
		19	28.72	22.98
		25	26.96	21.57
ii)	Toughened	4	97.47	77.97
		5	93.61	74.97
		6	90.49	72.39
		8	85.82	68.65
		10	82.01	65.61
		12	78.91	63.13
		15	75.37	60.3
		19	71.81	57.45
iii)	Heat strengthened	25	67.41	53.93
		3	65.6	52.48
		4	62.38	49.9
		5	59.91	47.93
		6	57.91	46.33
		8	54.92	43.94
		10	52.48	41.99
		12	50.51	40.4

Table 26 Ultimate Limit State Design Stresses for Glass Subjected to Wind Loading
(Clause 6.3.7)

Sl No.	Glass Type	Standard Nominal Thickness 'T' mm	Ultimate Limit State Design Stress at Given Location	
			Away from Edge MPa	At Edge MPa
(1)	(2)	(3)	(4)	(5)
iv)	Annealed laminated	5	37.73	30.18
		6	36.43	29.14
		8	34.41	27.53
		10	32.87	26.3
		12	31.62	25.3
		16	29.75	23.8
		20	28.23	22.58
		24	26.99	21.59

6.3.8 Insulating Glass Units

6.3.8.1 Determining bite of structural silicone

The structural bite requirement is directly proportional to the wind load and the dimension of glass. Higher the wind load and larger the dimensions of the glass are, the greater the amount of structural bite required. The controlling variables which affect the structural bite requirement are the maximum short span dimension of glass and the design wind load which the structural glazing system shall be designed to accommodate.

a) Structural bite calculation for wind load and glass dimension

$$\text{Minimum Structural Bite} = \frac{\text{Glass Short span dimension} \times \text{wind load} \times 0.5}{\text{Maximum allowable design stress}}$$

NOTES

- 1 Glass' short span dimension (SSD) is the shorter of the two dimensions (in m) of the rectangular glass panel.
- 2 Wind load is the maximum wind pressure, in Pascal, for a return period of 10 years based on local regulations.
- 3 The maximum allowable design stress for type of the structural sealant is selected as per manufacturer's recommendations.
- 4 The minimum structural bite shall be, not less than 6 mm for single glass and for double-glazed units (DGU) with equal glass thickness.

b) *Determination of thickness of structural silicone* — With a sealed IGU, there may be load sharing between the two panes of glass. If so, and if both panes are of the same thickness, the lateral load (P) is shared almost equally; thus the secondary seal bite is calculated as one half of that of the structural sealant bite, but should meet the minimum requirement. If of unequal thickness, the load shared by each pane will vary, depending on the difference in thicknesses. Further details are to be obtained from the manufacturer.

c) Structural bite calculation for dead load

$$\begin{aligned} \text{Minimum Bite (m)} &= \frac{\text{Weight of glass (kg)}}{\text{Perimeter of the glass (m)} \times \text{Sealant dead load design strength (kg/m}^2\text{)}} \\ &= \frac{\rho g T A}{P_s} \end{aligned}$$

where

- ρ = 2 500 kg/m³ is the specific mass of flat glass corresponding to approximately of specific weight;
- g = 9.81 m/s²;
- s = Allowable design stress for dead load (DL) for selected sealant considered for design, Pa;
- T = thickness of the bite, m;
- P = perimeter (if the horizontal frame members do not support the glass or deflect under the dead load of the glass, consider 2 times of its height only); and
- A = area of glass panel, m².

6.3.8.2 Maximum allowable span for glass and type

Maximum allowable span is dependent on the thickness of the glass selected and type of glass used (for example annealed, toughened or heat strengthened glass). The maximum span is calculated in accordance with [6.1.9](#) (see [Fig. 18](#) for details).

6.3.9 Safety During Installation

Glass and the coating layer (if any) should not be exposed to/in direct contact with aggressive materials such as acid, solvent, cement mortar, etc.

The fixing system of the panel should, in no case, cause shear stresses in the panel, and particularly the bonding interface between the glass sheet and the frame, either under the effect of external forces or due to differential expansion of components.

The protection of glass panel by means of a polyethylene sheet is recommended and shall provide ventilation for the glass in order to prevent thermal breakage.

Workers involved in the installation process shall be provided with adequate training and awareness about safety practices related to the handling and installation of glass panels. They shall be equipped with appropriate personal protective equipment (PPE), including but not limited to safety gloves, helmets, safety goggles, and footwear designed for the work environment, to minimize risks of injury during installation activities.

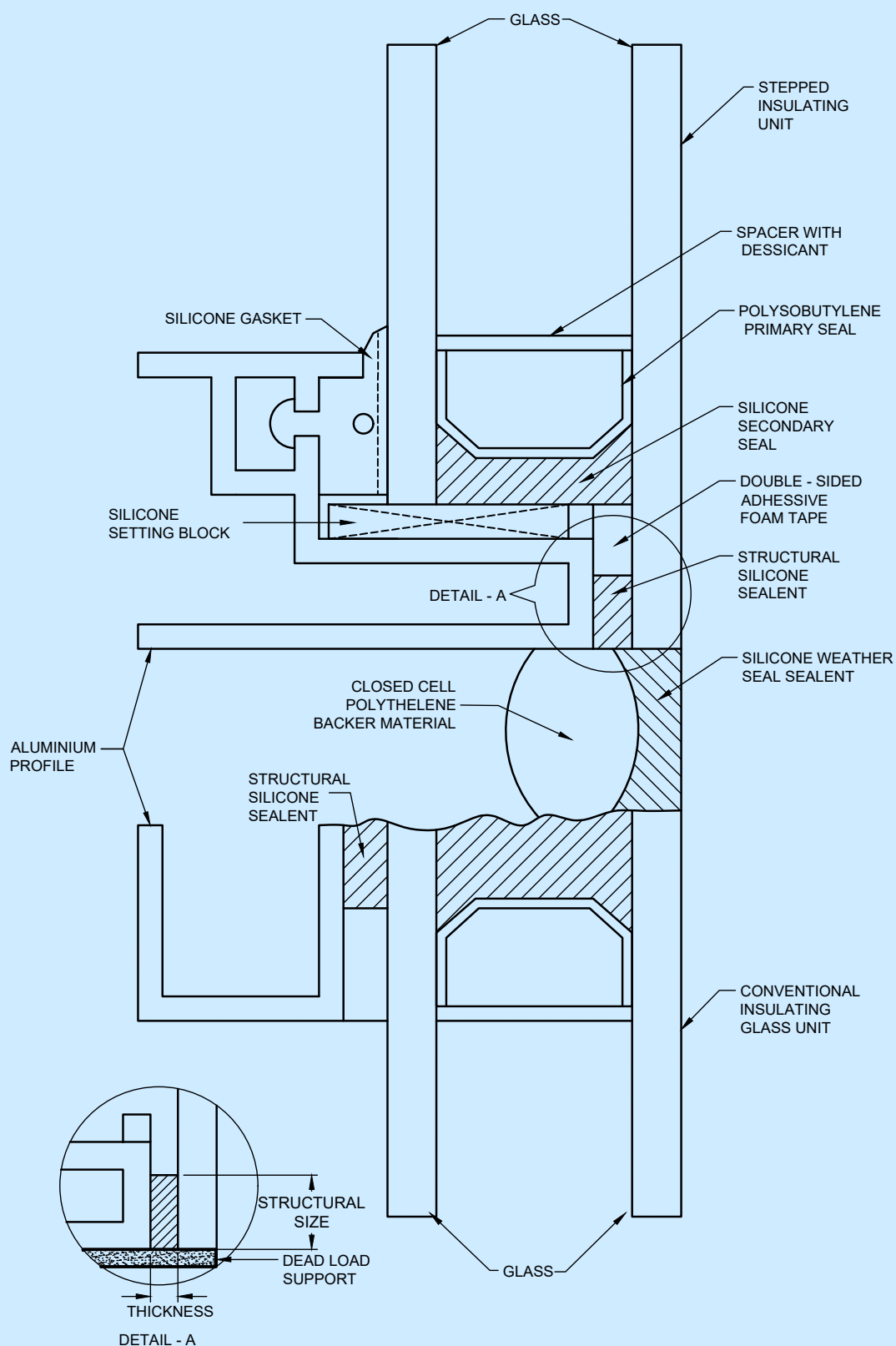


FIG. 18 TYPICAL STRUCTURAL GLAZING DETAIL

NOTE — Dead load support for the glass or infill shall be determined based on [6.3.8.1\(c\)](#). Minimum 50 percent of thickness of external glass pane shall be supported at the bottom, in the case of structural glazed systems.

FIG. 18 TYPICAL STRUCTURAL GLAZING DETAIL

6.3.10 Handling and Transportation

6.3.10.1 Transportation

During the transportation of the flat glass (clear, tinted), coated glass (solar control and mirror), lacquered or painted glass the following methods (if other standard packaging specifications are not available) of storage, packing and interleaving shall be followed:

- a) Flat glass shall be transported vertically;
- b) Glass panes should not come into direct contact with each other by using appropriate interleaving such as powder or foam or paper. The lacquer opacified/enamel coated/screen printed glasses shall be separated by paper or foam; and
- c) The packing and packing material shall be protected from water and if the glass is wrapped and sealed, the seal should remain closed until the product is used and precautions taken to not damage the packing while handling with a hoisting apparatus.

Workers involved in the transportation of glass shall be provided with adequate training on transportation safety practices. They shall be equipped with appropriate personal protective equipment (PPE), including but not limited to safety gloves, helmets, and safety footwear to minimize risks of injury during transportation activities.

6.3.10.2 Handling

During the handling of the flat glass (clear, tinted), coated glass (solar control and mirror), lacquered or painted glass, the following guidelines shall be followed:

- a) Glass shall be handled with dry, clean gloves, specifically after washing to prevent stain marks on them;
- b) Care shall be taken to avoid contact or friction of the coating or the opacified/ enamel/screen printed side with any rough surface or hard objects;
- c) better practice is to handle glass with vacuum cups, and ensure that the vacuum cups and the glass are clean; and
- d) It is recommended to keep the glass sheets clean if these are to be stored again before their processing or installation in the facade.

6.3.11 Storage at Site

To prevent staining, the storage area at site shall be,

- a) a dry, well ventilated location at a sufficiently constant temperature;
- b) protected from rain and running water;
- c) protected from wide changes in temperature or humidity;
- d) protected from alkaline material;
- e) protected from direct contact of aggressive products such as corrosive vapours, acids, etc; and
- f) glass panes are stored vertically on a soft surface such as wooden boards/battens.

6.3.12 Storage after Cutting

6.3.12.1 While stacking of coated glass/ mirror in the unpacked condition, the glass surface shall be facing towards the front (paint/coating backing towards the back).

6.3.12.2 Direct stacking of the mirror/coated glass one above the other should be prohibited and suitable interleaving material as mentioned in [6.3.10.1](#) should be used between two glass panes.

6.3.12.3 The support material for glass storage should not have affinity towards moisture absorption.

6.4 Fire Safety

6.4.1 This Section covers the requirement and selection of fire safety glass for buildings. General fire safety provisions for buildings may be drawn from Part F 'Fire and Life Safety' of this NBCS. It is important for any fire rated glass to provide sufficient amount of human impact safety feature even during fire or in general situations. The level of human impact safety of glass in buildings shall be in accordance with [7.1](#). It is necessary that the fire rated system shall withstand the standard time-temperature curve during a fire test and shall give adequate time to the occupants for safe evacuation. This Section does not deal with the safety and security of people or goods in relation to risks of,

- a) vandalism, riots, burglary or break in protection;
- b) protection from explosion (terrorist attack);
- c) natural disasters like earthquakes, cyclone, etc; and
- d) plastic glazing, safety and security glazing, etc.

The requirements given here are minimum fire resistance requirements. In circumstances wherein the requirements of other Indian Standards exceed the requirements of this Section, the higher requirements shall be applicable.

6.4.2 Glass for Fire Resistance

6.4.2.1 Glazing system shall be so constructed, treated, or combined with other materials as to reduce loss of life or property during an event of fire, in comparison with ordinary sheet or flat glass, thereby preventing the passage of flame, hot gases, and smoke.

6.4.2.2 The following are the products in the scope of this Section that may be used for fire resistance provided they meet the requirements of accepted standard [C-8(18)]:

- a) Wired fire resistant glass;
- b) Glass-ceramic fire resistant glass;
- c) Borosilicate fire resistant glass;
- d) Thermally super toughened soda lime silicate fire resistant glass;
- e) Laminated fire resistant glass;
- f) Intumescent interlayer laminated fire resistant glass;
- g) Gel-filled fire resistant glass; and
- h) Low-E coated fire resistant glass.

6.4.2.3 Testing procedures

Any fire rated glazed building element needs to be tested as per accepted standard [C-8(19)]. The fire rating criteria of any building element is given in Part F 'Fire and Life Safety' of this NBCS.

6.4.2.4 Identification of glass used for fire resistance

- a) All glass used for resisting fire shall conform to and preferably certified against available Indian Standards;
- b) Since a fire resistant glass is not easily distinguishable from normal types of glass, it is imperative that every fire rated glass panel/unit should preferably have an indelible mark on all the panels/units of fire resistant glass showing the name of the manufacturer and the name of the product; and
- c) Rating of the glass panel may also be provided, although the rating of glazing system is of primary concern.

The glass used for fire resistance should be in accordance with the requirements of Part F 'Fire and Life Safety' of this NBCS.

6.4.3 Precautions

6.4.3.1 Fire resistant glass alone shall not be enough to protect occupants from fire. Glass is only one component of the complete system.

6.4.3.2 Fire test report conducted for one application shall not be substituted for any other application. For example, a report for a partition shall not be used for a door application although the glass used and the fire ratings are the same. Similarly, approval in vertical orientation cannot be considered for acceptance in horizontal or inclined orientation.

6.4.3.3 The maximum glass size as mentioned in the test report should not be exceeded in practice with respect to the bigger linear dimension or area, unless otherwise supported with valid third-party tested evidence. The aspect ratio may be modified provided that none of the linear sides has a dimension greater than that of the largest dimension of originally tested glass.

6.4.3.4 The dimensions of the profile need to be the same as that tested. The sheet thickness of the profile can be increased with no change in the tested fire rating.

6.4.3.5 In case of any change in hardware, the new assembly should be retested as per the same standard and/or specification requirements. In the case of replacement of hardware, the same specification and make of the hardware can be used without retesting of the whole assembly.

6.4.3.6 Factors like edge cover, types of gaskets and type of fixtures also play a very vital role in achieving the fire performance of the building element. These have to be used and installed as per the test report/available standards.

6.4.3.7 Buildings having non-structural joints between floor slabs and external perimeter walls, pose an associated risk of smoke and fire traversing between adjacent floors is not uncommon. The use of suitable materials to seal such joints needs careful consideration towards fire safety as the same is part of the compartment/fire barrier.

All gaps between floor-slabs and facade assembly shall be sealed at all levels by tested system of equal fire rating as that of floor slab to prevent fire and smoke propagation from one floor to another. Accepted standard [C-8(20)] specifies the test method to determine the fire resistance performance of perimeter fire barrier joint system based on the length of time the system being tested to resist the fire before development of through openings or flaming and transmission of heat on the unexposed surface.

6.4.4 Classifications of Fire Resistant Glass

See 4 of accepted standard [C-8(18)].

6.4.5 Cleaning and Maintenance of Fire-Rated Glazing System

Routine maintenance and inspection are essential for fire rated glazing to ensure their optimal performance during a fire emergency. For more details in this regard, Part F 'Fire and Life Safety' of this NBCS may be referred.

7 SAFETY RELATED TO HUMAN IMPACT, ANTI-FALLOUT, WINDOW RESTRICTOR FOR CHILDREN AND DYNAMIC WIND RESTRAINT

7.1 Safety Related to Human Impact

7.1.1 General

The provisions herein are limited to the behaviour of glass when subjected to various kinds of human impact, precautions against risk of fall and falling glass. Conditions outside of 'human impact' are not covered under the scope of this Section.

This Section does not assume that the glass used in accordance to this Section will not break under all human impact conditions, rather,

- a) it will not break under most likely forms of human impact; and

- b) even if it breaks, the likelihood of cutting or piercing injuries will be minimised by virtue of the protection given to the glass, or by the limited size, or by the fracture characteristics of the glass.

Further, this Section does not deal with the safety and security of people or goods in relation to risks of,

- a) vandalism, riots, burglary or break in protection;
- b) fire arm protection;
- c) protection from explosion (terrorist attack);
- d) natural disasters like earthquakes, cyclone, fire, etc; and
- e) plastic glazing material, safety and security glazing, etc.

In circumstances wherein the requirements of any referred standard(s) exceeds the requirements of this Section, the higher requirement shall be applicable.

The use of this Section shall also be in conformity with all other relevant standards on fire safety, structural stability, natural disasters, safety and security, etc in force.

7.1.2 Safety Glass

Glazing material that are constructed, treated, or combined with other materials shall reduce, in comparison with ordinary sheet/ float/ plate glass, the likelihood of injury to persons by objects from exterior sources or by these safety glasses when they may be cracked or broken.

The following are the products in the scope of this Section that may be used in safety glazing provided they meet the conformity requirements of accepted standard [C-8(7)]. Safety glass according to this Section shall be of following types:

- a) Toughened safety (tempered) glass; and
- b) Laminated safety glass.

7.1.2.1 All safety glasses shall meet the requirement of tests specified in accepted standard [C-8(7)].

7.1.2.2 In atria where there is no impact loads above 10 m, suitable toughened glass is adequate. See [4.2.9](#), in case of use of glass fins as a structural member.

7.1.3 Critical Location

Where any glazing is within 1.5 m above the floor level of a building, it is considered likely to be subjected to human impact and hence, shall comply with the human impact safety requirements of this Section. Safety glazing should also be used,

- a) where there is danger of falling infill glass material(s) from overhead glazing;
- b) the danger of falling due to a change in floor level; and
- c) in case of balustrades, stairs and floors.

Based on typical accidents in glazed buildings, certain locations in buildings are found to be more vulnerable to human impact than others. Some of such critical locations are given below and shown in [Fig. 19](#), where necessary precautions have to be followed:

- 1) in-and-around doors, low windows;
- 2) door side panels;
- 3) panels mistaken for a doorway or opening;
- 4) panels at low levels in walls and partitions;
- 5) bathrooms;
- 6) building associated with special activities, for example, gymnasias, enclosed swimming pools, etc;

- 7) schools and child care facilities; and
- 8) nursing homes and care facilities for the aged and infirmed.

Suitable precautions should be taken to reduce the injuries that can result from glass breakage by,

- i) selecting glass of a suitable type, thickness and size;
- ii) enhancing a person's awareness of presence of glass by making glass visible (manifestation of glass); and
- iii) minimizing manual handling of large pieces of glass during installation.

Based on the above, a comprehensive suggestion on type of glass to be used at different critical location/cases is provided in [Table 27](#) for information.

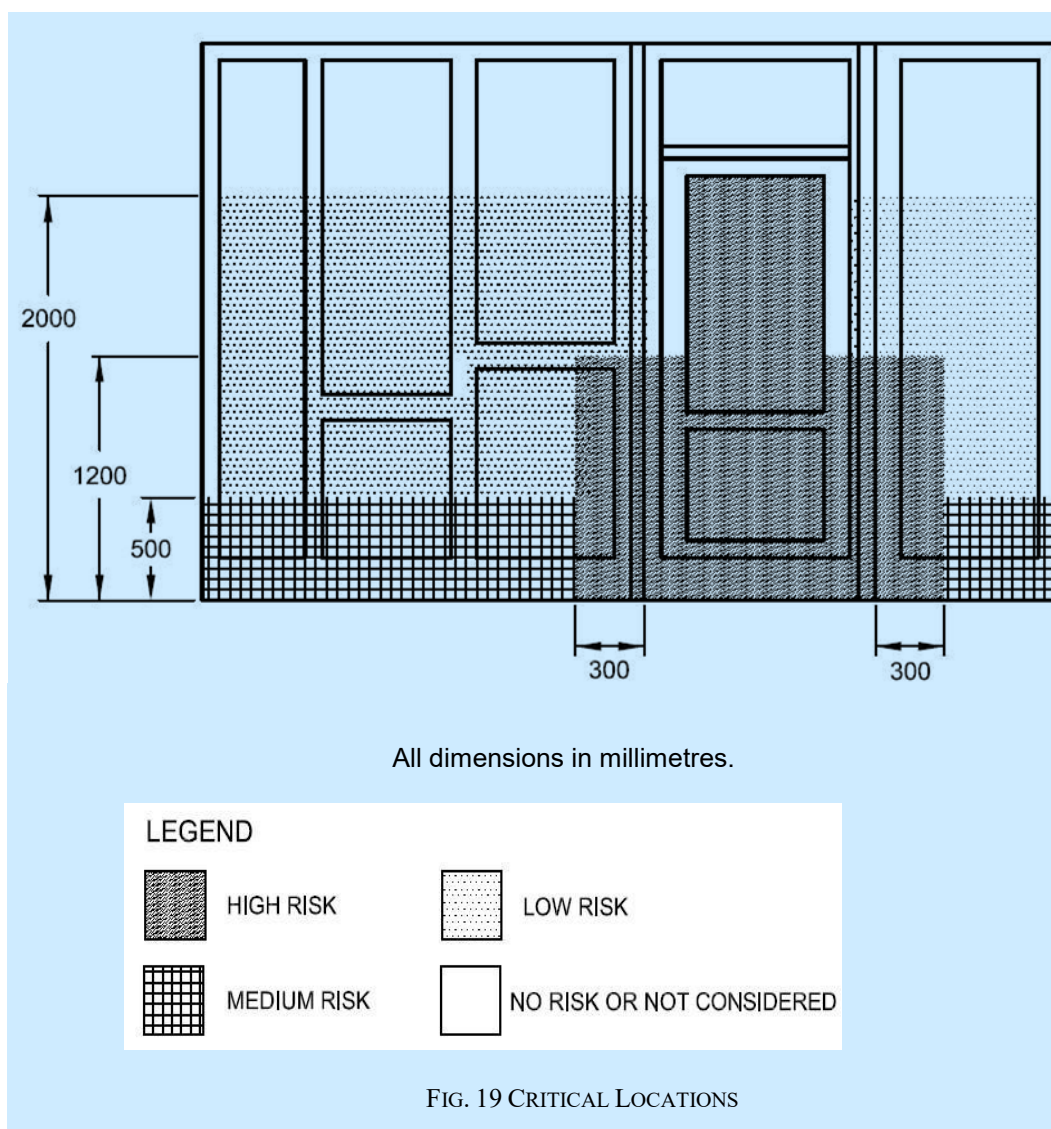


Table 27 Type of Glass Suggested for Use at Different Critical Locations/Cases in Buildings

(Clause 7.1.3)

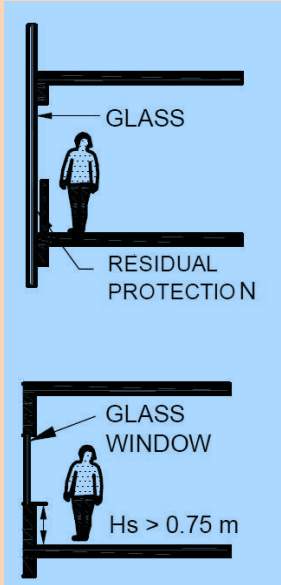
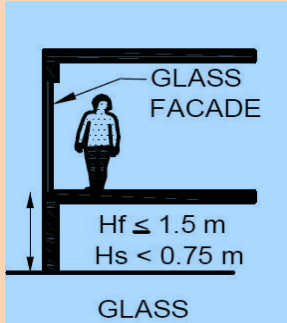
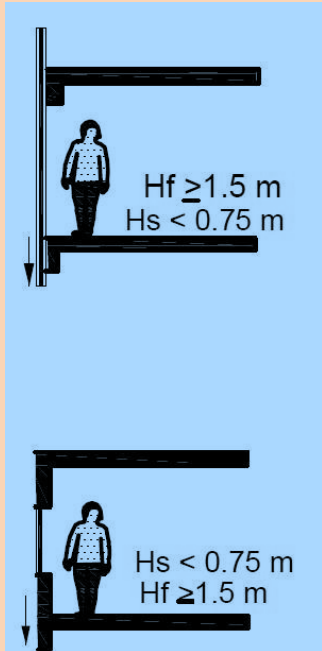
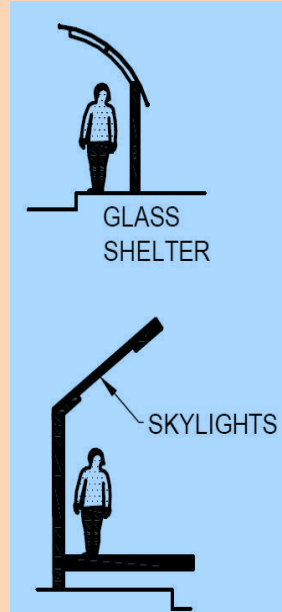
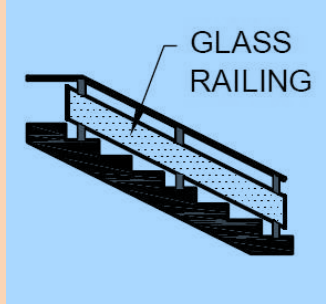
CASE 1	CASE 2	CASE 3	CASE 4	CASE 5
Vertical Walls with Residual Protection or $H_s \geq 0.75$ m (not likely to be Subjected to human impact)	Vertical Walls $H_s < 0.75$ m and $H_f \leq 1.5$ m (human impact but no risk of fall)	Vertical Walls $H_s < 0.75$ m and $H_f > 1.5$ m (human impact and risk of fall)	Horizontal or Sloped Glazing Glass (risk of fall)	Glass Acting as a Balustrade/Railing (human impact and risk of fall)
(1)	(2)	(3)	(4)	(5)
				
Type of glass	Any glass ¹⁾	Safety glass ²⁾	Safety glass ²⁾	Laminated safety glass

Table 27 (Continued)

Examples	Residual protection is the safeguard provided to avoid the impact of human being on glass. It is provided on the side of the glass where there are chances of human impact. It can be achieved by providing protection in the form of a sill structure or transom, chair rail or grill work inside.	a) Doors b) Side panels c) Curtain walls d) Glazed area e) Doors in bathroom 1) Fully framed 2) Partially framed 3) Frameless f) Facade g) Windows h) Internal partitions and doors j) External facade and doors on ground floor, above floor with terrace outside.	a) Curtain walls b) Facade c) Spandrels d) High activity area e) High risk area f) To avoid risk	a) Roof (Skylights) b) Ceilings c) Bus shelters d) Floors e) Stairs f) Sloped facade	a) Balustrades b) Balcony c) Railings

¹⁾ Safety glass is not mandatory.

²⁾ Laminated safety glass is preferred.

NOTES

1 'H_f' corresponds to height of fall of human being or glass in case of change in level and 'H_s' corresponds to the sill height with reference to floor height.

2 In case of mirror glazing, it should conform to the requirements of other safety glasses unless it is fully backed by a solid material.

3 In Case 2 and Case 3, if the smaller dimension of the pane is 250 mm or less and its area is 0.5 m² or less, glass other than safety glass may be used, provided that its nominal thickness is not less than 6 mm (applicable to vertical glazing).

4 Toughened or laminated safety glass should meet respective test requirements as given in respective Indian Standards.

5 The effective toughened safety glass thickness and/or laminated safety glass configuration shall be determined case by case with regard to,

- other solicitations (wind load, snow load, dead load, and human load);
- the overall dimension (length, width of surface);
- the aspect ratio of the glass; and
- the glazing fixing type (framing, bolted system, structural system, etc).

Table 27 (Concluded)

6 Precautions against chances of injuries due to broken glass falling on people include:

- a) Broken annealed glass falling on people can cause grievous or even fatal injuries; hence it is recommended to use safety glass in locations other than defined in Case 1 where the risk of people getting hurt by falling glass is high.
- b) Toughened safety (tempered) glass has a safe breakage pattern, as it breaks and disintegrates into small and relatively harmless particles. However, thick toughened glass particles may stay interlocked, and fall as lumps of these multiple particles and can cause a minor or medium injury mainly due to the weight of the cluster.
- c) Laminated safety glass shall generally not fall out of fixing. However, where laminated glass with both glasses toughened, used for horizontal or sloped glazing, in case of failure of both toughened glasses, it may crumble as a blanket and fall out of fixing. This factor needs to be considered while designing horizontal and sloped glazing. Further, when the slope is acute, inner pane facing the floor should be laminated; and when the slope is obtuse, the outer pane facing the ground/floor should be laminated and all obtuse angle sloped glazing shall be continuously capped for safety reasons.
- d) Any broken glass in any glazing should be removed immediately on breakage.
- e) Strength of the glazing system should be such that it has the ability to hold glass in place and prevent it from falling out as a whole.
- f) For over-head applications like skylights and canopies, for the safety of maintenance workers and people underneath, the broken laminated glass shall be capable of sustaining a load of 100 kg for a minimum duration 30 min.

7 In case of external laminated glass facades, openable portions have to be left at regular distances for firefighting and smoke exhaust. This portion should be of toughened (safety tempered) glass and clearly indicated by suitable visible marking. *See also* Part F 'Fire and Life Safety' of this NBCS.

8 If insulating glass unit (IGU) is used in situations mentioned in this Section, then one of the following shall apply:

- a) If IGU is installed in areas subjected to human impact on either side, then both the panes of the unit shall meet the requirements of this Section.
- b) In situations where access is restricted to one side of the unit, then only the accessible side should meet the requirements of this Section.

9 Awareness of presence of glass may be by making visible manifestation.

10 For specific provisions relating to smoke evacuation, Part F 'Fire and Life Safety' of this NBCS may be referred.

7.1.4 Manifestation (Making Glass Visible)

7.1.4.1 Presence of glass in a door, side panel or a panel capable of being mistaken for a doorway or opening, which is not made apparent by transoms, colonial bars, door frames, large door handles, stall or other components of glazing system, should be made apparent by some form of manifestation. Decorative treatment, such as being opaque, or patterned are the other common manifestation options.

The manifestation employed should be of a sufficient size to make it immediately obvious. The manifestation should preferably be permanent, for example etching of the glazing; alternatively, if applied materials are used, they should be durable and not easily removed.

7.1.4.2 A band or marking may not be required, where any one of the following apply:

- Width of the glass is not greater than 500 mm at any part (this applies to overall panel assembly – not individual glass pieces as in faceted glazing).
- Glass is provided with at least one fixed glazing bar, firmly attached to the styles to locate and protect each face of the glass. At least one transom (glazing bar) shall be located with its upper edge not less than 500 mm and its bottom edge not more than 1 000 mm above the floor level. The glazing bar shall have a face of width 40 ± 3 mm; and
- Alternatively, patterns may be used as an acceptable form of marking provided it meets the other criteria in this Section.

7.1.5 Identification

All safety glasses shall be procured from certified manufacturers and the product shall conform to relevant Indian Standards and shall carry all relevant information through the approved label/permanent (indelible) markings on the glass surface.

7.1.6 Test Requirements

7.1.6.1 Safety glass shall confirm to all the test requirements mentioned in the accepted standard [C-8(7)].

7.1.6.2 Mechanical strength

The required mechanical strength values (see [Table 28](#)) apply to quasi-static loading over a short time like wind loading relate to 5 percent probability of breakage at the lower limit and 95 percent confidence intervals.

If the glass satisfies the relevant impact test performance requirements (or fragmentation test for toughened glass), in addition to all other appropriate tests mentioned in available Indian Standards or as listed in [Table 28](#), these materials can be classified as safety glass.

Table 28 Minimum Mechanical Strength Required for Fully Toughened Glass		
(Clause 7.1.6.2)		
Sl No.	Type of Glass	Mechanical Strength N/mm ²
(1)	(2)	(3)
i)	Float (clear/ tinted/ coated/ mirror)	120
ii)	Enamelled flat (based on enamelled surface in tension)	75
iii)	Patterned glass	90

7.2 Anti-Fallout

The following shall apply:

- For all open out windows in buildings up to 5 storey height (< 15 m) anti-fallout secondary safety device is optional; and

- b) For all open out windows in buildings more than 5 storey height (> 15 m) anti-fallout secondary safety device is mandatory.

7.2.1 In the case of a fall out due to the sash getting completely disengaged from the hinges/stay arms/restrictors, the anti-fallout device shall be capable of withstanding the self-weight of the sash. The drop height for the test shall be 100 mm. Design and testing for the mechanism to take into consideration jerk load with a factor of safety 2.0. Both static tensile test and a drop test with the mechanism installed in a window to be carried out to validate the compliance of the hardware to the design requirements.

7.3 Window Restrictor for Children

For windows which are non-emergency exits, projected top hung window, parallel opening window, vertical sliding hung window, side hung window and pivot window (except when used as an access door), the window restrictor shall be located such that the clear opening shall not exceed 100 mm (for human life safety especially for anti-fallout for children). When a force of 250 N is applied in opening condition to any part of the sash a sphere of 100 mm shall not pass through any part of the opening. Key releasable window restrictor can be used for the purpose of cleaning/maintenance. The side hung opening shall be restricted to be used in low rise buildings $\{< 15$ m, *see* good practice [C-8(15)]}. For buildings above ≥ 15 m height, projected top hung or parallel opening with an opening restriction of 100 mm shall be used.

NOTES

- 1 For fire access panels/window, the local authority regulations will be applied and same will supersede the above.
- 2 The above requirement of window restrictor for children is mandatory for residential apartments and optional for commercial establishment. However, it is recommended to limit the opening to 100 mm for commercial establishments as well, especially for high rise buildings.

7.4 Dynamic Wind Restraint

This is applicable for open out windows in buildings more than 5 storey height (> 15 m) and where the dynamic wind load is high due to the adjacent structures or terrain. Installed hardware shall be capable of self-locking the open window upon first time closure of the window due to wind conditions; in order to avoid fatigue failure on the hinges and stays. The mechanism can be releasable either using existing handle or additional release trigger. Design shall be to serviceability conditions.

8 GLAZING SYSTEM

8.1 The provisions relating to glazing system are covered hereunder. The details and procedure for selection, design, fabrication, installation, testing and maintenance of glazing system and its materials, considering the performance factors, including structural, seismic, acoustic insulation, fire resistance and energy efficiency are covered here under.

For classification of glazing system, reference may be made to 6 of accepted standard [C-8(14)].

8.2 Design and Selection

The design considerations and parameters for the design and selection of a glazing system and the components involved, along with its associated materials used in buildings are given below.

8.2.1 Design Considerations

[Table 29](#) covers the various design considerations for glazing systems.

Table 29 Design Considerations

(Clause [8.2.1](#))

Sl No.	Design Considerations/Criteria	Reference Standards
(1)	(2)	(3)
i)	Structural aspects and parameters:	
	a) Design load and wind load classification	As per good practice [C-8(15)] and accepted standard [C-8(14)]
	b) Other Loads — Live load, dead load, etc	As per good practice [C-8(21)]
	c) Air infiltration/Exfiltration and its classification	As per accepted standard [C-8(14)]
	d) Deflection	As per 6 and accepted standards [C-8(14)] and [C-8(12)]
	e) Seismic performance	As per 6.1.6 and good practice [C-8(16)]
	f) Water tightness and its classification	As per accepted standard [C-8(14)]
	g) Glass thickness	As per good practice [C-8(22)]
	h) Structural sealant	As per 4.4.1
	j) Frame/profile specifications:	
	1) Aluminium	As per accepted standard [C-8(12)]
	2) uPVC	As per accepted standard [C-8(23)]
	3) Steel	As per accepted standard [C-8(24)]
	k) Hardware and fixing	As per 5.4 of accepted standard [C-8(12)]
ii)	Energy aspects and parameters:	
	a) Window to wall ratio	As per relevant guidelines/best practices (Handbook on ‘Sustainability in Buildings’)
	b) Building orientation	
	c) Window to floor ratio	
	d) Thermal performance (roofs and fenestration):	
	1) U-Value of glazing	As per relevant guidelines/best practices (Handbook on ‘Sustainability in Buildings’)
	2) Solar heat gain Co-efficient (SHGC)/Solar factor of glazing	
	3) Visible light transmittance (VLT)	
	4) Glass – Thermo – Optical properties: U value, solar heat gain co-efficient, visible light transmittance, internal and external (heat and light) reflectance and transmittance, shading co-efficient	As per 5

Table 29 (Concluded)

Sl No.	Design Considerations/Criteria	Reference Standards
(1)	(2)	(3)
	e) Lighting	As per Part D 'Building Services, Section 1, Lighting and Natural Ventilation' of this NBCS
iii)	Safety	
	a) Safety related to human impact	As per 7.1
	b) Window restrictor for children	As per 7.3
	c) Anti-fallout	As per 7.2
	d) Fire safety	As per 6.4 and Part F 'Fire and Life Safety' of this NBCS
	e) Security glazing:	
	1) Bullet resistance	As per 9
	2) Ballistics resistance	As per 9
	3) Resistance to explosion	As per 9
	4) Burglar resistance	As per 9
iv)	Acoustics parameters	
	a) Airborne sound insulation	As per accepted standard [C-8(25)]
	b) Impact sound insulation	As per accepted standard [C-8(26)]

8.2.2 Design Parameters

To achieve the performance criteria, care should be taken so that components in the system are selected and designed for the following parameters, such that the necessary performance criteria is met both individually and as a system:

- Loading;
- Deflections and allowable stress;
- Movements;
- Impact resistance;
- Acoustic and sound insulation; and
- Energy performance.

8.2.2.1 Loading

Wind load, live load, dead load, load combinations as per good practice [C-8(27)], seismic load as per good practice [C-8(16)], and other consequential loading may be considered.

8.2.2.2 Deflections and allowable stress

The deflection of framing members, members supporting brittle materials, metal panels, glass panels and the allowable stress shall be in accordance with [6.3.6.2](#). The allowable stress shall be in accordance with good practice [C-8(27)].

8.2.2.3 Movements

The glazing system (and its framing elements) shall be able to accommodate the movements caused due to wind loads, seismic loads, live loads, dead load deflection, thermal expansion and contraction of members, slab deflection, etc.

8.2.2.4 Impact resistance

Any glazing (in the glazing system) within 1 500 mm above the floor level of a building is considered likely to be subjected to human impact and hence, shall comply with the human impact safety requirements given in [7.1](#).

8.2.2.5 Acoustics and sound insulation

The recommendations on acoustics and sound insulation given in Part D 'Building Services, Section 4 Acoustics, Sound Insulation and Noise Control' of this NBCS shall be applicable.

8.2.2.6 Energy performance

The recommendations on the principal energy performance characteristics of the glazing system shall meet the provisions given in [5](#).

The primary characteristics required in general are reducing thermal losses and solar gains. For additional reference to visible light transmittance (VLT), solar heat gain coefficient (SHGC) and thermal transmittance (U-factor) features, relevant guidelines/best practices (Handbook on 'Sustainability in Buildings') may be referred.

8.2.3 Types of Structural Frame Materials

For the framed/semi-framed systems, materials such as aluminium, stainless steel, uPVC and timber, which are commonly employed are given hereunder.

8.2.3.1 Aluminium alloy

Extruded aluminium alloy used for structural framing members, shall be fabricated from the most appropriate grade of alloy complying with the relevant standard(s). The thickness of the aluminium sections should be in accordance with structural and hardware fastening requirements and shall be capable enough to meet the design requirements and satisfy the performance requirements.

Such extrusions in framing members and trims shall preferably be extruded aluminium grade 6063-T5, 6063-T6, 6060-T66, similar or stronger. Where 6063-T5 is to be used in shaped structures, supporting structural calculations particularly the stress criteria be ensured. Extrusion alloy 6061 or 6105 are commonly used for higher strength. Aluminium brackets shall be extruded aluminium grade 6061-T6 or approved equivalent. All aluminium alloys used as extruded rods/bars, tubes and profiles shall be compliant with accepted standard [C-8(28)].

Sheets and plates may be obtained from the following alloys of aluminium along with their primary characteristics:

- a) 1100 — A low strength alloy suitable for applications requiring a high degree of formability.
- b) 1135 — Brighter finish when anodized.
- c) 3003 — General purpose sheet alloy with fair mechanical properties and excellent formability, but susceptible to structural streaking when anodized (noticeable in large flat areas).
- d) 5005 — Excellent workability, weldability and corrosion resistance.
- e) 5052 — Slightly better mechanical properties and excellent corrosion resistance and weldability.

Other alloys can be used provided that they meet the required physical properties and the specified performance requirements.

8.2.3.2 Stainless steel

Mullions, panels, fascia, column covers, windows, doors, trim, roofing, gutters, flashing, hardware and other items where minimum maintenance is anticipated, shall be of austenitic stainless steels of Type 301, 302 and 304 complying with accepted standard [C-8(29)] and employed for their mechanical properties as they are highly corrosion resistant for all normal exposures.

For interior decorative applications, where regular maintenance will be provided, Type 430 may be used as it has lesser corrosion resistance properties. For external applications like in coastal regions where maximum resistance to corrosion is required, stainless steel shall be of Type 316. Particular attention shall be paid to the direction of the rolling grain on self-finished stainless steel components where the finish is aesthetically important. All screws, nuts, bolts, washers, self-tapping screws and other fastenings, including those used for the hardware, shall be of stainless steel Type 304, 316, 430 complying with accepted standard [C-8(29)]. Type 316 based steel fasteners, etc shall be preferable when they are used in coastal regions on the exposed side.

8.2.3.3 uPVC

A relatively flexible material, PVC which is internally reinforced with steel or aluminium to give it the required strength and stiffness may also be used. The inherent ability to make a variety of profiles, easier maintenance, corrosion resistance and heat resistance when used may be considered when choosing uPVC sections.

8.2.3.4 Timber

Timber frames are relatively stiff and particularly treated timber combined with good design and workmanship give an acceptable life. They are adopted for their inherent thermal insulation property, light weight and improved sound insulation. Tenderness to absorb moisture, if not carefully treated prior to use can lead to possible rotting or even warping resulting in air and water infiltration and regular painting and maintenance are the factors to be considered while choosing timber as the framing material.

8.2.4 Surface Finishes

The commonly employed surface finishes are listed hereunder:

- a) *Aluminium frames* — The main types of aluminium frame coatings are as follows:
 - 1) Anodized finish,
 - 2) Powder paint finish, and
 - 3) PVDF coating.
- b) *Steel frames* — Surface finish on stainless steel frames plays a major role in its resistance to corrosion and hence it is an important design aspect to be clearly specified. The types of surface finishes listed here shall be in accordance with applicable relevant standards:
 - 1) Mill,
 - 2) Ground,
 - 3) Brushed,
 - 4) Polished,
 - 5) Bead blasted,
 - 6) Electro polished,
 - 7) Coloured, and
 - 8) Patterned.

Electro polishing of steel frames shall be done in accordance with available standards.

- c) *Timber frames* — The function of finishes on timber is two-fold, first to improve the durability of the frame and second to add to the aesthetics. Unfinished, unprotected timber weathers as a result of gradual changes to its physio-chemical structure brought about by temperature and moisture content variations. The finishes may be of the following types:
- 1) *Conventional opaque systems* — Water and solvent borne;
 - 2) *Natural finishes* — Semi-transparent and opaque; and
 - 3) Water repellent preservatives, wood primers, undercoats and finishing coats (gloss, semi-gloss, flat and low sheen).

Based on the application, the appropriate type of finish shall be chosen.

8.2.5 Types of Anchorages

Post-installed anchors are used primarily to join the mullions and transoms to the beam or slab through an appropriate bracket. The most common types of post-installed anchors are torque controlled expansion anchors and adhesive anchors. The selection of the appropriate anchor depends on the load requirement. For details on the types of post-installed anchors, reference may be made to good practice [C-8(30)].

There are different factors which influence the selection of anchors and to ensure proper performance of the anchor during its service life. These include:

- a) *Diameter* — The following parameters shall be calculated for the selection of the anchor diameter:
 - 1) Dead load acting on the anchor;
 - 2) Live load incident to the anchor (wind load- tensile or shear);
 - 3) Weight of bracket, aluminium channels and other associated components; and
 - 4) Performance against earthquake load.
- b) *Length* — The overall length of the anchor is based on the effective embedment length into the concrete and the total thickness of the glazing components which have to be fastened to the base material. Insufficient embedment length shall lead to pull-out failure of the anchor;
- c) *Material* — Stainless steel and galvanized iron are the two main types of materials used in the manufacturing of anchors. The selection of anchor material shall be based on the location of the anchors, considering factors such as interior or exterior application, exposure to weather, or enclosed conditions;
- d) *Coating* — The following type of coating shall be used based on the specific environmental condition met with:

<i>Sl No.</i> (1)	<i>Environmental Condition</i> (2)	<i>Coating to be Used</i> (3)
i)	Particular influence of moisture	Zinc plated
ii)	Occasional exposure to condensation and in coastal areas/slightly corrosive	Hot-dip galvanized
iii)	Heavy condensation/corrosion and in coastal areas:	
	a) areas with high corrosion risk	Stainless steel (SS 316)
	b) risk of exposure to corrosion is medium	Stainless steel (SS 304)

- e) *Load Direction* — Based on the orientation of the anchor, they are incident to shear loading, tensile loading and combined loading;

- f) *Base material* — The material characteristic of the base material (concrete) also plays a major role in avoiding failure in the anchorage. The material shall be of suitable grade, have minimum thickness, shall be properly cured and have the necessary compressive stress. Unless otherwise mentioned by the concerned structural engineer, the concrete shall be considered as cracked for all design purposes. Reference shall be made to good practice [C-8(30)] for more details in this regard;
- g) *Anchor spacing and edge distance* — Anchor spacing is the centre to centre distance between the anchors. Edge distance is the distance from the centre of the anchor perpendicular to the respective edge of the concrete member. Reference shall be made to good practice [C-8(30)] for more details in this regard;
- h) *Effective embedment depth* — It is the effective depth which actually transfers the forces passing through the anchor into the concrete. Reference shall be made to good practice [C-8(30)] for more details in this regard;
- j) *Environmental condition* — Environmental conditions like temperature, humidity, salinity, etc will affect the performance of the anchors and may lead to pre-mature failure. Hence, careful selection of the material and coating should be done when selecting anchors which are to be installed in location subjected to adverse environmental impact such as coastal region; and
- k) All the anchors shall be designed as per the recommendations of good practice [C-8(30)].

8.2.6 Hardware Materials

The material, geometry and finish of all the hardware materials shall be as per the requirement and should not affect the performance of the glazing system. All materials and finishes shall be as per relevant Indian Standards and tested before installing into the system in accordance to relevant Indian Standards.

8.3 Fabrication

8.3.1 Factory

The manufacturer shall ensure that proper internal control and documentation including records of test results performed be maintained about the production. The above shall enable quality assurance of the materials and process involved.

NOTE — For fabrication tolerance of aluminium systems, reference shall be made to 6 of accepted standard [C-8(12)].

8.3.2 Handling of Materials on Site

The following lists among others, the important features to be considered:

- a) All materials shall be handled by workers who have been provided appropriate training and awareness on the techniques to handle the materials and shall operate with a full set of personal protection equipment (PPE);
- b) Large glass panes or panels shall be handled with vacuum cups. Care shall be taken to ensure that the vacuum cups and the glass are perfectly clean. An additional rope or belt is recommended to support the dead load as a fall arrestor;
- c) All materials shall be handled with dry, clean, gloves and shall not come in contact with rough surfaces or hard objects, in order to avoid scratches, dents or breakage; and
- d) Labels and marking shall not be put on the coated side of glass, and any relative movement of the coating with the next glass pane shall be avoided.

8.3.3 Transportation and Logistics

While transporting materials from vendor to factory, within the factory, to and from surface finish coater, finished goods to site and moving finished goods at site, care should be taken to ensure the following:

- a) Finished surface is adequately protected with a protective tape which can be easily removed without leaving any residue;

NOTE — Most protective tapes either peel off or adhere to the substrate when exposed to UV for more than three months.

- b) Suitable blocking, edge protection, crating, etc are done to avoid dents and gouges;
- c) Hardware is secured and not damaged in transit;
- d) Material is adequately secured in the truck/container especially when the goods are sent as loose container load (LCL) or partial shipment, avoiding any damages;
- e) Lifting equipment shall be adequately engineered for capacity, safety and balance (especially when units are very heavy);
- f) Suitable 'A' frames or customized transport support shall be used; and
- g) Recommendations from glass supplier towards the use of protective film shall be followed.

8.4 Installation

8.4.1 Site Survey of Building

- a) Measuring the elevation width and height, ascertaining the grid pattern including position of vision and spandrel frames, the levels of concrete and masonry in x, y and z axis, checking and recording the presence of any difference in plumbness of jambs, beam and column fascia and steel structures are the key activities necessary in deciding the final fabrication measurements;
- b) Levelling (setting x-axis, common sill level for all elevation and adjustment in bracket depth) and surveying (measuring x-axis distance and setting the width and height for all elevation) are involved in the process of site survey of building. The key processes are listed below:
 - 1) *Processes in levelling:*
 - i) The sill bottom level of all the glazing elevations are to be checked by fixing a common benchmark and the reading should be recorded at minimum one metre distance;
 - ii) After taking the reading the highest and lowest point can be located and a decision on the common level of all glazing elements made;
 - iii) The benchmark shall be a permanent mark, using oil paint mark in form of upside triangle;
 - iv) The levels of all beam bottoms and up stand, if any, coming across the glazing elevations at all floors are to be measured; and
 - v) Any other point that can affect fabrication drawings is to be noted.
 - 2) *Processes in surveying:*
 - i) The total width shall be measured using a theodolite and recorded at one plane;
 - ii) It shall be cross checked by measuring existing column to column (at other plan) distance; and
 - iii) The two end point RCC shall be measured for plumbness. If any difference is observed, the readings of first two steps shall be adjusted.

8.4.2 Installation Tolerance

The effect of installation tolerance is vital for any type of glazed system to perform as per the requirement and the specified performance. For structural glazing, the installation tolerances shall be strictly followed in order to assure that the sealant joint opening meets the specified minimum and maximum joint opening dimensions.

NOTE — For installation tolerance of aluminium systems, reference shall be made to 5.2 of accepted standard [C-8(14)].

8.4.3 Good Installation Practices

The good practices involved in installing a glazing system have been categorized hereunder.

8.4.3.1 Anchorage

The anchor installation shall be undertaken by installers who have been adequately trained by the manufacturer to install the specific type of anchor. The installer should be familiar with the correct installation procedures and ensure that the correct drilling and setting tools are available for the proper installation of the anchor.

For details with respect to the right installation procedure, reference shall be made to good practice [C-8(31)].

8.4.3.2 Sub-structures

- a) In order to avoid physical contact with the panels and frames during sealant application, which might cause adhesion loss, powder-free latex gloves shall be used during panel placement;
- b) Proper alignment and setting out of the sub-frame is critical in ensuring the ease of operation of the window/door;
- c) Anchor/bolt heads and joints between external wall and sub-frame should be sealed with sealant for effective water tightness;
- d) Prior to fixing the main frame, the sub frame should be checked for any physical damages and any damaged sub-frame should be repaired or replaced;
- e) Setting out of the sub-frame should also be verified before the installation of the main frame; and
- f) The sub-frame shall be properly cleaned before assembly as dust and debris accumulated in the sub-frame could affect the alignment and fixing of the main frame and may also cause blockage to the weep holes, resulting in overflow of any incidental water into the interior.

8.4.3.3 Sealants

- a) All components (glass and framing member such as aluminium) receiving the structural glazing silicone sealant shall be wiped thoroughly with a clean cloth dampened with a cleaner recommended by the manufacturer, and immediately followed by a dry wipe, using clean cloth;
- b) All surfaces once cleaned/primed shall be handled carefully so as not to contaminate the surface;
- c) The substrates shall be kept free from moisture, condensation, dust and other impurities;
- d) An application temperature range of + 10 °C to + 40 °C (or as specified by the manufacturer) is recommended and shall be adhered to;
- e) Substrate temperatures in excess of + 50 °C will adversely affect curing process and also the adhesion of the sealant to the substrate;
- f) Minimal time shall be taken for cleaning, placement and sealing of panels since environment conditions at job site cannot be controlled;
- g) Temporary fasteners may be used to keep the joints stable during the curing process of the sealants and they shall be removed only after complete cure and adhesion has been achieved by the sealant;
- h) Tooling of the sealant shall be done with light pressure, within 5 to 10 min, before a skin forms on the sealant;
- j) The use of wet tooling aids such as soaps and solvents shall be avoided. Dry tooling is appropriate;
- k) Masking tape shall be used in ensuring neat straight-line application and avoiding smears;
- m) The masking tape shall be removed immediately after tooling has taken place and before the sealant starts to cure;
- n) The general recommendation for the application of structural sealant required for a 4-sided system, bonding the glass to the aluminium frame should be done in a factory atmosphere;
- p) Complete quality control process with respect to surface preparation, storage, sealant application and quality check are to be implemented in the factory with full traceability. In most cases sealant

manufacturers have quality requirements and documentation process related to warranties which need to be followed; and

- q) *In-situ* application may be required for 2-sided system, retrofit, repair and small areas in a facade. Sealant manufacturers should be contacted to get specific recommendation on sealant application for *in-situ* application.

8.4.3.4 Installation

The framing and fixing system of the panel should, in no case, cause shear stresses in the panel, either under the impact of external forces or due to differential expansion of components. The following shall be taken care of:

- a) All framing shall be checked prior to glazing to assure that the opening is square, plumb and secure so that the uniform sealant bite, face and edge clearances are maintained;
- b) Inspection of all butt and mitre joints shall be carried out and if found open, they shall be sealed using sealants prior to glazing;
- c) Minimum edge clearances shall be maintained between glass and sash, as mentioned by the manufacturer;
- d) Setting blocks shall be used to support the glass unit which shall prevent the addition of stress applied on the structural tensile bead due to the self-weight of glass; and
- e) Adhesion tests shall be performed by the contractor/agency at the beginning and during the installation process.

8.5 Testing

8.5.1 Laboratory Testing

The glazing system is subjected to various adverse environmental conditions and impacts during its service life. A properly designed and installed glazing system may be able to resist any major damage to its stability or performance under any condition within its effective life time. The glazing system and its component materials shall be tested for the required parameters and performance.

8.5.2 Field/On-Site Testing

Testing of the glazing system as a whole unit in site (the field) subject to loads and effects such as water infiltration, wind (air) infiltration, acoustic insulation, fire insulation, thermal performance, structural performance, etc may be carried out for specific cases as mutually agreed to between the parties concerned, to ensure homogeneity.

The [Table 30](#) lists the tests to be carried out for glazing systems including curtain walls, windows, skylights, sliders, doors, etc. The frequency of these tests shall be mutually agreed upon by the parties concerned.

Table 30 Testing of Glazing Systems including Curtain Walls, Windows, Skylights, Sliders, Doors, etc
(Clause [8.5.2](#))

Sl No. (1)	Test Parameters (2)	Testing Method (3)
i)	Air permeability	As per accepted standard [C-8(32)]
ii)	Water penetration — Static air pressure difference	As per accepted standard [C-8(33)]
iii)	Water penetration — Dynamic air pressure difference	As per accepted standard [C-8(34)]
iv)	Standard operating forces	As per accepted standard [C-8(35)]
v)	Window and doors resistance to repeated opening and closing	As per accepted standard [C-8(36)]
vi)	Wind resistance for curtain walls, windows, sliders and skylights	As per accepted standard [C-8(37)]

Table 30 (Concluded)

Sl No. (1)	Test Parameters (2)	Testing Method (3)
vii)	Structural load test (100 percent design wind load)	As per accepted standard [C-8(37)]
viii)	Structural proof load test (150 percent design wind load and seismic force, if applicable)	As per accepted standard [C-8(37)]
ix)	Seismic racking test	As per Annex F
x)	Water penetration — Field	As per accepted standard [C-8(32)]
xi)	Thermal performance of windows, doors and shading devices — Detailed calculations	As per accepted standard [C-8(38)]
xii)	Calculation of thermal transmittance — Numerical method of frames	As per accepted standard [C-8(39)]
xiii)	Mechanical tests for standard windows and door height windows	As per accepted standard [C-8(40)]
xiv)	Fire resistance of perimeter fire barrier joint system	As per accepted standard [C-8(20)]
xv)	Fire resistance tests — Door and shutter assemblies	As per accepted standard [C-8(41)]
xvi)	Fire resistance test for glass walls	As per accepted standard [C-8(42)]
xvii)	Fire resistance test for doors with glass panes, openable glass windows and sliding glass doors	As per accepted standard [C-8(43)]
xviii)	Forced entry security glazing	As per accepted standard [C-8(44)]
xix)	Bullet resistant glass	As per accepted standard [C-8(45)]
xx)	Acoustics — Sound insulation	As per accepted standard [C-8(25)] and [C-8(26)]
xxi)	Blast resistant security glass	As per 9.4.3

8.6 Maintenance

8.6.1 Maintenance Infrastructure

Regular maintenance of the building facades plays an important role in the holistic building maintenance. The access systems shall provide sufficient approach which enables all parts of the facade to be reached safely for the purpose of cleaning, inspection and maintenance and also offer high work efficiency to the maintenance workers, without also hindering the aesthetics of the building. The provisions of relevant guidelines/best practices (Handbook on 'Asset and Facility Management in Buildings') may also be referred.

The means of access and supporting infrastructure should be established at the design stage so that the loads are properly envisaged for.

Facade access requirements come in many varieties and may present challenges. Powered access cradles for facade access, cleaning of the glass directly from the ground, hoists based access and window based cleaning/access from inside the building are some of the common techniques employed. Recently, access techniques using ropes have been used for facade access and are generally considered to be safe, provided the workers have been trained and are using all the correct safety equipment and precautions.

In nearly all cases, facade access and maintenance requires a worker to be exposed to working at height with its associated risks. Therefore, use of height safety equipment, anchor point attachments (like the fall restraint or fall arrest system) and those given below should be employed:

- a) *Roof guardrails* — When choosing equipment for working at heights, collective protection like roof guardrails may also be considered. Roof guardrails ensure that all users are protected from falling without the need for specialised training and/or equipment. They also eliminate the need for rescue

procedures to be put in place. Roof guardrails are available as fixed, free standing, straight upright, cranked upright or collapsible; and

- b) *Fall arrest systems and fall prevention cable systems* — Fall arrest systems allow safe access to roof areas for maintenance and inspections to be carried out by workers. Users of the fall arrest systems require wearing of full body harnesses and attachment to the cable/fall arrest system via a lanyard (rope) and shuttle device which allows them to walk the full length of the system without having to detach from the system. Fall arrest systems shall be of horizontal, overhead, sloping and vertical lifeline type and shall comply to the relevant/available standards.

Depending on the project requirement, the maintenance infrastructure shall consist of the following systems:

- 1) *Roof trolley/Cradles* — Fixed or adjustable jib, telescopic jib and cradle with pantograph.
- 2) *Rail systems* — Hooked, anchored, turntables and rotary plates.
- 3) *Other systems* — Additional winches for glass replacement and parking positions/lifting platforms.

8.6.2 Maintenance of Materials

The glazing system and the associated materials require a suitable degree of maintenance, if they are to fulfil their intended working life. Early detection of defects can mitigate expensive repairs or replacement later. The degree of inspection and maintenance required will depend on the type of glazing system and its intended design life.

8.6.3 Common Practices for Maintenance

The common principle maintenance practices to be followed shall be as given below:

- a) When using the cleaning equipment or carrying out maintenance on the exterior of the building, the risk to personnel and property, prior to the start of work shall be assessed at first;
- b) Where applicable, qualified personnel to carry out routine or complex maintenance shall be employed;
- c) Power supplies may need to be isolated prior to maintaining electrically operated equipment such as motors to opening vents, roller blinds to atrium roof and main entrance pass doors;
- d) While accessing the external facade, fall arrest systems shall always be used even while working in the house cradle;
- e) Manufacturer's instructions shall be followed when using cleaning fluids and shall not be mixed up with drinking water or the like. To deal with instances of contact with eyes, skin, etc necessary eyewash arrangement and first aid equipment shall be in place;
- f) Loosened fixing or bolts shall be identified, reported and attended to immediately, and shall never be tampered with;
- g) Unauthorized loosening or removing of the fixings may prove dangerous and can result in structural damage and displacement of the system;
- h) When lifting equipment such as hoists and sucker machines are used, it shall be ensured that they are capable of lifting the loads required. Further, the recommended safe working loads of any equipment shall never be exceeded;
- j) When handling any insulation of firebreak material, it shall be ensured that gloves, goggles and masks are worn as fibrous materials (if any) may cause irritation;
- k) Prior to the commencement of work involving glues or sealants (particularly in confined spaces), manufacturer's advice/special instructions shall be adhered to before exposing the product to air. Such materials shall preferably be worked with, in a well-ventilated area;
- m) In the case of coated glass, the following precautions shall particularly be adhered to:
 - 1) Any scratching shall be avoided as this may penetrate the surface of the coating and cannot be repaired;
 - 2) Any excessive mechanical treatment shall be avoided as this might remove the coating in localised areas;

- 3) Contact with metal objects shall be avoided; and
- 4) Use of chemicals that may attack the surface and damage it irreparably shall be avoided.

9 SECURITY GLASS

9.1 This clause covers the requirements and selection criteria for security glass in buildings. The general methodology for selecting types of security glass shall be in accordance with [4](#). It is essential for any security glass in addition to its required function to provide adequate human impact safety features. The level of human impact safety for glass in buildings shall be in accordance with [7.1](#). Furthermore, the security glass shall meet the classification standards to which it has been tested, ensuring the safety and security of occupants and goods. This clause deals with the safety and security of people or goods in relation to risks of,

- a) vandalism, riots, burglary or break in protection;
- b) protection from firearm shooting;
- c) protection from explosion (terrorist attack); and
- d) protection from ballistic attacks.

The requirements given below pertain to glass used in areas with high-security needs. In circumstances wherein the requirements of other Indian Standards exceed the requirements of this clause, the higher requirements shall become applicable.

9.2 Glass and Related Material for Security

Glass used for security purpose are laminated glass with different combination of glass panes including annealed glass, heat strengthened glass, toughened glass, and interlayers (including polyvinyl butyral, ethylene vinyl acetate, plastic sheets, polycarbonate sheets, etc) in accordance with [4](#).

9.3 Selection Criteria

The selection criteria is based on the threat levels/risks (firearm shooting, ballistic attacks, explosion, burglary, etc) and stages of protection as per the requirements of the buildings and the user.

9.4 Testing procedures

9.4.1 Burglary Resistance of Glazing Systems

The framing member along with its hardware and glass should comply for the burglar resistance requirements as a whole. The glass to be used should comply with the testing requirements of accepted standard [C-8(44)]. The following [Table 31](#) lists the minimum requirements for various resistance classes of glazing.

Table 31 Minimum Requirements for Glazing (Clause 9.4.1)		
Sl No.	Resistance Class	Minimum Resistance Class of Glazing as per Accepted Standard [C-8(46)]
(1)	(2)	(3)
i)	RC 1 N	Any glass (test is for hardware manipulation for entry)
ii)	RC 1	P4A
iii)	RC 2 N	Any glass (test is for hardware manipulation for entry)
iv)	RC 2	P4A
v)	RC 3	P5A
vi)	RC 4	P1B
vii)	RC 5	P2B
viii)	RC 6	P3B

If a glass unit with a higher security level is used within the specimen used for the tests, it may not be possible to assess the use of glass units with a lower grade within those products without conducting further tests. This is because higher grades of glass can increase the rigidity of the product.

The following [Table 32](#) lists the criteria for the selection of the resistance class (offender type, offender behavior, location, risk and recommendation for use).

Table 32 Criteria for the Selection of the Resistance Class					
(Clause 9.4.1)					
Sl No.	Resistance Class	Expected Offender Type, Suspected Offender Behaviour	Recommended Place of Use of the Burglar-Resistant Component		
			A Residential properties	B Commercial properties, public properties	C Commercial properties, public properties (High risk)
(1)	(2)	(3)	(4)	(5)	(6)
i)	RC 1 N/ RC 1	Components of resistance class RC 1 N/RC 1 provide basic protection against attempts to break in with physical force such as kicking in, jumping, shoulder slamming, pushing up and tearing out (mainly vandalism). Components of resistance class RC 1 N/RC 1 only have a low level of protection against the use of lever tools	If burglar resistance is required, the use of resistance class RC 1 N/RC 1 is only recommended for building components where no direct access (not ground level access) is possible		
ii)	RC 2 N/ RC 2	The occasional offender tries in addition, to break open the component with simple tools such as screwdrivers, pliers and wedges. (RC 2 N is only recommended for building components where no direct attack on the glazing used is to be expected)	Average risk attack on the glazing used is to be expected	Average risk	Not applicable

Table 32 (Concluded)

Sl No.	Resistance Class	Expected Offender Type, Suspected Offender Behaviour	Recommended Place of Use of the Burglar-Resistant Component		
			A Residential properties	B Commercial properties, public properties	C Commercial properties, public properties (High risk)
(1)	(2)	(3)	(4)	(5)	(6)
iii)	RC 3		High risk	The offender tries in addition, to break open the component with a second screwdriver and a crow bar.	Not applicable
iv)	RC 4	The experienced burglar uses in addition, tools such as a hand saw, heavy hammer, axe, chisels, and a battery-powered drill.	Not applicable	Not applicable	Low risk
v)	RC 5	The experienced burglar uses in addition, electric tools such as an electric drill, jig saw, or sabre saw and an angle	Not applicable	Not applicable	Average risk
vi)	RC 6	The experienced burglar uses in addition, heavy-duty electric tools, such as a heavy-duty drill, jig saw or sabre saw and angle grinder	Not applicable	Not applicable	High risk
NOTES 1 This table is only an approximate guide. Expert advice, for example, from the local police advice centres, is essential. The assessment of the risk should be on one's own responsibility, taking into account the location of the building (protected/unprotected), use and the value of the goods inside. In case of high risk, tested and certified burglar alarm systems should be used in addition. 2 When selecting burglar-resistant elements of resistance classes RC 4 to RC 6, it should be noted that when selecting such elements in escape and rescue routes, the use of tools by the fire brigade is made more difficult and shall therefore be taken into account. 3 External sockets, for example, in the hallway, in the garden or in the patio (open space) area should be de-energized to prevent their use by the burglar.					

In addition, parties to agreement may mutually decide on testing arrangement based on specialist literature.

NOTE — Specialist literature may be EN 1630 : 2021 'Pedestrian doorsets, windows, curtain walling, grilles and shutters - Burglar resistance - Test method for the determination of resistance to manual burglary attempts'.

9.4.2 Bullet Resistance Glazing

The framing member along with its hardware and glass shall comply for the bullet resistance requirements as a whole. The glass to be used should comply with the testing requirements of accepted standard [C-8(45)].

The testing of framing member shall follow the same parameters of spall, no spall as well as threat levels (see [Table 33](#)) described in accepted standard [C-8(45)] but with penetration points in the framing member. Glazing system classification will hold good only if both the framing member and glass complies. In case, either glass or frame achieves a lower classification, then the whole glazing system shall be classified to the lower class that has been achieved.

Table 33 Threat Levels

(Clause [9.4.2](#))

SI No.	Threat Level	Ammunition	Bullet weight	Bullet Type	Strike Velocity	Distance of Impact	No. of Strikes	Remarks
(1)	(2)	(3)	g (4)	(5)	m/s (6)	m (7)	(8)	(9)
i)	1	9 mm $\times$ 19 mm	7.4 to 8.2	FMJ/Pb	430 $\pm$ 15	5 $\pm$ 0.5	5	–
ii)	2	7.62 mm $\times$ 39 mm	7.45 to 8.05	FMJ/ MSC	710 $\pm$ 15	10 $\pm$ 0.5	5	–
iii)	3	7.62 mm $\times$ 51 mm	9.4 to 9.6	FMJ/Pb	838 $\pm$ 15	10 $\pm$ 0.5	5	plus compliance with Threat Level 2
iv)	4	5.56 mm $\times$ 45 mm	3.5 to 4.0	FMJ/(SI + Pb)	890 $\pm$ 15	10 $\pm$ 0.5	5	plus compliance with Threat Level 3
v)	5	7.62 mm $\times$ 39 mm	7.45 to 8.05	HSC	700 $\pm$ 15	10 $\pm$ 0.5	5	plus compliance with Threat Level 3
vi)	6	7.62 mm $\times$ 54 mm	10.3 to 10.5	API	830 $\pm$ 15	10 $\pm$ 0.5	1	–
vii)	Special (see Notes)	Any other requirement by the user. Complete details of ammunition shall be stored for future upgradation of this NBCS						
NOTES								
1 For special threat levels, the bullet velocity specified for the test should be representative of that developed by the weapon and ammunition, preferably at the higher end of the range of possible velocities. The range selected should be sufficient to enable the bullet flight to stabilize.								
2 FMJ/Pb = Full metal jacket – Lead core, FMJ/MSJ = Full metal jacket – Mild steel core, HSC = Hard steel core, API = Armour-piercing incendiary.								

9.4.3 Blast Resistance Glazing

The framing member, along with its hardware and glass, shall collectively comply with the blast resistance requirements specified in [Annex G](#). Glazing system classification will hold good only if both the framing member and glass complies. In case either glass or frame achieves a lower classification, then the whole glazing system shall be classified to the lower class that has been achieved.

The test ammunitions specified in this NBCS represent the most frequently faced threats and are classified into six threat levels given in [Table 34](#).

Table 34 Classification, Charge (TNT/Equivalent Charge) Mass, Distance (*d*) and Height of Burst (HOB)

(Clauses [9.4.3](#), [G-4.7](#) and [G-4.8](#))

Sl No.	Threat Level	TNT/Equivalent Charge Mass	Distance of Explosive Charge (TNT/Equivalent Charge) from the Test Specimen (<i>d</i>)	Height of Burst (HOB)
		kg	m	m
(1)	(2)	(3)	(4)	(5)
i)	1	2	4.5	0
ii)	2	3	5.0	0.5 ± 0.05
iii)	3	3	3.0	0.5 ± 0.05
iv)	4	12	5.5	0.8 ± 0.05
v)	5	12	4.0	0.8 ± 0.05
vi)	6	20	4.0	0.8 ± 0.05
vii)	Special	Any other requirement by the user.		

The structural integrity of glass system test specimen is evaluated by visual inspection in the following categories of outcome:

- Category A* — No damage to the glass and the frame of glass system;
- Category B* — The entire system is bent or deformed leading to the overall jamming or dis-functioning of the system;
- Category C* — Formation of cracks on the glass system and no damage to the frame of the glass system;
- Category D* — Cracking/shattering of different glass layers and damage to frame of the glass system;
- Category E* — Formation of splinters from the glass system or the frame within a distance of 3 m. The size of perforations in the glass or the frame shall not be more than 10 mm; and
- Category F* — Total collapse of the glass or frame of the glass system or both.

ANNEX A

(Clause [2.15](#))

TEST METHOD FOR ASSESSING CURTAIN WALLS AND STOREFRONT SYSTEMS UNDER INTER-STOREY DRIFT CONDITIONS

A-1 OBJECTIVE

This test method provides a means of evaluating the performance of window wall, curtain walls and storefront systems when subjected to specified horizontal displacements in the plane of the wall. This test is applicable to window wall, curtain wall and/or storefront mock-ups consisting of supports and components, with glass panels fixed across more than one storey (hereinafter referred to as exterior wall systems). This test method is applicable to window systems only when included as part of a full-size wall mock-up. This test method describes the apparatus and the procedure to be used for displacing a specimen horizontally. The displacement cycle time is not limited and is intended to duplicate the inter-storey displacement rates associated with dynamic building sway or seismic or wind events. This test shall be applicable for seismic Zone IV and Zone V. However, depending upon application and for tall structures, it shall be performed for other seismic zones as well.

This annexure does not support to address all of the safety problems, if any, associated with its use. It is the responsibility of the user of this test method to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

A-2 TERMINOLOGY

A-2.1 Lateral Movement — A sideways displacement of a structure or part of the structure.

A-2.2 Test Specimen — The test element is referred to as the specimen. The test specimen for a wall shall be of sufficient size to determine the required performance of all typical parts of the wall system.

A-2.3 Specimen Area – The area calculated from the overall dimensions of the test specimen.

A-2.4 Floor Height – Height of one floor of the sample to be tested.

A-2.5 Horizontal Displacement – The relative displacement of the curtain wall in the horizontal direction when subjected to horizontal movement.

A-2.6 Window Wall – It sits between floor slabs and are anchored at the sill and head.

A-3 APPARATUS

A-3.1 The description of the apparatus in this section is general. Any suitable apparatus arrangement during the laboratory accreditation process is permitted.

A-3.2 The displacement(s) may be accomplished using hydraulic or pneumatic cylinders or jacks of sufficient load and stroke capacity, or mechanical devices, such as ratchet pulleys or a chain hoist.

A-3.3 Test chamber members that are representative of the building structure shall be designed to accomplish the specified displacement(s). This may be done through a system of sleeves, fixtures, pins, or any system capable of accomplishing the displacement(s) directed by the specifier.

A-3.4 The displacement measuring system shall be accurate to within 1.5 mm (0.06 inch).

A-4 CALIBRATION

All test equipment shall be calibrated periodically, at least once every six months. Calibration should be done from an accredited laboratory providing traceability of calibration to national standards established through a chain of calibration certificates.

A-5 TEST SPECIMEN

A-5.1 The exterior wall system test specimen shall be of sufficient size and configuration to determine the performance of all typical parts of the exterior wall system. A minimum of three vertical bays in two adjoining floors of the exterior wall system shall be tested. If the wall has inside and/or outside corner, one set of corner bays shall also be included in the testing specimen. For glazing that is anchored within the floor/storey such as windows, doors, sliders and horizontal strip facade, the test specimen can be of a single floor.

A-5.2 All parts of the exterior wall system test specimen shall be full size, using the same materials, type of glass, detailing, methods of construction and anchorage as those used on the actual building.

A-5.3 The test chamber structure shall simulate the main structural supports of the actual building. However, the test chamber and/or support structure may differ from the actual building, as required, to perform the required displacement. For multi-storey mock-ups, the test chamber shall be constructed so that the anchorage at the simulated floor structure at an intermediate level of the test specimen is moveable in the horizontal direction(s). For single storey mock-ups, the test chamber shall be constructed so that the anchorage at the top or bottom is moveable in the horizontal direction(s).

A-5.4 The test laboratory shall minimize unintended torsional or flexural movements of chamber and/or support structure that could affect test results.

A-6 PROCEDURE

A-6.1 Prior to conducting the displacement tests, the test specimen shall be, at a minimum, characterized for serviceability by conducting air leakage and water penetration tests. The air leakage and water penetration tests shall be conducted at the differential pressures specified for the project.

A-6.2 Conduct Specified Displacement Test(s)

A-6.2.1 Test chamber elements representing the primary building structure shall be displaced to produce the specified movements. Each test shall consist of minimum three full cycles. A cycle is defined as a full displacement in one direction, back to the originating point, full displacement in the opposite direction, and back to the originating point (*see Fig. 20*).

A-6.2.2 Hold the specimen at displaced position not more than 60 s at each direction.

A-6.2.3 An inspection to be conducted during and after every cycle and recorded.

A-6.2.4 At the conclusion of the test, technicians and witnesses shall visually inspect the mock-up for evidences of failure.

A-6.2.5 The test agency shall record all areas of visual distress, such as disengagement, metal distortion, sealant or glazing failure, gap between framing members, or permanent deformation.

A-6.2.5.1 If glass breakage occurs during displacement tests, the test agency personnel shall carefully examine the test specimen and document the extent of this glass breakage.

A-6.2.6 Elements representing the building structure shall be displaced in a manner and method as determined by the test agency and may be subject to review by the design professional. Movement shall be parallel to the primary plane of the wall.

A-6.2.7 Changes in displacement direction(s) shall allow for movement of equipment, stops, measuring devices, etc. The test agency shall displace the chamber elements in a manner that is not intended to induce sudden acceleration and deceleration.

A-6.2.8 Additional cycles and/or displacement direction(s) may be specified by the design professional. The chamber elements shall be returned to their nominal position after completion of each cycling test, before further tests are conducted.

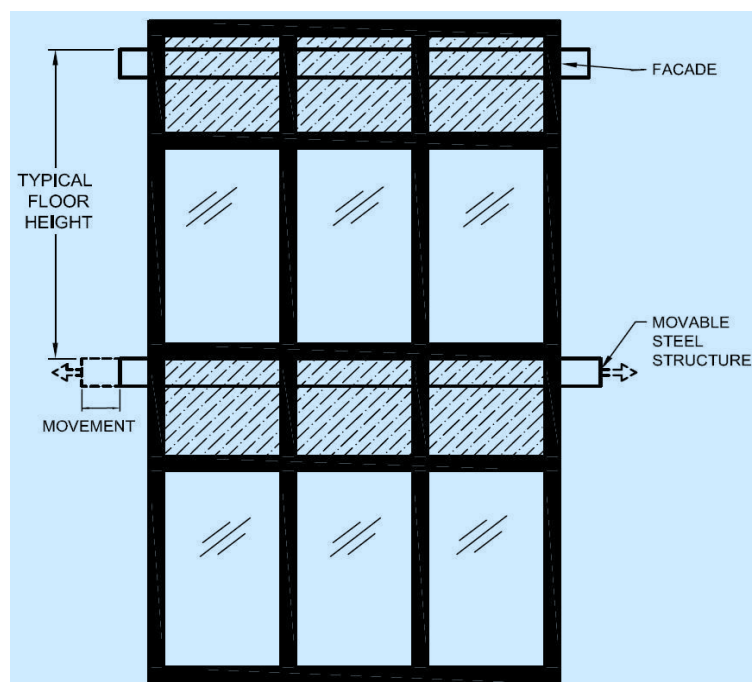


FIG. 20 ILLUSTRATION OF MOVABLE STRUCTURAL ELEMENTS IN CURTAIN WALL SYSTEMS UNDER INTER-STOREY DRIFT CONDITIONS

A-6.2.9 The elastic design displacement shall be as provided by the building engineer of record, determined by elastic analysis or other suitable method for a specific building. The displacement shall be measured at the movable floor element, not at the test specimen. For multi-storey mock-ups, the displacement between levels may vary due to different storey heights. If unspecified in contract documents, for testing, elastic design displacement in each direction shall be $H/500$, for curtainwall systems that span from floor-to-floor and primarily accommodate movement through racking or tipping.

A-6.3 Serviceability tests for air leakage as per accepted standard [C-8(47)] and static water penetration as per accepted standard [C-8(33)] may be conducted after the static elastic design displacement tests. The air leakage and static water penetration tests should be conducted at the pressures specified for the project.

A-6.4 The displacement shall be measured at the movable chamber member that represents the floor element, not at the test specimen or its anchorage.

A-7 CRITERIA

A-7.1 The project specifications shall state detailed performance criteria for the windows, window wall, curtain wall or storefront system. If detailed performance criteria are not included in the project specification, the criteria outlined in this test method shall be utilized.

A-7.2 Unless otherwise specified, an exterior wall system test specimen subjected to the elastic design displacement test shall be considered as passing the inter-storey drift provisions of this test method if the specimen meets the following criteria:

- No visible damage to framing or trim components or assemblies is allowed.
- No glass breakage or glass fallout is allowed.
- Full disengagement of gaskets or weather seals is not allowed at any location.
- Any gap between framing members shall be notified.
- Air infiltration and static water penetration resistance shall remain within specified allowable limits without adjustments or repair for serviceability conditions.

NOTE — If any of air infiltration or static water penetration test fails, then seismic test has to be repeated before proceeding with air infiltration or static water penetration retests.

- f) No wall components may fall off.

NOTE — If an inter-storey drift ($1.5 \times$ storey height) test is conducted, the results shall be recorded and mutually agreed upon with the specifier. In this case, compliance with the above failure criteria is not required.

A-8 REPORTING

The information to be duly reported after testing are given hereunder.

A-8.1 General

Testing agency, date and time of test, and date of report.

A-8.2 Specimen Details

A-8.2.1 Specimen Description

Manufacturer, model, operation type, materials, and other pertinent information; description of the locking and operating mechanisms if applicable; glass thickness, type and method of glazing; weather seal dimensions, type, weep holes, and material; and crack perimeter or specimen area, or both.

A-8.2.2 Drawings of Specimen

Detailed drawings of the specimen showing dimensioned section profiles, framing location, panel arrangement, installation and spacing of anchorage, weather stripping, locking arrangement, hardware, sealants, glazing details, weep holes, drainage path detail, and any other pertinent construction details. Any modifications made on the specimen to obtain the reported test values shall be noted.

A-8.3 For window and door components, a description of the locking and operating mechanism.

A-8.4 Identification of glass thickness and type and method of glazing.

A-8.5 Test Parameter

A-8.6 List of Equipment Used

List the equipment name, identification number and calibration validity of all the equipment used at the time of testing.

A-8.7 Test Observation

A visual record of the test observations.

A-8.8 Photographs or drawings of the test chamber and the wall system test specimen mock-up indicating location of measuring devices and movement devices.

H-8.9 Compliance Statement

A statement that the tests were conducted in accordance with this test method, or a complete description of any deviation from this test method. When the tests are conducted to check for conformity of the specimen to a particular performance specification, the specification shall be identified.

A-9 INFORMATION OF DESIGN LOADS

In order to design and detail exterior wall systems and their attachments to meet the inter storey drift requirements, the design load shall be as per good practice [C-8(48)].

Inter-storey drift can result from either seismic loads or by wind loads.

ANNEX B

(Clauses [5.1](#), [5.2.11](#) and [5.3.3.6](#))

MINIMUM PERFORMANCE REQUIREMENT OF GLAZING FOR EXTERIORS

B-1 The panel size design of the exterior glass for the building is done in accordance with this Section considering the effect due to wind load, other imposed load and seismic design. The type of glazing to be installed is determined considering human safety in [7.1](#).

In the determination of the energy and thermal performance of the glazing, the three compliance approaches prescriptive in nature, defines the minimum glazing/glazing system requirement to be met.

The prescriptive values are specified with respect to the geographic climatic zone and glazing area of the building.

B-2 BUILDING FENESTRATIONS

The fenestration may comply with the maximum glazing area weighted on *U*-value and solar factor as given in relevant guidelines/best practices (Handbook on ‘Sustainability in Buildings’).

The window wall ratio (WWR) is defined by the total fenestration area to the total gross area of the building envelope which is limited to 70 percent. The corresponding minimum visible light transmission is given below. For fenestration area greater than 70 percent, a reference may be made to other specialist literature.

<i>Sl No.</i>	<i>WWR</i>	<i>Minimum Visible Light Transmission Percent</i>
(1)	(2)	(3)
i)	0 to 0.3	27
ii)	0.31 to 0.4	20
iii)	0.41 to 0.5	16
iv)	0.51 to 0.6	13
v)	0.61 to 0.7	11

B-3 WEATHER PERFORMANCE

Air infiltration from the fix litres shall be permitted within 1.5 m³/h.m² and from operable sash 2 m³/h.m.

Water should be contained in the gutter and drained back to the exterior. Any uncontrolled water in excess of 15 ml or more on the top surface of any exposed interior shall be considered as leakage.

B-4 In glass facade buildings, it is recommended to have provision for smoke/heat exhaust ventilation that automatically or by manual trigger expels smoke, heat and gases during a fire or emergency. These vents should be integrated into the building's smoke control system to improve evacuation safety, firefighting efforts, and overall air quality. These vents should be part of building facade and may also be located at suitable locations for effective exhaust. For further details in this regard, Part F ‘Fire and Life Safety’ of this NBCS may be referred.

In glass facade buildings, where airtight construction can restrict natural airflow, it is recommended to have make-up air vent also as a part of ventilation system that automatically or by manual trigger, introduces fresh air into the building to compensate for the air exhausted by smoke vents, HVAC systems, or other ventilation processes. These vents help maintain air balance and indoor air quality which ideally should be 85 percent to 90 percent of vented volume of smoke/heat.

ANNEX C

[Clause 5.3.2(b)]

TEST REQUIREMENTS FOR COATED GLASS

C-1 The coated glass that is used for the exterior of commercial and non-commercial (residential building) building should meet the test requirement listed in this Annex.

C-2 MEASUREMENT OF PERFORMANCE

The performance factors and spectral details of the coated glass shall be determined using standard accepted spectrophotometer equipment. The actual size shall be dependent on the type of equipment being used for the measurements. The transmittance of the measurement sample shall be measured with radiation of normal incidence at the following wavelengths:

- a) 550 nm (representative wavelength for light and solar transmittance); and
- b) 900 nm (representative wavelength for solar transmittance).

For glass claiming to have a low emissivity coating, a measurement of the reflectance shall be made at 8 μm using radiation of nearly normal incidence.

C-3 CONDENSATION RESISTANCE TEST

C-3.1 This test consists of subjecting the coated glass to a water saturated atmosphere at constant temperature. The samples have condensation continually forming on them and it is this condensation that may cause surface degradation.

C-3.2 Test apparatus should have a provision of four test pieces and the materials used for the inner walls shall be corrosion-resistant and shall not affect the test pieces.

C-3.3 Procedure

- a) The water tank shall be filled with demineralized water, having conductivity lower than 30 μS and a pH higher than 5. The internal temperature of the cabinet shall be controlled by means of the reference thermocouple keeping a temperature of the reference glass piece of $40 \pm 1.5^\circ\text{C}$;
- b) The test cabinet shall be in a room with an ambient temperature of $23 \pm 3^\circ\text{C}$. Care shall be taken to ensure that draughts, dust, moisture and solar radiation do not interfere with the test cabinet; and
- c) The reference temperature shall be reached within 2 h of commencing heating. Condensation shall be seen to form on the glass pieces. The test is continued without interruption for the required time. Both the internal reference and the external air temperature shall be regularly checked.

C-3.4 Requirements

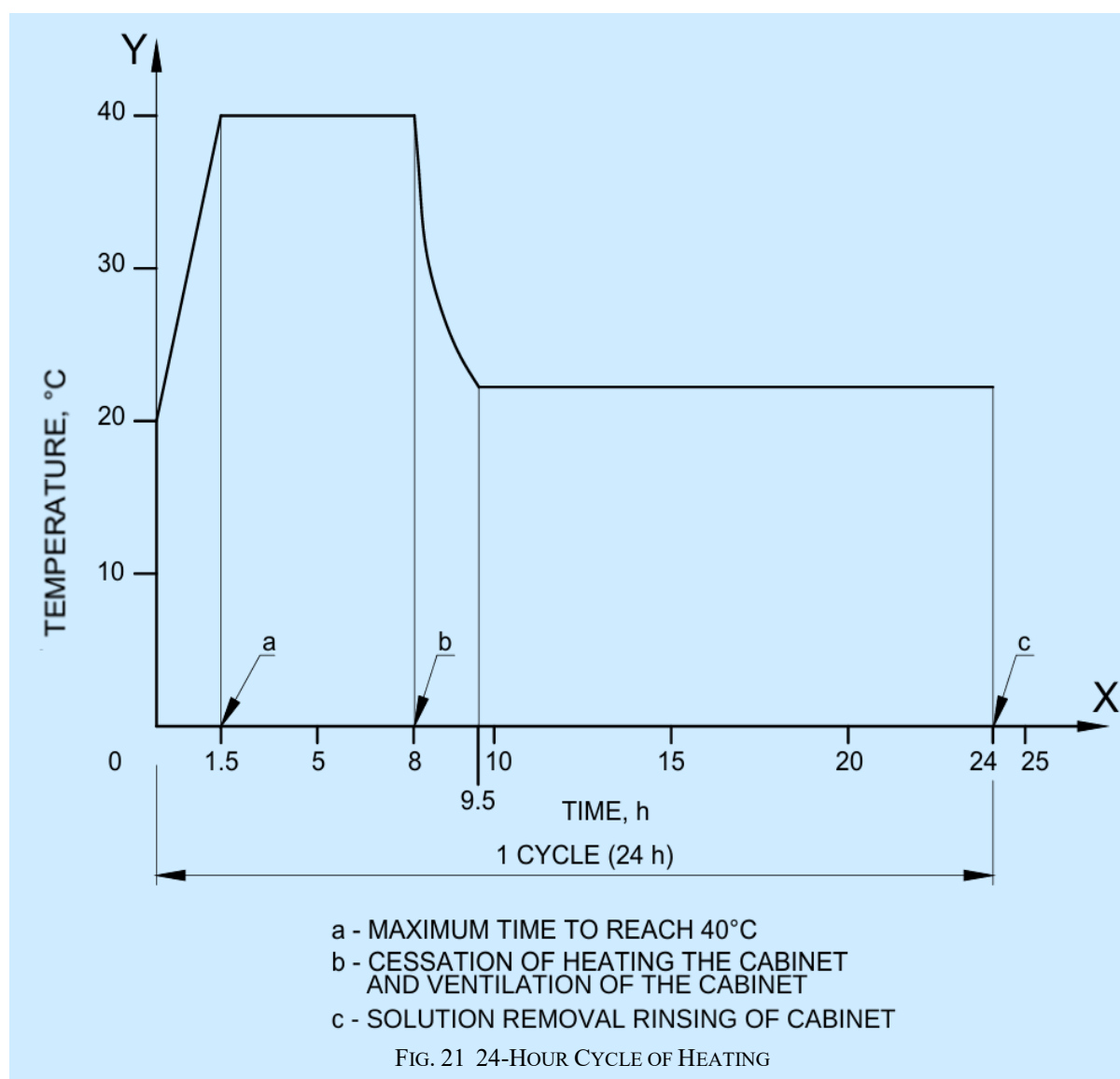
- a) No defect, greater than 3 mm length;
- b) Maximum one defect between 2 mm and 3 mm length;
- c) Maximum five defects between 1 mm and 2 mm length;
- d) No scratches, staining of the coating or clusters of pinholes greater than 1 mm;
- e) When compared with the reference test piece, in both reflection and transmission, there shall be no significant colour change. This observation shall be made within 20 s; and
- f) The transmittance measured at 550 nm and 900 nm shall differ by no more than ± 0.03 from the corresponding measured value on the reference test piece.

C-4 ACID RESISTANCE TEST

C-4.1 This test consists of subjecting the coated glass to a sulphur dioxide saturated atmosphere at constant temperature. The samples shall have condensation continually forming on them. It is this condensation together with the quantity of sulphur dioxide that may cause surface degradation. Test apparatus should have a provision of four test pieces and the material used for the inner walls shall be corrosion-resistant and shall not affect the test pieces.

C-4.2 Procedure

- The test cabinet contains 2 litre of demineralized water having conductivity lower than 30 μS . When the cabinet is closed 0.2 litre of SO_2 shall be added and the heating system switched on;
- The test consists of a repetition of 24 h cycles (*see Fig. 21*);
- Each cycle consists of high temperature plus condensation period and an ambient temperature without condensation period; and
- The temperature shall increase to $40^\circ\text{C} \pm 1.5^\circ\text{C}$ in less than 1.5 h. During the next 6.5 h the test pieces shall be subjected to condensation in the SO_2 atmosphere.



C-4.3 Requirement

The transmittance measured at 550 nm and 900 nm shall differ by no more than ± 0.03 nm from the corresponding measured value on the reference test piece.

C-5 NEUTRAL SALT SPRAY

C-5.1 This test consists of subjecting the coated glass to neutral, water saline atmosphere at constant temperature. It is the water saline spray that may cause surface degradation.

Test apparatus should have a provision of four test pieces and the material used for the inner walls shall be corrosion-resistant and shall not affect the test pieces.

C-5.2 Procedure

- a) The test cabinet shall be prepared and run for a minimum period of 24 h before the test pieces are placed within it. The neutral salt solution is made up by dissolving NaCl in demineralized water having conductivity lower than 30 μ S, to produce a concentration of 50 ± 5 g/l at 25 ± 2 °C; and
- b) The compressed air supplied to the spray nozzle shall be passed through a filter to remove all traces of oil or solid matter and shall be at an absolute pressure of 70 kPa to 170 kPa through a saturator at 40 ± 1.5 °C. The spray nozzle shall be made of inert material, with baffles to prevent direct impact of spray on the test pieces.

C-5.3 Requirements

- a) No defect, greater than 3 mm length;
- b) Maximum one defect between 2 mm and 3 mm length;
- c) Maximum five defects between 1 mm and 2 mm length;
- d) No scratches, staining of the coating or clusters of pinholes greater than 1 mm;
- e) When compared with the reference test piece, in both reflection and transmission, there shall be no significant colour change. This observation shall be made within 20 s; and
- f) The transmittance measured at 550 nm and 900 nm shall differ by no more than ± 0.03 nm from the corresponding measured value on the reference test piece.

C-6 ABRASION RESISTANCE TEST

C-6.1 This test consists of subjecting the coated surface of the glass to rubbing with a felt pad in dry condition. It is the type of pad, the loading on it and the number of strokes that may cause surface degradation.

C-6.2 Procedure

- a) The metal finger shall be approximately 15 mm to 20 mm in diameter and shall be driven so as to produce a frequency of 60 strokes/min $\pm$ 6 strokes/min, alternating forwards and backwards. The stroke length shall be $120 \text{ mm} \pm 5 \text{ mm}$. The strokes shall be parallel and ensure a constant pressure over the zone to be tested;
- b) A circle shape with a diameter of $14.5 \text{ mm} \pm 0.5 \text{ mm}$;
- c) The test sequence shall commence within 30 min of the sample being cleaned; and
- d) The metal finger containing the felt pad shall be lowered on to the glass surface and a load of 4 N applied perpendicular to the glass surface via the felt pad.

C-6.3 Requirements

- a) Ensure that the abraded area is uniform; and

- b) The transmittance measured at 550 nm and 900 nm shall differ by no more than ± 0.03 nm from the corresponding measured value on the reference test piece.

C-7 DURATION OF TESTS

The duration of tests shall be as per the following:

Sl No.	Test	Test Duration for Class		
		A	B	S
(1)	(2)	(3)	(4)	(5)
i)	Condensation resistance	21 days	4 days	14 days
ii)	Acid resistance	5 cycles	1 cycle	5 cycle
iii)	Neutral salt spray	21 days	10 days	—
iv)	Abrasion resistance	500 stokes	50 stokes	500 stokes
NOTES				
1 Class A – The coated surface of the glass can be positioned on the outer or the inner face of the building.				
2 Class B – The coated glass can be used as monolithic glazing but the coated surface shall be on the inner face of the building.				
3 Class S – The coated surface of the glass can be positioned on the outer or the inner face of the building but these types of coated glasses can only be used in specifically defined applications.				

ANNEX D

[Clauses 6.1.9 and E-1.1.1(4)]

PROCEDURE FOR CALCULATING THE APPROXIMATE DEFLECTION AT CENTRE OF GLASS PANEL

D-1 Maximum glass deflection as a function of plate geometry and load may be calculated using the following:

$$\text{Deflection, } w = t \cdot \exp(r_0 + r_1 X + r_2 X^2)$$

where

$$X = \ln\{\ln[q(a.b)^2/Et^4]\}$$

$$r_0 = 0.553 - 3.83(a/b) + 1.11(a/b)^2 - 0.0969(a/b)^3$$

$$r_1 = -2.29 + 5.83(a/b) - 2.17(a/b)^2 + 0.2067(a/b)^3$$

$$r_2 = 1.485 - 1.908(a/b) + 0.815(a/b)^2 - 0.0822(a/b)^3$$

where

E = Young's modulus of glass, N/mm²;

q = net pressure on the pane, N/mm²;

a = longer dimension, mm;

b = shorter dimension, mm; and

t = thickness of the glass pane calculated.

Example:

Lateral Deflection Calculation — Determine the maximum lateral deflection (w) of a vertical 1 200 mm × 1 500 mm × 6 mm rectangular glass plate subjected to a uniform lateral load of 1.80 kPa. The actual thickness of the glass is 5.60 mm as determined through direct measurement.

$$a = 1\,500;$$

$$b = 1\,200;$$

$$r_0 = -2.689;$$

$$r_1 = 2.011;$$

$$r_2 = 0.213;$$

$$q = 1.80 \times 10^{-3} \text{ N/mm}^2;$$

$$E = 71.73 \times 10^3 \text{ N/mm}^2;$$

$$t = 5.60; \text{ and}$$

$$x = 1.490.$$

Therefore, maximum deflection at the centre of glass is:

$$w = 5.6 \times \exp(-2.689 + 2.111 \times 1.490 + 0.213 \times 1.490^2)$$

$$w = 12.2 \text{ mm}$$

ANNEX E

(Clause 6.1.9)

MODEL CALCULATION OF GLASS THICKNESS

E-1 MODEL CALCULATION

E-1.1 Glass panel dimension width 1 200 mm and height 1 800 mm for location Chennai for annealed glass type.

E-1.1.1 *General Checks have to be done for Safety Performance*

- The maximum area of glass panel is restricted to 15 m²;
- The maximum span of window is restricted to 4 m;
- Applicable to normal, reflective, laminated, tempered and insulating glass; and
- Applicable to rectangular panels properly fixed.

The procedure for calculating glass thickness is described below:

- Step 1* — Calculate aspect ratio ($AR_{\max}$) from Table 15.

Aspect ratio ($AR_{\max}$) = Longer side/shorter side of glass

Aspect ratio = 1 800/1 200 = 1.5

- Step 2* — Wind load calculation

For wind load calculation, reference shall be made to good practice [C-8(15)].

$$p_z = 0.6 V_z^2$$

$$V_z = V_b \cdot k_1 \cdot k_2 \cdot k_3 \cdot k_4$$

where

V_b = basic wind speed based on location {see Fig. 1 of good practice [C-8(15)]}.

k_1 = probability factor (risk co-efficient) {see 6.3.1 of good practice [C-8(15)]}.

k_2 = terrain roughness and height factor {see 6.3.2 of good practice [C-8(15)]}.

k_3 = topography factor {see 6.3.3 of good practice [C-8(15)]}.

k_4 = importance factor for the cyclonic region {see 6.3.4 of good practice [C-8(15)]}.

V_z = design wind speed at height z , in m/s.

p_z = wind pressure at height z , in N/m².

The design wind pressure (p_d) can be obtained as:

$$p_d = K_d \cdot K_a \cdot K_c \cdot p_z$$

where

K_d = wind directionality factor {see 7.2.1 of good practice [C-8(15)]}.

K_a = area averaging factor {see 7.2.2 of good practice [C-8(15)]}.

K_c = Combination factor {see 7.3.3.13 of good practice [C-8(15)]}.

The value of p_d , however shall not be taken less than 0.70 p_z .

Design wind speed for Chennai location {Basic wind speed = 50 m/s taken from good practice [C-8(15)]}
= 1 750 N/m².

3) Step 3 — Strength factor

Based on the types of glass (laminated/tempered/insulating glass) pressure factor is calculated. To determine the thickness of laminated/ tempered/insulating glass, the design wind pressure P_{net} is modified as below:

$$P_{net} = p_d / P_f$$

where

P_f = pressure factor dependent on the type of glass.

The values of the pressure factor P_f can be taken from [Table 14](#).

For annealed units, $P_f = 1.0$, therefore,

$$P_{net} = 1\,750 / 1.0 = 1\,750 \text{ N/m}^2$$

At this stage the glass thickness value shall be calculated from the table and then the deflection and stress check be done as described below:

4) Deflection Check

The deflection of the glass can be calculated using the formula in [Annex D](#). Actual deflection for the pane shall be obtained as below:

Deflection, $w = t \cdot \exp(r_0 + r_1 X + r_2 X^2)$

$$X = \ln \{ \ln [q(a*b)^2 / Et^4] \}$$

$$r_0 = 0.553 - 3.83(a/b) + 1.11(a/b)^2 - 0.0969(a/b)^3$$

$$r_1 = -2.29 + 5.83(a/b) - 2.17(a/b)^2 + 0.2067(a/b)^3$$

$$r_2 = 1.485 - 1.908(a/b) + 0.815(a/b)^2 - 0.0822(a/b)^3$$

where

E = Young's modulus ($71.7 \times 10^3 \text{ N/mm}^2$);

Q = net pressure on the pane (N/mm^2);

A = longer dimension (mm);

B = shorter dimension (mm); and

T = thickness of the glass pane calculated.

The actual deflection is, $4 \text{ mm} < 19 \text{ mm}$ maximum allowable.

As per [6.3.6.2](#), deflection at the centre of the glass for monolithic glass shall be the lesser of shortest span/60 or 19 mm.

Result: As $1200/60 = 20 \text{ mm}$, hence, 19 mm is the maximum allowable deflection.

Hence the glass thickness calculated is safe against deflection.

Glass Thickness Calculation:

$$P_{net} \cdot A = 200.0 \text{ T}^k \text{ (let } T \leq 6 \text{ mm)}$$

$$P_{\text{net}} = 1750$$

$$A = 2.16 \text{ m}^2$$

$$K = 1.765$$

$$T = 5.28 \text{ mm, therefore choose the thickness of glass as 6 mm.}$$

ANNEX F

([Table 30](#))

PERFORMANCE TESTS FOR GLAZING SYSTEM

F-1 The following enlists the performance tests for glazing systems. The general test requirements shall be as per 7.1 of accepted standard [C-8(14)].

F-2 SEISMIC RACKING TEST

Criteria shall be the movement of the sub-structure in the horizontal (+/-) directions. Optional testing for vertical movements (+/-) shall be based on sub-structure movement to accommodate live load, long term creep, column shortening and thermal elongation or contraction. No glass breakage or fall out is allowed. No wall component fallout is allowed. The system shall comply with the requirements of air permeability and static water penetration tests as per the designed performance criteria. Post proof load testing, air permeability testing and static water penetration testing shall not be required unless specified otherwise.

F-3 BUILDING MAINTENANCE UNIT (BMU) PULL OUT TEST (WHEN APPLICABLE)

A tensile load of minimum 1.5 kN shall be applied on the BMU restraint in the following directions:

- a) Outward and held for a period of 1 min;
- b) Horizontally to the right and held for a period of 1 min;
- c) Horizontally to the left and held for a period of 1 min;
- d) Vertically upward and held for a period of 1 min; and
- e) Vertically downward and held for a period of 1 min.

After each step the BMU restrain should not fail or have permanent deformation.

NOTE — If the load on the BMU restraints required by the BMU supplier is higher, the higher load shall apply.

ANNEX G

(Clause [9.4.3](#))

TESTING REQUIREMENTS FOR BLAST RESISTANT GLAZING SYSTEMS

G-1 OBJECTIVE

G-1.1 The purpose of this clause is to provide general guidelines about equipment, range conditions and procedures used to conduct the blast evaluation of glazing systems against the designed blast load and to verify that it meets the defined blast pressure load.

G-1.2 The glazing system is evaluated for its structural integrity by visual inspection only.

G-1.3 The safety precautions and regulations undertaken during blast resistant tests of glazing systems shall be as specified by the test agency.

G-1.4 Blast resistant test of glazing systems shall be carried out only by the accredited institutions/laboratories authorized by the Government of India.

G-2 TERMINOLOGY

G-2.1 Test Sponsor — Party requesting and sponsoring the test. The test sponsor may be its manufacturer or user (or his representative).

G-2.2 Test Agency — Party performing the testing and documenting the test results.

G-2.3 Test Specimen — The test specimen is provided as a complete assembly by test sponsor to the test agency. It may include glazing systems like glass door, glass windows/frames, glass panels, etc.

G-2.4 Trial Officer — Individual identified by the test agency responsible to complete the specified tests as required and to document the results in accordance with this test method. The Test Director shall sign all of the test reports.

G-2.5 Trial Safety Officer — Individual identified by the test agency nominated by the means of trial plan approved by the competent authority. The Trial Safety Officer is responsible for all the safety aspects during the trial time and safe conduct of trials.

G-3 APPARATUS

G-3.1 High Explosive

The high explosive charge (prepared by using TNT/equivalent charge) as mentioned by the test sponsor is used to generate the desired blast parameters for the performance evaluation of test specimen.

G-3.2 Instrumentation

The instrumentation system used for measurement of blast pressure comprises of piezoelectric based pressure sensors, signal conditioner, low noise anti-microphonic cables and data acquisition systems. The details are as follows:

G-3.2.1 Pressure Sensor

Pressure sensor is used to record pressure-time history of blast wave. The sensor contains piezoelectric crystal as a sensing element which produces an electric charge proportional to impinging pressure. The pressure sensor has an in-built circuitry that converts pressure to voltage signal. It requires installation control panel (ICP) constant current power supply which is provided by compatible signal conditioner.

G-3.2.2 Low Noise Cables

A special low noise anti-microphonic cable is used to transmit the signal from sensor to recording system. These cables are immune to tribo-electric effect and electromagnetic interference.

G-3.2.3 Data Acquisition System (DAS)

The DAS consists of an analog or digital recording system with a sufficient number of data channels to accommodate the pressure sensor(s) and any other electronic measuring devices. The DAS shall operate at a sufficiently high frequency to record reliably positive pressure and positive phase duration. The triggering of recording system is synchronized with explosion by using transistor-transistor logic system.

G-3.3 Photographic Equipment

Each test is documented with still photography. Video or high speed photography of the testing is deployed, if required.

G-4 PROCEDURE

G-4.1 The Trial Officer along with all team members will be responsible for the execution of the test. The Trial Safety Officer along with the range safety team will take care of the safety of personnel and property during the conduct of test.

G-4.2 The trial site location and range safety distance shall be decided by the test agency.

G-4.3 The parameters (identification number, shape, weight, dimensions, etc) are verified before trial. The documentary proof for the same shall be submitted by the test sponsor to the test agency.

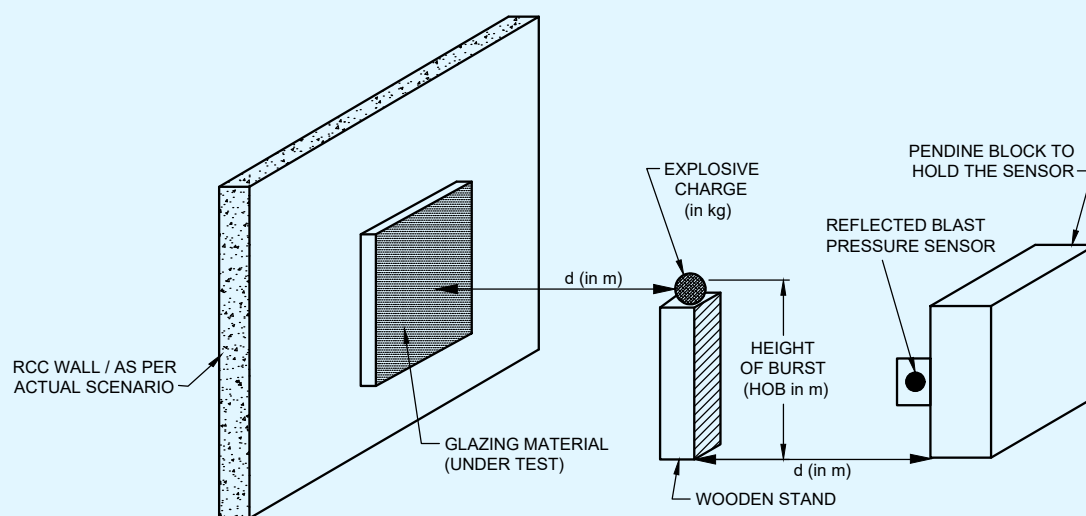
G-4.4 The test specimen is held on to the wall or as per actual scenario like wall having duct with or without room with the help of supporting frame such that no additional pressure gets generated on the glazing system when it is fixed to the frame using nuts and bolts.

G-4.5 The attack face of the test specimen will be towards the explosive charge (TNT/equivalent charge).

G-4.6 Photographic records of details of the glazing assembly and the test configuration are taken prior to the test. If a video photographic record of the response of glazing assembly during the test is desired, video cameras should be deployed.

G-4.7 The explosive charge (TNT/equivalent charge) is prepared as per the threat levels based on weight of explosive and standoff (see [Table 34](#)) or as defined by the test sponsor.

G-4.8 The explosive charge (TNT/equivalent charge) is positioned at a certain height of burst (HOB) and at distance d from the test specimen [as shown in [Fig. 22](#) (isometric view) and [Fig. 23](#) (top view)] (see [Table 34](#)).



NOTE — The height of burst (HOB) shall be referenced from the hard stand on which the test specimen support (like RCC wall/test specimen support) stands.

FIG. 22 ISOMETRIC VIEW — SCHEMATIC OF TEST SETUP

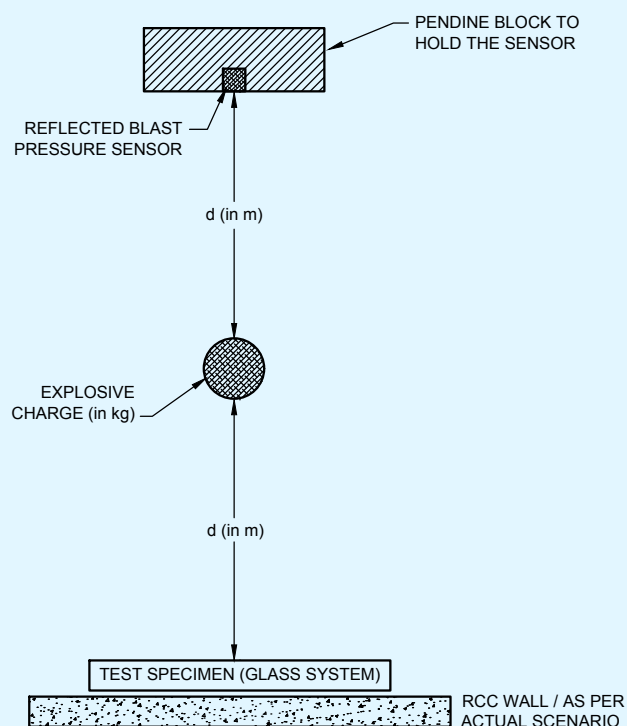


FIG. 23 TOP VIEW — SCHEMATIC OF TEST SETUP

G-4.9 The pressure sensor to measure the reflected blast pressure is deployed at distance d (as shown in [Fig. 22](#) and [Fig. 23](#)). The pressure sensor is deployed in such a way that its sensing element is directed towards the explosive charge (TNT/equivalent charge).

G-4.10 The distance d and height of burst (HOB) varies depending on the threat level.

G-4.11 The pressure sensor is then connected to the data acquisition system through signal conditioner via low noise cables and tested to verify proper functioning.

G-4.12 The explosive charge (prepared by using TNT/equivalent) is then detonated by following all safety precautions ensured by trial safety officer and range safety team.

G-4.13 After the test, the visual inspection of the glazing assembly is done to evaluate its structural integrity. Adequate photography is taken after each test.

G-4.14 In case of misfire, the second attempt is done as per misfire drill of the test agency.

G-5 OUTCOMES

The structural integrity of glass system test specimen after the trial is evaluated by visual inspection according to the outcome categories specified in [9.4.3](#).

G-6 REPORT GENERATION

G-6.1 Upon completion of the testing, the Test Director assigns a test number and record the test measurement and observations obtained from all test specimen(s) for preparation of test report.

G-6.2 The Trial Officer and the Test Director signs the test report and gets it approved from competent authority of the test agency.

ANNEX H

(Foreword)

INSTALLATION OF MIRRORS

H-1 This annex provides general guidelines, which is to be followed during the installation of mirror. Mirrors (Piece of glass silvered on one side, with a protective paint coating) are used in shops, gymnasiums, etc, where it is predominantly used as wall cladding. The mirror panels in such application use large panel size and it is recommended to follow the installation procedure in accordance with this Section.

H-1.1 The mirror should always be mounted on a perfectly flat, clean and dry surface free from acids and aggressive substances such as acid, solvent and cement mortar.

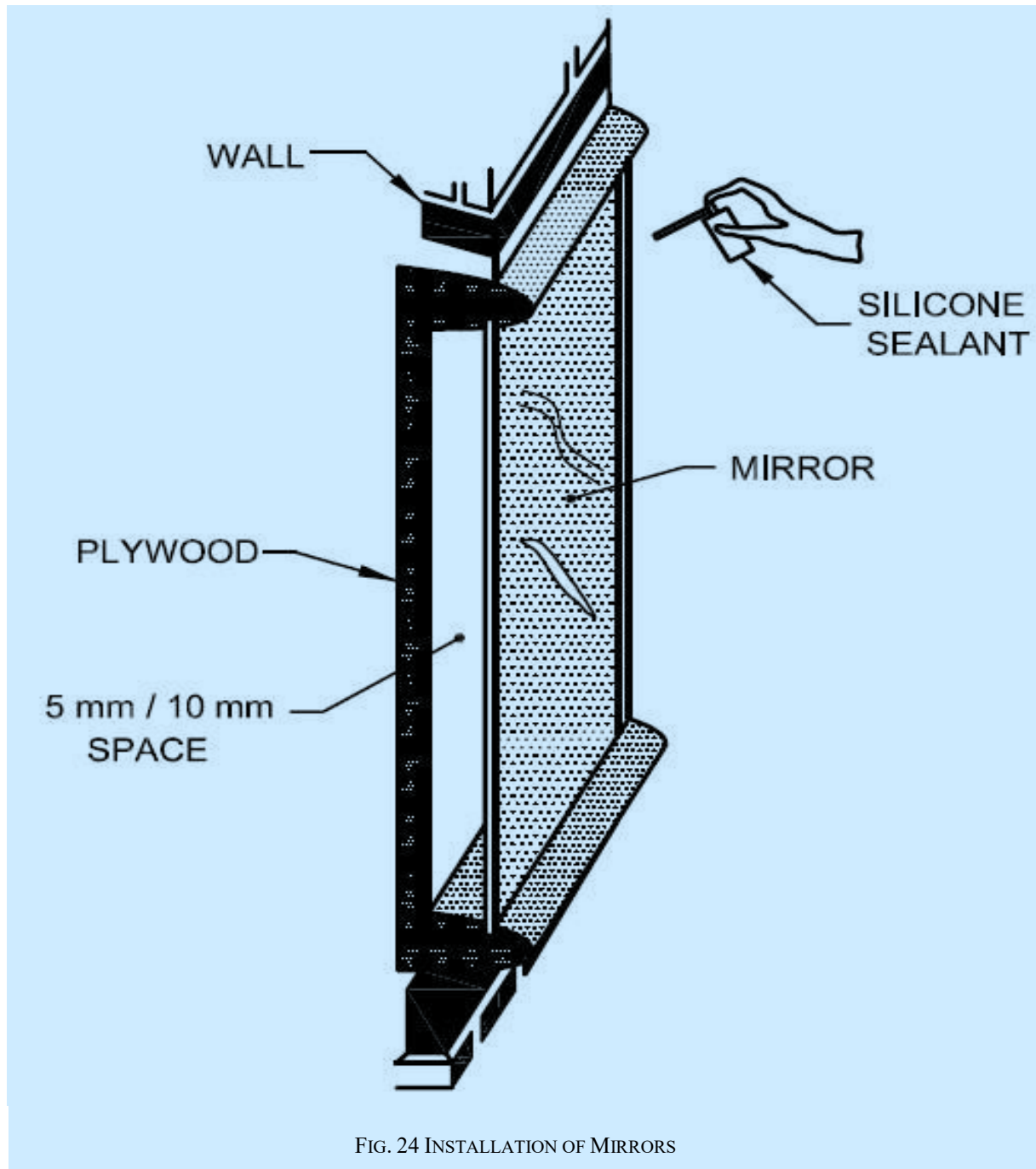
H-1.2 The mirror shall be mounted on wall or plywood and care is taken that the mirror is never stuck over unstable support (painted paper, paint or old plaster). It is recommended that the mirror installation is not done directly on the mirror supports. In order to have good ventilation, the following spacing behind the mirror is recommended depending on the height of the mirror panel:

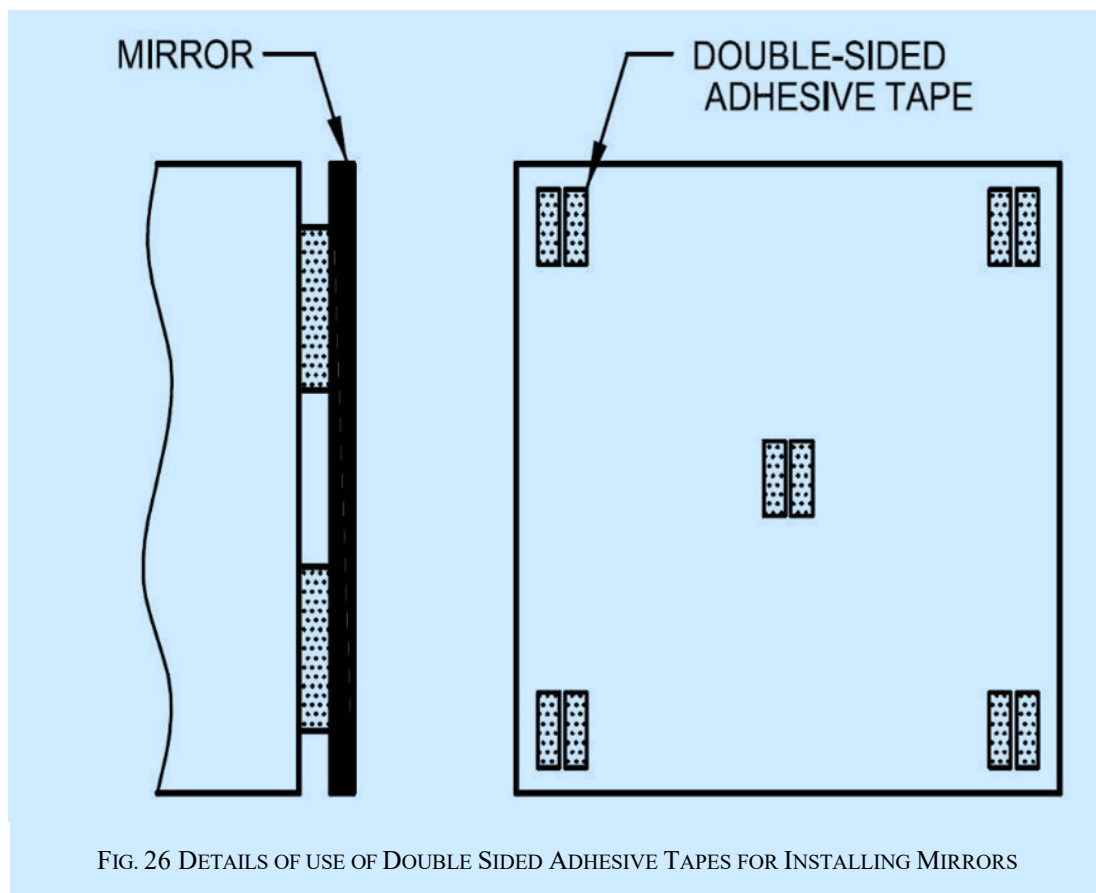
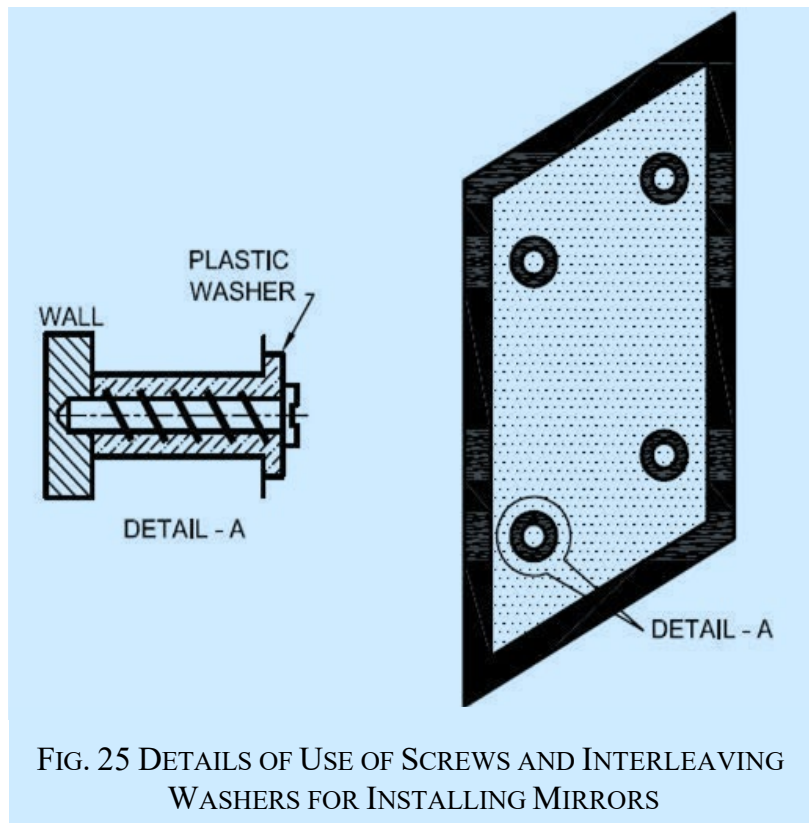
- a) A space of 5 mm for a mirror less than 1 m height; and
- b) A space of 10 mm for a mirror greater than 1 m height.

[Figure 24](#) gives typical details of the installation of mirrors. During the installation of mirror panels at several levels that is, one above the other, it is recommended to provide a space of 10 mm at the top and at the bottom for circulation of air. The side of the mirror is designed with a minimum 1 mm or 2 mm of interleaving space. When mirror is fixed in a profile (frame), care is to be taken to maintain perfectly dry and clean frame. It is recommended that mirror is supported on hard non-metallic/plastic wedges of at least 3 mm, to raise the mirror and thus avoid contact with the condensed water which could accumulate in the profile. If the mirror is screwed on, the screws of a suitable dimension is designed, with the interleaving and washers (*see* [Fig. 25](#)) made of plastic to avoid any direct glass-metal contact. To minimize the risk of breakage, cracks and splinters during screw-mounting the mirror, it is recommended to use synthetic protective collars.

H-1.3 For the adhesive used to mount mirrors, care should be taken to ensure that the adhesive is compatible with the protective coatings on the mirror. It is recommended to use only neutral binding agents such as alkoxy silicone, oxime silicone, etc. Avoid use of acid silicones, such as acetoxysilicones which contain acetic acid or water/rubber/polyurethane-based glues.

H-1.4 In the high humidity areas (for example bathrooms), it is recommended to seal the edges with neutral silicone to protect the backing paint from damage due to moisture and enhanced protection from moisture. It is recommended to use double sided adhesive tapes to fix the mirror on the mounting surface (*see* [Fig. 26](#)). The adhesive tapes used for installation shall be certified to be compatible with the mirror and the adhesive or the adhesive tapes shall always be fixed vertically.





ANNEX J

(Foreword)

GLASS RAILINGS

J-1 GENERAL

Glass railings are generally sought in institutional, commercial, residential buildings and special structures like stadiums, foot over bridges, etc. Glass of appropriate thickness when used as a railing can provide transparency and when engineered properly can provide structural performance too. In public places such as stadiums, foot over bridges, shopping malls, etc, use of frameless glass railing shall be avoided and framed systems with vertical balustrades and glass infill panel may be used with proper design as per the design provisions.

The basic requirement of a parapet, which a glass railing (if so used) intends to substitute, shall be met with. This includes protecting people from falling and in some cases to control movement. Careful consideration, including in design, shall be made/determined where glass railings are used.

NOTE — Notwithstanding the above, the potential benefits including safety (through psychological aspects) of regular railings shall be borne in mind.

J-2 PRELIMINARY CONSIDERATIONS

In the assessment of the need for a railing and the type of railing to be provided, the designer should give consideration to the likely hazards, the building use and the risks to building users. Where, in a building, more than one use of the building is anticipated, either the railing design should be chosen to suit the worst case, or more than one type of railing should be provided, as appropriate, to the location. Generally, when there is a change in adjacent levels (600 mm in dwellings, 380 mm in other buildings) a railing is needed to restrict movement of people.

The railing adopted should be designed so as to minimize the risk of persons falling, rolling, sliding or slipping through gaps in the railing. In dwellings and other buildings which can be accessed by children, gaps in railing or infill should not be large enough to permit a sphere of 100 mm diameter to pass through, making due allowance for deflection under load.

The designer should ensure all the elements of the railing system satisfy the criteria given in available relevant standards.

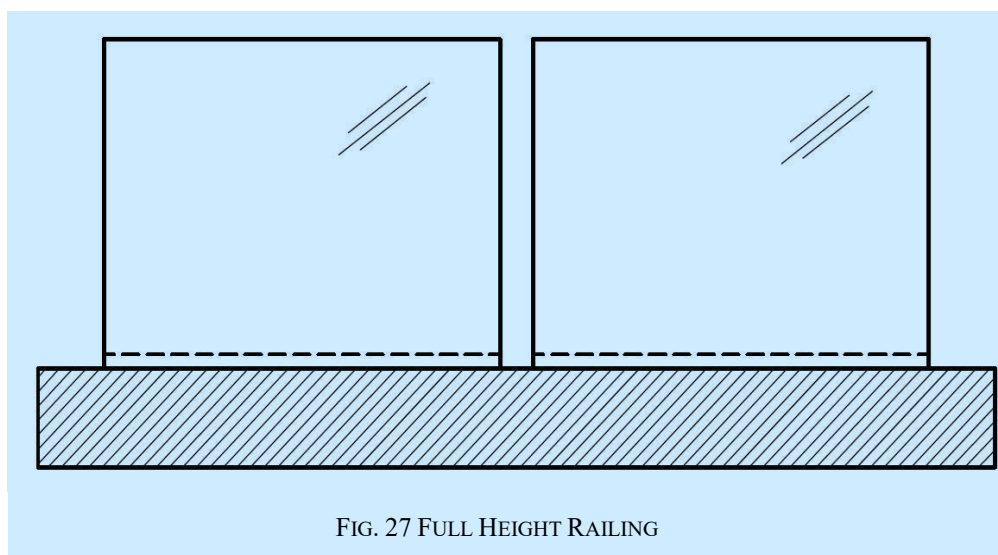
For general safety, the finished barrier should have no sharp edges or projections that could cause injury to persons or clothing. Infill panels and balusters are intended to provide support and protection to users, and should be designed to restrain people without causing additional injury from sharp edges, thin sections, open-ended tubes, projecting details, etc.

The design shall be based on the horizontal uniformly distributed load (UDL) applied at a notional height of 1.1 m above finished floor level, even though the minimum permissible height of the railing may be less (the notional design height of 1.1 m is chosen as being representative of the height of the centre of gravity of a human being above the datum level).

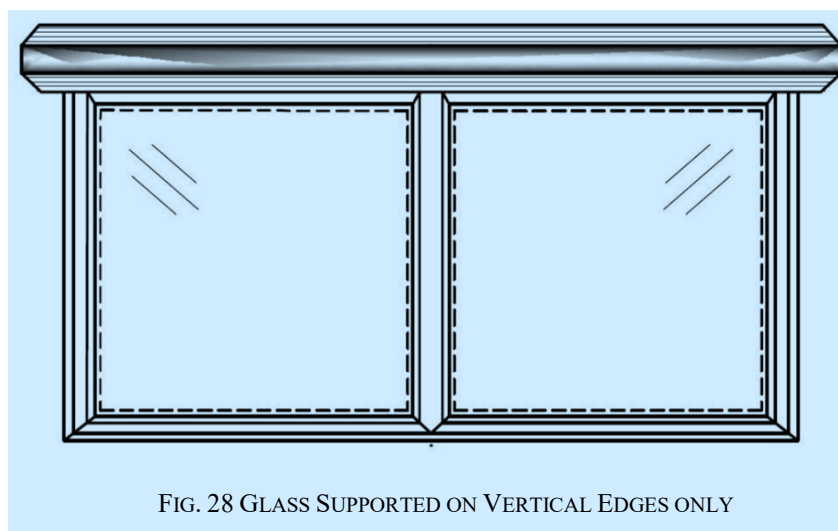
J-3 GLASS RAILING TYPES

There are four common types of railings that are used to fix glass.

- a) *Full Height Railing* — Where glass forms part or whole of a wall element, the railing should be carefully designed as per the relevant standards and as per this Section, if any part of the glass comes below the minimum railing height (see [Table 35](#) and [Fig. 27](#)).



- b) *Railings with a Glass Infill Panel* — In this type of railing, the main frame of the railing (that is, top rail and baluster) should be designed to withstand the loads applied to the top rail and the glass should be used to form infill panels. It should be noted that the glass provides no support to the main frame. The designer shall ensure that, irrespective of the fixing type, in the event of glass breakage, the broken glass remains securely attached to its framing or supporting structure, thereby preventing safety hazards. This can be further sub-categorized into 4 types:
- 1) *Glass supported on vertical edges only* — This may involve the vertical edges of the glass being supported in a rebate into the side of each vertical post. The top and bottom edges of the glass are exposed. The glass may also be supported by patch fittings fixed to the posts. In all cases a handrail is required (see [Fig. 28](#)).



- 2) *Glass supported on horizontal edges only* — This may involve a handrail at the top, and a channel or a rail at the base of the glass. In this case the handrail should be designed to carry the railing load with the glass designed to carry the infill load (see [Fig. 29](#)).

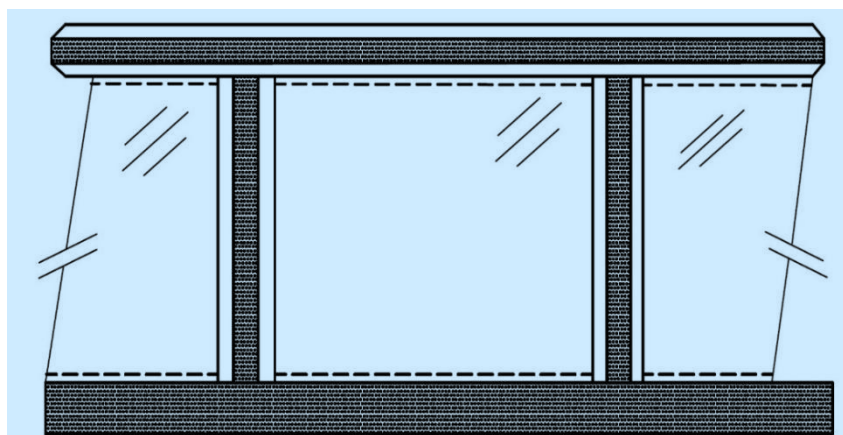


FIG. 29 GLASS SUPPORTED ON HORIZONTAL EDGES ONLY

- 3) *Glass supported on all four edges* — (See [Fig. 30](#))

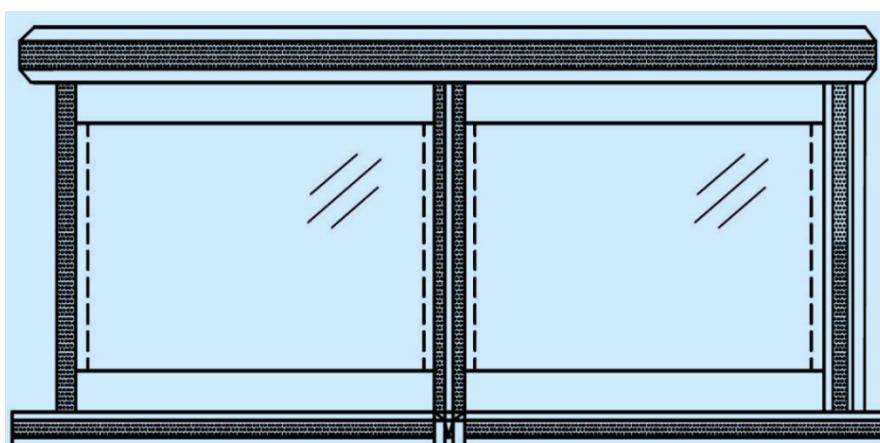


FIG. 30 GLASS SUPPORTED ON ALL FOUR EDGES

- 4) *Glass fixed by patch/bolted fittings* — This system requires the use of two rows of bolts (commonly) to provide a stiff fixing which minimizes panel deflection. There are a variety of bolt fixing types available in the market, so design and performance information should be sought for each type. The purpose of these fixings is to transfer the force applied to the railing into the support structure. These fixings occupy a small surface area on the glass which can produce high stresses in the glass, so careful analysis is required to determine suitable glass thickness, fixing spacing and size. Glass fixings shall be designed to avoid high local stresses and minimize the risk of failure. Minimum fixing sizes shall be used, and for large panels exceeding 2 m in dimension or under high wind pressure exceeding 2.5 kPa, fixings shall accommodate rotations by using articulated fixings (see [Fig. 31](#)).

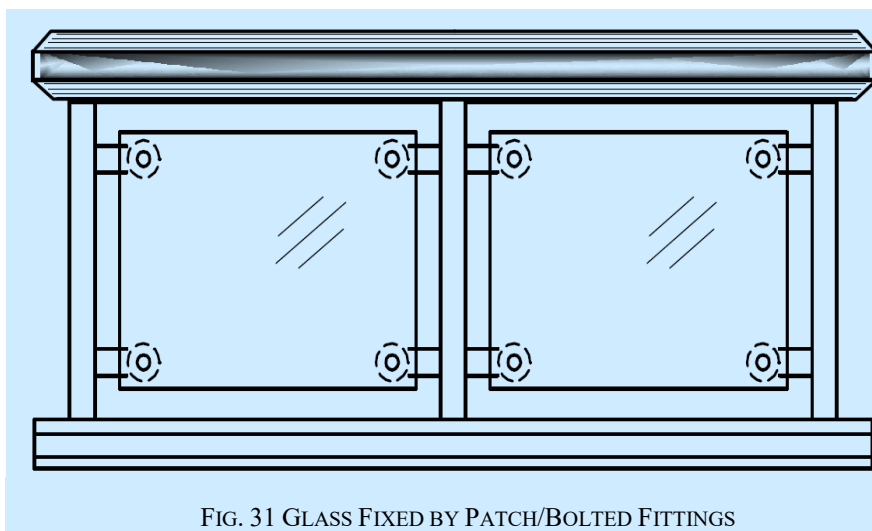


FIG. 31 GLASS FIXED BY PATCH/BOLTED FITTINGS

- c) *Free Standing Glass Protective Railing* — In this type of railing, the glass should be designed to withstand the design loads. Each glass plate should be rigidly clamped to the structure along its bottom edge, the handrail attached to the top edge of the glass and there should be no balusters. In the event that a free standing railing is supplied without a handrail, each panel should be able to withstand the entire appropriate design load. This applies to vertical glazing and glass within 15° of vertical. Adequate provision for drainage of water should be made in the base structure to avoid any water ponding which may cause delamination issue. Laminated safety glass shall be used for this application. Clamping and brackets shall be designed with adequate anchorage, and the main supporting building elements (beam/slab) shall be verified to meet the required minimum thickness. The bottom glass clamping embedment shall be a minimum of 100 mm, regardless of railing size. The bottom clamping member shall provide continuous support along the glass bottom edge. In the event of glass breakage, the glass panel shall remain securely in place without risk of detachment or falling (see [Fig. 32](#)).

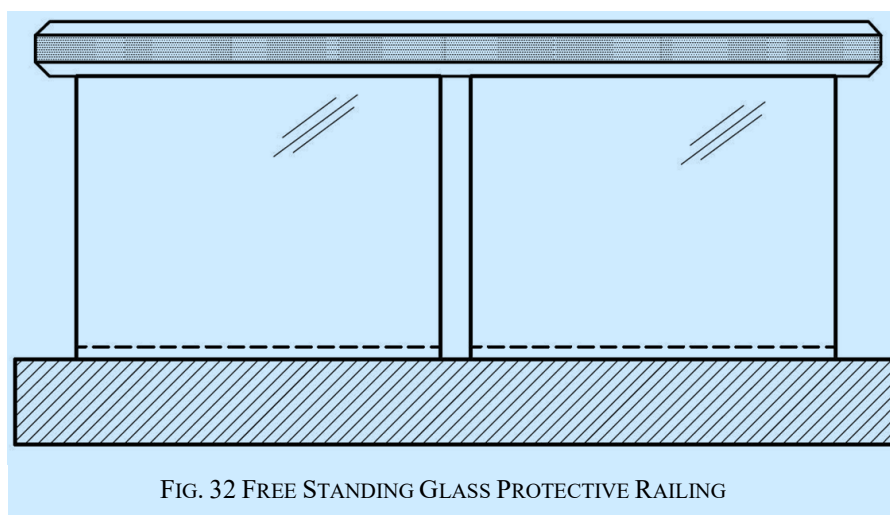
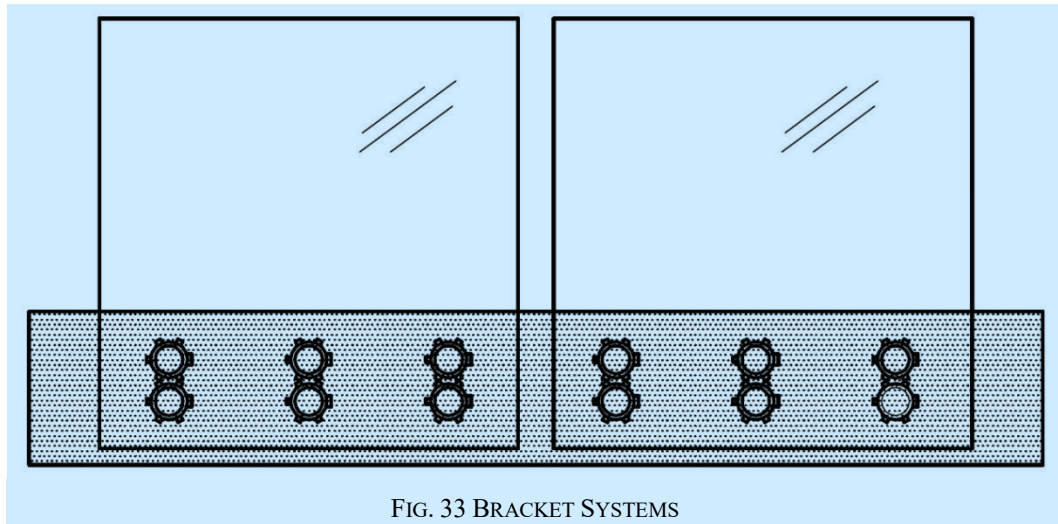


FIG. 32 FREE STANDING GLASS PROTECTIVE RAILING

- d) *Bracket Systems* — These systems are an extension of the bolt fixed system where the bolts are part of a bracket and the bracket is fixed to the floor or the face of a concrete slab, steel or timber structure. The use of these systems shall be investigated carefully as they can develop very high stresses in the glass where the brackets are located. The design shall balance glass thickness and bracket location to produce an adequate system. Some systems may be designed to produce test data, to justify their use. This data shall be studied carefully as there have been some cases where such data has been shown to be inadequate. Proper interpretation of these test data shall also become necessary (see [Fig. 33](#)).



- e) *Bracket Systems* — Bottom supported with surface attached/bolted Handrail — Glazing as structural member (see [Fig. 34](#)).

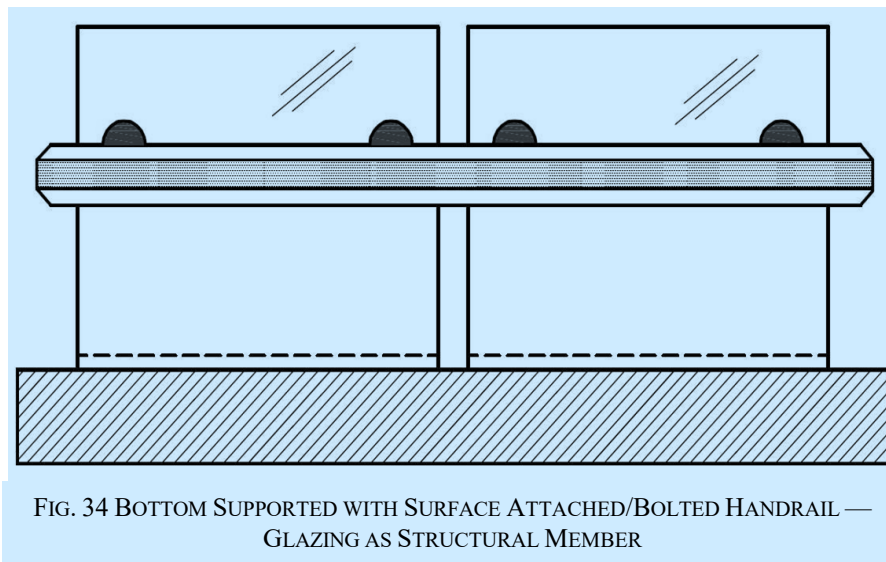


Table 35 Minimum Barrier Heights

[Clauses [J-3 \(a\)](#), [J-4.1](#), [J-5.4.1](#) and [J-6.1](#)]

SI No.	Use	Position	Height mm
(1)	(2)	(3)	(4)
i)	Single-family dwelling	a) Barriers in front of a window up to 10 floors	800
		b) Barriers in front of a window from 11 th floor to 20 th floor	1 100
		c) Barriers in front of a window above 21 st floor	1 200
		d) Stairs, landings, ramps, edges of internal floors	900
		e) External balconies including <i>balconet</i> , edges of roofs:	
		1) up to 20 th floor	1 100
ii)	All other uses	2) 21 st floor and above	1 200
		a) Barrier in front of a window	800
		b) Stairs	900
		c) Balconies and stands, etc having fixed seating within 530 mm of the barrier	800
		d) Balconies and stands, etc having fixed seating within 530 mm of the barrier, provided the sum of the barrier width and the barrier height is greater than 975 mm	750
		e) Other positions including balconets	1 100

NOTE — Height shall be measured from the top edge of the railing to the highest point of the floor where occupants can stand. This measurement shall be from the top of any ledges, up-stands, or toe walls, if present at the base.

J-4 DESIGN CONSIDERATIONS

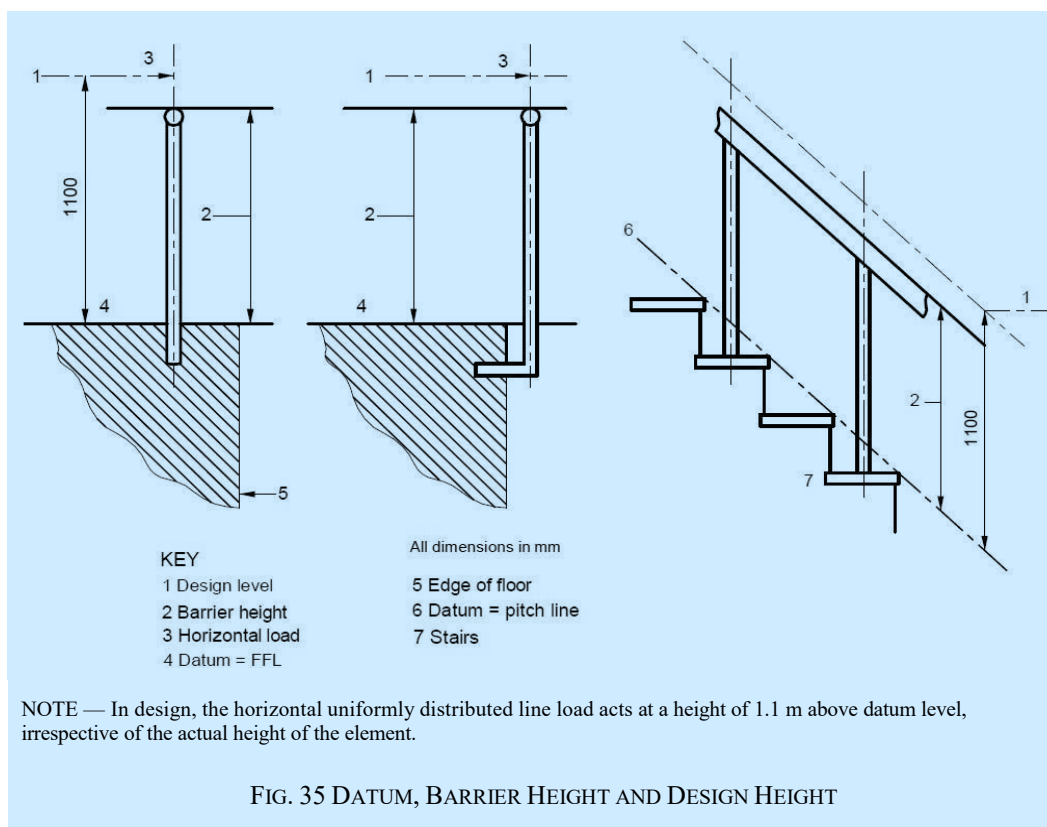
The designer shall first determine the purpose of the railing. Once this is done, the appropriate load requirement can be derived. Typical values (minimum requirements) of railing loads to be chosen are given in [Table 36](#) (If so desired by the designer, higher values than that in the [Table 36](#) may also be chosen).

J-4.1 General

Railings should be designed to resist the most unfavourable likely imposed loads and wind loads separately without unacceptable deflections or distortions. Heights of railings shall be same as that for parapet walls as per other Parts of this NBCS. The railing height above datum is defined in [Fig. 35](#). The minimum heights for each railing location shall be as given in [Table 35](#).

The height of glass railings installed on top of low parapet walls should be measured from the top of the parapet and not at walk level.

For railings, either permissible stress or limit state design procedure should be used, according to the recommended procedures. When using limit state design, the partial safety factors for loads and materials should be those recommended by the appropriate standard for relevant materials. The deflection of the railing shall be determined as per serviceability limit state.



J-4.2 Loading

Minimum horizontal imposed loads appropriate to the design of parapets, railings, balustrades and other elements of structure intended to retain, stop or guide people, should be determined in accordance with [Table 36](#), which recommends a uniformly distributed line load for the railing and a uniformly distributed and point load applied to the infill.

NOTE — These are not additive and should be considered as three separate load cases, all loads being determined according to the type of occupancy which reflects the possible in-service conditions.

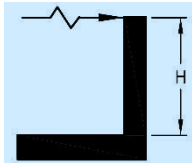
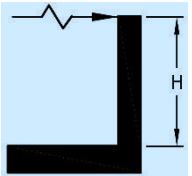
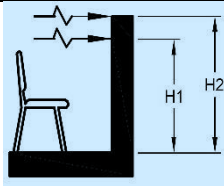
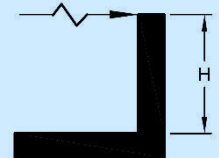
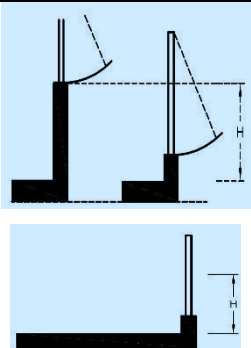
Horizontal uniformly distributed line load should be applied at design height as presented in [Table 36](#) or at design level (1 100 mm) for railings higher than the design height.

Uniformly distributed load should be applied to the area below the design height.

Point load should be applied at the most onerous point anywhere on the railing structure.

NOTE — For clarity, the design level (the level at which the horizontal uniformly distributed line load ought to be considered to act) is shown in Fig. 36.

Sl No.	Building	Element	Strength	Height (H)
(1)	(2)	(3)	(4)	(5)
i)	Single family dwelling	Stairs landings, ramps edges of internal floors	0.36 kN/m	900 mm for all elements
		External balconies and edges of roofs	0.74 kN/m	1 100 mm

Sl No.	Building	Element	Strength	Height (H)	
(1)	(2)	(3)	(4)	(5)	(6)
ii)	Factories and warehouses (light traffic)	Stairs, ramps	0.36 kN/m	900 mm	
		Landing and edges of floor	0.36 kN/m	1 100 mm	
iii)	Residential institutional, educational, office and public building	All location	0.74 kN/m	900 mm for flights otherwise 1 100 mm	
iv)	Assembly	Within 530 mm in front of fixed seating	1.5 kN/m	800 mm (H1)	
		All other locations	3.0 kN/m	900 mm for flights elsewhere 1 100 mm (H2)	
v)	Retail	All locations	1.5 kN/m	900 mm for flights otherwise 1 100 mm	
vi)	All buildings	At opening windows		800 mm	
		At glazing to changes of level	To provide containment	Below 800 mm	

NOTE — H is the height at which the applied load (horizontal design load) is located and it need not be the height of the barrier itself.

FIG. 36 HEIGHT LIMIT FOR THE UNIFORM LOADING FOR VARIOUS CATEGORY OF BUILDINGS

Table 36 Minimum Horizontal Imposed Loads for Parapets, Barriers and Balustrades
[Clauses [J-4](#), [J-4.2](#), [J-4.4](#) and [J-4.6](#)]

SI No.	Type of Occupancy for Part of the Building or Structure	Examples of Specific Use	Horizontal Uniformly Distributed Line load	Uniformly Distributed Load Applied to the Infill	A Point Load Applied to Part of the Infill
(1)	(2)	(3)	kN/m	kN/m ²	kN
i)	Domestic and residential activities	a) All areas within or serving exclusively one single family dwelling including stairs, landings, etc but excluding external balconies and edges of roofs b) Other residential, that is houses of multiple occupancy and balconies, including Juliette balconies and edges of roofs in single family dwellings	0.36	0.5	0.25
ii)	Offices and work areas not included elsewhere, including storage areas	a) Light access stairs and gangways not more than 600 mm wide b) Light pedestrian traffic routes in industrial and storage buildings except designated escape routes c) Area not susceptible to overcrowding in office and institutional buildings, also industrial and storage buildings except as given above	0.22	—	—
			0.36	0.5	0.25
			0.74	1.0	0.5
iii)	Areas where people might congregate	Areas having fixed seating within 530 mm of the barrier, balustrade or parapet	1.5	1.5	1.5

Table 36 (Concluded)

Table 36 Minimum Horizontal Imposed Loads for Parapets, Barriers and Balustrades [Clauses J-4 , J-4.2 , J-4.4 and J-4.6]					
SI No.	Type of Occupancy for Part of the Building or Structure	Examples of Specific Use	Horizontal Uniformly Distributed Line load	Uniformly Distributed Load Applied to the Infill	A Point Load Applied to Part of the Infill
(1)	(2)	(3)	kN/m	kN/m ²	kN
iv)	Area with tables or fixed seatings	Restaurants and bars	1.5	1.5	1.5
v)	Areas without obstacles for moving people and not susceptible to overcrowding	a) Stairs, landings, corridors, ramps	0.74	1.0	0.5
		b) External balconies including Juliette balconies and edges of roofs. Footways and pavements within building curtilage adjacent to basement/sunken areas	0.74	1.0	0.5
vi)	Areas susceptible to overcrowding	a) Footways or pavements less than 3 m wide adjacent to sunken areas	1.5	1.5	1.5
		b) Theatres, cinemas, discotheques, bars, auditoria, shopping malls, assembly areas, studio. Footways or pavements greater than 3 m wide adjacent to sunken areas.	3.0	1.5	1.5
vii)	Retail areas	All retail areas including public areas of banks/building societies or betting shops	1.5	1.5	1.5
viii)	Vehicular	Pedestrian areas in car parks, including stairs, landings, ramps, edges or internal floors, footways, edges of roofs	1.5	1.5	1.5

In general, the following load combinations as described in [J-4.2.1](#) to [J-4.2.3](#) should be applied.

J-4.2.1 Free Standing Glass Protective Railing (Without any Balusters)

It requires four design loadings to be taken into account:

- a) Infill load, in kN;
- b) Uniform load, in kN/m;
- c) Concentrated load, in kN; and
- d) Shot bag test load, in Joules.

J-4.2.2 Railing with Glass Infill Panel

It requires four design loadings to be taken into account:

- a) Infill load, in kN;
- b) Uniform load, in kN/m;
- c) Concentrated load, in kN; and
- d) Shot bag test load, in Joules.

J-4.2.3 Full Height railing

It requires four design loadings to be taken into account:

- a) Infill load, in kN;
- b) Uniform load, in kN/m;
- c) Concentrated load, in kN; and
- d) Shot bag test load, in Joules.

J-4.3 Wind Loads

External railings should be designed to resist the wind loads given in good practice [C-8(15)].

J-4.4 Deflection

Railings for the protection of people should be of adequate strength and stiffness to sustain the applied loads given in [Table 36](#). In addition, a railing that is structurally safe should not possess flexibility to such an extent so as to create a false alarm to building users when subject to normal service conditions. Therefore, for serviceability considerations, the limiting condition for deflection appropriate for a railing for the protection of people is that the total horizontal displacement of the railing at any point from its original unloaded position should not exceed the deflection limits determined from the relevant structural design code (where applicable) for the material used, or 25 mm, whichever is the smaller. Where the infill of a railing is subjected to the imposed loads given in [Table 36](#), or if appropriate, other calculated design loads, the displacement of any point of the railing should not exceed $H/65$ or 25 mm, whichever is lower. These deflection limits shall apply to the deflection of the cantilevered top edge of the railing, measured perpendicular to the glass plane or the direction of the wind load.

J-4.5 Fixings, Attachments and Anchorage

The strength of fixings, attachments and anchorage securing the railing to a substrate should be adequate to sustain a loading greater than that to which the railing will be subjected. All joints should be designed to provide the full strength of the members being joined. To that end, where any uncertainty exists with regard to the strength of any component in the fixing, the design loading factors should be increased by 50 percent.

While designing fixings, particular care should be taken to account for the material into which the fixing is placed, the spacing between fixings, the edge distance, and where the substrate is concrete, the position of reinforcement. Building elements (beam/slab/up-stand) shall have adequate thickness and shall be composed of structural

reinforced cement concrete (RCC) or steel. The designer shall exercise caution when the thickness of the concrete element is less than 100 mm, as adequate embedment may not be achievable. The designer shall ensure that the concrete element is cast with high quality and shall refrain from fixing to any loose concrete. Design of anchors shall conform to the requirements of accepted standard [C-8(30)].

NOTES

- 1 The recommendation of an additional load factor of 1.5 is intended to ensure that under an extreme load condition, railings give an indication of failure by deflection or distortion and not by total collapse, as may be brought about by a failure of the fixing, attachment or anchorage system. Where the design strength of a proposed system of fixing to an existing substrate cannot be determined with reasonable accuracy by theoretical consideration, load testing should be used to validate the design.
- 2 A factor greater than 1.5 on railing load design might also be chosen based on the importance desired by the designer/parties concerned.

J-4.6 Testing

It is recommended to conduct the tests as detailed in this clause, on railings provided for the type of occupancy classified as 'Areas susceptible to overcrowding', as specified in [Table 36](#). For all other occupancy types, the tests to be conducted on railings for performance evaluation, shall be mutually decided between the parties concerned.

J-4.6.1 Test Selection

Load test methods for the uniform load test and the concentrated load test are selected based on the performance specification.

J-4.6.2 Sequence of Test Method

Specimens tested for two or more of the loading conditions are subjected first to the in-fill load test, followed by the uniform load test, and then the concentrated load test and shot bag glazing impact test, steel nose pendulum impact, and sphere penetration evaluation, in that order, unless directed otherwise by the performance specification.

J-4.6.2.1 Apparatus and load application

- a) *Assembly support fixture* — An assembly support fixture shall ensure the rigidity normally provided to an assembly in a building by the ceiling, floor, and walls. The support fixture for the specimen shall consist of a vertical wall section constructed from nominal steel or 2 by 4 wood studs, 400 mm on centre, with a rough opening of sufficient size to support the test specimen in accordance with the manufacturer's specifications. The limiting deflection of the wall shall be $L/175$ (based on the anticipated loads);
- b) *Load attachments* — Brackets, fasteners, or other devices used in performing these tests shall be designed and attached so as to minimize their influence on the test results;
- c) *Shot bag, traction and release system* — The shot bag shall be of weight 45.4 kg as described in **6.4.2.5** and **6.4.6** of accepted standard [C-8(3)]. The suspension device shall conform to **6.4.2.6** of accepted standard [C-8(3)] and the Traction Release system shall conform to **6.4.2.7** of accepted standard [C-8(3)];
- d) *Load measuring device* — Load and time-measuring devices with an accuracy of ± 2 percent of the full scale shall be incorporated in the test setups. The scale ranges used shall assure that the performance levels are within an accuracy of ± 5 percent; and
- e) *Rate of loading* — The rate of loading shall be such that the maximum specified load is achieved in no less than 10 s and no more than 5 min. Maintain the maximum specified load for 1 min before releasing. [Warnings: 1) Possible injury to personnel and damage to the test equipment and instrumentation prior to, during, and after load application by any unexpected release of potential strain energy accumulated during testing can occur and shall be given consideration. 2) If tests are conducted in a structure and not in a testing laboratory, exercise caution against unwanted damage to the building, its components, and its finish].

For in-fill testing, use a bearing plate with appropriate cushioning between the plate and the glazing. The bearing plate should be 0.09 m².

Each load test to be performed on a new specimen or the undamaged specimen from the previously applied test. Perform only one test per glazing in-fill type on each of the three specimens at the most critical point of the system.

J-4.6.3 Sampling

A minimum of three representative replicate specimens of each type of system shall be tested. Sequential testing of the same specimen shall be permissible provided no breakage of any component occurs during the previously performed tests.

J-4.6.4 Test Specimens

The specimen installation, including the post spacing, shall be the same as the actual field installation. The specimen shall have a minimum of three posts or minimum two largest glass panels. Each unique attachment scenario shall be tested.

Multiple types of infill panels shall be qualified within one test, provided each panel is individually glazed into the system prior to the start of the test. No substitution of infill product shall be permitted once testing has commenced.

J-4.6.5 Installation

The glass rail, guard, or balustrade system being tested shall be installed in accordance with the manufacturer's or designer's specifications.

J-4.6.6 Test Procedure

J-4.6.6.1 In-fill load test

The test specimen shall be capable of satisfactorily resisting the required test load applied over a 0.09 m² area normal to the in-fill. The test load shall be applied at the intersection of the horizontal and vertical centre lines of the in-fill, unless specified otherwise by the performance specification or the purposes of the test. If it is determined that another location will yield lesser results, that position should be used instead of the indicated intersection point of the centre lines. Record deflection of the in-fill at the point of maximum deflection of the in-fill.

J-4.6.6.2 Uniform load test

Subject the top rail or other rail of the test specimen to the required maximum uniform test load applied vertically and horizontally, as required by the performance specification. Required vertical and horizontal test loads shall be applied separately and sequentially, unless otherwise required by the performance specification or to meet the purpose of the test. Record the deflection at mid-span during each test.

J-4.6.6.3 Concentrated load test

The top rail or other subject rail of the test specimen shall be subjected to the concentrated test load applied vertically and horizontally, as required by the performance specification. Apply required vertical and horizontal test separately and sequentially, unless otherwise required by the performance specification or to meet the purpose of the test. The required concentrated test load shall be applied at the following three critical locations – rail mid-span between posts, top rail adjacent to a post (maximum 75 mm from post) and top of a single post. Test loads are to be applied and deflection recorded sequentially, one location at a time. The load applied on the rail adjacent to the post can be applied to the opposite post from which the top-of-post-test was performed. In the case of swing gates, if the structure of the gate requires locating the test load more than 75 mm from a post due to hinge or latch construction, the load should be located as close to the 75 mm maximum distance from the post as is possible. Record the deflection at the point of application of the load.

J-4.6.6.4 Shot bag impact test

Any protective masking or decorative components shall be removed from the glazing material.

To position the impactor at the selected drop height, a traction force shall be applied to raise the impactor such that the axis of the impactor shall be aligned with the suspension cable, with the cable remaining taut. To ensure this, the top and bottom ends of the impactor shall be connected to the release device by a suitable link.

To reduce bag deformation during testing, the bag shall be rotated about the axis of its suspension device before each specimen or sample set, by no less than 30°, and by no more than 90°.

To reduce bag damage during testing, a thin homogeneous or non-woven plastic film no more than 0.13 mm thick or a loosely draped woven cloth towel weighing no more than 0.05 g/cm² shall be permitted to be suspended vertically from its top edge directly in front of the surface of the specimen at a distance no more than 10 mm.

The impactor shall be suspended from an overhead support, located so when at rest it will, at its maximum diameter, be located no more than 12 mm from the surface of the specimen and no more than 50 mm from the impact location of the glazing infill specimen.

Raise the impactor to the required height (460 mm, 1 220 mm or 1 525 mm) and stabilize it. The suspension device shall be taut and the axes of the impactor and cable shall be aligned.

The impactor, stabilized in the launch position in a vertical plane normal to the test specimen, shall be released and falls without initial velocity or axial rotation.

If the glazing does not break, the impactor is to be raised to the next designated height and glazing impacted from the new height. Glazing found to be unbroken and remaining completely in the frame after impact shall be recorded as intact.

J-5 GLASS

The use of glass in railings, the type and thickness of glass required for various support conditions shall be as given below. If designed properly, glass can form a structural component of the railing, or the railing may be designed to have a structural component made from another material such as steel or aluminium, with glass acting as an infill panel only. This places a lesser load on the glass as the major load is taken by the structural component.

NOTE — There are certain situations, particularly under some lighting conditions, where the presence of transparent glass is not readily evident and suitable manifestation should be provided.

J-5.1 Types of Glass

It is recommended to use laminated safety glass for all types of glass railing applications, whether framed or frameless. In the event of glass breakage, the glass panel shall remain in place without falling from its position.

J-5.2 Working of Glass

When toughened glass or heat-strengthened glass is used in laminated glass, it shall conform to accepted standard [C-8(7)] for toughened glass and accepted standard [C-8(8)] for heat-strengthened glass.

J-5.3 Fixing of Glass

Contact between glass and any other hard material (including other glass parts) should be prevented. Rubber gaskets or other glazing materials should be used with frame sections. The frame section should give a minimum of 15 mm edge cover to the glass unless it can be shown by test or calculation that the frame will retain the glass under load while conforming to design requirements.

The frame and/or fastenings should be designed so that they do not tend to distort the glass panel. This is particularly important with bolted connections, which can exert considerable forces on the glass, where particular attention should be paid to the alignment and position of the fasteners in order to avoid unnecessary stresses being developed.

J-5.3.1 Bolt Fixing of Infill Panels

At the bolted connections there should be clamping plates and gaskets on both sides of the glass that provide a minimum of 50 mm diameter cover to the glass. These plates should be not less than 6 mm thick in steel or 10 mm thick in other suitable metals.

Where the length of a glass pane is greater than the span between the bolted connectors, giving rise to a cantilevered portion of the pane, the cantilevered portion should be less than one-quarter of the span between the bolted connectors.

Under the design loads, the railing should be designed so that the relative in-plane movement of the bolted connections in the same panel is not greater than 2 mm.

The glass, framing system and connections of railings and infill panels should be capable of sustaining and safely transmitting the design loads to the supporting structure.

J-5.4 Design of Glass

J-5.4.1 Design of Glass in Full Height Protective Railings

Glass partly or totally below the minimum railing height should be designed to satisfy the appropriate design criteria. Any part of a glass pane below the minimum railing height should sustain the infill loads. Where there is glass at the minimum railing height given in [Table 35](#), the glass should also sustain the line load applied at the design level.

For any specific information on the glass size relative to the glass type, support system, loadings and thickness, available standards or the manufacturer recommendations should be sought.

J-5.4.2 Design of Glass Infill Panels

Infill panels should be designed to satisfy the appropriate design criteria as below:

- a) *Fully framed infill panels* — The deflection of the glass in fully framed infill panels should be as per recommended deflections;
- b) *Two-edge framed infill panels* — The deflection of the glass in two-edge framed infill panels should be as recommended under [J-4.4](#) taking L as the distance between the supports;
- c) *Clipped infill panels* — The clips should be positioned around the periphery of the infill panel, at a maximum spacing of 600 mm. Each clip should be not less than 50 mm in length and should give a minimum depth of cover to the glass of 25 mm;
- d) *Bolt fixing of glass infill panels* — Where glass is supported by bolted connections through holes in the glass, toughened/toughened laminated glass should be used; and
- e) *Position of infill panels relative to the main frame* — Infill panels should be fully contained within the supporting structure. In order not to apply unintended loads to the infill panels, they should have handrails above the glass or attached to the side to which the public have access.

J-5.4.3 Design of Free-Standing Glass Protective Railings

The glass should be designed to satisfy the appropriate design criteria given in [J-4](#). The deflection of the glass should not be more than 25 mm at any point.

J-5.4.3.1 Handrail attachment

Where the railing protects a difference in level greater than 600 mm, a handrail shall be used unless a laminated safety glass construction is used that remains *in-situ*, if a panel fails. Continuous fixing should be used for fixing the handrail to the glass, or individual fixings where calculations or tests demonstrate that component failure will not occur.

Where handrail is not desired, the laminated safety glass should stay in place in the event of any glass panel breakage.

The handrail should be attached to the glass in such a manner that, should a glass panel gets fractured, the handrail,

- a) will remain in position; and
- b) will not fail if the design load is applied across the resulting gap.

NOTE — The above condition (b) can be relaxed where the glass pane is an end pane and protects a difference in level of 600 mm or less, for example at the foot of a flight of stairs. In cases where an end pane protects a difference in level greater than 600 mm, there may normally be some adjacent structure to which the handrail could be attached, thus enabling it to meet condition (b).

J-5.4.4 Base Fixing of Free-Standing Railings

J-5.4.4.1 Point of fixing clamps

The fixing clamps on each side of the glass should be not less than 100 mm to 150 mm and should be made of a suitable metal of minimum thickness 12 mm. The number and design of fixing clamps shall be in accordance with structural requirements.

J-5.4.4.2 Continuous fixing clamps

The fixing clamps on each side of the glass should be not less than 100 mm wide and should be made of a suitable metal of minimum thickness 12 mm. The clamps should be continuous for the entire length of the glass pane and have a maximum bolt spacing of 500 mm.

J-5.4.4.3 Other clamping systems

Different clamping methods could be used, provided they cater for an effective continuous clamping over the length of the glass pane. Where a clamping system that does not rely on bolts through the glass is used, the depth over which the clamping force operates should be not less than 75 mm, unless specific tests have been carried out to prove the integrity of the system and that it meets the design criteria given.

J-5.4.4.5 Structural movement

The structural movement due to live or dead loads or creep of the structure to which a railing pane is clamped should be not greater than 2 mm over the length of any railing panel, if more than two bolts are used to clamp that panel.

J-5.4.4.6 Fixing clamps

The attachment of the fixing clamps to the structure should be capable of withstanding the turning moment induced at the fixing clamps.

J-6 IMPACT RESISTANCE AND CONTAINMENT

J-6.1 General

In addition to resisting the design loads given, glass below the minimum railing height (see [Table 35](#)) should also be able to resist impact forces appropriate to whether the railing indicates a route or protects people from a hazard.

J-6.2 Glass in Full Height Railings

The glass in full height railings should be selected to resist the appropriate design loads given and for its impact performance in accordance with the safety glazing recommendations.

J-6.3 Railing with Glass In-fills or Free-Standing Balustrades

The glass in railings with glass in-fills or free-standing should be selected to resist the appropriate design loads given, and to provide containment, that is, it should meet the recommended impact class without penetration.

J-7 INSTALLATION

The installation of a railing should be supervised by a suitably qualified person or persons, who should ensure that the design assumptions have been effectively implemented. Inspections and investigations should be carried out as necessary to establish the integrity of the materials and the elements of construction used.

J-7.1 Fasteners and Fittings

Fasteners and fittings should be of stainless steel, or should be hot-dip galvanized. Thin zinc plated or cadmium plated coatings should not be used for external exposure without additional protection.

J-8 MAINTENANCE

All glass should be regularly cleaned and fixings should be checked for corrosion and loosening. Maintenance recommendations, and the frequency at which railings should be inspected, are governed by environment, usage, incidence of malicious damage and the protection applied. Protective railings assembled entirely from metal-coated steel and situated in normal environments should not require any maintenance for a very long period. They should, however, be inspected regularly and the residual coating thickness should be measured.

ANNEX K

(Informative)

(Foreword)

CONFORMITY ASSESMENT SHEET FOR GLASS AND GLAZING SYSTEM

K-1 For the purpose of conformity assessment by building regulator for glass and glazing aspects, following sheet shall be used.

Sl No.	Parameters	Details
(1)	(2)	(3)
i) GENERAL		
	a) Building typology	
	b) Location (City, State)	
	c) Orientation	
	d) Climate zone	
	e) Window wall ratio (WWR)	
ii) GLASS		
	a) Component of building (Glass on roof/floor/wall/railing/glass fin)	
	b) Type of Glass:	
	Combination (if needed)	
	c) Tinted/Clear/Coated:	
	if yes, Side (Outer/Inner)	
	Description	
	d) Patterned/Non- patterned:	
	if yes, Side (Outer/Inner)	
	Dimension of Glass (L × B) (mm)	
	e) Thickness and Deflection:	
	Calculated Deflection as per this NBCS	
	Calculated Thickness (mm) as per this NBCS	
	Actual Thickness (mm)	
	Glass specifications as per this NBCS	Yes/No
	f) Safety:	

SI No.	Parameters	Details
(1)	(2)	(3)
	Glass Height from Ground	
	Presence of Residual Protection? (Yes / No)	
	Presence of Window Restrictor for children?	
	Anti-Fallout norms followed? (Yes / No)	
	Safety glass norms followed as per this NBCS	Yes/No
	g) Energy Performance:	
	U-value	
	SHGC/Solar factor	
	Visible light transmittance (VLT)	
	Shading co-efficient	
iii) FRAME		
	a) Thickness	
	b) Dimension	
	c) Frame Material	
	d) Frame supported sides (Glass inside structures can also be considered as framed)	
	e) Deflection of frames	
	f) Face width	
	g) U-value of frame	
	h) Depth of the profile	
	Frame specifications as per this NBCS	Yes/No
iv) GLAZING SYSTEM		
	a) Thickness and Dimension	
	Dimension	
	Thickness	
	b) Energy	
	U-factor	

SI No.	Parameters	Details
(1)	(2)	(3)
	Solar Factor / SHGC	
	Visible Light Transmittance (VLT)	
	Energy efficiency specifications as per this NBCS	Yes/No
	c) Structural parameters	
	Design load and wind load	
	Other loads	
	Air Permeability	
	Deflection	
	Seismic parameters	
	Water tightness (classification)	
	Structural aspects as per this NBCS	Yes/No
	d) Glazing Material	
	Front and back clearance	
	Edge clearance	
	Edge cover	
	Rebate depth	
	e) Fire Resistant Glazing system:	
	If Fire Resistant, Category (E/EW/EI)	
	Duration of resistance to fire (in min)	
	Thickness of glazing system	
	f) Acoustics	
	Airborne Sound Insulation	
	Impact Sound Insulation	
v) THIRD PARTY TEST CERTIFICATES (FOR ALL PRESCRIBED PROPERTIES)		

SI No.	Parameters	Details
(1)	(2)	(3)
	a) Glass	Yes/No
	b) Frames	Yes/No
	c) Glazing systems — Energy performance	Yes/No
	d) Glazing systems — Structural aspects	Yes/No
	e) Glazing systems — Fire resistance (if applicable)	Yes/No
	f) Glazing systems — Acoustics	Yes/No

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Section:

	<i>IS No.</i>	<i>Title</i>
(1)	IS 14900 : 2018	Transparent float glass — Specification (<i>first revision</i>)
	IS 2553 (Part 1) : 2018	Safety glass — Specification: Part 1 Architectural, building and general uses (<i>fourth revision</i>)
(2)	IS 3808 : 1979	Method of test for non-combustibility of building materials (<i>first revision</i>)
(3)	IS 17004 : 2018	Testing methods for processed glass
(4)	IS 19331 : 2025	Tinted float glass — Specification
(5)	IS 2835 : 1987	Specification for flat transparent sheet glass (<i>third revision</i>)
(6)	IS 14900 : 2018	Specification for Transparent float glass (<i>first revision</i>)
(7)	IS 2553 (Part 1) : 2018	Safety glass — Specification: Part 1 Architectural, building and general uses (<i>fourth revision</i>)
(8)	IS 16982 : 2018	Heat strengthened glass — Specification
(9)	IS 17346 : 2020	Insulating glazing unit — Specification
(10)	IS 5437 : 2024	Rolled glass — Patterned, extra clear patterned, wired and wired-patterned glass — Specification (<i>second revision</i>)
(11)	Doc no. CED 13 (27189)	Structural glazing silicone sealants – Specification (<i>under preparation</i>)
(12)	IS 1948 : 2024	Aluminium framed doors, windows and sliders — Specification (<i>first revision</i>)
(13)	SP 32 : 1986	Handbook on functional requirements of industrial buildings (lighting and ventilation)
	SP 41 : 1987	Handbook on functional requirements of buildings (other than industrial buildings)
(14)	IS 18648 : 2024	Classification and performance requirements for doors, windows and sliders — Specification
(15)	IS 875 (Part 3) : 2015	Design loads (other than earthquake) for buildings and structures — Code of practice: Part 3 Wind loads (<i>third revision</i>)
(16)	IS 1893 (Part 1) : 2016	Criteria for earthquake resistant design of structures: Part 1 General provisions and buildings (<i>sixth revision</i>)
(17)	IS 875 (Part 2) : 1987	Code of Practice for design loads (other than earthquake) for buildings and structures: Part 2 Imposed loads (<i>second revision</i>)
(18)	IS 19328 : 2025	Fire resistant glass — Specification
(19)	IS/ISO 834-1 : 1999	Fire-resistance tests — Elements of building construction: Part 1 General requirements
	IS/ISO 834-4 : 2000	Fire-resistance tests — Elements of building construction: Part 4 Specific requirements for load bearing vertical separating elements

	<i>IS No.</i>	<i>Title</i>
	IS/ISO 834-5 : 2000	Fire-resistance tests — Elements of building construction: Part 5 Specific requirements for load bearing horizontal separating elements
	IS/ISO 834-6 : 2000	Fire-resistance tests — Elements of building construction: Part 6 Specific requirements for beams
	IS/ISO 834-7 : 2000	Fire-resistance tests — Elements of building construction: Part 7 Specific requirements for columns
	IS/ISO 834-8 : 2002	Fire-resistance tests — Elements of building construction: Part 8 Specific requirements for non-load bearing vertical separating elements
	IS/ISO 834-9 : 2003	Fire-resistance tests — Elements of building construction: Part 9 Specific requirements for non-load bearing ceiling elements
	IS 17518 (Part 1) : 2022/ ISO 3008-1 : 2019	Fire resistance tests — Door and shutter assemblies: Part 1 General requirements
	IS/ISO 3009 : 2003	Fire-resistance tests — Elements of building construction — Glazed elements
	IS 16945 : 2018	Fire resistance test for glass walls
	IS 16947 : 2018	Fire resistance tests for doors with glass panes, openable glass windows and sliding glass doors
(20)	IS 18190 : 2023	Fire resistance of perimeter fire barrier joint system — Method of tests
(21)	IS 16231 (Part 3) : 2019	Use of glass in buildings — Code of practice: Part 3 Fire and loading (<i>first revision</i>)
	IS 875	Code of Practice for design loads (other than earthquake) for buildings and structures:
	(Part 1) : 2026	Dead loads — Unit weights of building, construction and stored materials (<i>third revision</i>)
	(Part 2) : 1987	Imposed loads (<i>second revision</i>)
(22)	IS 16231 (Part 3) : 2019	Use of glass in buildings — Code of practice: Part 3 Fire and loading (<i>first revision</i>)
(23)	IS 17953 : 2023	uPVC profiles for windows and doors — Specification
(24)	IS 1038 : 1983	Specification for steel doors, windows and ventilators (<i>third revision</i>)
(25)	IS 11050 (Part 1) : 2023/ ISO 717-1 : 2020	Rating of sound insulation in buildings and of building elements: Part 1 Airborne sound insulation (<i>first revision</i>)
(26)	IS 11050 (Part 2) : 2023/ ISO 717-2 : 2020	Rating of sound insulation in buildings and of building elements: Part 2 Impact sound insulation (<i>first revision</i>)
(27)	IS 875	Code of practice for design loads (other than earthquake) for buildings and structures:
	(Part 1) : 2026	Dead loads — Unit weights of building, construction and stored materials (<i>third revision</i>)
	(Part 2) : 1987	Imposed loads (<i>second revision</i>)
	(Part 3) : 2015	Wind loads (<i>third revision</i>)
	(Part 5) : 1987	Special loads and load combinations (<i>second revision</i>)

	<i>IS No.</i>	<i>Title</i>
(28)	IS 733 : 2025	Wrought aluminium and aluminium alloy bars, rods, sections, extruded tubes, hollow sections and profiles for general engineering purposes — Specification (<i>fourth revision</i>)
	IS 738 : 1994	Specification for wrought aluminium alloys, drawn tube for general engineering purposes (<i>third revision</i>)
(29)	IS 6911 : 2026	Stainless steel plate, sheet and strip – Specification (<i>third revision</i>)
(30)	IS 1946 (Part 2) : 2025	Design of post-installed anchorage to concrete — Code of practice
(31)	IS 1946 (Part 5) : 20xx	Installation and on-site inspection of post-installed anchorage to concrete — Code of practice (<i>under preparation</i>)
(32)	IS 18434 : 2023	Field measurement of air permeability and water penetration through installed curtain wall, windows, doors, sliders and skylights under static air pressure difference — Method of test
(33)	IS 18459 : 2024	Water penetration of curtain walls, windows, sliders, doors and skylights by uniform static air pressure difference — Method of test
(34)	IS 18647 : 2024	Water penetration of curtain walls, windows, sliders and doors by dynamic air pressure — Method of test
(35)	IS 18268 : 2023/ ISO 9379 : 2005	Operating forces — Test methods — Doors
(36)	IS 17909 : 2022/ ISO 8274 : 2005	Windows and doors — Resistance to repeated opening and closing — Test method
(37)	IS 18473 : 2024	Wind resistance for curtain walls, windows, sliders, doors and skylights — Method of test
(38)	IS 17911 : 2022/ ISO 15099 : 2003	Thermal performance of windows, doors and shading devices — Detailed calculations
(39)	IS 17920 : 2022/ ISO 10077-2 : 2017	Thermal performance of windows, doors and shutters — Calculation of thermal transmittance — Numerical method for frames
(40)	IS 19006 : 2023/ ISO 8248 : 1985	Windows and door height windows — Mechanical tests
(41)	IS 17518 (Part 1) : 2022/ ISO 3008-1 : 2019	Fire resistance tests — Door and shutter assemblies: Part 1 General requirements
(42)	IS 16945 : 2018	Fire resistance test for glass walls
(43)	IS 16947 : 2018	Fire resistance tests for doors with glass panes, openable glass windows and sliding glass doors
(44)	IS 16978	Glass in building — Forced-entry security glazing: Part 1 Test and classification by repetitive ball drop (<i>first revision</i>)
	(Part 1) : 2022/ ISO 16936-1 : 2020	Test and classification by repetitive ball drop (<i>first revision</i>)
	(Part 2) : 2018/ ISO 16936-2 : 2005	Test and classification by repetitive impact of a hammer and axe at room temperature
	(Part 3) : 2018/ ISO 16936-3 : 2005	Test and classification by manual attack
	(Part 4) : 2018/ ISO 16936-4 : 2005	Test and classification by pendulum impact under thermally and fire stressed conditions
(45)	IS 18518 : 2023	Bullet resistant security glass — Quality and performance requirements

	<i>IS No.</i>	<i>Title</i>
(46)	IS 16978	Glass in building — Forced-entry security glazing:
	(Part 1) : 2022/ ISO 16936-1 : 2020	Test and classification by repetitive ball drop (<i>first revision</i>)
	(Part 2) : 2018/ ISO 16936-2 : 2005	Test and classification by repetitive impact of a hammer and axe at room temperature
(47)	IS 18472 : 2024	Air permeability for curtain walls, windows, sliders, doors and skylights — Method of test
(48)	IS 875	Code of Practice for design loads (other than earthquake) for buildings and structures:
	(Part 1) : 2026	Dead loads — Unit weights of building, construction and stored materials (<i>third revision</i>)
	(Part 2) : 1987	Imposed loads (<i>second revision</i>)
	(Part 3) : 2015	Wind loads (<i>third revision</i>)
	(Part 4) : 2021	Snow loads (<i>third revision</i>)
	(Part 5) : 1987	Special loads and load combinations (<i>second revision</i>)
	IS 1893 (Part 1) : 2016	Criteria for earthquake resistant design of structures: Part 1 General provisions and buildings (<i>sixth revision</i>)
	IS 16700 : 2023	Criteria for structural safety of tall concrete buildings (<i>first revision</i>)

INDEX OF TERMS

(Lists the terms appearing in the Part/Section of the NBCS)

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